

ISC Class XII Mathematics

Chapter 7 Integrals

595 Solved Problems – Numericals & Proofs

Step-by-Step ISC Board Solutions

Topics Covered

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| I. 7.1 – Standard Integrals (Power Rule) | II. 7.2 – Simplification Before Integrating |
| III. 7.3 – Substitution: Linear/Algebraic Forms | IV. 7.4 – Substitution: Algebraic Powers |
| V. 7.5 – Substitution: Composite Trig Forms | VI. 7.6 – Substitution: Trigonometric Functions |
| VII. 7.7 – Standard Forms (Difference/Sum of Squares) | VIII. 7.8 – Standard Forms (Completing the Square) |
| IX. 7.9 – Rational Trig Functions (Weierstrass Sub.) | X. 7.10 – Integration by Partial Fractions |
| XI. 7.11 – Integration by Parts | XII. 7.12 – By Parts: Special Exponential Form |
| XIII. 7.13 – Integrals Involving $\sqrt{a^2 - x^2}$ | XIV. 7.14 – Integrals Involving $\sqrt{ax^2 + bx + c}$ |
| XV. 7.15 – Definite Integrals (Fundamental Theorem) | XVI. 7.16 – Definite Integrals via Substitution |
| XVII. 7.17 – Properties of Definite Integrals | |

Compiled Answer Key – Step-by-step ISC board-style solutions

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Exercise 7.1 – Standard Integrals (Power Rule and Direct Forms)

1	(see parts below)	1	Numerical	7.1
(i)	Evaluate the integral: $\int x^5 dx$	1	Numerical	7.1
(ii)	Evaluate the integral: $\int \frac{1}{x^3} dx$	1	Numerical	7.1
(iii)	Evaluate the integral: $\int \frac{1}{\sqrt{y}} dy$	1	Numerical	7.1
2	(see parts below)	1	Numerical	7.1
(i)	Evaluate the integral: $\int \frac{1}{\sqrt[3]{x}} dx$	1	Numerical	7.1
(ii)	Evaluate the integral: $\int \sin^2 x \csc^2 x dx$	2	Numerical	7.1
(iii)	Evaluate the integral: $\int \sin x \sec^2 x dx$	2	Numerical	7.1
3	(see parts below)	3	Numerical	7.1
(i)	Evaluate the integral: $\int 5^x dx$	3	Numerical	7.1
(ii)	Evaluate the integral: $\int 7^x e^x dx$	3	Numerical	7.1
(iii)	Evaluate the integral: $\int \frac{4^x}{9^x} dx$	3	Numerical	7.1
4	(see parts below)	2	Numerical	7.1
(i)	Evaluate the integral: $\int \frac{\sin x}{1 - \sin^2 x} dx$	2	Numerical	7.1
(ii)	Evaluate the integral: $\int \frac{\cos 2x + 2 \sin^2 x}{\cos^2 x} dx$	2	Numerical	7.1
(iii)	Evaluate the integral: $\int \sqrt{\frac{1}{2}(1 + \cos 2x)} dx$	2	Numerical	7.1
5	(see parts below)	4	Numerical	7.1
(i)	Evaluate the integral: $\int e^{2 \log x} x^{-3} dx$	4	Numerical	7.1
(ii)	Evaluate the integral: $\int 5^{2 \log_5 x} dx$	4	Numerical	7.1
(iii)	Evaluate the integral: $\int 2^{\log_4 x} dx$	4	Numerical	7.1

Exercise 7.2 – Simplification Before Integrating

1	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int (ax^2 + bx + c) dx$	1	Numerical	7.2
(ii)	Evaluate $\int (x^{2/3} + 1) dx$	1	Numerical	7.2
2	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int (\sqrt{3x} + \frac{1}{\sqrt{x}}) dx$	1	Numerical	7.2
(ii)	Evaluate ...	1	Numerical	7.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
3	(see parts below)	2	Numerical	7.2
(i)	Evaluate $\int \frac{2 \cos x}{3 \sin^2 x} dx$	2	Numerical	7.2
(ii)	Evaluate $\int (2x^2 - 3 \sin x + 5\sqrt{x}) dx$	1	Numerical	7.2
4	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int (2x - 3 \cos x + e^x) dx$	1	Numerical	7.2
(ii)	Evaluate $\int x^2(1 - \frac{1}{x^2}) dx$	1	Numerical	7.2
5	(see parts below)	4	Numerical	7.2
(i)	Evaluate $\int (x^{3/2} + 2e^x - \frac{1}{x}) dx$	4	Numerical	7.2
(ii)	Evaluate $\int (\sqrt{x} + \frac{1}{\sqrt{x}})^2 dx$	1	Numerical	7.2
6	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int (1 - x)\sqrt{x} dx$	1	Numerical	7.2
(ii)	Evaluate $\int \sqrt{x}(3x^2 + 2x + 3) dx$	1	Numerical	7.2
7	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int \frac{x^3+5x^2+4x+1}{x^2} dx$	1	Numerical	7.2
(ii)	Evaluate $\int \frac{x^3+3x+4}{\sqrt{x}} dx$	1	Numerical	7.2
8	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int x^2(2x - \frac{1}{x})^2 dx$	1	Numerical	7.2
(ii)	Evaluate $\int (x + \sqrt{x})^3 dx$	1	Numerical	7.2
9	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int \frac{(x+a)^2}{\sqrt{x}} dx$	1	Numerical	7.2
(ii)	Evaluate $\int \frac{x^3-x^2+x-1}{x-1} dx$	1	Numerical	7.2
10	(see parts below)	2	Numerical	7.2
(i)	Evaluate $\int \frac{1-\sin x}{\cos^2 x} dx$	2	Numerical	7.2
(ii)	Evaluate ...	4	Numerical	7.2
11	(see parts below)	2	Numerical	7.2
(i)	Evaluate $\int \sec x(\sec x + \tan x) dx$	2	Numerical	7.2
(ii)	Evaluate $\int \frac{\sec^2 x}{\csc^2 x} dx$	2	Numerical	7.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
12	(see parts below)	2	Numerical	7.2
(i)	Evaluate ...	2	Numerical	7.2
(ii)	Evaluate $\int \frac{3-5\sin x}{\cos^2 x} dx$	2	Numerical	7.2
13	(see parts below)	5	Numerical	7.2
(i)	Evaluate $\int \frac{1}{1+\cos x} dx$	5	Numerical	7.2
(ii)	Evaluate $\int \frac{1}{1-\sin x} dx$	5	Numerical	7.2
14	(see parts below)	5	Numerical	7.2
(i)	Evaluate $\int \frac{1}{1+\csc x} dx$	5	Numerical	7.2
(ii)	Evaluate $\int \frac{\sin^2 x}{1+\cos x} dx$	2	Numerical	7.2
15	(see parts below)	2	Numerical	7.2
(i)	Evaluate $\int (\sqrt{x} - \sin \frac{x}{2} \cos \frac{x}{2} + 5) dx$	2	Numerical	7.2
(ii)	Evaluate $\int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx$	2	Numerical	7.2
16	(see parts below)	5	Numerical	7.2
(i)	Evaluate $\int \frac{\sec x}{\sec x + \tan x} dx$	5	Numerical	7.2
(ii)	Evaluate $\int \frac{\csc x}{\csc x - \cot x} dx$	5	Numerical	7.2
17	(see parts below)	1	Numerical	7.2
(i)	Evaluate $\int \frac{x^6+1}{x^2+1} dx$	1	Numerical	7.2
(ii)	Evaluate $\int \frac{6\sin^3 x + 5\cos^3 x}{\sin^2 x \cos^2 x} dx$	2	Numerical	7.2
18	(see parts below)	2	Numerical	7.2
(i)	Evaluate $\int \sqrt{1+\cos 2x} dx, \quad 0 < x < \frac{\pi}{2}$	2	Numerical	7.2
(ii)	Evaluate $\int \sqrt{1-\cos 2x} dx, \quad 0 < x < \pi$	2	Numerical	7.2
19	(see parts below)	6	Numerical	7.2
(i)	Evaluate $\int \cot^{-1}\left(\frac{\sin 2x}{1-\cos 2x}\right) dx$	6	Numerical	7.2
(ii)	Evaluate $\int \tan^{-1}\left(\sqrt{\frac{1-\cos 2x}{1+\cos 2x}}\right) dx$	6	Numerical	7.2
20	(see parts below)	6	Numerical	7.2
(i)	Evaluate $\int \tan^{-1}(\csc x - \cot x) dx$	6	Numerical	7.2

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Evaluate $\int \cos^{-1}\left(\frac{1-\tan^2 x}{1+\tan^2 x}\right) dx$	6	Numerical	7.2
21	If $f'(x) = 4x^3 - 6$ and $f(0) = 3$, find $f(x)$.	7	Numerical	7.2

Exercise 7.3 – Integration by Substitution: Linear and Algebraic Forms

1	(see parts below)	12a	Numerical	7.3
(i)	Find all the anti-derivatives of $(ax + b)^3$.	12a	Numerical	7.3
(ii)	Find all the anti-derivatives of $4e^{3x} + 1$.	12c	Numerical	7.3
2	(see parts below)	12b	Numerical	7.3
(i)	Find all the anti-derivatives of $\frac{1}{\sqrt{7-2x}}$.	12b	Numerical	7.3
(ii)	Find all the anti-derivatives of $\sqrt{ax + b}$.	12b	Numerical	7.3
3	(see parts below)	2	Numerical	7.3
(i)	Find all the anti-derivatives of $\tan^2(2x - 3)$.	2	Numerical	7.3
(ii)	Find all the anti-derivatives of $5 \csc^2(2 - 7x)$.	12a	Numerical	7.3
4	(see parts below)	12b	Numerical	7.3
(i)	Find all the anti-derivatives of $\sqrt{e^x}$.	12b	Numerical	7.3
(ii)	Find all the anti-derivatives of $\csc(3x + 2) \cot(3x + 2)$.	12a	Numerical	7.3
5	(see parts below)	12c	Numerical	7.3
(i)	Evaluate $\int e^{2x+3} dx$.	12c	Numerical	7.3
(ii)	Evaluate $\int (e^{3a \log x} + e^{3x \log a}) dx$.	4	Numerical	7.3
6	(see parts below)	2	Numerical	7.3
(i)	Evaluate $\int \sqrt{1 + \cos x} dx, \quad 0 < x < \pi$.	2	Numerical	7.3
(ii)	Evaluate $\int \sqrt{1 + \sin x} dx, \quad 0 < x < \pi$.	2	Numerical	7.3
7	(see parts below)	12a	Numerical	7.3
(i)	Evaluate $\int \sec^2(7 - x) dx$.	12a	Numerical	7.3
(ii)	Evaluate $\int a^{3x+5} dx, \quad a > 0$.	12a	Numerical	7.3
8	(see parts below)	2	Numerical	7.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int \frac{1-\tan^2 x}{1+\tan^2 x} dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \sec^2 x \csc^2 x dx$.	2	Numerical	7.3
9	(see parts below)	1	Numerical	7.3
(i)	Evaluate $\int \frac{x}{2x-3} dx$.	1	Numerical	7.3
(ii)	Evaluate $\int \frac{x+2}{3x^2+5x-2} dx$.	1	Numerical	7.3
10	(see parts below)	1	Numerical	7.3
(i)	Evaluate $\int \frac{x}{(x+1)^2} dx$.	1	Numerical	7.3
(ii)	Evaluate $\int \frac{x^2+2}{x+1} dx$.	1	Numerical	7.3
11	(see parts below)	1	Numerical	7.3
(i)	Evaluate $\int \frac{x^3}{2x+1} dx$.	1	Numerical	7.3
(ii)	Evaluate $\int \frac{x^3+4x^2-3x-2}{x+2} dx$.	1	Numerical	7.3
12	(see parts below)	5	Numerical	7.3
(i)	Evaluate $\int \frac{1}{\sqrt{x+1}+\sqrt{x+2}} dx$.	5	Numerical	7.3
(ii)	Evaluate $\int \frac{1}{\sqrt{2x+3}-\sqrt{2x}} dx$.	5	Numerical	7.3
13	(see parts below)	12b	Numerical	7.3
(i)	Evaluate $\int \frac{x+1}{\sqrt{2x-1}} dx$.	12b	Numerical	7.3
(ii)	Evaluate $\int x\sqrt{3x-2} dx$.	12b	Numerical	7.3
14	(see parts below)	2	Numerical	7.3
(i)	Evaluate $\int \cos^2 x dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \sin^4 x dx$.	2	Numerical	7.3
15	(see parts below)	2	Numerical	7.3
(i)	Evaluate $\int \sin^2(2x+5) dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \cos^4 2x dx$.	2	Numerical	7.3
16	(see parts below)	2	Numerical	7.3
(i)	Evaluate $\int \frac{1}{1-\sin \frac{x}{2}} dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \sin x \sqrt{1-\cos 2x} dx$.	2	Numerical	7.3
17	(see parts below)	2	Numerical	7.3

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int \cos 3x \cos 5x \, dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \sin 2x \cos 3x \, dx$.	2	Numerical	7.3
18	(see parts below)	2	Numerical	7.3
(i)	Evaluate $\int \frac{\sin 4x}{\cos x} \, dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \sin x \sin 2x \sin 3x \, dx$.	2	Numerical	7.3
19	(see parts below)	2	Numerical	7.3
(i)	Evaluate $\int \sin^6 x \, dx$.	2	Numerical	7.3
(ii)	Evaluate $\int \sin^4 x \cos^2 x \, dx$.	2	Numerical	7.3
20	Evaluate $\int \cos mx \cos nx \, dx$, where m, n are positive integers ...	2	Numerical	7.3

Exercise 7.4 – Integration by Substitution: Algebraic Powers

1	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int x^3(1+x^4)^3 \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int (2x+4)\sqrt{x^2+4x+3} \, dx$	12b	Numerical	7.4
2	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int x(x^2+3)^{3/2} \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{4x+1}{\sqrt{2x^2+x-7}} \, dx$	12b	Numerical	7.4
3	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \frac{3x+1}{(3x^2+2x-5)^3} \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{2x}{\sqrt[3]{x^2+1}} \, dx$	12b	Numerical	7.4
4	(see parts below)	12c	Numerical	7.4
(i)	Evaluate: $\int e^x(a+be^x)^6 \, dx$	12c	Numerical	7.4
(ii)	Evaluate: $\int \frac{(\log x)^2}{x} \, dx$	12a	Numerical	7.4
(iii)	Evaluate: $\int \frac{\log x}{x} \, dx$	12a	Numerical	7.4
(iv)	Evaluate: $\int \frac{x^2}{1+x^3} \, dx$	12a	Numerical	7.4
5	(see parts below)	12b	Numerical	7.4
(i)	Evaluate: $\int \frac{\sqrt{3+\log x}}{x} \, dx$	12b	Numerical	7.4
(ii)	Evaluate: $\int \frac{3(x+1)(x+\log x)^2}{x} \, dx$	12a	Numerical	7.4
6	(see parts below)	12b	Numerical	7.4
(i)	Evaluate: $\int \frac{\sqrt{\tan^{-1} x}}{1+x^2} \, dx$	12b	Numerical	7.4
(ii)	Evaluate: $\int \frac{(\sin^{-1} 2x)^3}{\sqrt{1-4x^2}} \, dx$	12b	Numerical	7.4
7	(see parts below)	12a	Numerical	7.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate: $\int \sin x \cos^5 x \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \cos^3(ax+b) \sin(ax+b) \, dx$	12a	Numerical	7.4
8	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \sec^3 x \tan x \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \cot^3 x \csc^2 x \, dx$	12a	Numerical	7.4
9	(see parts below)	12b	Numerical	7.4
(i)	Evaluate: $\int \sqrt{\tan x} (1 + \tan^2 x) \, dx$	12b	Numerical	7.4
(ii)	Evaluate: $\int \frac{\sin x}{1 + \cos x} \, dx$	12e	Numerical	7.4
(iii)	Evaluate: $\int \frac{5-3x^2}{6-x^3+5x} \, dx$	12a	Numerical	7.4
(iv)	Evaluate: $\int x^2 e^{x^3} \, dx$	12c	Numerical	7.4
10	(see parts below)	12b	Numerical	7.4
(i)	Evaluate: $\int \frac{\cot x}{\sqrt{\sin x}} \, dx$	12b	Numerical	7.4
(ii)	Evaluate: $\int \frac{\tan x}{\sqrt{\cos x}} \, dx$	12b	Numerical	7.4
11	(see parts below)	12e	Numerical	7.4
(i)	Evaluate: $\int \frac{\sin x}{\sqrt{3+2\cos x}} \, dx$	12e	Numerical	7.4
(ii)	Evaluate: $\int \frac{1+\sin x}{\sqrt{x-\cos x}} \, dx$	12e	Numerical	7.4
12	(see parts below)	12e	Numerical	7.4
(i)	Evaluate: $\int \frac{\sin x}{(1+\cos x)^2} \, dx$	12e	Numerical	7.4
(ii)	Evaluate: $\int \frac{(3\tan^2 x + 2)\sec^2 x}{(\tan^3 x + 2\tan x + 5)^2} \, dx$	12a	Numerical	7.4
13	(see parts below)	12c	Numerical	7.4
(i)	Evaluate: $\int (e^{3x} + 1)^2 e^{3x} \, dx$	12c	Numerical	7.4
(ii)	Evaluate: $\int \frac{e^{4x}}{(1+e^{4x})^2} \, dx$	12c	Numerical	7.4
14	(see parts below)	2	Numerical	7.4
(i)	Evaluate: $\int \tan^4 x \, dx$	2	Numerical	7.4
(ii)	Evaluate: $\int \cot^4 x \, dx$	2	Numerical	7.4
15	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \frac{1+\tan x}{x+\log \sec x} \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{1+\sin 2x}{x+\sin^2 x} \, dx$	12a	Numerical	7.4
16	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \frac{\tan x \sec^2 x}{1-\tan^2 x} \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{x^5}{1+x^6} \, dx$	12a	Numerical	7.4
17	(see parts below)	12e	Numerical	7.4
(i)	Evaluate: $\int \frac{2\cos x - 3\sin x}{6\cos x + 4\sin x} \, dx$	12e	Numerical	7.4
(ii)	Evaluate: $\int \frac{\sin 2x - \sin x}{\sin^2 x + \cos x + 7} \, dx$	12a	Numerical	7.4
18	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \frac{x+\tan x}{x^2-2\log \cos x} \, dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{1}{x \log x} \, dx$	12a	Numerical	7.4

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
19	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \frac{2x^2-1}{x(x^2-\log x)} dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{e^{2x}+1}{e^{2x}-1} dx$	12c	Numerical	7.4
(iii)	Evaluate: $\int \frac{e^{4x}-1}{e^{4x}+1} dx$	12c	Numerical	7.4
20	(see parts below)	12c	Numerical	7.4
(i)	Evaluate: $\int \frac{1}{1+e^{-x}} dx$	12c	Numerical	7.4
(ii)	Evaluate: $\int \frac{\tan x}{a+b \tan^2 x} dx$	12a	Numerical	7.4
21	(see parts below)	12a	Numerical	7.4
(i)	Evaluate: $\int \frac{1-\cot x}{1+\cot x} dx$	12a	Numerical	7.4
(ii)	Evaluate: $\int \frac{\cos 2x}{(\cos x + \sin x)^2} dx$	2	Numerical	7.4

Exercise 7.5 – Integration by Substitution: Composite Trigonometric Forms

1	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int 2x \sin(x^2 + 1) dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int x^3 \cos(x^4) dx$.	12a	Numerical	7.5
2	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$.	12b	Numerical	7.5
(ii)	Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.	12b	Numerical	7.5
(iii)	Evaluate $\int \frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx$.	12b	Numerical	7.5
(iv)	Evaluate $\int \frac{\csc^2 \sqrt{x}}{\sqrt{x}} dx$.	12b	Numerical	7.5
3	(see parts below)	12c	Numerical	7.5
(i)	Evaluate $\int \sin x \cdot e^{\cos x} dx$.	12c	Numerical	7.5
(ii)	Evaluate $\int \frac{e^{2x}}{1+e^x} dx$.	12c	Numerical	7.5
4	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \sin x \sin(\cos x) dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int x^3 \sec^2(x^4 + 3) dx$.	12a	Numerical	7.5
5	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \frac{1}{x^2} \tan^2\left(\frac{1}{x}\right) dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \frac{1}{x^2} \sin^2\left(\frac{1}{x}\right) dx$.	12a	Numerical	7.5
(iii)	Evaluate $\int 2x \sec^3(x^2 + 5) \tan(x^2 + 5) dx$.	12a	Numerical	7.5
6	(see parts below)	12a	Numerical	7.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int \frac{\sin(\log x)}{x} dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \frac{\csc^2(\log x)}{x} dx$.	12a	Numerical	7.5
7	(see parts below)	12c	Numerical	7.5
(i)	Evaluate $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$.	12c	Numerical	7.5
(ii)	Evaluate $\int \frac{\sin(2 \tan^{-1} x)}{1+x^2} dx$.	12a	Numerical	7.5
8	(see parts below)	12c	Numerical	7.5
(i)	Evaluate $\int \frac{e^{-1/x} + 1}{x^2} dx$.	12c	Numerical	7.5
(ii)	Evaluate $\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$.	12c	Numerical	7.5
9	(see parts below)	12c	Numerical	7.5
(i)	Evaluate $\int e^x \csc^2(e^x) dx$.	12c	Numerical	7.5
(ii)	Evaluate $\int \frac{(x+1)e^x}{\cot^2(xe^x)} dx$.	12c	Numerical	7.5
10	(see parts below)	12c	Numerical	7.5
(i)	Evaluate $\int x^2 e^{x^3} \sin(e^{x^3}) dx$.	12c	Numerical	7.5
(ii)	Evaluate $\int \frac{\sin^2(\log x)}{x} dx$.	12a	Numerical	7.5
(iii)	Evaluate $\int \frac{\sqrt{\tan x}}{\sin^2 x} dx$.	12b	Numerical	7.5
11	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int x \sin^3(x^2) \cos(x^2) dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \tan^3 2x \sec 2x dx$.	12a	Numerical	7.5
12	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \sin^3(2x + 1) dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \sin^3 x \cos^4 x dx$.	12a	Numerical	7.5
13	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \sin^7 x dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \sin x \cot^3 x dx$.	12a	Numerical	7.5
14	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int \frac{\cos^3 x}{\sqrt{\sin x}} dx$.	12b	Numerical	7.5
(ii)	Evaluate $\int \frac{\sin x \cos^3 x}{1+\cos^2 x} dx$.	12a	Numerical	7.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
15	(see parts below)	12e	Numerical	7.5
(i)	Evaluate $\int \csc x \log(\csc x - \cot x) dx$.	12e	Numerical	7.5
(ii)	Evaluate $\int \frac{\log(\tan \frac{x}{2})}{\sin x} dx$.	12a	Numerical	7.5
16	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \sec^4 x dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \tan^2 x \sec^4 x dx$.	12a	Numerical	7.5
17	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int \frac{\tan^5 \sqrt{x} \sec^2 \sqrt{x}}{\sqrt{x}} dx$.	12b	Numerical	7.5
(ii)	Evaluate $\int \frac{\cos x - \sin x}{\sqrt{1 + \sin 2x}} dx$.	12e	Numerical	7.5
(iii)	Evaluate ...	12e	Numerical	7.5
18	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int \frac{x}{\sqrt{x+4}} dx$.	12b	Numerical	7.5
(ii)	Evaluate $\int x\sqrt{x+2} dx$.	12b	Numerical	7.5
19	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \frac{x}{(x^2+1)^2} dx$.	12a	Numerical	7.5
(ii)	Evaluate $\int \frac{x^2}{(4+x)^{3/2}} dx$.	12a	Numerical	7.5
20	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int 4x^3 \sqrt{5-x^2} dx$.	12b	Numerical	7.5
(ii)	Evaluate $\int \frac{dx}{x-\sqrt{x}}$.	12b	Numerical	7.5
21	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int \frac{x}{1+\sqrt{x}} dx$.	12b	Numerical	7.5
(ii)	Evaluate $\int \frac{x^{1/2}}{1+x^{3/4}} dx$.	12a	Numerical	7.5
22	(see parts below)	12b	Numerical	7.5
(i)	Evaluate $\int \frac{dx}{\sqrt{x+1} + \sqrt[3]{x+1}}$.	12b	Numerical	7.5
(ii)	Evaluate $\int \frac{dx}{\sqrt{1+\sqrt{x}}}$.	12b	Numerical	7.5
23	(see parts below)	12a	Numerical	7.5
(i)	Evaluate $\int \frac{\sin 2x}{(a^2+b^2 \sin^2 x)^2} dx$.	12a	Numerical	7.5

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Evaluate $\int \frac{\sqrt{1+x^2}}{x^4} dx$.	12b	Numerical	7.5
(iii)	Evaluate $\int \frac{1}{x^2\sqrt{1+x^2}} dx$.	12b	Numerical	7.5
24	If $\frac{d}{dx}(F(x)) = \frac{\sec^4 x}{\csc^4 x}$ and ...	12a	Numerical	7.5
25	If $\int x^5 \cos(x^6) dx = k \sin(x^6) + C$, find the value of k .	12a	Numerical	7.5

Exercise 7.6 – Integration by Substitution: Trigonometric Functions

1	(see parts below)	12a	Numerical	7.6
(i)	Evaluate $\int \tan(2-3x) dx$.	12a	Numerical	7.6
(ii)	Evaluate $\int \cot(7x-4) dx$.	12a	Numerical	7.6
2	(see parts below)	2	Numerical	7.6
(i)	Evaluate $\int \frac{1+\tan^2 x}{2 \tan x} dx$.	2	Numerical	7.6
(ii)	Evaluate $\int \frac{1}{\sin x \cos x} dx$.	2	Numerical	7.6
3	(see parts below)	2	Numerical	7.6
(i)	Evaluate $\int \frac{\cos 2x}{\sin x} dx$.	2	Numerical	7.6
(ii)	Evaluate $\int \frac{\sin 2x}{\sin 4x} dx$.	2	Numerical	7.6
4	(see parts below)	12a	Numerical	7.6
(i)	Evaluate $\int \frac{\sin x}{\sin(x-\alpha)} dx$.	12a	Numerical	7.6
(ii)	Evaluate $\int \frac{\cos x}{\cos(x+\alpha)} dx$.	12a	Numerical	7.6
5	(see parts below)	2	Numerical	7.6
(i)	Evaluate $\int \frac{dx}{\sin(x-\alpha) \sin(x-\beta)}$.	2	Numerical	7.6
(ii)	Evaluate $\int \frac{dx}{\cos(x-a) \cos(x-b)}$.	2	Numerical	7.6
6	(see parts below)	2	Numerical	7.6
(i)	Evaluate $\int \frac{dx}{\sin(x-a) \cos(x-b)}$.	2	Numerical	7.6
(ii)	Evaluate $\int \tan^3 x dx$. (ISC 2016)	12a	Numerical	7.6
7	(see parts below)	2	Numerical	7.6
(i)	Evaluate $\int \frac{\sin 2x}{\sin 5x \sin 3x} dx$.	2	Numerical	7.6
(ii)	Evaluate $\int \frac{1}{\sin x \cos^2 x} dx$.	2	Numerical	7.6

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
8	(see parts below)	2	Numerical	7.6
(i)	Evaluate $\int \frac{dx}{\sqrt{1-\sin x}}$.	2	Numerical	7.6
(ii)	Evaluate $\int \frac{\sin x + \cos x}{1 + \sin 2x} dx$.	2	Numerical	7.6
9	If ...	12b	Numerical	7.6

Exercise 7.7 – Integrals Reducible to Standard Forms (Difference/Sum of Squares)

1	Evaluate the following integrals:	1	Numerical	7.7
(i)	$\int \frac{1}{x^2-16} dx$	1	Numerical	7.7
(ii)	$\int \frac{1}{9-x^2} dx$	1	Numerical	7.7
(iii)	$\int \frac{dx}{\sqrt{1-x^2}}$	1	Numerical	7.7
(iv)	$\int \frac{dx}{x^2+16}$	1	Numerical	7.7
2	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{dx}{\sqrt{9-4x^2}}$	12b	Numerical	7.7
(ii)	$\int \frac{dx}{25+9x^2}$	12b	Numerical	7.7
3	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{dx}{x\sqrt{9x^2-16}}$	12b	Numerical	7.7
(ii)	$\int \frac{dx}{\sqrt{x^2+5}}$	12b	Numerical	7.7
4	Evaluate the following integrals:	1	Numerical	7.7
(i)	$\int \frac{dx}{9x^2-1}$	1	Numerical	7.7
(ii)	$\int \frac{dx}{32-2x^2}$	1	Numerical	7.7
5	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{dx}{\sqrt{15-8x^2}}$	12b	Numerical	7.7
(ii)	$\int \frac{dx}{\sqrt{1+4x^2}}$	12b	Numerical	7.7
6	Evaluate the following integrals:	1	Numerical	7.7
(i)	$\int \frac{2y^2}{y^2+4} dy$ (ISC 2015)	1	Numerical	7.7
(ii)	$\int \frac{x^4+1}{x^2+1} dx$	1	Numerical	7.7
7	Evaluate the following integrals:	12a	Numerical	7.7
(i)	$\int \frac{x}{x^4-1} dx$ (Exemplar)	12a	Numerical	7.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	$\int \frac{3x^2}{1+x^6} dx$ (NCERT)	12a	Numerical	7.7
8	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{x^2}{\sqrt{x^6+a^6}} dx$ (NCERT)	12b	Numerical	7.7
(ii)	$\int \frac{x^3}{\sqrt{1-x^8}} dx$ (NCERT)	12b	Numerical	7.7
9	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{\sqrt{x}}{\sqrt{a^3-x^3}} dx$ (Exemplar)	12b	Numerical	7.7
(ii)	$\int \frac{x(1+x^2)}{1+x^4} dx$ (ISC 2012)	12b	Numerical	7.7
10	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{\cos x}{\sqrt{4-\sin^2 x}} dx$ (NCERT)	12b	Numerical	7.7
(ii)	$\int \frac{\sec^2 x}{\sqrt{\tan^2 x+4}} dx$ (NCERT)	12b	Numerical	7.7
11	Evaluate the following integrals:	12a	Numerical	7.7
(i)	$\int \frac{\sin x}{16-9\cos^2 x} dx$	12a	Numerical	7.7
(ii)	$\int \frac{\tan^2 x \sec^2 x}{1-\tan^6 x} dx$	12a	Numerical	7.7
12	Evaluate the following integrals:	2	Numerical	7.7
(i)	$\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$	2	Numerical	7.7
(ii)	$\int \frac{dx}{2\sin^2 x + 5\cos^2 x}$	2	Numerical	7.7
13	Evaluate the following integrals:	2	Numerical	7.7
(i)	$\int \frac{\cos x}{\cos 3x} dx$	2	Numerical	7.7
(ii)	$\int \frac{dx}{2-3\cos 2x}$	2	Numerical	7.7
14	Evaluate the following integrals:	5	Numerical	7.7
(i)	$\int \frac{dx}{e^x - e^{-x}}$	5	Numerical	7.7
(ii)	$\int \sqrt{\frac{1+x}{1-x}} dx$ (Exemplar)	5	Numerical	7.7
15	Evaluate the following integrals:	12b	Numerical	7.7
(i)	$\int \frac{x-1}{\sqrt{x^2-1}} dx$ (NCERT)	12b	Numerical	7.7
(ii)	$\int \frac{x+2}{\sqrt{x^2-1}} dx$ (NCERT)	12b	Numerical	7.7
16	Evaluate the following integrals:	5	Numerical	7.7
(i)	$\int \sqrt{\frac{a+x}{a-x}} dx$ (Exemplar)	5	Numerical	7.7

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	$\int \frac{\sqrt{x^2-a^2}}{x} dx$	5	Numerical	7.7

Exercise 7.8 – Integrals Reducible to Standard Forms (Completing the Square)

1	(see parts below)	12a	Numerical	7.8
(i)	Evaluate $\int \frac{dx}{x^2+4x+8}$.	12a	Numerical	7.8
(ii)	Evaluate $\int \frac{dx}{x^2+2x+2}$.	12a	Numerical	7.8
2	(see parts below)	12a	Numerical	7.8
(i)	Evaluate $\int \frac{dx}{x^2-6x+13}$.	12a	Numerical	7.8
(ii)	Evaluate $\int \frac{x+3}{x^2-2x-5} dx$.	12a	Numerical	7.8
3	(see parts below)	12a	Numerical	7.8
(i)	Evaluate $\int \frac{x+2}{2x^2+6x+5} dx$.	12a	Numerical	7.8
(ii)	Evaluate $\int \frac{2x+1}{18-4x-x^2} dx$.	12a	Numerical	7.8
4	(see parts below)	12c	Numerical	7.8
(i)	Evaluate $\int \frac{e^x}{2e^{2x}+3e^x+5} dx$.	12c	Numerical	7.8
(ii)	Evaluate $\int \frac{\cos x}{6+4 \sin x - \cos^2 x} dx$.	12e	Numerical	7.8
5	(see parts below)	12b	Numerical	7.8
(i)	Evaluate $\int \frac{dx}{\sqrt{4x-x^2}}$.	12b	Numerical	7.8
(ii)	Evaluate $\int \frac{dx}{\sqrt{7-6x-x^2}}$.	12b	Numerical	7.8
6	(see parts below)	12b	Numerical	7.8
(i)	Evaluate $\int \frac{dx}{\sqrt{(x-1)(x-2)}}$.	12b	Numerical	7.8
(ii)	Evaluate $\int \frac{dx}{\sqrt{8+3x-x^2}}$.	12b	Numerical	7.8
7	(see parts below)	12b	Numerical	7.8
(i)	Evaluate $\int \frac{dx}{\sqrt{5-4x-2x^2}}$.	12b	Numerical	7.8
(ii)	Evaluate $\int \frac{dx}{\sqrt{5x-4x^2}}$.	12b	Numerical	7.8
8	(see parts below)	12b	Numerical	7.8
(i)	Evaluate $\int \frac{x-1}{\sqrt{x^2-x}} dx$.	12b	Numerical	7.8
(ii)	Evaluate $\int \frac{5x+3}{\sqrt{x^2+4x+10}} dx$.	12b	Numerical	7.8
9	(see parts below)	12b	Numerical	7.8

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int \frac{6x+7}{\sqrt{(x-5)(x-4)}} dx$.	12b	Numerical	7.8
(ii)	Evaluate $\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$.	12b	Numerical	7.8
10	(see parts below)	12b	Numerical	7.8
(i)	Evaluate $\int \frac{3x+5}{\sqrt{x^2-8x+7}} dx$.	12b	Numerical	7.8
(ii)	Evaluate $\int \frac{x^2}{\sqrt{x^6-2x^3-3}} dx$.	12b	Numerical	7.8
11	Evaluate $\int \frac{1}{x[(\log x)^2-3\log x-4]} dx$.	12a	Numerical	7.8

Exercise 7.9 – Integration of Rational Trigonometric Functions (Weierstrass Substitution)

1	(see parts below)	12e	Numerical	7.9
(i)	Evaluate $\int \frac{dx}{2+\cos x}$.	12e	Numerical	7.9
(ii)	Evaluate $\int \frac{dx}{5+4\cos x}$.	12e	Numerical	7.9
2	(see parts below)	12e	Numerical	7.9
(i)	Evaluate $\int \frac{dx}{1+2\cos x}$.	12e	Numerical	7.9
(ii)	Evaluate $\int \frac{dx}{4\cos x-1}$.	12a	Numerical	7.9
3	(see parts below)	12e	Numerical	7.9
(i)	Evaluate $\int \frac{dx}{4+5\sin x}$.	12e	Numerical	7.9
(ii)	Evaluate $\int \frac{dx}{1-\sin x+\cos x}$.	12e	Numerical	7.9
4	(see parts below)	12a	Numerical	7.9
(i)	Evaluate $\int \frac{dx}{1-\tan x}$. (NCERT)	12a	Numerical	7.9
(ii)	Evaluate $\int \frac{dx}{1+\cot x}$. (NCERT)	12a	Numerical	7.9
5	If ...	12e	Numerical	7.9

Exercise 7.10 – Integration by Partial Fractions

1	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{x-1}{(x-2)(x-3)} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{3x+5}{x^2+3x-18} dx$.	14	Numerical	7.10
2	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{2x+7}{x^2-x-2} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2+1}{x^2-5x+6} dx$.	14	Numerical	7.10

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
3	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{x^3+x+1}{x^2-1} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^3+1}{x^3-x} dx$.	14	Numerical	7.10
4	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{x}{(x-1)(x-2)(x-3)} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{3x-1}{(x-1)(x-2)(x-3)} dx$.	14	Numerical	7.10
5	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{5x}{(x+1)(x^2-4)} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x}{(x-1)^2(x+2)} dx$.	14	Numerical	7.10
6	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{3x-1}{(x-2)^2} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{3x+5}{x^3-x^2-x+1} dx$.	14	Numerical	7.10
7	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{2}{(1-x)(1+x^2)} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{4}{(x-2)(x^2+4)} dx$.	14	Numerical	7.10
8	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{x^2+x}{x^3-x^2+x-1} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^3}{(x-1)(x^2+1)} dx$.	14	Numerical	7.10
9	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{dx}{1-x^3}$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{dx}{(x+1)^2(x^2+1)}$.	14	Numerical	7.10
10	(see parts below)	12a	Numerical	7.10
(i)	Evaluate $\int \frac{2x}{(x^2+1)(x^2+3)} dx$.	12a	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2}{(1+x^3)(2+x^3)} dx$.	12a	Numerical	7.10
11	(see parts below)	12a	Numerical	7.10
(i)	Evaluate $\int \frac{dx}{x(x^4-1)}$.	12a	Numerical	7.10
(ii)	Evaluate $\int \frac{dx}{x(x^2+1)}$.	14	Numerical	7.10

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
12	(see parts below)	12a	Numerical	7.10
(i)	Evaluate $\int \frac{dx}{x(x^n+1)}$.	12a	Numerical	7.10
(ii)	Evaluate $\int \frac{dx}{x(x^3+8)}$.	12a	Numerical	7.10
13	(see parts below)	12c	Numerical	7.10
(i)	Evaluate $\int \frac{e^x}{(1+e^x)(2+e^x)} dx$.	12c	Numerical	7.10
(ii)	Evaluate $\int \frac{dx}{e^{2x}+1}$.	12c	Numerical	7.10
14	(see parts below)	12e	Numerical	7.10
(i)	Evaluate $\int \frac{\cos x}{(1+\sin x)(2+\sin x)} dx$.	12e	Numerical	7.10
(ii)	Evaluate $\int \frac{\sin 2x}{(1+\sin x)(2+\sin x)} dx$.	15	Numerical	7.10
15	(see parts below)	15	Numerical	7.10
(i)	Evaluate $\int \frac{2 \cos x}{(1-\sin x)(2-\cos^2 x)} dx$.	15	Numerical	7.10
(ii)	Evaluate $\int \frac{\sin 2x}{(\sin^2 x+1)(\sin^2 x+3)} dx$.	15	Numerical	7.10
16	(see parts below)	12a	Numerical	7.10
(i)	Evaluate $\int \frac{\sec^2 x}{(2+\tan x)(3+\tan x)} dx$.	12a	Numerical	7.10
(ii)	Evaluate $\int \frac{dx}{x(6(\log x)^2+7 \log x+2)}$.	12a	Numerical	7.10
17	(see parts below)	15	Numerical	7.10
(i)	Evaluate $\int \frac{dx}{\sin x - \sin 2x}$.	15	Numerical	7.10
(ii)	Evaluate $\int \frac{dx}{(x^2+1)(x^2+4)}$.	14	Numerical	7.10
18	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{dx}{(x^2+1)(x^2+4)}$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2}{(x^2+1)(x^2+4)} dx$.	14	Numerical	7.10
19	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{x^2+1}{(x^2+4)(x^2+25)} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2}{x^4-x^2-12} dx$.	14	Numerical	7.10
20	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{dx}{x^4-1}$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2}{1-x^4} dx$.	14	Numerical	7.10

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
21	(see parts below)	12b	Numerical	7.10
(i)	Evaluate $\int \frac{\sqrt{\cos x}}{\sin x} dx$.	12b	Numerical	7.10
(ii)	Evaluate $\int \frac{\sin x}{\sin 4x} dx$.	15	Numerical	7.10
22	(see parts below)	12a	Numerical	7.10
(i)	Evaluate $\int \frac{1+x^2}{1+x^4} dx$.	12a	Numerical	7.10
(ii)	Evaluate $\int \frac{1-x^2}{1+x^4} dx$.	12a	Numerical	7.10
23	(see parts below)	12a	Numerical	7.10
(i)	Evaluate $\int \frac{x^2-1}{x^4+x^2+1} dx$.	12a	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2+4}{x^4+x^2+16} dx$.	12a	Numerical	7.10
24	(see parts below)	14	Numerical	7.10
(i)	Evaluate $\int \frac{1}{x^4+1} dx$.	14	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2}{x^4+1} dx$.	14	Numerical	7.10
25	(see parts below)	1	Numerical	7.10
(i)	Evaluate $\int \frac{x^2}{x^4+x^2+1} dx$.	1	Numerical	7.10
(ii)	Evaluate $\int \frac{x^2-3x+1}{x^4-x^2+1} dx$.	1	Numerical	7.10
26	(see parts below)	15	Numerical	7.10
(i)	Evaluate $\int \frac{\sin x + \cos x}{\sin^2 x + \cos^4 x} dx$.	15	Numerical	7.10
(ii)	Evaluate $\int (\sqrt{\cot x} + \frac{1}{\sqrt{\cot x}}) dx$.	15	Numerical	7.10
27	Evaluate $\int \frac{x+7}{x^2+4x+7} dx$.	7	Numerical	7.10

Exercise 7.11 – Integration by Parts

1	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x e^x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x \sin x dx$.	13	Numerical	7.11
2	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x \sec^2 x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x \sin 3x dx$.	13	Numerical	7.11
3	(see parts below)	13	Numerical	7.11

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int x e^{2x} dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x \cos 2x dx$.	13	Numerical	7.11
4	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x \sec^2 x \tan x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int (e^{\log x} + \sin x) \cos x dx$.	13	Numerical	7.11
5	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x \log x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x \log 2x dx$.	13	Numerical	7.11
6	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x^4 \log x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int \log x dx$.	13	Numerical	7.11
7	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int (x^2 + 1) \log x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x^2 \log(1 + x) dx$.	13	Numerical	7.11
(iii)	Evaluate $\int \sin x \cdot \log(\cos x) dx$.	12a	Numerical	7.11
8	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int \tan^{-1} x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int \cos^{-1}(\frac{1}{x}) dx$.	13	Numerical	7.11
9	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x^3 \tan^{-1} x dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x^2 \sin^{-1} x dx$.	13	Numerical	7.11
10	(see parts below)	12b	Numerical	7.11
(i)	Evaluate $\int \frac{\sin^{-1} x}{\sqrt{1-x}} dx$.	12b	Numerical	7.11
(ii)	Evaluate $\int x^2 e^x dx$.	13	Numerical	7.11
11	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int x^2 e^{3x} dx$.	13	Numerical	7.11

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Evaluate $\int x^2 \cos x \, dx$.	13	Numerical	7.11
12	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int (\log x)^2 \, dx$.	13	Numerical	7.11
(ii)	Evaluate $\int x^3 \sin(x^2) \, dx$.	12a	Numerical	7.11
13	(see parts below)	12c	Numerical	7.11
(i)	Evaluate $\int 2x^3 e^{x^2} \, dx$.	12c	Numerical	7.11
(ii)	Evaluate $\int \cos \sqrt{x} \, dx$.	12b	Numerical	7.11
14	(see parts below)	12b	Numerical	7.11
(i)	Evaluate $\int \tan^{-1} \sqrt{x} \, dx$.	12b	Numerical	7.11
(ii)	Evaluate $\int x^3 \tan^{-1}(x^2) \, dx$.	12a	Numerical	7.11
15	(see parts below)	12b	Numerical	7.11
(i)	Evaluate $\int \sin^3 \sqrt{x} \, dx$.	12b	Numerical	7.11
(ii)	Evaluate $\int \frac{x \cos^{-1} x}{\sqrt{1-x^2}} \, dx$.	12b	Numerical	7.11
16	(see parts below)	6	Numerical	7.11
(i)	Evaluate $\int \sin^{-1}(3x - 4x^3) \, dx$.	6	Numerical	7.11
(ii)	Evaluate $\int \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right) dx$.	6	Numerical	7.11
17	(see parts below)	12a	Numerical	7.11
(i)	Evaluate $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} \, dx$.	12a	Numerical	7.11
(ii)	Evaluate $\int \frac{x^2 \tan^{-1} x}{1+x^2} \, dx$.	13	Numerical	7.11
18	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int e^x \sin 2x \, dx$.	13	Numerical	7.11
(ii)	Evaluate $\int e^{2x} \sin x \, dx$.	15	Numerical	7.11
19	(see parts below)	13	Proof	7.11
(i)	Evaluate $\int e^{ax} \sin bx \, dx$.	13	Proof	7.11
(ii)	Evaluate $\int e^{ax} \cos(bx + c) \, dx$.	13	Numerical	7.11
20	(see parts below)	12c	Numerical	7.11
(i)	Evaluate $\int x^2 e^{x^3} \cos x^3 \, dx$.	12c	Numerical	7.11

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Evaluate $\int e^x \sin^2 x \, dx$.	15	Numerical	7.11
21	(see parts below)	12c	Numerical	7.11
(i)	Evaluate $\int \frac{e^m \tan^{-1} x}{(1+x^2)^{3/2}} \, dx$.	12c	Numerical	7.11
(ii)	Evaluate $\int \csc^3 x \, dx$.	13	Numerical	7.11
22	(see parts below)	13	Numerical	7.11
(i)	Evaluate $\int \cos(\log x) \, dx$.	13	Numerical	7.11
(ii)	Evaluate $\int \frac{\sin^{-1} x}{x^2} \, dx$.	13	Numerical	7.11

Exercise 7.12 – Integration by Parts: Special Exponential Form

1	(see parts below)	12e	Numerical	7.12
(i)	Evaluate $\int e^x (\sin x + \cos x) \, dx$.	12e	Numerical	7.12
(ii)	Evaluate $\int e^x (\tan x + \sec^2 x) \, dx$.	12c	Numerical	7.12
2	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int e^x (\log x + \frac{1}{x}) \, dx$.	12c	Numerical	7.12
(ii)	Evaluate $\int e^x (\tan x + 1) \sec x \, dx$.	12c	Numerical	7.12
3	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int e^x (x^2 + 2x) \, dx$.	12c	Numerical	7.12
(ii)	Evaluate $\int (x+1)e^x \, dx$.	12c	Numerical	7.12
4	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int e^x (\frac{1}{x} - \frac{1}{x^2}) \, dx$.	12c	Numerical	7.12
(ii)	Evaluate $\int (1 + \log x) \, dx$.	13	Numerical	7.12
5	(see parts below)	12b	Numerical	7.12
(i)	Evaluate $\int e^x (\sin^{-1} x + \frac{1}{\sqrt{1-x^2}}) \, dx$.	12b	Numerical	7.12
(ii)	Evaluate $\int e^x (x \cos x + \sin x) \, dx$.	12e	Numerical	7.12
6	(see parts below)	12c	Numerical	7.12
(i)	If $\int (\cot x - \csc^2 x) e^x \, dx = e^x f(x) + C$, then write the value ...	12c	Numerical	7.12

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	If $\int (\frac{x-1}{x^2})e^x dx = f(x)e^x + C$, then write the value ...	12c	Numerical	7.12
7	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int e^x(\tan x + \log \sec x) dx.$	12c	Numerical	7.12
(ii)	Evaluate $\int \frac{2x-1}{4x^2} e^{2x} dx.$	12c	Numerical	7.12
8	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int \frac{\sin x \cos x - 1}{\sin^2 x} e^x dx.$	12c	Numerical	7.12
(ii)	Evaluate $\int e^x(\cot x + \log \sin x) dx.$	12c	Numerical	7.12
9	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int \frac{xe^x}{(x+1)^2} dx.$ (NCERT)	12c	Numerical	7.12
(ii)	Evaluate $\int \frac{x-1}{(x+1)^3} e^x dx.$	12c	Numerical	7.12
10	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int \frac{x-3}{(x-1)^3} e^x dx.$ (NCERT)	12c	Numerical	7.12
(ii)	Evaluate $\int \frac{xe^{2x}}{(1+2x)^2} dx.$	12c	Numerical	7.12
11	(see parts below)	12e	Numerical	7.12
(i)	Evaluate $\int \frac{1+\sin x}{1+\cos x} e^x dx. \quad \dots$	12e	Numerical	7.12
(ii)	Evaluate $\int \frac{2+\sin 2x}{1+\cos 2x} e^x dx. \quad \dots$	12c	Numerical	7.12
12	(see parts below)	12c	Numerical	7.12
(i)	Evaluate $\int \frac{\sin 4x-4}{1-\cos 4x} e^x dx.$	12c	Numerical	7.12
(ii)	Evaluate $\int (\cos x + 3 \sin x) e^{3x} dx.$	13	Numerical	7.12
13	(see parts below)	12a	Numerical	7.12
(i)	Evaluate $\int (\sin(\log x) + \cos(\log x)) dx.$	12a	Numerical	7.12
(ii)	Evaluate $\int \frac{\log x-3}{(\log x)^4} dx.$	12a	Numerical	7.12
(iii)	Evaluate $\int e^{\cot^{-1} x} (\frac{1-x+x^2}{1+x^2}) dx.$	12c	Numerical	7.12

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
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Exercise 7.13 – Integrals Involving the Square Root of $a^2 - x^2$ and Related Forms

1	(see parts below)	1	Numerical	7.13
(i)	Evaluate $\int \sqrt{4 - x^2} dx$.	1	Numerical	7.13
(ii)	Evaluate $\int \sqrt{x^2 - 9} dx$.	1	Numerical	7.13
2	(see parts below)	1	Numerical	7.13
(i)	Evaluate $\int \sqrt{1 + x^2} dx$.	1	Numerical	7.13
(ii)	Evaluate $\int \sqrt{1 - 4x^2} dx$.	12b	Numerical	7.13
3	(see parts below)	12b	Numerical	7.13
(i)	Evaluate $\int \sqrt{4x^2 - 9} dx$.	12b	Numerical	7.13
(ii)	Evaluate $\int \sqrt{4 - 9x^2} dx$.	12b	Numerical	7.13
4	(see parts below)	1	Numerical	7.13
(i)	Evaluate $\int \frac{x^2}{\sqrt{x^2+6}} dx$.	1	Numerical	7.13
(ii)	Evaluate $\int \frac{1}{x-\sqrt{x^2-1}} dx$.	5	Numerical	7.13
5	(see parts below)	1	Numerical	7.13
(i)	Evaluate $\int \frac{x^2+3}{\sqrt{x^2-25}} dx$.	1	Numerical	7.13
(ii)	Evaluate $\int (x-3)\sqrt{\frac{x+2}{x-2}} dx$.	1	Numerical	7.13
6	(see parts below)	12b	Numerical	7.13
(i)	Evaluate $\int x\sqrt{x^4-1} dx$.	12b	Numerical	7.13
(ii)	Evaluate $\int \frac{\sqrt{9-(\log x)^2}}{x} dx$.	12b	Numerical	7.13
7	(see parts below)	12b	Numerical	7.13
(i)	Evaluate $\int \sin x \cos x \sqrt{\sin^4 x + 4} dx$.	12b	Numerical	7.13
(ii)	Evaluate $\int \frac{\log x}{x} \sqrt{(\log x)^4 - 1} dx$.	12b	Numerical	7.13
8	(see parts below)	13	Numerical	7.13
(i)	Evaluate $\int x \cos^{-1} x dx$.	13	Numerical	7.13
(ii)	Evaluate $\int \cos^{-1} \sqrt{x} dx$.	13	Numerical	7.13

Exercise 7.14 – Integrals Involving the Square Root of a General Quadratic

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
1	(see parts below)	12b	Numerical	7.14
(i)	Evaluate $\int \sqrt{x^2 + 4x + 6} dx$.	12b	Numerical	7.14
(ii)	Evaluate $\int \sqrt{5 - 2x + x^2} dx$.	12b	Numerical	7.14
2	(see parts below)	12b	Numerical	7.14
(i)	Evaluate $\int \sqrt{3 - 2x - x^2} dx$.	12b	Numerical	7.14
(ii)	Evaluate $\int \sqrt{1 + 3x - x^2} dx$.	12b	Numerical	7.14
3	(see parts below)	12b	Numerical	7.14
(i)	Evaluate $\int (x + 3)\sqrt{3 - 4x - x^2} dx$.	12b	Numerical	7.14
(ii)	Evaluate $\int (3x + 1)\sqrt{4 - 3x - 2x^2} dx$.	12b	Numerical	7.14

Exercise 7.15 – Definite Integrals via the Fundamental Theorem

1	(see parts below)	1	Numerical	7.15
(i)	Evaluate $\int_2^3 x^2 dx$.	1	Numerical	7.15
(ii)	Evaluate $\int_{-1}^1 (x + 1) dx$.	1	Numerical	7.15
(iii)	Evaluate $\int_2^3 \frac{1}{x} dx$.	4	Numerical	7.15
2	(see parts below)	4	Numerical	7.15
(i)	Evaluate $\int_{-4}^{-1} \frac{1}{x} dx$.	4	Numerical	7.15
(ii)	Evaluate $\int_0^1 \frac{1}{2x-3} dx$.	4	Numerical	7.15
(iii)	Evaluate $\int_0^1 \sqrt{5x + 4} dx$.	1	Numerical	7.15
3	(see parts below)	1	Numerical	7.15
(i)	Evaluate $\int_2^3 3^x dx$.	1	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/4} \tan x dx$.	2	Numerical	7.15
(iii)	Evaluate $\int_{\pi/4}^{\pi/2} \cot x dx$.	2	Numerical	7.15
4	(see parts below)	2	Numerical	7.15
(i)	Evaluate $\int_0^{\pi/4} \sin 2x dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/2} \cos 2x dx$.	2	Numerical	7.15

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(iii)	Evaluate $\int_0^{\pi} (\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2}) dx$.	2	Numerical	7.15
5	(see parts below)	6	Numerical	7.15
(i)	Evaluate $\int_0^{1/\sqrt{2}} \frac{dx}{\sqrt{1-x^2}}$.	6	Numerical	7.15
(ii)	Evaluate $\int_0^3 \frac{dx}{9+x^2}$.	6	Numerical	7.15
(iii)	Evaluate $\int_1^{\sqrt{3}} \frac{dx}{1+x^2}$.	6	Numerical	7.15
6	(see parts below)	14	Numerical	7.15
(i)	Evaluate $\int_2^3 \frac{dx}{x^2-1}$.	14	Numerical	7.15
(ii)	Evaluate $\int_0^{2\pi} \frac{1}{1+\sin x} dx$.	5	Numerical	7.15
(iii)	Evaluate $\int_{-\pi/2}^{\pi/2} \frac{1}{1+\cos x} dx$.	10	Numerical	7.15
7	(see parts below)	8	Numerical	7.15
(i)	Evaluate $\int_0^6 x+3 dx$.	8	Numerical	7.15
(ii)	Evaluate $\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx$.	1	Numerical	7.15
(iii)	Evaluate $\int_0^8 (\sqrt{8x} - \frac{x^2}{8}) dx$.	1	Numerical	7.15
8	(see parts below)	2	Numerical	7.15
(i)	Evaluate $\int_{\pi/6}^{\pi/4} \csc x dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/4} (2 \sec^2 x + x^3 + 2) dx$.	1	Numerical	7.15
9	(see parts below)	5	Numerical	7.15
(i)	Evaluate $\int_0^1 \frac{dx}{\sqrt{1+x-\sqrt{x}}}$.	5	Numerical	7.15
(ii)	Evaluate $\int_0^1 \frac{1-x}{1+x} dx$.	1	Numerical	7.15
10	(see parts below)	12d	Numerical	7.15
(i)	Evaluate $\int_0^{\pi/2} \sin^3 x dx$.	12d	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/4} (\tan x + \cot x)^{-1} dx$.	2	Numerical	7.15
11	(see parts below)	2	Numerical	7.15

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int_0^{\pi/2} \sqrt{1 - \cos 2x} dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi} \sqrt{1 + \sin x} dx$.	2	Numerical	7.15
12	(see parts below)	2	Numerical	7.15
(i)	Evaluate $\int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/4} \frac{\cos 3x}{\cos x} dx$.	2	Numerical	7.15
13	(see parts below)	15	Numerical	7.15
(i)	Evaluate $\int_0^{\pi/4} \sin 2x \sin 3x dx$.	15	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/2} (a^2 \cos^2 x + b^2 \sin^2 x) dx$.	15	Numerical	7.15
14	(see parts below)	2	Numerical	7.15
(i)	Evaluate $\int_{\pi/4}^{\pi/3} (\tan x + \cot x)^2 dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_1^2 \frac{dx}{(x+1)(x+2)}$.	14	Numerical	7.15
15	(see parts below)	2	Numerical	7.15
(i)	Evaluate $\int_{-\pi/4}^0 \frac{1 - \tan x}{1 + \tan x} dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx$.	7	Numerical	7.15
16	(see parts below)	14	Numerical	7.15
(i)	Evaluate $\int_1^2 \frac{x+3}{x(x+2)} dx$.	14	Numerical	7.15
(ii)	Evaluate $\int_1^3 \frac{dx}{x^2(x+1)}$.	14	Numerical	7.15
17	(see parts below)	2	Numerical	7.15
(i)	Evaluate $\int_{-\pi/4}^0 \frac{1 + \tan x}{1 - \tan x} dx$.	2	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx$.	7	Numerical	7.15
18	(see parts below)	13	Numerical	7.15
(i)	Evaluate $\int_0^1 x e^x dx$.	13	Numerical	7.15
(ii)	Evaluate $\int_0^1 (x e^x + \sin \frac{\pi x}{4}) dx$.	13	Numerical	7.15
19	(see parts below)	13	Numerical	7.15

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int_0^1 (xe^{2x} + \sin \frac{\pi x}{4}) dx$.	13	Numerical	7.15
(ii)	Evaluate $\int_0^{\pi/2} x^2 \cos 2x dx$.	13	Numerical	7.15
20	(see parts below)	7	Numerical	7.15
(i)	Evaluate $\int_1^2 e^x (\frac{1}{x} - \frac{1}{x^2}) dx$.	7	Numerical	7.15
(ii)	Evaluate $\int_{\pi/2}^{1/\pi} e^x (\frac{1-\sin x}{1-\cos x}) dx$.	7	Numerical	7.15
21	(see parts below)	1	Numerical	7.15
(i)	If $\int_0^a 3x^2 dx = 8$, find the value of a .	1	Numerical	7.15
(ii)	If $\int_1^a (3x^2 + 2x + 1) dx = 11$, find the value(s) of a .	1	Numerical	7.15
(iii)	If $\int_0^a \frac{dx}{1+4x^2} = \frac{\pi}{8}$, then find the value ...	6	Numerical	7.15
22	(see parts below)	1	Numerical	7.15
(i)	If $\int_0^a \sqrt{x} dx = 4a \int_0^{\pi/4} \sin 2x dx$, find the ...	1	Numerical	7.15
(ii)	If $\int_a^b x^3 dx = 0$ and ...	1	Numerical	7.15

Exercise 7.16 – Definite Integrals via Substitution

1	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_{-1}^1 x^2(x^3 + 1)^3 dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^1 xe^{x^2} dx$.	12d	Numerical	7.16
2	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^1 \frac{x}{x^2+1} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^{\pi/3} \frac{\cos x}{3+4 \sin x} dx$.	12d	Numerical	7.16
3	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_1^2 \frac{3x}{9x^2-1} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_a^b \frac{\log x}{x} dx$.	12d	Numerical	7.16
4	(see parts below)	12d	Numerical	7.16

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int_0^2 \frac{6x+3}{x^2+4} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^{\pi/2} \frac{\sin x}{\sqrt{1+\cos x}} dx$.	12d	Numerical	7.16
5	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^1 \frac{x^5}{1+x^6} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^1 \frac{x}{\sqrt{1+x^2}} dx$.	12d	Numerical	7.16
6	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_1^e \frac{1+\log x}{2x} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^1 \frac{x}{1+x^4} dx$.	12d	Numerical	7.16
7	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^1 \frac{e^{2x}}{1+e^{4x}} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^{\pi/2} \sqrt{\cos x} \sin^3 x dx$.	12d	Numerical	7.16
8	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^{\pi/4} \sin^3 2x \cos 2x dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^{\pi/2} \sqrt{\cos x} \sin^5 x dx$.	12d	Numerical	7.16
9	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_1^3 \frac{\cos(\log x)}{x} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_1^2 \frac{dx}{x(1+\log x)^2}$.	12d	Numerical	7.16
10	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^{\pi/3} \frac{\sec x \tan x}{1+\sec^2 x} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^1 \tan^{-1} x dx$.	13	Numerical	7.16
11	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^1 \sin^{-1} x dx$.	13	Numerical	7.16
12	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^4 \frac{dx}{\sqrt{x^2+2x+3}}$.	12d	Numerical	7.16

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Evaluate $\int_0^a \frac{dx}{\sqrt{ax-x^2}}$.	12d	Numerical	7.16
13	(see parts below)	14	Numerical	7.16
(i)	Evaluate $\int_0^{\pi/2} \frac{\cos x}{(1+\sin x)(4+\sin x)} dx$.	14	Numerical	7.16
(ii)	Evaluate $\int_0^1 \frac{2x}{(x^2+1)(x^2+2)} dx$.	14	Numerical	7.16
14	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^{\pi/2} \frac{\sin x \cos x}{1+\sin^4 x} dx$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^1 \frac{dx}{2e^x-1}$.	14	Numerical	7.16
(iii)	Evaluate $\int_0^{\pi/4} \frac{\sin x + \cos x}{16+9 \sin 2x} dx$.	12d	Numerical	7.16
(iv)	Evaluate $\int_0^a \sin^{-1}\left(\frac{x}{a+x}\right) dx$.	12d	Numerical	7.16
15	(see parts below)	6	Numerical	7.16
(i)	Evaluate $\int_0^1 \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) dx$.	6	Numerical	7.16
(ii)	Evaluate $\int_0^1 \tan^{-1}\left(\frac{2x}{1-x^2}\right) dx$.	6	Numerical	7.16
16	(see parts below)	13	Numerical	7.16
(i)	Evaluate $\int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^2} dx$.	13	Numerical	7.16
(ii)	Evaluate $\int_0^1 x(\tan^{-1} x)^2 dx$.	13	Numerical	7.16
17	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^1 \frac{5x}{(4+x^2)^2} dx$.	12d	Numerical	7.16
(ii)	Evaluate ...	15	Numerical	7.16
18	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^{\pi/2} \frac{dx}{5+4 \sin x}$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^{\pi/2} \frac{dx}{4 \cos x + 2 \sin x}$.	12d	Numerical	7.16
19	(see parts below)	12d	Numerical	7.16
(i)	Evaluate $\int_0^{\pi} \frac{dx}{5+3 \cos x}$.	12d	Numerical	7.16
(ii)	Evaluate $\int_0^{\pi} \frac{dx}{6-\cos x}$.	12d	Numerical	7.16
20	(see parts below)	1	Numerical	7.16
(i)	Evaluate $\int_0^1 \frac{1-x^2}{x^4+x^2+1} dx$.	1	Numerical	7.16

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	Evaluate $\int_0^1 \frac{x^3}{(1+x^2)^4} dx$.	12d	Numerical	7.16
21	(see parts below)	11	Numerical	7.16
(i)	Evaluate $\int_0^{\pi/4} \frac{1}{1+\tan x} dx$.	11	Numerical	7.16
(ii)	Evaluate ...	12d	Numerical	7.16
(iii)	Evaluate ...	1	Numerical	7.16
22	(see parts below)	5	Numerical	7.16
(i)	Evaluate $\int_{-2}^2 \sqrt{\frac{2-x}{2+x}} dx$.	5	Numerical	7.16
(ii)	Evaluate $\int_0^{a^3} \frac{x^2}{x^6+a^6} dx$.	12d	Numerical	7.16

Exercise 7.17 – Properties of Definite Integrals

1	(see parts below)	8	Numerical	7.17
(i)	Evaluate the definite integral: $\int_{-1}^1 x dx$	8	Numerical	7.17
(ii)	Evaluate the definite integral: $\int_1^4 x-2 dx$ (ISC 2022)	8	Numerical	7.17
(iii)	Evaluate the definite integral: $\int_0^2 \frac{x-1}{ x-1 } dx$	8	Numerical	7.17
(iv)	Evaluate the definite integral: $\int_{-1}^1 \frac{ x+2 }{x+2} dx$	8	Numerical	7.17
(v)	Evaluate the definite integral: $\int_{-3}^0 \frac{x+3}{ x+3 } dx$	8	Numerical	7.17
(vi)	Evaluate the definite integral: $\int_0^3 [x] dx$	9	Numerical	7.17
(vii)	Evaluate the definite integral: ...	9	Numerical	7.17
2	(see parts below)	8	Numerical	7.17
(i)	Evaluate the definite integral: $\int_{-3}^1 \frac{ x+2 }{x+2} dx$	8	Numerical	7.17
(ii)	Evaluate the definite integral: $\int_{-1}^1 [x] dx$	9	Numerical	7.17
(iii)	Evaluate the definite integral: $\int_0^{1.5} [x] dx$	9	Numerical	7.17
3	(see parts below)	10	Numerical	7.17
(i)	Evaluate the definite integral: $\int_{-\pi/2}^{\pi/2} \sin^5 x dx$	10	Numerical	7.17
(ii)	Evaluate the definite integral: $\int_{-\pi/2}^{\pi/2} x \cos^2 x dx$	10	Numerical	7.17
(iii)	Evaluate the definite integral: $\int_{-\pi/3}^{\pi/3} (x^2 \sin x + \tan^3 x) dx$	10	Numerical	7.17

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(iv)	Evaluate the definite integral: $\int_{-\pi}^{\pi} (1 - x^2) \sin x \cos^2 x \, dx$	10	Numerical	7.17
4	(see parts below)	10	Numerical	7.17
(i)	Evaluate the definite integral: $\int_{-1}^1 \log\left(\frac{2-x}{2+x}\right) dx$	10	Numerical	7.17
(ii)	Evaluate the definite integral: $\int_{-\pi}^{\pi} \frac{2x}{1+\cos x} dx$	10	Numerical	7.17
(iii)	Evaluate the definite integral: $\int_0^{\pi} e^{\cos^2 x} \cos x \, dx$	11	Numerical	7.17
5	(see parts below)	11	Numerical	7.17
(i)	Evaluate the definite integral: $\int_0^{\pi/2} \log(\tan x) \, dx$	11	Numerical	7.17
(ii)	Evaluate the definite integral: ...	8	Numerical	7.17
(iii)	Evaluate the definite integral: $\int_0^{2\pi} \cos^2 x \sin x \, dx$	12a	Numerical	7.17
6	(see parts below)	11	Numerical	7.17
(i)	Evaluate the definite integral: $\int_0^{\pi} \sin^2 x \cos^3 x \, dx$	11	Numerical	7.17
(ii)	Evaluate the definite integral: $\int_0^{2\pi} \cos^5 x \, dx$	11	Numerical	7.17
(iii)	Evaluate the definite integral: ...	10	Numerical	7.17
7	Prove that: $\int_0^{2a} f(x) \, dx = \int_0^{2a} f(2a - x) \, dx$	11	Proof	7.17
8	(see parts below)	11	Proof	7.17
(i)	Prove that $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$ and hence prove ...	11	Proof	7.17
(ii)	Prove that $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$. Hence, evaluate: ...	11	Numerical	7.17
9	(see parts below)	8	Numerical	7.17
(i)	By using properties of definite integrals, evaluate: $\int_0^4 x - 1 \, dx$	8	Numerical	7.17
(ii)	By using properties of definite integrals, evaluate: $\int_2^8 x - 5 \, dx$	8	Numerical	7.17
(iii)	By using properties of definite integrals, evaluate: ...	8	Numerical	7.17
10	(see parts below)	8	Numerical	7.17
(i)	By using properties of definite integrals, evaluate: $\int_0^1 2x - 1 \, dx$	8	Numerical	7.17

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(ii)	By using properties of definite integrals, evaluate: $\int_{-2}^2 2x + 3 dx$	8	Numerical	7.17
(iii)	By using properties of definite integrals, evaluate: $\int_1^3 x^2 - 2x dx$	8	Numerical	7.17
11	(see parts below)	10	Numerical	7.17
(i)	By using properties of definite integrals, evaluate: ...	10	Numerical	7.17
(ii)	By using properties of definite integrals, evaluate: $\int_0^\pi \cos x dx$	8	Numerical	7.17
(iii)	By using properties of definite integrals, evaluate: $\int_0^{\pi/2} \cos 2x dx$	8	Numerical	7.17
12	(see parts below)	8	Numerical	7.17
(i)	By using properties of definite integrals, evaluate: $\int_0^{2\pi} \sin x dx$	8	Numerical	7.17
(ii)	By using properties of definite integrals, evaluate: $\int_{-1}^1 \sqrt{ x - x} dx$	8	Numerical	7.17
(iii)	By using properties of definite integrals, evaluate: ...	8	Numerical	7.17
13	(see parts below)	8	Numerical	7.17
(i)	By using properties of definite integrals, evaluate: ...	8	Numerical	7.17
(ii)	By using properties of definite integrals, evaluate: ...	8	Numerical	7.17
(iii)	By using properties of definite integrals, evaluate: ...	8	Numerical	7.17
(iv)	By using properties of definite integrals, evaluate: ...	8	Numerical	7.17
(v)	By using properties of definite integrals, evaluate: ...	10	Numerical	7.17
(vi)	By using properties of definite integrals, evaluate: $\int_0^{3/2} x \sin \pi x dx$	13	Numerical	7.17
14	Evaluate: $\int_1^4 f(x) dx$, where $f(x) =$ (piecewise definition)	8	Numerical	7.17
15	Evaluate: $\int_0^9 f(x) dx$, where $f(x) =$ (piecewise definition)	8	Numerical	7.17
16	(see parts below)	10	Numerical	7.17

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	Evaluate $\int_{-\pi/2}^{\pi/2} \cos^2 x \, dx$.	10	Numerical	7.17
(ii)	Evaluate $\int_{-\pi/2}^{\pi/2} \sin^2 x \, dx$.	10	Numerical	7.17
(iii)	Evaluate $\int_{-\pi/2}^{\pi/2} \cos^4 x \, dx$.	10	Numerical	7.17
17	(see parts below)	11	Numerical	7.17
(i)	Evaluate $\int_0^1 x(1-x)^{5/2} \, dx$.	11	Numerical	7.17
(ii)	Evaluate $\int_0^2 x\sqrt{2-x} \, dx$.	11	Numerical	7.17
(iii)	Evaluate $\int_0^3 x^2\sqrt{3-x} \, dx$.	11	Numerical	7.17
18	(see parts below)	11	Numerical	7.17
(i)	Evaluate ...	11	Numerical	7.17
(ii)	Evaluate $\int_0^{\pi/2} \frac{dx}{1+(\tan x)^{2/3}}$.	11	Numerical	7.17
(iii)	Evaluate $\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} \, dx$.	11	Numerical	7.17
(iv)	Evaluate $\int_0^{\pi/2} \frac{\cos^5 x}{\cos^5 x + \sin^5 x} \, dx$.	11	Numerical	7.17
(v)	Evaluate $\int_0^{\pi/2} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} \, dx$.	11	Numerical	7.17
(vi)	Evaluate ...	11	Numerical	7.17
19	(see parts below)	11	Numerical	7.17
(i)	Evaluate $\int_0^{\pi/2} \frac{\cos^2 x}{\sin x + \cos x} \, dx$.	11	Numerical	7.17
(ii)	Evaluate $\int_0^{\pi} \frac{4x \sin x}{1 + \cos^2 x} \, dx$.	11	Numerical	7.17
(iii)	Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + 3 \cos^2 x} \, dx$.	11	Numerical	7.17
20	(see parts below)	11	Numerical	7.17
(i)	Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} \, dx$.	11	Numerical	7.17
(ii)	Evaluate ...	11	Numerical	7.17
(iii)	Evaluate ...	11	Numerical	7.17
(iv)	Evaluate $\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}}$.	11	Numerical	7.17
(v)	Evaluate $\int_1^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{4-x}} \, dx$.	11	Numerical	7.17
21	Prove that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. Hence ...	11	Both	7.17

Q. No.	Topic / Statement Summary	Type No.	Kind	Section
(i)	$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$	11	Both	7.17
(ii)	$\int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx$	11	Both	7.17
(iii)	$\int_0^1 x^2(1-x)^n dx$	11	Both	7.17
22	Find the value of $\int_0^1 \tan^{-1}\left(\frac{1-2x}{1+x-x^2}\right) dx$.	11	Numerical	7.17
23	Prove that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$. Hence, evaluate ...	11	Both	7.17
24	(see parts below)	11	Proof	7.17
(i)	If a function f is continuous on $[0, a]$, then show that ...	11	Proof	7.17
(ii)	If a function f is continuous on $[a, b]$, then show that ...	11	Proof	7.17
25	Prove that ...	11	Proof	7.17
26	(see parts below)	11	Numerical	7.17
(i)	Evaluate $\int_0^{\pi/4} \log(\sin 2x) dx$.	11	Numerical	7.17
(ii)	Evaluate $\int_0^1 \frac{\log x}{\sqrt{1-x^2}} dx$.	11	Numerical	7.17
(iii)	Evaluate $\int_0^{\pi/2} [\log(\sin x) - \log(2 \cos x)] dx$.	11	Numerical	7.17
27	Evaluate $\int_0^{\pi} x \log(\sin x) dx$.	11	Numerical	7.17
28	Evaluate $\int_{-3}^3 \frac{x^4}{1+e^x} dx$.	11	Numerical	7.17
29	Evaluate $\int_{-a}^a f(x) dx$, where $f(x) = \frac{9^x}{1+9^x}$.	11	Numerical	7.17

Formula & Standard Results Sheet

I. Standard Integrals and Simplification Before Integrating (Ex 7.1 – 7.2)

Power Rule

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

The most basic and most frequently used integration rule

Standard Trigonometric Integrals

$$\begin{aligned} \int \sin x dx &= -\cos x + C, & \int \cos x dx &= \sin x + C, \\ \int \sec^2 x dx &= \tan x + C, & \int \csc^2 x dx &= -\cot x + C \end{aligned}$$

Learn these directly – do not derive them from first principles every time

Standard Exponential and Logarithmic Integrals

$$\int e^x dx = e^x + C, \quad \int a^x dx = \frac{a^x}{\log a} + C, \quad \int \frac{1}{x} dx = \log |x| + C$$

The modulus in $\log |x|$ is required since x may be negative

Integral of a Sum

$$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$$

Integration is linear – split before integrating term-by-term

Simplify Before Integrating

Expand algebraically, apply a trigonometric identity, or use $e^{\log f(x)} = f(x)$ before attempting to integrate

A large fraction of "hard-looking" integrals become trivial power-rule integrals once simplified

II. Integration by Substitution (Ex 7.3 – 7.6)

Substitution Method

Let $t = g(x)$. Then $dt = g'(x) dx$, and $\int f(g(x))g'(x) dx =$

$$\int f(t) dt$$

Look for an expression and its derivative both appearing in the integrand

Linear Substitution

$$\int f(ax + b) dx =$$

$$\frac{1}{a}F(ax + b) + C, \quad \text{where } F \text{ is an antiderivative of } f$$

A shortcut for any function of a linear expression $ax + b$

Recognizing the $f'(x)/f(x)$ Form

$$\int \frac{f'(x)}{f(x)} dx = \log |f(x)| + C$$

One of the most common substitution patterns – always check if the numerator is the derivative of the denominator

Recognizing the $f'(x)$ times e to the $f(x)$ Form

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + C$$

Substitute $t = f(x)$ to reduce directly to $\int e^t dt$

III. Integrals Reducible to Standard Forms (Ex 7.7 – 7.8)

Standard Form – Difference of Squares

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + C, \quad \int \frac{dx}{a^2 - x^2} =$$

$$\frac{1}{2a} \log \left| \frac{a + x}{a - x} \right| + C$$

Note the sign difference between the two forms – easy to mix up

Standard Form – Sum of Squares

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

Also gives $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$ and $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + C$

Completing the Square

$$ax^2 + bx + c = a \left(x + \frac{b}{2a} \right)^2 + \left(c - \frac{b^2}{4a} \right)$$

Converts a general quadratic under a root or in a denominator into one of the standard forms above

IV. Integration of Rational Trigonometric Functions (Ex 7.9)**Weierstrass Substitution**

$$\text{Let } \tan \frac{x}{2} = t, \text{ so } \sin x = \frac{2t}{1+t^2}, \cos x = \frac{1-t^2}{1+t^2}, dx = \frac{2dt}{1+t^2}$$

Converts any rational function of $\sin x$ and $\cos x$ into a rational function of t

V. Partial Fractions (Ex 7.10)**Partial Fractions – Distinct Linear Factors**

$$\frac{px + q}{(x - a)(x - b)} = \frac{A}{x - a} + \frac{B}{x - b}$$

Multiply out and compare coefficients, or substitute $x = a$ and $x = b$ to isolate A and B

Partial Fractions – Repeated Linear Factor

$$\frac{px + q}{(x - a)^2} = \frac{A}{x - a} + \frac{B}{(x - a)^2}$$

The repeated factor needs one term for each power up to the multiplicity

Partial Fractions – Irreducible Quadratic Factor

$$\frac{px + q}{(x - a)(x^2 + bx + c)} = \frac{A}{x - a} + \frac{Bx + C}{x^2 + bx + c}$$

An irreducible quadratic factor takes a linear numerator, not just a constant

Improper Rational Function

If degree of numerator $\geq$
degree of denominator, divide first to get a polynomial plus a proper fraction
Partial fractions only apply to the proper-fraction remainder

VI. Integration by Parts (Ex 7.11 – 7.12)**Integration by Parts**

$$\int u v dx = u \int v dx - \int \left[\frac{du}{dx} \int v dx \right] dx$$

Choose u using ILATE (Inverse, Log, Algebraic, Trig, Exponential) as a priority order for which function to differentiate

ILATE Rule

Prefer choosing u in this order: Inverse trig $\rightarrow$ Log $\rightarrow$
Algebraic $\rightarrow$ Trig $\rightarrow$ Exponential

The earlier a function appears in this list, the better a choice it is for u

Special Integral – e to the x times f(x) plus f'(x)

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

Recognize this pattern instantly – it avoids a lengthy by-parts computation

VII. Integrals Involving Square Roots of Quadratics (Ex 7.13 – 7.14)**Standard Integral – Root of a-squared minus x-squared**

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$$

One of three standard root-of-quadratic integrals to memorize directly

Standard Integral – Root of x-squared minus/plus a-squared

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + C$$

The $\sqrt{x^2 + a^2}$ case follows the same pattern with log replacing $\sin^{-1}$ and a sign change

General Quadratic Under a Root

$$\int \sqrt{ax^2 + bx + c} dx :$$

complete the square first, then apply the matching standard form above

Always reduce to the form $\sqrt{a^2 \pm u^2}$ or $\sqrt{u^2 - a^2}$ before applying the standard result

VIII. Definite Integrals (Ex 7.15 – 7.16)**Fundamental Theorem of Calculus**

$$\int_a^b f(x) dx = F(b) - F(a), \quad \text{where } F'(x) = f(x)$$

Find the indefinite integral first, then evaluate at the two limits and subtract

Definite Integral via Substitution

When substituting $t =$

$g(x)$ in a definite integral, change the limits to $g(a)$ and $g(b)$ instead of re

Avoids having to back-substitute before evaluating

IX. Properties of Definite Integrals (Ex 7.17)**Reversing the Limits**

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

Splitting the Interval

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx, \quad a < c < b$$

Especially useful when f is defined piecewise

Reflection Property

$$\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

The single most useful property for definite integrals – always try this substitution first

Symmetric Interval About the Origin

$$\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & f \text{ even } (f(-x) = f(x)) \\ 0, & f \text{ odd } (f(-x) = -f(x)) \end{cases}$$

Always check odd/even symmetry first on a symmetric interval before evaluating directly

Symmetric Interval About a Point

$$\int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & f(2a-x) = f(x) \\ 0, & f(2a-x) = -f(x) \end{cases}$$

The general version of the odd/even symmetry property, centred at $x = a$ instead of $x = 0$

7.1 – Standard Integrals (Power Rule and Direct Forms)

Exercise 7.1 — Question 1 (i)

Evaluate the integral:

$$\int x^5 dx$$

TYPE 1 Power Rule of Integration

Given: Indefinite integral $I = \int x^5 dx$

Concept / Formula Used: Using the standard power rule of integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

Solution:

Given the integral:

$$I = \int x^5 dx$$

Applying the power rule with $n = 5$:

$$\begin{aligned} I &= \frac{x^{5+1}}{5+1} + C \\ &= \frac{x^6}{6} + C \end{aligned}$$

Final Answer

$$I = \frac{x^6}{6} + C$$

Exercise 7.1 — Question 1 (ii)

Evaluate the integral:

$$\int \frac{1}{x^3} dx$$

TYPE 1 Power Rule of Integration

Given: Indefinite integral $I = \int \frac{1}{x^3} dx$

Concept / Formula Used: Using the standard power rule of integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

Solution:

Expressing the integrand in exponent form:

$$\frac{1}{x^3} = x^{-3}$$

Thus, the integral becomes:

$$I = \int x^{-3} dx$$

Applying the power rule with $n = -3$:

$$\begin{aligned} I &= \frac{x^{-3+1}}{-3+1} + C \\ &= \frac{x^{-2}}{-2} + C \\ &= -\frac{1}{2x^2} + C \end{aligned}$$

Final Answer

$$I = -\frac{1}{2x^2} + C$$

Common Mistake: Writing $\int \frac{1}{x^3} dx = \log(x^3)$ is incorrect; $\int \frac{1}{x^n} dx = \log|x|$ holds only for $n = 1$.

Exercise 7.1 — Question 1 (iii)

Evaluate the integral:

$$\int \frac{1}{\sqrt{y}} dy$$

TYPE 1 Power Rule of Integration

Given: Indefinite integral $I = \int \frac{1}{\sqrt{y}} dy$

Concept / Formula Used: Using the standard power rule of integration:

$$\int y^n dy = \frac{y^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

Solution:

Expressing the integrand in exponent form:

$$\frac{1}{\sqrt{y}} = y^{-1/2}$$

Thus, the integral becomes:

$$I = \int y^{-1/2} dy$$

Applying the power rule with $n = -\frac{1}{2}$:

$$\begin{aligned} I &= \frac{y^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C \\ &= \frac{y^{1/2}}{\frac{1}{2}} + C \\ &= 2\sqrt{y} + C \end{aligned}$$

Final Answer

$$I = 2\sqrt{y} + C$$

Exercise 7.1 — Question 2 (i)

Evaluate the integral:

$$\int \frac{1}{\sqrt[3]{x}} dx$$

TYPE 1 Power Rule of Integration

Given: Indefinite integral $I = \int \frac{1}{\sqrt[3]{x}} dx$

Concept / Formula Used: Using the standard power rule of integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (\text{for } n \neq -1)$$

Solution:

Expressing the integrand in exponent form:

$$\frac{1}{\sqrt[3]{x}} = \frac{1}{x^{1/3}} = x^{-1/3}$$

Thus, the integral becomes:

$$I = \int x^{-1/3} dx$$

Applying the power rule with $n = -\frac{1}{3}$:

$$\begin{aligned} I &= \frac{x^{-\frac{1}{3}+1}}{-\frac{1}{3}+1} + C \\ &= \frac{x^{2/3}}{\frac{2}{3}} + C \\ &= \frac{3}{2}x^{2/3} + C \end{aligned}$$

Final Answer

$$I = \frac{3}{2}x^{2/3} + C$$

Exercise 7.1 — Question 2 (ii)

Evaluate the integral:

$$\int \sin^2 x \csc^2 x \, dx$$

TYPE 2 Trigonometric Identity Simplification

Given: Indefinite integral $I = \int \sin^2 x \csc^2 x \, dx$

Concept / Formula Used: Using the reciprocal trigonometric identity:

$$\csc x = \frac{1}{\sin x} \implies \sin x \csc x = 1$$

Solution:

Simplifying the integrand using the reciprocal identity:

$$\begin{aligned} \sin^2 x \csc^2 x &= (\sin x \csc x)^2 \\ &= 1^2 \\ &= 1 \end{aligned}$$

Substituting this into the integral:

$$\begin{aligned} I &= \int 1 \, dx \\ &= x + C \end{aligned}$$

Final Answer

$$I = x + C$$

Exercise 7.1 — Question 2 (iii)

Evaluate the integral:

$$\int \sin x \sec^2 x \, dx$$

TYPE 2 Trigonometric Identity Simplification**Given:** Indefinite integral $I = \int \sin x \sec^2 x \, dx$ **Concept / Formula Used:** Using the trigonometric definitions:

$$\sec x = \frac{1}{\cos x} \quad \text{and} \quad \tan x = \frac{\sin x}{\cos x}$$

and the standard integral result:

$$\int \sec x \tan x \, dx = \sec x + C$$

Solution:

Rewriting the integrand in terms of sine and cosine:

$$\begin{aligned} \sin x \sec^2 x &= \sin x \cdot \frac{1}{\cos^2 x} \\ &= \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} \\ &= \tan x \sec x \\ &= \sec x \tan x \end{aligned}$$

Substituting this into the integral:

$$I = \int \sec x \tan x \, dx$$

Using the standard result $\int \sec x \tan x \, dx = \sec x + C$:

$$I = \sec x + C$$

Final Answer

$$I = \sec x + C$$

Exercise 7.1 — Question 3 (i)

Evaluate the integral:

$$\int 5^x \, dx$$

TYPE 3 Exponential Integration

Given: Indefinite integral $I = \int 5^x dx$

Concept / Formula Used: Using the standard exponential integral rule:

$$\int a^x dx = \frac{a^x}{\log a} + C \quad (a > 0, a \neq 1)$$

Solution:

Given the integral:

$$I = \int 5^x dx$$

Applying the standard result with $a = 5$:

$$I = \frac{5^x}{\log 5} + C$$

Final Answer

$$I = \frac{5^x}{\log 5} + C$$

Exercise 7.1 — Question 3 (ii)

Evaluate the integral:

$$\int 7^x e^x dx$$

TYPE 3 Exponential Integration

Given: Indefinite integral $I = \int 7^x e^x dx$

Concept / Formula Used: Using the law of exponents:

$$a^x b^x = (ab)^x$$

and the standard exponential integral rule:

$$\int k^x dx = \frac{k^x}{\log k} + C$$

Solution:

Combining the base terms of the integrand:

$$7^x e^x = (7e)^x$$

Thus, the integral becomes:

$$I = \int (7e)^x dx$$

Applying the standard exponential formula with $k = 7e$:

$$I = \frac{(7e)^x}{\log(7e)} + C$$

Final Answer

$$I = \frac{(7e)^x}{\log(7e)} + C$$

Exercise 7.1 — Question 3 (iii)

Evaluate the integral:

$$\int \frac{4^x}{9^x} dx$$

TYPE 3 Exponential Integration

Given: Indefinite integral $I = \int \frac{4^x}{9^x} dx$

Concept / Formula Used: Using the law of exponents:

$$\frac{a^x}{b^x} = \left(\frac{a}{b}\right)^x$$

and the standard exponential integral rule:

$$\int a^x dx = \frac{a^x}{\log a} + C$$

Solution:

Rewriting the integrand under a single exponent:

$$\frac{4^x}{9^x} = \left(\frac{4}{9}\right)^x$$

Thus, the integral becomes:

$$I = \int \left(\frac{4}{9}\right)^x dx$$

Applying the standard exponential formula with $a = \frac{4}{9}$:

$$I = \frac{\left(\frac{4}{9}\right)^x}{\log\left(\frac{4}{9}\right)} + C$$

Final Answer

$$I = \frac{\left(\frac{4}{9}\right)^x}{\log\left(\frac{4}{9}\right)} + C$$

Exercise 7.1 — Question 4 (i)

Evaluate the integral:

$$\int \frac{\sin x}{1 - \sin^2 x} dx$$

TYPE 2 Trigonometric Identity Simplification**Given:** Indefinite integral $I = \int \frac{\sin x}{1 - \sin^2 x} dx$ **Concept / Formula Used:** Using the fundamental Pythagorean identity:

$$\cos^2 x + \sin^2 x = 1 \implies 1 - \sin^2 x = \cos^2 x$$

and the standard integral result:

$$\int \sec x \tan x dx = \sec x + C$$

Solution:Applying the identity $1 - \sin^2 x = \cos^2 x$ in the denominator:

$$\frac{\sin x}{1 - \sin^2 x} = \frac{\sin x}{\cos^2 x}$$

Decomposing the fraction into standard trigonometric ratios:

$$\begin{aligned} \frac{\sin x}{\cos^2 x} &= \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} \\ &= \tan x \sec x \\ &= \sec x \tan x \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} I &= \int \sec x \tan x dx \\ &= \sec x + C \end{aligned}$$

Final Answer

$$I = \sec x + C$$

Exercise 7.1 — Question 4 (ii)

Evaluate the integral:

$$\int \frac{\cos 2x + 2 \sin^2 x}{\cos^2 x} dx$$

TYPE 2 Trigonometric Identity Simplification

Given: Indefinite integral $I = \int \frac{\cos 2x + 2 \sin^2 x}{\cos^2 x} dx$

Concept / Formula Used: Using the double-angle identity for cosine:

$$\cos 2x = 1 - 2 \sin^2 x$$

and the standard integral result:

$$\int \sec^2 x dx = \tan x + C$$

Solution:

Substituting the identity $\cos 2x = 1 - 2 \sin^2 x$ into the numerator:

$$\begin{aligned} \cos 2x + 2 \sin^2 x &= (1 - 2 \sin^2 x) + 2 \sin^2 x \\ &= 1 \end{aligned}$$

The integrand simplifies to:

$$\begin{aligned} \frac{\cos 2x + 2 \sin^2 x}{\cos^2 x} &= \frac{1}{\cos^2 x} \\ &= \sec^2 x \end{aligned}$$

Substituting this into the integral:

$$I = \int \sec^2 x dx$$

Using the standard result $\int \sec^2 x dx = \tan x + C$:

$$I = \tan x + C$$

Final Answer

$$I = \tan x + C$$

Exercise 7.1 — Question 4 (iii)

Evaluate the integral:

$$\int \sqrt{\frac{1}{2}(1 + \cos 2x)} dx$$

TYPE 2 Trigonometric Identity Simplification

Given: Indefinite integral $I = \int \sqrt{\frac{1}{2}(1 + \cos 2x)} dx$

Concept / Formula Used: Using the half-angle trigonometric identity:

$$1 + \cos 2x = 2 \cos^2 x$$

and the standard integral result:

$$\int \cos x \, dx = \sin x + C$$

Solution:

Simplifying the expression under the square root:

$$\begin{aligned} \frac{1}{2}(1 + \cos 2x) &= \frac{1}{2}(2 \cos^2 x) \\ &= \cos^2 x \end{aligned}$$

Taking the square root:

$$\begin{aligned} \sqrt{\frac{1}{2}(1 + \cos 2x)} &= \sqrt{\cos^2 x} \\ &= \cos x \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} I &= \int \cos x \, dx \\ &= \sin x + C \end{aligned}$$

Final Answer

$$I = \sin x + C$$

Exercise 7.1 — Question 5 (i)

Evaluate the integral:

$$\int e^{2 \log x} x^{-3} \, dx$$

TYPE 4 Logarithmic Identity Simplification

Given: Indefinite integral $I = \int e^{2 \log x} x^{-3} \, dx$

Concept / Formula Used: Using the properties of logarithms and exponentials:

$$n \log x = \log(x^n) \quad \text{and} \quad e^{\log f(x)} = f(x)$$

and the standard integral result:

$$\int \frac{1}{x} \, dx = \log x + C$$

Solution:

Simplifying the exponential factor using power and inverse properties:

$$\begin{aligned} e^{2\log x} &= e^{\log(x^2)} \\ &= x^2 \end{aligned}$$

Multiplying with the remaining factor x^{-3} :

$$\begin{aligned} e^{2\log x} \cdot x^{-3} &= x^2 \cdot x^{-3} \\ &= x^{2-3} \\ &= x^{-1} \\ &= \frac{1}{x} \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} I &= \int \frac{1}{x} dx \\ &= \log x + C \end{aligned}$$

Final Answer

$$I = \log x + C$$

Common Mistake: Applying the power rule to $\int x^{-1} dx$ to get $\frac{x^0}{0}$ is a division by zero error; always recall that $\int x^{-1} dx = \log |x| + C$.

Exercise 7.1 — Question 5 (ii)

Evaluate the integral:

$$\int 5^{2\log_5 x} dx$$

TYPE 4 Logarithmic Identity Simplification

Given: Indefinite integral $I = \int 5^{2\log_5 x} dx$

Concept / Formula Used: Using the properties of logarithms and powers:

$$k \log_a x = \log_a(x^k) \quad \text{and} \quad a^{\log_a f(x)} = f(x)$$

and the power rule of integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Solution:

Simplifying the integrand using logarithmic power property:

$$\begin{aligned} 5^{2\log_5 x} &= 5^{\log_5(x^2)} \\ &= x^2 \end{aligned}$$

Substituting this simplified form into the integral:

$$I = \int x^2 dx$$

Applying the power rule with $n = 2$:

$$\begin{aligned} I &= \frac{x^{2+1}}{2+1} + C \\ &= \frac{x^3}{3} + C \\ &= \frac{1}{3}x^3 + C \end{aligned}$$

Final Answer

$$I = \frac{1}{3}x^3 + C$$

Exercise 7.1 — Question 5 (iii)

Evaluate the integral:

$$\int 2^{\log_4 x} dx$$

TYPE 4 Logarithmic Identity Simplification

Given: Indefinite integral $I = \int 2^{\log_4 x} dx$

Concept / Formula Used: Using the base-change formula for logarithms:

$$\log_b a = \frac{\log_c a}{\log_c b} \implies \log_4 x = \frac{\log_2 x}{\log_2 4} = \frac{\log_2 x}{2} = \log_2(x^{1/2})$$

and the identity $a^{\log_a u} = u$.

Solution:

Rewriting $\log_4 x$ in base 2:

$$\begin{aligned} \log_4 x &= \frac{\log_2 x}{\log_2(2^2)} \\ &= \frac{\log_2 x}{2} \\ &= \frac{1}{2} \log_2 x \\ &= \log_2(x^{1/2}) \end{aligned}$$

Now substituting this into the integrand:

$$\begin{aligned} 2^{\log_4 x} &= 2^{\log_2(x^{1/2})} \\ &= x^{1/2} \end{aligned}$$

Thus, the integral becomes:

$$I = \int x^{1/2} dx$$

Applying the power rule with $n = \frac{1}{2}$:

$$\begin{aligned} I &= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C \\ &= \frac{x^{3/2}}{\frac{3}{2}} + C \\ &= \frac{2}{3}x^{3/2} + C \end{aligned}$$

Final Answer

$$I = \frac{2}{3}x^{3/2} + C$$

7.2 – Simplification Before Integrating

Question 1 (i)

Evaluate $\int (ax^2 + bx + c) dx$

TYPE 1 Power Rule of Integration

To Find: $\int (ax^2 + bx + c) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

Solution:

Using the linearity property of indefinite integrals:

$$\int (ax^2 + bx + c) dx = a \int x^2 dx + b \int x dx + c \int 1 dx$$

Using the standard power rule result $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$a \int x^2 dx = a \cdot \frac{x^{2+1}}{2+1} = \frac{ax^3}{3}$$

$$b \int x^1 dx = b \cdot \frac{x^{1+1}}{1+1} = \frac{bx^2}{2}$$

$$c \int 1 dx = cx$$

Summing the individual terms and adding the constant of integration C :

$$= \frac{ax^3}{3} + \frac{bx^2}{2} + cx + C$$

Final Answer

$$\int (ax^2 + bx + c) dx = \frac{ax^3}{3} + \frac{bx^2}{2} + cx + C$$

Question 1 (ii)

Evaluate $\int (x^{2/3} + 1) dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int (x^{2/3} + 1) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Splitting the integral term-by-term:

$$\int (x^{2/3} + 1) dx = \int x^{2/3} dx + \int 1 dx$$

Using the standard result $\int x^n dx = \frac{x^{n+1}}{n+1}$ with $n = \frac{2}{3}$:

$$\frac{2}{3} + 1 = \frac{5}{3}$$

$$\int x^{2/3} dx = \frac{x^{5/3}}{\frac{5}{3}} = \frac{3}{5}x^{5/3}$$

$$\int 1 dx = x$$

Combining the integrated terms:

$$= \frac{3}{5}x^{5/3} + x + C$$

Final Answer

$$\int (x^{2/3} + 1) dx = \frac{3}{5}x^{5/3} + x + C$$

Question 2 (i)

Evaluate $\int \left(\sqrt{3x} + \frac{1}{\sqrt{x}} \right) dx$

TYPE 1 Power Rule of Integration

To Find: $\int \left(\sqrt{3x} + \frac{1}{\sqrt{x}} \right) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Expressing terms with fractional exponents:

$$\begin{aligned}\sqrt{3x} &= \sqrt{3} \cdot x^{1/2} \\ \frac{1}{\sqrt{x}} &= x^{-1/2}\end{aligned}$$

Rewrite the integrand:

$$\int \left(\sqrt{3}x^{1/2} + x^{-1/2} \right) dx = \sqrt{3} \int x^{1/2} dx + \int x^{-1/2} dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned}\int x^{1/2} dx &= \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2} \\ \int x^{-1/2} dx &= \frac{x^{1/2}}{\frac{1}{2}} = 2\sqrt{x}\end{aligned}$$

Substituting these back into the expression:

$$\begin{aligned}&= \sqrt{3} \left(\frac{2}{3}x^{3/2} \right) + 2\sqrt{x} + C \\ &= \frac{2\sqrt{3}}{3}x^{3/2} + 2\sqrt{x} + C \\ &= \frac{2}{\sqrt{3}}x^{3/2} + 2\sqrt{x} + C\end{aligned}$$

Final Answer

$$\int \left(\sqrt{3x} + \frac{1}{\sqrt{x}} \right) dx = \frac{2}{\sqrt{3}} x^{3/2} + 2\sqrt{x} + C$$

Question 2 (ii)

Evaluate $\int \left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2} \right) dx$

TYPE 1 Power Rule of Integration**★ Likely Exam Question**

To Find: $\int \left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2} \right) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Rewrite each component using exponent rules:

$$\begin{aligned} \frac{2a}{\sqrt{x}} &= 2ax^{-1/2} \\ \frac{b}{x^2} &= bx^{-2} \\ 3c\sqrt[3]{x^2} &= 3cx^{2/3} \end{aligned}$$

Rewrite the given integral:

$$2a \int x^{-1/2} dx - b \int x^{-2} dx + 3c \int x^{2/3} dx$$

Using the standard result $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned} \int x^{-1/2} dx &= \frac{x^{1/2}}{\frac{1}{2}} = 2\sqrt{x} \\ \int x^{-2} dx &= \frac{x^{-1}}{-1} = -\frac{1}{x} \\ \int x^{2/3} dx &= \frac{x^{5/3}}{\frac{5}{3}} = \frac{3}{5}x^{5/3} \end{aligned}$$

Substitute these integrated terms back:

$$\begin{aligned} &= 2a(2\sqrt{x}) - b \left(-\frac{1}{x} \right) + 3c \left(\frac{3}{5}x^{5/3} \right) + C \\ &= 4a\sqrt{x} + \frac{b}{x} + \frac{9c}{5}x^{5/3} + C \end{aligned}$$

Final Answer

$$\int \left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2} \right) dx = 4a\sqrt{x} + \frac{b}{x} + \frac{9c}{5}x^{5/3} + C$$

Question 3 (i)

Evaluate $\int \frac{2 \cos x}{3 \sin^2 x} dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \frac{2 \cos x}{3 \sin^2 x} dx$

Concept / Formula Used: $\int \csc x \cot x dx = -\csc x + C$

Solution:

Rewrite the trigonometric ratio inside the integral:

$$\frac{2 \cos x}{3 \sin^2 x} = \frac{2}{3} \cdot \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = \frac{2}{3} \csc x \cot x$$

The integral becomes:

$$= \frac{2}{3} \int \csc x \cot x dx$$

Using the standard result $\int \csc x \cot x dx = -\csc x + C$:

$$\begin{aligned} &= \frac{2}{3}(-\csc x) + C \\ &= -\frac{2}{3} \csc x + C \end{aligned}$$

Final Answer

$$\int \frac{2 \cos x}{3 \sin^2 x} dx = -\frac{2}{3} \csc x + C$$

Question 3 (ii)

Evaluate $\int (2x^2 - 3 \sin x + 5\sqrt{x}) dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int (2x^2 - 3 \sin x + 5\sqrt{x}) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, $\int \sin x dx = -\cos x + C$

Solution:

Split into three separate integrals:

$$2 \int x^2 dx - 3 \int \sin x dx + 5 \int x^{1/2} dx$$

Using standard integration results:

$$\begin{aligned} \int x^2 dx &= \frac{x^3}{3} \\ \int \sin x dx &= -\cos x \\ \int x^{1/2} dx &= \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2} \end{aligned}$$

Combining these terms:

$$\begin{aligned} &= 2 \left(\frac{x^3}{3} \right) - 3(-\cos x) + 5 \left(\frac{2}{3}x^{3/2} \right) + C \\ &= \frac{2}{3}x^3 + 3 \cos x + \frac{10}{3}x^{3/2} + C \end{aligned}$$

Final Answer

$$\int (2x^2 - 3 \sin x + 5\sqrt{x}) dx = \frac{2}{3}x^3 + 3 \cos x + \frac{10}{3}x^{3/2} + C$$

Question 4 (i)

Evaluate $\int (2x - 3 \cos x + e^x) dx$

TYPE 1 Power Rule of Integration

★ **Likely Exam Question**

To Find: $\int (2x - 3 \cos x + e^x) dx$

Concept / Formula Used: $\int x dx = \frac{x^2}{2}$, $\int \cos x dx = \sin x$, $\int e^x dx = e^x$

Solution:

Separate into term-by-term integrals:

$$2 \int x \, dx - 3 \int \cos x \, dx + \int e^x \, dx$$

Applying standard elementary integrals:

$$\begin{aligned} \int x \, dx &= \frac{x^2}{2} \\ \int \cos x \, dx &= \sin x \\ \int e^x \, dx &= e^x \end{aligned}$$

Substituting these into the expression:

$$\begin{aligned} &= 2 \left(\frac{x^2}{2} \right) - 3 \sin x + e^x + C \\ &= x^2 - 3 \sin x + e^x + C \end{aligned}$$

Final Answer

$$\int (2x - 3 \cos x + e^x) \, dx = x^2 - 3 \sin x + e^x + C$$

Question 4 (ii)

Evaluate $\int x^2 \left(1 - \frac{1}{x^2} \right) \, dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int x^2 \left(1 - \frac{1}{x^2} \right) \, dx$

Concept / Formula Used: $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Expand the integrand by multiplying x^2 inside the parentheses:

$$x^2 \left(1 - \frac{1}{x^2} \right) = x^2 - \frac{x^2}{x^2} = x^2 - 1$$

The integral becomes:

$$\int (x^2 - 1) \, dx = \int x^2 \, dx - \int 1 \, dx$$

Using the standard power rule result $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\int x^2 dx = \frac{x^3}{3}$$

$$\int 1 dx = x$$

Combining the integrated terms:

$$= \frac{1}{3}x^3 - x + C$$

Final Answer

$$\int x^2 \left(1 - \frac{1}{x^2}\right) dx = \frac{1}{3}x^3 - x + C$$

Question 5 (i)

Evaluate $\int \left(x^{3/2} + 2e^x - \frac{1}{x}\right) dx$

TYPE 4 Logarithmic Identity Simplification

★ Likely Exam Question

To Find: $\int \left(x^{3/2} + 2e^x - \frac{1}{x}\right) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, $\int e^x dx = e^x$, $\int \frac{1}{x} dx = \log |x| + C$

Solution:

Split into separate integrals:

$$\int x^{3/2} dx + 2 \int e^x dx - \int \frac{1}{x} dx$$

Evaluating each integral term using standard results:

$$\int x^{3/2} dx = \frac{x^{5/2}}{\frac{5}{2}} = \frac{2}{5}x^{5/2}$$

$$\int e^x dx = e^x$$

$$\int \frac{1}{x} dx = \log |x|$$

Combining all terms:

$$= \frac{2}{5}x^{5/2} + 2e^x - \log|x| + C$$

Final Answer

$$\int \left(x^{3/2} + 2e^x - \frac{1}{x} \right) dx = \frac{2}{5}x^{5/2} + 2e^x - \log|x| + C$$

Question 5 (ii)

Evaluate $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx$

Concept / Formula Used: $(a+b)^2 = a^2 + 2ab + b^2$, $\int \frac{1}{x} dx = \log|x| + C$

Solution:

Expand the integrand using identity $(a+b)^2 = a^2 + 2ab + b^2$:

$$\begin{aligned} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 &= (\sqrt{x})^2 + 2(\sqrt{x}) \left(\frac{1}{\sqrt{x}} \right) + \left(\frac{1}{\sqrt{x}} \right)^2 \\ &= x + 2 + \frac{1}{x} \end{aligned}$$

Write the integral as a sum:

$$\int \left(x + 2 + \frac{1}{x} \right) dx = \int x dx + 2 \int 1 dx + \int \frac{1}{x} dx$$

Using standard integration formulas:

$$\begin{aligned} \int x dx &= \frac{x^2}{2} \\ 2 \int 1 dx &= 2x \\ \int \frac{1}{x} dx &= \log|x| \end{aligned}$$

Combining the integrated terms:

$$= \frac{1}{2}x^2 + 2x + \log|x| + C$$

Final Answer

$$\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx = \frac{1}{2}x^2 + 2x + \log|x| + C$$

Question 6 (i)

Evaluate $\int (1-x)\sqrt{x} dx$

TYPE 1 Power Rule of Integration

To Find: $\int (1-x)\sqrt{x} dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Distribute $\sqrt{x} = x^{1/2}$ into the parentheses:

$$(1-x)x^{1/2} = x^{1/2} - x \cdot x^{1/2} = x^{1/2} - x^{3/2}$$

The integral becomes:

$$\int x^{1/2} dx - \int x^{3/2} dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned} \int x^{1/2} dx &= \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2} \\ \int x^{3/2} dx &= \frac{x^{5/2}}{\frac{5}{2}} = \frac{2}{5}x^{5/2} \end{aligned}$$

Subtracting the terms:

$$= \frac{2}{3}x^{3/2} - \frac{2}{5}x^{5/2} + C$$

Final Answer

$$\int (1-x)\sqrt{x} dx = \frac{2}{3}x^{3/2} - \frac{2}{5}x^{5/2} + C$$

Question 6 (ii)

Evaluate $\int \sqrt{x}(3x^2 + 2x + 3) dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int \sqrt{x}(3x^2 + 2x + 3) dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Multiply $\sqrt{x} = x^{1/2}$ across each term in the polynomial:

$$3x^2 \cdot x^{1/2} = 3x^{2+1/2} = 3x^{5/2}$$

$$2x \cdot x^{1/2} = 2x^{1+1/2} = 2x^{3/2}$$

$$3 \cdot x^{1/2} = 3x^{1/2}$$

Rewrite the integrand:

$$\int (3x^{5/2} + 2x^{3/2} + 3x^{1/2}) dx = 3 \int x^{5/2} dx + 2 \int x^{3/2} dx + 3 \int x^{1/2} dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\int x^{5/2} dx = \frac{x^{7/2}}{\frac{7}{2}} = \frac{2}{7}x^{7/2}$$

$$\int x^{3/2} dx = \frac{x^{5/2}}{\frac{5}{2}} = \frac{2}{5}x^{5/2}$$

$$\int x^{1/2} dx = \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2}$$

Multiplying coefficients and combining:

$$\begin{aligned} &= 3 \left(\frac{2}{7}x^{7/2} \right) + 2 \left(\frac{2}{5}x^{5/2} \right) + 3 \left(\frac{2}{3}x^{3/2} \right) + C \\ &= \frac{6}{7}x^{7/2} + \frac{4}{5}x^{5/2} + 2x^{3/2} + C \end{aligned}$$

Final Answer

$$\int \sqrt{x}(3x^2 + 2x + 3) dx = \frac{6}{7}x^{7/2} + \frac{4}{5}x^{5/2} + 2x^{3/2} + C$$

Question 7 (i)

Evaluate $\int \frac{x^3 + 5x^2 + 4x + 1}{x^2} dx$

★ Likely Exam Question

To Find: $\int \frac{x^3 + 5x^2 + 4x + 1}{x^2} dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad \int \frac{1}{x} dx = \log |x| + C$

Solution:

Divide each term of the numerator individually by x^2 :

$$\begin{aligned} \frac{x^3 + 5x^2 + 4x + 1}{x^2} &= \frac{x^3}{x^2} + \frac{5x^2}{x^2} + \frac{4x}{x^2} + \frac{1}{x^2} \\ &= x + 5 + \frac{4}{x} + x^{-2} \end{aligned}$$

Write as a sum of integrals:

$$\int x dx + 5 \int 1 dx + 4 \int \frac{1}{x} dx + \int x^{-2} dx$$

Applying standard integration rules:

$$\begin{aligned} \int x dx &= \frac{x^2}{2} \\ 5 \int 1 dx &= 5x \\ 4 \int \frac{1}{x} dx &= 4 \log |x| \\ \int x^{-2} dx &= \frac{x^{-1}}{-1} = -\frac{1}{x} \end{aligned}$$

Combining the integrated terms:

$$= \frac{1}{2}x^2 + 5x + 4 \log |x| - \frac{1}{x} + C$$

Final Answer

$$\int \frac{x^3 + 5x^2 + 4x + 1}{x^2} dx = \frac{1}{2}x^2 + 5x + 4 \log |x| - \frac{1}{x} + C$$

Question 7 (ii)

Evaluate $\int \frac{x^3 + 3x + 4}{\sqrt{x}} dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int \frac{x^3 + 3x + 4}{\sqrt{x}} dx$

Concept / Formula Used: $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Divide each term in the numerator by $\sqrt{x} = x^{1/2}$:

$$\begin{aligned}\frac{x^3}{x^{1/2}} &= x^{3-1/2} = x^{5/2} \\ \frac{3x}{x^{1/2}} &= 3x^{1-1/2} = 3x^{1/2} \\ \frac{4}{x^{1/2}} &= 4x^{-1/2}\end{aligned}$$

Rewrite the given integral:

$$\int (x^{5/2} + 3x^{1/2} + 4x^{-1/2}) dx = \int x^{5/2} dx + 3 \int x^{1/2} dx + 4 \int x^{-1/2} dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned}\int x^{5/2} dx &= \frac{x^{7/2}}{\frac{7}{2}} = \frac{2}{7}x^{7/2} \\ \int x^{1/2} dx &= \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2} \\ \int x^{-1/2} dx &= \frac{x^{1/2}}{\frac{1}{2}} = 2x^{1/2} = 2\sqrt{x}\end{aligned}$$

Substitute these back into the expression:

$$\begin{aligned}&= \frac{2}{7}x^{7/2} + 3 \left(\frac{2}{3}x^{3/2} \right) + 4(2\sqrt{x}) + C \\ &= \frac{2}{7}x^{7/2} + 2x^{3/2} + 8\sqrt{x} + C\end{aligned}$$

Final Answer

$$\int \frac{x^3 + 3x + 4}{\sqrt{x}} dx = \frac{2}{7}x^{7/2} + 2x^{3/2} + 8\sqrt{x} + C$$

Question 8 (i)

Evaluate $\int x^2 \left(2x - \frac{1}{x} \right)^2 dx$

TYPE 1 Power Rule of Integration

To Find: $\int x^2 \left(2x - \frac{1}{x}\right)^2 dx$

Concept / Formula Used: $(a - b)^2 = a^2 - 2ab + b^2$, $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

First expand the squared binomial $\left(2x - \frac{1}{x}\right)^2$:

$$\left(2x - \frac{1}{x}\right)^2 = (2x)^2 - 2(2x)\left(\frac{1}{x}\right) + \left(\frac{1}{x}\right)^2 = 4x^2 - 4 + \frac{1}{x^2}$$

Multiply the expanded result by x^2 :

$$x^2 \left(4x^2 - 4 + \frac{1}{x^2}\right) = 4x^4 - 4x^2 + 1$$

Rewrite the integrand:

$$\int (4x^4 - 4x^2 + 1) dx = 4 \int x^4 dx - 4 \int x^2 dx + \int 1 dx$$

Using the standard result $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\int x^4 dx = \frac{x^5}{5}$$

$$\int x^2 dx = \frac{x^3}{3}$$

$$\int 1 dx = x$$

Combining terms:

$$= \frac{4}{5}x^5 - \frac{4}{3}x^3 + x + C$$

Final Answer

$$\int x^2 \left(2x - \frac{1}{x}\right)^2 dx = \frac{4}{5}x^5 - \frac{4}{3}x^3 + x + C$$

Question 8 (ii)

Evaluate $\int (x + \sqrt{x})^3 dx$

TYPE 1 Power Rule of Integration

To Find: $\int (x + \sqrt{x})^3 dx$

Concept / Formula Used: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Expand $(x + \sqrt{x})^3$ using the binomial identity with $a = x$ and $b = \sqrt{x} = x^{1/2}$:

$$\begin{aligned}(x + \sqrt{x})^3 &= x^3 + 3x^2(\sqrt{x}) + 3x(\sqrt{x})^2 + (\sqrt{x})^3 \\ &= x^3 + 3x^2 \cdot x^{1/2} + 3x \cdot x + x^{3/2} \\ &= x^3 + 3x^{5/2} + 3x^2 + x^{3/2}\end{aligned}$$

The integral becomes:

$$\int x^3 dx + 3 \int x^{5/2} dx + 3 \int x^2 dx + \int x^{3/2} dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned}\int x^3 dx &= \frac{x^4}{4} \\ \int x^{5/2} dx &= \frac{x^{7/2}}{\frac{7}{2}} = \frac{2}{7}x^{7/2} \\ \int x^2 dx &= \frac{x^3}{3} \\ \int x^{3/2} dx &= \frac{x^{5/2}}{\frac{5}{2}} = \frac{2}{5}x^{5/2}\end{aligned}$$

Substitute back and combine:

$$\begin{aligned}&= \frac{x^4}{4} + 3 \left(\frac{2}{7}x^{7/2} \right) + 3 \left(\frac{x^3}{3} \right) + \frac{2}{5}x^{5/2} + C \\ &= \frac{1}{4}x^4 + \frac{6}{7}x^{7/2} + x^3 + \frac{2}{5}x^{5/2} + C\end{aligned}$$

Final Answer

$$\int (x + \sqrt{x})^3 dx = \frac{1}{4}x^4 + \frac{6}{7}x^{7/2} + x^3 + \frac{2}{5}x^{5/2} + C$$

Question 9 (i)

Evaluate $\int \frac{(x+a)^2}{\sqrt{x}} dx$

TYPE 1 Power Rule of Integration

To Find: $\int \frac{(x+a)^2}{\sqrt{x}} dx$

Concept / Formula Used: $(x + a)^2 = x^2 + 2ax + a^2$, $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

Solution:

Expand the numerator $(x + a)^2$:

$$(x + a)^2 = x^2 + 2ax + a^2$$

Divide each term in the numerator by $\sqrt{x} = x^{1/2}$:

$$\begin{aligned}\frac{x^2}{x^{1/2}} &= x^{2-1/2} = x^{3/2} \\ \frac{2ax}{x^{1/2}} &= 2ax^{1-1/2} = 2ax^{1/2} \\ \frac{a^2}{x^{1/2}} &= a^2x^{-1/2}\end{aligned}$$

Rewrite the given integral:

$$\int x^{3/2} dx + 2a \int x^{1/2} dx + a^2 \int x^{-1/2} dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned}\int x^{3/2} dx &= \frac{x^{5/2}}{\frac{5}{2}} = \frac{2}{5}x^{5/2} \\ \int x^{1/2} dx &= \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2} \\ \int x^{-1/2} dx &= \frac{x^{1/2}}{\frac{1}{2}} = 2x^{1/2} = 2\sqrt{x}\end{aligned}$$

Substitute these integrated terms back:

$$\begin{aligned}&= \frac{2}{5}x^{5/2} + 2a \left(\frac{2}{3}x^{3/2} \right) + a^2(2\sqrt{x}) + C \\ &= \frac{2}{5}x^{5/2} + \frac{4a}{3}x^{3/2} + 2a^2\sqrt{x} + C\end{aligned}$$

Final Answer

$$\int \frac{(x + a)^2}{\sqrt{x}} dx = \frac{2}{5}x^{5/2} + \frac{4a}{3}x^{3/2} + 2a^2\sqrt{x} + C$$

Question 9 (ii)

Evaluate $\int \frac{x^3 - x^2 + x - 1}{x - 1} dx$

TYPE 1 Power Rule of Integration

★ Likely Exam Question

To Find: $\int \frac{x^3 - x^2 + x - 1}{x - 1} dx$

Concept / Formula Used: Factorization by grouping terms in the numerator

Solution:

Factorize the cubic numerator by grouping terms:

$$\begin{aligned} x^3 - x^2 + x - 1 &= x^2(x - 1) + 1(x - 1) \\ &= (x^2 + 1)(x - 1) \end{aligned}$$

Substitute the factored expression into the integrand:

$$\frac{(x^2 + 1)(x - 1)}{x - 1} = x^2 + 1 \quad (x \neq 1)$$

The integral simplifies to:

$$\int (x^2 + 1) dx = \int x^2 dx + \int 1 dx$$

Using standard integration rules:

$$\begin{aligned} \int x^2 dx &= \frac{x^3}{3} \\ \int 1 dx &= x \end{aligned}$$

Combining terms:

$$= \frac{1}{3}x^3 + x + C$$

Final Answer

$$\int \frac{x^3 - x^2 + x - 1}{x - 1} dx = \frac{1}{3}x^3 + x + C$$

Question 10 (i)

Evaluate $\int \frac{1 - \sin x}{\cos^2 x} dx$

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

To Find: $\int \frac{1 - \sin x}{\cos^2 x} dx$

Concept / Formula Used: $\int \sec^2 x \, dx = \tan x + C$, $\int \sec x \tan x \, dx = \sec x + C$

Solution:

Split the integrand into two separate fractions:

$$\frac{1 - \sin x}{\cos^2 x} = \frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x}$$

Using trigonometric reciprocal definitions:

$$\begin{aligned} \frac{1}{\cos^2 x} &= \sec^2 x \\ \frac{\sin x}{\cos^2 x} &= \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x \end{aligned}$$

Rewrite the integral:

$$\int (\sec^2 x - \sec x \tan x) \, dx = \int \sec^2 x \, dx - \int \sec x \tan x \, dx$$

Using standard trigonometric integral results:

$$\begin{aligned} \int \sec^2 x \, dx &= \tan x \\ \int \sec x \tan x \, dx &= \sec x \end{aligned}$$

Combining the integrated terms:

$$= \tan x - \sec x + C$$

Final Answer

$$\int \frac{1 - \sin x}{\cos^2 x} \, dx = \tan x - \sec x + C$$

Question 10 (ii)

Evaluate $\int \frac{e^{6 \log x} - e^{5 \log x}}{e^{4 \log x} - e^{3 \log x}} \, dx$

TYPE 4 Logarithmic Identity Simplification

★ Likely Exam Question

To Find: $\int \frac{e^{6 \log x} - e^{5 \log x}}{e^{4 \log x} - e^{3 \log x}} \, dx$

Concept / Formula Used: $e^{n \log x} = e^{\log(x^n)} = x^n$

Solution:

Simplify each exponential-logarithmic term using identity $e^{\log a} = a$:

$$e^{6 \log x} = e^{\log(x^6)} = x^6$$

$$e^{5 \log x} = e^{\log(x^5)} = x^5$$

$$e^{4 \log x} = e^{\log(x^4)} = x^4$$

$$e^{3 \log x} = e^{\log(x^3)} = x^3$$

Substitute these back into the integrand:

$$\frac{x^6 - x^5}{x^4 - x^3}$$

Factor out common powers from numerator and denominator:

$$\frac{x^5(x-1)}{x^3(x-1)} = \frac{x^5}{x^3} = x^2 \quad (x \neq 1, x > 0)$$

The integral simplifies to:

$$\int x^2 dx$$

Using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$= \frac{1}{3}x^3 + C$$

Final Answer

$$\int \frac{e^{6 \log x} - e^{5 \log x}}{e^{4 \log x} - e^{3 \log x}} dx = \frac{1}{3}x^3 + C$$

Question 11 (i)

Evaluate $\int \sec x(\sec x + \tan x) dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \sec x(\sec x + \tan x) dx$

Concept / Formula Used: $\int \sec^2 x dx = \tan x + C$, $\int \sec x \tan x dx = \sec x + C$

Solution:

Expand the integrand:

$$\sec x(\sec x + \tan x) = \sec^2 x + \sec x \tan x$$

Rewrite as two separate integrals:

$$\int \sec^2 x \, dx + \int \sec x \tan x \, dx$$

Applying standard trigonometric integrals:

$$\begin{aligned}\int \sec^2 x \, dx &= \tan x \\ \int \sec x \tan x \, dx &= \sec x\end{aligned}$$

Combining the integrated terms:

$$= \tan x + \sec x + C$$

Final Answer

$$\int \sec x (\sec x + \tan x) \, dx = \tan x + \sec x + C$$

Question 11 (ii)

Evaluate $\int \frac{\sec^2 x}{\csc^2 x} \, dx$

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

To Find: $\int \frac{\sec^2 x}{\csc^2 x} \, dx$

Concept / Formula Used: $\sec^2 x = \frac{1}{\cos^2 x}$, $\csc^2 x = \frac{1}{\sin^2 x}$, $\tan^2 x = \sec^2 x - 1$

Solution:

Express $\sec^2 x$ and $\csc^2 x$ in terms of $\sin x$ and $\cos x$:

$$\frac{\sec^2 x}{\csc^2 x} = \frac{\frac{1}{\cos^2 x}}{\frac{1}{\sin^2 x}} = \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$$

Using Pythagorean identity $\tan^2 x = \sec^2 x - 1$:

$$\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \int \sec^2 x \, dx - \int 1 \, dx$$

Using standard integration results:

$$\begin{aligned}\int \sec^2 x \, dx &= \tan x \\ \int 1 \, dx &= x\end{aligned}$$

Combining terms:

$$= \tan x - x + C$$

Final Answer

$$\int \frac{\sec^2 x}{\csc^2 x} dx = \tan x - x + C$$

Question 12 (i)

Evaluate $\int \left(\frac{\cot x}{\sin x} - \frac{\tan x}{\cos x} - 3 \tan^2 x \right) dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \left(\frac{\cot x}{\sin x} - \frac{\tan x}{\cos x} - 3 \tan^2 x \right) dx$

Concept / Formula Used: $\frac{\cot x}{\sin x} = \csc x \cot x$, $\frac{\tan x}{\cos x} = \sec x \tan x$, $\tan^2 x = \sec^2 x - 1$

Solution:

Rewrite each term using trigonometric identities:

$$\begin{aligned} \frac{\cot x}{\sin x} &= \csc x \cot x \\ \frac{\tan x}{\cos x} &= \sec x \tan x \\ 3 \tan^2 x &= 3(\sec^2 x - 1) = 3 \sec^2 x - 3 \end{aligned}$$

Substitute back into the integrand:

$$\int (\csc x \cot x - \sec x \tan x - 3 \sec^2 x + 3) dx$$

Separate into individual integrals:

$$\int \csc x \cot x dx - \int \sec x \tan x dx - 3 \int \sec^2 x dx + 3 \int 1 dx$$

Using standard trigonometric integral results:

$$\begin{aligned} \int \csc x \cot x dx &= -\csc x \\ \int \sec x \tan x dx &= \sec x \\ 3 \int \sec^2 x dx &= 3 \tan x \\ 3 \int 1 dx &= 3x \end{aligned}$$

Combining all integrated terms:

$$= -\csc x - \sec x - 3 \tan x + 3x + C$$

Final Answer

$$\int \left(\frac{\cot x}{\sin x} - \frac{\tan x}{\cos x} - 3 \tan^2 x \right) dx = -\csc x - \sec x - 3 \tan x + 3x + C$$

Question 12 (ii)

Evaluate $\int \frac{3 - 5 \sin x}{\cos^2 x} dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \frac{3 - 5 \sin x}{\cos^2 x} dx$

Concept / Formula Used: $\frac{1}{\cos^2 x} = \sec^2 x$, $\frac{\sin x}{\cos^2 x} = \sec x \tan x$

Solution:

Split into two fractions:

$$\frac{3 - 5 \sin x}{\cos^2 x} = \frac{3}{\cos^2 x} - \frac{5 \sin x}{\cos^2 x} = 3 \sec^2 x - 5 \sec x \tan x$$

Rewrite as separate integrals:

$$3 \int \sec^2 x dx - 5 \int \sec x \tan x dx$$

Using standard integration formulas:

$$3 \int \sec^2 x dx = 3 \tan x$$

$$5 \int \sec x \tan x dx = 5 \sec x$$

Combining the integrated terms:

$$= 3 \tan x - 5 \sec x + C$$

Final Answer

$$\int \frac{3 - 5 \sin x}{\cos^2 x} dx = 3 \tan x - 5 \sec x + C$$

Question 13 (i)

Evaluate $\int \frac{1}{1 + \cos x} dx$

TYPE 5 Conjugate Multiplication

★ Likely Exam Question

To Find: $\int \frac{1}{1 + \cos x} dx$

Concept / Formula Used: Multiply numerator and denominator by $(1 - \cos x)$

Solution:

Multiply numerator and denominator by the conjugate $(1 - \cos x)$:

$$\frac{1}{1 + \cos x} \cdot \frac{1 - \cos x}{1 - \cos x} = \frac{1 - \cos x}{1 - \cos^2 x}$$

Using Pythagorean identity $1 - \cos^2 x = \sin^2 x$:

$$= \frac{1 - \cos x}{\sin^2 x}$$

Split into two fractions:

$$\frac{1}{\sin^2 x} - \frac{\cos x}{\sin^2 x} = \csc^2 x - \csc x \cot x$$

Rewrite as separate integrals:

$$\int \csc^2 x dx - \int \csc x \cot x dx$$

Using standard trigonometric integral results:

$$\begin{aligned} \int \csc^2 x dx &= -\cot x \\ \int \csc x \cot x dx &= -\csc x \end{aligned}$$

Combining terms:

$$\begin{aligned} &= -\cot x - (-\csc x) + C \\ &= -\cot x + \csc x + C \end{aligned}$$

Final Answer

$$\int \frac{1}{1 + \cos x} dx = -\cot x + \csc x + C$$

Question 13 (ii)

Evaluate $\int \frac{1}{1 - \sin x} dx$

TYPE 5 Conjugate Multiplication

To Find: $\int \frac{1}{1 - \sin x} dx$

Concept / Formula Used: Multiply numerator and denominator by $(1 + \sin x)$

Solution:

Multiply numerator and denominator by the conjugate $(1 + \sin x)$:

$$\frac{1}{1 - \sin x} \cdot \frac{1 + \sin x}{1 + \sin x} = \frac{1 + \sin x}{1 - \sin^2 x}$$

Using Pythagorean identity $1 - \sin^2 x = \cos^2 x$:

$$= \frac{1 + \sin x}{\cos^2 x}$$

Split into two separate fractions:

$$\frac{1}{\cos^2 x} + \frac{\sin x}{\cos^2 x} = \sec^2 x + \sec x \tan x$$

Rewrite as separate integrals:

$$\int \sec^2 x dx + \int \sec x \tan x dx$$

Using standard integration formulas:

$$\begin{aligned} \int \sec^2 x dx &= \tan x \\ \int \sec x \tan x dx &= \sec x \end{aligned}$$

Combining terms:

$$= \tan x + \sec x + C$$

Final Answer

$$\int \frac{1}{1 - \sin x} dx = \tan x + \sec x + C$$

Question 14 (i)

Evaluate $\int \frac{1}{1 + \csc x} dx$

TYPE 5 Conjugate Multiplication

To Find: $\int \frac{1}{1 + \csc x} dx$

Concept / Formula Used: $\csc x = \frac{1}{\sin x}$

Solution:

Express $\csc x$ in terms of $\sin x$:

$$\frac{1}{1 + \csc x} = \frac{1}{1 + \frac{1}{\sin x}} = \frac{\sin x}{\sin x + 1}$$

Add and subtract 1 in the numerator:

$$\frac{\sin x + 1 - 1}{\sin x + 1} = \frac{\sin x + 1}{\sin x + 1} - \frac{1}{1 + \sin x} = 1 - \frac{1}{1 + \sin x}$$

Simplify $\frac{1}{1 + \sin x}$ using conjugate multiplication:

$$\frac{1}{1 + \sin x} \cdot \frac{1 - \sin x}{1 - \sin x} = \frac{1 - \sin x}{1 - \sin^2 x} = \frac{1 - \sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x$$

Substitute this back into the integrand:

$$1 - (\sec^2 x - \sec x \tan x) = 1 - \sec^2 x + \sec x \tan x$$

Integrate term-by-term:

$$\int 1 dx - \int \sec^2 x dx + \int \sec x \tan x dx$$

Using standard integration results:

$$= x - \tan x + \sec x + C$$

Final Answer

$$\int \frac{1}{1 + \csc x} dx = x - \tan x + \sec x + C$$

Question 14 (ii)

Evaluate $\int \frac{\sin^2 x}{1 + \cos x} dx$

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

To Find: $\int \frac{\sin^2 x}{1 + \cos x} dx$

Concept / Formula Used: $\sin^2 x = 1 - \cos^2 x = (1 - \cos x)(1 + \cos x)$

Solution:

Use the Pythagorean identity $\sin^2 x = 1 - \cos^2 x$:

$$\frac{\sin^2 x}{1 + \cos x} = \frac{1 - \cos^2 x}{1 + \cos x}$$

Factor the numerator using difference of two squares $a^2 - b^2 = (a - b)(a + b)$:

$$\frac{(1 - \cos x)(1 + \cos x)}{1 + \cos x} = 1 - \cos x \quad (\text{for } \cos x \neq -1)$$

The integral simplifies to:

$$\int (1 - \cos x) dx = \int 1 dx - \int \cos x dx$$

Evaluating standard elementary integrals:

$$\begin{aligned} \int 1 dx &= x \\ \int \cos x dx &= \sin x \end{aligned}$$

Combining terms:

$$= x - \sin x + C$$

Final Answer

$$\int \frac{\sin^2 x}{1 + \cos x} dx = x - \sin x + C$$

Question 15 (i)

Evaluate $\int \left(\sqrt{x} - \sin \frac{x}{2} \cos \frac{x}{2} + 5 \right) dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \left(\sqrt{x} - \sin \frac{x}{2} \cos \frac{x}{2} + 5 \right) dx$

Concept / Formula Used: $2 \sin \frac{x}{2} \cos \frac{x}{2} = \sin x$

Solution:

Use double angle identity $\sin \frac{x}{2} \cos \frac{x}{2} = \frac{1}{2} \sin x$: Rewrite the integrand:

$$x^{1/2} - \frac{1}{2} \sin x + 5$$

Split into separate integrals:

$$\int x^{1/2} dx - \frac{1}{2} \int \sin x dx + 5 \int 1 dx$$

Applying standard integration rules:

$$\begin{aligned}\int x^{1/2} dx &= \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3}x^{3/2} \\ \int \sin x dx &= -\cos x \\ \int 1 dx &= x\end{aligned}$$

Combining all terms:

$$\begin{aligned}&= \frac{2}{3}x^{3/2} - \frac{1}{2}(-\cos x) + 5x + C \\ &= \frac{2}{3}x^{3/2} + \frac{1}{2}\cos x + 5x + C\end{aligned}$$

Final Answer

$$\int \left(\sqrt{x} - \sin \frac{x}{2} \cos \frac{x}{2} + 5 \right) dx = \frac{2}{3}x^{3/2} + \frac{1}{2}\cos x + 5x + C$$

Question 15 (ii)

Evaluate $\int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx$

Concept / Formula Used: $\cos 2x = \cos^2 x - \sin^2 x$

Solution:

Substitute double-angle identity $\cos 2x = \cos^2 x - \sin^2 x$:

$$\frac{\cos 2x}{\sin^2 x \cos^2 x} = \frac{\cos^2 x - \sin^2 x}{\sin^2 x \cos^2 x}$$

Split into two fractions:

$$\frac{\cos^2 x}{\sin^2 x \cos^2 x} - \frac{\sin^2 x}{\sin^2 x \cos^2 x} = \frac{1}{\sin^2 x} - \frac{1}{\cos^2 x} = \csc^2 x - \sec^2 x$$

Rewrite as separate integrals:

$$\int \csc^2 x dx - \int \sec^2 x dx$$

Using standard results $\int \csc^2 x dx = -\cot x$ and $\int \sec^2 x dx = \tan x$:

$$= -\cot x - \tan x + C$$

Final Answer

$$\int \frac{\cos 2x}{\sin^2 x \cos^2 x} dx = -\cot x - \tan x + C$$

Question 16 (i)

Evaluate $\int \frac{\sec x}{\sec x + \tan x} dx$

TYPE 5 Conjugate Multiplication

To Find: $\int \frac{\sec x}{\sec x + \tan x} dx$

Concept / Formula Used: Multiply numerator and denominator by $(\sec x - \tan x)$ **Solution:**Multiply numerator and denominator by $(\sec x - \tan x)$:

$$\frac{\sec x(\sec x - \tan x)}{(\sec x + \tan x)(\sec x - \tan x)} = \frac{\sec^2 x - \sec x \tan x}{\sec^2 x - \tan^2 x}$$

Using identity $\sec^2 x - \tan^2 x = 1$:

$$= \sec^2 x - \sec x \tan x$$

Rewrite as separate integrals:

$$\int \sec^2 x dx - \int \sec x \tan x dx$$

Using standard integration results:

$$\begin{aligned} \int \sec^2 x dx &= \tan x \\ \int \sec x \tan x dx &= \sec x \end{aligned}$$

Combining terms:

$$= \tan x - \sec x + C$$

Final Answer

$$\int \frac{\sec x}{\sec x + \tan x} dx = \tan x - \sec x + C$$

Question 16 (ii)

Evaluate $\int \frac{\csc x}{\csc x - \cot x} dx$

TYPE 5 Conjugate Multiplication

To Find: $\int \frac{\csc x}{\csc x - \cot x} dx$

Concept / Formula Used: Multiply numerator and denominator by $(\csc x + \cot x)$

Solution:

Multiply numerator and denominator by $(\csc x + \cot x)$:

$$\frac{\csc x(\csc x + \cot x)}{(\csc x - \cot x)(\csc x + \cot x)} = \frac{\csc^2 x + \csc x \cot x}{\csc^2 x - \cot^2 x}$$

Using identity $\csc^2 x - \cot^2 x = 1$:

$$= \csc^2 x + \csc x \cot x$$

Rewrite as separate integrals:

$$\int \csc^2 x dx + \int \csc x \cot x dx$$

Using standard integration formulas:

$$\begin{aligned} \int \csc^2 x dx &= -\cot x \\ \int \csc x \cot x dx &= -\csc x \end{aligned}$$

Combining terms:

$$= -\cot x - \csc x + C$$

Final Answer

$$\int \frac{\csc x}{\csc x - \cot x} dx = -\cot x - \csc x + C$$

Question 17 (i)

Evaluate $\int \frac{x^6 + 1}{x^2 + 1} dx$

TYPE 1 Power Rule of Integration

To Find: $\int \frac{x^6 + 1}{x^2 + 1} dx$

Concept / Formula Used: $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$

Solution:

Factorize $x^6 + 1$ as sum of cubes $(x^2)^3 + 1^3$:

$$x^6 + 1 = (x^2 + 1)((x^2)^2 - x^2 \cdot 1 + 1^2) = (x^2 + 1)(x^4 - x^2 + 1)$$

Substitute into the integrand:

$$\frac{(x^2 + 1)(x^4 - x^2 + 1)}{x^2 + 1} = x^4 - x^2 + 1$$

The integral simplifies to:

$$\int (x^4 - x^2 + 1) dx = \int x^4 dx - \int x^2 dx + \int 1 dx$$

Integrating term-by-term using power rule:

$$\begin{aligned}\int x^4 dx &= \frac{x^5}{5} \\ \int x^2 dx &= \frac{x^3}{3} \\ \int 1 dx &= x\end{aligned}$$

Combining terms:

$$= \frac{1}{5}x^5 - \frac{1}{3}x^3 + x + C$$

Final Answer

$$\int \frac{x^6 + 1}{x^2 + 1} dx = \frac{1}{5}x^5 - \frac{1}{3}x^3 + x + C$$

Question 17 (ii)

Evaluate $\int \frac{6 \sin^3 x + 5 \cos^3 x}{\sin^2 x \cos^2 x} dx$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \frac{6 \sin^3 x + 5 \cos^3 x}{\sin^2 x \cos^2 x} dx$

Concept / Formula Used: $\frac{\sin x}{\cos^2 x} = \sec x \tan x$, $\frac{\cos x}{\sin^2 x} = \csc x \cot x$

Solution:

Split into two terms:

$$\frac{6 \sin^3 x + 5 \cos^3 x}{\sin^2 x \cos^2 x} = \frac{6 \sin^3 x}{\sin^2 x \cos^2 x} + \frac{5 \cos^3 x}{\sin^2 x \cos^2 x} = \frac{6 \sin x}{\cos^2 x} + \frac{5 \cos x}{\sin^2 x}$$

Rewrite using trigonometric identities:

$$\frac{6 \sin x}{\cos^2 x} = 6 \sec x \tan x$$

$$\frac{5 \cos x}{\sin^2 x} = 5 \csc x \cot x$$

Rewrite the integral:

$$6 \int \sec x \tan x \, dx + 5 \int \csc x \cot x \, dx$$

Applying standard trigonometric integrals:

$$\int \sec x \tan x \, dx = \sec x$$

$$\int \csc x \cot x \, dx = -\csc x$$

Combining terms:

$$= 6 \sec x - 5 \csc x + C$$

Final Answer

$$\int \frac{6 \sin^3 x + 5 \cos^3 x}{\sin^2 x \cos^2 x} \, dx = 6 \sec x - 5 \csc x + C$$

Question 18 (i)

Evaluate $\int \sqrt{1 + \cos 2x} \, dx$, $0 < x < \frac{\pi}{2}$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \sqrt{1 + \cos 2x} \, dx$, $0 < x < \frac{\pi}{2}$

Concept / Formula Used: $1 + \cos 2x = 2 \cos^2 x$

Solution:

Use trigonometric identity $1 + \cos 2x = 2 \cos^2 x$:

$$\sqrt{1 + \cos 2x} = \sqrt{2 \cos^2 x} = \sqrt{2} |\cos x|$$

Since $0 < x < \frac{\pi}{2}$, x lies in the first quadrant where $\cos x > 0$, so $|\cos x| = \cos x$. The integral becomes:

$$\sqrt{2} \int \cos x \, dx$$

Using standard result $\int \cos x \, dx = \sin x + C$:

$$= \sqrt{2} \sin x + C$$

Final Answer

$$\int \sqrt{1 + \cos 2x} \, dx = \sqrt{2} \sin x + C$$

Question 18 (ii)

Evaluate $\int \sqrt{1 - \cos 2x} \, dx$, $0 < x < \pi$

TYPE 2 Trigonometric Identity Simplification

To Find: $\int \sqrt{1 - \cos 2x} \, dx$, $0 < x < \pi$

Concept / Formula Used: $1 - \cos 2x = 2 \sin^2 x$

Solution:

Use trigonometric identity $1 - \cos 2x = 2 \sin^2 x$:

$$\sqrt{1 - \cos 2x} = \sqrt{2 \sin^2 x} = \sqrt{2} |\sin x|$$

Since $0 < x < \pi$, x lies in the first or second quadrant where $\sin x > 0$, so $|\sin x| = \sin x$.
The integral becomes:

$$\sqrt{2} \int \sin x \, dx$$

Using standard result $\int \sin x \, dx = -\cos x + C$:

$$= -\sqrt{2} \cos x + C$$

Final Answer

$$\int \sqrt{1 - \cos 2x} \, dx = -\sqrt{2} \cos x + C$$

Question 19 (i)

Evaluate $\int \cot^{-1} \left(\frac{\sin 2x}{1 - \cos 2x} \right) dx$

TYPE 6 Inverse Trigonometric Simplification

To Find: $\int \cot^{-1} \left(\frac{\sin 2x}{1 - \cos 2x} \right) dx$

Concept / Formula Used: $\sin 2x = 2 \sin x \cos x$, $1 - \cos 2x = 2 \sin^2 x$

Solution:

Simplify the fraction inside $\cot^{-1}$:

$$\frac{\sin 2x}{1 - \cos 2x} = \frac{2 \sin x \cos x}{2 \sin^2 x} = \frac{\cos x}{\sin x} = \cot x$$

Substitute back into inverse trigonometric function:

$$\cot^{-1}(\cot x) = x$$

The integral simplifies to:

$$\int x \, dx$$

Using the standard power rule $\int x \, dx = \frac{x^2}{2} + C$:

$$= \frac{1}{2}x^2 + C$$

Final Answer

$$\int \cot^{-1} \left(\frac{\sin 2x}{1 - \cos 2x} \right) dx = \frac{1}{2}x^2 + C$$

Question 19 (ii)

Evaluate $\int \tan^{-1} \left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \right) dx$

TYPE 6 Inverse Trigonometric Simplification

To Find: $\int \tan^{-1} \left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \right) dx$

Concept / Formula Used: $1 - \cos 2x = 2 \sin^2 x$, $1 + \cos 2x = 2 \cos^2 x$

Solution:

Simplify the fraction inside the square root:

$$\frac{1 - \cos 2x}{1 + \cos 2x} = \frac{2 \sin^2 x}{2 \cos^2 x} = \tan^2 x$$

Taking the square root:

$$\sqrt{\tan^2 x} = \tan x$$

Substitute into the inverse trigonometric function:

$$\tan^{-1}(\tan x) = x$$

The integral simplifies to:

$$\int x \, dx$$

Using the standard power rule $\int x \, dx = \frac{x^2}{2} + C$:

$$= \frac{1}{2}x^2 + C$$

Final Answer

$$\int \tan^{-1} \left(\sqrt{\frac{1 - \cos 2x}{1 + \cos 2x}} \right) dx = \frac{1}{2}x^2 + C$$

Question 20 (i)

Evaluate $\int \tan^{-1}(\csc x - \cot x) \, dx$

TYPE 6 Inverse Trigonometric Simplification

To Find: $\int \tan^{-1}(\csc x - \cot x) \, dx$

Concept / Formula Used: $\csc x - \cot x = \frac{1 - \cos x}{\sin x} = \tan \left(\frac{x}{2} \right)$

Solution:

Express $\csc x - \cot x$ in terms of sine and cosine:

$$\csc x - \cot x = \frac{1}{\sin x} - \frac{\cos x}{\sin x} = \frac{1 - \cos x}{\sin x}$$

Using half-angle identities $1 - \cos x = 2 \sin^2 \left(\frac{x}{2} \right)$ and $\sin x = 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)$:

$$\frac{1 - \cos x}{\sin x} = \frac{2 \sin^2 \left(\frac{x}{2} \right)}{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)} = \frac{\sin \left(\frac{x}{2} \right)}{\cos \left(\frac{x}{2} \right)} = \tan \left(\frac{x}{2} \right)$$

Substitute into inverse tangent function:

$$\tan^{-1} \left(\tan \left(\frac{x}{2} \right) \right) = \frac{x}{2}$$

The integral becomes:

$$\int \frac{x}{2} \, dx = \frac{1}{2} \int x \, dx$$

Using the standard power rule $\int x \, dx = \frac{x^2}{2}$:

$$= \frac{1}{2} \left(\frac{x^2}{2} \right) + C = \frac{1}{4}x^2 + C$$

Final Answer

$$\int \tan^{-1}(\csc x - \cot x) \, dx = \frac{1}{4}x^2 + C$$

Question 20 (ii)

Evaluate $\int \cos^{-1} \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) dx$

TYPE 6 Inverse Trigonometric Simplification

To Find: $\int \cos^{-1} \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) dx$

Concept / Formula Used: $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$

Solution:

Use double angle identity for cosine:

$$\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$$

Substitute into inverse cosine function:

$$\cos^{-1}(\cos 2x) = 2x$$

The integral becomes:

$$\int 2x \, dx = 2 \int x \, dx$$

Using the standard power rule $\int x \, dx = \frac{x^2}{2}$:

$$= 2 \left(\frac{x^2}{2} \right) + C = x^2 + C$$

Final Answer

$$\int \cos^{-1} \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) dx = x^2 + C$$

Question 21

If $f'(x) = 4x^3 - 6$ and $f(0) = 3$, find $f(x)$.

TYPE 7 Integration From Derivative

★ Likely Exam Question

Given: $f'(x) = 4x^3 - 6$, $f(0) = 3$

To Find: $f(x)$

Concept / Formula Used: $f(x) = \int f'(x) dx$

Solution:

By definition of anti-derivative:

$$f(x) = \int f'(x) dx = \int (4x^3 - 6) dx$$

Integrating term-by-term:

$$4 \int x^3 dx = 4 \cdot \frac{x^4}{4} = x^4$$

$$6 \int 1 dx = 6x$$

So,

$$f(x) = x^4 - 6x + C$$

Substitute initial condition $f(0) = 3$ to find constant C :

$$f(0) = (0)^4 - 6(0) + C = 3$$

$$0 - 0 + C = 3 \implies C = 3$$

Substitute $C = 3$ back into the function $f(x)$:

$$f(x) = x^4 - 6x + 3$$

Final Answer

$$f(x) = x^4 - 6x + 3$$

7.3 – Integration by Substitution: Linear and Algebraic Forms

Question 1 (i)

Find all the anti-derivatives of $(ax + b)^3$.

TYPE 12a Direct/Algebraic Substitution

Given: Function to integrate: $f(x) = (ax + b)^3$

Concept / Formula Used: Power rule for integration: $\int u^n du = \frac{u^{n+1}}{n+1} + C \quad (n \neq -1)$

Solution:

We evaluate the integral:

$$I = \int (ax + b)^3 dx$$

Let $u = ax + b$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= a \\ du &= a dx \\ dx &= \frac{du}{a}\end{aligned}$$

Substituting u and dx into the integral:

$$\begin{aligned}I &= \int u^3 \left(\frac{du}{a} \right) \\ &= \frac{1}{a} \int u^3 du\end{aligned}$$

Using the power rule:

$$\begin{aligned}I &= \frac{1}{a} \left[\frac{u^{3+1}}{3+1} \right] + C \\ &= \frac{1}{a} \left[\frac{u^4}{4} \right] + C \\ &= \frac{u^4}{4a} + C\end{aligned}$$

Back-substituting $u = ax + b$:

$$I = \frac{(ax + b)^4}{4a} + C$$

Final Answer

$$\int (ax + b)^3 dx = \frac{1}{4a}(ax + b)^4 + C$$

Question 1 (ii)

Find all the anti-derivatives of $4e^{3x} + 1$.

TYPE 12c Substitution Involving Exponential Functions

★ Likely Exam Question

Given: Function to integrate: $f(x) = 4e^{3x} + 1$

Concept / Formula Used: Exponential integral rule: $\int e^{kx} dx = \frac{1}{k}e^{kx} + C$

Solution:

We evaluate the integral:

$$I = \int (4e^{3x} + 1) dx = 4 \int e^{3x} dx + \int 1 dx$$

For the integral $\int e^{3x} dx$, let $u = 3x$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= 3 \\ du &= 3 dx \\ dx &= \frac{du}{3}\end{aligned}$$

Substituting into the integral:

$$\begin{aligned}\int e^{3x} dx &= \int e^u \left(\frac{du}{3} \right) \\ &= \frac{1}{3} \int e^u du \\ &= \frac{1}{3} e^u \\ &= \frac{1}{3} e^{3x}\end{aligned}$$

Substituting this back into I :

$$\begin{aligned}I &= 4 \left(\frac{1}{3} e^{3x} \right) + x + C \\ &= \frac{4}{3} e^{3x} + x + C\end{aligned}$$

Final Answer

$$\int (4e^{3x} + 1) dx = \frac{4}{3} e^{3x} + x + C$$

Question 2 (i)

Find all the anti-derivatives of $\frac{1}{\sqrt{7-2x}}$.

TYPE 12b Substitution Involving Radical Expressions

Given: Function to integrate: $f(x) = \frac{1}{\sqrt{7-2x}}$

Concept / Formula Used: Power rule for integration: $\int u^{-1/2} du = 2\sqrt{u} + C$

Solution:

We evaluate the integral:

$$I = \int \frac{dx}{\sqrt{7-2x}}$$

Let $u = 7 - 2x$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= -2 \\ du &= -2 dx \\ dx &= -\frac{du}{2}\end{aligned}$$

Substituting u and dx into the integral:

$$\begin{aligned}I &= \int \frac{1}{\sqrt{u}} \left(-\frac{du}{2}\right) \\ &= -\frac{1}{2} \int u^{-1/2} du\end{aligned}$$

Using the power rule:

$$\begin{aligned}I &= -\frac{1}{2} \left[\frac{u^{(-1/2)+1}}{-\frac{1}{2}+1} \right] + C \\ &= -\frac{1}{2} \left[\frac{u^{1/2}}{1/2} \right] + C \\ &= -u^{1/2} + C \\ &= -\sqrt{u} + C\end{aligned}$$

Back-substituting $u = 7 - 2x$:

$$I = -\sqrt{7-2x} + C$$

Final Answer

$$\int \frac{dx}{\sqrt{7-2x}} = -\sqrt{7-2x} + C$$

Question 2 (ii)

Find all the anti-derivatives of $\sqrt{ax+b}$.

TYPE 12b Substitution Involving Radical Expressions

★ Likely Exam Question

Given: Function to integrate: $f(x) = \sqrt{ax+b}$

Concept / Formula Used: Power rule for integration: $\int u^{1/2} du = \frac{2}{3}u^{3/2} + C$

Solution:

We evaluate the integral:

$$I = \int \sqrt{ax+b} dx$$

Let $u = ax + b$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= a \\ du &= a dx \\ dx &= \frac{du}{a}\end{aligned}$$

Substituting u and dx into the integral:

$$\begin{aligned}I &= \int \sqrt{u} \left(\frac{du}{a} \right) \\ &= \frac{1}{a} \int u^{1/2} du\end{aligned}$$

Integrating:

$$\begin{aligned}I &= \frac{1}{a} \left[\frac{u^{3/2}}{3/2} \right] + C \\ &= \frac{2}{3a} u^{3/2} + C\end{aligned}$$

Back-substituting $u = ax + b$:

$$I = \frac{2}{3a} (ax+b)^{3/2} + C$$

Final Answer

$$\int \sqrt{ax+b} dx = \frac{2}{3a} (ax+b)^{3/2} + C$$

Question 3 (i)

Find all the anti-derivatives of $\tan^2(2x - 3)$.

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

Given: Function to integrate: $f(x) = \tan^2(2x - 3)$

Concept / Formula Used: Trigonometric identity: $\tan^2 \theta = \sec^2 \theta - 1$

Solution:

Using the identity $\tan^2 \theta = \sec^2 \theta - 1$, we rewrite the function:

$$f(x) = \sec^2(2x - 3) - 1$$

Now evaluating the integral:

$$\begin{aligned} I &= \int [\sec^2(2x - 3) - 1] dx \\ &= \int \sec^2(2x - 3) dx - \int 1 dx \end{aligned}$$

For $\int \sec^2(2x - 3) dx$, let $u = 2x - 3$.

Differentiating with respect to x :

$$du = 2 dx$$

$$dx = \frac{du}{2}$$

Substituting into the integral:

$$\begin{aligned} \int \sec^2(2x - 3) dx &= \int \sec^2 u \left(\frac{du}{2} \right) \\ &= \frac{1}{2} \int \sec^2 u du \\ &= \frac{1}{2} \tan u \\ &= \frac{1}{2} \tan(2x - 3) \end{aligned}$$

Therefore, combining the results:

$$I = \frac{1}{2} \tan(2x - 3) - x + C$$

Final Answer

$$\int \tan^2(2x - 3) dx = \frac{1}{2} \tan(2x - 3) - x + C$$

Question 3 (ii)

Find all the anti-derivatives of $5 \csc^2(2 - 7x)$.

TYPE 12a Direct/Algebraic Substitution

Given: Function to integrate: $f(x) = 5 \csc^2(2 - 7x)$

Concept / Formula Used: Standard integral: $\int \csc^2 u \, du = -\cot u + C$

Solution:

We evaluate the integral:

$$I = \int 5 \csc^2(2 - 7x) \, dx = 5 \int \csc^2(2 - 7x) \, dx$$

Let $u = 2 - 7x$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= -7 \\ du &= -7 \, dx \\ dx &= -\frac{du}{7}\end{aligned}$$

Substituting u and dx into the integral:

$$\begin{aligned}I &= 5 \int \csc^2 u \left(-\frac{du}{7}\right) \\ &= -\frac{5}{7} \int \csc^2 u \, du\end{aligned}$$

Using the standard result $\int \csc^2 u \, du = -\cot u$:

$$\begin{aligned}I &= -\frac{5}{7}(-\cot u) + C \\ &= \frac{5}{7} \cot u + C\end{aligned}$$

Back-substituting $u = 2 - 7x$:

$$I = \frac{5}{7} \cot(2 - 7x) + C$$

Final Answer

$$\int 5 \csc^2(2 - 7x) \, dx = \frac{5}{7} \cot(2 - 7x) + C$$

Question 4 (i)

Find all the anti-derivatives of $\sqrt{e^x}$.

TYPE 12b Substitution Involving Radical Expressions

Given: Function to integrate: $f(x) = \sqrt{e^x}$

Concept / Formula Used: Exponent property: $\sqrt{e^x} = (e^x)^{1/2} = e^{x/2}$

Solution:

Rewriting the integrand using index laws:

$$f(x) = e^{x/2}$$

Evaluating the integral:

$$I = \int e^{x/2} dx$$

Let $u = \frac{x}{2}$.

Differentiating with respect to x :

$$\begin{aligned} du &= \frac{1}{2} dx \\ dx &= 2 du \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} I &= \int e^u (2 du) \\ &= 2 \int e^u du \\ &= 2e^u + C \end{aligned}$$

Back-substituting $u = \frac{x}{2}$:

$$I = 2e^{x/2} + C = 2\sqrt{e^x} + C$$

Final Answer

$$\int \sqrt{e^x} dx = 2\sqrt{e^x} + C$$

Question 4 (ii)

Find all the anti-derivatives of $\csc(3x+2) \cot(3x+2)$.

TYPE 12a Direct/Algebraic Substitution

Given: Function to integrate: $f(x) = \csc(3x + 2) \cot(3x + 2)$

Concept / Formula Used: Standard integral: $\int \csc u \cot u \, du = -\csc u + C$

Solution:

We evaluate the integral:

$$I = \int \csc(3x + 2) \cot(3x + 2) \, dx$$

Let $u = 3x + 2$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= 3 \\ du &= 3 \, dx \\ dx &= \frac{du}{3}\end{aligned}$$

Substituting u and dx into the integral:

$$\begin{aligned}I &= \int \csc u \cot u \left(\frac{du}{3} \right) \\ &= \frac{1}{3} \int \csc u \cot u \, du\end{aligned}$$

Using the standard integral result:

$$\begin{aligned}I &= \frac{1}{3}(-\csc u) + C \\ &= -\frac{1}{3} \csc u + C\end{aligned}$$

Back-substituting $u = 3x + 2$:

$$I = -\frac{1}{3} \csc(3x + 2) + C$$

Final Answer

$$\int \csc(3x + 2) \cot(3x + 2) \, dx = -\frac{1}{3} \csc(3x + 2) + C$$

Question 5 (i)

Evaluate $\int e^{2x+3} \, dx$.

TYPE 12c Substitution Involving Exponential Functions

★ Likely Exam Question

Given: Integral: $I = \int e^{2x+3} dx$

Concept / Formula Used: Exponential integral rule: $\int e^u du = e^u + C$

Solution:

Let $u = 2x + 3$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= 2 \\ du &= 2 dx \\ dx &= \frac{du}{2}\end{aligned}$$

Substituting u and dx :

$$\begin{aligned}I &= \int e^u \left(\frac{du}{2} \right) \\ &= \frac{1}{2} \int e^u du \\ &= \frac{1}{2} e^u + C\end{aligned}$$

Back-substituting $u = 2x + 3$:

$$I = \frac{1}{2} e^{2x+3} + C$$

Final Answer

$$\int e^{2x+3} dx = \frac{1}{2} e^{2x+3} + C$$

Question 5 (ii)

Evaluate $\int (e^{3a \log x} + e^{3x \log a}) dx$.

TYPE 4 Logarithmic Identity Simplification

Given: Integral: $I = \int (e^{3a \log x} + e^{3x \log a}) dx$

Concept / Formula Used: Logarithm property: $e^{\log(y^k)} = y^k$

Solution:

Simplifying each exponential term using logarithm rules:

$$e^{3a \log x} = e^{\log(x^{3a})} = x^{3a}$$

$$e^{3x \log a} = e^{\log(a^{3x})} = a^{3x} = (a^3)^x$$

Now rewriting the integral:

$$I = \int (x^{3a} + a^{3x}) dx$$

$$= \int x^{3a} dx + \int a^{3x} dx$$

Using the integration rules $\int x^n dx = \frac{x^{n+1}}{n+1}$ and $\int a^{kx} dx = \frac{a^{kx}}{k \log a}$:

$$I = \frac{x^{3a+1}}{3a+1} + \frac{a^{3x}}{3 \log a} + C$$

Final Answer

$$\int (e^{3a \log x} + e^{3x \log a}) dx = \frac{x^{3a+1}}{3a+1} + \frac{a^{3x}}{3 \log a} + C$$

Question 6 (i)

Evaluate $\int \sqrt{1 + \cos x} dx$, $0 < x < \pi$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sqrt{1 + \cos x} dx$ for $x \in (0, \pi)$

Concept / Formula Used: Half-angle identity: $1 + \cos x = 2 \cos^2 \frac{x}{2}$

Solution:

Using the identity $1 + \cos x = 2 \cos^2 \frac{x}{2}$:

$$\sqrt{1 + \cos x} = \sqrt{2 \cos^2 \frac{x}{2}} = \sqrt{2} \left| \cos \frac{x}{2} \right|$$

Since $0 < x < \pi$, we have $0 < \frac{x}{2} < \frac{\pi}{2}$. In the first quadrant, $\cos \frac{x}{2} > 0$, so $\left| \cos \frac{x}{2} \right| = \cos \frac{x}{2}$.

Substituting this into the integral:

$$I = \int \sqrt{2} \cos \frac{x}{2} dx$$

$$= \sqrt{2} \int \cos \frac{x}{2} dx$$

Let $u = \frac{x}{2}$, so $du = \frac{1}{2}dx \implies dx = 2 du$:

$$\begin{aligned} I &= \sqrt{2} \int \cos u (2 du) \\ &= 2\sqrt{2} \sin u + C \end{aligned}$$

Back-substituting $u = \frac{x}{2}$:

$$I = 2\sqrt{2} \sin \frac{x}{2} + C$$

Final Answer

$$\int \sqrt{1 + \cos x} dx = 2\sqrt{2} \sin \frac{x}{2} + C$$

Question 6 (ii)

Evaluate $\int \sqrt{1 + \sin x} dx$, $0 < x < \pi$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sqrt{1 + \sin x} dx$ for $x \in (0, \pi)$

Concept / Formula Used: Trigonometric identity: $1 + \sin x = \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)^2$

Solution:

Using $\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} = 1$ and $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$:

$$1 + \sin x = \sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2} = \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right)^2$$

For $0 < x < \pi$, $0 < \frac{x}{2} < \frac{\pi}{2}$, both $\sin \frac{x}{2} > 0$ and $\cos \frac{x}{2} > 0$. Thus:

$$\sqrt{1 + \sin x} = \sin \frac{x}{2} + \cos \frac{x}{2}$$

Evaluating the integral:

$$\begin{aligned} I &= \int \left(\sin \frac{x}{2} + \cos \frac{x}{2} \right) dx \\ &= \int \sin \frac{x}{2} dx + \int \cos \frac{x}{2} dx \end{aligned}$$

Using the integration results $\int \sin(kx) dx = -\frac{1}{k} \cos(kx)$ and $\int \cos(kx) dx = \frac{1}{k} \sin(kx)$:

$$\begin{aligned} I &= -2 \cos \frac{x}{2} + 2 \sin \frac{x}{2} + C \\ &= 2 \left(\sin \frac{x}{2} - \cos \frac{x}{2} \right) + C \end{aligned}$$

Final Answer

$$\int \sqrt{1 + \sin x} \, dx = 2 \left(\sin \frac{x}{2} - \cos \frac{x}{2} \right) + C$$

Question 7 (i)

Evaluate $\int \sec^2(7 - x) \, dx$.

TYPE 12a Direct/Algebraic Substitution

Given: Integral: $I = \int \sec^2(7 - x) \, dx$

Concept / Formula Used: Standard integral: $\int \sec^2 u \, du = \tan u + C$

Solution:

Let $u = 7 - x$.

Differentiating with respect to x :

$$\begin{aligned} \frac{du}{dx} &= -1 \\ du &= -dx \\ dx &= -du \end{aligned}$$

Substituting u and dx :

$$\begin{aligned} I &= \int \sec^2 u (-du) \\ &= - \int \sec^2 u \, du \\ &= -\tan u + C \end{aligned}$$

Back-substituting $u = 7 - x$:

$$I = -\tan(7 - x) + C$$

Final Answer

$$\int \sec^2(7 - x) \, dx = -\tan(7 - x) + C$$

Question 7 (ii)

Evaluate $\int a^{3x+5} \, dx$, $a > 0$.

TYPE 12a Direct/Algebraic Substitution

Given: Integral: $I = \int a^{3x+5} dx$

Concept / Formula Used: General exponential integral: $\int a^u du = \frac{a^u}{\log a} + C$

Solution:

Let $u = 3x + 5$.

Differentiating with respect to x :

$$\begin{aligned}\frac{du}{dx} &= 3 \\ du &= 3 dx \\ dx &= \frac{du}{3}\end{aligned}$$

Substituting u and dx :

$$\begin{aligned}I &= \int a^u \left(\frac{du}{3}\right) \\ &= \frac{1}{3} \int a^u du\end{aligned}$$

Using the formula $\int a^u du = \frac{a^u}{\log a}$:

$$\begin{aligned}I &= \frac{1}{3} \left(\frac{a^u}{\log a} \right) + C \\ &= \frac{a^u}{3 \log a} + C\end{aligned}$$

Back-substituting $u = 3x + 5$:

$$I = \frac{a^{3x+5}}{3 \log a} + C$$

Final Answer

$$\int a^{3x+5} dx = \frac{a^{3x+5}}{3 \log a} + C$$

Question 8 (i)

Evaluate $\int \frac{1 - \tan^2 x}{1 + \tan^2 x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \frac{1 - \tan^2 x}{1 + \tan^2 x} dx$

Concept / Formula Used: Double-angle identity: $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$

Solution:

Applying the double-angle formula for cosine:

$$\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$$

Substituting into the integral:

$$I = \int \cos 2x \, dx$$

Let $u = 2x$, so $du = 2 \, dx \implies dx = \frac{du}{2}$:

$$\begin{aligned} I &= \int \cos u \left(\frac{du}{2} \right) \\ &= \frac{1}{2} \int \cos u \, du \\ &= \frac{1}{2} \sin u + C \end{aligned}$$

Back-substituting $u = 2x$:

$$I = \frac{1}{2} \sin 2x + C$$

Final Answer

$$\int \frac{1 - \tan^2 x}{1 + \tan^2 x} \, dx = \frac{1}{2} \sin 2x + C$$

Question 8 (ii)

Evaluate $\int \sec^2 x \csc^2 x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sec^2 x \csc^2 x \, dx$

Concept / Formula Used: Pythagorean identity: $\sin^2 x + \cos^2 x = 1$

Solution:

Expressing secant and cosecant in terms of sine and cosine:

$$\sec^2 x \csc^2 x = \frac{1}{\cos^2 x \sin^2 x}$$

Replacing 1 with $\sin^2 x + \cos^2 x$:

$$\begin{aligned}\sec^2 x \csc^2 x &= \frac{\sin^2 x + \cos^2 x}{\cos^2 x \sin^2 x} \\ &= \frac{\sin^2 x}{\cos^2 x \sin^2 x} + \frac{\cos^2 x}{\cos^2 x \sin^2 x} \\ &= \frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} \\ &= \sec^2 x + \csc^2 x\end{aligned}$$

Now evaluating the integral term-by-term:

$$\begin{aligned}I &= \int (\sec^2 x + \csc^2 x) dx \\ &= \int \sec^2 x dx + \int \csc^2 x dx \\ &= \tan x - \cot x + C\end{aligned}$$

Final Answer

$$\int \sec^2 x \csc^2 x dx = \tan x - \cot x + C$$

Question 9 (i)

Evaluate $\int \frac{x}{2x-3} dx$.

TYPE 1 Power Rule of Integration

Given: Integral: $I = \int \frac{x}{2x-3} dx$

Concept / Formula Used: Algebraic splitting: $\frac{x}{2x-3} = \frac{1}{2} \left(\frac{2x-3+3}{2x-3} \right)$

Solution:

Rewriting the numerator x in terms of the denominator $(2x-3)$:

$$x = \frac{1}{2}(2x-3) + \frac{3}{2}$$

Dividing by $(2x-3)$:

$$\frac{x}{2x-3} = \frac{\frac{1}{2}(2x-3) + \frac{3}{2}}{2x-3} = \frac{1}{2} + \frac{3}{2(2x-3)}$$

Now evaluating the integral:

$$\begin{aligned}I &= \int \left(\frac{1}{2} + \frac{3}{2(2x-3)} \right) dx \\ &= \frac{1}{2} \int 1 dx + \frac{3}{2} \int \frac{dx}{2x-3}\end{aligned}$$

For $\int \frac{dx}{2x-3}$, let $u = 2x - 3 \implies du = 2 dx \implies dx = \frac{du}{2}$:

$$\int \frac{dx}{2x-3} = \frac{1}{2} \log |2x-3|$$

Thus:

$$\begin{aligned} I &= \frac{1}{2}x + \frac{3}{2} \left(\frac{1}{2} \log |2x-3| \right) + C \\ &= \frac{1}{2}x + \frac{3}{4} \log |2x-3| + C \end{aligned}$$

Final Answer

$$\int \frac{x}{2x-3} dx = \frac{1}{2}x + \frac{3}{4} \log |2x-3| + C$$

Question 9 (ii)

Evaluate $\int \frac{x+2}{3x^2+5x-2} dx$.

TYPE 1 Power Rule of Integration

Given: Integral: $I = \int \frac{x+2}{3x^2+5x-2} dx$

Concept / Formula Used: Factorization of quadratic expression: $3x^2 + 5x - 2 = (3x-1)(x+2)$

Solution:

Factorizing the quadratic expression in the denominator:

$$3x^2 + 5x - 2 = 3x^2 + 6x - x - 2 = 3x(x+2) - 1(x+2) = (3x-1)(x+2)$$

Simplifying the integrand for $x \neq -2$:

$$\frac{x+2}{3x^2+5x-2} = \frac{x+2}{(3x-1)(x+2)} = \frac{1}{3x-1}$$

Evaluating the integral:

$$I = \int \frac{dx}{3x-1}$$

Let $u = 3x - 1$.

Differentiating with respect to x :

$$\begin{aligned} du &= 3 dx \\ dx &= \frac{du}{3} \end{aligned}$$

Substituting:

$$\begin{aligned} I &= \int \frac{1}{u} \left(\frac{du}{3} \right) \\ &= \frac{1}{3} \int \frac{du}{u} \\ &= \frac{1}{3} \log |u| + C \end{aligned}$$

Back-substituting $u = 3x - 1$:

$$I = \frac{1}{3} \log |3x - 1| + C$$

Final Answer

$$\int \frac{x+2}{3x^2+5x-2} dx = \frac{1}{3} \log |3x-1| + C$$

Question 10 (i)

Evaluate $\int \frac{x}{(x+1)^2} dx$.

TYPE 1 Power Rule of Integration

Given: Integral: $I = \int \frac{x}{(x+1)^2} dx$

Concept / Formula Used: Numerator adjustment: $x = (x+1) - 1$

Solution:

Rewriting the numerator:

$$x = (x+1) - 1$$

Splitting the integrand:

$$\frac{x}{(x+1)^2} = \frac{(x+1) - 1}{(x+1)^2} = \frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} = \frac{1}{x+1} - \frac{1}{(x+1)^2}$$

Evaluating the integral term-by-term:

$$\begin{aligned} I &= \int \frac{1}{x+1} dx - \int (x+1)^{-2} dx \\ &= \log |x+1| - \left[\frac{(x+1)^{-1}}{-1} \right] + C \\ &= \log |x+1| + \frac{1}{x+1} + C \end{aligned}$$

Final Answer

$$\int \frac{x}{(x+1)^2} dx = \log|x+1| + \frac{1}{x+1} + C$$

Question 10 (ii)

Evaluate $\int \frac{x^2+2}{x+1} dx$.

TYPE 1 Power Rule of Integration**★ Likely Exam Question**

Given: Integral: $I = \int \frac{x^2+2}{x+1} dx$

Concept / Formula Used: Polynomial division: $\frac{x^2+2}{x+1} = x - 1 + \frac{3}{x+1}$

Solution:

Performing polynomial division on $\frac{x^2+2}{x+1}$:

$$\begin{aligned} x^2 + 2 &= x^2 - 1 + 3 \\ &= (x-1)(x+1) + 3 \end{aligned}$$

Dividing by $(x+1)$:

$$\frac{x^2+2}{x+1} = \frac{(x-1)(x+1)}{x+1} + \frac{3}{x+1} = x - 1 + \frac{3}{x+1}$$

Evaluating the integral:

$$\begin{aligned} I &= \int \left(x - 1 + \frac{3}{x+1} \right) dx \\ &= \int x dx - \int 1 dx + 3 \int \frac{dx}{x+1} \\ &= \frac{x^2}{2} - x + 3 \log|x+1| + C \end{aligned}$$

Final Answer

$$\int \frac{x^2+2}{x+1} dx = \frac{1}{2}x^2 - x + 3 \log|x+1| + C$$

Question 11 (i)

Evaluate $\int \frac{x^3}{2x+1} dx$.

TYPE 1 Power Rule of Integration

Given: Integral: $I = \int \frac{x^3}{2x+1} dx$

Concept / Formula Used: Polynomial long division

Solution:

Dividing polynomial x^3 by $(2x + 1)$:

$$\begin{aligned} x^3 &= \frac{x^2}{2}(2x + 1) - \frac{x^2}{2} \\ &= \frac{x^2}{2}(2x + 1) - \frac{x}{4}(2x + 1) + \frac{x}{4} \\ &= \frac{x^2}{2}(2x + 1) - \frac{x}{4}(2x + 1) + \frac{1}{8}(2x + 1) - \frac{1}{8} \end{aligned}$$

Thus, splitting the fraction:

$$\frac{x^3}{2x+1} = \frac{x^2}{2} - \frac{x}{4} + \frac{1}{8} - \frac{1}{8(2x+1)}$$

Integrating term-by-term:

$$\begin{aligned} I &= \int \left(\frac{x^2}{2} - \frac{x}{4} + \frac{1}{8} - \frac{1}{8(2x+1)} \right) dx \\ &= \frac{1}{2} \int x^2 dx - \frac{1}{4} \int x dx + \frac{1}{8} \int 1 dx - \frac{1}{8} \int \frac{dx}{2x+1} \\ &= \frac{1}{2} \left(\frac{x^3}{3} \right) - \frac{1}{4} \left(\frac{x^2}{2} \right) + \frac{1}{8}x - \frac{1}{8} \left(\frac{\log |2x+1|}{2} \right) + C \\ &= \frac{x^3}{6} - \frac{x^2}{8} + \frac{x}{8} - \frac{1}{16} \log |2x+1| + C \end{aligned}$$

Final Answer

$$\int \frac{x^3}{2x+1} dx = \frac{x^3}{6} - \frac{x^2}{8} + \frac{x}{8} - \frac{1}{16} \log |2x+1| + C$$

Question 11 (ii)

Evaluate $\int \frac{x^3 + 4x^2 - 3x - 2}{x+2} dx$.

TYPE 1 Power Rule of Integration

Given: Integral: $I = \int \frac{x^3 + 4x^2 - 3x - 2}{x+2} dx$

Concept / Formula Used: Polynomial long division

Solution:

Dividing $(x^3 + 4x^2 - 3x - 2)$ by $(x + 2)$:

$$\begin{aligned} x^3 + 4x^2 - 3x - 2 &= x^2(x + 2) + 2x^2 - 3x - 2 \\ &= x^2(x + 2) + 2x(x + 2) - 7x - 2 \\ &= x^2(x + 2) + 2x(x + 2) - 7(x + 2) + 12 \end{aligned}$$

Thus:

$$\frac{x^3 + 4x^2 - 3x - 2}{x + 2} = x^2 + 2x - 7 + \frac{12}{x + 2}$$

Integrating term-by-term:

$$\begin{aligned} I &= \int \left(x^2 + 2x - 7 + \frac{12}{x + 2} \right) dx \\ &= \int x^2 dx + 2 \int x dx - 7 \int 1 dx + 12 \int \frac{dx}{x + 2} \\ &= \frac{x^3}{3} + 2 \left(\frac{x^2}{2} \right) - 7x + 12 \log |x + 2| + C \\ &= \frac{1}{3}x^3 + x^2 - 7x + 12 \log |x + 2| + C \end{aligned}$$

Final Answer

$$\int \frac{x^3 + 4x^2 - 3x - 2}{x + 2} dx = \frac{1}{3}x^3 + x^2 - 7x + 12 \log |x + 2| + C$$

Question 12 (i)

Evaluate $\int \frac{1}{\sqrt{x+1} + \sqrt{x+2}} dx$.

TYPE 5 Conjugate Multiplication

Given: Integral: $I = \int \frac{1}{\sqrt{x+1} + \sqrt{x+2}} dx$

Concept / Formula Used: Rationalization identity: $(A + B)(A - B) = A^2 - B^2$

Solution:

Rationalizing the denominator by multiplying numerator and denominator by $(\sqrt{x+2} -$

$\sqrt{x+1}$):

$$\begin{aligned}\frac{1}{\sqrt{x+1} + \sqrt{x+2}} &= \frac{\sqrt{x+2} - \sqrt{x+1}}{(\sqrt{x+2} + \sqrt{x+1})(\sqrt{x+2} - \sqrt{x+1})} \\ &= \frac{\sqrt{x+2} - \sqrt{x+1}}{(x+2) - (x+1)} \\ &= \frac{\sqrt{x+2} - \sqrt{x+1}}{1} \\ &= (x+2)^{1/2} - (x+1)^{1/2}\end{aligned}$$

Integrating term-by-term:

$$\begin{aligned}I &= \int (x+2)^{1/2} dx - \int (x+1)^{1/2} dx \\ &= \frac{(x+2)^{3/2}}{3/2} - \frac{(x+1)^{3/2}}{3/2} + C \\ &= \frac{2}{3}(x+2)^{3/2} - \frac{2}{3}(x+1)^{3/2} + C \\ &= \frac{2}{3}[(x+2)^{3/2} - (x+1)^{3/2}] + C\end{aligned}$$

Final Answer

$$\int \frac{1}{\sqrt{x+1} + \sqrt{x+2}} dx = \frac{2}{3}[(x+2)^{3/2} - (x+1)^{3/2}] + C$$

Question 12 (ii)

Evaluate $\int \frac{1}{\sqrt{2x+3} - \sqrt{2x}} dx$.

TYPE 5 Conjugate Multiplication

Given: Integral: $I = \int \frac{1}{\sqrt{2x+3} - \sqrt{2x}} dx$

Concept / Formula Used: Rationalization identity: $(A - B)(A + B) = A^2 - B^2$

Solution:

Rationalizing the denominator by multiplying numerator and denominator by $(\sqrt{2x+3} +$

$\sqrt{2x}$):

$$\begin{aligned}\frac{1}{\sqrt{2x+3}-\sqrt{2x}} &= \frac{\sqrt{2x+3}+\sqrt{2x}}{(\sqrt{2x+3}-\sqrt{2x})(\sqrt{2x+3}+\sqrt{2x})} \\ &= \frac{\sqrt{2x+3}+\sqrt{2x}}{(2x+3)-2x} \\ &= \frac{\sqrt{2x+3}+\sqrt{2x}}{3} \\ &= \frac{1}{3} \left((2x+3)^{1/2} + (2x)^{1/2} \right)\end{aligned}$$

Integrating:

$$\begin{aligned}I &= \frac{1}{3} \left[\int (2x+3)^{1/2} dx + \int (2x)^{1/2} dx \right] \\ &= \frac{1}{3} \left[\frac{(2x+3)^{3/2}}{(3/2) \cdot 2} + \frac{(2x)^{3/2}}{(3/2) \cdot 2} \right] + C \\ &= \frac{1}{3} \left[\frac{(2x+3)^{3/2}}{3} + \frac{(2x)^{3/2}}{3} \right] + C \\ &= \frac{1}{9} [(2x+3)^{3/2} + (2x)^{3/2}] + C\end{aligned}$$

Final Answer

$$\int \frac{1}{\sqrt{2x+3}-\sqrt{2x}} dx = \frac{1}{9} [(2x+3)^{3/2} + (2x)^{3/2}] + C$$

Question 13 (i)

Evaluate $\int \frac{x+1}{\sqrt{2x-1}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: Integral: $I = \int \frac{x+1}{\sqrt{2x-1}} dx$

Concept / Formula Used: Linear substitution method

Solution:

Let $u = 2x - 1$.

Isolating x :

$$\begin{aligned}2x &= u + 1 \\ x &= \frac{u+1}{2}\end{aligned}$$

Differentiating with respect to x :

$$\begin{aligned} du &= 2 dx \\ dx &= \frac{du}{2} \end{aligned}$$

Expressing numerator $(x + 1)$ in terms of u :

$$x + 1 = \frac{u + 1}{2} + 1 = \frac{u + 3}{2}$$

Substituting into the integral:

$$\begin{aligned} I &= \int \frac{\frac{u+3}{2}}{\sqrt{u}} \left(\frac{du}{2} \right) \\ &= \frac{1}{4} \int \left(\frac{u+3}{u^{1/2}} \right) du \\ &= \frac{1}{4} \int (u^{1/2} + 3u^{-1/2}) du \end{aligned}$$

Integrating term-by-term:

$$\begin{aligned} I &= \frac{1}{4} \left[\frac{u^{3/2}}{3/2} + 3 \left(\frac{u^{1/2}}{1/2} \right) \right] + C \\ &= \frac{1}{4} \left[\frac{2}{3} u^{3/2} + 6u^{1/2} \right] + C \\ &= \frac{1}{6} u^{3/2} + \frac{3}{2} u^{1/2} + C \end{aligned}$$

Back-substituting $u = 2x - 1$:

$$I = \frac{1}{6} (2x - 1)^{3/2} + \frac{3}{2} \sqrt{2x - 1} + C$$

Final Answer

$$\int \frac{x+1}{\sqrt{2x-1}} dx = \frac{1}{6} (2x-1)^{3/2} + \frac{3}{2} \sqrt{2x-1} + C$$

Question 13 (ii)

Evaluate $\int x\sqrt{3x-2} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: Integral: $I = \int x\sqrt{3x-2} dx$

Concept / Formula Used: Linear substitution method**Solution:**Let $u = 3x - 2$.Isolating x :

$$3x = u + 2$$
$$x = \frac{u + 2}{3}$$

Differentiating with respect to x :

$$du = 3 dx$$
$$dx = \frac{du}{3}$$

Substituting into the integral:

$$I = \int \left(\frac{u + 2}{3} \right) u^{1/2} \left(\frac{du}{3} \right)$$
$$= \frac{1}{9} \int (u + 2) u^{1/2} du$$
$$= \frac{1}{9} \int (u^{3/2} + 2u^{1/2}) du$$

Integrating term-by-term:

$$I = \frac{1}{9} \left[\frac{u^{5/2}}{5/2} + 2 \left(\frac{u^{3/2}}{3/2} \right) \right] + C$$
$$= \frac{1}{9} \left[\frac{2}{5} u^{5/2} + \frac{4}{3} u^{3/2} \right] + C$$
$$= \frac{2}{45} u^{5/2} + \frac{4}{27} u^{3/2} + C$$

Back-substituting $u = 3x - 2$:

$$I = \frac{2}{45} (3x - 2)^{5/2} + \frac{4}{27} (3x - 2)^{3/2} + C$$

Final Answer

$$\int x\sqrt{3x-2} dx = \frac{2}{45} (3x-2)^{5/2} + \frac{4}{27} (3x-2)^{3/2} + C$$

Question 14 (i)Evaluate $\int \cos^2 x dx$.

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

Given: Integral: $I = \int \cos^2 x \, dx$

Concept / Formula Used: Half-angle identity: $\cos^2 x = \frac{1 + \cos 2x}{2}$

Solution:

Rewriting $\cos^2 x$ using the half-angle formula:

$$\cos^2 x = \frac{1 + \cos 2x}{2}$$

Substituting into the integral:

$$\begin{aligned} I &= \int \frac{1 + \cos 2x}{2} \, dx \\ &= \frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos 2x \, dx \\ &= \frac{1}{2}x + \frac{1}{2} \left(\frac{\sin 2x}{2} \right) + C \\ &= \frac{1}{2}x + \frac{1}{4} \sin 2x + C \\ &= \frac{1}{4}(2x + \sin 2x) + C \end{aligned}$$

Final Answer

$$\int \cos^2 x \, dx = \frac{1}{4}(2x + \sin 2x) + C$$

Question 14 (ii)

Evaluate $\int \sin^4 x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sin^4 x \, dx$

Concept / Formula Used: Half-angle identity: $\sin^2 x = \frac{1 - \cos 2x}{2}$

Solution:

Expressing $\sin^4 x$ as $(\sin^2 x)^2$:

$$\sin^4 x = \left(\frac{1 - \cos 2x}{2} \right)^2 = \frac{1 - 2 \cos 2x + \cos^2 2x}{4}$$

Using $\cos^2 2x = \frac{1 + \cos 4x}{2}$:

$$\begin{aligned} \sin^4 x &= \frac{1}{4} \left[1 - 2 \cos 2x + \frac{1 + \cos 4x}{2} \right] \\ &= \frac{1}{8} [2 - 4 \cos 2x + 1 + \cos 4x] \\ &= \frac{1}{8} [3 - 4 \cos 2x + \cos 4x] \end{aligned}$$

Evaluating the integral:

$$\begin{aligned} I &= \frac{1}{8} \int (3 - 4 \cos 2x + \cos 4x) dx \\ &= \frac{1}{8} \left[3x - 4 \left(\frac{\sin 2x}{2} \right) + \frac{\sin 4x}{4} \right] + C \\ &= \frac{1}{8} \left[3x - 2 \sin 2x + \frac{1}{4} \sin 4x \right] + C \\ &= \frac{1}{32} (12x - 8 \sin 2x + \sin 4x) + C \end{aligned}$$

Final Answer

$$\int \sin^4 x dx = \frac{1}{32} (12x - 8 \sin 2x + \sin 4x) + C$$

Question 15 (i)

Evaluate $\int \sin^2(2x + 5) dx$.

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

Given: Integral: $I = \int \sin^2(2x + 5) dx$

Concept / Formula Used: Half-angle identity: $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$

Solution:

Applying the identity with $\theta = 2x + 5$:

$$2\theta = 2(2x + 5) = 4x + 10$$

$$\sin^2(2x + 5) = \frac{1 - \cos(4x + 10)}{2}$$

Substituting into the integral:

$$\begin{aligned} I &= \int \frac{1 - \cos(4x + 10)}{2} dx \\ &= \frac{1}{2} \int 1 dx - \frac{1}{2} \int \cos(4x + 10) dx \\ &= \frac{1}{2}x - \frac{1}{2} \left(\frac{\sin(4x + 10)}{4} \right) + C \\ &= \frac{1}{2}x - \frac{1}{8} \sin(4x + 10) + C \\ &= \frac{1}{8}(4x - \sin(4x + 10)) + C \end{aligned}$$

Final Answer

$$\int \sin^2(2x + 5) dx = \frac{1}{8}(4x - \sin(4x + 10)) + C$$

Question 15 (ii)

Evaluate $\int \cos^4 2x dx$.

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

Given: Integral: $I = \int \cos^4 2x dx$

Concept / Formula Used: Half-angle identity: $\cos^2 2x = \frac{1 + \cos 4x}{2}$

Solution:

Expressing $\cos^4 2x$ as $(\cos^2 2x)^2$:

$$\cos^4 2x = \left(\frac{1 + \cos 4x}{2} \right)^2 = \frac{1 + 2 \cos 4x + \cos^2 4x}{4}$$

Using $\cos^2 4x = \frac{1 + \cos 8x}{2}$:

$$\begin{aligned} \cos^4 2x &= \frac{1}{4} \left[1 + 2 \cos 4x + \frac{1 + \cos 8x}{2} \right] \\ &= \frac{1}{8} [2 + 4 \cos 4x + 1 + \cos 8x] \\ &= \frac{1}{8} [3 + 4 \cos 4x + \cos 8x] \end{aligned}$$

Evaluating the integral:

$$\begin{aligned}
 I &= \frac{1}{8} \int (3 + 4 \cos 4x + \cos 8x) dx \\
 &= \frac{1}{8} \left[3x + 4 \left(\frac{\sin 4x}{4} \right) + \frac{\sin 8x}{8} \right] + C \\
 &= \frac{1}{8} \left[3x + \sin 4x + \frac{1}{8} \sin 8x \right] + C \\
 &= \frac{1}{64} (24x + 8 \sin 4x + \sin 8x) + C
 \end{aligned}$$

Final Answer

$$\int \cos^4 2x dx = \frac{1}{64} (24x + 8 \sin 4x + \sin 8x) + C$$

Question 16 (i)

Evaluate $\int \frac{1}{1 - \sin \frac{x}{2}} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \frac{1}{1 - \sin \frac{x}{2}} dx$

Concept / Formula Used: Pythagorean identity: $1 - \sin^2 \theta = \cos^2 \theta$

Solution:

Multiplying numerator and denominator by $(1 + \sin \frac{x}{2})$:

$$\begin{aligned}
 \frac{1}{1 - \sin \frac{x}{2}} &= \frac{1 + \sin \frac{x}{2}}{(1 - \sin \frac{x}{2})(1 + \sin \frac{x}{2})} \\
 &= \frac{1 + \sin \frac{x}{2}}{1 - \sin^2 \frac{x}{2}} \\
 &= \frac{1 + \sin \frac{x}{2}}{\cos^2 \frac{x}{2}} \\
 &= \frac{1}{\cos^2 \frac{x}{2}} + \frac{\sin \frac{x}{2}}{\cos^2 \frac{x}{2}} \\
 &= \sec^2 \frac{x}{2} + \sec \frac{x}{2} \tan \frac{x}{2}
 \end{aligned}$$

Evaluating the integral term-by-term:

$$\begin{aligned}
 I &= \int \sec^2 \frac{x}{2} dx + \int \sec \frac{x}{2} \tan \frac{x}{2} dx \\
 &= \frac{\tan \frac{x}{2}}{1/2} + \frac{\sec \frac{x}{2}}{1/2} + C \\
 &= 2 \tan \frac{x}{2} + 2 \sec \frac{x}{2} + C \\
 &= 2 \left(\tan \frac{x}{2} + \sec \frac{x}{2} \right) + C
 \end{aligned}$$

Final Answer

$$\int \frac{1}{1 - \sin \frac{x}{2}} dx = 2 \left(\tan \frac{x}{2} + \sec \frac{x}{2} \right) + C$$

Question 16 (ii)

Evaluate $\int \sin x \sqrt{1 - \cos 2x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sin x \sqrt{1 - \cos 2x} dx$

Concept / Formula Used: Double-angle identity: $1 - \cos 2x = 2 \sin^2 x$

Solution:

Using the identity $1 - \cos 2x = 2 \sin^2 x$:

$$\sqrt{1 - \cos 2x} = \sqrt{2 \sin^2 x} = \sqrt{2} \sin x$$

Substituting into the integral:

$$\begin{aligned}
 I &= \int \sin x (\sqrt{2} \sin x) dx \\
 &= \sqrt{2} \int \sin^2 x dx
 \end{aligned}$$

Using $\sin^2 x = \frac{1 - \cos 2x}{2}$:

$$\begin{aligned}
 I &= \sqrt{2} \int \frac{1 - \cos 2x}{2} dx \\
 &= \frac{\sqrt{2}}{2} \int (1 - \cos 2x) dx \\
 &= \frac{1}{\sqrt{2}} \left[x - \frac{\sin 2x}{2} \right] + C \\
 &= \frac{1}{2\sqrt{2}} (2x - \sin 2x) + C
 \end{aligned}$$

Final Answer

$$\int \sin x \sqrt{1 - \cos 2x} \, dx = \frac{1}{2\sqrt{2}}(2x - \sin 2x) + C$$

Question 17 (i)

Evaluate $\int \cos 3x \cos 5x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \cos 3x \cos 5x \, dx$

Concept / Formula Used: Product-to-sum formula: $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$

Solution:

Using $A = 5x$ and $B = 3x$:

$$\begin{aligned} \cos 5x \cos 3x &= \frac{1}{2} [\cos(5x + 3x) + \cos(5x - 3x)] \\ &= \frac{1}{2} [\cos 8x + \cos 2x] \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} I &= \frac{1}{2} \int (\cos 8x + \cos 2x) \, dx \\ &= \frac{1}{2} \left[\frac{\sin 8x}{8} + \frac{\sin 2x}{2} \right] + C \\ &= \frac{1}{16} \sin 8x + \frac{1}{4} \sin 2x + C \\ &= \frac{1}{16} (\sin 8x + 4 \sin 2x) + C \end{aligned}$$

Final Answer

$$\int \cos 3x \cos 5x \, dx = \frac{1}{16} (\sin 8x + 4 \sin 2x) + C$$

Question 17 (ii)

Evaluate $\int \sin 2x \cos 3x \, dx$.

TYPE 2 Trigonometric Identity Simplification

★ Likely Exam Question

Given: Integral: $I = \int \sin 2x \cos 3x \, dx$

Concept / Formula Used: Product-to-sum formula: $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$

Solution:

Using $A = 2x$ and $B = 3x$:

$$\begin{aligned} \sin 2x \cos 3x &= \frac{1}{2} [\sin(2x + 3x) + \sin(2x - 3x)] \\ &= \frac{1}{2} [\sin 5x + \sin(-x)] \\ &= \frac{1}{2} [\sin 5x - \sin x] \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} I &= \frac{1}{2} \int (\sin 5x - \sin x) \, dx \\ &= \frac{1}{2} \left[-\frac{\cos 5x}{5} - (-\cos x) \right] + C \\ &= \frac{1}{2} \left[\cos x - \frac{\cos 5x}{5} \right] + C \\ &= \frac{1}{10} (5 \cos x - \cos 5x) + C \end{aligned}$$

Final Answer

$$\int \sin 2x \cos 3x \, dx = \frac{1}{10} (5 \cos x - \cos 5x) + C$$

Question 18 (i)

Evaluate $\int \frac{\sin 4x}{\cos x} \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \frac{\sin 4x}{\cos x} \, dx$

Concept / Formula Used: Double-angle formula: $\sin 2\theta = 2 \sin \theta \cos \theta$

Solution:

Expanding $\sin 4x$:

$$\begin{aligned}\sin 4x &= 2 \sin 2x \cos 2x \\ &= 2(2 \sin x \cos x) \cos 2x \\ &= 4 \sin x \cos x \cos 2x\end{aligned}$$

Simplifying the integrand:

$$\frac{\sin 4x}{\cos x} = \frac{4 \sin x \cos x \cos 2x}{\cos x} = 4 \sin x \cos 2x$$

Using the product-to-sum formula $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$:

$$\begin{aligned}4 \sin x \cos 2x &= 2[2 \sin x \cos 2x] \\ &= 2[\sin(x + 2x) + \sin(x - 2x)] \\ &= 2[\sin 3x + \sin(-x)] \\ &= 2[\sin 3x - \sin x]\end{aligned}$$

Evaluating the integral:

$$\begin{aligned}I &= 2 \int (\sin 3x - \sin x) dx \\ &= 2 \left[-\frac{\cos 3x}{3} - (-\cos x) \right] + C \\ &= 2 \left[\cos x - \frac{\cos 3x}{3} \right] + C \\ &= \frac{2}{3}(3 \cos x - \cos 3x) + C\end{aligned}$$

Final Answer

$$\int \frac{\sin 4x}{\cos x} dx = \frac{2}{3}(3 \cos x - \cos 3x) + C$$

Question 18 (ii)

Evaluate $\int \sin x \sin 2x \sin 3x dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sin x \sin 2x \sin 3x dx$

Concept / Formula Used: Product-to-sum formula: $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

Solution:

First combining $\sin x \sin 2x$:

$$\begin{aligned}\sin x \sin 2x &= \frac{1}{2}[\cos(2x - x) - \cos(2x + x)] \\ &= \frac{1}{2}[\cos x - \cos 3x]\end{aligned}$$

Multiplying by $\sin 3x$:

$$\begin{aligned}\sin x \sin 2x \sin 3x &= \frac{1}{2}(\cos x - \cos 3x) \sin 3x \\ &= \frac{1}{2}[\sin 3x \cos x - \sin 3x \cos 3x]\end{aligned}$$

Using $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$ and $\sin 3x \cos 3x = \frac{1}{2} \sin 6x$:

$$\begin{aligned}\sin 3x \cos x &= \frac{1}{2}[\sin 4x + \sin 2x] \\ \sin 3x \cos 3x &= \frac{1}{2} \sin 6x\end{aligned}$$

Substituting these into the expression:

$$\sin x \sin 2x \sin 3x = \frac{1}{4}[\sin 4x + \sin 2x - \sin 6x]$$

Evaluating the integral:

$$\begin{aligned}I &= \frac{1}{4} \int (\sin 2x + \sin 4x - \sin 6x) dx \\ &= \frac{1}{4} \left[-\frac{\cos 2x}{2} - \frac{\cos 4x}{4} - \left(-\frac{\cos 6x}{6} \right) \right] + C \\ &= \frac{1}{4} \left[\frac{\cos 6x}{6} - \frac{\cos 4x}{4} - \frac{\cos 2x}{2} \right] + C \\ &= \frac{1}{48}(2 \cos 6x - 3 \cos 4x - 6 \cos 2x) + C\end{aligned}$$

Final Answer

$$\int \sin x \sin 2x \sin 3x dx = \frac{1}{48}(2 \cos 6x - 3 \cos 4x - 6 \cos 2x) + C$$

Question 19 (i)

Evaluate $\int \sin^6 x dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sin^6 x dx$

Concept / Formula Used: Half-angle identity: $\sin^2 x = \frac{1 - \cos 2x}{2}$

Solution:

Expressing $\sin^6 x$ as $(\sin^2 x)^3$:

$$\sin^6 x = \left(\frac{1 - \cos 2x}{2} \right)^3 = \frac{1}{8} [1 - 3 \cos 2x + 3 \cos^2 2x - \cos^3 2x]$$

Using $\cos^2 2x = \frac{1 + \cos 4x}{2}$ and $\cos^3 2x = \frac{\cos 6x + 3 \cos 2x}{4}$:

$$\begin{aligned} \sin^6 x &= \frac{1}{8} \left[1 - 3 \cos 2x + 3 \left(\frac{1 + \cos 4x}{2} \right) - \left(\frac{\cos 6x + 3 \cos 2x}{4} \right) \right] \\ &= \frac{1}{32} [4 - 12 \cos 2x + 6(1 + \cos 4x) - (\cos 6x + 3 \cos 2x)] \\ &= \frac{1}{32} [4 - 12 \cos 2x + 6 + 6 \cos 4x - \cos 6x - 3 \cos 2x] \\ &= \frac{1}{32} [10 - 15 \cos 2x + 6 \cos 4x - \cos 6x] \end{aligned}$$

Evaluating the integral term-by-term:

$$\begin{aligned} I &= \frac{1}{32} \int (10 - 15 \cos 2x + 6 \cos 4x - \cos 6x) dx \\ &= \frac{1}{32} \left[10x - 15 \left(\frac{\sin 2x}{2} \right) + 6 \left(\frac{\sin 4x}{4} \right) - \frac{\sin 6x}{6} \right] + C \\ &= \frac{1}{32} \left[10x - \frac{15}{2} \sin 2x + \frac{3}{2} \sin 4x - \frac{1}{6} \sin 6x \right] + C \end{aligned}$$

Final Answer

$$\int \sin^6 x dx = \frac{1}{32} \left(10x - \frac{15}{2} \sin 2x + \frac{3}{2} \sin 4x - \frac{1}{6} \sin 6x \right) + C$$

Question 19 (ii)

Evaluate $\int \sin^4 x \cos^2 x dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \sin^4 x \cos^2 x dx$

Concept / Formula Used: Double-angle identity: $\sin x \cos x = \frac{\sin 2x}{2}$

Solution:

Rewriting the integrand:

$$\begin{aligned}\sin^4 x \cos^2 x &= \sin^2 x (\sin x \cos x)^2 \\ &= \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{\sin 2x}{2} \right)^2 \\ &= \frac{1}{8} (1 - \cos 2x) \sin^2 2x \\ &= \frac{1}{8} \sin^2 2x - \frac{1}{8} \sin^2 2x \cos 2x\end{aligned}$$

Evaluating $I_1 = \int \sin^2 2x \, dx$:

$$\int \sin^2 2x \, dx = \int \frac{1 - \cos 4x}{2} \, dx = \frac{x}{2} - \frac{\sin 4x}{8}$$

Evaluating $I_2 = \int \sin^2 2x \cos 2x \, dx$: Let $u = \sin 2x \implies du = 2 \cos 2x \, dx \implies \cos 2x \, dx = \frac{du}{2}$:

$$\int u^2 \left(\frac{du}{2} \right) = \frac{u^3}{6} = \frac{\sin^3 2x}{6}$$

Combining the terms:

$$\begin{aligned}I &= \frac{1}{8} \left(\frac{x}{2} - \frac{\sin 4x}{8} \right) - \frac{1}{8} \left(\frac{\sin^3 2x}{6} \right) + C \\ &= \frac{x}{16} - \frac{\sin 4x}{64} - \frac{\sin^3 2x}{48} + C\end{aligned}$$

Final Answer

$$\int \sin^4 x \cos^2 x \, dx = \frac{x}{16} - \frac{\sin 4x}{64} - \frac{\sin^3 2x}{48} + C$$

Question 20

Evaluate $\int \cos mx \cos nx \, dx$, where m, n are positive integers, $m \neq n$. What happens if $m = n$?

TYPE 2 Trigonometric Identity Simplification

Given: Integral: $I = \int \cos mx \cos nx \, dx$, with $m, n \in \mathbb{Z}^+$

Concept / Formula Used: Product-to-sum identity: $2 \cos mx \cos nx = \cos(m + n)x + \cos(m - n)x$

Solution:

Case 1: When $m \neq n$

Using the product-to-sum identity:

$$\cos mx \cos nx = \frac{1}{2} [\cos(m+n)x + \cos(m-n)x]$$

Since $m \neq n$, $m - n \neq 0$. Integrating term-by-term:

$$\begin{aligned} I &= \frac{1}{2} \int [\cos(m+n)x + \cos(m-n)x] dx \\ &= \frac{1}{2} \left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n} \right] + C \end{aligned}$$

Case 2: When $m = n$

If $m = n$, the integral becomes:

$$I = \int \cos^2 mx \, dx$$

Using the identity $\cos^2 mx = \frac{1+\cos 2mx}{2}$:

$$\begin{aligned} I &= \int \frac{1 + \cos 2mx}{2} dx \\ &= \frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos 2mx \, dx \\ &= \frac{x}{2} + \frac{\sin 2mx}{4m} + C \end{aligned}$$

Final Answer

For $m \neq n$:

$$\int \cos mx \cos nx \, dx = \frac{1}{2} \left[\frac{\sin(m+n)x}{m+n} + \frac{\sin(m-n)x}{m-n} \right] + C$$

For $m = n$:

$$\int \cos^2 mx \, dx = \frac{x}{2} + \frac{\sin 2mx}{4m} + C$$

7.4 – Integration by Substitution: Algebraic Powers

Question 1 (i)

Evaluate:

$$\int x^3(1+x^4)^3 \, dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int x^3(1+x^4)^3 dx$

Concept / Formula Used: Integration by substitution: let $1+x^4 = t$, so $4x^3 dx = dt$

Solution:

Let $1+x^4 = t$.

Differentiating both sides:

$$4x^3 dx = dt$$

which gives:

$$x^3 dx = \frac{1}{4} dt$$

The substituted integral is:

$$\int t^3 \left(\frac{1}{4}\right) dt$$

Simplifying the constant factor:

$$= \frac{1}{4} \int t^3 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 3$:

$$\begin{aligned} &= \frac{1}{4} \cdot \frac{t^4}{4} + C \\ &= \frac{1}{16} t^4 + C \end{aligned}$$

Now back-substitute $t = 1+x^4$:

$$= \frac{1}{16} (1+x^4)^4 + C$$

Final Answer

$$\int x^3(1+x^4)^3 dx = \frac{1}{16} (1+x^4)^4 + C$$

Question 1 (ii)

Evaluate:

$$\int (2x+4)\sqrt{x^2+4x+3} dx$$

TYPE 12b Substitution Involving Radical Expressions

To Find: The indefinite integral $\int (2x+4)\sqrt{x^2+4x+3} dx$

Concept / Formula Used: Integration by substitution: let $x^2+4x+3 = t$, so $(2x+4) dx = dt$

Solution:

Let $x^2 + 4x + 3 = t$.

Differentiating both sides:

$$(2x + 4) dx = dt$$

The substituted integral is:

$$\int \sqrt{t} dt = \int t^{1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = \frac{1}{2}$:

$$\begin{aligned} &= \frac{t^{3/2}}{3/2} + C \\ &= \frac{2}{3} t^{3/2} + C \end{aligned}$$

Now back-substitute $t = x^2 + 4x + 3$:

$$= \frac{2}{3} (x^2 + 4x + 3)^{3/2} + C$$

Final Answer

$$\int (2x + 4) \sqrt{x^2 + 4x + 3} dx = \frac{2}{3} (x^2 + 4x + 3)^{3/2} + C$$

Question 2 (i)

Evaluate:

$$\int x(x^2 + 3)^{3/2} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int x(x^2 + 3)^{3/2} dx$

Solution:

Let $x^2 + 3 = t$.

Differentiating both sides:

$$2x dx = dt$$

which gives:

$$x dx = \frac{1}{2} dt$$

The substituted integral is:

$$\int t^{3/2} \left(\frac{1}{2} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{2} \int t^{3/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = \frac{3}{2}$:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{t^{5/2}}{5/2} \right] + C \\ &= \frac{1}{2} \cdot \frac{2}{5} t^{5/2} + C \\ &= \frac{1}{5} t^{5/2} + C \end{aligned}$$

Now back-substitute $t = x^2 + 3$:

$$= \frac{1}{5} (x^2 + 3)^{5/2} + C$$

Final Answer

$$\int x(x^2 + 3)^{3/2} dx = \frac{1}{5} (x^2 + 3)^{5/2} + C$$

Question 2 (ii)

Evaluate:

$$\int \frac{4x + 1}{\sqrt{2x^2 + x - 7}} dx$$

TYPE 12b Substitution Involving Radical Expressions

To Find: The indefinite integral $\int \frac{4x+1}{\sqrt{2x^2+x-7}} dx$

Solution:

Let $2x^2 + x - 7 = t$.

Differentiating both sides:

$$(4x + 1) dx = dt$$

The substituted integral is:

$$\int \frac{1}{\sqrt{t}} dt$$

Simplifying the exponent:

$$= \int t^{-1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -\frac{1}{2}$:

$$\begin{aligned} &= \frac{t^{1/2}}{1/2} + C \\ &= 2\sqrt{t} + C \end{aligned}$$

Now back-substitute $t = 2x^2 + x - 7$:

$$= 2\sqrt{2x^2 + x - 7} + C$$

Final Answer

$$\int \frac{4x+1}{\sqrt{2x^2+x-7}} dx = 2\sqrt{2x^2+x-7} + C$$

Question 3 (i)

Evaluate:

$$\int \frac{3x+1}{(3x^2+2x-5)^3} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{3x+1}{(3x^2+2x-5)^3} dx$ **Solution:**Let $3x^2 + 2x - 5 = t$.

Differentiating both sides:

$$(6x + 2) dx = dt$$

Factoring out 2:

$$2(3x + 1) dx = dt$$

which gives:

$$(3x + 1) dx = \frac{1}{2} dt$$

The substituted integral is:

$$\int \frac{1}{t^3} \left(\frac{1}{2} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{2} \int t^{-3} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -3$:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{t^{-2}}{-2} \right] + C \\ &= -\frac{1}{4t^2} + C \end{aligned}$$

Now back-substitute $t = 3x^2 + 2x - 5$:

$$= -\frac{1}{4(3x^2 + 2x - 5)^2} + C$$

Final Answer

$$\int \frac{3x+1}{(3x^2+2x-5)^3} dx = -\frac{1}{4(3x^2+2x-5)^2} + C$$

Question 3 (ii)

Evaluate:

$$\int \frac{2x}{\sqrt[3]{x^2+1}} dx$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The indefinite integral $\int \frac{2x}{\sqrt[3]{x^2+1}} dx$ **Solution:**Let $x^2 + 1 = t$.

Differentiating both sides:

$$2x dx = dt$$

The substituted integral is:

$$\int \frac{1}{t^{1/3}} dt$$

Simplifying the exponent:

$$= \int t^{-1/3} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -\frac{1}{3}$:

$$\begin{aligned} &= \frac{t^{2/3}}{2/3} + C \\ &= \frac{3}{2} t^{2/3} + C \end{aligned}$$

Now back-substitute $t = x^2 + 1$:

$$= \frac{3}{2} (x^2 + 1)^{2/3} + C$$

Final Answer

$$\int \frac{2x}{\sqrt[3]{x^2+1}} dx = \frac{3}{2} (x^2 + 1)^{2/3} + C$$

Question 4 (i)

Evaluate:

$$\int e^x (a + be^x)^6 dx$$

TYPE 12c Substitution Involving Exponential Functions**To Find:** The indefinite integral $\int e^x (a + be^x)^6 dx$ **Solution:**Let $a + be^x = t$.

Differentiating both sides:

$$be^x dx = dt$$

which gives:

$$e^x dx = \frac{1}{b} dt$$

The substituted integral is:

$$\int t^6 \left(\frac{1}{b}\right) dt$$

Simplifying the constant factor:

$$= \frac{1}{b} \int t^6 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 6$:

$$\begin{aligned} &= \frac{1}{b} \left[\frac{t^7}{7} \right] + C \\ &= \frac{1}{7b} t^7 + C \end{aligned}$$

Now back-substitute $t = a + be^x$:

$$= \frac{1}{7b} (a + be^x)^7 + C$$

Final Answer

$$\int e^x (a + be^x)^6 dx = \frac{1}{7b} (a + be^x)^7 + C$$

Question 4 (ii)

Evaluate:

$$\int \frac{(\log x)^2}{x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{(\log x)^2}{x} dx$

Solution:

Let $\log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

The substituted integral is:

$$\int t^2 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 2$:

$$\begin{aligned} &= \frac{t^3}{3} + C \\ &= \frac{1}{3}t^3 + C \end{aligned}$$

Now back-substitute $t = \log x$:

$$= \frac{1}{3}(\log x)^3 + C$$

Final Answer

$$\int \frac{(\log x)^2}{x} dx = \frac{1}{3}(\log x)^3 + C$$

Question 4 (iii)

Evaluate:

$$\int \frac{\log x}{x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{\log x}{x} dx$

Solution:

Let $\log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

The substituted integral is:

$$\int t dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 1$:

$$\begin{aligned} &= \frac{t^2}{2} + C \\ &= \frac{1}{2}t^2 + C \end{aligned}$$

Now back-substitute $t = \log x$:

$$= \frac{1}{2}(\log x)^2 + C$$

Final Answer

$$\int \frac{\log x}{x} dx = \frac{1}{2}(\log x)^2 + C$$

Question 4 (iv)

Evaluate:

$$\int \frac{x^2}{1+x^3} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{x^2}{1+x^3} dx$ **Solution:**Let $1 + x^3 = t$.

Differentiating both sides:

$$3x^2 dx = dt$$

which gives:

$$x^2 dx = \frac{1}{3} dt$$

The substituted integral is:

$$\int \frac{1}{t} \left(\frac{1}{3} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{3} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \frac{1}{3} \log |t| + C$$

Now back-substitute $t = 1 + x^3$:

$$= \frac{1}{3} \log |1 + x^3| + C$$

Final Answer

$$\int \frac{x^2}{1+x^3} dx = \frac{1}{3} \log |1 + x^3| + C$$

Question 5 (i)

Evaluate:

$$\int \frac{\sqrt{3+\log x}}{x} dx$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The indefinite integral $\int \frac{\sqrt{3+\log x}}{x} dx$ **Solution:**

Let $3 + \log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

The substituted integral is:

$$\int \sqrt{t} dt$$

Simplifying the exponent:

$$= \int t^{1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = \frac{1}{2}$:

$$\begin{aligned} &= \frac{t^{3/2}}{3/2} + C \\ &= \frac{2}{3} t^{3/2} + C \end{aligned}$$

Now back-substitute $t = 3 + \log x$:

$$= \frac{2}{3} (3 + \log x)^{3/2} + C$$

Final Answer

$$\int \frac{\sqrt{3 + \log x}}{x} dx = \frac{2}{3} (3 + \log x)^{3/2} + C$$

Question 5 (ii)

Evaluate:

$$\int \frac{3(x+1)(x+\log x)^2}{x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{3(x+1)(x+\log x)^2}{x} dx$

Solution:

Let $x + \log x = t$.

Differentiating both sides:

$$\left(1 + \frac{1}{x}\right) dx = dt$$

which simplifies to:

$$\left(\frac{x+1}{x}\right) dx = dt$$

The substituted integral is:

$$\int 3t^2 dt$$

Simplifying the constant factor:

$$= 3 \int t^2 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 2$:

$$\begin{aligned} &= 3 \left[\frac{t^3}{3} \right] + C \\ &= t^3 + C \end{aligned}$$

Now back-substitute $t = x + \log x$:

$$= (x + \log x)^3 + C$$

Final Answer

$$\int \frac{3(x+1)(x+\log x)^2}{x} dx = (x + \log x)^3 + C$$

Question 6 (i)

Evaluate:

$$\int \frac{\sqrt{\tan^{-1} x}}{1+x^2} dx$$

TYPE 12b Substitution Involving Radical Expressions

To Find: The indefinite integral $\int \frac{\sqrt{\tan^{-1} x}}{1+x^2} dx$

Solution:

Let $\tan^{-1} x = t$.

Differentiating both sides:

$$\frac{1}{1+x^2} dx = dt$$

The substituted integral is:

$$\int \sqrt{t} dt$$

Simplifying the exponent:

$$= \int t^{1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = \frac{1}{2}$:

$$\begin{aligned} &= \frac{t^{3/2}}{3/2} + C \\ &= \frac{2}{3} t^{3/2} + C \end{aligned}$$

Now back-substitute $t = \tan^{-1} x$:

$$= \frac{2}{3} (\tan^{-1} x)^{3/2} + C$$

Final Answer

$$\int \frac{\sqrt{\tan^{-1} x}}{1+x^2} dx = \frac{2}{3}(\tan^{-1} x)^{3/2} + C$$

Question 6 (ii)

Evaluate:

$$\int \frac{(\sin^{-1} 2x)^3}{\sqrt{1-4x^2}} dx$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The indefinite integral $\int \frac{(\sin^{-1} 2x)^3}{\sqrt{1-4x^2}} dx$ **Solution:**Let $\sin^{-1} 2x = t$.

Differentiating both sides:

$$\frac{1}{\sqrt{1-(2x)^2}} \cdot 2 dx = dt$$

which simplifies to:

$$\frac{2}{\sqrt{1-4x^2}} dx = dt$$

giving:

$$\frac{1}{\sqrt{1-4x^2}} dx = \frac{1}{2} dt$$

The substituted integral is:

$$\int t^3 \left(\frac{1}{2} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{2} \int t^3 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 3$:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{t^4}{4} \right] + C \\ &= \frac{1}{8} t^4 + C \end{aligned}$$

Now back-substitute $t = \sin^{-1} 2x$:

$$= \frac{1}{8} (\sin^{-1} 2x)^4 + C$$

Final Answer

$$\int \frac{(\sin^{-1} 2x)^3}{\sqrt{1-4x^2}} dx = \frac{1}{8} (\sin^{-1} 2x)^4 + C$$

Question 7 (i)

Evaluate:

$$\int \sin x \cos^5 x \, dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \sin x \cos^5 x \, dx$ **Solution:**Let $\cos x = t$.

Differentiating both sides:

$$-\sin x \, dx = dt$$

which gives:

$$\sin x \, dx = -dt$$

The substituted integral is:

$$\int t^5 (-dt)$$

Simplifying the constant factor:

$$= - \int t^5 \, dt$$

Using the standard result $\int t^n \, dt = \frac{t^{n+1}}{n+1} + C$ with $n = 5$:

$$= -\frac{t^6}{6} + C$$

Now back-substitute $t = \cos x$:

$$= -\frac{1}{6} \cos^6 x + C$$

Final Answer

$$\int \sin x \cos^5 x \, dx = -\frac{1}{6} \cos^6 x + C$$

Question 7 (ii)

Evaluate:

$$\int \cos^3(ax + b) \sin(ax + b) \, dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \cos^3(ax + b) \sin(ax + b) \, dx$ **Solution:**Let $\cos(ax + b) = t$.

Differentiating both sides:

$$-\sin(ax + b) \cdot a \, dx = dt$$

which gives:

$$\sin(ax + b) \, dx = -\frac{1}{a} \, dt$$

The substituted integral is:

$$\int t^3 \left(-\frac{1}{a}\right) dt$$

Simplifying the constant factor:

$$= -\frac{1}{a} \int t^3 \, dt$$

Using the standard result $\int t^n \, dt = \frac{t^{n+1}}{n+1} + C$ with $n = 3$:

$$\begin{aligned} &= -\frac{1}{a} \left[\frac{t^4}{4} \right] + C \\ &= -\frac{1}{4a} t^4 + C \end{aligned}$$

Now back-substitute $t = \cos(ax + b)$:

$$= -\frac{1}{4a} \cos^4(ax + b) + C$$

Final Answer

$$\int \cos^3(ax + b) \sin(ax + b) \, dx = -\frac{1}{4a} \cos^4(ax + b) + C$$

Question 8 (i)

Evaluate:

$$\int \sec^3 x \tan x \, dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \sec^3 x \tan x \, dx$

Solution:

We can rewrite the integrand as:

$$\sec^3 x \tan x = \sec^2 x (\sec x \tan x)$$

Let $\sec x = t$.

Differentiating both sides:

$$\sec x \tan x \, dx = dt$$

The substituted integral is:

$$\int t^2 \, dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 2$:

$$= \frac{t^3}{3} + C$$

Now back-substitute $t = \sec x$:

$$= \frac{1}{3} \sec^3 x + C$$

Final Answer

$$\int \sec^3 x \tan x dx = \frac{1}{3} \sec^3 x + C$$

Question 8 (ii)

Evaluate:

$$\int \cot^3 x \csc^2 x dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \cot^3 x \csc^2 x dx$

Solution:

Let $\cot x = t$.

Differentiating both sides:

$$- \csc^2 x dx = dt$$

which gives:

$$\csc^2 x dx = -dt$$

The substituted integral is:

$$\int t^3 (-dt)$$

Simplifying the constant factor:

$$= - \int t^3 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 3$:

$$= -\frac{t^4}{4} + C$$

Now back-substitute $t = \cot x$:

$$= -\frac{1}{4} \cot^4 x + C$$

Final Answer

$$\int \cot^3 x \csc^2 x dx = -\frac{1}{4} \cot^4 x + C$$

Question 9 (i)

Evaluate:

$$\int \sqrt{\tan x}(1 + \tan^2 x) dx$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The indefinite integral $\int \sqrt{\tan x}(1 + \tan^2 x) dx$ **Solution:**Using the standard trigonometric identity $1 + \tan^2 x = \sec^2 x$, the integral becomes:

$$\int \sqrt{\tan x} \sec^2 x dx$$

Let $\tan x = t$.

Differentiating both sides:

$$\sec^2 x dx = dt$$

The substituted integral is:

$$\int \sqrt{t} dt$$

Simplifying the exponent:

$$= \int t^{1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = \frac{1}{2}$:

$$\begin{aligned} &= \frac{t^{3/2}}{3/2} + C \\ &= \frac{2}{3} t^{3/2} + C \end{aligned}$$

Now back-substitute $t = \tan x$:

$$= \frac{2}{3} (\tan x)^{3/2} + C$$

Final Answer

$$\int \sqrt{\tan x}(1 + \tan^2 x) dx = \frac{2}{3} (\tan x)^{3/2} + C$$

Question 9 (ii)

Evaluate:

$$\int \frac{\sin x}{1 + \cos x} dx$$

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: The indefinite integral $\int \frac{\sin x}{1+\cos x} dx$

Solution:

Let $1 + \cos x = t$.

Differentiating both sides:

$$-\sin x dx = dt$$

which gives:

$$\sin x dx = -dt$$

The substituted integral is:

$$\int \frac{-dt}{t}$$

Simplifying the constant factor:

$$= - \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= -\log |t| + C$$

Now back-substitute $t = 1 + \cos x$:

$$= -\log |1 + \cos x| + C$$

Final Answer

$$\int \frac{\sin x}{1 + \cos x} dx = -\log |1 + \cos x| + C$$

Question 9 (iii)

Evaluate:

$$\int \frac{5 - 3x^2}{6 - x^3 + 5x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{5-3x^2}{6-x^3+5x} dx$

Solution:

Let $6 - x^3 + 5x = t$.

Differentiating both sides:

$$(-3x^2 + 5) dx = dt$$

which is:

$$(5 - 3x^2) dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = 6 - x^3 + 5x$:

$$= \log |6 - x^3 + 5x| + C$$

Final Answer

$$\int \frac{5 - 3x^2}{6 - x^3 + 5x} dx = \log |6 - x^3 + 5x| + C$$

Question 9 (iv)

Evaluate:

$$\int x^2 e^{x^3} dx$$

TYPE 12c Substitution Involving Exponential Functions

To Find: The indefinite integral $\int x^2 e^{x^3} dx$

Solution:

Let $x^3 = t$.

Differentiating both sides:

$$3x^2 dx = dt$$

which gives:

$$x^2 dx = \frac{1}{3} dt$$

The substituted integral is:

$$\int e^t \left(\frac{1}{3} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{3} \int e^t dt$$

Using the standard result $\int e^t dt = e^t + C$:

$$= \frac{1}{3} e^t + C$$

Now back-substitute $t = x^3$:

$$= \frac{1}{3} e^{x^3} + C$$

Final Answer

$$\int x^2 e^{x^3} dx = \frac{1}{3} e^{x^3} + C$$

Question 10 (i)

Evaluate:

$$\int \frac{\cot x}{\sqrt{\sin x}} dx$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The indefinite integral $\int \frac{\cot x}{\sqrt{\sin x}} dx$ **Solution:**

We can rewrite the integrand using trigonometric definitions:

$$\frac{\cot x}{\sqrt{\sin x}} = \frac{\cos x}{\sin x \sqrt{\sin x}} = (\sin x)^{-3/2} \cos x$$

Let $\sin x = t$.

Differentiating both sides:

$$\cos x dx = dt$$

The substituted integral is:

$$\int t^{-3/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -\frac{3}{2}$:

$$\begin{aligned} &= \frac{t^{-1/2}}{-1/2} + C \\ &= -\frac{2}{\sqrt{t}} + C \end{aligned}$$

Now back-substitute $t = \sin x$:

$$= -\frac{2}{\sqrt{\sin x}} + C$$

Final Answer

$$\int \frac{\cot x}{\sqrt{\sin x}} dx = -\frac{2}{\sqrt{\sin x}} + C$$

Question 10 (ii)

Evaluate:

$$\int \frac{\tan x}{\sqrt{\cos x}} dx$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The indefinite integral $\int \frac{\tan x}{\sqrt{\cos x}} dx$ **Solution:**

We can rewrite the integrand using trigonometric definitions:

$$\frac{\tan x}{\sqrt{\cos x}} = \frac{\sin x}{\cos x \sqrt{\cos x}} = (\cos x)^{-3/2} \sin x$$

Let $\cos x = t$.

Differentiating both sides:

$$-\sin x \, dx = dt$$

which gives:

$$\sin x \, dx = -dt$$

The substituted integral is:

$$\int t^{-3/2} (-dt)$$

Simplifying the constant factor:

$$= - \int t^{-3/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -\frac{3}{2}$:

$$\begin{aligned} &= - \left[\frac{t^{-1/2}}{-1/2} \right] + C \\ &= \frac{2}{\sqrt{t}} + C \end{aligned}$$

Now back-substitute $t = \cos x$:

$$= \frac{2}{\sqrt{\cos x}} + C$$

Final Answer

$$\int \frac{\tan x}{\sqrt{\cos x}} dx = \frac{2}{\sqrt{\cos x}} + C$$

Question 11 (i)

Evaluate:

$$\int \frac{\sin x}{\sqrt{3+2\cos x}} dx$$

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: The indefinite integral $\int \frac{\sin x}{\sqrt{3+2\cos x}} dx$

Solution:

Let $3 + 2\cos x = t$.

Differentiating both sides:

$$-2\sin x \, dx = dt$$

which gives:

$$\sin x \, dx = -\frac{1}{2} dt$$

The substituted integral is:

$$\int \frac{1}{\sqrt{t}} \left(-\frac{1}{2}\right) dt$$

Simplifying the constant factor:

$$= -\frac{1}{2} \int t^{-1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -\frac{1}{2}$:

$$\begin{aligned} &= -\frac{1}{2} \left[\frac{t^{1/2}}{1/2} \right] + C \\ &= -\sqrt{t} + C \end{aligned}$$

Now back-substitute $t = 3 + 2 \cos x$:

$$= -\sqrt{3 + 2 \cos x} + C$$

Final Answer

$$\int \frac{\sin x}{\sqrt{3 + 2 \cos x}} dx = -\sqrt{3 + 2 \cos x} + C$$

Question 11 (ii)

Evaluate:

$$\int \frac{1 + \sin x}{\sqrt{x - \cos x}} dx$$

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: The indefinite integral $\int \frac{1 + \sin x}{\sqrt{x - \cos x}} dx$

Solution:

Let $x - \cos x = t$.

Differentiating both sides:

$$(1 - (-\sin x)) dx = dt$$

which simplifies to:

$$(1 + \sin x) dx = dt$$

The substituted integral is:

$$\int \frac{1}{\sqrt{t}} dt$$

Simplifying the exponent:

$$= \int t^{-1/2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -\frac{1}{2}$:

$$\begin{aligned} &= \frac{t^{1/2}}{1/2} + C \\ &= 2\sqrt{t} + C \end{aligned}$$

Now back-substitute $t = x - \cos x$:

$$= 2\sqrt{x - \cos x} + C$$

Final Answer

$$\int \frac{1 + \sin x}{\sqrt{x - \cos x}} dx = 2\sqrt{x - \cos x} + C$$

Question 12 (i)

Evaluate:

$$\int \frac{\sin x}{(1 + \cos x)^2} dx$$

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: The indefinite integral $\int \frac{\sin x}{(1 + \cos x)^2} dx$

Solution:

Let $1 + \cos x = t$.

Differentiating both sides:

$$-\sin x dx = dt$$

which gives:

$$\sin x dx = -dt$$

The substituted integral is:

$$\int \frac{-dt}{t^2}$$

Simplifying the constant factor:

$$= - \int t^{-2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -2$:

$$\begin{aligned} &= - \left[\frac{t^{-1}}{-1} \right] + C \\ &= \frac{1}{t} + C \end{aligned}$$

Now back-substitute $t = 1 + \cos x$:

$$= \frac{1}{1 + \cos x} + C$$

Final Answer

$$\int \frac{\sin x}{(1 + \cos x)^2} dx = \frac{1}{1 + \cos x} + C$$

Question 12 (ii)

Evaluate:

$$\int \frac{(3 \tan^2 x + 2) \sec^2 x}{(\tan^3 x + 2 \tan x + 5)^2} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{(3 \tan^2 x + 2) \sec^2 x}{(\tan^3 x + 2 \tan x + 5)^2} dx$ **Solution:**Let $\tan^3 x + 2 \tan x + 5 = t$.

Differentiating both sides:

$$(3 \tan^2 x \cdot \sec^2 x + 2 \sec^2 x) dx = dt$$

Factoring out $\sec^2 x$:

$$(3 \tan^2 x + 2) \sec^2 x dx = dt$$

The substituted integral is:

$$\int \frac{1}{t^2} dt$$

Simplifying the exponent:

$$= \int t^{-2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -2$:

$$\begin{aligned} &= \frac{t^{-1}}{-1} + C \\ &= -\frac{1}{t} + C \end{aligned}$$

Now back-substitute $t = \tan^3 x + 2 \tan x + 5$:

$$= -\frac{1}{\tan^3 x + 2 \tan x + 5} + C$$

Final Answer

$$\int \frac{(3 \tan^2 x + 2) \sec^2 x}{(\tan^3 x + 2 \tan x + 5)^2} dx = -\frac{1}{\tan^3 x + 2 \tan x + 5} + C$$

Question 13 (i)

Evaluate:

$$\int (e^{3x} + 1)^2 e^{3x} dx$$

TYPE 12c Substitution Involving Exponential Functions

To Find: The indefinite integral $\int (e^{3x} + 1)^2 e^{3x} dx$

Solution:

Let $e^{3x} + 1 = t$.

Differentiating both sides:

$$3e^{3x} dx = dt$$

which gives:

$$e^{3x} dx = \frac{1}{3} dt$$

The substituted integral is:

$$\int t^2 \left(\frac{1}{3}\right) dt$$

Simplifying the constant factor:

$$= \frac{1}{3} \int t^2 dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = 2$:

$$\begin{aligned} &= \frac{1}{3} \left[\frac{t^3}{3} \right] + C \\ &= \frac{1}{9} t^3 + C \end{aligned}$$

Now back-substitute $t = e^{3x} + 1$:

$$= \frac{1}{9} (e^{3x} + 1)^3 + C$$

Final Answer

$$\int (e^{3x} + 1)^2 e^{3x} dx = \frac{1}{9} (e^{3x} + 1)^3 + C$$

Question 13 (ii)

Evaluate:

$$\int \frac{e^{4x}}{(1 + e^{4x})^2} dx$$

TYPE 12c Substitution Involving Exponential Functions

To Find: The indefinite integral $\int \frac{e^{4x}}{(1 + e^{4x})^2} dx$

Solution:

Let $1 + e^{4x} = t$.

Differentiating both sides:

$$4e^{4x} dx = dt$$

which gives:

$$e^{4x} dx = \frac{1}{4} dt$$

The substituted integral is:

$$\int \frac{1}{t^2} \left(\frac{1}{4} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{4} \int t^{-2} dt$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -2$:

$$\begin{aligned} &= \frac{1}{4} \left[\frac{t^{-1}}{-1} \right] + C \\ &= -\frac{1}{4t} + C \end{aligned}$$

Now back-substitute $t = 1 + e^{4x}$:

$$= -\frac{1}{4(1 + e^{4x})} + C$$

Final Answer

$$\int \frac{e^{4x}}{(1 + e^{4x})^2} dx = -\frac{1}{4(1 + e^{4x})} + C$$

Question 14 (i)

Evaluate:

$$\int \tan^4 x dx$$

TYPE 2 Trigonometric Identity Simplification

To Find: The indefinite integral $\int \tan^4 x dx$

Solution:

We can rewrite the integrand as:

$$\tan^4 x = \tan^2 x \cdot \tan^2 x$$

Using the standard trigonometric identity $\tan^2 x = \sec^2 x - 1$:

$$\begin{aligned} \tan^4 x &= \tan^2 x (\sec^2 x - 1) \\ &= \tan^2 x \sec^2 x - \tan^2 x \end{aligned}$$

Applying the identity $\tan^2 x = \sec^2 x - 1$ again to the second term:

$$\begin{aligned} \tan^4 x &= \tan^2 x \sec^2 x - (\sec^2 x - 1) \\ &= \tan^2 x \sec^2 x - \sec^2 x + 1 \end{aligned}$$

Thus, the integral splits into three parts:

$$\int \tan^4 x dx = \int \tan^2 x \sec^2 x dx - \int \sec^2 x dx + \int 1 dx$$

For the first integral $\int \tan^2 x \sec^2 x \, dx$, let $\tan x = t$.

Differentiating both sides:

$$\sec^2 x \, dx = dt$$

The substituted integral is:

$$\int t^2 \, dt = \frac{t^3}{3} = \frac{1}{3} \tan^3 x$$

For the remaining terms, we use the standard results $\int \sec^2 x \, dx = \tan x$ and $\int 1 \, dx = x$:

$$\int \tan^4 x \, dx = \frac{1}{3} \tan^3 x - \tan x + x + C$$

Final Answer

$$\int \tan^4 x \, dx = \frac{1}{3} \tan^3 x - \tan x + x + C$$

Question 14 (ii)

Evaluate:

$$\int \cot^4 x \, dx$$

TYPE 2 Trigonometric Identity Simplification

To Find: The indefinite integral $\int \cot^4 x \, dx$

Solution:

We can rewrite the integrand as:

$$\cot^4 x = \cot^2 x \cdot \cot^2 x$$

Using the standard trigonometric identity $\cot^2 x = \csc^2 x - 1$:

$$\begin{aligned} \cot^4 x &= \cot^2 x (\csc^2 x - 1) \\ &= \cot^2 x \csc^2 x - \cot^2 x \end{aligned}$$

Applying the identity $\cot^2 x = \csc^2 x - 1$ again to the second term:

$$\begin{aligned} \cot^4 x &= \cot^2 x \csc^2 x - (\csc^2 x - 1) \\ &= \cot^2 x \csc^2 x - \csc^2 x + 1 \end{aligned}$$

Thus, the integral splits into three parts:

$$\int \cot^4 x \, dx = \int \cot^2 x \csc^2 x \, dx - \int \csc^2 x \, dx + \int 1 \, dx$$

For the first integral $\int \cot^2 x \csc^2 x \, dx$, let $\cot x = t$.

Differentiating both sides:

$$-\csc^2 x \, dx = dt$$

which gives:

$$\csc^2 x \, dx = -dt$$

The substituted integral is:

$$\int t^2(-dt) = -\frac{t^3}{3} = -\frac{1}{3} \cot^3 x$$

For the remaining terms, we use the standard results $\int \csc^2 x \, dx = -\cot x$ and $\int 1 \, dx = x$:

$$\begin{aligned} \int \cot^4 x \, dx &= -\frac{1}{3} \cot^3 x - (-\cot x) + x + C \\ &= -\frac{1}{3} \cot^3 x + \cot x + x + C \end{aligned}$$

Final Answer

$$\int \cot^4 x \, dx = -\frac{1}{3} \cot^3 x + \cot x + x + C$$

Question 15 (i)

Evaluate:

$$\int \frac{1 + \tan x}{x + \log \sec x} \, dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{1+\tan x}{x+\log \sec x} \, dx$

Solution:

Let $x + \log \sec x = t$.

Differentiating both sides:

$$\left(1 + \frac{1}{\sec x} \cdot \sec x \tan x\right) dx = dt$$

which simplifies to:

$$(1 + \tan x) \, dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} \, dt$$

Using the standard result $\int \frac{1}{t} \, dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = x + \log \sec x$:

$$= \log |x + \log \sec x| + C$$

Final Answer

$$\int \frac{1 + \tan x}{x + \log \sec x} \, dx = \log |x + \log \sec x| + C$$

Question 15 (ii)

Evaluate:

$$\int \frac{1 + \sin 2x}{x + \sin^2 x} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{1+\sin 2x}{x+\sin^2 x} dx$ **Solution:**Let $x + \sin^2 x = t$.

Differentiating both sides:

$$(1 + 2 \sin x \cos x) dx = dt$$

Using the standard double-angle identity $2 \sin x \cos x = \sin 2x$:

$$(1 + \sin 2x) dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = x + \sin^2 x$:

$$= \log |x + \sin^2 x| + C$$

Final Answer

$$\int \frac{1 + \sin 2x}{x + \sin^2 x} dx = \log |x + \sin^2 x| + C$$

Question 16 (i)

Evaluate:

$$\int \frac{\tan x \sec^2 x}{1 - \tan^2 x} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{\tan x \sec^2 x}{1 - \tan^2 x} dx$ **Solution:**Let $1 - \tan^2 x = t$.

Differentiating both sides:

$$-2 \tan x \sec^2 x dx = dt$$

which gives:

$$\tan x \sec^2 x \, dx = -\frac{1}{2} dt$$

The substituted integral is:

$$\int \frac{1}{t} \left(-\frac{1}{2} \right) dt$$

Simplifying the constant factor:

$$= -\frac{1}{2} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= -\frac{1}{2} \log |t| + C$$

Now back-substitute $t = 1 - \tan^2 x$:

$$= -\frac{1}{2} \log |1 - \tan^2 x| + C$$

Final Answer

$$\int \frac{\tan x \sec^2 x}{1 - \tan^2 x} dx = -\frac{1}{2} \log |1 - \tan^2 x| + C$$

Question 16 (ii)

Evaluate:

$$\int \frac{x^5}{1 + x^6} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{x^5}{1+x^6} dx$

Solution:

Let $1 + x^6 = t$.

Differentiating both sides:

$$6x^5 dx = dt$$

which gives:

$$x^5 dx = \frac{1}{6} dt$$

The substituted integral is:

$$\int \frac{1}{t} \left(\frac{1}{6} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{6} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \frac{1}{6} \log |t| + C$$

Now back-substitute $t = 1 + x^6$:

$$= \frac{1}{6} \log(1 + x^6) + C$$

Final Answer

$$\int \frac{x^5}{1 + x^6} dx = \frac{1}{6} \log(1 + x^6) + C$$

Question 17 (i)

Evaluate:

$$\int \frac{2 \cos x - 3 \sin x}{6 \cos x + 4 \sin x} dx$$

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: The indefinite integral $\int \frac{2 \cos x - 3 \sin x}{6 \cos x + 4 \sin x} dx$

Solution:

We can factor out 2 from the denominator:

$$\int \frac{2 \cos x - 3 \sin x}{2(3 \cos x + 2 \sin x)} dx = \frac{1}{2} \int \frac{2 \cos x - 3 \sin x}{3 \cos x + 2 \sin x} dx$$

Let $3 \cos x + 2 \sin x = t$.

Differentiating both sides:

$$(-3 \sin x + 2 \cos x) dx = dt$$

which is:

$$(2 \cos x - 3 \sin x) dx = dt$$

The substituted integral is:

$$\frac{1}{2} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \frac{1}{2} \log |t| + C$$

Now back-substitute $t = 3 \cos x + 2 \sin x$:

$$= \frac{1}{2} \log |3 \cos x + 2 \sin x| + C$$

Final Answer

$$\int \frac{2 \cos x - 3 \sin x}{6 \cos x + 4 \sin x} dx = \frac{1}{2} \log |3 \cos x + 2 \sin x| + C$$

Question 17 (ii)

Evaluate:

$$\int \frac{\sin 2x - \sin x}{\sin^2 x + \cos x + 7} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{\sin 2x - \sin x}{\sin^2 x + \cos x + 7} dx$ **Solution:**Let $\sin^2 x + \cos x + 7 = t$.

Differentiating both sides:

$$(2 \sin x \cos x - \sin x) dx = dt$$

Using the standard double-angle identity $2 \sin x \cos x = \sin 2x$:

$$(\sin 2x - \sin x) dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = \sin^2 x + \cos x + 7$:

$$= \log |\sin^2 x + \cos x + 7| + C$$

Final Answer

$$\int \frac{\sin 2x - \sin x}{\sin^2 x + \cos x + 7} dx = \log |\sin^2 x + \cos x + 7| + C$$

Question 18 (i)

Evaluate:

$$\int \frac{x + \tan x}{x^2 - 2 \log \cos x} dx$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The indefinite integral $\int \frac{x + \tan x}{x^2 - 2 \log \cos x} dx$ **Solution:**Let $x^2 - 2 \log \cos x = t$.

Differentiating both sides:

$$\left(2x - 2 \cdot \frac{1}{\cos x} \cdot (-\sin x)\right) dx = dt$$

which simplifies to:

$$(2x + 2 \tan x) dx = dt$$

Factoring out 2:

$$2(x + \tan x) dx = dt$$

which gives:

$$(x + \tan x) dx = \frac{1}{2} dt$$

The substituted integral is:

$$\int \frac{1}{t} \left(\frac{1}{2}\right) dt$$

Simplifying the constant factor:

$$= \frac{1}{2} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \frac{1}{2} \log |t| + C$$

Now back-substitute $t = x^2 - 2 \log \cos x$:

$$= \frac{1}{2} \log |x^2 - 2 \log \cos x| + C$$

Final Answer

$$\int \frac{x + \tan x}{x^2 - 2 \log \cos x} dx = \frac{1}{2} \log |x^2 - 2 \log \cos x| + C$$

Question 18 (ii)

Evaluate:

$$\int \frac{1}{x \log x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{1}{x \log x} dx$

Solution:

Let $\log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = \log x$:

$$= \log |\log x| + C$$

Final Answer

$$\int \frac{1}{x \log x} dx = \log |\log x| + C$$

Question 19 (i)

Evaluate:

$$\int \frac{2x^2 - 1}{x(x^2 - \log x)} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{2x^2 - 1}{x(x^2 - \log x)} dx$

Solution:

We can rewrite the integrand by dividing the numerator and denominator by x :

$$\frac{2x^2 - 1}{x(x^2 - \log x)} = \frac{2x - \frac{1}{x}}{x^2 - \log x}$$

Let $x^2 - \log x = t$.

Differentiating both sides:

$$\left(2x - \frac{1}{x}\right) dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = x^2 - \log x$:

$$= \log |x^2 - \log x| + C$$

Final Answer

$$\int \frac{2x^2 - 1}{x(x^2 - \log x)} dx = \log |x^2 - \log x| + C$$

Question 19 (ii)

Evaluate:

$$\int \frac{e^{2x} + 1}{e^{2x} - 1} dx$$

TYPE 12c Substitution Involving Exponential Functions**To Find:** The indefinite integral $\int \frac{e^{2x} + 1}{e^{2x} - 1} dx$ **Solution:**We can divide both the numerator and the denominator by e^x :

$$\frac{e^{2x} + 1}{e^{2x} - 1} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

Let $e^x - e^{-x} = t$.

Differentiating both sides:

$$(e^x - e^{-x}(-1)) dx = dt$$

which simplifies to:

$$(e^x + e^{-x}) dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = e^x - e^{-x}$:

$$= \log |e^x - e^{-x}| + C$$

Final Answer

$$\int \frac{e^{2x} + 1}{e^{2x} - 1} dx = \log |e^x - e^{-x}| + C$$

Question 19 (iii)

Evaluate:

$$\int \frac{e^{4x} - 1}{e^{4x} + 1} dx$$

TYPE 12c Substitution Involving Exponential Functions**To Find:** The indefinite integral $\int \frac{e^{4x} - 1}{e^{4x} + 1} dx$ **Solution:**

We can divide both the numerator and the denominator by e^{2x} :

$$\frac{e^{4x} - 1}{e^{4x} + 1} = \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}}$$

Let $e^{2x} + e^{-2x} = t$.

Differentiating both sides:

$$(2e^{2x} - 2e^{-2x}) dx = dt$$

Factoring out 2:

$$2(e^{2x} - e^{-2x}) dx = dt$$

which gives:

$$(e^{2x} - e^{-2x}) dx = \frac{1}{2} dt$$

The substituted integral is:

$$\int \frac{1}{t} \left(\frac{1}{2} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{2} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \frac{1}{2} \log |t| + C$$

Now back-substitute $t = e^{2x} + e^{-2x}$:

$$= \frac{1}{2} \log |e^{2x} + e^{-2x}| + C$$

Final Answer

$$\int \frac{e^{4x} - 1}{e^{4x} + 1} dx = \frac{1}{2} \log |e^{2x} + e^{-2x}| + C$$

Question 20 (i)

Evaluate:

$$\int \frac{1}{1 + e^{-x}} dx$$

TYPE 12c Substitution Involving Exponential Functions

To Find: The indefinite integral $\int \frac{1}{1+e^{-x}} dx$

Solution:

We can rewrite the integrand by multiplying both the numerator and the denominator by e^x :

$$\frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1}$$

Let $e^x + 1 = t$.

Differentiating both sides:

$$e^x dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = e^x + 1$:

$$= \log(e^x + 1) + C$$

Final Answer

$$\int \frac{1}{1 + e^{-x}} dx = \log(e^x + 1) + C$$

Question 20 (ii)

Evaluate:

$$\int \frac{\tan x}{a + b \tan^2 x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{\tan x}{a + b \tan^2 x} dx$

Solution:

We can express the integrand in terms of $\sin x$ and $\cos x$:

$$\begin{aligned} \frac{\tan x}{a + b \tan^2 x} &= \frac{\frac{\sin x}{\cos x}}{a + b \frac{\sin^2 x}{\cos^2 x}} \\ &= \frac{\sin x \cos x}{a \cos^2 x + b \sin^2 x} \end{aligned}$$

Let $a \cos^2 x + b \sin^2 x = t$.

Differentiating both sides:

$$(2a \cos x(-\sin x) + 2b \sin x \cos x) dx = dt$$

Factoring out $2 \sin x \cos x$:

$$2(b - a) \sin x \cos x dx = dt$$

which gives:

$$\sin x \cos x dx = \frac{1}{2(b - a)} dt$$

The substituted integral is:

$$\int \frac{1}{t} \left(\frac{1}{2(b - a)} \right) dt$$

Simplifying the constant factor:

$$= \frac{1}{2(b-a)} \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \frac{1}{2(b-a)} \log |t| + C$$

Now back-substitute $t = a \cos^2 x + b \sin^2 x$:

$$= \frac{1}{2(b-a)} \log |a \cos^2 x + b \sin^2 x| + C$$

Final Answer

$$\int \frac{\tan x}{a + b \tan^2 x} dx = \frac{1}{2(b-a)} \log |a \cos^2 x + b \sin^2 x| + C$$

Question 21 (i)

Evaluate:

$$\int \frac{1 - \cot x}{1 + \cot x} dx$$

TYPE 12a Direct/Algebraic Substitution

To Find: The indefinite integral $\int \frac{1 - \cot x}{1 + \cot x} dx$

Solution:

We can express $\cot x$ in terms of $\sin x$ and $\cos x$:

$$\begin{aligned} \frac{1 - \cot x}{1 + \cot x} &= \frac{1 - \frac{\cos x}{\sin x}}{1 + \frac{\cos x}{\sin x}} \\ &= \frac{\sin x - \cos x}{\sin x + \cos x} \end{aligned}$$

Let $\sin x + \cos x = t$.

Differentiating both sides:

$$(\cos x - \sin x) dx = dt$$

which gives:

$$(\sin x - \cos x) dx = -dt$$

The substituted integral is:

$$\int \frac{-dt}{t}$$

Simplifying the constant factor:

$$= - \int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= -\log |t| + C$$

Now back-substitute $t = \sin x + \cos x$:

$$= -\log |\sin x + \cos x| + C$$

Final Answer

$$\int \frac{1 - \cot x}{1 + \cot x} dx = -\log |\sin x + \cos x| + C$$

Question 21 (ii)

Evaluate:

$$\int \frac{\cos 2x}{(\cos x + \sin x)^2} dx$$

TYPE 2 Trigonometric Identity Simplification

To Find: The indefinite integral $\int \frac{\cos 2x}{(\cos x + \sin x)^2} dx$

Solution:

Using the standard trigonometric identity $\cos 2x = \cos^2 x - \sin^2 x = (\cos x - \sin x)(\cos x + \sin x)$, we can simplify the integrand:

$$\begin{aligned} \frac{\cos 2x}{(\cos x + \sin x)^2} &= \frac{(\cos x - \sin x)(\cos x + \sin x)}{(\cos x + \sin x)^2} \\ &= \frac{\cos x - \sin x}{\cos x + \sin x} \end{aligned}$$

Let $\cos x + \sin x = t$.

Differentiating both sides:

$$(-\sin x + \cos x) dx = dt$$

which is:

$$(\cos x - \sin x) dx = dt$$

The substituted integral is:

$$\int \frac{1}{t} dt$$

Using the standard result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = \cos x + \sin x$:

$$= \log |\cos x + \sin x| + C$$

Final Answer

$$\int \frac{\cos 2x}{(\cos x + \sin x)^2} dx = \log |\cos x + \sin x| + C$$

7.5 – Integration by Substitution: Composite Trigonometric Forms

Question 1 (i)

Evaluate $\int 2x \sin(x^2 + 1) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int 2x \sin(x^2 + 1) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$ and $\int f(g(x))g'(x) dx = \int f(u) du$

Solution:

Let $x^2 + 1 = t$.

Differentiating both sides:

$$2x dx = dt$$

Substituting these into the integral:

$$\int \sin(t) dt$$

Using the standard integration result $\int \sin(t) dt = -\cos(t) + C$:

$$= -\cos(t) + C$$

Now back-substitute $t = x^2 + 1$:

$$= -\cos(x^2 + 1) + C$$

Final Answer

$$\int 2x \sin(x^2 + 1) dx = -\cos(x^2 + 1) + C$$

Question 1 (ii)

Evaluate $\int x^3 \cos(x^4) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int x^3 \cos(x^4) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x^4 = t$.

Differentiating both sides:

$$4x^3 dx = dt$$

Rearranging to isolate the term in the integrand:

$$x^3 dx = \frac{1}{4} dt$$

Substituting these into the integral:

$$\int \cos(t) \cdot \frac{1}{4} dt$$

Factoring out the constant:

$$= \frac{1}{4} \int \cos(t) dt$$

Using the standard integration result $\int \cos(t) dt = \sin(t) + C$:

$$= \frac{1}{4} \sin(t) + C$$

Now back-substitute $t = x^4$:

$$= \frac{1}{4} \sin(x^4) + C$$

Final Answer

$$\int x^3 \cos(x^4) dx = \frac{1}{4} \sin(x^4) + C$$

Question 2 (i)

Evaluate $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\sqrt{x} = t$.

Differentiating both sides:

$$\frac{1}{2\sqrt{x}} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{1}{\sqrt{x}} dx = 2 dt$$

Substituting these into the integral:

$$\int \cos(t) \cdot 2 dt$$

Factoring out the constant:

$$= 2 \int \cos(t) dt$$

Using the standard integration result $\int \cos(t) dt = \sin(t) + C$:

$$= 2 \sin(t) + C$$

Now back-substitute $t = \sqrt{x}$:

$$= 2 \sin(\sqrt{x}) + C$$

Final Answer

$$\int \frac{\cos \sqrt{x}}{\sqrt{x}} dx = 2 \sin(\sqrt{x}) + C$$

Question 2 (ii)

Evaluate $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\sqrt{x} = t$.

Differentiating both sides:

$$\frac{1}{2\sqrt{x}} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{1}{\sqrt{x}} dx = 2 dt$$

Substituting these into the integral:

$$\int \sin(t) \cdot 2 dt$$

Factoring out the constant:

$$= 2 \int \sin(t) dt$$

Using the standard integration result $\int \sin(t) dt = -\cos(t) + C$:

$$= 2(-\cos(t)) + C$$

$$= -2\cos(t) + C$$

Now back-substitute $t = \sqrt{x}$:

$$= -2\cos(\sqrt{x}) + C$$

Final Answer

$$\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx = -2\cos(\sqrt{x}) + C$$

Question 2 (iii)

Evaluate $\int \frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\sqrt{x} = t$.

Differentiating both sides:

$$\frac{1}{2\sqrt{x}} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{1}{\sqrt{x}} dx = 2 dt$$

Substituting these into the integral:

$$\int \sec^2(t) \cdot 2 dt$$

Factoring out the constant:

$$= 2 \int \sec^2(t) dt$$

Using the standard integration result $\int \sec^2(t) dt = \tan(t) + C$:

$$= 2 \tan(t) + C$$

Now back-substitute $t = \sqrt{x}$:

$$= 2 \tan(\sqrt{x}) + C$$

Final Answer

$$\int \frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx = 2 \tan(\sqrt{x}) + C$$

Question 2 (iv)

Evaluate $\int \frac{\csc^2 \sqrt{x}}{\sqrt{x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\csc^2 \sqrt{x}}{\sqrt{x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\sqrt{x} = t$.

Differentiating both sides:

$$\frac{1}{2\sqrt{x}} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{1}{\sqrt{x}} dx = 2 dt$$

Substituting these into the integral:

$$\int \csc^2(t) \cdot 2 dt$$

Factoring out the constant:

$$= 2 \int \csc^2(t) dt$$

Using the standard integration result $\int \csc^2(t) dt = -\cot(t) + C$:

$$= 2(-\cot(t)) + C$$

$$= -2\cot(t) + C$$

Now back-substitute $t = \sqrt{x}$:

$$= -2\cot(\sqrt{x}) + C$$

Final Answer

$$\int \frac{\csc^2 \sqrt{x}}{\sqrt{x}} dx = -2\cot(\sqrt{x}) + C$$

Question 3 (i)

Evaluate $\int \sin x \cdot e^{\cos x} dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int \sin x \cdot e^{\cos x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\cos x = t$.

Differentiating both sides:

$$-\sin x dx = dt$$

Rearranging to isolate the term in the integrand:

$$\sin x dx = -dt$$

Substituting these into the integral:

$$\int e^t \cdot (-dt)$$

Factoring out the negative sign:

$$= - \int e^t dt$$

Using the standard integration result $\int e^t dt = e^t + C$:

$$= -e^t + C$$

Now back-substitute $t = \cos x$:

$$= -e^{\cos x} + C$$

Final Answer

$$\int \sin x \cdot e^{\cos x} dx = -e^{\cos x} + C$$

Question 3 (ii)

Evaluate $\int \frac{e^{2x}}{1+e^x} dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int \frac{e^{2x}}{1+e^x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $e^x = t$.

Differentiating both sides:

$$e^x dx = dt$$

We rewrite the numerator of the integrand to match our substitution:

$$e^{2x} dx = e^x \cdot e^x dx = t dt$$

Substituting these into the integral:

$$\int \frac{t}{1+t} dt$$

Rewriting the numerator to simplify the fraction:

$$\begin{aligned} &= \int \frac{t+1-1}{1+t} dt \\ &= \int \left(\frac{t+1}{1+t} - \frac{1}{1+t} \right) dt \\ &= \int \left(1 - \frac{1}{1+t} \right) dt \end{aligned}$$

Integrating term by term:

$$= \int 1 dt - \int \frac{1}{1+t} dt$$

Using the standard integration results $\int 1 dt = t$ and $\int \frac{1}{1+t} dt = \log |1+t| + C$:

$$= t - \log |1+t| + C$$

Now back-substitute $t = e^x$:

$$= e^x - \log(1+e^x) + C \quad (\text{since } 1+e^x > 0)$$

Final Answer

$$\int \frac{e^{2x}}{1+e^x} dx = e^x - \log(1+e^x) + C$$

Question 4 (i)

Evaluate $\int \sin x \sin(\cos x) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \sin x \sin(\cos x) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\cos x = t$.

Differentiating both sides:

$$-\sin x dx = dt$$

Rearranging to isolate the term in the integrand:

$$\sin x dx = -dt$$

Substituting these into the integral:

$$\int \sin(t) \cdot (-dt)$$

Factoring out the negative sign:

$$= - \int \sin(t) dt$$

Using the standard integration result $\int \sin(t) dt = -\cos(t) + C$:

$$= -(-\cos(t)) + C$$

$$= \cos(t) + C$$

Now back-substitute $t = \cos x$:

$$= \cos(\cos x) + C$$

Final Answer

$$\int \sin x \sin(\cos x) dx = \cos(\cos x) + C$$

Question 4 (ii)

Evaluate $\int x^3 \sec^2(x^4 + 3) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int x^3 \sec^2(x^4 + 3) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x^4 + 3 = t$.

Differentiating both sides:

$$4x^3 dx = dt$$

Rearranging to isolate the term in the integrand:

$$x^3 dx = \frac{1}{4} dt$$

Substituting these into the integral:

$$\int \sec^2(t) \cdot \frac{1}{4} dt$$

Factoring out the constant:

$$= \frac{1}{4} \int \sec^2(t) dt$$

Using the standard integration result $\int \sec^2(t) dt = \tan(t) + C$:

$$= \frac{1}{4} \tan(t) + C$$

Now back-substitute $t = x^4 + 3$:

$$= \frac{1}{4} \tan(x^4 + 3) + C$$

Final Answer

$$\int x^3 \sec^2(x^4 + 3) dx = \frac{1}{4} \tan(x^4 + 3) + C$$

Question 5 (i)

Evaluate $\int \frac{1}{x^2} \tan^2 \left(\frac{1}{x} \right) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{1}{x^2} \tan^2 \left(\frac{1}{x} \right) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\frac{1}{x} = t$.

Differentiating both sides:

$$-\frac{1}{x^2} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{1}{x^2} dx = -dt$$

Substituting these into the integral:

$$\int \tan^2(t) \cdot (-dt)$$

Factoring out the negative sign:

$$= - \int \tan^2(t) dt$$

Using the trigonometric identity $\tan^2(t) = \sec^2(t) - 1$:

$$= - \int (\sec^2(t) - 1) dt$$

Integrating term by term:

$$= - \int \sec^2(t) dt + \int 1 dt$$

Using the standard integration result $\int \sec^2(t) dt = \tan(t) + C$:

$$= -\tan(t) + t + C$$

Now back-substitute $t = \frac{1}{x}$:

$$= -\tan \left(\frac{1}{x} \right) + \frac{1}{x} + C$$

Final Answer

$$\int \frac{1}{x^2} \tan^2 \left(\frac{1}{x} \right) dx = -\tan \left(\frac{1}{x} \right) + \frac{1}{x} + C$$

Question 5 (ii)

Evaluate $\int \frac{1}{x^2} \sin^2 \left(\frac{1}{x} \right) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{1}{x^2} \sin^2 \left(\frac{1}{x} \right) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\frac{1}{x} = t$.

Differentiating both sides:

$$-\frac{1}{x^2} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{1}{x^2} dx = -dt$$

Substituting these into the integral:

$$\int \sin^2(t) \cdot (-dt)$$

Factoring out the negative sign:

$$= - \int \sin^2(t) dt$$

Using the trigonometric identity $\sin^2(t) = \frac{1 - \cos(2t)}{2}$:

$$\begin{aligned} &= - \int \frac{1 - \cos(2t)}{2} dt \\ &= -\frac{1}{2} \int (1 - \cos(2t)) dt \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= -\frac{1}{2} \left[t - \frac{\sin(2t)}{2} \right] + C \\ &= -\frac{t}{2} + \frac{\sin(2t)}{4} + C \end{aligned}$$

Now back-substitute $t = \frac{1}{x}$:

$$= -\frac{1}{2x} + \frac{1}{4} \sin \left(\frac{2}{x} \right) + C$$

Final Answer

$$\int \frac{1}{x^2} \sin^2 \left(\frac{1}{x} \right) dx = -\frac{1}{2x} + \frac{1}{4} \sin \left(\frac{2}{x} \right) + C$$

Question 5 (iii)

Evaluate $\int 2x \sec^3(x^2 + 5) \tan(x^2 + 5) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int 2x \sec^3(x^2 + 5) \tan(x^2 + 5) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x^2 + 5 = u$.

Differentiating both sides:

$$2x dx = du$$

Substituting these into the integral:

$$\int \sec^3(u) \tan(u) du$$

Rewriting the integrand to isolate $\sec(u) \tan(u) du$:

$$\int \sec^2(u) \cdot (\sec(u) \tan(u)) du$$

Now, let $\sec(u) = t$.

Differentiating both sides:

$$\sec(u) \tan(u) du = dt$$

Substituting these into the integral:

$$\int t^2 dt$$

Using the standard power rule $\int t^2 dt = \frac{t^3}{3} + C$:

$$= \frac{t^3}{3} + C$$

Now back-substitute $t = \sec(u)$:

$$= \frac{\sec^3(u)}{3} + C$$

Now back-substitute $u = x^2 + 5$:

$$= \frac{1}{3} \sec^3(x^2 + 5) + C$$

Final Answer

$$\int 2x \sec^3(x^2 + 5) \tan(x^2 + 5) dx = \frac{1}{3} \sec^3(x^2 + 5) + C$$

Question 6 (i)

Evaluate $\int \frac{\sin(\log x)}{x} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\sin(\log x)}{x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

Substituting these into the integral:

$$\int \sin(t) dt$$

Using the standard integration result $\int \sin(t) dt = -\cos(t) + C$:

$$= -\cos(t) + C$$

Now back-substitute $t = \log x$:

$$= -\cos(\log x) + C$$

Final Answer

$$\int \frac{\sin(\log x)}{x} dx = -\cos(\log x) + C$$

Question 6 (ii)

Evaluate $\int \frac{\csc^2(\log x)}{x} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\csc^2(\log x)}{x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

Substituting these into the integral:

$$\int \csc^2(t) dt$$

Using the standard integration result $\int \csc^2(t) dt = -\cot(t) + C$:

$$= -\cot(t) + C$$

Now back-substitute $t = \log x$:

$$= -\cot(\log x) + C$$

Final Answer

$$\int \frac{\csc^2(\log x)}{x} dx = -\cot(\log x) + C$$

Question 7 (i)

Evaluate $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\tan^{-1} x = t$.

Differentiating both sides:

$$\frac{1}{1+x^2} dx = dt$$

Substituting these into the integral:

$$\int e^t dt$$

Using the standard integration result $\int e^t dt = e^t + C$:

$$= e^t + C$$

Now back-substitute $t = \tan^{-1} x$:

$$= e^{\tan^{-1} x} + C$$

Final Answer

$$\int \frac{e^{\tan^{-1} x}}{1+x^2} dx = e^{\tan^{-1} x} + C$$

Question 7 (ii)

Evaluate $\int \frac{\sin(2 \tan^{-1} x)}{1 + x^2} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\sin(2 \tan^{-1} x)}{1 + x^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\tan^{-1} x = t$.

Differentiating both sides:

$$\frac{1}{1 + x^2} dx = dt$$

Substituting these into the integral:

$$\int \sin(2t) dt$$

Using the standard integration result $\int \sin(at) dt = -\frac{\cos(at)}{a} + C$ with $a = 2$:

$$= -\frac{\cos(2t)}{2} + C$$

Now back-substitute $t = \tan^{-1} x$:

$$= -\frac{1}{2} \cos(2 \tan^{-1} x) + C$$

Final Answer

$$\int \frac{\sin(2 \tan^{-1} x)}{1 + x^2} dx = -\frac{1}{2} \cos(2 \tan^{-1} x) + C$$

Question 8 (i)

Evaluate $\int \frac{e^{-1/x} + 1}{x^2} dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int \frac{e^{-1/x} + 1}{x^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

We split the integral into two parts:

$$\int \frac{e^{-1/x} + 1}{x^2} dx = \int \frac{e^{-1/x}}{x^2} dx + \int \frac{1}{x^2} dx$$

For the first integral, let $-\frac{1}{x} = t$.

Differentiating both sides:

$$\frac{1}{x^2} dx = dt$$

Substituting these into the first integral:

$$\int e^t dt = e^t + C_1$$

Back-substituting $t = -\frac{1}{x}$:

$$= e^{-1/x} + C_1$$

For the second integral, using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ with $n = -2$:

$$\int x^{-2} dx = \frac{x^{-1}}{-1} + C_2 = -\frac{1}{x} + C_2$$

Combining both parts:

$$= e^{-1/x} - \frac{1}{x} + C$$

Final Answer

$$\int \frac{e^{-1/x} + 1}{x^2} dx = e^{-1/x} - \frac{1}{x} + C$$

Question 8 (ii)

Evaluate $\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\tan^{-1} x = t$.

Differentiating both sides:

$$\frac{1}{1+x^2} dx = dt$$

Substituting these into the integral:

$$\int e^{mt} dt$$

Using the standard integration result $\int e^{mt} dt = \frac{e^{mt}}{m} + C$:

$$= \frac{e^{mt}}{m} + C$$

Now back-substitute $t = \tan^{-1} x$:

$$= \frac{e^{m \tan^{-1} x}}{m} + C$$

Final Answer

$$\int \frac{e^{m \tan^{-1} x}}{1+x^2} dx = \frac{e^{m \tan^{-1} x}}{m} + C$$

Question 9 (i)

Evaluate $\int e^x \csc^2(e^x) dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int e^x \csc^2(e^x) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $e^x = t$.

Differentiating both sides:

$$e^x dx = dt$$

Substituting these into the integral:

$$\int \csc^2(t) dt$$

Using the standard integration result $\int \csc^2(t) dt = -\cot(t) + C$:

$$= -\cot(t) + C$$

Now back-substitute $t = e^x$:

$$= -\cot(e^x) + C$$

Final Answer

$$\int e^x \csc^2(e^x) dx = -\cot(e^x) + C$$

Question 9 (ii)

Evaluate $\int \frac{(x+1)e^x}{\cot^2(xe^x)} dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int \frac{(x+1)e^x}{\cot^2(xe^x)} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $xe^x = t$.

Differentiating both sides using the product rule:

$$(1 \cdot e^x + x \cdot e^x) dx = dt$$

$$(x+1)e^x dx = dt$$

Substituting these into the integral:

$$\int \frac{dt}{\cot^2(t)}$$

Using the trigonometric identity $\frac{1}{\cot^2(t)} = \tan^2(t)$:

$$= \int \tan^2(t) dt$$

Using the identity $\tan^2(t) = \sec^2(t) - 1$:

$$= \int (\sec^2(t) - 1) dt$$

Integrating term by term:

$$= \int \sec^2(t) dt - \int 1 dt$$

Using the standard integration result $\int \sec^2(t) dt = \tan(t) + C$:

$$= \tan(t) - t + C$$

Now back-substitute $t = xe^x$:

$$= \tan(xe^x) - xe^x + C$$

Final Answer

$$\int \frac{(x+1)e^x}{\cot^2(xe^x)} dx = \tan(xe^x) - xe^x + C$$

Question 10 (i)

Evaluate $\int x^2 e^{x^3} \sin(e^{x^3}) dx$.

TYPE 12c Substitution Involving Exponential Functions

To Find: $\int x^2 e^{x^3} \sin(e^{x^3}) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $e^{x^3} = t$.

Differentiating both sides using the chain rule:

$$e^{x^3} \cdot 3x^2 dx = dt$$

Rearranging to isolate the term in the integrand:

$$x^2 e^{x^3} dx = \frac{1}{3} dt$$

Substituting these into the integral:

$$\int \sin(t) \cdot \frac{1}{3} dt$$

Factoring out the constant:

$$= \frac{1}{3} \int \sin(t) dt$$

Using the standard integration result $\int \sin(t) dt = -\cos(t) + C$:

$$= \frac{1}{3} (-\cos(t)) + C$$

$$= -\frac{1}{3} \cos(t) + C$$

Now back-substitute $t = e^{x^3}$:

$$= -\frac{1}{3} \cos(e^{x^3}) + C$$

Final Answer

$$\int x^2 e^{x^3} \sin(e^{x^3}) dx = -\frac{1}{3} \cos(e^{x^3}) + C$$

Question 10 (ii)

Evaluate $\int \frac{\sin^2(\log x)}{x} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\sin^2(\log x)}{x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\log x = t$.

Differentiating both sides:

$$\frac{1}{x} dx = dt$$

Substituting these into the integral:

$$\int \sin^2(t) dt$$

Using the trigonometric identity $\sin^2(t) = \frac{1 - \cos(2t)}{2}$:

$$\begin{aligned} &= \int \frac{1 - \cos(2t)}{2} dt \\ &= \frac{1}{2} \int (1 - \cos(2t)) dt \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= \frac{1}{2} \left[t - \frac{\sin(2t)}{2} \right] + C \\ &= \frac{t}{2} - \frac{\sin(2t)}{4} + C \end{aligned}$$

Now back-substitute $t = \log x$:

$$= \frac{\log x}{2} - \frac{\sin(2 \log x)}{4} + C$$

Final Answer

$$\int \frac{\sin^2(\log x)}{x} dx = \frac{\log x}{2} - \frac{\sin(2 \log x)}{4} + C$$

Question 10 (iii)

Evaluate $\int \frac{\sqrt{\tan x}}{\sin^2 x} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\sqrt{\tan x}}{\sin^2 x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\tan x = t$.

Differentiating both sides:

$$\sec^2 x dx = dt \implies dx = \frac{dt}{\sec^2 x} = \frac{dt}{1+t^2}$$

Also, we express the denominator $\sin^2 x$ in terms of t :

$$\sin^2 x = \frac{\tan^2 x}{1 + \tan^2 x} = \frac{t^2}{1 + t^2}$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{\sqrt{\tan x}}{\sin^2 x} dx &= \int \frac{\sqrt{t}}{\left(\frac{t^2}{1+t^2}\right)} \cdot \frac{dt}{1+t^2} \\ &= \int \frac{\sqrt{t}(1+t^2)}{t^2(1+t^2)} dt \\ &= \int \frac{\sqrt{t}}{t^2} dt \\ &= \int t^{-3/2} dt \end{aligned}$$

Using the power rule $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ with $n = -3/2$:

$$\begin{aligned} &= \frac{t^{-1/2}}{-1/2} + C \\ &= -\frac{2}{\sqrt{t}} + C \end{aligned}$$

Now back-substitute $t = \tan x$:

$$= -\frac{2}{\sqrt{\tan x}} + C$$

Final Answer

$$\int \frac{\sqrt{\tan x}}{\sin^2 x} dx = -\frac{2}{\sqrt{\tan x}} + C$$

Question 11 (i)

Evaluate $\int x \sin^3(x^2) \cos(x^2) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int x \sin^3(x^2) \cos(x^2) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x^2 = u$.

Differentiating both sides:

$$2x dx = du \implies x dx = \frac{1}{2} du$$

Substituting these into the integral:

$$\int \sin^3(u) \cos(u) \cdot \frac{1}{2} du = \frac{1}{2} \int \sin^3(u) \cos(u) du$$

Now, let $\sin(u) = t$.

Differentiating both sides:

$$\cos(u) du = dt$$

Substituting these into the integral:

$$\frac{1}{2} \int t^3 dt$$

Using the power rule $\int t^3 dt = \frac{t^4}{4} + C$:

$$\begin{aligned} &= \frac{1}{2} \left(\frac{t^4}{4} \right) + C \\ &= \frac{t^4}{8} + C \end{aligned}$$

Now back-substitute $t = \sin(u)$:

$$= \frac{\sin^4(u)}{8} + C$$

Now back-substitute $u = x^2$:

$$= \frac{\sin^4(x^2)}{8} + C$$

Final Answer

$$\int x \sin^3(x^2) \cos(x^2) dx = \frac{\sin^4(x^2)}{8} + C$$

Question 11 (ii)

Evaluate $\int \tan^3 2x \sec 2x \, dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \tan^3 2x \sec 2x \, dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) \, dx$

Solution:

We rewrite the integrand to isolate a factor of $\sec 2x \tan 2x$:

$$\tan^3 2x \sec 2x = \tan^2 2x \cdot (\sec 2x \tan 2x)$$

Using the trigonometric identity $\tan^2 2x = \sec^2 2x - 1$:

$$= (\sec^2 2x - 1) \sec 2x \tan 2x$$

Let $\sec 2x = t$.

Differentiating both sides:

$$2 \sec 2x \tan 2x \, dx = dt \implies \sec 2x \tan 2x \, dx = \frac{1}{2} dt$$

Substituting these into the integral:

$$\begin{aligned} \int \tan^3 2x \sec 2x \, dx &= \int (t^2 - 1) \cdot \frac{1}{2} dt \\ &= \frac{1}{2} \int (t^2 - 1) dt \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= \frac{1}{2} \left(\frac{t^3}{3} - t \right) + C \\ &= \frac{t^3}{6} - \frac{t}{2} + C \end{aligned}$$

Now back-substitute $t = \sec 2x$:

$$= \frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$$

Final Answer

$$\int \tan^3 2x \sec 2x \, dx = \frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$$

Question 12 (i)

Evaluate $\int \sin^3(2x + 1) dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \sin^3(2x + 1) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $2x + 1 = u$.

Differentiating both sides:

$$2 dx = du \implies dx = \frac{1}{2} du$$

Substituting these into the integral:

$$\int \sin^3(u) \cdot \frac{1}{2} du = \frac{1}{2} \int \sin^3(u) du$$

Rewriting $\sin^3(u)$ as $\sin^2(u) \sin(u)$ and using the identity $\sin^2(u) = 1 - \cos^2(u)$:

$$= \frac{1}{2} \int (1 - \cos^2(u)) \sin(u) du$$

Now, let $\cos(u) = t$.

Differentiating both sides:

$$-\sin(u) du = dt \implies \sin(u) du = -dt$$

Substituting these into the integral:

$$\begin{aligned} &= \frac{1}{2} \int (1 - t^2)(-dt) \\ &= \frac{1}{2} \int (t^2 - 1) dt \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= \frac{1}{2} \left(\frac{t^3}{3} - t \right) + C \\ &= \frac{t^3}{6} - \frac{t}{2} + C \end{aligned}$$

Now back-substitute $t = \cos(u)$:

$$= \frac{1}{6} \cos^3(u) - \frac{1}{2} \cos(u) + C$$

Now back-substitute $u = 2x + 1$:

$$= \frac{1}{6} \cos^3(2x + 1) - \frac{1}{2} \cos(2x + 1) + C$$

Final Answer

$$\int \sin^3(2x + 1) dx = \frac{1}{6} \cos^3(2x + 1) - \frac{1}{2} \cos(2x + 1) + C$$

Question 12 (ii)

Evaluate $\int \sin^3 x \cos^4 x \, dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \sin^3 x \cos^4 x \, dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) \, dx$

Solution:

We rewrite the integrand to isolate a factor of $\sin x$:

$$\sin^3 x \cos^4 x = \sin^2 x \cos^4 x \sin x$$

Using the trigonometric identity $\sin^2 x = 1 - \cos^2 x$:

$$= (1 - \cos^2 x) \cos^4 x \sin x$$

Let $\cos x = t$.

Differentiating both sides:

$$-\sin x \, dx = dt \implies \sin x \, dx = -dt$$

Substituting these into the integral:

$$\begin{aligned} \int \sin^3 x \cos^4 x \, dx &= \int (1 - t^2)t^4(-dt) \\ &= \int (t^6 - t^4) \, dt \end{aligned}$$

Integrating term by term using the power rule:

$$= \frac{t^7}{7} - \frac{t^5}{5} + C$$

Now back-substitute $t = \cos x$:

$$= \frac{1}{7} \cos^7 x - \frac{1}{5} \cos^5 x + C$$

Final Answer

$$\int \sin^3 x \cos^4 x \, dx = \frac{1}{7} \cos^7 x - \frac{1}{5} \cos^5 x + C$$

Question 13 (i)

Evaluate $\int \sin^7 x \, dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \sin^7 x \, dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) \, dx$

Solution:

We rewrite the integrand to isolate a factor of $\sin x$:

$$\sin^7 x = \sin^6 x \sin x = (\sin^2 x)^3 \sin x$$

Using the trigonometric identity $\sin^2 x = 1 - \cos^2 x$:

$$= (1 - \cos^2 x)^3 \sin x$$

Let $\cos x = t$.

Differentiating both sides:

$$-\sin x \, dx = dt \implies \sin x \, dx = -dt$$

Substituting these into the integral:

$$\int (1 - t^2)^3 (-dt)$$

Expanding the cubic term $(1 - t^2)^3 = 1 - 3t^2 + 3t^4 - t^6$:

$$\begin{aligned} &= - \int (1 - 3t^2 + 3t^4 - t^6) \, dt \\ &= \int (t^6 - 3t^4 + 3t^2 - 1) \, dt \end{aligned}$$

Integrating term by term using the power rule:

$$= \frac{t^7}{7} - \frac{3t^5}{5} + t^3 - t + C$$

Now back-substitute $t = \cos x$:

$$= \frac{1}{7} \cos^7 x - \frac{3}{5} \cos^5 x + \cos^3 x - \cos x + C$$

Final Answer

$$\int \sin^7 x \, dx = \frac{1}{7} \cos^7 x - \frac{3}{5} \cos^5 x + \cos^3 x - \cos x + C$$

Question 13 (ii)

Evaluate $\int \sin x \cot^3 x \, dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \sin x \cot^3 x \, dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) \, dx$

Solution:

We rewrite the integrand using the definition $\cot x = \frac{\cos x}{\sin x}$:

$$\sin x \cot^3 x = \sin x \cdot \frac{\cos^3 x}{\sin^3 x} = \frac{\cos^3 x}{\sin^2 x}$$

Isolating a factor of $\cos x$ in the numerator:

$$= \frac{\cos^2 x \cos x}{\sin^2 x}$$

Using the trigonometric identity $\cos^2 x = 1 - \sin^2 x$:

$$= \frac{(1 - \sin^2 x) \cos x}{\sin^2 x}$$

Let $\sin x = t$.

Differentiating both sides:

$$\cos x \, dx = dt$$

Substituting these into the integral:

$$\begin{aligned} \int \sin x \cot^3 x \, dx &= \int \frac{1 - t^2}{t^2} \, dt \\ &= \int \left(\frac{1}{t^2} - 1 \right) \, dt \\ &= \int (t^{-2} - 1) \, dt \end{aligned}$$

Integrating term by term:

$$= -\frac{1}{t} - t + C$$

Now back-substitute $t = \sin x$:

$$\begin{aligned} &= -\frac{1}{\sin x} - \sin x + C \\ &= -\csc x - \sin x + C \end{aligned}$$

Final Answer

$$\int \sin x \cot^3 x \, dx = -\csc x - \sin x + C$$

Question 14 (i)

Evaluate $\int \frac{\cos^3 x}{\sqrt{\sin x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\cos^3 x}{\sqrt{\sin x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

We rewrite the integrand to isolate a factor of $\cos x$:

$$\frac{\cos^3 x}{\sqrt{\sin x}} = \frac{\cos^2 x \cos x}{\sqrt{\sin x}}$$

Using the trigonometric identity $\cos^2 x = 1 - \sin^2 x$:

$$= \frac{(1 - \sin^2 x) \cos x}{\sqrt{\sin x}}$$

Let $\sin x = t$.

Differentiating both sides:

$$\cos x dx = dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{\cos^3 x}{\sqrt{\sin x}} dx &= \int \frac{1 - t^2}{\sqrt{t}} dt \\ &= \int (t^{-1/2} - t^{3/2}) dt \end{aligned}$$

Integrating term by term using the power rule:

$$\begin{aligned} &= \frac{t^{1/2}}{1/2} - \frac{t^{5/2}}{5/2} + C \\ &= 2\sqrt{t} - \frac{2}{5}t^{5/2} + C \end{aligned}$$

Now back-substitute $t = \sin x$:

$$= 2\sqrt{\sin x} - \frac{2}{5}(\sin x)^{5/2} + C$$

Final Answer

$$\int \frac{\cos^3 x}{\sqrt{\sin x}} dx = 2\sqrt{\sin x} - \frac{2}{5}(\sin x)^{5/2} + C$$

Question 14 (ii)

Evaluate $\int \frac{\sin x \cos^3 x}{1 + \cos^2 x} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\sin x \cos^3 x}{1 + \cos^2 x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\cos x = t$.

Differentiating both sides:

$$-\sin x dx = dt \implies \sin x dx = -dt$$

Substituting these into the integral:

$$\int \frac{t^3}{1 + t^2} (-dt) = - \int \frac{t^3}{1 + t^2} dt$$

Using polynomial division or rewriting the numerator $t^3 = t(t^2 + 1) - t$:

$$\begin{aligned} &= - \int \frac{t(t^2 + 1) - t}{1 + t^2} dt \\ &= - \int \left(t - \frac{t}{1 + t^2} \right) dt \\ &= - \int t dt + \int \frac{t}{1 + t^2} dt \end{aligned}$$

For the second integral, let $1 + t^2 = u \implies 2t dt = du \implies t dt = \frac{1}{2} du$:

$$\begin{aligned} &= -\frac{t^2}{2} + \frac{1}{2} \int \frac{1}{u} du \\ &= -\frac{t^2}{2} + \frac{1}{2} \log |u| + C \\ &= -\frac{t^2}{2} + \frac{1}{2} \log(1 + t^2) + C \end{aligned}$$

Now back-substitute $t = \cos x$:

$$= -\frac{1}{2} \cos^2 x + \frac{1}{2} \log(1 + \cos^2 x) + C$$

Final Answer

$$\int \frac{\sin x \cos^3 x}{1 + \cos^2 x} dx = -\frac{1}{2} \cos^2 x + \frac{1}{2} \log(1 + \cos^2 x) + C$$

Question 15 (i)

Evaluate $\int \csc x \log(\csc x - \cot x) dx$.

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: $\int \csc x \log(\csc x - \cot x) dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\log(\csc x - \cot x) = t$.

Differentiating both sides using the chain rule:

$$\begin{aligned} \frac{1}{\csc x - \cot x} \cdot \frac{d}{dx}(\csc x - \cot x) dx &= dt \\ \frac{1}{\csc x - \cot x} \cdot (-\csc x \cot x - (-\csc^2 x)) dx &= dt \\ \frac{\csc^2 x - \csc x \cot x}{\csc x - \cot x} dx &= dt \end{aligned}$$

Factoring out $\csc x$ in the numerator:

$$\begin{aligned} \frac{\csc x(\csc x - \cot x)}{\csc x - \cot x} dx &= dt \\ \csc x dx &= dt \end{aligned}$$

Substituting these into the integral:

$$\int t dt$$

Using the power rule $\int t dt = \frac{t^2}{2} + C$:

$$= \frac{t^2}{2} + C$$

Now back-substitute $t = \log(\csc x - \cot x)$:

$$= \frac{1}{2}[\log(\csc x - \cot x)]^2 + C$$

Final Answer

$$\int \csc x \log(\csc x - \cot x) dx = \frac{1}{2}[\log(\csc x - \cot x)]^2 + C$$

Question 15 (ii)

Evaluate $\int \frac{\log \left(\tan \frac{x}{2} \right)}{\sin x} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\log \left(\tan \frac{x}{2} \right)}{\sin x} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\log \left(\tan \frac{x}{2} \right) = t$.

Differentiating both sides using the chain rule:

$$\frac{1}{\tan \frac{x}{2}} \cdot \sec^2 \left(\frac{x}{2} \right) \cdot \frac{1}{2} dx = dt$$

Expressing in terms of sine and cosine:

$$\frac{\cos \frac{x}{2}}{\sin \frac{x}{2}} \cdot \frac{1}{\cos^2 \frac{x}{2}} \cdot \frac{1}{2} dx = dt$$

$$\frac{1}{2 \sin \frac{x}{2} \cos \frac{x}{2}} dx = dt$$

Using the double-angle identity $2 \sin \theta \cos \theta = \sin 2\theta$:

$$\frac{1}{\sin x} dx = dt$$

Substituting these into the integral:

$$\int t dt$$

Using the power rule $\int t dt = \frac{t^2}{2} + C$:

$$= \frac{t^2}{2} + C$$

Now back-substitute $t = \log \left(\tan \frac{x}{2} \right)$:

$$= \frac{1}{2} \left[\log \left(\tan \frac{x}{2} \right) \right]^2 + C$$

Final Answer

$$\int \frac{\log \left(\tan \frac{x}{2} \right)}{\sin x} dx = \frac{1}{2} \left[\log \left(\tan \frac{x}{2} \right) \right]^2 + C$$

Question 16 (i)

Evaluate $\int \sec^4 x \, dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \sec^4 x \, dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) \, dx$

Solution:

We rewrite the integrand to isolate a factor of $\sec^2 x$:

$$\sec^4 x = \sec^2 x \sec^2 x$$

Using the trigonometric identity $\sec^2 x = 1 + \tan^2 x$:

$$= (1 + \tan^2 x) \sec^2 x$$

Let $\tan x = t$.

Differentiating both sides:

$$\sec^2 x \, dx = dt$$

Substituting these into the integral:

$$\int (1 + t^2) \, dt$$

Integrating term by term:

$$= t + \frac{t^3}{3} + C$$

Now back-substitute $t = \tan x$:

$$= \tan x + \frac{1}{3} \tan^3 x + C$$

Final Answer

$$\int \sec^4 x \, dx = \tan x + \frac{1}{3} \tan^3 x + C$$

Question 16 (ii)

Evaluate $\int \tan^2 x \sec^4 x \, dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \tan^2 x \sec^4 x \, dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) \, dx$

Solution:

We rewrite the integrand to isolate a factor of $\sec^2 x$:

$$\tan^2 x \sec^4 x = \tan^2 x \sec^2 x \sec^2 x$$

Using the trigonometric identity $\sec^2 x = 1 + \tan^2 x$:

$$= \tan^2 x (1 + \tan^2 x) \sec^2 x$$

Let $\tan x = t$.

Differentiating both sides:

$$\sec^2 x \, dx = dt$$

Substituting these into the integral:

$$\begin{aligned} \int \tan^2 x \sec^4 x \, dx &= \int t^2 (1 + t^2) \, dt \\ &= \int (t^2 + t^4) \, dt \end{aligned}$$

Integrating term by term using the power rule:

$$= \frac{t^3}{3} + \frac{t^5}{5} + C$$

Now back-substitute $t = \tan x$:

$$= \frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C$$

Final Answer

$$\int \tan^2 x \sec^4 x \, dx = \frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C$$

Question 17 (i)

Evaluate $\int \frac{\tan^5 \sqrt{x} \sec^2 \sqrt{x}}{\sqrt{x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\tan^5 \sqrt{x} \sec^2 \sqrt{x}}{\sqrt{x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\tan \sqrt{x} = t$.

Differentiating both sides using the chain rule:

$$\sec^2 \sqrt{x} \cdot \frac{1}{2\sqrt{x}} dx = dt$$

Rearranging to isolate the term in the integrand:

$$\frac{\sec^2 \sqrt{x}}{\sqrt{x}} dx = 2 dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{\tan^5 \sqrt{x} \sec^2 \sqrt{x}}{\sqrt{x}} dx &= \int t^5 \cdot 2 dt \\ &= 2 \int t^5 dt \end{aligned}$$

Using the power rule $\int t^5 dt = \frac{t^6}{6} + C$:

$$\begin{aligned} &= 2 \left(\frac{t^6}{6} \right) + C \\ &= \frac{t^6}{3} + C \end{aligned}$$

Now back-substitute $t = \tan \sqrt{x}$:

$$= \frac{1}{3} \tan^6 \sqrt{x} + C$$

Final Answer

$$\int \frac{\tan^5 \sqrt{x} \sec^2 \sqrt{x}}{\sqrt{x}} dx = \frac{1}{3} \tan^6 \sqrt{x} + C$$

Question 17 (ii)

Evaluate $\int \frac{\cos x - \sin x}{\sqrt{1 + \sin 2x}} dx$.

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: $\int \frac{\cos x - \sin x}{\sqrt{1 + \sin 2x}} dx$

Concept / Formula Used: Trigonometric identity: $1 + \sin 2x = (\sin x + \cos x)^2$

Solution:

Using the identity $1 + \sin 2x = \sin^2 x + \cos^2 x + 2 \sin x \cos x = (\sin x + \cos x)^2$:

$$\sqrt{1 + \sin 2x} = \sin x + \cos x$$

The integral becomes:

$$\int \frac{\cos x - \sin x}{\sin x + \cos x} dx$$

Let $\sin x + \cos x = t$.

Differentiating both sides:

$$(\cos x - \sin x) dx = dt$$

Substituting these into the integral:

$$\int \frac{1}{t} dt$$

Using the standard integration result $\int \frac{1}{t} dt = \log |t| + C$:

$$= \log |t| + C$$

Now back-substitute $t = \sin x + \cos x$:

$$= \log |\sin x + \cos x| + C$$

Final Answer

$$\int \frac{\cos x - \sin x}{\sqrt{1 + \sin 2x}} dx = \log |\sin x + \cos x| + C$$

Question 17 (iii)

Evaluate $\int \frac{\sin x - \cos x}{\sqrt{1 + \sin 2x}} dx$, $0 < x < \frac{\pi}{2}$.

TYPE 12e Substitution for Rational Trigonometric Integrals

To Find: $\int \frac{\sin x - \cos x}{\sqrt{1 + \sin 2x}} dx$

Concept / Formula Used: Trigonometric identity: $1 + \sin 2x = (\sin x + \cos x)^2$

Solution:

Since $0 < x < \frac{\pi}{2}$, both $\sin x > 0$ and $\cos x > 0$, so $\sin x + \cos x > 0$. Thus, $\sqrt{1 + \sin 2x} = \sin x + \cos x$. The integral becomes:

$$\int \frac{\sin x - \cos x}{\sin x + \cos x} dx$$

Let $\sin x + \cos x = t$.

Differentiating both sides:

$$(\cos x - \sin x) dx = dt \implies (\sin x - \cos x) dx = -dt$$

Substituting these into the integral:

$$\int \frac{-dt}{t} = - \int \frac{1}{t} dt$$

Using the standard integration result $\int \frac{1}{t} dt = \log |t| + C$:

$$= -\log |t| + C$$

Now back-substitute $t = \sin x + \cos x$:

$$= -\log(\sin x + \cos x) + C \quad (\text{since } \sin x + \cos x > 0)$$

Final Answer

$$\int \frac{\sin x - \cos x}{\sqrt{1 + \sin 2x}} dx = -\log(\sin x + \cos x) + C$$

Question 18 (i)

Evaluate $\int \frac{x}{\sqrt{x+4}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{x}{\sqrt{x+4}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x + 4 = t^2 \implies x = t^2 - 4$.

Differentiating both sides:

$$dx = 2t dt$$

Also, $\sqrt{x+4} = \sqrt{t^2} = t$. Substituting these into the integral:

$$\begin{aligned} \int \frac{x}{\sqrt{x+4}} dx &= \int \frac{t^2 - 4}{t} \cdot 2t dt \\ &= 2 \int (t^2 - 4) dt \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= 2 \left(\frac{t^3}{3} - 4t \right) + C \\ &= \frac{2}{3}t^3 - 8t + C \end{aligned}$$

Now back-substitute $t = \sqrt{x+4}$:

$$= \frac{2}{3}(x+4)^{3/2} - 8\sqrt{x+4} + C$$

Factoring out $\frac{2}{3}\sqrt{x+4}$:

$$\begin{aligned} &= \frac{2}{3}\sqrt{x+4} [(x+4) - 12] + C \\ &= \frac{2}{3}(x-8)\sqrt{x+4} + C \end{aligned}$$

Final Answer

$$\int \frac{x}{\sqrt{x+4}} dx = \frac{2}{3}(x-8)\sqrt{x+4} + C$$

Question 18 (ii)

Evaluate $\int x\sqrt{x+2} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int x\sqrt{x+2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x + 2 = t^2 \implies x = t^2 - 2$.

Differentiating both sides:

$$dx = 2t dt$$

Also, $\sqrt{x+2} = t$. Substituting these into the integral:

$$\begin{aligned}\int x\sqrt{x+2} dx &= \int (t^2 - 2) \cdot t \cdot 2t dt \\ &= 2 \int (t^4 - 2t^2) dt\end{aligned}$$

Integrating term by term using the power rule:

$$\begin{aligned}&= 2 \left(\frac{t^5}{5} - \frac{2t^3}{3} \right) + C \\ &= \frac{2}{5}t^5 - \frac{4}{3}t^3 + C\end{aligned}$$

Now back-substitute $t = \sqrt{x+2}$:

$$= \frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$$

Final Answer

$$\int x\sqrt{x+2} dx = \frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$$

Question 19 (i)

Evaluate $\int \frac{x}{(x^2 + 1)^2} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{x}{(x^2 + 1)^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $x^2 + 1 = t$.

Differentiating both sides:

$$2x dx = dt \implies x dx = \frac{1}{2} dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{x}{(x^2 + 1)^2} dx &= \int \frac{1}{t^2} \cdot \frac{1}{2} dt \\ &= \frac{1}{2} \int t^{-2} dt \end{aligned}$$

Using the power rule $\int t^{-2} dt = -\frac{1}{t} + C$:

$$\begin{aligned} &= \frac{1}{2} \left(-\frac{1}{t} \right) + C \\ &= -\frac{1}{2t} + C \end{aligned}$$

Now back-substitute $t = x^2 + 1$:

$$= -\frac{1}{2(x^2 + 1)} + C$$

Final Answer

$$\int \frac{x}{(x^2 + 1)^2} dx = -\frac{1}{2(x^2 + 1)} + C$$

Question 19 (ii)

Evaluate $\int \frac{x^2}{(4+x)^{3/2}} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{x^2}{(4+x)^{3/2}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $4 + x = t \implies x = t - 4$.

Differentiating both sides:

$$dx = dt$$

Substituting these into the integral:

$$\int \frac{(t-4)^2}{t^{3/2}} dt$$

Expanding the numerator $(t-4)^2 = t^2 - 8t + 16$:

$$\begin{aligned} &= \int \frac{t^2 - 8t + 16}{t^{3/2}} dt \\ &= \int (t^{1/2} - 8t^{-1/2} + 16t^{-3/2}) dt \end{aligned}$$

Integrating term by term using the power rule:

$$\begin{aligned} &= \frac{t^{3/2}}{3/2} - 8 \left(\frac{t^{1/2}}{1/2} \right) + 16 \left(\frac{t^{-1/2}}{-1/2} \right) + C \\ &= \frac{2}{3} t^{3/2} - 16t^{1/2} - 32t^{-1/2} + C \end{aligned}$$

Now back-substitute $t = 4 + x$:

$$= \frac{2}{3} (4+x)^{3/2} - 16\sqrt{4+x} - \frac{32}{\sqrt{4+x}} + C$$

Final Answer

$$\int \frac{x^2}{(4+x)^{3/2}} dx = \frac{2}{3} (4+x)^{3/2} - 16\sqrt{4+x} - \frac{32}{\sqrt{4+x}} + C$$

Question 20 (i)

Evaluate $\int 4x^3\sqrt{5-x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int 4x^3\sqrt{5-x^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $5 - x^2 = t^2 \implies x^2 = 5 - t^2$.

Differentiating both sides:

$$-2x dx = 2t dt \implies x dx = -t dt$$

We rewrite the integrand to match our substitution:

$$4x^3\sqrt{5-x^2} dx = 4x^2\sqrt{5-x^2} \cdot (x dx)$$

Substituting $x^2 = 5 - t^2$, $\sqrt{5-x^2} = t$, and $x dx = -t dt$:

$$\begin{aligned} &= 4(5 - t^2) \cdot t \cdot (-t dt) \\ &= -4t^2(5 - t^2) dt \\ &= (4t^4 - 20t^2) dt \end{aligned}$$

Integrating term by term using the power rule:

$$\begin{aligned} \int (4t^4 - 20t^2) dt &= 4 \left(\frac{t^5}{5} \right) - 20 \left(\frac{t^3}{3} \right) + C \\ &= \frac{4}{5}t^5 - \frac{20}{3}t^3 + C \end{aligned}$$

Now back-substitute $t = \sqrt{5-x^2}$:

$$= \frac{4}{5}(5-x^2)^{5/2} - \frac{20}{3}(5-x^2)^{3/2} + C$$

Final Answer

$$\int 4x^3\sqrt{5-x^2} dx = \frac{4}{5}(5-x^2)^{5/2} - \frac{20}{3}(5-x^2)^{3/2} + C$$

Question 20 (ii)

Evaluate $\int \frac{dx}{x - \sqrt{x}}$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{dx}{x - \sqrt{x}}$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\sqrt{x} = t \implies x = t^2$.

Differentiating both sides:

$$dx = 2t dt$$

Substituting these into the integral:

$$\begin{aligned}\int \frac{dx}{x - \sqrt{x}} &= \int \frac{2t dt}{t^2 - t} \\ &= \int \frac{2t}{t(t-1)} dt \\ &= \int \frac{2}{t-1} dt\end{aligned}$$

Using the standard integration result $\int \frac{1}{u} du = \log |u| + C$:

$$= 2 \log |t - 1| + C$$

Now back-substitute $t = \sqrt{x}$:

$$= 2 \log |\sqrt{x} - 1| + C$$

Final Answer

$$\int \frac{dx}{x - \sqrt{x}} = 2 \log |\sqrt{x} - 1| + C$$

Question 21 (i)

Evaluate $\int \frac{x}{1 + \sqrt{x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{x}{1 + \sqrt{x}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $\sqrt{x} = t \implies x = t^2$.

Differentiating both sides:

$$dx = 2t dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{x}{1 + \sqrt{x}} dx &= \int \frac{t^2}{1 + t} \cdot 2t dt \\ &= 2 \int \frac{t^3}{1 + t} dt \end{aligned}$$

Using polynomial division to rewrite $\frac{t^3}{t + 1}$:

$$t^3 = (t + 1)(t^2 - t + 1) - 1 \implies \frac{t^3}{t + 1} = t^2 - t + 1 - \frac{1}{t + 1}$$

The integral becomes:

$$= 2 \int \left(t^2 - t + 1 - \frac{1}{t + 1} \right) dt$$

Integrating term by term:

$$\begin{aligned} &= 2 \left(\frac{t^3}{3} - \frac{t^2}{2} + t - \log |t + 1| \right) + C \\ &= \frac{2}{3} t^3 - t^2 + 2t - 2 \log |t + 1| + C \end{aligned}$$

Now back-substitute $t = \sqrt{x}$:

$$= \frac{2}{3} x^{3/2} - x + 2\sqrt{x} - 2 \log(1 + \sqrt{x}) + C \quad (\text{since } 1 + \sqrt{x} > 0)$$

Final Answer

$$\int \frac{x}{1 + \sqrt{x}} dx = \frac{2}{3} x^{3/2} - x + 2\sqrt{x} - 2 \log(1 + \sqrt{x}) + C$$

Question 21 (ii)

Evaluate $\int \frac{x^{1/2}}{1+x^{3/4}} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{x^{1/2}}{1+x^{3/4}} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

To clear the fractional exponents, we find the LCM of the denominators of the exponents $1/2$ and $3/4$, which is 4. Let $x = t^4 \implies x^{1/2} = t^2$ and $x^{3/4} = t^3$.

Differentiating both sides:

$$dx = 4t^3 dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{x^{1/2}}{1+x^{3/4}} dx &= \int \frac{t^2}{1+t^3} \cdot 4t^3 dt \\ &= 4 \int \frac{t^5}{1+t^3} dt \end{aligned}$$

Using polynomial division to rewrite $\frac{t^5}{t^3+1}$:

$$t^5 = t^2(t^3+1) - t^2 \implies \frac{t^5}{t^3+1} = t^2 - \frac{t^2}{t^3+1}$$

The integral becomes:

$$\begin{aligned} &= 4 \int \left(t^2 - \frac{t^2}{t^3+1} \right) dt \\ &= 4 \int t^2 dt - 4 \int \frac{t^2}{t^3+1} dt \end{aligned}$$

For the second integral, let $t^3+1 = u \implies 3t^2 dt = du \implies t^2 dt = \frac{1}{3} du$:

$$\begin{aligned} &= 4 \left(\frac{t^3}{3} \right) - 4 \left(\frac{1}{3} \log |u| \right) + C \\ &= \frac{4}{3} t^3 - \frac{4}{3} \log |t^3+1| + C \end{aligned}$$

Now back-substitute $t = x^{1/4}$:

$$= \frac{4}{3} x^{3/4} - \frac{4}{3} \log(1+x^{3/4}) + C \quad (\text{since } 1+x^{3/4} > 0)$$

Final Answer

$$\int \frac{x^{1/2}}{1+x^{3/4}} dx = \frac{4}{3} x^{3/4} - \frac{4}{3} \log(1+x^{3/4}) + C$$

Question 22 (i)

Evaluate $\int \frac{dx}{\sqrt{x+1} + \sqrt[3]{x+1}}.$

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{dx}{\sqrt{x+1} + \sqrt[3]{x+1}}$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

To clear the fractional exponents $1/2$ and $1/3$, we find the LCM of 2 and 3, which is 6. Let $x + 1 = t^6$.

Differentiating both sides:

$$dx = 6t^5 dt$$

Also, $\sqrt{x+1} = t^3$ and $\sqrt[3]{x+1} = t^2$. Substituting these into the integral:

$$\begin{aligned} \int \frac{dx}{\sqrt{x+1} + \sqrt[3]{x+1}} &= \int \frac{6t^5 dt}{t^3 + t^2} \\ &= 6 \int \frac{t^5}{t^2(t+1)} dt \\ &= 6 \int \frac{t^3}{t+1} dt \end{aligned}$$

Using polynomial division to rewrite $\frac{t^3}{t+1}$:

$$\frac{t^3}{t+1} = t^2 - t + 1 - \frac{1}{t+1}$$

The integral becomes:

$$= 6 \int \left(t^2 - t + 1 - \frac{1}{t+1} \right) dt$$

Integrating term by term:

$$\begin{aligned} &= 6 \left(\frac{t^3}{3} - \frac{t^2}{2} + t - \log|t+1| \right) + C \\ &= 2t^3 - 3t^2 + 6t - 6 \log|t+1| + C \end{aligned}$$

Now back-substitute $t = (x+1)^{1/6}$:

$$= 2(x+1)^{1/2} - 3(x+1)^{1/3} + 6(x+1)^{1/6} - 6 \log(1 + (x+1)^{1/6}) + C$$

Final Answer

$$\int \frac{dx}{\sqrt{x+1} + \sqrt[3]{x+1}} = 2\sqrt{x+1} - 3\sqrt[3]{x+1} + 6(x+1)^{1/6} - 6 \log(1 + (x+1)^{1/6}) + C$$

Question 22 (ii)

Evaluate $\int \frac{dx}{\sqrt{1+\sqrt{x}}}$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{dx}{\sqrt{1+\sqrt{x}}}$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

$$\text{Let } 1 + \sqrt{x} = t^2 \implies \sqrt{x} = t^2 - 1 \implies x = (t^2 - 1)^2.$$

Differentiating both sides using the chain rule:

$$dx = 2(t^2 - 1) \cdot 2t dt = 4t(t^2 - 1) dt$$

Also, $\sqrt{1 + \sqrt{x}} = t$. Substituting these into the integral:

$$\begin{aligned} \int \frac{dx}{\sqrt{1+\sqrt{x}}} &= \int \frac{4t(t^2 - 1) dt}{t} \\ &= 4 \int (t^2 - 1) dt \end{aligned}$$

Integrating term by term:

$$\begin{aligned} &= 4 \left(\frac{t^3}{3} - t \right) + C \\ &= \frac{4}{3}t^3 - 4t + C \end{aligned}$$

Now back-substitute $t = \sqrt{1 + \sqrt{x}}$:

$$= \frac{4}{3}(1 + \sqrt{x})^{3/2} - 4\sqrt{1 + \sqrt{x}} + C$$

Final Answer

$$\int \frac{dx}{\sqrt{1+\sqrt{x}}} = \frac{4}{3}(1 + \sqrt{x})^{3/2} - 4\sqrt{1 + \sqrt{x}} + C$$

Question 23 (i)

Evaluate $\int \frac{\sin 2x}{(a^2 + b^2 \sin^2 x)^2} dx$.

TYPE 12a Direct/Algebraic Substitution

To Find: $\int \frac{\sin 2x}{(a^2 + b^2 \sin^2 x)^2} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

Let $a^2 + b^2 \sin^2 x = t$.

Differentiating both sides using the chain rule and the identity $\sin 2x = 2 \sin x \cos x$:

$$b^2 \cdot 2 \sin x \cos x dx = dt$$

$$b^2 \sin 2x dx = dt \implies \sin 2x dx = \frac{1}{b^2} dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{\sin 2x}{(a^2 + b^2 \sin^2 x)^2} dx &= \int \frac{1}{t^2} \cdot \frac{1}{b^2} dt \\ &= \frac{1}{b^2} \int t^{-2} dt \end{aligned}$$

Using the power rule $\int t^{-2} dt = -\frac{1}{t} + C$:

$$\begin{aligned} &= \frac{1}{b^2} \left(-\frac{1}{t} \right) + C \\ &= -\frac{1}{b^2 t} + C \end{aligned}$$

Now back-substitute $t = a^2 + b^2 \sin^2 x$:

$$= -\frac{1}{b^2(a^2 + b^2 \sin^2 x)} + C$$

Final Answer

$$\int \frac{\sin 2x}{(a^2 + b^2 \sin^2 x)^2} dx = -\frac{1}{b^2(a^2 + b^2 \sin^2 x)} + C$$

Question 23 (ii)

Evaluate $\int \frac{\sqrt{1+x^2}}{x^4} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{\sqrt{1+x^2}}{x^4} dx$

Concept / Formula Used: Substitution rule: if $u = g(x)$, then $du = g'(x) dx$

Solution:

We rewrite the integrand by factoring out x^2 from the square root:

$$\frac{\sqrt{1+x^2}}{x^4} = \frac{\sqrt{x^2 \left(\frac{1}{x^2} + 1\right)}}{x^4} = \frac{x\sqrt{1+\frac{1}{x^2}}}{x^4} = \frac{\sqrt{1+\frac{1}{x^2}}}{x^3}$$

Let $1 + \frac{1}{x^2} = t \implies 1 + x^{-2} = t$.

Differentiating both sides:

$$-2x^{-3} dx = dt \implies \frac{1}{x^3} dx = -\frac{1}{2} dt$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{\sqrt{1+x^2}}{x^4} dx &= \int \sqrt{t} \cdot \left(-\frac{1}{2}\right) dt \\ &= -\frac{1}{2} \int t^{1/2} dt \end{aligned}$$

Using the power rule $\int t^{1/2} dt = \frac{2}{3} t^{3/2} + C$:

$$\begin{aligned} &= -\frac{1}{2} \left(\frac{2}{3} t^{3/2} \right) + C \\ &= -\frac{1}{3} t^{3/2} + C \end{aligned}$$

Now back-substitute $t = 1 + \frac{1}{x^2}$:

$$\begin{aligned} &= -\frac{1}{3} \left(1 + \frac{1}{x^2} \right)^{3/2} + C \\ &= -\frac{(x^2 + 1)^{3/2}}{3x^3} + C \end{aligned}$$

Final Answer

$$\int \frac{\sqrt{1+x^2}}{x^4} dx = -\frac{(x^2 + 1)^{3/2}}{3x^3} + C$$

Question 23 (iii)

Evaluate $\int \frac{1}{x^2\sqrt{1+x^2}} dx$.

TYPE 12b Substitution Involving Radical Expressions

To Find: $\int \frac{1}{x^2\sqrt{1+x^2}} dx$

Concept / Formula Used: Trigonometric substitution: let $x = \tan \theta$, so $dx = \sec^2 \theta d\theta$ and $\sqrt{1+x^2} = \sec \theta$

Solution:

Let $x = \tan \theta$.

Differentiating both sides:

$$dx = \sec^2 \theta d\theta$$

Also, using the identity $\sqrt{1+\tan^2 \theta} = \sec \theta$:

$$\sqrt{1+x^2} = \sec \theta$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{1}{x^2\sqrt{1+x^2}} dx &= \int \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sec \theta} \\ &= \int \frac{\sec \theta}{\tan^2 \theta} d\theta \end{aligned}$$

Simplifying the trigonometric expression:

$$\frac{\sec \theta}{\tan^2 \theta} = \frac{1}{\cos \theta} \cdot \frac{\cos^2 \theta}{\sin^2 \theta} = \frac{\cos \theta}{\sin^2 \theta} = \csc \theta \cot \theta$$

The integral becomes:

$$\int \csc \theta \cot \theta d\theta$$

Using the standard integration result $\int \csc \theta \cot \theta d\theta = -\csc \theta + C$:

$$= -\csc \theta + C$$

Since $x = \tan \theta$, we construct a right triangle where the opposite side is x , adjacent side is 1, and hypotenuse is $\sqrt{1+x^2}$. Thus:

$$\csc \theta = \frac{\sqrt{1+x^2}}{x}$$

Substituting this back:

$$= -\frac{\sqrt{1+x^2}}{x} + C$$

Final Answer

$$\int \frac{1}{x^2\sqrt{1+x^2}} dx = -\frac{\sqrt{1+x^2}}{x} + C$$

Question 24

If $\frac{d}{dx}(F(x)) = \frac{\sec^4 x}{\csc^4 x}$ and $F\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$, then find $F(x)$.

TYPE 12a Direct/Algebraic Substitution

Given: $\frac{d}{dx}(F(x)) = \frac{\sec^4 x}{\csc^4 x}$ and $F\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$

To Find: $F(x)$

Solution:

First, simplify the given derivative:

$$\frac{\sec^4 x}{\csc^4 x} = \frac{\left(\frac{1}{\cos^4 x}\right)}{\left(\frac{1}{\sin^4 x}\right)} = \frac{\sin^4 x}{\cos^4 x} = \tan^4 x$$

Thus, $F(x)$ is the antiderivative of $\tan^4 x$:

$$F(x) = \int \tan^4 x \, dx$$

We rewrite $\tan^4 x$ using trigonometric identities:

$$\begin{aligned} \tan^4 x &= \tan^2 x \tan^2 x \\ &= \tan^2 x (\sec^2 x - 1) \\ &= \tan^2 x \sec^2 x - \tan^2 x \\ &= \tan^2 x \sec^2 x - (\sec^2 x - 1) \\ &= \tan^2 x \sec^2 x - \sec^2 x + 1 \end{aligned}$$

Now, integrate term by term:

$$F(x) = \int \tan^2 x \sec^2 x \, dx - \int \sec^2 x \, dx + \int 1 \, dx$$

For the first integral, let $\tan x = t \implies \sec^2 x \, dx = dt$:

$$\int \tan^2 x \sec^2 x \, dx = \int t^2 \, dt = \frac{t^3}{3} = \frac{\tan^3 x}{3}$$

For the remaining terms:

$$\begin{aligned} - \int \sec^2 x \, dx &= -\tan x \\ \int 1 \, dx &= x \end{aligned}$$

Combining these results:

$$F(x) = \frac{1}{3} \tan^3 x - \tan x + x + C$$

Using the given initial condition $F\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$:

$$\frac{\pi}{4} = \frac{1}{3} \tan^3\left(\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{4}\right) + \frac{\pi}{4} + C$$

$$\frac{\pi}{4} = \frac{1}{3}(1)^3 - 1 + \frac{\pi}{4} + C$$

$$0 = \frac{1}{3} - 1 + C$$

$$0 = -\frac{2}{3} + C \implies C = \frac{2}{3}$$

Substituting C back into the expression for $F(x)$:

$$F(x) = \frac{1}{3} \tan^3 x - \tan x + x + \frac{2}{3}$$

Final Answer

$$F(x) = \frac{1}{3} \tan^3 x - \tan x + x + \frac{2}{3}$$

Question 25

If $\int x^5 \cos(x^6) dx = k \sin(x^6) + C$, find the value of k .

TYPE 12a Direct/Algebraic Substitution

Given: $\int x^5 \cos(x^6) dx = k \sin(x^6) + C$

To Find: k

Solution:

Let us evaluate the integral on the left-hand side:

$$I = \int x^5 \cos(x^6) dx$$

Let $x^6 = t$.

Differentiating both sides:

$$6x^5 dx = dt \implies x^5 dx = \frac{1}{6} dt$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \cos(t) \cdot \frac{1}{6} dt \\ &= \frac{1}{6} \int \cos(t) dt \end{aligned}$$

Using the standard integration result $\int \cos(t) dt = \sin(t) + C$:

$$= \frac{1}{6} \sin(t) + C$$

Now back-substitute $t = x^6$:

$$= \frac{1}{6} \sin(x^6) + C$$

Comparing this with the given equation:

$$\frac{1}{6} \sin(x^6) + C = k \sin(x^6) + C$$

By comparing the coefficients of $\sin(x^6)$, we get:

$$k = \frac{1}{6}$$

Final Answer

$$k = \frac{1}{6}$$

7.6 – Integration by Substitution: Trigonometric Functions

Question 1 (i)

Evaluate $\int \tan(2 - 3x) dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \tan(2 - 3x) dx$

Concept / Formula Used: Integration of standard trigonometric functions and linear substitution

Solution:

Let $2 - 3x = t$.

Differentiating both sides with respect to x :

$$-3 dx = dt$$

So, the differential substitution is:

$$dx = -\frac{1}{3} dt$$

Substituting $2 - 3x = t$ and $dx = -\frac{1}{3} dt$ into the integral:

$$\begin{aligned} I &= \int \tan(t) \left(-\frac{1}{3} dt \right) \\ &= -\frac{1}{3} \int \tan(t) dt \end{aligned}$$

Using the standard result $\int \tan(t) dt = \ln |\sec t| + c$:

$$I = -\frac{1}{3} \ln |\sec t| + c$$

Now back-substitute $t = 2 - 3x$:

$$I = -\frac{1}{3} \ln |\sec(2 - 3x)| + c$$

Final Answer

$$\int \tan(2 - 3x) dx = -\frac{1}{3} \ln |\sec(2 - 3x)| + c$$

Common Mistake: Forgetting to divide by the coefficient of x (-3) after integrating $\tan(2 - 3x)$ is the most common error.

Question 1 (ii)

Evaluate $\int \cot(7x - 4) dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \cot(7x - 4) dx$

Concept / Formula Used: Integration of standard trigonometric functions and linear substitution

Solution:

Let $7x - 4 = t$.

Differentiating both sides with respect to x :

$$7 dx = dt$$

So, the differential substitution is:

$$dx = \frac{1}{7} dt$$

Substituting $7x - 4 = t$ and $dx = \frac{1}{7} dt$ into the integral:

$$\begin{aligned} I &= \int \cot(t) \left(\frac{1}{7} dt \right) \\ &= \frac{1}{7} \int \cot(t) dt \end{aligned}$$

Using the standard result $\int \cot(t) dt = \ln |\sin t| + c$:

$$I = \frac{1}{7} \ln |\sin t| + c$$

Now back-substitute $t = 7x - 4$:

$$I = \frac{1}{7} \ln |\sin(7x - 4)| + c$$

Final Answer

$$\int \cot(7x - 4) dx = \frac{1}{7} \ln |\sin(7x - 4)| + c$$

Common Mistake: Using $\ln |\cos(7x - 4)|$ instead of $\ln |\sin(7x - 4)|$ as the integral of $\cot t$.

Question 2 (i)

Evaluate $\int \frac{1 + \tan^2 x}{2 \tan x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{1 + \tan^2 x}{2 \tan x} dx$

Concept / Formula Used: Trigonometric identity $\sec^2 x = 1 + \tan^2 x$ and substitution

Solution:

Using the trigonometric identity $1 + \tan^2 x = \sec^2 x$, the integral becomes:

$$I = \int \frac{\sec^2 x}{2 \tan x} dx$$

Let $\tan x = t$.

Differentiating both sides with respect to x :

$$\sec^2 x dx = dt$$

Substituting $\tan x = t$ and $\sec^2 x dx = dt$ into the integral:

$$\begin{aligned} I &= \int \frac{1}{2t} dt \\ &= \frac{1}{2} \int \frac{1}{t} dt \end{aligned}$$

Using the standard result $\int \frac{1}{t} dt = \ln |t| + c$:

$$I = \frac{1}{2} \ln |t| + c$$

Now back-substitute $t = \tan x$:

$$I = \frac{1}{2} \ln |\tan x| + c$$

Final Answer

$$\int \frac{1 + \tan^2 x}{2 \tan x} dx = \frac{1}{2} \ln |\tan x| + c$$

Common Mistake: Failing to recognize that $1 + \tan^2 x$ is the derivative of $\tan x$ up to a constant multiplier.

Question 2 (ii)

Evaluate $\int \frac{1}{\sin x \cos x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{1}{\sin x \cos x} dx$

Concept / Formula Used: Trigonometric identities and substitution

Solution:

Multiply and divide the numerator and denominator by $\cos^2 x$, or multiply the numerator and denominator by 2:

$$\begin{aligned} I &= \int \frac{2}{2 \sin x \cos x} dx \\ &= \int \frac{2}{\sin 2x} dx \\ &= 2 \int \csc 2x dx \end{aligned}$$

Let $2x = t$, so $2 dx = dt \implies dx = \frac{1}{2} dt$:

$$\begin{aligned} I &= 2 \int \csc(t) \left(\frac{1}{2} dt \right) \\ &= \int \csc(t) dt \end{aligned}$$

Using the standard result $\int \csc(t) dt = \ln |\csc t - \cot t| + c$:

$$I = \ln |\csc t - \cot t| + c$$

Now back-substitute $t = 2x$:

$$I = \ln |\csc 2x - \cot 2x| + c$$

Alternatively, using $\tan x$ substitution:

$$I = \int \frac{\sec^2 x}{\tan x} dx = \ln |\tan x| + c$$

Final Answer

$$\int \frac{1}{\sin x \cos x} dx = \ln |\tan x| + c$$

Common Mistake: Not simplifying $\frac{1}{\sin x \cos x}$ using the double-angle formula $\sin 2x = 2 \sin x \cos x$ before integrating.

Question 3 (i)

Evaluate $\int \frac{\cos 2x}{\sin x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{\cos 2x}{\sin x} dx$

Concept / Formula Used: Trigonometric identity $\cos 2x = 1 - 2\sin^2 x$

Solution:

Using the double-angle identity $\cos 2x = 1 - 2\sin^2 x$, rewrite the numerator:

$$\begin{aligned} I &= \int \frac{1 - 2\sin^2 x}{\sin x} dx \\ &= \int \left(\frac{1}{\sin x} - \frac{2\sin^2 x}{\sin x} \right) dx \\ &= \int (\csc x - 2\sin x) dx \end{aligned}$$

Integrating term by term using standard formulas:

$$\begin{aligned} I &= \int \csc x dx - 2 \int \sin x dx \\ &= \ln |\csc x - \cot x| - 2(-\cos x) + c \\ &= \ln |\csc x - \cot x| + 2\cos x + c \end{aligned}$$

Final Answer

$$\int \frac{\cos 2x}{\sin x} dx = \ln |\csc x - \cot x| + 2\cos x + c$$

Common Mistake: Using $\cos 2x = 2\cos^2 x - 1$, which introduces $\cos^2 x$ over $\sin x$ and complicates further integration.

Question 3 (ii)

Evaluate $\int \frac{\sin 2x}{\sin 4x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{\sin 2x}{\sin 4x} dx$

Concept / Formula Used: Trigonometric identity $\sin 4x = 2 \sin 2x \cos 2x$

Solution:

Using the double-angle identity $\sin 4x = 2 \sin 2x \cos 2x$:

$$\begin{aligned} I &= \int \frac{\sin 2x}{2 \sin 2x \cos 2x} dx \\ &= \int \frac{1}{2 \cos 2x} dx \\ &= \frac{1}{2} \int \sec 2x dx \end{aligned}$$

Using the standard result $\int \sec(ax + b) dx = \frac{1}{a} \ln |\sec(ax + b) + \tan(ax + b)| + c$:

$$\begin{aligned} I &= \frac{1}{2} \times \frac{1}{2} \ln |\sec 2x + \tan 2x| + c \\ &= \frac{1}{4} \ln |\sec 2x + \tan 2x| + c \end{aligned}$$

Final Answer

$$\int \frac{\sin 2x}{\sin 4x} dx = \frac{1}{4} \ln |\sec 2x + \tan 2x| + c$$

Common Mistake: Forgetting to divide by 2 twice: once from the denominator expansion of $\sin 4x$ and once from the integration of $\sec 2x$.

Question 4 (i)

Evaluate $\int \frac{\sin x}{\sin(x - \alpha)} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{\sin x}{\sin(x - \alpha)} dx$

Concept / Formula Used: Substitution by shifting the variable of integration

Solution:

Let $x - \alpha = t$, which gives $x = t + \alpha$.

Differentiating both sides with respect to x :

$$dx = dt$$

Substituting $x = t + \alpha$, $x - \alpha = t$, and $dx = dt$ into the integral:

$$I = \int \frac{\sin(t + \alpha)}{\sin t} dt$$

Expanding the numerator using the compound angle formula $\sin(A + B) = \sin A \cos B + \cos A \sin B$:

$$\begin{aligned} I &= \int \frac{\sin t \cos \alpha + \cos t \sin \alpha}{\sin t} dt \\ &= \int \left(\frac{\sin t \cos \alpha}{\sin t} + \frac{\cos t \sin \alpha}{\sin t} \right) dt \\ &= \int (\cos \alpha + \cot t \sin \alpha) dt \end{aligned}$$

Since α is a constant with respect to t , $\cos \alpha$ and $\sin \alpha$ are constants:

$$\begin{aligned} I &= \cos \alpha \int 1 dt + \sin \alpha \int \cot t dt \\ &= (\cos \alpha)t + (\sin \alpha) \ln |\sin t| + c \end{aligned}$$

Now back-substitute $t = x - \alpha$:

$$\begin{aligned} I &= \cos \alpha(x - \alpha) + \sin \alpha \ln |\sin(x - \alpha)| + c \\ &= x \cos \alpha - \alpha \cos \alpha + \sin \alpha \ln |\sin(x - \alpha)| + c \end{aligned}$$

Absorbing the constant $-\alpha \cos \alpha$ into the arbitrary constant c :

$$I = x \cos \alpha + \sin \alpha \ln |\sin(x - \alpha)| + c$$

Final Answer

$$\int \frac{\sin x}{\sin(x - \alpha)} dx = x \cos \alpha + \sin \alpha \ln |\sin(x - \alpha)| + c$$

Common Mistake: Treating α as a variable of integration rather than a constant.

Question 4 (ii)

Evaluate $\int \frac{\cos x}{\cos(x + \alpha)} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{\cos x}{\cos(x + \alpha)} dx$

Concept / Formula Used: Substitution by shifting the variable of integration

Solution:

Let $x + \alpha = t$, which gives $x = t - \alpha$.

Differentiating both sides with respect to x :

$$dx = dt$$

Substituting $x = t - \alpha$, $x + \alpha = t$, and $dx = dt$ into the integral:

$$I = \int \frac{\cos(t - \alpha)}{\cos t} dt$$

Expanding the numerator using the compound angle formula $\cos(A - B) = \cos A \cos B + \sin A \sin B$:

$$\begin{aligned} I &= \int \frac{\cos t \cos \alpha + \sin t \sin \alpha}{\cos t} dt \\ &= \int \left(\frac{\cos t \cos \alpha}{\cos t} + \frac{\sin t \sin \alpha}{\cos t} \right) dt \\ &= \int (\cos \alpha + \tan t \sin \alpha) dt \end{aligned}$$

Integrating term by term with respect to t :

$$\begin{aligned} I &= \cos \alpha \int 1 dt + \sin \alpha \int \tan t dt \\ &= (\cos \alpha)t + (\sin \alpha) \ln |\sec t| + c \end{aligned}$$

Now back-substitute $t = x + \alpha$:

$$\begin{aligned} I &= \cos \alpha(x + \alpha) + \sin \alpha \ln |\sec(x + \alpha)| + c \\ &= x \cos \alpha + \alpha \cos \alpha + \sin \alpha \ln |\sec(x + \alpha)| + c \end{aligned}$$

Absorbing the constant $\alpha \cos \alpha$ into the arbitrary constant c :

$$I = x \cos \alpha + \sin \alpha \ln |\sec(x + \alpha)| + c$$

Final Answer

$$\int \frac{\cos x}{\cos(x + \alpha)} dx = x \cos \alpha + \sin \alpha \ln |\sec(x + \alpha)| + c$$

Common Mistake: Applying the wrong compound angle expansion for $\cos(t - \alpha)$, using minus instead of plus.

Question 5 (i)

Evaluate $\int \frac{dx}{\sin(x - \alpha) \sin(x - \beta)}$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{dx}{\sin(x - \alpha) \sin(x - \beta)}$

Concept / Formula Used: Trigonometric identity manipulation for product of sines

Solution:

Multiply and divide the integrand by $\sin(\alpha - \beta)$:

$$I = \frac{1}{\sin(\alpha - \beta)} \int \frac{\sin(\alpha - \beta)}{\sin(x - \alpha) \sin(x - \beta)} dx$$

Rewrite the numerator $\sin(\alpha - \beta)$ in terms of $(x - \beta) - (x - \alpha)$:

$$\alpha - \beta = (x - \beta) - (x - \alpha)$$

Substitute this back into the numerator:

$$I = \frac{1}{\sin(\alpha - \beta)} \int \frac{\sin[(x - \beta) - (x - \alpha)]}{\sin(x - \alpha) \sin(x - \beta)} dx$$

Expand the numerator using the sine subtraction formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

$$\begin{aligned} I &= \frac{1}{\sin(\alpha - \beta)} \int \frac{\sin(x - \beta) \cos(x - \alpha) - \cos(x - \beta) \sin(x - \alpha)}{\sin(x - \alpha) \sin(x - \beta)} dx \\ &= \frac{1}{\sin(\alpha - \beta)} \int \left(\frac{\sin(x - \beta) \cos(x - \alpha)}{\sin(x - \alpha) \sin(x - \beta)} - \frac{\cos(x - \beta) \sin(x - \alpha)}{\sin(x - \alpha) \sin(x - \beta)} \right) dx \\ &= \frac{1}{\sin(\alpha - \beta)} \int \left(\frac{\cos(x - \alpha)}{\sin(x - \alpha)} - \frac{\cos(x - \beta)}{\sin(x - \beta)} \right) dx \\ &= \frac{1}{\sin(\alpha - \beta)} \int (\cot(x - \alpha) - \cot(x - \beta)) dx \end{aligned}$$

Integrating term by term using $\int \cot(kx + p) dx = \frac{1}{k} \ln |\sin(kx + p)|$:

$$I = \frac{1}{\sin(\alpha - \beta)} [\ln |\sin(x - \alpha)| - \ln |\sin(x - \beta)|] + c$$

Using logarithm property $\ln A - \ln B = \ln \left(\frac{A}{B} \right)$:

$$I = \frac{1}{\sin(\alpha - \beta)} \ln \left| \frac{\sin(x - \alpha)}{\sin(x - \beta)} \right| + c$$

Final Answer

$$\int \frac{dx}{\sin(x - \alpha) \sin(x - \beta)} = \frac{1}{\sin(\alpha - \beta)} \ln \left| \frac{\sin(x - \alpha)}{\sin(x - \beta)} \right| + c$$

Common Mistake: Forgetting to multiply and divide by $\sin(\alpha - \beta)$ at the start to balance the integral.

Question 5 (ii)

Evaluate $\int \frac{dx}{\cos(x - a) \cos(x - b)}$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{dx}{\cos(x-a)\cos(x-b)}$

Concept / Formula Used: Trigonometric identity manipulation for product of cosines

Solution:

Multiply and divide the integrand by $\sin(a-b)$:

$$I = \frac{1}{\sin(a-b)} \int \frac{\sin(a-b)}{\cos(x-a)\cos(x-b)} dx$$

Express the numerator $\sin(a-b)$ in terms of $(x-b) - (x-a)$:

$$a-b = (x-b) - (x-a)$$

Substitute this into the numerator:

$$I = \frac{1}{\sin(a-b)} \int \frac{\sin[(x-b) - (x-a)]}{\cos(x-a)\cos(x-b)} dx$$

Expand the numerator using the sine subtraction formula $\sin(A-B) = \sin A \cos B - \cos A \sin B$:

$$\begin{aligned} I &= \frac{1}{\sin(a-b)} \int \frac{\sin(x-b)\cos(x-a) - \cos(x-b)\sin(x-a)}{\cos(x-a)\cos(x-b)} dx \\ &= \frac{1}{\sin(a-b)} \int \left(\frac{\sin(x-b)\cos(x-a)}{\cos(x-a)\cos(x-b)} - \frac{\cos(x-b)\sin(x-a)}{\cos(x-a)\cos(x-b)} \right) dx \\ &= \frac{1}{\sin(a-b)} \int \left(\frac{\sin(x-b)}{\cos(x-b)} - \frac{\sin(x-a)}{\cos(x-a)} \right) dx \\ &= \frac{1}{\sin(a-b)} \int (\tan(x-b) - \tan(x-a)) dx \end{aligned}$$

Integrating term by term using $\int \tan(kx+p) dx = -\frac{1}{k} \ln |\cos(kx+p)|$:

$$I = \frac{1}{\sin(a-b)} [-\ln |\cos(x-b)| - (-\ln |\cos(x-a)|)] + c$$

Using logarithm property $\ln A - \ln B = \ln \left(\frac{A}{B}\right)$:

$$I = \frac{1}{\sin(a-b)} \ln \left| \frac{\cos(x-a)}{\cos(x-b)} \right| + c$$

Final Answer

$$\int \frac{dx}{\cos(x-a)\cos(x-b)} = \frac{1}{\sin(a-b)} \ln \left| \frac{\cos(x-a)}{\cos(x-b)} \right| + c$$

Common Mistake: Swapping the order of terms inside the logarithm, yielding an inverted fraction argument.

Question 6 (i)

Evaluate $\int \frac{dx}{\sin(x-a) \cos(x-b)}$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{dx}{\sin(x-a) \cos(x-b)}$

Concept / Formula Used: Trigonometric identity manipulation for mixed sine-cosine product

Solution:

Multiply and divide the integrand by $\cos(a-b)$:

$$I = \frac{1}{\cos(a-b)} \int \frac{\cos(a-b)}{\sin(x-a) \cos(x-b)} dx$$

Express the numerator $\cos(a-b)$ in terms of $(x-a) - (x-b)$:

$$a-b = (x-b) - (x-a) \implies \cos(a-b) = \cos[(x-b) - (x-a)]$$

Using the cosine subtraction formula $\cos(A-B) = \cos A \cos B + \sin A \sin B$:

$$\cos(a-b) = \cos(x-b) \cos(x-a) + \sin(x-b) \sin(x-a)$$

Substitute this back into the integral:

$$\begin{aligned} I &= \frac{1}{\cos(a-b)} \int \frac{\cos(x-b) \cos(x-a) + \sin(x-b) \sin(x-a)}{\sin(x-a) \cos(x-b)} dx \\ &= \frac{1}{\cos(a-b)} \int \left(\frac{\cos(x-b) \cos(x-a)}{\sin(x-a) \cos(x-b)} + \frac{\sin(x-b) \sin(x-a)}{\sin(x-a) \cos(x-b)} \right) dx \\ &= \frac{1}{\cos(a-b)} \int \left(\frac{\cos(x-a)}{\sin(x-a)} + \frac{\sin(x-b)}{\cos(x-b)} \right) dx \\ &= \frac{1}{\cos(a-b)} \int (\cot(x-a) + \tan(x-b)) dx \end{aligned}$$

Integrating term by term:

$$I = \frac{1}{\cos(a-b)} [\ln |\sin(x-a)| - \ln |\cos(x-b)|] + c$$

Using logarithm property $\ln A - \ln B = \ln \left(\frac{A}{B} \right)$:

$$I = \frac{1}{\cos(a-b)} \ln \left| \frac{\sin(x-a)}{\cos(x-b)} \right| + c$$

Final Answer

$$\int \frac{dx}{\sin(x-a) \cos(x-b)} = \frac{1}{\cos(a-b)} \ln \left| \frac{\sin(x-a)}{\cos(x-b)} \right| + c$$

Common Mistake: Multiplying and dividing by $\sin(a - b)$ instead of $\cos(a - b)$, which leaves unresolvable products.

Question 6 (ii) ★ Likely Exam Question

Evaluate $\int \tan^3 x \, dx$. (ISC 2016)

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \tan^3 x \, dx$

Concept / Formula Used: Trigonometric identity $\tan^2 x = \sec^2 x - 1$

Solution:

Split $\tan^3 x$ as $\tan x \cdot \tan^2 x$:

$$I = \int \tan x \cdot \tan^2 x \, dx$$

Using the identity $\tan^2 x = \sec^2 x - 1$:

$$\begin{aligned} I &= \int \tan x (\sec^2 x - 1) \, dx \\ &= \int (\tan x \sec^2 x - \tan x) \, dx \\ &= \int \tan x \sec^2 x \, dx - \int \tan x \, dx \end{aligned}$$

For the first integral $\int \tan x \sec^2 x \, dx$, let $\tan x = t$, so $\sec^2 x \, dx = dt$:

$$\begin{aligned} \int \tan x \sec^2 x \, dx &= \int t \, dt \\ &= \frac{t^2}{2} \\ &= \frac{\tan^2 x}{2} \end{aligned}$$

For the second integral $\int \tan x \, dx$, using the standard result:

$$\int \tan x \, dx = \ln |\sec x|$$

Combining both parts:

$$I = \frac{\tan^2 x}{2} - \ln |\sec x| + c$$

Final Answer

$$\int \tan^3 x \, dx = \frac{\tan^2 x}{2} - \ln |\sec x| + c$$

Common Mistake: Attempting to integrate $\tan^3 x$ directly as $\frac{\tan^4 x}{4}$ without using substitution or identities.

Question 7 (i)

Evaluate $\int \frac{\sin 2x}{\sin 5x \sin 3x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{\sin 2x}{\sin 5x \sin 3x} dx$

Concept / Formula Used: Trigonometric identity $\sin 2x = \sin(5x - 3x)$

Solution:

Express the numerator $\sin 2x$ as $\sin(5x - 3x)$:

$$I = \int \frac{\sin(5x - 3x)}{\sin 5x \sin 3x} dx$$

Expand the numerator using the sine subtraction formula $\sin(A - B) = \sin A \cos B - \cos A \sin B$:

$$\begin{aligned} I &= \int \frac{\sin 5x \cos 3x - \cos 5x \sin 3x}{\sin 5x \sin 3x} dx \\ &= \int \left(\frac{\sin 5x \cos 3x}{\sin 5x \sin 3x} - \frac{\cos 5x \sin 3x}{\sin 5x \sin 3x} \right) dx \\ &= \int \left(\frac{\cos 3x}{\sin 3x} - \frac{\cos 5x}{\sin 5x} \right) dx \\ &= \int (\cot 3x - \cot 5x) dx \end{aligned}$$

Integrating term by term using $\int \cot(kx) dx = \frac{1}{k} \ln |\sin(kx)|$:

$$I = \frac{1}{3} \ln |\sin 3x| - \frac{1}{5} \ln |\sin 5x| + c$$

Final Answer

$$\int \frac{\sin 2x}{\sin 5x \sin 3x} dx = \frac{1}{3} \ln |\sin 3x| - \frac{1}{5} \ln |\sin 5x| + c$$

Common Mistake: Failing to divide by the coefficients 3 and 5 when integrating $\cot 3x$ and $\cot 5x$.

Question 7 (ii)

Evaluate $\int \frac{1}{\sin x \cos^2 x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{1}{\sin x \cos^2 x} dx$

Concept / Formula Used: Trigonometric identity $1 = \sin^2 x + \cos^2 x$ and substitution

Solution:

Multiply and divide the numerator by $\sin^2 x + \cos^2 x = 1$:

$$\begin{aligned} I &= \int \frac{\sin^2 x + \cos^2 x}{\sin x \cos^2 x} dx \\ &= \int \left(\frac{\sin^2 x}{\sin x \cos^2 x} + \frac{\cos^2 x}{\sin x \cos^2 x} \right) dx \\ &= \int \left(\frac{\sin x}{\cos^2 x} + \frac{1}{\sin x} \right) dx \\ &= \int (\tan x \sec x + \csc x) dx \end{aligned}$$

Integrating term by term using standard formulas:

$$\begin{aligned} I &= \int \sec x \tan x dx + \int \csc x dx \\ &= \sec x + \ln |\csc x - \cot x| + c \end{aligned}$$

Final Answer

$$\int \frac{1}{\sin x \cos^2 x} dx = \sec x + \ln |\csc x - \cot x| + c$$

Common Mistake: Attempting substitution with $t = \cos x$ without expanding the numerator first, leaving an unmanageable fraction.

Question 8 (i)

Evaluate $\int \frac{dx}{\sqrt{1 - \sin x}}$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{dx}{\sqrt{1 - \sin x}}$

Concept / Formula Used: Rationalization using trigonometric half-angle identities

Solution:

Rationalize the denominator by multiplying the numerator and denominator by $\sqrt{1 + \sin x}$:

$$\begin{aligned} I &= \int \frac{\sqrt{1 + \sin x}}{\sqrt{1 - \sin x} \sqrt{1 + \sin x}} dx \\ &= \int \frac{\sqrt{1 + \sin x}}{\sqrt{1 - \sin^2 x}} dx \end{aligned}$$

Using the fundamental identity $1 - \sin^2 x = \cos^2 x$:

$$\begin{aligned} I &= \int \frac{\sqrt{1 + \sin x}}{\sqrt{\cos^2 x}} dx \\ &= \int \frac{\sqrt{1 + \sin x}}{\cos x} dx \end{aligned}$$

Rewrite 1 as $\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)$ and $\sin x$ as $2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$:

$$\begin{aligned} I &= \int \frac{\sqrt{\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right) + 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)}}{\cos x} dx \\ &= \int \frac{\sqrt{\left(\sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right)\right)^2}}{\cos x} dx \\ &= \int \frac{\sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right)}{\cos x} dx \end{aligned}$$

Using $\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right) = \left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right) \left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right)$:

$$\begin{aligned} I &= \int \frac{\sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right)}{\left(\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)\right) \left(\cos\left(\frac{x}{2}\right) + \sin\left(\frac{x}{2}\right)\right)} dx \\ &= \int \frac{1}{\cos\left(\frac{x}{2}\right) - \sin\left(\frac{x}{2}\right)} dx \end{aligned}$$

Alternatively, separating terms directly:

$$I = \int \left(\frac{\sin\left(\frac{x}{2}\right)}{\cos x} + \frac{\cos\left(\frac{x}{2}\right)}{\cos x} \right) dx$$

Using standard substitution or half-angle formulas: Let $\frac{x}{2} = u$, $dx = 2 du$:

$$I = \int \frac{1}{\cos u - \sin u} (2 du) = 2 \int \frac{du}{\cos u - \sin u}$$

Divide numerator and denominator by $\sqrt{2}$:

$$\begin{aligned}
 &= 2 \int \frac{du}{\frac{1}{\sqrt{2}} \cos u - \frac{1}{\sqrt{2}} \sin u} \\
 &= 2 \int \frac{du}{\cos\left(\frac{\pi}{4}\right) \cos u - \sin\left(\frac{\pi}{4}\right) \sin u} \\
 &= 2 \int \frac{du}{\cos\left(u + \frac{\pi}{4}\right)} \\
 &= 2 \int \sec\left(u + \frac{\pi}{4}\right) du \\
 &= 2 \ln \left| \sec\left(u + \frac{\pi}{4}\right) + \tan\left(u + \frac{\pi}{4}\right) \right| + c
 \end{aligned}$$

Now back-substitute $u = \frac{x}{2}$:

$$I = 2 \ln \left| \sec\left(\frac{x}{2} + \frac{\pi}{4}\right) + \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) \right| + c$$

Final Answer

$$\int \frac{dx}{\sqrt{1 - \sin x}} = 2 \ln \left| \sec\left(\frac{x}{2} + \frac{\pi}{4}\right) + \tan\left(\frac{x}{2} + \frac{\pi}{4}\right) \right| + c$$

Common Mistake: Omitting the absolute value signs in the logarithmic answer.

Question 8 (ii)

Evaluate $\int \frac{\sin x + \cos x}{1 + \sin 2x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: $I = \int \frac{\sin x + \cos x}{1 + \sin 2x} dx$

Concept / Formula Used: Trigonometric identity $1 + \sin 2x = (\sin x + \cos x)^2$ and substitution

Solution:

Using the trigonometric identity $1 + \sin 2x = \sin^2 x + \cos^2 x + 2 \sin x \cos x = (\sin x + \cos x)^2$:

$$\begin{aligned}
 I &= \int \frac{\sin x + \cos x}{(\sin x + \cos x)^2} dx \\
 &= \int \frac{1}{\sin x + \cos x} dx
 \end{aligned}$$

Let $\sin x + \cos x = t$.

Differentiating both sides with respect to x :

$$(\cos x - \sin x) dx = dt \implies dx = \frac{dt}{\cos x - \sin x}$$

Alternatively, use the substitution method for $\frac{1}{a \sin x + b \cos x}$: Divide numerator and denominator by $\sqrt{1^2 + 1^2} = \sqrt{2}$:

$$\begin{aligned} I &= \int \frac{1}{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x} dx \\ &= \frac{1}{\sqrt{2}} \int \frac{1}{\sin x \cos\left(\frac{\pi}{4}\right) + \cos x \sin\left(\frac{\pi}{4}\right)} dx \\ &= \frac{1}{\sqrt{2}} \int \frac{1}{\sin\left(x + \frac{\pi}{4}\right)} dx \\ &= \frac{1}{\sqrt{2}} \int \csc\left(x + \frac{\pi}{4}\right) dx \end{aligned}$$

Using the standard result $\int \csc(u) dx = \ln |\csc u - \cot u| + c$:

$$I = \frac{1}{\sqrt{2}} \ln \left| \csc\left(x + \frac{\pi}{4}\right) - \cot\left(x + \frac{\pi}{4}\right) \right| + c$$

Final Answer

$$\int \frac{\sin x + \cos x}{1 + \sin 2x} dx = \frac{1}{\sqrt{2}} \ln \left| \csc\left(x + \frac{\pi}{4}\right) - \cot\left(x + \frac{\pi}{4}\right) \right| + c$$

Common Mistake: Failing to simplify $1 + \sin 2x$ into $(\sin x + \cos x)^2$ before integrating.

Question 9 ★ Likely Exam Question

If $\int \frac{\sqrt{2} \sin x}{\sin\left(x - \frac{\pi}{4}\right)} dx = ax + b \ln \left| \sin\left(x - \frac{\pi}{4}\right) \right| + C$, find the values of a and b .

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \frac{\sqrt{2} \sin x}{\sin\left(x - \frac{\pi}{4}\right)} dx = ax + b \ln \left| \sin\left(x - \frac{\pi}{4}\right) \right| + C$

Concept / Formula Used: Integration by substitution and comparing coefficients

Solution:

Let $t = x - \frac{\pi}{4}$, which gives $x = t + \frac{\pi}{4}$.

Differentiating both sides with respect to x :

$$dx = dt$$

Substitute $x = t + \frac{\pi}{4}$ and $dx = dt$ into the integral:

$$I = \int \frac{\sqrt{2} \sin\left(t + \frac{\pi}{4}\right)}{\sin t} dt$$

Expand the numerator using the compound angle formula $\sin(A + B) = \sin A \cos B + \cos A \sin B$:

$$\begin{aligned}\sin\left(t + \frac{\pi}{4}\right) &= \sin t \cos\left(\frac{\pi}{4}\right) + \cos t \sin\left(\frac{\pi}{4}\right) \\ &= \sin t \left(\frac{1}{\sqrt{2}}\right) + \cos t \left(\frac{1}{\sqrt{2}}\right) \\ &= \frac{1}{\sqrt{2}}(\sin t + \cos t)\end{aligned}$$

Substitute this back into the integral:

$$\begin{aligned}I &= \int \frac{\sqrt{2} \cdot \frac{1}{\sqrt{2}}(\sin t + \cos t)}{\sin t} dt \\ &= \int \frac{\sin t + \cos t}{\sin t} dt \\ &= \int \left(1 + \frac{\cos t}{\sin t}\right) dt \\ &= \int (1 + \cot t) dt\end{aligned}$$

Integrating term by term with respect to t :

$$I = t + \ln |\sin t| + C$$

Now back-substitute $t = x - \frac{\pi}{4}$:

$$\begin{aligned}I &= \left(x - \frac{\pi}{4}\right) + \ln \left|\sin\left(x - \frac{\pi}{4}\right)\right| + C \\ &= x + \ln \left|\sin\left(x - \frac{\pi}{4}\right)\right| + \left(-\frac{\pi}{4} + C\right)\end{aligned}$$

Comparing this result with the given form $ax + b \ln \left|\sin\left(x - \frac{\pi}{4}\right)\right| + C$:

$$\begin{aligned}ax &= 1 \cdot x \implies a = 1 \\ b \ln \left|\sin\left(x - \frac{\pi}{4}\right)\right| &= 1 \cdot \ln \left|\sin\left(x - \frac{\pi}{4}\right)\right| \implies b = 1\end{aligned}$$

Final Answer

$$a = 1, \quad b = 1$$

Common Mistake: Incorrectly expanding $\sin\left(x - \frac{\pi}{4}\right)$ or misidentifying the constants a and b after integration.

7.7 – Integrals Reducible to Standard Forms (Difference/Sum of Squares)

Question 1

Evaluate the following integrals:

- (i) $\int \frac{1}{x^2 - 16} dx$
- (ii) $\int \frac{1}{9 - x^2} dx$
- (iii) $\int \frac{dx}{\sqrt{1 - x^2}}$
- (iv) $\int \frac{dx}{x^2 + 16}$

TYPE 1 Power Rule of Integration

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of** $\int \frac{1}{x^2 - 16} dx$:

$$I = \int \frac{1}{x^2 - 4^2} dx$$

Using the standard result $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + C$:

$$\begin{aligned} I &= \frac{1}{2(4)} \log \left| \frac{x - 4}{x + 4} \right| + C \\ &= \frac{1}{8} \log \left| \frac{x - 4}{x + 4} \right| + C \end{aligned}$$

(ii) **Evaluation of** $\int \frac{1}{9 - x^2} dx$:

$$I = \int \frac{1}{3^2 - x^2} dx$$

Using the standard result $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a + x}{a - x} \right| + C$:

$$\begin{aligned} I &= \frac{1}{2(3)} \log \left| \frac{3 + x}{3 - x} \right| + C \\ &= \frac{1}{6} \log \left| \frac{3 + x}{3 - x} \right| + C \end{aligned}$$

(iii) **Evaluation of** $\int \frac{dx}{\sqrt{1 - x^2}}$: Using the standard result $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$:

$$I = \sin^{-1}(x) + C$$

(iv) Evaluation of $\int \frac{dx}{x^2 + 16}$:

$$I = \int \frac{dx}{x^2 + 4^2}$$

Using the standard result $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$:

$$I = \frac{1}{4} \tan^{-1} \left(\frac{x}{4} \right) + C$$

Final Answer

$$(i) \int \frac{1}{x^2 - 16} dx = \frac{1}{8} \log \left| \frac{x-4}{x+4} \right| + C$$

$$(ii) \int \frac{1}{9 - x^2} dx = \frac{1}{6} \log \left| \frac{3+x}{3-x} \right| + C$$

$$(iii) \int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}(x) + C$$

$$(iv) \int \frac{dx}{x^2 + 16} = \frac{1}{4} \tan^{-1} \left(\frac{x}{4} \right) + C$$

Question 2

Evaluate the following integrals:

(i) $\int \frac{dx}{\sqrt{9-4x^2}}$

(ii) $\int \frac{dx}{25+9x^2}$

TYPE 12b Substitution Involving Radical Expressions

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of** $\int \frac{dx}{\sqrt{9-4x^2}}$:

$$\begin{aligned} I &= \int \frac{dx}{\sqrt{4\left(\frac{9}{4}-x^2\right)}} \\ &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{3}{2}\right)^2-x^2}} \end{aligned}$$

Using the standard result $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$:

$$\begin{aligned} I &= \frac{1}{2} \sin^{-1}\left(\frac{x}{3/2}\right) + C \\ &= \frac{1}{2} \sin^{-1}\left(\frac{2x}{3}\right) + C \end{aligned}$$

(ii) **Evaluation of** $\int \frac{dx}{25+9x^2}$:

$$\begin{aligned} I &= \int \frac{dx}{9\left(\frac{25}{9}+x^2\right)} \\ &= \frac{1}{9} \int \frac{dx}{x^2+\left(\frac{5}{3}\right)^2} \end{aligned}$$

Using the standard result $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$:

$$\begin{aligned} I &= \frac{1}{9} \cdot \frac{1}{5/3} \tan^{-1}\left(\frac{x}{5/3}\right) + C \\ &= \frac{1}{9} \cdot \frac{3}{5} \tan^{-1}\left(\frac{3x}{5}\right) + C \\ &= \frac{1}{15} \tan^{-1}\left(\frac{3x}{5}\right) + C \end{aligned}$$

Final Answer

$$(i) \int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \sin^{-1} \left(\frac{2x}{3} \right) + C$$

$$(ii) \int \frac{dx}{25+9x^2} = \frac{1}{15} \tan^{-1} \left(\frac{3x}{5} \right) + C$$

Question 3

Evaluate the following integrals:

(i) $\int \frac{dx}{x\sqrt{9x^2 - 16}}$

(ii) $\int \frac{dx}{\sqrt{x^2 + 5}}$

TYPE 12b Substitution Involving Radical Expressions

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of** $\int \frac{dx}{x\sqrt{9x^2 - 16}}$:

$$\begin{aligned} I &= \int \frac{dx}{x\sqrt{9\left(x^2 - \frac{16}{9}\right)}} \\ &= \frac{1}{3} \int \frac{dx}{x\sqrt{x^2 - \left(\frac{4}{3}\right)^2}} \end{aligned}$$

Using the standard result $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \sec^{-1} \left(\frac{x}{a} \right) + C$:

$$\begin{aligned} I &= \frac{1}{3} \cdot \frac{1}{4/3} \sec^{-1} \left(\frac{x}{4/3} \right) + C \\ &= \frac{1}{3} \cdot \frac{3}{4} \sec^{-1} \left(\frac{3x}{4} \right) + C \\ &= \frac{1}{4} \sec^{-1} \left(\frac{3x}{4} \right) + C \end{aligned}$$

(ii) **Evaluation of** $\int \frac{dx}{\sqrt{x^2 + 5}}$:

$$I = \int \frac{dx}{\sqrt{x^2 + (\sqrt{5})^2}}$$

Using the standard result $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + C$:

$$I = \log \left| x + \sqrt{x^2 + 5} \right| + C$$

Final Answer

(i) $\int \frac{dx}{x\sqrt{9x^2 - 16}} = \frac{1}{4} \sec^{-1} \left(\frac{3x}{4} \right) + C$

(ii) $\int \frac{dx}{\sqrt{x^2 + 5}} = \log \left| x + \sqrt{x^2 + 5} \right| + C$

Question 4

Evaluate the following integrals:

- (i) $\int \frac{dx}{9x^2 - 1}$
 (ii) $\int \frac{dx}{32 - 2x^2}$

TYPE 1 Power Rule of Integration

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of** $\int \frac{dx}{9x^2 - 1}$:

$$\begin{aligned} I &= \frac{1}{9} \int \frac{dx}{x^2 - \frac{1}{9}} \\ &= \frac{1}{9} \int \frac{dx}{x^2 - \left(\frac{1}{3}\right)^2} \end{aligned}$$

Using the standard result $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right| + C$:

$$\begin{aligned} I &= \frac{1}{9} \cdot \frac{1}{2(1/3)} \log \left| \frac{x - 1/3}{x + 1/3} \right| + C \\ &= \frac{1}{9} \cdot \frac{3}{2} \log \left| \frac{3x - 1}{3x + 1} \right| + C \\ &= \frac{1}{6} \log \left| \frac{3x - 1}{3x + 1} \right| + C \end{aligned}$$

(ii) **Evaluation of** $\int \frac{dx}{32 - 2x^2}$:

$$\begin{aligned} I &= \frac{1}{2} \int \frac{dx}{16 - x^2} \\ &= \frac{1}{2} \int \frac{dx}{4^2 - x^2} \end{aligned}$$

Using the standard result $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a + x}{a - x} \right| + C$:

$$\begin{aligned} I &= \frac{1}{2} \cdot \frac{1}{2(4)} \log \left| \frac{4 + x}{4 - x} \right| + C \\ &= \frac{1}{16} \log \left| \frac{4 + x}{4 - x} \right| + C \end{aligned}$$

Final Answer

$$(i) \int \frac{dx}{9x^2 - 1} = \frac{1}{6} \log \left| \frac{3x - 1}{3x + 1} \right| + C$$

$$(ii) \int \frac{dx}{32 - 2x^2} = \frac{1}{16} \log \left| \frac{4 + x}{4 - x} \right| + C$$

Question 5

Evaluate the following integrals:

(i) $\int \frac{dx}{\sqrt{15-8x^2}}$

(ii) $\int \frac{dx}{\sqrt{1+4x^2}}$

TYPE 12b Substitution Involving Radical Expressions

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of** $\int \frac{dx}{\sqrt{15-8x^2}}$:

$$\begin{aligned} I &= \int \frac{dx}{\sqrt{8\left(\frac{15}{8}-x^2\right)}} \\ &= \frac{1}{2\sqrt{2}} \int \frac{dx}{\sqrt{\left(\frac{\sqrt{15}}{2\sqrt{2}}\right)^2 - x^2}} \end{aligned}$$

Using the standard result $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$:

$$\begin{aligned} I &= \frac{1}{2\sqrt{2}} \sin^{-1}\left(\frac{x}{\frac{\sqrt{15}}{2\sqrt{2}}}\right) + C \\ &= \frac{1}{2\sqrt{2}} \sin^{-1}\left(\frac{2\sqrt{2}x}{\sqrt{15}}\right) + C \\ &= \frac{1}{2\sqrt{2}} \sin^{-1}\left(\sqrt{\frac{8}{15}}x\right) + C \end{aligned}$$

(ii) **Evaluation of** $\int \frac{dx}{\sqrt{1+4x^2}}$:

$$\begin{aligned} I &= \int \frac{dx}{\sqrt{4\left(\frac{1}{4}+x^2\right)}} \\ &= \frac{1}{2} \int \frac{dx}{\sqrt{x^2 + \left(\frac{1}{2}\right)^2}} \end{aligned}$$

Using the standard result $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + C$:

$$\begin{aligned} I &= \frac{1}{2} \log \left| x + \sqrt{x^2 + \frac{1}{4}} \right| + C' \\ &= \frac{1}{2} \log \left| x + \frac{\sqrt{4x^2 + 1}}{2} \right| + C' \\ &= \frac{1}{2} \log \left| \frac{2x + \sqrt{4x^2 + 1}}{2} \right| + C' \\ &= \frac{1}{2} \log \left| 2x + \sqrt{1 + 4x^2} \right| - \frac{1}{2} \log 2 + C' \end{aligned}$$

Letting $C = C' - \frac{1}{2} \log 2$:

$$I = \frac{1}{2} \log \left| 2x + \sqrt{1 + 4x^2} \right| + C$$

Final Answer

- (i) $\int \frac{dx}{\sqrt{15 - 8x^2}} = \frac{1}{2\sqrt{2}} \sin^{-1} \left(\sqrt{\frac{8}{15}} x \right) + C$
- (ii) $\int \frac{dx}{\sqrt{1 + 4x^2}} = \frac{1}{2} \log \left| 2x + \sqrt{1 + 4x^2} \right| + C$

Question 6

Evaluate the following integrals:

$$(i) \int \frac{2y^2}{y^2 + 4} dy \quad (ISC 2015)$$

$$(ii) \int \frac{x^4 + 1}{x^2 + 1} dx$$

TYPE 1 Power Rule of Integration**★ Likely Exam Question****To Find:** The evaluated values of the given integrals.**Solution:****(i) Evaluation of $\int \frac{2y^2}{y^2 + 4} dy$:** We rewrite the numerator to perform division:

$$2y^2 = 2(y^2 + 4) - 8$$

Substituting this into the integral:

$$\begin{aligned} I &= \int \frac{2(y^2 + 4) - 8}{y^2 + 4} dy \\ &= \int \left(2 - \frac{8}{y^2 + 4} \right) dy \\ &= 2 \int dy - 8 \int \frac{dy}{y^2 + 2^2} \end{aligned}$$

Using the standard result $\int \frac{dy}{y^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{y}{a} \right) + C$:

$$\begin{aligned} I &= 2y - 8 \cdot \frac{1}{2} \tan^{-1} \left(\frac{y}{2} \right) + C \\ &= 2y - 4 \tan^{-1} \left(\frac{y}{2} \right) + C \end{aligned}$$

(ii) Evaluation of $\int \frac{x^4 + 1}{x^2 + 1} dx$: We rewrite the numerator to perform division:

$$\begin{aligned} x^4 + 1 &= (x^4 - 1) + 2 \\ &= (x^2 - 1)(x^2 + 1) + 2 \end{aligned}$$

Substituting this into the integral:

$$\begin{aligned} I &= \int \frac{(x^2 - 1)(x^2 + 1) + 2}{x^2 + 1} dx \\ &= \int \left(x^2 - 1 + \frac{2}{x^2 + 1} \right) dx \\ &= \int x^2 dx - \int 1 dx + 2 \int \frac{dx}{x^2 + 1} \end{aligned}$$

Using the standard result $\int \frac{dx}{x^2+1} = \tan^{-1}(x) + C$ and the power rule:

$$I = \frac{x^3}{3} - x + 2 \tan^{-1}(x) + C$$

Final Answer

(i) $\int \frac{2y^2}{y^2+4} dy = 2y - 4 \tan^{-1}\left(\frac{y}{2}\right) + C$

(ii) $\int \frac{x^4+1}{x^2+1} dx = \frac{x^3}{3} - x + 2 \tan^{-1}(x) + C$

Question 7

Evaluate the following integrals:

$$(i) \int \frac{x}{x^4 - 1} dx \quad \text{(Exemplar)}$$

$$(ii) \int \frac{3x^2}{1 + x^6} dx \quad \text{(NCERT)}$$

TYPE 12a Direct/Algebraic Substitution**To Find:** The evaluated values of the given integrals.**Solution:****(i) Evaluation of $\int \frac{x}{x^4 - 1} dx$:** Let $x^2 = t$.

Differentiating:

$$\begin{aligned} 2x dx &= dt \\ x dx &= \frac{1}{2} dt \end{aligned}$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{\frac{1}{2} dt}{t^2 - 1} \\ &= \frac{1}{2} \int \frac{dt}{t^2 - 1^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 - a^2} = \frac{1}{2a} \log \left| \frac{t - a}{t + a} \right| + C$:

$$\begin{aligned} I &= \frac{1}{2} \cdot \frac{1}{2(1)} \log \left| \frac{t - 1}{t + 1} \right| + C \\ &= \frac{1}{4} \log \left| \frac{t - 1}{t + 1} \right| + C \end{aligned}$$

Back-substituting $t = x^2$:

$$I = \frac{1}{4} \log \left| \frac{x^2 - 1}{x^2 + 1} \right| + C$$

(ii) Evaluation of $\int \frac{3x^2}{1 + x^6} dx$: Let $x^3 = t$.

Differentiating:

$$3x^2 dx = dt$$

Substituting these into the integral:

$$I = \int \frac{dt}{1 + t^2}$$

Using the standard result $\int \frac{dt}{1+t^2} = \tan^{-1}(t) + C$:

$$I = \tan^{-1}(t) + C$$

Back-substituting $t = x^3$:

$$I = \tan^{-1}(x^3) + C$$

Final Answer

$$(i) \int \frac{x}{x^4 - 1} dx = \frac{1}{4} \log \left| \frac{x^2 - 1}{x^2 + 1} \right| + C$$

$$(ii) \int \frac{3x^2}{1 + x^6} dx = \tan^{-1}(x^3) + C$$

Question 8

Evaluate the following integrals:

$$(i) \int \frac{x^2}{\sqrt{x^6 + a^6}} dx \quad (\text{NCERT})$$

$$(ii) \int \frac{x^3}{\sqrt{1 - x^8}} dx \quad (\text{NCERT})$$

TYPE 12b Substitution Involving Radical Expressions

To Find: The evaluated values of the given integrals.

Solution:

(i) Evaluation of $\int \frac{x^2}{\sqrt{x^6 + a^6}} dx$: Let $x^3 = t$.

Differentiating:

$$3x^2 dx = dt$$

$$x^2 dx = \frac{1}{3} dt$$

Substituting these into the integral:

$$I = \int \frac{\frac{1}{3} dt}{\sqrt{t^2 + (a^3)^2}}$$

$$= \frac{1}{3} \int \frac{dt}{\sqrt{t^2 + (a^3)^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{t^2 + A^2}} = \log |t + \sqrt{t^2 + A^2}| + C$:

$$I = \frac{1}{3} \log |t + \sqrt{t^2 + a^6}| + C$$

Back-substituting $t = x^3$:

$$I = \frac{1}{3} \log |x^3 + \sqrt{x^6 + a^6}| + C$$

(ii) Evaluation of $\int \frac{x^3}{\sqrt{1 - x^8}} dx$: Let $x^4 = t$.

Differentiating:

$$4x^3 dx = dt$$

$$x^3 dx = \frac{1}{4} dt$$

Substituting these into the integral:

$$I = \int \frac{\frac{1}{4} dt}{\sqrt{1 - t^2}}$$

$$= \frac{1}{4} \int \frac{dt}{\sqrt{1 - t^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1}(t) + C$:

$$I = \frac{1}{4} \sin^{-1}(t) + C$$

Back-substituting $t = x^4$:

$$I = \frac{1}{4} \sin^{-1}(x^4) + C$$

Final Answer

$$(i) \int \frac{x^2}{\sqrt{x^6 + a^6}} dx = \frac{1}{3} \log \left| x^3 + \sqrt{x^6 + a^6} \right| + C$$

$$(ii) \int \frac{x^3}{\sqrt{1-x^8}} dx = \frac{1}{4} \sin^{-1}(x^4) + C$$

Question 9

Evaluate the following integrals:

$$(i) \int \frac{\sqrt{x}}{\sqrt{a^3 - x^3}} dx \quad (\text{Exemplar})$$

$$(ii) \int \frac{x(1+x^2)}{1+x^4} dx \quad (\text{ISC 2012})$$

TYPE 12b Substitution Involving Radical Expressions

★ Likely Exam Question

To Find: The evaluated values of the given integrals.

Solution:

(i) Evaluation of $\int \frac{\sqrt{x}}{\sqrt{a^3 - x^3}} dx$: Let $x^{3/2} = t$.

Differentiating:

$$\begin{aligned} \frac{3}{2} x^{1/2} dx &= dt \\ \sqrt{x} dx &= \frac{2}{3} dt \end{aligned}$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{\frac{2}{3} dt}{\sqrt{(a^{3/2})^2 - t^2}} \\ &= \frac{2}{3} \int \frac{dt}{\sqrt{(a^{3/2})^2 - t^2}} \end{aligned}$$

Using the standard result $\int \frac{dt}{\sqrt{A^2 - t^2}} = \sin^{-1} \left(\frac{t}{A} \right) + C$:

$$I = \frac{2}{3} \sin^{-1} \left(\frac{t}{a^{3/2}} \right) + C$$

Back-substituting $t = x^{3/2}$:

$$\begin{aligned} I &= \frac{2}{3} \sin^{-1} \left(\frac{x^{3/2}}{a^{3/2}} \right) + C \\ &= \frac{2}{3} \sin^{-1} \left(\sqrt{\frac{x^3}{a^3}} \right) + C \end{aligned}$$

(ii) Evaluation of $\int \frac{x(1+x^2)}{1+x^4} dx$: We expand the numerator:

$$\begin{aligned} I &= \int \frac{x + x^3}{1 + x^4} dx \\ &= \int \frac{x}{1 + x^4} dx + \int \frac{x^3}{1 + x^4} dx \end{aligned}$$

Let $I_1 = \int \frac{x}{1+x^4} dx$ and $I_2 = \int \frac{x^3}{1+x^4} dx$.

For I_1 : Let $x^2 = u$.

Differentiating:

$$\begin{aligned} 2x dx &= du \\ x dx &= \frac{1}{2} du \end{aligned}$$

Substituting these into I_1 :

$$\begin{aligned} I_1 &= \int \frac{\frac{1}{2} du}{1+u^2} \\ &= \frac{1}{2} \tan^{-1}(u) + C_1 \end{aligned}$$

Back-substituting $u = x^2$:

$$I_1 = \frac{1}{2} \tan^{-1}(x^2) + C_1$$

For I_2 : Let $1 + x^4 = v$.

Differentiating:

$$\begin{aligned} 4x^3 dx &= dv \\ x^3 dx &= \frac{1}{4} dv \end{aligned}$$

Substituting these into I_2 :

$$\begin{aligned} I_2 &= \int \frac{\frac{1}{4} dv}{v} \\ &= \frac{1}{4} \log |v| + C_2 \end{aligned}$$

Back-substituting $v = 1 + x^4$:

$$I_2 = \frac{1}{4} \log(1 + x^4) + C_2$$

Combining I_1 and I_2 :

$$I = \frac{1}{2} \tan^{-1}(x^2) + \frac{1}{4} \log(1 + x^4) + C$$

Final Answer

$$\begin{aligned} \text{(i)} \quad & \int \frac{\sqrt{x}}{\sqrt{a^3 - x^3}} dx = \frac{2}{3} \sin^{-1} \left(\sqrt{\frac{x^3}{a^3}} \right) + C \\ \text{(ii)} \quad & \int \frac{x(1+x^2)}{1+x^4} dx = \frac{1}{2} \tan^{-1}(x^2) + \frac{1}{4} \log(1+x^4) + C \end{aligned}$$

Question 10

Evaluate the following integrals:

$$(i) \int \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx \quad (\text{NCERT})$$

$$(ii) \int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx \quad (\text{NCERT})$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The evaluated values of the given integrals.**Solution:****(i) Evaluation of** $\int \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx$: Let $\sin x = t$.

Differentiating:

$$\cos x \, dx = dt$$

Substituting these into the integral:

$$I = \int \frac{dt}{\sqrt{2^2 - t^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{a^2 - t^2}} = \sin^{-1} \left(\frac{t}{a} \right) + C$:

$$I = \sin^{-1} \left(\frac{t}{2} \right) + C$$

Back-substituting $t = \sin x$:

$$I = \sin^{-1} \left(\frac{\sin x}{2} \right) + C$$

(ii) Evaluation of $\int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx$: Let $\tan x = t$.

Differentiating:

$$\sec^2 x \, dx = dt$$

Substituting these into the integral:

$$I = \int \frac{dt}{\sqrt{t^2 + 2^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{t^2 + a^2}} = \log \left| t + \sqrt{t^2 + a^2} \right| + C$:

$$I = \log \left| t + \sqrt{t^2 + 4} \right| + C$$

Back-substituting $t = \tan x$:

$$I = \log \left| \tan x + \sqrt{\tan^2 x + 4} \right| + C$$

Final Answer

$$(i) \int \frac{\cos x}{\sqrt{4 - \sin^2 x}} dx = \sin^{-1} \left(\frac{\sin x}{2} \right) + C$$

$$(ii) \int \frac{\sec^2 x}{\sqrt{\tan^2 x + 4}} dx = \log \left| \tan x + \sqrt{\tan^2 x + 4} \right| + C$$

Question 11

Evaluate the following integrals:

- (i) $\int \frac{\sin x}{16 - 9 \cos^2 x} dx$
 (ii) $\int \frac{\tan^2 x \sec^2 x}{1 - \tan^6 x} dx$

TYPE 12a Direct/Algebraic Substitution

To Find: The evaluated values of the given integrals.

Solution:

(i) Evaluation of $\int \frac{\sin x}{16 - 9 \cos^2 x} dx$: Let $\cos x = t$.

Differentiating:

$$\begin{aligned} -\sin x \, dx &= dt \\ \sin x \, dx &= -dt \end{aligned}$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{-dt}{16 - 9t^2} \\ &= \int \frac{dt}{9t^2 - 16} \\ &= \frac{1}{9} \int \frac{dt}{t^2 - \frac{16}{9}} \\ &= \frac{1}{9} \int \frac{dt}{t^2 - \left(\frac{4}{3}\right)^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 - a^2} = \frac{1}{2a} \log \left| \frac{t - a}{t + a} \right| + C$:

$$\begin{aligned} I &= \frac{1}{9} \cdot \frac{1}{2(4/3)} \log \left| \frac{t - 4/3}{t + 4/3} \right| + C \\ &= \frac{1}{24} \log \left| \frac{3t - 4}{3t + 4} \right| + C \\ &= -\frac{1}{24} \log \left| \frac{3t + 4}{3t - 4} \right| + C \\ &= -\frac{1}{24} \log \left| \frac{4 + 3t}{4 - 3t} \right| + C \end{aligned}$$

Back-substituting $t = \cos x$:

$$I = -\frac{1}{24} \log \left| \frac{4 + 3 \cos x}{4 - 3 \cos x} \right| + C$$

(ii) **Evaluation of** $\int \frac{\tan^2 x \sec^2 x}{1 - \tan^6 x} dx$: Let $\tan^3 x = t$.

Differentiating:

$$3 \tan^2 x \sec^2 x dx = dt$$

$$\tan^2 x \sec^2 x dx = \frac{1}{3} dt$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{\frac{1}{3} dt}{1 - t^2} \\ &= \frac{1}{3} \int \frac{dt}{1 - t^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{a^2 - t^2} = \frac{1}{2a} \log \left| \frac{a+t}{a-t} \right| + C$:

$$\begin{aligned} I &= \frac{1}{3} \cdot \frac{1}{2(1)} \log \left| \frac{1+t}{1-t} \right| + C \\ &= \frac{1}{6} \log \left| \frac{1+t}{1-t} \right| + C \end{aligned}$$

Back-substituting $t = \tan^3 x$:

$$I = \frac{1}{6} \log \left| \frac{1 + \tan^3 x}{1 - \tan^3 x} \right| + C$$

Final Answer

$$\begin{aligned} \text{(i)} \quad & \int \frac{\sin x}{16 - 9 \cos^2 x} dx = -\frac{1}{24} \log \left| \frac{4 + 3 \cos x}{4 - 3 \cos x} \right| + C \\ \text{(ii)} \quad & \int \frac{\tan^2 x \sec^2 x}{1 - \tan^6 x} dx = \frac{1}{6} \log \left| \frac{1 + \tan^3 x}{1 - \tan^3 x} \right| + C \end{aligned}$$

Common Mistake: In part (i), forgetting the negative sign when differentiating $\cos x$ is a very common error. Always write out $d(\cos x) = -\sin x dx$ explicitly.

Question 12

Evaluate the following integrals:

(i) $\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$

(ii) $\int \frac{dx}{2 \sin^2 x + 5 \cos^2 x}$

TYPE 2 Trigonometric Identity Simplification

To Find: The evaluated values of the given integrals.

Solution:

(i) Evaluation of $\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx$: We use the algebraic identity:

$$\begin{aligned} (\sin x - \cos x)^2 &= \sin^2 x + \cos^2 x - 2 \sin x \cos x \\ &= 1 - \sin 2x \end{aligned}$$

This gives:

$$\sin 2x = 1 - (\sin x - \cos x)^2$$

Let $\sin x - \cos x = t$.

Differentiating:

$$(\cos x + \sin x) dx = dt$$

Substituting these into the integral:

$$I = \int \frac{dt}{\sqrt{1 - t^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{1 - t^2}} = \sin^{-1}(t) + C$:

$$I = \sin^{-1}(t) + C$$

Back-substituting $t = \sin x - \cos x$:

$$I = \sin^{-1}(\sin x - \cos x) + C$$

(ii) Evaluation of $\int \frac{dx}{2 \sin^2 x + 5 \cos^2 x}$: We divide the numerator and denominator by $\cos^2 x$:

$$\begin{aligned} I &= \int \frac{\frac{1}{\cos^2 x}}{\frac{2 \sin^2 x + 5 \cos^2 x}{\cos^2 x}} dx \\ &= \int \frac{\sec^2 x}{2 \tan^2 x + 5} dx \end{aligned}$$

Let $\tan x = t$.

Differentiating:

$$\sec^2 x \, dx = dt$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{dt}{2t^2 + 5} \\ &= \frac{1}{2} \int \frac{dt}{t^2 + \frac{5}{2}} \\ &= \frac{1}{2} \int \frac{dt}{t^2 + \left(\sqrt{\frac{5}{2}}\right)^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + C$:

$$\begin{aligned} I &= \frac{1}{2} \cdot \frac{1}{\sqrt{5/2}} \tan^{-1} \left(\frac{t}{\sqrt{5/2}} \right) + C \\ &= \frac{1}{2} \cdot \sqrt{\frac{2}{5}} \tan^{-1} \left(\sqrt{\frac{2}{5}} t \right) + C \\ &= \frac{1}{\sqrt{10}} \tan^{-1} \left(\sqrt{\frac{2}{5}} t \right) + C \end{aligned}$$

Back-substituting $t = \tan x$:

$$I = \frac{1}{\sqrt{10}} \tan^{-1} \left(\sqrt{\frac{2}{5}} \tan x \right) + C$$

Final Answer

$$\begin{aligned} \text{(i)} \quad & \int \frac{\sin x + \cos x}{\sqrt{\sin 2x}} \, dx = \sin^{-1}(\sin x - \cos x) + C \\ \text{(ii)} \quad & \int \frac{dx}{2 \sin^2 x + 5 \cos^2 x} = \frac{1}{\sqrt{10}} \tan^{-1} \left(\sqrt{\frac{2}{5}} \tan x \right) + C \end{aligned}$$

Common Mistake: In part (ii), attempting to substitute directly without dividing by $\cos^2 x$ makes the integral extremely complicated. Dividing by $\cos^2 x$ is the standard technique for integrals of the form $\int \frac{dx}{a \sin^2 x + b \cos^2 x}$.

Question 13

Evaluate the following integrals:

- (i) $\int \frac{\cos x}{\cos 3x} dx$
 (ii) $\int \frac{dx}{2 - 3 \cos 2x}$

TYPE 2 Trigonometric Identity Simplification

To Find: The evaluated values of the given integrals.

Solution:

(i) Evaluation of $\int \frac{\cos x}{\cos 3x} dx$: Using the triple-angle identity $\cos 3x = \cos x(4 \cos^2 x - 3)$:

$$\begin{aligned} I &= \int \frac{\cos x}{\cos x(4 \cos^2 x - 3)} dx \\ &= \int \frac{dx}{4 \cos^2 x - 3} \end{aligned}$$

Dividing the numerator and denominator by $\cos^2 x$:

$$I = \int \frac{\sec^2 x}{4 - 3 \sec^2 x} dx$$

Using the identity $\sec^2 x = 1 + \tan^2 x$:

$$\begin{aligned} I &= \int \frac{\sec^2 x}{4 - 3(1 + \tan^2 x)} dx \\ &= \int \frac{\sec^2 x}{1 - 3 \tan^2 x} dx \end{aligned}$$

Let $\tan x = t$.

Differentiating:

$$\sec^2 x dx = dt$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{dt}{1 - 3t^2} \\ &= \frac{1}{3} \int \frac{dt}{\frac{1}{3} - t^2} \\ &= \frac{1}{3} \int \frac{dt}{\left(\frac{1}{\sqrt{3}}\right)^2 - t^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{a^2 - t^2} = \frac{1}{2a} \log \left| \frac{a+t}{a-t} \right| + C$:

$$\begin{aligned} I &= \frac{1}{3} \cdot \frac{1}{2(1/\sqrt{3})} \log \left| \frac{\frac{1}{\sqrt{3}} + t}{\frac{1}{\sqrt{3}} - t} \right| + C \\ &= \frac{1}{2\sqrt{3}} \log \left| \frac{1 + \sqrt{3}t}{1 - \sqrt{3}t} \right| + C \end{aligned}$$

Back-substituting $t = \tan x$:

$$I = \frac{1}{2\sqrt{3}} \log \left| \frac{1 + \sqrt{3} \tan x}{1 - \sqrt{3} \tan x} \right| + C$$

(ii) **Evaluation of** $\int \frac{dx}{2 - 3 \cos 2x}$: Using the identity $\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$:

$$\begin{aligned} 2 - 3 \cos 2x &= 2 - 3 \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) \\ &= \frac{2(1 + \tan^2 x) - 3(1 - \tan^2 x)}{1 + \tan^2 x} \\ &= \frac{5 \tan^2 x - 1}{1 + \tan^2 x} \end{aligned}$$

Substituting this into the integral:

$$\begin{aligned} I &= \int \frac{1 + \tan^2 x}{5 \tan^2 x - 1} dx \\ &= \int \frac{\sec^2 x}{5 \tan^2 x - 1} dx \end{aligned}$$

Let $\tan x = t$.

Differentiating:

$$\sec^2 x dx = dt$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{dt}{5t^2 - 1} \\ &= \frac{1}{5} \int \frac{dt}{t^2 - \frac{1}{5}} \\ &= \frac{1}{5} \int \frac{dt}{t^2 - \left(\frac{1}{\sqrt{5}}\right)^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 - a^2} = \frac{1}{2a} \log \left| \frac{t-a}{t+a} \right| + C$:

$$\begin{aligned} I &= \frac{1}{5} \cdot \frac{1}{2(1/\sqrt{5})} \log \left| \frac{t - 1/\sqrt{5}}{t + 1/\sqrt{5}} \right| + C \\ &= \frac{1}{2\sqrt{5}} \log \left| \frac{\sqrt{5}t - 1}{\sqrt{5}t + 1} \right| + C \end{aligned}$$

Back-substituting $t = \tan x$:

$$I = \frac{1}{2\sqrt{5}} \log \left| \frac{\sqrt{5} \tan x - 1}{\sqrt{5} \tan x + 1} \right| + C$$

Final Answer

$$(i) \int \frac{\cos x}{\cos 3x} dx = \frac{1}{2\sqrt{3}} \log \left| \frac{1 + \sqrt{3} \tan x}{1 - \sqrt{3} \tan x} \right| + C$$

$$(ii) \int \frac{dx}{2 - 3 \cos 2x} = \frac{1}{2\sqrt{5}} \log \left| \frac{\sqrt{5} \tan x - 1}{\sqrt{5} \tan x + 1} \right| + C$$

Question 14

Evaluate the following integrals:

- (i) $\int \frac{dx}{e^x - e^{-x}}$
- (ii) $\int \sqrt{\frac{1+x}{1-x}} dx$ (Exemplar)

TYPE 5 Conjugate Multiplication

★ Likely Exam Question

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of $\int \frac{dx}{e^x - e^{-x}}$:** We rewrite the integrand:

$$\begin{aligned} I &= \int \frac{dx}{e^x - \frac{1}{e^x}} \\ &= \int \frac{e^x dx}{e^{2x} - 1} \end{aligned}$$

Let $e^x = t$.

Differentiating:

$$e^x dx = dt$$

Substituting these into the integral:

$$I = \int \frac{dt}{t^2 - 1}$$

Using the standard result $\int \frac{dt}{t^2 - a^2} = \frac{1}{2a} \log \left| \frac{t-a}{t+a} \right| + C$:

$$I = \frac{1}{2} \log \left| \frac{t-1}{t+1} \right| + C$$

Back-substituting $t = e^x$:

$$I = \frac{1}{2} \log \left| \frac{e^x - 1}{e^x + 1} \right| + C$$

(ii) **Evaluation of $\int \sqrt{\frac{1+x}{1-x}} dx$:** We rationalize the numerator inside the square root:

$$\begin{aligned} I &= \int \sqrt{\frac{(1+x)(1+x)}{(1-x)(1+x)}} dx \\ &= \int \frac{1+x}{\sqrt{1-x^2}} dx \\ &= \int \frac{1}{\sqrt{1-x^2}} dx + \int \frac{x}{\sqrt{1-x^2}} dx \end{aligned}$$

Let $I_1 = \int \frac{1}{\sqrt{1-x^2}} dx$ and $I_2 = \int \frac{x}{\sqrt{1-x^2}} dx$.

For I_1 : Using the standard result $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1}(x) + C$:

$$I_1 = \sin^{-1}(x) + C_1$$

For I_2 : Let $1 - x^2 = u$.

Differentiating:

$$\begin{aligned} -2x dx &= du \\ x dx &= -\frac{1}{2} du \end{aligned}$$

Substituting these into I_2 :

$$\begin{aligned} I_2 &= \int \frac{-\frac{1}{2} du}{\sqrt{u}} \\ &= -\frac{1}{2} \int u^{-1/2} du \\ &= -\frac{1}{2} \cdot 2\sqrt{u} + C_2 \\ &= -\sqrt{u} + C_2 \end{aligned}$$

Back-substituting $u = 1 - x^2$:

$$I_2 = -\sqrt{1-x^2} + C_2$$

Combining I_1 and I_2 :

$$I = \sin^{-1}(x) - \sqrt{1-x^2} + C$$

Final Answer

- (i) $\int \frac{dx}{e^x - e^{-x}} = \frac{1}{2} \log \left| \frac{e^x - 1}{e^x + 1} \right| + C$
- (ii) $\int \sqrt{\frac{1+x}{1-x}} dx = \sin^{-1}(x) - \sqrt{1-x^2} + C$

Question 15

Evaluate the following integrals:

$$(i) \int \frac{x-1}{\sqrt{x^2-1}} dx \quad (\text{NCERT})$$

$$(ii) \int \frac{x+2}{\sqrt{x^2-1}} dx \quad (\text{NCERT})$$

TYPE 12b Substitution Involving Radical Expressions**To Find:** The evaluated values of the given integrals.**Solution:****(i) Evaluation of $\int \frac{x-1}{\sqrt{x^2-1}} dx$:** We split the integral into two parts:

$$I = \int \frac{x}{\sqrt{x^2-1}} dx - \int \frac{1}{\sqrt{x^2-1}} dx$$

Let $I_1 = \int \frac{x}{\sqrt{x^2-1}} dx$ and $I_2 = \int \frac{1}{\sqrt{x^2-1}} dx$.For I_1 : Let $x^2 - 1 = t$.

Differentiating:

$$\begin{aligned} 2x dx &= dt \\ x dx &= \frac{1}{2} dt \end{aligned}$$

Substituting these into I_1 :

$$\begin{aligned} I_1 &= \int \frac{\frac{1}{2} dt}{\sqrt{t}} \\ &= \sqrt{t} + C_1 \end{aligned}$$

Back-substituting $t = x^2 - 1$:

$$I_1 = \sqrt{x^2-1} + C_1$$

For I_2 : Using the standard result $\int \frac{dx}{\sqrt{x^2-a^2}} = \log |x + \sqrt{x^2-a^2}| + C$:

$$I_2 = \log |x + \sqrt{x^2-1}| + C_2$$

Combining I_1 and I_2 :

$$I = \sqrt{x^2-1} - \log |x + \sqrt{x^2-1}| + C$$

(ii) Evaluation of $\int \frac{x+2}{\sqrt{x^2-1}} dx$: We split the integral into two parts:

$$I = \int \frac{x}{\sqrt{x^2-1}} dx + 2 \int \frac{1}{\sqrt{x^2-1}} dx$$

Using the evaluations from part (i):

$$I = \sqrt{x^2 - 1} + 2 \log \left| x + \sqrt{x^2 - 1} \right| + C$$

Final Answer

$$(i) \int \frac{x-1}{\sqrt{x^2-1}} dx = \sqrt{x^2-1} - \log \left| x + \sqrt{x^2-1} \right| + C$$

$$(ii) \int \frac{x+2}{\sqrt{x^2-1}} dx = \sqrt{x^2-1} + 2 \log \left| x + \sqrt{x^2-1} \right| + C$$

Question 16

Evaluate the following integrals:

- (i) $\int \sqrt{\frac{a+x}{a-x}} dx$ (Exemplar)
- (ii) $\int \frac{\sqrt{x^2 - a^2}}{x} dx$

TYPE 5 Conjugate Multiplication

★ Likely Exam Question

To Find: The evaluated values of the given integrals.

Solution:

(i) **Evaluation of $\int \sqrt{\frac{a+x}{a-x}} dx$:** We rationalize the numerator inside the square root:

$$\begin{aligned} I &= \int \sqrt{\frac{(a+x)(a+x)}{(a-x)(a+x)}} dx \\ &= \int \frac{a+x}{\sqrt{a^2-x^2}} dx \\ &= a \int \frac{1}{\sqrt{a^2-x^2}} dx + \int \frac{x}{\sqrt{a^2-x^2}} dx \end{aligned}$$

Let $I_1 = a \int \frac{1}{\sqrt{a^2-x^2}} dx$ and $I_2 = \int \frac{x}{\sqrt{a^2-x^2}} dx$.

For I_1 : Using the standard result $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$:

$$I_1 = a \sin^{-1} \left(\frac{x}{a} \right) + C_1$$

For I_2 : Let $a^2 - x^2 = t$.

Differentiating:

$$\begin{aligned} -2x dx &= dt \\ x dx &= -\frac{1}{2} dt \end{aligned}$$

Substituting these into I_2 :

$$\begin{aligned} I_2 &= \int \frac{-\frac{1}{2} dt}{\sqrt{t}} \\ &= -\sqrt{t} + C_2 \end{aligned}$$

Back-substituting $t = a^2 - x^2$:

$$I_2 = -\sqrt{a^2 - x^2} + C_2$$

Combining I_1 and I_2 :

$$I = a \sin^{-1} \left(\frac{x}{a} \right) - \sqrt{a^2 - x^2} + C$$

(ii) **Evaluation of** $\int \frac{\sqrt{x^2 - a^2}}{x} dx$: Let $x = a \sec \theta$.

Differentiating:

$$dx = a \sec \theta \tan \theta d\theta$$

Also, using the identity $\sec^2 \theta - 1 = \tan^2 \theta$:

$$\begin{aligned} \sqrt{x^2 - a^2} &= \sqrt{a^2 \sec^2 \theta - a^2} \\ &= a \tan \theta \end{aligned}$$

Substituting these into the integral:

$$\begin{aligned} I &= \int \frac{a \tan \theta}{a \sec \theta} \cdot a \sec \theta \tan \theta d\theta \\ &= a \int \tan^2 \theta d\theta \\ &= a \int (\sec^2 \theta - 1) d\theta \\ &= a(\tan \theta - \theta) + C \end{aligned}$$

Now, we back-substitute to the original variable x : Since $\sec \theta = \frac{x}{a}$, we have:

$$\begin{aligned} \theta &= \sec^{-1} \left(\frac{x}{a} \right) \\ \tan \theta &= \frac{\sqrt{x^2 - a^2}}{a} \end{aligned}$$

Substituting these back:

$$\begin{aligned} I &= a \left(\frac{\sqrt{x^2 - a^2}}{a} - \sec^{-1} \left(\frac{x}{a} \right) \right) + C \\ &= \sqrt{x^2 - a^2} - a \sec^{-1} \left(\frac{x}{a} \right) + C \end{aligned}$$

Final Answer

$$\begin{aligned} \text{(i)} \quad & \int \sqrt{\frac{a+x}{a-x}} dx = a \sin^{-1} \left(\frac{x}{a} \right) - \sqrt{a^2 - x^2} + C \\ \text{(ii)} \quad & \int \frac{\sqrt{x^2 - a^2}}{x} dx = \sqrt{x^2 - a^2} - a \sec^{-1} \left(\frac{x}{a} \right) + C \end{aligned}$$

7.8 – Integrals Reducible to Standard Forms (Completing the Square)

Question 1 (i)

Evaluate $\int \frac{dx}{x^2 + 4x + 8}$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{dx}{x^2 + 4x + 8}$

Concept / Formula Used: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Completing the square in the denominator:

$$\begin{aligned} x^2 + 4x + 8 &= x^2 + 4x + 4 + 4 \\ &= (x + 2)^2 + 2^2 \end{aligned}$$

Substituting this back into the integral:

$$I = \int \frac{dx}{(x + 2)^2 + 2^2}$$

Let $u = x + 2$.

Differentiating with respect to x :

$$du = dx$$

Substituting u and dx into the integral:

$$I = \int \frac{du}{u^2 + 2^2}$$

Using the standard result $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$ with $a = 2$:

$$I = \frac{1}{2} \tan^{-1} \left(\frac{u}{2} \right) + C$$

Back-substituting $u = x + 2$:

$$I = \frac{1}{2} \tan^{-1} \left(\frac{x + 2}{2} \right) + C$$

Final Answer

$$\int \frac{dx}{x^2 + 4x + 8} = \frac{1}{2} \tan^{-1} \left(\frac{x+2}{2} \right) + C$$

Common Mistake: A common error is writing $\tan^{-1}(x+2)$ without dividing by $a = 2$ both outside and inside the inverse tangent function.

Question 1 (ii)

Evaluate $\int \frac{dx}{x^2 + 2x + 2}$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{dx}{x^2 + 2x + 2}$

Concept / Formula Used: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Completing the square in the denominator:

$$\begin{aligned} x^2 + 2x + 2 &= x^2 + 2x + 1 + 1 \\ &= (x+1)^2 + 1^2 \end{aligned}$$

Substituting this back into the integral:

$$I = \int \frac{dx}{(x+1)^2 + 1^2}$$

Let $u = x + 1$.

Differentiating with respect to x :

$$du = dx$$

Substituting u and dx into the integral:

$$I = \int \frac{du}{u^2 + 1^2}$$

Using the standard result $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$ with $a = 1$:

$$I = \tan^{-1}(u) + C$$

Back-substituting $u = x + 1$:

$$I = \tan^{-1}(x+1) + C$$

Final Answer

$$\int \frac{dx}{x^2 + 2x + 2} = \tan^{-1}(x + 1) + C$$

Question 2 (i)

Evaluate $\int \frac{dx}{x^2 - 6x + 13}$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{dx}{x^2 - 6x + 13}$

Concept / Formula Used: $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Completing the square in the denominator:

$$\begin{aligned} x^2 - 6x + 13 &= x^2 - 6x + 9 + 4 \\ &= (x - 3)^2 + 2^2 \end{aligned}$$

Substituting this back into the integral:

$$I = \int \frac{dx}{(x - 3)^2 + 2^2}$$

Let $u = x - 3$.

Differentiating with respect to x :

$$du = dx$$

Substituting u and dx into the integral:

$$I = \int \frac{du}{u^2 + 2^2}$$

Using the standard result $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$ with $a = 2$:

$$I = \frac{1}{2} \tan^{-1} \left(\frac{u}{2} \right) + C$$

Back-substituting $u = x - 3$:

$$I = \frac{1}{2} \tan^{-1} \left(\frac{x - 3}{2} \right) + C$$

Final Answer

$$\int \frac{dx}{x^2 - 6x + 13} = \frac{1}{2} \tan^{-1} \left(\frac{x - 3}{2} \right) + C$$

Question 2 (ii)

Evaluate $\int \frac{x+3}{x^2-2x-5} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{x+3}{x^2-2x-5} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{denominator}) + B$

Solution:

We write the numerator in terms of the derivative of the denominator:

$$\begin{aligned} x+3 &= A \frac{d}{dx}(x^2-2x-5) + B \\ x+3 &= A(2x-2) + B \\ x+3 &= 2Ax + (-2A+B) \end{aligned}$$

Equating coefficients of x on both sides:

$$\begin{aligned} 2A &= 1 \\ A &= \frac{1}{2} \end{aligned}$$

Equating constant terms on both sides:

$$-2A + B = 3$$

Substituting $A = \frac{1}{2}$:

$$\begin{aligned} -2\left(\frac{1}{2}\right) + B &= 3 \\ -1 + B &= 3 \\ B &= 4 \end{aligned}$$

Thus, $x+3 = \frac{1}{2}(2x-2) + 4$.

Splitting the integral into two parts:

$$\begin{aligned} I &= \int \frac{\frac{1}{2}(2x-2) + 4}{x^2-2x-5} dx \\ &= \frac{1}{2} \int \frac{2x-2}{x^2-2x-5} dx + 4 \int \frac{dx}{x^2-2x-5} \\ &= \frac{1}{2} I_1 + 4I_2 \end{aligned}$$

For $I_1 = \int \frac{2x-2}{x^2-2x-5} dx$: Let $t = x^2 - 2x - 5$.

Differentiating with respect to x :

$$dt = (2x - 2) dx$$

Substituting t and dt :

$$\begin{aligned} I_1 &= \int \frac{dt}{t} \\ &= \log |t| \\ &= \log |x^2 - 2x - 5| \end{aligned}$$

For $I_2 = \int \frac{dx}{x^2 - 2x - 5}$: Completing the square in the denominator:

$$\begin{aligned} x^2 - 2x - 5 &= (x - 1)^2 - 1 - 5 \\ &= (x - 1)^2 - 6 \\ &= (x - 1)^2 - (\sqrt{6})^2 \end{aligned}$$

Substituting into I_2 :

$$I_2 = \int \frac{dx}{(x - 1)^2 - (\sqrt{6})^2}$$

Let $v = x - 1$, so $dv = dx$.

$$I_2 = \int \frac{dv}{v^2 - (\sqrt{6})^2}$$

Using the standard result $\int \frac{dv}{v^2 - a^2} = \frac{1}{2a} \log \left| \frac{v - a}{v + a} \right| + C$ with $a = \sqrt{6}$:

$$I_2 = \frac{1}{2\sqrt{6}} \log \left| \frac{v - \sqrt{6}}{v + \sqrt{6}} \right|$$

Back-substituting $v = x - 1$:

$$I_2 = \frac{1}{2\sqrt{6}} \log \left| \frac{x - 1 - \sqrt{6}}{x - 1 + \sqrt{6}} \right|$$

Combining I_1 and I_2 :

$$\begin{aligned} I &= \frac{1}{2} \log |x^2 - 2x - 5| + 4 \cdot \frac{1}{2\sqrt{6}} \log \left| \frac{x - 1 - \sqrt{6}}{x - 1 + \sqrt{6}} \right| + C \\ &= \frac{1}{2} \log |x^2 - 2x - 5| + \frac{2}{\sqrt{6}} \log \left| \frac{x - 1 - \sqrt{6}}{x - 1 + \sqrt{6}} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{x + 3}{x^2 - 2x - 5} dx = \frac{1}{2} \log |x^2 - 2x - 5| + \frac{2}{\sqrt{6}} \log \left| \frac{x - 1 - \sqrt{6}}{x - 1 + \sqrt{6}} \right| + C$$

Question 3 (i)

Evaluate $\int \frac{x+2}{2x^2+6x+5} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{x+2}{2x^2+6x+5} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{denominator}) + B$

Solution:

Express the numerator in terms of the derivative of $2x^2 + 6x + 5$:

$$x + 2 = A \frac{d}{dx}(2x^2 + 6x + 5) + B$$

$$x + 2 = A(4x + 6) + B$$

$$x + 2 = 4Ax + (6A + B)$$

Equating coefficients of x :

$$4A = 1$$

$$A = \frac{1}{4}$$

Equating constant terms:

$$6A + B = 2$$

Substituting $A = \frac{1}{4}$:

$$6\left(\frac{1}{4}\right) + B = 2$$

$$\frac{3}{2} + B = 2$$

$$B = 2 - \frac{3}{2}$$

$$B = \frac{1}{2}$$

Thus, $x + 2 = \frac{1}{4}(4x + 6) + \frac{1}{2}$.

Splitting the integral:

$$\begin{aligned} I &= \int \frac{\frac{1}{4}(4x + 6) + \frac{1}{2}}{2x^2 + 6x + 5} dx \\ &= \frac{1}{4} \int \frac{4x + 6}{2x^2 + 6x + 5} dx + \frac{1}{2} \int \frac{dx}{2x^2 + 6x + 5} \\ &= \frac{1}{4} I_1 + \frac{1}{2} I_2 \end{aligned}$$

For $I_1 = \int \frac{4x+6}{2x^2+6x+5} dx$: Let $t = 2x^2 + 6x + 5$.

Differentiating with respect to x :

$$dt = (4x + 6) dx$$

Substituting t and dt :

$$\begin{aligned} I_1 &= \int \frac{dt}{t} \\ &= \log |2x^2 + 6x + 5| \end{aligned}$$

For $I_2 = \int \frac{dx}{2x^2+6x+5}$: Factoring 2 out of the denominator:

$$2x^2 + 6x + 5 = 2 \left(x^2 + 3x + \frac{5}{2} \right)$$

Completing the square for $x^2 + 3x + \frac{5}{2}$:

$$\begin{aligned} x^2 + 3x + \frac{5}{2} &= \left(x + \frac{3}{2} \right)^2 + \frac{5}{2} - \frac{9}{4} \\ &= \left(x + \frac{3}{2} \right)^2 + \frac{1}{4} \\ &= \left(x + \frac{3}{2} \right)^2 + \left(\frac{1}{2} \right)^2 \end{aligned}$$

So,

$$I_2 = \frac{1}{2} \int \frac{dx}{\left(x + \frac{3}{2} \right)^2 + \left(\frac{1}{2} \right)^2}$$

Let $v = x + \frac{3}{2}$, so $dv = dx$.

$$I_2 = \frac{1}{2} \int \frac{dv}{v^2 + \left(\frac{1}{2} \right)^2}$$

Using the standard result $\int \frac{dv}{v^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{v}{a} \right) + C$ with $a = \frac{1}{2}$:

$$\begin{aligned} I_2 &= \frac{1}{2} \cdot \frac{1}{\frac{1}{2}} \tan^{-1} \left(\frac{v}{\frac{1}{2}} \right) \\ &= \tan^{-1}(2v) \end{aligned}$$

Back-substituting $v = x + \frac{3}{2}$:

$$\begin{aligned} I_2 &= \tan^{-1} \left(2 \left(x + \frac{3}{2} \right) \right) \\ &= \tan^{-1}(2x + 3) \end{aligned}$$

Combining I_1 and I_2 :

$$I = \frac{1}{4} \log |2x^2 + 6x + 5| + \frac{1}{2} \tan^{-1}(2x + 3) + C$$

Final Answer

$$\int \frac{x+2}{2x^2+6x+5} dx = \frac{1}{4} \log |2x^2 + 6x + 5| + \frac{1}{2} \tan^{-1}(2x + 3) + C$$

Question 3 (ii)

Evaluate $\int \frac{2x+1}{18-4x-x^2} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{2x+1}{18-4x-x^2} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{denominator}) + B$

Solution:

Express the numerator in terms of the derivative of $18 - 4x - x^2$:

$$2x + 1 = A \frac{d}{dx}(18 - 4x - x^2) + B$$

$$2x + 1 = A(-4 - 2x) + B$$

$$2x + 1 = -2Ax + (-4A + B)$$

Equating coefficients of x :

$$-2A = 2$$

$$A = -1$$

Equating constant terms:

$$-4A + B = 1$$

Substituting $A = -1$:

$$-4(-1) + B = 1$$

$$4 + B = 1$$

$$B = -3$$

Thus, $2x + 1 = -1(-4 - 2x) - 3$.

Splitting the integral:

$$\begin{aligned} I &= \int \frac{-1(-4 - 2x) - 3}{18 - 4x - x^2} dx \\ &= - \int \frac{-4 - 2x}{18 - 4x - x^2} dx - 3 \int \frac{dx}{18 - 4x - x^2} \\ &= -I_1 - 3I_2 \end{aligned}$$

For $I_1 = \int \frac{-4-2x}{18-4x-x^2} dx$: Let $t = 18 - 4x - x^2$.

Differentiating with respect to x :

$$dt = (-4 - 2x) dx$$

Substituting t and dt :

$$\begin{aligned} I_1 &= \int \frac{dt}{t} \\ &= \log |18 - 4x - x^2| \end{aligned}$$

For $I_2 = \int \frac{dx}{18-4x-x^2}$: Completing the square in the denominator:

$$\begin{aligned} 18 - 4x - x^2 &= 18 - (x^2 + 4x) \\ &= 18 - [(x + 2)^2 - 4] \\ &= 22 - (x + 2)^2 \\ &= (\sqrt{22})^2 - (x + 2)^2 \end{aligned}$$

Substituting into I_2 :

$$I_2 = \int \frac{dx}{(\sqrt{22})^2 - (x + 2)^2}$$

Let $v = x + 2$, so $dv = dx$.

$$I_2 = \int \frac{dv}{(\sqrt{22})^2 - v^2}$$

Using the standard result $\int \frac{dv}{a^2 - v^2} = \frac{1}{2a} \log \left| \frac{a+v}{a-v} \right| + C$ with $a = \sqrt{22}$:

$$I_2 = \frac{1}{2\sqrt{22}} \log \left| \frac{\sqrt{22} + v}{\sqrt{22} - v} \right|$$

Back-substituting $v = x + 2$:

$$I_2 = \frac{1}{2\sqrt{22}} \log \left| \frac{\sqrt{22} + x + 2}{\sqrt{22} - x - 2} \right|$$

Combining $-I_1$ and $-3I_2$:

$$I = -\log |18 - 4x - x^2| - \frac{3}{2\sqrt{22}} \log \left| \frac{\sqrt{22} + x + 2}{\sqrt{22} - x - 2} \right| + C$$

Final Answer

$$\int \frac{2x+1}{18-4x-x^2} dx = -\log |18 - 4x - x^2| - \frac{3}{2\sqrt{22}} \log \left| \frac{\sqrt{22} + x + 2}{\sqrt{22} - x - 2} \right| + C$$

Question 4 (i)

Evaluate $\int \frac{e^x}{2e^{2x} + 3e^x + 5} dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $I = \int \frac{e^x}{2e^{2x} + 3e^x + 5} dx$

Concept / Formula Used: Substitution $t = e^x$ followed by completing the square

Solution:

Let $e^x = t$.

Differentiating with respect to x :

$$\begin{aligned} e^x dx &= dt \\ dx &= \frac{dt}{t} \end{aligned}$$

Substituting $e^x = t$ and $dx = \frac{dt}{t}$ into I :

$$\begin{aligned} I &= \int \frac{t}{2t^2 + 3t + 5} \cdot \frac{dt}{t} \\ &= \int \frac{dt}{2t^2 + 3t + 5} \end{aligned}$$

Factoring 2 out of the denominator:

$$I = \frac{1}{2} \int \frac{dt}{t^2 + \frac{3}{2}t + \frac{5}{2}}$$

Completing the square in the denominator:

$$\begin{aligned} t^2 + \frac{3}{2}t + \frac{5}{2} &= \left(t + \frac{3}{4}\right)^2 + \frac{5}{2} - \frac{9}{16} \\ &= \left(t + \frac{3}{4}\right)^2 + \frac{31}{16} \\ &= \left(t + \frac{3}{4}\right)^2 + \left(\frac{\sqrt{31}}{4}\right)^2 \end{aligned}$$

So,

$$I = \frac{1}{2} \int \frac{dt}{\left(t + \frac{3}{4}\right)^2 + \left(\frac{\sqrt{31}}{4}\right)^2}$$

Let $u = t + \frac{3}{4}$, so $du = dt$.

$$I = \frac{1}{2} \int \frac{du}{u^2 + \left(\frac{\sqrt{31}}{4}\right)^2}$$

Using the standard result $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$ with $a = \frac{\sqrt{31}}{4}$:

$$\begin{aligned} I &= \frac{1}{2} \cdot \frac{1}{\frac{\sqrt{31}}{4}} \tan^{-1} \left(\frac{u}{\frac{\sqrt{31}}{4}} \right) + C \\ &= \frac{2}{\sqrt{31}} \tan^{-1} \left(\frac{4u}{\sqrt{31}} \right) + C \end{aligned}$$

Back-substituting $u = t + \frac{3}{4}$:

$$\begin{aligned} I &= \frac{2}{\sqrt{31}} \tan^{-1} \left(\frac{4 \left(t + \frac{3}{4} \right)}{\sqrt{31}} \right) + C \\ &= \frac{2}{\sqrt{31}} \tan^{-1} \left(\frac{4t + 3}{\sqrt{31}} \right) + C \end{aligned}$$

Back-substituting $t = e^x$:

$$I = \frac{2}{\sqrt{31}} \tan^{-1} \left(\frac{4e^x + 3}{\sqrt{31}} \right) + C$$

Final Answer

$$\int \frac{e^x}{2e^{2x} + 3e^x + 5} dx = \frac{2}{\sqrt{31}} \tan^{-1} \left(\frac{4e^x + 3}{\sqrt{31}} \right) + C$$

Question 4 (ii)

Evaluate $\int \frac{\cos x}{6 + 4 \sin x - \cos^2 x} dx$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Given: $I = \int \frac{\cos x}{6 + 4 \sin x - \cos^2 x} dx$

Concept / Formula Used: Trigonometric identity $\cos^2 x = 1 - \sin^2 x$

Solution:

Using the Pythagorean identity $\cos^2 x = 1 - \sin^2 x$ in the denominator:

$$\begin{aligned} 6 + 4 \sin x - \cos^2 x &= 6 + 4 \sin x - (1 - \sin^2 x) \\ &= \sin^2 x + 4 \sin x + 5 \end{aligned}$$

Thus, the integral becomes:

$$I = \int \frac{\cos x}{\sin^2 x + 4 \sin x + 5} dx$$

Let $\sin x = t$.

Differentiating with respect to x :

$$\cos x dx = dt$$

Substituting into I :

$$I = \int \frac{dt}{t^2 + 4t + 5}$$

Completing the square in the denominator:

$$t^2 + 4t + 5 = (t + 2)^2 + 1^2$$

So,

$$I = \int \frac{dt}{(t + 2)^2 + 1^2}$$

Let $u = t + 2$, so $du = dt$.

$$I = \int \frac{du}{u^2 + 1^2}$$

Using the standard result $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$ with $a = 1$:

$$I = \tan^{-1}(u) + C$$

Back-substituting $u = t + 2$:

$$I = \tan^{-1}(t + 2) + C$$

Back-substituting $t = \sin x$:

$$I = \tan^{-1}(\sin x + 2) + C$$

Final Answer

$$\int \frac{\cos x}{6 + 4 \sin x - \cos^2 x} dx = \tan^{-1}(\sin x + 2) + C$$

Question 5 (i)

Evaluate $\int \frac{dx}{\sqrt{4x - x^2}}$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{dx}{\sqrt{4x - x^2}}$

Concept / Formula Used: $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Completing the square for $4x - x^2$:

$$\begin{aligned} 4x - x^2 &= -(x^2 - 4x) \\ &= -[(x - 2)^2 - 4] \\ &= 4 - (x - 2)^2 \\ &= 2^2 - (x - 2)^2 \end{aligned}$$

Substituting this into the integral:

$$I = \int \frac{dx}{\sqrt{2^2 - (x - 2)^2}}$$

Let $u = x - 2$.

Differentiating with respect to x :

$$du = dx$$

Substituting u and dx :

$$I = \int \frac{du}{\sqrt{2^2 - u^2}}$$

Using the standard result $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$ with $a = 2$:

$$I = \sin^{-1} \left(\frac{u}{2} \right) + C$$

Back-substituting $u = x - 2$:

$$I = \sin^{-1} \left(\frac{x - 2}{2} \right) + C$$

Final Answer

$$\int \frac{dx}{\sqrt{4x - x^2}} = \sin^{-1} \left(\frac{x - 2}{2} \right) + C$$

Question 5 (ii)

Evaluate $\int \frac{dx}{\sqrt{7 - 6x - x^2}}$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{dx}{\sqrt{7-6x-x^2}}$

Concept / Formula Used: $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

Solution:

Completing the square for $7-6x-x^2$:

$$\begin{aligned} 7-6x-x^2 &= 7-(x^2+6x) \\ &= 7-[(x+3)^2-9] \\ &= 16-(x+3)^2 \\ &= 4^2-(x+3)^2 \end{aligned}$$

Substituting this into the integral:

$$I = \int \frac{dx}{\sqrt{4^2-(x+3)^2}}$$

Let $u = x + 3$.

Differentiating with respect to x :

$$du = dx$$

Substituting u and dx :

$$I = \int \frac{du}{\sqrt{4^2-u^2}}$$

Using the standard result $\int \frac{du}{\sqrt{a^2-u^2}} = \sin^{-1}\left(\frac{u}{a}\right) + C$ with $a = 4$:

$$I = \sin^{-1}\left(\frac{u}{4}\right) + C$$

Back-substituting $u = x + 3$:

$$I = \sin^{-1}\left(\frac{x+3}{4}\right) + C$$

Final Answer

$$\int \frac{dx}{\sqrt{7-6x-x^2}} = \sin^{-1}\left(\frac{x+3}{4}\right) + C$$

Question 6 (i)

Evaluate $\int \frac{dx}{\sqrt{(x-1)(x-2)}}$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{dx}{\sqrt{(x-1)(x-2)}}$

Concept / Formula Used: $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log \left| x + \sqrt{x^2 - a^2} \right| + C$

Solution:

Expanding the expression under the square root:

$$(x-1)(x-2) = x^2 - 3x + 2$$

Completing the square for $x^2 - 3x + 2$:

$$\begin{aligned} x^2 - 3x + 2 &= \left(x - \frac{3}{2}\right)^2 + 2 - \frac{9}{4} \\ &= \left(x - \frac{3}{2}\right)^2 - \frac{1}{4} \\ &= \left(x - \frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \end{aligned}$$

Substituting this into the integral:

$$I = \int \frac{dx}{\sqrt{\left(x - \frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2}}$$

Let $u = x - \frac{3}{2}$, so $du = dx$.

$$I = \int \frac{du}{\sqrt{u^2 - \left(\frac{1}{2}\right)^2}}$$

Using the standard result $\int \frac{du}{\sqrt{u^2 - a^2}} = \log \left| u + \sqrt{u^2 - a^2} \right| + C$ with $a = \frac{1}{2}$:

$$I = \log \left| u + \sqrt{u^2 - \left(\frac{1}{2}\right)^2} \right| + C$$

Back-substituting $u = x - \frac{3}{2}$:

$$\begin{aligned} I &= \log \left| x - \frac{3}{2} + \sqrt{\left(x - \frac{3}{2}\right)^2 - \frac{1}{4}} \right| + C \\ &= \log \left| x - \frac{3}{2} + \sqrt{x^2 - 3x + 2} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{\sqrt{(x-1)(x-2)}} = \log \left| x - \frac{3}{2} + \sqrt{x^2 - 3x + 2} \right| + C$$

Question 6 (ii)

Evaluate $\int \frac{dx}{\sqrt{8+3x-x^2}}$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{dx}{\sqrt{8+3x-x^2}}$

Concept / Formula Used: $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:Completing the square for $8+3x-x^2$:

$$\begin{aligned} 8+3x-x^2 &= 8-(x^2-3x) \\ &= 8-\left[\left(x-\frac{3}{2}\right)^2-\frac{9}{4}\right] \\ &= 8+\frac{9}{4}-\left(x-\frac{3}{2}\right)^2 \\ &= \frac{41}{4}-\left(x-\frac{3}{2}\right)^2 \\ &= \left(\frac{\sqrt{41}}{2}\right)^2-\left(x-\frac{3}{2}\right)^2 \end{aligned}$$

Substituting this into the integral:

$$I = \int \frac{dx}{\sqrt{\left(\frac{\sqrt{41}}{2}\right)^2 - \left(x-\frac{3}{2}\right)^2}}$$

Let $u = x - \frac{3}{2}$, so $du = dx$.

$$I = \int \frac{du}{\sqrt{\left(\frac{\sqrt{41}}{2}\right)^2 - u^2}}$$

Using the standard result $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$ with $a = \frac{\sqrt{41}}{2}$:

$$\begin{aligned} I &= \sin^{-1} \left(\frac{u}{\frac{\sqrt{41}}{2}} \right) + C \\ &= \sin^{-1} \left(\frac{2u}{\sqrt{41}} \right) + C \end{aligned}$$

Back-substituting $u = x - \frac{3}{2}$:

$$\begin{aligned} I &= \sin^{-1} \left(\frac{2 \left(x - \frac{3}{2} \right)}{\sqrt{41}} \right) + C \\ &= \sin^{-1} \left(\frac{2x - 3}{\sqrt{41}} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{\sqrt{8 + 3x - x^2}} = \sin^{-1} \left(\frac{2x - 3}{\sqrt{41}} \right) + C$$

Question 7 (i)

Evaluate $\int \frac{dx}{\sqrt{5 - 4x - 2x^2}}$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{dx}{\sqrt{5 - 4x - 2x^2}}$

Concept / Formula Used: $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Factoring 2 out of the expression inside the square root:

$$\begin{aligned} 5 - 4x - 2x^2 &= 2 \left[\frac{5}{2} - 2x - x^2 \right] \\ &= 2 \left[\frac{5}{2} - (x^2 + 2x) \right] \\ &= 2 \left[\frac{5}{2} - ((x + 1)^2 - 1) \right] \\ &= 2 \left[\frac{5}{2} + 1 - (x + 1)^2 \right] \\ &= 2 \left[\frac{7}{2} - (x + 1)^2 \right] \end{aligned}$$

Substituting this back into the integral:

$$\begin{aligned} I &= \int \frac{dx}{\sqrt{2 \left[\frac{7}{2} - (x+1)^2 \right]}} \\ &= \frac{1}{\sqrt{2}} \int \frac{dx}{\sqrt{\left(\sqrt{\frac{7}{2}} \right)^2 - (x+1)^2}} \end{aligned}$$

Let $u = x + 1$, so $du = dx$.

$$I = \frac{1}{\sqrt{2}} \int \frac{du}{\sqrt{\left(\sqrt{\frac{7}{2}} \right)^2 - u^2}}$$

Using the standard result $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$ with $a = \sqrt{\frac{7}{2}}$:

$$\begin{aligned} I &= \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{u}{\sqrt{\frac{7}{2}}} \right) + C \\ &= \frac{1}{\sqrt{2}} \sin^{-1} \left(\sqrt{\frac{2}{7}} u \right) + C \end{aligned}$$

Back-substituting $u = x + 1$:

$$I = \frac{1}{\sqrt{2}} \sin^{-1} \left(\sqrt{\frac{2}{7}} (x+1) \right) + C$$

Final Answer

$$\int \frac{dx}{\sqrt{5-4x-2x^2}} = \frac{1}{\sqrt{2}} \sin^{-1} \left(\sqrt{\frac{2}{7}} (x+1) \right) + C$$

Question 7 (ii)

Evaluate $\int \frac{dx}{\sqrt{5x-4x^2}}$.

TYPE 12b Substitution Involving Radical Expressions

★ Likely Exam Question

Given: $I = \int \frac{dx}{\sqrt{5x-4x^2}}$

Concept / Formula Used: $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Factoring 4 out of the quadratic expression under the square root:

$$\begin{aligned}
 5x - 4x^2 &= 4 \left(\frac{5}{4}x - x^2 \right) \\
 &= 4 \left[- \left(x^2 - \frac{5}{4}x \right) \right] \\
 &= 4 \left[- \left(\left(x - \frac{5}{8} \right)^2 - \frac{25}{64} \right) \right] \\
 &= 4 \left[\frac{25}{64} - \left(x - \frac{5}{8} \right)^2 \right] \\
 &= 4 \left[\left(\frac{5}{8} \right)^2 - \left(x - \frac{5}{8} \right)^2 \right]
 \end{aligned}$$

Substituting into the integral:

$$\begin{aligned}
 I &= \int \frac{dx}{\sqrt{4 \left[\left(\frac{5}{8} \right)^2 - \left(x - \frac{5}{8} \right)^2 \right]}} \\
 &= \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{5}{8} \right)^2 - \left(x - \frac{5}{8} \right)^2}}
 \end{aligned}$$

Let $u = x - \frac{5}{8}$, so $du = dx$.

$$I = \frac{1}{2} \int \frac{du}{\sqrt{\left(\frac{5}{8} \right)^2 - u^2}}$$

Using the standard result $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + C$ with $a = \frac{5}{8}$:

$$\begin{aligned}
 I &= \frac{1}{2} \sin^{-1} \left(\frac{u}{\frac{5}{8}} \right) + C \\
 &= \frac{1}{2} \sin^{-1} \left(\frac{8u}{5} \right) + C
 \end{aligned}$$

Back-substituting $u = x - \frac{5}{8}$:

$$\begin{aligned}
 I &= \frac{1}{2} \sin^{-1} \left(\frac{8 \left(x - \frac{5}{8} \right)}{5} \right) + C \\
 &= \frac{1}{2} \sin^{-1} \left(\frac{8x - 5}{5} \right) + C
 \end{aligned}$$

Final Answer

$$\int \frac{dx}{\sqrt{5x - 4x^2}} = \frac{1}{2} \sin^{-1} \left(\frac{8x - 5}{5} \right) + C$$

Question 8 (i)

Evaluate $\int \frac{x-1}{\sqrt{x^2-x}} dx$.

TYPE 12b Substitution Involving Radical Expressions

★ Likely Exam Question

Given: $I = \int \frac{x-1}{\sqrt{x^2-x}} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{quadratic}) + B$

Solution:

Express the numerator in terms of the derivative of $x^2 - x$:

$$x - 1 = A \frac{d}{dx}(x^2 - x) + B$$

$$x - 1 = A(2x - 1) + B$$

$$x - 1 = 2Ax + (-A + B)$$

Equating coefficients of x :

$$2A = 1$$

$$A = \frac{1}{2}$$

Equating constant terms:

$$-A + B = -1$$

Substituting $A = \frac{1}{2}$:

$$-\frac{1}{2} + B = -1$$

$$B = -1 + \frac{1}{2}$$

$$B = -\frac{1}{2}$$

Thus, $x - 1 = \frac{1}{2}(2x - 1) - \frac{1}{2}$.

Splitting the integral:

$$\begin{aligned} I &= \int \frac{\frac{1}{2}(2x - 1) - \frac{1}{2}}{\sqrt{x^2 - x}} dx \\ &= \frac{1}{2} \int \frac{2x - 1}{\sqrt{x^2 - x}} dx - \frac{1}{2} \int \frac{dx}{\sqrt{x^2 - x}} \\ &= \frac{1}{2} I_1 - \frac{1}{2} I_2 \end{aligned}$$

For $I_1 = \int \frac{2x-1}{\sqrt{x^2-x}} dx$: Let $u = x^2 - x$.

Differentiating with respect to x :

$$du = (2x - 1) dx$$

Substituting u and du :

$$\begin{aligned} I_1 &= \int \frac{du}{\sqrt{u}} \\ &= \int u^{-1/2} du \\ &= \frac{u^{1/2}}{1/2} \\ &= 2\sqrt{u} \\ &= 2\sqrt{x^2 - x} \end{aligned}$$

For $I_2 = \int \frac{dx}{\sqrt{x^2-x}}$: Completing the square for $x^2 - x$:

$$\begin{aligned} x^2 - x &= \left(x - \frac{1}{2}\right)^2 - \frac{1}{4} \\ &= \left(x - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \end{aligned}$$

Substituting into I_2 :

$$I_2 = \int \frac{dx}{\sqrt{\left(x - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2}}$$

Let $v = x - \frac{1}{2}$, so $dv = dx$.

$$I_2 = \int \frac{dv}{\sqrt{v^2 - \left(\frac{1}{2}\right)^2}}$$

Using the standard result $\int \frac{dv}{\sqrt{v^2 - a^2}} = \log \left| v + \sqrt{v^2 - a^2} \right| + C$:

$$I_2 = \log \left| v + \sqrt{v^2 - \left(\frac{1}{2}\right)^2} \right|$$

Back-substituting $v = x - \frac{1}{2}$:

$$I_2 = \log \left| x - \frac{1}{2} + \sqrt{x^2 - x} \right|$$

Combining $\frac{1}{2}I_1$ and $-\frac{1}{2}I_2$:

$$\begin{aligned} I &= \frac{1}{2} \left(2\sqrt{x^2 - x} \right) - \frac{1}{2} \log \left| x - \frac{1}{2} + \sqrt{x^2 - x} \right| + C \\ &= \sqrt{x^2 - x} - \frac{1}{2} \log \left| x - \frac{1}{2} + \sqrt{x^2 - x} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{x-1}{\sqrt{x^2-x}} dx = \sqrt{x^2-x} - \frac{1}{2} \log \left| x - \frac{1}{2} + \sqrt{x^2-x} \right| + C$$

Question 8 (ii)

Evaluate $\int \frac{5x+3}{\sqrt{x^2+4x+10}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{5x+3}{\sqrt{x^2+4x+10}} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{quadratic}) + B$ **Solution:**Express the numerator in terms of the derivative of $x^2 + 4x + 10$:

$$5x + 3 = A \frac{d}{dx}(x^2 + 4x + 10) + B$$

$$5x + 3 = A(2x + 4) + B$$

$$5x + 3 = 2Ax + (4A + B)$$

Equating coefficients of x :

$$2A = 5$$

$$A = \frac{5}{2}$$

Equating constant terms:

$$4A + B = 3$$

Substituting $A = \frac{5}{2}$:

$$4 \left(\frac{5}{2} \right) + B = 3$$

$$10 + B = 3$$

$$B = -7$$

Thus, $5x + 3 = \frac{5}{2}(2x + 4) - 7$.

Splitting the integral:

$$\begin{aligned} I &= \int \frac{\frac{5}{2}(2x+4) - 7}{\sqrt{x^2+4x+10}} dx \\ &= \frac{5}{2} \int \frac{2x+4}{\sqrt{x^2+4x+10}} dx - 7 \int \frac{dx}{\sqrt{x^2+4x+10}} \\ &= \frac{5}{2} I_1 - 7 I_2 \end{aligned}$$

For $I_1 = \int \frac{2x+4}{\sqrt{x^2+4x+10}} dx$: Let $u = x^2 + 4x + 10$.

Differentiating with respect to x :

$$du = (2x + 4) dx$$

Substituting u and du :

$$\begin{aligned} I_1 &= \int \frac{du}{\sqrt{u}} \\ &= 2\sqrt{u} \\ &= 2\sqrt{x^2 + 4x + 10} \end{aligned}$$

For $I_2 = \int \frac{dx}{\sqrt{x^2+4x+10}}$: Completing the square for $x^2 + 4x + 10$:

$$\begin{aligned} x^2 + 4x + 10 &= (x + 2)^2 + 6 \\ &= (x + 2)^2 + (\sqrt{6})^2 \end{aligned}$$

Substituting into I_2 :

$$I_2 = \int \frac{dx}{\sqrt{(x + 2)^2 + (\sqrt{6})^2}}$$

Let $v = x + 2$, so $dv = dx$.

$$I_2 = \int \frac{dv}{\sqrt{v^2 + (\sqrt{6})^2}}$$

Using the standard result $\int \frac{dv}{\sqrt{v^2 + a^2}} = \log |v + \sqrt{v^2 + a^2}| + C$:

$$I_2 = \log |v + \sqrt{v^2 + 6}|$$

Back-substituting $v = x + 2$:

$$I_2 = \log |x + 2 + \sqrt{x^2 + 4x + 10}|$$

Combining $\frac{5}{2}I_1$ and $-7I_2$:

$$\begin{aligned} I &= \frac{5}{2} \left(2\sqrt{x^2 + 4x + 10} \right) - 7 \log |x + 2 + \sqrt{x^2 + 4x + 10}| + C \\ &= 5\sqrt{x^2 + 4x + 10} - 7 \log |x + 2 + \sqrt{x^2 + 4x + 10}| + C \end{aligned}$$

Final Answer

$$\int \frac{5x + 3}{\sqrt{x^2 + 4x + 10}} dx = 5\sqrt{x^2 + 4x + 10} - 7 \log |x + 2 + \sqrt{x^2 + 4x + 10}| + C$$

Question 9 (i)

Evaluate $\int \frac{6x+7}{\sqrt{(x-5)(x-4)}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{6x+7}{\sqrt{(x-5)(x-4)}} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{quadratic}) + B$

Solution:

Expanding the denominator expression:

$$(x-5)(x-4) = x^2 - 9x + 20$$

So,

$$I = \int \frac{6x+7}{\sqrt{x^2 - 9x + 20}} dx$$

Express the numerator in terms of the derivative of $x^2 - 9x + 20$:

$$6x+7 = A \frac{d}{dx}(x^2 - 9x + 20) + B$$

$$6x+7 = A(2x-9) + B$$

$$6x+7 = 2Ax + (-9A+B)$$

Equating coefficients of x :

$$2A = 6$$

$$A = 3$$

Equating constant terms:

$$-9A + B = 7$$

Substituting $A = 3$:

$$-9(3) + B = 7$$

$$-27 + B = 7$$

$$B = 34$$

Thus, $6x+7 = 3(2x-9) + 34$.

Splitting the integral:

$$\begin{aligned} I &= \int \frac{3(2x-9) + 34}{\sqrt{x^2 - 9x + 20}} dx \\ &= 3 \int \frac{2x-9}{\sqrt{x^2 - 9x + 20}} dx + 34 \int \frac{dx}{\sqrt{x^2 - 9x + 20}} \\ &= 3I_1 + 34I_2 \end{aligned}$$

For $I_1 = \int \frac{2x-9}{\sqrt{x^2-9x+20}} dx$: Let $u = x^2 - 9x + 20$, so $du = (2x - 9) dx$.

$$\begin{aligned} I_1 &= \int \frac{du}{\sqrt{u}} \\ &= 2\sqrt{u} \\ &= 2\sqrt{x^2 - 9x + 20} \end{aligned}$$

For $I_2 = \int \frac{dx}{\sqrt{x^2-9x+20}}$: Completing the square for $x^2 - 9x + 20$:

$$\begin{aligned} x^2 - 9x + 20 &= \left(x - \frac{9}{2}\right)^2 + 20 - \frac{81}{4} \\ &= \left(x - \frac{9}{2}\right)^2 - \frac{1}{4} \\ &= \left(x - \frac{9}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \end{aligned}$$

Let $v = x - \frac{9}{2}$, so $dv = dx$.

$$I_2 = \int \frac{dv}{\sqrt{v^2 - \left(\frac{1}{2}\right)^2}}$$

Using the standard result $\int \frac{dv}{\sqrt{v^2 - a^2}} = \log \left| v + \sqrt{v^2 - a^2} \right| + C$:

$$\begin{aligned} I_2 &= \log \left| v + \sqrt{v^2 - \left(\frac{1}{2}\right)^2} \right| \\ &= \log \left| x - \frac{9}{2} + \sqrt{x^2 - 9x + 20} \right| \end{aligned}$$

Combining $3I_1$ and $34I_2$:

$$\begin{aligned} I &= 3 \left(2\sqrt{x^2 - 9x + 20} \right) + 34 \log \left| x - \frac{9}{2} + \sqrt{x^2 - 9x + 20} \right| + C \\ &= 6\sqrt{x^2 - 9x + 20} + 34 \log \left| x - \frac{9}{2} + \sqrt{x^2 - 9x + 20} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{6x+7}{\sqrt{(x-5)(x-4)}} dx = 6\sqrt{x^2 - 9x + 20} + 34 \log \left| x - \frac{9}{2} + \sqrt{x^2 - 9x + 20} \right| + C$$

Question 9 (ii)

Evaluate $\int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{2x+3}{\sqrt{x^2+4x+1}} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{quadratic}) + B$

Solution:

Express the numerator in terms of the derivative of $x^2 + 4x + 1$:

$$\begin{aligned} 2x + 3 &= A \frac{d}{dx}(x^2 + 4x + 1) + B \\ 2x + 3 &= A(2x + 4) + B \\ 2x + 3 &= 2Ax + (4A + B) \end{aligned}$$

Equating coefficients of x :

$$\begin{aligned} 2A &= 2 \\ A &= 1 \end{aligned}$$

Equating constant terms:

$$4A + B = 3$$

Substituting $A = 1$:

$$\begin{aligned} 4(1) + B &= 3 \\ B &= -1 \end{aligned}$$

Thus, $2x + 3 = 1(2x + 4) - 1$.

Splitting the integral:

$$\begin{aligned} I &= \int \frac{(2x+4) - 1}{\sqrt{x^2+4x+1}} dx \\ &= \int \frac{2x+4}{\sqrt{x^2+4x+1}} dx - \int \frac{dx}{\sqrt{x^2+4x+1}} \\ &= I_1 - I_2 \end{aligned}$$

For $I_1 = \int \frac{2x+4}{\sqrt{x^2+4x+1}} dx$: Let $u = x^2 + 4x + 1$, so $du = (2x + 4) dx$.

$$\begin{aligned} I_1 &= \int \frac{du}{\sqrt{u}} \\ &= 2\sqrt{u} \\ &= 2\sqrt{x^2+4x+1} \end{aligned}$$

For $I_2 = \int \frac{dx}{\sqrt{x^2+4x+1}}$: Completing the square for $x^2 + 4x + 1$:

$$\begin{aligned} x^2 + 4x + 1 &= (x+2)^2 - 3 \\ &= (x+2)^2 - (\sqrt{3})^2 \end{aligned}$$

Let $v = x + 2$, so $dv = dx$.

$$I_2 = \int \frac{dv}{\sqrt{v^2 - (\sqrt{3})^2}}$$

Using the standard result $\int \frac{dv}{\sqrt{v^2 - a^2}} = \log \left| v + \sqrt{v^2 - a^2} \right| + C$:

$$\begin{aligned} I_2 &= \log \left| v + \sqrt{v^2 - 3} \right| \\ &= \log \left| x + 2 + \sqrt{x^2 + 4x + 1} \right| \end{aligned}$$

Combining I_1 and $-I_2$:

$$I = 2\sqrt{x^2 + 4x + 1} - \log \left| x + 2 + \sqrt{x^2 + 4x + 1} \right| + C$$

Final Answer

$$\int \frac{2x + 3}{\sqrt{x^2 + 4x + 1}} dx = 2\sqrt{x^2 + 4x + 1} - \log \left| x + 2 + \sqrt{x^2 + 4x + 1} \right| + C$$

Question 10 (i)

Evaluate $\int \frac{3x + 5}{\sqrt{x^2 - 8x + 7}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{3x + 5}{\sqrt{x^2 - 8x + 7}} dx$

Concept / Formula Used: Expressing numerator as $A \cdot \frac{d}{dx}(\text{quadratic}) + B$

Solution:

Express the numerator in terms of the derivative of $x^2 - 8x + 7$:

$$\begin{aligned} 3x + 5 &= A \frac{d}{dx}(x^2 - 8x + 7) + B \\ 3x + 5 &= A(2x - 8) + B \\ 3x + 5 &= 2Ax + (-8A + B) \end{aligned}$$

Equating coefficients of x :

$$\begin{aligned} 2A &= 3 \\ A &= \frac{3}{2} \end{aligned}$$

Equating constant terms:

$$-8A + B = 5$$

Substituting $A = \frac{3}{2}$:

$$\begin{aligned}-8\left(\frac{3}{2}\right) + B &= 5 \\ -12 + B &= 5 \\ B &= 17\end{aligned}$$

Thus, $3x + 5 = \frac{3}{2}(2x - 8) + 17$.

Splitting the integral:

$$\begin{aligned}I &= \int \frac{\frac{3}{2}(2x - 8) + 17}{\sqrt{x^2 - 8x + 7}} dx \\ &= \frac{3}{2} \int \frac{2x - 8}{\sqrt{x^2 - 8x + 7}} dx + 17 \int \frac{dx}{\sqrt{x^2 - 8x + 7}} \\ &= \frac{3}{2}I_1 + 17I_2\end{aligned}$$

For $I_1 = \int \frac{2x-8}{\sqrt{x^2-8x+7}} dx$: Let $u = x^2 - 8x + 7$, so $du = (2x - 8) dx$.

$$\begin{aligned}I_1 &= \int \frac{du}{\sqrt{u}} \\ &= 2\sqrt{u} \\ &= 2\sqrt{x^2 - 8x + 7}\end{aligned}$$

For $I_2 = \int \frac{dx}{\sqrt{x^2-8x+7}}$: Completing the square for $x^2 - 8x + 7$:

$$\begin{aligned}x^2 - 8x + 7 &= (x - 4)^2 - 9 \\ &= (x - 4)^2 - 3^2\end{aligned}$$

Let $v = x - 4$, so $dv = dx$.

$$I_2 = \int \frac{dv}{\sqrt{v^2 - 3^2}}$$

Using the standard result $\int \frac{dv}{\sqrt{v^2 - a^2}} = \log |v + \sqrt{v^2 - a^2}| + C$:

$$\begin{aligned}I_2 &= \log |v + \sqrt{v^2 - 9}| \\ &= \log |x - 4 + \sqrt{x^2 - 8x + 7}|\end{aligned}$$

Combining $\frac{3}{2}I_1$ and $17I_2$:

$$\begin{aligned}I &= \frac{3}{2} \left(2\sqrt{x^2 - 8x + 7} \right) + 17 \log |x - 4 + \sqrt{x^2 - 8x + 7}| + C \\ &= 3\sqrt{x^2 - 8x + 7} + 17 \log |x - 4 + \sqrt{x^2 - 8x + 7}| + C\end{aligned}$$

Final Answer

$$\int \frac{3x + 5}{\sqrt{x^2 - 8x + 7}} dx = 3\sqrt{x^2 - 8x + 7} + 17 \log |x - 4 + \sqrt{x^2 - 8x + 7}| + C$$

Question 10 (ii)

Evaluate $\int \frac{x^2}{\sqrt{x^6 - 2x^3 - 3}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $I = \int \frac{x^2}{\sqrt{x^6 - 2x^3 - 3}} dx$

Concept / Formula Used: Substitution $t = x^3$ followed by completing the square

Solution:

Expressing x^6 as $(x^3)^2$:

$$I = \int \frac{x^2}{\sqrt{(x^3)^2 - 2(x^3) - 3}} dx$$

Let $x^3 = t$.

Differentiating with respect to x :

$$\begin{aligned} 3x^2 dx &= dt \\ x^2 dx &= \frac{1}{3} dt \end{aligned}$$

Substituting into I :

$$I = \frac{1}{3} \int \frac{dt}{\sqrt{t^2 - 2t - 3}}$$

Completing the square in the denominator:

$$\begin{aligned} t^2 - 2t - 3 &= (t - 1)^2 - 1 - 3 \\ &= (t - 1)^2 - 4 \\ &= (t - 1)^2 - 2^2 \end{aligned}$$

So,

$$I = \frac{1}{3} \int \frac{dt}{\sqrt{(t - 1)^2 - 2^2}}$$

Let $u = t - 1$, so $du = dt$.

$$I = \frac{1}{3} \int \frac{du}{\sqrt{u^2 - 2^2}}$$

Using the standard result $\int \frac{du}{\sqrt{u^2 - a^2}} = \log \left| u + \sqrt{u^2 - a^2} \right| + C$ with $a = 2$:

$$I = \frac{1}{3} \log \left| u + \sqrt{u^2 - 2^2} \right| + C$$

Back-substituting $u = t - 1$:

$$\begin{aligned} I &= \frac{1}{3} \log \left| t - 1 + \sqrt{(t-1)^2 - 4} \right| + C \\ &= \frac{1}{3} \log \left| t - 1 + \sqrt{t^2 - 2t - 3} \right| + C \end{aligned}$$

Back-substituting $t = x^3$:

$$I = \frac{1}{3} \log \left| x^3 - 1 + \sqrt{x^6 - 2x^3 - 3} \right| + C$$

Final Answer

$$\int \frac{x^2}{\sqrt{x^6 - 2x^3 - 3}} dx = \frac{1}{3} \log \left| x^3 - 1 + \sqrt{x^6 - 2x^3 - 3} \right| + C$$

Question 11

Evaluate $\int \frac{1}{x[(\log x)^2 - 3 \log x - 4]} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $I = \int \frac{1}{x[(\log x)^2 - 3 \log x - 4]} dx$

Concept / Formula Used: Substitution $t = \log x$ followed by standard integral form

Solution:

Let $\log x = t$.

Differentiating with respect to x :

$$\frac{1}{x} dx = dt$$

Substituting t and dt into I :

$$I = \int \frac{dt}{t^2 - 3t - 4}$$

Completing the square in the denominator:

$$\begin{aligned} t^2 - 3t - 4 &= \left(t - \frac{3}{2}\right)^2 - 4 - \frac{9}{4} \\ &= \left(t - \frac{3}{2}\right)^2 - \frac{25}{4} \\ &= \left(t - \frac{3}{2}\right)^2 - \left(\frac{5}{2}\right)^2 \end{aligned}$$

So,

$$I = \int \frac{dt}{\left(t - \frac{3}{2}\right)^2 - \left(\frac{5}{2}\right)^2}$$

Let $u = t - \frac{3}{2}$, so $du = dt$.

$$I = \int \frac{du}{u^2 - \left(\frac{5}{2}\right)^2}$$

Using the standard result $\int \frac{du}{u^2 - a^2} = \frac{1}{2a} \log \left| \frac{u - a}{u + a} \right| + C$ with $a = \frac{5}{2}$:

$$\begin{aligned} I &= \frac{1}{2\left(\frac{5}{2}\right)} \log \left| \frac{u - \frac{5}{2}}{u + \frac{5}{2}} \right| + C \\ &= \frac{1}{5} \log \left| \frac{u - \frac{5}{2}}{u + \frac{5}{2}} \right| + C \end{aligned}$$

Back-substituting $u = t - \frac{3}{2}$:

$$\begin{aligned} I &= \frac{1}{5} \log \left| \frac{t - \frac{3}{2} - \frac{5}{2}}{t - \frac{3}{2} + \frac{5}{2}} \right| + C \\ &= \frac{1}{5} \log \left| \frac{t - 4}{t + 1} \right| + C \end{aligned}$$

Back-substituting $t = \log x$:

$$I = \frac{1}{5} \log \left| \frac{\log x - 4}{\log x + 1} \right| + C$$

Final Answer

$$\int \frac{1}{x[(\log x)^2 - 3 \log x - 4]} dx = \frac{1}{5} \log \left| \frac{\log x - 4}{\log x + 1} \right| + C$$

Common Mistake: Forgetting to perform both back-substitution steps ($u \rightarrow t \rightarrow \log x$) leaves the final answer in terms of an intermediate variable.

7.9 – Integration of Rational Trigonometric Functions (Weierstrass Substitution)

Question 1 (i)

Evaluate $\int \frac{dx}{2 + \cos x}$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\cos x = \frac{1 - t^2}{1 + t^2}$ and $dx = \frac{2 dt}{1 + t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1 + t^2) dx = 2 dt$, giving

$$dx = \frac{2 dt}{1 + t^2}$$

Also, using the standard identity $\cos x = \frac{1 - t^2}{1 + t^2}$, the integral becomes

$$\int \frac{\frac{2 dt}{1 + t^2}}{2 + \frac{1 - t^2}{1 + t^2}} = \int \frac{2 dt}{(1 + t^2) \left[2 + \frac{1 - t^2}{1 + t^2} \right]}$$

Simplifying the bracket (combine over the common denominator $1 + t^2$):

$$\begin{aligned} &= \int \frac{2 dt}{2(1 + t^2) + (1 - t^2)} \\ &= \int \frac{2 dt}{2 + 2t^2 + 1 - t^2} \\ &= \int \frac{2 dt}{t^2 + 3} \\ &= 2 \int \frac{dt}{(t)^2 + (\sqrt{3})^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + C$ with $a = \sqrt{3}$:

$$\begin{aligned} &= 2 \times \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{t}{\sqrt{3}} \right) + C \\ &= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t}{\sqrt{3}} \right) + C \end{aligned}$$

Now back-substitute $t = \tan \frac{x}{2}$:

$$= \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\tan \frac{x}{2}}{\sqrt{3}} \right) + C$$

Final Answer

$$\int \frac{dx}{2 + \cos x} = \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \tan \frac{x}{2} \right) + C$$

Common Mistake: Forgetting to convert dx using $(1 + t^2) dx = 2 dt$ before substituting is the most common slip.

Question 1 (ii)

Evaluate $\int \frac{dx}{5 + 4 \cos x}$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\cos x = \frac{1 - t^2}{1 + t^2}$ and $dx = \frac{2 dt}{1 + t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1 + t^2) dx = 2 dt$, giving

$$dx = \frac{2 dt}{1 + t^2}$$

Also, using the standard identity $\cos x = \frac{1 - t^2}{1 + t^2}$, the integral becomes

$$\int \frac{\frac{2 dt}{1 + t^2}}{5 + 4 \left(\frac{1 - t^2}{1 + t^2} \right)} = \int \frac{2 dt}{(1 + t^2) \left[5 + 4 \left(\frac{1 - t^2}{1 + t^2} \right) \right]}$$

Simplifying the bracket:

$$\begin{aligned} &= \int \frac{2 dt}{5(1 + t^2) + 4(1 - t^2)} \\ &= \int \frac{2 dt}{5 + 5t^2 + 4 - 4t^2} \\ &= \int \frac{2 dt}{t^2 + 9} \\ &= 2 \int \frac{dt}{(t)^2 + (3)^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{t}{a}\right) + C$ with $a = 3$:

$$\begin{aligned} &= 2 \times \frac{1}{3} \tan^{-1}\left(\frac{t}{3}\right) + C \\ &= \frac{2}{3} \tan^{-1}\left(\frac{t}{3}\right) + C \end{aligned}$$

Now back-substitute $t = \tan \frac{x}{2}$:

$$= \frac{2}{3} \tan^{-1}\left(\frac{\tan \frac{x}{2}}{3}\right) + C$$

Final Answer

$$\int \frac{dx}{5 + 4 \cos x} = \frac{2}{3} \tan^{-1}\left(\frac{1}{3} \tan \frac{x}{2}\right) + C$$

Question 2 (i)

Evaluate $\int \frac{dx}{1 + 2 \cos x}$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\cos x = \frac{1 - t^2}{1 + t^2}$ and $dx = \frac{2 dt}{1 + t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1 + t^2) dx = 2 dt$, giving

$$dx = \frac{2 dt}{1 + t^2}$$

Also, using the standard identity $\cos x = \frac{1 - t^2}{1 + t^2}$, the integral becomes

$$\int \frac{\frac{2 dt}{1 + t^2}}{1 + 2 \left(\frac{1 - t^2}{1 + t^2} \right)} = \int \frac{2 dt}{(1 + t^2) \left[1 + 2 \left(\frac{1 - t^2}{1 + t^2} \right) \right]}$$

Simplifying the denominator:

$$\begin{aligned}
 &= \int \frac{2 \, dt}{(1+t^2) + 2(1-t^2)} \\
 &= \int \frac{2 \, dt}{1+t^2+2-2t^2} \\
 &= \int \frac{2 \, dt}{3-t^2} \\
 &= 2 \int \frac{dt}{(\sqrt{3})^2 - (t)^2}
 \end{aligned}$$

Using the standard result $\int \frac{dt}{a^2 - t^2} = \frac{1}{2a} \log \left| \frac{a+t}{a-t} \right| + C$ with $a = \sqrt{3}$:

$$\begin{aligned}
 &= 2 \times \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3}+t}{\sqrt{3}-t} \right| + C \\
 &= \frac{1}{\sqrt{3}} \log \left| \frac{\sqrt{3}+t}{\sqrt{3}-t} \right| + C
 \end{aligned}$$

Now back-substitute $t = \tan \frac{x}{2}$:

$$= \frac{1}{\sqrt{3}} \log \left| \frac{\sqrt{3} + \tan \frac{x}{2}}{\sqrt{3} - \tan \frac{x}{2}} \right| + C$$

Final Answer

$$\int \frac{dx}{1+2\cos x} = \frac{1}{\sqrt{3}} \log \left| \frac{\sqrt{3} + \tan \frac{x}{2}}{\sqrt{3} - \tan \frac{x}{2}} \right| + C$$

Question 2 (ii)

Evaluate $\int \frac{dx}{4\cos x - 1}$.

TYPE 12a Direct/Algebraic Substitution

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\cos x = \frac{1-t^2}{1+t^2}$ and $dx = \frac{2 \, dt}{1+t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1+t^2) dx = 2 \, dt$, giving

$$dx = \frac{2 \, dt}{1+t^2}$$

Also, using the standard identity $\cos x = \frac{1-t^2}{1+t^2}$, the integral becomes

$$\int \frac{\frac{2 dt}{1+t^2}}{4\left(\frac{1-t^2}{1+t^2}\right) - 1} = \int \frac{2 dt}{(1+t^2) \left[4\left(\frac{1-t^2}{1+t^2}\right) - 1\right]}$$

Simplifying the denominator:

$$\begin{aligned} &= \int \frac{2 dt}{4(1-t^2) - (1+t^2)} \\ &= \int \frac{2 dt}{4 - 4t^2 - 1 - t^2} \\ &= \int \frac{2 dt}{3 - 5t^2} \\ &= 2 \int \frac{dt}{(\sqrt{3})^2 - (\sqrt{5}t)^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{a^2 - (bt)^2} = \frac{1}{b} \cdot \frac{1}{2a} \log \left| \frac{a+bt}{a-bt} \right| + C$ with $a = \sqrt{3}$ and $b = \sqrt{5}$:

$$\begin{aligned} &= 2 \times \frac{1}{\sqrt{5}} \times \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sqrt{5}t}{\sqrt{3} - \sqrt{5}t} \right| + C \\ &= \frac{1}{\sqrt{15}} \log \left| \frac{\sqrt{3} + \sqrt{5}t}{\sqrt{3} - \sqrt{5}t} \right| + C \end{aligned}$$

Now back-substitute $t = \tan \frac{x}{2}$:

$$= \frac{1}{\sqrt{15}} \log \left| \frac{\sqrt{3} + \sqrt{5} \tan \frac{x}{2}}{\sqrt{3} - \sqrt{5} \tan \frac{x}{2}} \right| + C$$

Final Answer

$$\int \frac{dx}{4 \cos x - 1} = \frac{1}{\sqrt{15}} \log \left| \frac{\sqrt{3} + \sqrt{5} \tan \frac{x}{2}}{\sqrt{3} - \sqrt{5} \tan \frac{x}{2}} \right| + C$$

Question 3 (i)

Evaluate $\int \frac{dx}{4 + 5 \sin x}$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\sin x = \frac{2t}{1+t^2}$
and $dx = \frac{2 dt}{1+t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1 + t^2) dx = 2 dt$, giving

$$dx = \frac{2 dt}{1 + t^2}$$

Also, using the standard identity $\sin x = \frac{2t}{1 + t^2}$, the integral becomes

$$\int \frac{\frac{2 dt}{1 + t^2}}{4 + 5 \left(\frac{2t}{1 + t^2} \right)} = \int \frac{2 dt}{(1 + t^2) \left[4 + 5 \left(\frac{2t}{1 + t^2} \right) \right]}$$

Simplifying the denominator:

$$\begin{aligned} &= \int \frac{2 dt}{4(1 + t^2) + 10t} \\ &= \int \frac{2 dt}{4 + 4t^2 + 10t} \\ &= \int \frac{2 dt}{4t^2 + 10t + 4} \\ &= \int \frac{dt}{2t^2 + 5t + 2} \end{aligned}$$

Factorizing the denominator quadratic $2t^2 + 5t + 2$:

$$2t^2 + 4t + t + 2 = 2t(t + 2) + 1(t + 2) = (2t + 1)(t + 2)$$

Now, using partial fractions:

$$\frac{1}{(2t + 1)(t + 2)} = \frac{A}{2t + 1} + \frac{B}{t + 2}$$

Multiplying by $(2t + 1)(t + 2)$ gives:

$$1 = A(t + 2) + B(2t + 1)$$

Setting $t = -2$:

$$1 = A(-2 + 2) + B(2(-2) + 1) \implies 1 = -3B \implies B = -\frac{1}{3}$$

Setting $t = -\frac{1}{2}$:

$$1 = A\left(-\frac{1}{2} + 2\right) + B\left(2\left(-\frac{1}{2}\right) + 1\right) \implies 1 = A\left(\frac{3}{2}\right) \implies A = \frac{2}{3}$$

Thus, the integral becomes:

$$\begin{aligned}
 &= \int \left(\frac{\frac{2}{3}}{2t+1} - \frac{\frac{1}{3}}{t+2} \right) dt \\
 &= \frac{2}{3} \int \frac{dt}{2t+1} - \frac{1}{3} \int \frac{dt}{t+2} \\
 &= \frac{2}{3} \cdot \frac{1}{2} \log |2t+1| - \frac{1}{3} \log |t+2| + C \\
 &= \frac{1}{3} \log |2t+1| - \frac{1}{3} \log |t+2| + C \\
 &= \frac{1}{3} \log \left| \frac{2t+1}{t+2} \right| + C
 \end{aligned}$$

Now back-substitute $t = \tan \frac{x}{2}$:

$$= \frac{1}{3} \log \left| \frac{2 \tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right| + C$$

Final Answer

$$\int \frac{dx}{4 + 5 \sin x} = \frac{1}{3} \log \left| \frac{2 \tan \frac{x}{2} + 1}{\tan \frac{x}{2} + 2} \right| + C$$

Question 3 (ii)

Evaluate $\int \frac{dx}{1 - \sin x + \cos x}$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, and $dx = \frac{2dt}{1+t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1+t^2) dx = 2 dt$, giving

$$dx = \frac{2 dt}{1+t^2}$$

Substituting $\sin x = \frac{2t}{1+t^2}$ and $\cos x = \frac{1-t^2}{1+t^2}$:

$$\int \frac{\frac{2 dt}{1+t^2}}{1 - \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} = \int \frac{2 dt}{(1+t^2) \left[1 - \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} \right]}$$

Simplifying the denominator:

$$\begin{aligned}
 &= \int \frac{2 \, dt}{(1+t^2) - 2t + (1-t^2)} \\
 &= \int \frac{2 \, dt}{1+t^2 - 2t + 1 - t^2} \\
 &= \int \frac{2 \, dt}{2-2t} \\
 &= \int \frac{2 \, dt}{2(1-t)} \\
 &= \int \frac{dt}{1-t}
 \end{aligned}$$

Using the standard result $\int \frac{dt}{a-t} = -\log|a-t| + C$:

$$= -\log|1-t| + C$$

Now back-substitute $t = \tan \frac{x}{2}$:

$$= -\log \left| 1 - \tan \frac{x}{2} \right| + C$$

Final Answer

$$\int \frac{dx}{1 - \sin x + \cos x} = -\log \left| 1 - \tan \frac{x}{2} \right| + C$$

Question 4 (i)

Evaluate $\int \frac{dx}{1 - \tan x}$. (NCERT)

TYPE 12a Direct/Algebraic Substitution

Concept / Formula Used: $\tan x = \frac{\sin x}{\cos x}$

Solution:

Rewrite $\tan x$ in terms of sine and cosine:

$$\begin{aligned}
 \int \frac{dx}{1 - \tan x} &= \int \frac{dx}{1 - \frac{\sin x}{\cos x}} \\
 &= \int \frac{\cos x}{\cos x - \sin x} dx
 \end{aligned}$$

Multiply the numerator and denominator by 2:

$$= \frac{1}{2} \int \frac{2 \cos x}{\cos x - \sin x} dx$$

Express the numerator $2 \cos x$ as a sum of $(\cos x - \sin x)$ and its derivative $(\cos x + \sin x)$:

$$2 \cos x = (\cos x - \sin x) + (\cos x + \sin x)$$

Substituting this back into the integral:

$$\begin{aligned} &= \frac{1}{2} \int \frac{(\cos x - \sin x) + (\cos x + \sin x)}{\cos x - \sin x} dx \\ &= \frac{1}{2} \int \left(1 + \frac{\cos x + \sin x}{\cos x - \sin x} \right) dx \\ &= \frac{1}{2} \int 1 dx + \frac{1}{2} \int \frac{\cos x + \sin x}{\cos x - \sin x} dx \end{aligned}$$

For the second integral, let $u = \cos x - \sin x$.

Differentiating: $du = (-\sin x - \cos x) dx = -(\cos x + \sin x) dx$, so $(\cos x + \sin x) dx = -du$.

Thus, the expression becomes:

$$\begin{aligned} &= \frac{1}{2}x + \frac{1}{2} \int \frac{-du}{u} \\ &= \frac{1}{2}x - \frac{1}{2} \log |u| + C \end{aligned}$$

Now back-substitute $u = \cos x - \sin x$:

$$= \frac{1}{2}x - \frac{1}{2} \log |\cos x - \sin x| + C$$

Final Answer

$$\int \frac{dx}{1 - \tan x} = \frac{1}{2}x - \frac{1}{2} \log |\cos x - \sin x| + C$$

Question 4 (ii)

Evaluate $\int \frac{dx}{1 + \cot x}$. (NCERT)

TYPE 12a Direct/Algebraic Substitution

Concept / Formula Used: $\cot x = \frac{\cos x}{\sin x}$

Solution:

Rewrite $\cot x$ in terms of sine and cosine:

$$\begin{aligned} \int \frac{dx}{1 + \cot x} &= \int \frac{dx}{1 + \frac{\cos x}{\sin x}} \\ &= \int \frac{\sin x}{\sin x + \cos x} dx \end{aligned}$$

Multiply the numerator and denominator by 2:

$$= \frac{1}{2} \int \frac{2 \sin x}{\sin x + \cos x} dx$$

Express the numerator $2 \sin x$ as $(\sin x + \cos x) - (\cos x - \sin x)$:

$$2 \sin x = (\sin x + \cos x) - (\cos x - \sin x)$$

Substituting this back into the integral:

$$\begin{aligned} &= \frac{1}{2} \int \frac{(\sin x + \cos x) - (\cos x - \sin x)}{\sin x + \cos x} dx \\ &= \frac{1}{2} \int \left(1 - \frac{\cos x - \sin x}{\sin x + \cos x} \right) dx \\ &= \frac{1}{2} \int 1 dx - \frac{1}{2} \int \frac{\cos x - \sin x}{\sin x + \cos x} dx \end{aligned}$$

For the second integral, let $u = \sin x + \cos x$.

Differentiating: $du = (\cos x - \sin x) dx$.

Thus, the expression becomes:

$$\begin{aligned} &= \frac{1}{2}x - \frac{1}{2} \int \frac{du}{u} \\ &= \frac{1}{2}x - \frac{1}{2} \log |u| + C \end{aligned}$$

Now back-substitute $u = \sin x + \cos x$:

$$= \frac{1}{2}x - \frac{1}{2} \log |\sin x + \cos x| + C$$

Final Answer

$$\int \frac{dx}{1 + \cot x} = \frac{1}{2}x - \frac{1}{2} \log |\sin x + \cos x| + C$$

Question 5

If $\int \frac{5 \tan x}{\tan x - 2} dx = ax + b \log |\sin x - 2 \cos x| + C$, then find the values of a and b .

★ Likely Exam Question

TYPE 12e Substitution for Rational Trigonometric Integrals

Concept / Formula Used: $\tan x = \frac{\sin x}{\cos x}$

Solution:

Rewrite the integrand in terms of sine and cosine:

$$\begin{aligned}\int \frac{5 \tan x}{\tan x - 2} dx &= \int \frac{5 \left(\frac{\sin x}{\cos x} \right)}{\frac{\sin x}{\cos x} - 2} dx \\ &= \int \frac{5 \sin x}{\sin x - 2 \cos x} dx\end{aligned}$$

We need to express the numerator $5 \sin x$ in the form:

$$5 \sin x = A \frac{d}{dx}(\sin x - 2 \cos x) + B(\sin x - 2 \cos x)$$

Compute the derivative:

$$\frac{d}{dx}(\sin x - 2 \cos x) = \cos x + 2 \sin x$$

So we set:

$$5 \sin x = A(\cos x + 2 \sin x) + B(\sin x - 2 \cos x)$$

Equate the coefficients of $\sin x$ and $\cos x$ on both sides: For $\cos x$: $0 = A - 2B \implies A = 2B$

For $\sin x$: $5 = 2A + B$ Substitute $A = 2B$ into the second equation:

$$5 = 2(2B) + B \implies 5 = 4B + B \implies 5 = 5B \implies B = 1$$

Since $A = 2B$, we get $A = 2(1) = 2$.

Thus, the numerator $5 \sin x$ can be rewritten as:

$$5 \sin x = 2(\cos x + 2 \sin x) + 1(\sin x - 2 \cos x)$$

Substitute this back into the integral:

$$\begin{aligned}\int \frac{5 \sin x}{\sin x - 2 \cos x} dx &= \int \frac{2(\cos x + 2 \sin x) + 1(\sin x - 2 \cos x)}{\sin x - 2 \cos x} dx \\ &= \int \left(2 \cdot \frac{\cos x + 2 \sin x}{\sin x - 2 \cos x} + 1 \right) dx \\ &= 2 \int \frac{\cos x + 2 \sin x}{\sin x - 2 \cos x} dx + \int 1 dx\end{aligned}$$

For the first integral, let $u = \sin x - 2 \cos x$, so $du = (\cos x + 2 \sin x) dx$.

$$\begin{aligned}&= 2 \int \frac{du}{u} + x + C \\ &= 2 \log |u| + x + C \\ &= x + 2 \log |\sin x - 2 \cos x| + C\end{aligned}$$

Comparing this with the given expression $ax + b \log |\sin x - 2 \cos x| + C$:

$$ax + b \log |\sin x - 2 \cos x| + C = x + 2 \log |\sin x - 2 \cos x| + C$$

Equating coefficients of corresponding terms:

$$a = 1, \quad b = 2$$

Final Answer

$$a = 1, \quad b = 2$$

7.10 – Integration by Partial Fractions

Question 1 (i)

Evaluate $\int \frac{x-1}{(x-2)(x-3)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x-1}{(x-2)(x-3)} dx$

Concept / Formula Used: Partial fractions decomposition: $\frac{x-1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{x-1}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$$

Multiplying both sides by $(x-2)(x-3)$:

$$x-1 = A(x-3) + B(x-2)$$

Putting $x = 2$:

$$2-1 = A(2-3) + B(2-2)$$

$$1 = -A$$

$$A = -1$$

Putting $x = 3$:

$$3-1 = A(3-3) + B(3-2)$$

$$2 = B$$

$$B = 2$$

Thus, the integrand becomes:

$$\frac{x-1}{(x-2)(x-3)} = \frac{-1}{x-2} + \frac{2}{x-3}$$

Integrating both sides with respect to x :

$$\begin{aligned} \int \frac{x-1}{(x-2)(x-3)} dx &= \int \left(\frac{-1}{x-2} + \frac{2}{x-3} \right) dx \\ &= - \int \frac{1}{x-2} dx + 2 \int \frac{1}{x-3} dx \end{aligned}$$

Using the standard result $\int \frac{1}{x} dx = \ln |x| + c$:

$$\begin{aligned} &= -\ln |x - 2| + 2 \ln |x - 3| + C \\ &= 2 \ln |x - 3| - \ln |x - 2| + C \\ &= \ln |x - 3|^2 - \ln |x - 2| + C \\ &= \ln \left| \frac{(x - 3)^2}{x - 2} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{x - 1}{(x - 2)(x - 3)} dx = 2 \ln |x - 3| - \ln |x - 2| + C$$

Common Mistake: Mixing up the signs of A and B when evaluating the cover-up method or linear equations.

Question 1 (ii)

Evaluate $\int \frac{3x + 5}{x^2 + 3x - 18} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{3x + 5}{x^2 + 3x - 18} dx$

Concept / Formula Used: Factorization and Partial Fractions: $x^2 + 3x - 18 = (x + 6)(x - 3)$

Solution:

Factorize the denominator:

$$x^2 + 3x - 18 = (x + 6)(x - 3)$$

Decompose the integrand into partial fractions:

$$\frac{3x + 5}{(x + 6)(x - 3)} = \frac{A}{x + 6} + \frac{B}{x - 3}$$

Multiplying both sides by $(x + 6)(x - 3)$:

$$3x + 5 = A(x - 3) + B(x + 6)$$

Putting $x = 3$:

$$3(3) + 5 = A(3 - 3) + B(3 + 6)$$

$$9 + 5 = 9B$$

$$14 = 9B$$

$$B = \frac{14}{9}$$

Putting $x = -6$:

$$\begin{aligned}3(-6) + 5 &= A(-6 - 3) + B(-6 + 6) \\-18 + 5 &= -9A \\-13 &= -9A \\A &= \frac{13}{9}\end{aligned}$$

Thus, the integral is:

$$\begin{aligned}\int \frac{3x + 5}{x^2 + 3x - 18} dx &= \int \left(\frac{13/9}{x + 6} + \frac{14/9}{x - 3} \right) dx \\&= \frac{13}{9} \int \frac{1}{x + 6} dx + \frac{14}{9} \int \frac{1}{x - 3} dx \\&= \frac{13}{9} \ln |x + 6| + \frac{14}{9} \ln |x - 3| + C\end{aligned}$$

Final Answer

$$\int \frac{3x + 5}{x^2 + 3x - 18} dx = \frac{13}{9} \ln |x + 6| + \frac{14}{9} \ln |x - 3| + C$$

Common Mistake: Forgetting to factorize the quadratic denominator completely before applying partial fractions.

Question 2 (i)

Evaluate $\int \frac{2x+7}{x^2-x-2} dx$.

TYPE 14 Partial Fractions

★ **Likely Exam Question**

Given: $\int \frac{2x+7}{x^2-x-2} dx$

Concept / Formula Used: Factorization and Partial Fractions: $x^2 - x - 2 = (x-2)(x+1)$

Solution:

Factorize the denominator:

$$x^2 - x - 2 = (x-2)(x+1)$$

Decompose into partial fractions:

$$\frac{2x+7}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$$

Multiplying both sides by $(x-2)(x+1)$:

$$2x+7 = A(x+1) + B(x-2)$$

Putting $x = 2$:

$$2(2)+7 = A(2+1) + B(2-2)$$

$$4+7 = 3A$$

$$11 = 3A$$

$$A = \frac{11}{3}$$

Putting $x = -1$:

$$2(-1)+7 = A(-1+1) + B(-1-2)$$

$$-2+7 = -3B$$

$$5 = -3B$$

$$B = -\frac{5}{3}$$

Thus, the integral is:

$$\int \frac{2x+7}{x^2-x-2} dx = \int \left(\frac{11/3}{x-2} - \frac{5/3}{x+1} \right) dx$$

$$\begin{aligned}
 &= \frac{11}{3} \int \frac{1}{x-2} dx - \frac{5}{3} \int \frac{1}{x+1} dx \\
 &= \frac{11}{3} \ln |x-2| - \frac{5}{3} \ln |x+1| + C
 \end{aligned}$$

Final Answer

$$\int \frac{2x+7}{x^2-x-2} dx = \frac{11}{3} \ln |x-2| - \frac{5}{3} \ln |x+1| + C$$

Common Mistake: Incorrectly assigning signs to constants A and B when substituting roots.

Question 2 (ii)

Evaluate $\int \frac{x^2+1}{x^2-5x+6} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2+1}{x^2-5x+6} dx$

Concept / Formula Used: Long Division for Improper Rational Functions: $\frac{P(x)}{Q(x)} = \text{Quotient} + \frac{\text{Remainder}}{Q(x)}$

Solution:

Since the degree of the numerator equals the degree of the denominator, perform polynomial long division:

$$\frac{x^2+1}{x^2-5x+6} = 1 + \frac{5x-5}{x^2-5x+6}$$

Factorize the denominator of the remainder term:

$$x^2 - 5x + 6 = (x-2)(x-3)$$

Decompose the proper rational part into partial fractions:

$$\frac{5x-5}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3}$$

Multiplying both sides by $(x-2)(x-3)$:

$$5x-5 = A(x-3) + B(x-2)$$

Putting $x = 2$:

$$5(2) - 5 = A(2-3) + B(2-2)$$

$$5 = -A$$

$$A = -5$$

Putting $x = 3$:

$$5(3) - 5 = A(3 - 3) + B(3 - 2)$$

$$10 = B$$

$$B = 10$$

Thus, the integral becomes:

$$\int \left(1 + \frac{-5}{x-2} + \frac{10}{x-3} \right) dx$$

Integrating term by term:

$$\begin{aligned} &= \int 1 \, dx - 5 \int \frac{1}{x-2} \, dx + 10 \int \frac{1}{x-3} \, dx \\ &= x - 5 \ln |x-2| + 10 \ln |x-3| + C \end{aligned}$$

Final Answer

$$\int \frac{x^2 + 1}{x^2 - 5x + 6} \, dx = x - 5 \ln |x-2| + 10 \ln |x-3| + C$$

Common Mistake: Forgetting to perform long division first when the degrees of numerator and denominator are equal.

Question 3 (i)

Evaluate $\int \frac{x^3 + x + 1}{x^2 - 1} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^3 + x + 1}{x^2 - 1} dx$

Concept / Formula Used: Polynomial Long Division: $\frac{x^3+x+1}{x^2-1} = x + \frac{2x+1}{x^2-1}$

Solution:

Since the degree of the numerator (3) is greater than the degree of the denominator (2), perform polynomial long division:

$$\frac{x^3 + x + 1}{x^2 - 1} = x + \frac{2x + 1}{x^2 - 1}$$

Now, resolve the proper rational part into partial fractions:

$$\frac{2x + 1}{x^2 - 1} = \frac{2x + 1}{(x - 1)(x + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1}$$

Multiplying both sides by $(x - 1)(x + 1)$:

$$2x + 1 = A(x + 1) + B(x - 1)$$

Putting $x = 1$:

$$2(1) + 1 = A(1 + 1) + B(1 - 1)$$

$$3 = 2A$$

$$A = \frac{3}{2}$$

Putting $x = -1$:

$$2(-1) + 1 = A(-1 + 1) + B(-1 - 1)$$

$$-1 = -2B$$

$$B = \frac{1}{2}$$

Thus, the integrand is:

$$x + \frac{3/2}{x - 1} + \frac{1/2}{x + 1}$$

Integrating term by term:

$$\begin{aligned} \int \left(x + \frac{3/2}{x - 1} + \frac{1/2}{x + 1} \right) dx &= \int x dx + \frac{3}{2} \int \frac{1}{x - 1} dx + \frac{1}{2} \int \frac{1}{x + 1} dx \\ &= \frac{x^2}{2} + \frac{3}{2} \ln |x - 1| + \frac{1}{2} \ln |x + 1| + C \end{aligned}$$

Final Answer

$$\int \frac{x^3 + x + 1}{x^2 - 1} dx = \frac{x^2}{2} + \frac{3}{2} \ln |x - 1| + \frac{1}{2} \ln |x + 1| + C$$

Common Mistake: Directly applying partial fractions without performing long division first when degree of numerator exceeds denominator.

Question 3 (ii)

Evaluate $\int \frac{x^3 + 1}{x^3 - x} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^3 + 1}{x^3 - x} dx$

Concept / Formula Used: Polynomial Long Division: $\frac{x^3+1}{x^3-x} = 1 + \frac{x+1}{x^3-x}$

Solution:

Since the degree of the numerator equals the degree of the denominator, perform polynomial long division:

$$\frac{x^3 + 1}{x^3 - x} = 1 + \frac{x + 1}{x^3 - x}$$

Simplify the remainder term:

$$\frac{x + 1}{x(x^2 - 1)} = \frac{x + 1}{x(x - 1)(x + 1)} = \frac{1}{x(x - 1)}$$

Decompose $\frac{1}{x(x-1)}$ into partial fractions:

$$\frac{1}{x(x - 1)} = \frac{A}{x} + \frac{B}{x - 1}$$

Multiplying both sides by $x(x - 1)$:

$$1 = A(x - 1) + Bx$$

Putting $x = 0$:

$$1 = A(0 - 1) + B(0)$$

$$1 = -A$$

$$A = -1$$

Putting $x = 1$:

$$1 = A(1 - 1) + B(1)$$

$$1 = B$$

$$B = 1$$

Thus, the integral is:

$$\int \left(1 - \frac{1}{x} + \frac{1}{x-1} \right) dx$$

Integrating term by term:

$$\begin{aligned} &= \int 1 \, dx - \int \frac{1}{x} \, dx + \int \frac{1}{x-1} \, dx \\ &= x - \ln |x| + \ln |x-1| + C \\ &= x + \ln \left| \frac{x-1}{x} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{x^3 + 1}{x^3 - x} \, dx = x + \ln \left| \frac{x-1}{x} \right| + C$$

Common Mistake: Failing to cancel the common factor $(x+1)$ in numerator and denominator of the remainder term.

Question 4 (i)

Evaluate $\int \frac{x}{(x-1)(x-2)(x-3)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x}{(x-1)(x-2)(x-3)} dx$

Concept / Formula Used: Partial fractions with three linear factors:
 $\frac{x}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{x}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

Multiplying both sides by $(x-1)(x-2)(x-3)$:

$$x = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

Putting $x = 1$:

$$1 = A(1-2)(1-3) + 0 + 0$$

$$1 = A(-1)(-2)$$

$$1 = 2A \implies A = \frac{1}{2}$$

Putting $x = 2$:

$$2 = 0 + B(2-1)(2-3) + 0$$

$$2 = B(1)(-1)$$

$$2 = -B \implies B = -2$$

Putting $x = 3$:

$$3 = 0 + 0 + C(3-1)(3-2)$$

$$3 = C(2)(1)$$

$$3 = 2C \implies C = \frac{3}{2}$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1/2}{x-1} - \frac{2}{x-2} + \frac{3/2}{x-3} \right) dx \\ &= \frac{1}{2} \int \frac{1}{x-1} dx - 2 \int \frac{1}{x-2} dx + \frac{3}{2} \int \frac{1}{x-3} dx \\ &= \frac{1}{2} \ln |x-1| - 2 \ln |x-2| + \frac{3}{2} \ln |x-3| + C \end{aligned}$$

Final Answer

$$\int \frac{x}{(x-1)(x-2)(x-3)} dx = \frac{1}{2} \ln|x-1| - 2 \ln|x-2| + \frac{3}{2} \ln|x-3| + C$$

Common Mistake: Arithmetic sign errors when evaluating constants A, B, C by substituting roots.

Question 4 (ii)

Evaluate $\int \frac{3x-1}{(x-1)(x-2)(x-3)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{3x-1}{(x-1)(x-2)(x-3)} dx$

Concept / Formula Used: Partial fractions decomposition for three distinct linear factors

Solution:

Decompose the integrand into partial fractions:

$$\frac{3x-1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

Multiplying both sides by $(x-1)(x-2)(x-3)$:

$$3x-1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

Putting $x = 1$:

$$\begin{aligned} 3(1) - 1 &= A(1-2)(1-3) \\ 2 &= 2A \implies A = 1 \end{aligned}$$

Putting $x = 2$:

$$\begin{aligned} 3(2) - 1 &= B(2-1)(2-3) \\ 5 &= -B \implies B = -5 \end{aligned}$$

Putting $x = 3$:

$$\begin{aligned} 3(3) - 1 &= C(3-1)(3-2) \\ 8 &= 2C \implies C = 4 \end{aligned}$$

Thus, the integral is:

$$\int \left(\frac{1}{x-1} - \frac{5}{x-2} + \frac{4}{x-3} \right) dx$$

$$\begin{aligned} &= \int \frac{1}{x-1} dx - 5 \int \frac{1}{x-2} dx + 4 \int \frac{1}{x-3} dx \\ &= \ln |x-1| - 5 \ln |x-2| + 4 \ln |x-3| + C \end{aligned}$$

Final Answer

$$\int \frac{3x-1}{(x-1)(x-2)(x-3)} dx = \ln |x-1| - 5 \ln |x-2| + 4 \ln |x-3| + C$$

Common Mistake: Calculating numerator values incorrectly when substituting $x = 1, 2, 3$.

Question 5 (i)

Evaluate $\int \frac{5x}{(x+1)(x^2-4)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{5x}{(x+1)(x^2-4)} dx$

Concept / Formula Used: Factorization and Partial Fractions: $x^2 - 4 = (x - 2)(x + 2)$

Solution:

Factorize the denominator:

$$(x+1)(x^2-4) = (x+1)(x-2)(x+2)$$

Decompose the integrand into partial fractions:

$$\frac{5x}{(x+1)(x-2)(x+2)} = \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{x+2}$$

Multiplying both sides by $(x+1)(x-2)(x+2)$:

$$5x = A(x-2)(x+2) + B(x+1)(x+2) + C(x+1)(x-2)$$

Putting $x = -1$:

$$\begin{aligned} 5(-1) &= A(-1-2)(-1+2) \\ -5 &= A(-3)(1) \\ -5 &= -3A \implies A = \frac{5}{3} \end{aligned}$$

Putting $x = 2$:

$$\begin{aligned} 5(2) &= B(2+1)(2+2) \\ 10 &= B(3)(4) \\ 10 &= 12B \implies B = \frac{5}{6} \end{aligned}$$

Putting $x = -2$:

$$\begin{aligned} 5(-2) &= C(-2+1)(-2-2) \\ -10 &= C(-1)(-4) \\ -10 &= 4C \implies C = -\frac{5}{2} \end{aligned}$$

Thus, the integral is:

$$\int \left(\frac{5/3}{x+1} + \frac{5/6}{x-2} - \frac{5/2}{x+2} \right) dx$$

$$\begin{aligned}
 &= \frac{5}{3} \int \frac{1}{x+1} dx + \frac{5}{6} \int \frac{1}{x-2} dx - \frac{5}{2} \int \frac{1}{x+2} dx \\
 &= \frac{5}{3} \ln|x+1| + \frac{5}{6} \ln|x-2| - \frac{5}{2} \ln|x+2| + C
 \end{aligned}$$

Final Answer

$$\int \frac{5x}{(x+1)(x^2-4)} dx = \frac{5}{3} \ln|x+1| + \frac{5}{6} \ln|x-2| - \frac{5}{2} \ln|x+2| + C$$

Common Mistake: Failing to factorize $(x^2 - 4)$ completely into $(x - 2)(x + 2)$.

Question 5 (ii)

Evaluate $\int \frac{x}{(x-1)^2(x+2)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x}{(x-1)^2(x+2)} dx$

Concept / Formula Used: Partial fractions for repeated linear factors: $\frac{x}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{x}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

Multiplying both sides by $(x-1)^2(x+2)$:

$$x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

Putting $x = 1$:

$$\begin{aligned}
 1 &= 0 + B(1+2) + 0 \\
 1 &= 3B \implies B = \frac{1}{3}
 \end{aligned}$$

Putting $x = -2$:

$$\begin{aligned}
 -2 &= 0 + 0 + C(-2-1)^2 \\
 -2 &= C(9) \implies C = -\frac{2}{9}
 \end{aligned}$$

To find A , equate coefficients of x^2 on both sides:

$$\begin{aligned}
 0 &= A + C \\
 A &= -C = -\left(-\frac{2}{9}\right) = \frac{2}{9}
 \end{aligned}$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{2/9}{x-1} + \frac{1/3}{(x-1)^2} - \frac{2/9}{x+2} \right) dx \\ &= \frac{2}{9} \int \frac{1}{x-1} dx + \frac{1}{3} \int (x-1)^{-2} dx - \frac{2}{9} \int \frac{1}{x+2} dx \\ &= \frac{2}{9} \ln|x-1| + \frac{1}{3} \left(\frac{(x-1)^{-1}}{-1} \right) - \frac{2}{9} \ln|x+2| + C \\ &= \frac{2}{9} \ln|x-1| - \frac{1}{3(x-1)} - \frac{2}{9} \ln|x+2| + C \end{aligned}$$

Final Answer

$$\int \frac{x}{(x-1)^2(x+2)} dx = \frac{2}{9} \ln|x-1| - \frac{1}{3(x-1)} - \frac{2}{9} \ln|x+2| + C$$

Common Mistake: Omitting the middle term with the repeated factor power $\frac{B}{(x-1)^2}$ in the partial fraction expansion.

Question 6 (i)

Evaluate $\int \frac{3x-1}{(x-2)^2} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{3x-1}{(x-2)^2} dx$

Concept / Formula Used: Partial fractions for repeated linear factor: $\frac{3x-1}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{3x-1}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$$

Multiplying both sides by $(x-2)^2$:

$$3x-1 = A(x-2) + B$$

Rearranging terms by powers of x :

$$3x-1 = Ax + (B-2A)$$

Equating coefficients of x and constant terms:

$$A = 3$$

$$B - 2A = -1$$

Substitute $A = 3$ into the second equation:

$$B - 2(3) = -1$$

$$B - 6 = -1$$

$$B = 5$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{3}{x-2} + \frac{5}{(x-2)^2} \right) dx \\ &= 3 \int \frac{1}{x-2} dx + 5 \int (x-2)^{-2} dx \\ &= 3 \ln |x-2| + 5 \left(\frac{(x-2)^{-1}}{-1} \right) + C \\ &= 3 \ln |x-2| - \frac{5}{x-2} + C \end{aligned}$$

Final Answer

$$\int \frac{3x-1}{(x-2)^2} dx = 3 \ln |x-2| - \frac{5}{x-2} + C$$

Common Mistake: Treating the repeated factor incorrectly as a single linear term without the second constant numerator.

Question 6 (ii)

Evaluate $\int \frac{3x+5}{x^3-x^2-x+1} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{3x+5}{x^3-x^2-x+1} dx$

Concept / Formula Used: Cubic Factorization and Partial Fractions: $x^3 - x^2 - x + 1 = (x-1)^2(x+1)$

Solution:

Factorize the cubic denominator by grouping:

$$x^3 - x^2 - x + 1 = x^2(x-1) - 1(x-1) = (x-1)(x^2-1) = (x-1)^2(x+1)$$

Decompose the integrand into partial fractions:

$$\frac{3x+5}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

Multiplying both sides by $(x-1)^2(x+1)$:

$$3x+5 = A(x-1)(x+1) + B(x+1) + C(x-1)^2$$

Putting $x = 1$:

$$\begin{aligned} 3(1) + 5 &= 0 + B(1+1) + 0 \\ 8 &= 2B \implies B = 4 \end{aligned}$$

Putting $x = -1$:

$$\begin{aligned} 3(-1) + 5 &= 0 + 0 + C(-1-1)^2 \\ 2 &= C(4) \implies C = \frac{1}{2} \end{aligned}$$

Equating coefficients of x^2 on both sides:

$$\begin{aligned} 0 &= A + C \\ A &= -C = -\frac{1}{2} \end{aligned}$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{-1/2}{x-1} + \frac{4}{(x-1)^2} + \frac{1/2}{x+1} \right) dx \\ &= -\frac{1}{2} \int \frac{1}{x-1} dx + 4 \int (x-1)^{-2} dx + \frac{1}{2} \int \frac{1}{x+1} dx \\ &= -\frac{1}{2} \ln|x-1| - \frac{4}{x-1} + \frac{1}{2} \ln|x+1| + C \\ &= \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| - \frac{4}{x-1} + C \end{aligned}$$

Final Answer

$$\int \frac{3x+5}{x^3-x^2-x+1} dx = \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| - \frac{4}{x-1} + C$$

Common Mistake: Incorrectly factorizing the cubic denominator as distinct linear factors instead of noting the repeated root $(x-1)^2$.

Question 7 (i)

Evaluate $\int \frac{2}{(1-x)(1+x^2)} dx$.

TYPE 14 Partial Fractions

★ **Likely Exam Question**

Given: $\int \frac{2}{(1-x)(1+x^2)} dx$

Concept / Formula Used: Partial fractions with irreducible quadratic factor:

$$\frac{2}{(1-x)(1+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x^2}$$

Solution:

Decompose the integrand into partial fractions:

$$\frac{2}{(1-x)(1+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x^2}$$

Multiplying both sides by $(1-x)(1+x^2)$:

$$2 = A(1+x^2) + (Bx+C)(1-x)$$

Putting $x = 1$:

$$2 = A(1+1^2) + (B(1)+C)(1-1)$$

$$2 = 2A$$

$$A = 1$$

Expanding the identity:

$$2 = A + Ax^2 + Bx - Bx^2 + C - Cx$$

Grouping coefficients of powers of x :

$$2 = (A+C) + (B-C)x + (A-B)x^2$$

Equating coefficients of x^2 , x , and constant terms on both sides:

$$A - B = 0 \implies B = A = 1$$

$$B - C = 0 \implies C = B = 1$$

$$A + C = 2 \implies 1 + 1 = 2 \text{ (consistent)}$$

Thus, the values are $A = 1, B = 1, C = 1$. The integral becomes:

$$\int \left(\frac{1}{1-x} + \frac{x+1}{1+x^2} \right) dx$$

Splitting the terms:

$$\begin{aligned} &= \int \frac{1}{1-x} dx + \int \frac{x}{1+x^2} dx + \int \frac{1}{1+x^2} dx \\ &= -\ln|1-x| + \frac{1}{2} \ln(1+x^2) + \tan^{-1}(x) + C \end{aligned}$$

Final Answer

$$\int \frac{2}{(1-x)(1+x^2)} dx = \frac{1}{2} \ln(1+x^2) - \ln|1-x| + \tan^{-1}(x) + C$$

Common Mistake: Forgetting to include a linear term ($Bx + C$) in the numerator over the irreducible quadratic factor ($1 + x^2$).

Question 7 (ii)

Evaluate $\int \frac{4}{(x-2)(x^2+4)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{4}{(x-2)(x^2+4)} dx$

Concept / Formula Used: Partial fractions: $\frac{4}{(x-2)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+4}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{4}{(x-2)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+4}$$

Multiplying both sides by $(x-2)(x^2+4)$:

$$4 = A(x^2+4) + (Bx+C)(x-2)$$

Putting $x = 2$:

$$\begin{aligned} 4 &= A(2^2+4) + 0 \\ 4 &= 8A \implies A = \frac{1}{2} \end{aligned}$$

Expanding the remaining expression:

$$4 = Ax^2 + 4A + Bx^2 - 2Bx + Cx - 2C$$

Grouping coefficients of powers of x :

$$4 = (4A - 2C) + (-2B + C)x + (A + B)x^2$$

Equating coefficients of x^2 , x , and constant terms:

$$\begin{aligned}A + B &= 0 \implies B = -A = -\frac{1}{2} \\4A - 2C &= 4 \implies 4\left(\frac{1}{2}\right) - 2C = 4 \implies 2 - 2C = 4 \implies C = -1\end{aligned}$$

Thus, $A = \frac{1}{2}$, $B = -\frac{1}{2}$, $C = -1$. The integral is:

$$\begin{aligned}&\int \left(\frac{1/2}{x-2} + \frac{-\frac{1}{2}x-1}{x^2+4} \right) dx \\&= \frac{1}{2} \int \frac{1}{x-2} dx - \frac{1}{2} \int \frac{x}{x^2+4} dx - \int \frac{1}{x^2+2^2} dx \\&= \frac{1}{2} \ln|x-2| - \frac{1}{4} \ln(x^2+4) - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C\end{aligned}$$

Final Answer

$$\int \frac{4}{(x-2)(x^2+4)} dx = \frac{1}{2} \ln|x-2| - \frac{1}{4} \ln(x^2+4) - \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C$$

Common Mistake: Failing to correctly compute coefficients of the linear numerator over the quadratic factor.

Question 8 (i)

Evaluate $\int \frac{x^2 + x}{x^3 - x^2 + x - 1} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2 + x}{x^3 - x^2 + x - 1} dx$

Concept / Formula Used: Factorization by grouping and partial fractions: $x^3 - x^2 + x - 1 = (x - 1)(x^2 + 1)$

Solution:

Factorize the cubic denominator by grouping:

$$x^3 - x^2 + x - 1 = x^2(x - 1) + 1(x - 1) = (x - 1)(x^2 + 1)$$

Decompose the integrand into partial fractions:

$$\frac{x^2 + x}{(x - 1)(x^2 + 1)} = \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 1}$$

Multiplying both sides by $(x - 1)(x^2 + 1)$:

$$x^2 + x = A(x^2 + 1) + (Bx + C)(x - 1)$$

Putting $x = 1$:

$$\begin{aligned} 1^2 + 1 &= A(1^2 + 1) + 0 \\ 2 &= 2A \implies A = 1 \end{aligned}$$

Expanding the right side:

$$x^2 + x = Ax^2 + A + Bx^2 - Bx + Cx - C$$

Grouping coefficients by powers of x :

$$x^2 + x = (A - C) + (-B + C)x + (A + B)x^2$$

Equating coefficients of x^2 and x :

$$\begin{aligned} A + B &= 1 \implies 1 + B = 1 \implies B = 0 \\ -B + C &= 1 \implies 0 + C = 1 \implies C = 1 \end{aligned}$$

Thus, $A = 1, B = 0, C = 1$. The integral becomes:

$$\int \left(\frac{1}{x - 1} + \frac{1}{x^2 + 1} \right) dx$$

Integrating term by term:

$$\begin{aligned} &= \int \frac{1}{x-1} dx + \int \frac{1}{x^2+1} dx \\ &= \ln|x-1| + \tan^{-1}(x) + C \end{aligned}$$

Final Answer

$$\int \frac{x^2+x}{x^3-x^2+x-1} dx = \ln|x-1| + \tan^{-1}(x) + C$$

Common Mistake: Incorrectly grouping terms during factorization of the cubic denominator.

Question 8 (ii)

Evaluate $\int \frac{x^3}{(x-1)(x^2+1)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^3}{(x-1)(x^2+1)} dx$

Concept / Formula Used: Polynomial Long Division followed by Partial Fractions

Solution:

Expand the denominator: $(x-1)(x^2+1) = x^3 - x^2 + x - 1$. Perform polynomial long division:

$$\frac{x^3}{x^3 - x^2 + x - 1} = 1 + \frac{x^2 - x + 1}{x^3 - x^2 + x - 1}$$

Rewrite the remainder using the factors:

$$1 + \frac{x^2 - x + 1}{(x-1)(x^2+1)}$$

Decompose the proper rational part into partial fractions:

$$\frac{x^2 - x + 1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

Multiplying both sides by $(x-1)(x^2+1)$:

$$x^2 - x + 1 = A(x^2+1) + (Bx+C)(x-1)$$

Putting $x = 1$:

$$\begin{aligned} 1^2 - 1 + 1 &= A(1^2 + 1) \\ 1 &= 2A \implies A = \frac{1}{2} \end{aligned}$$

Equating coefficients of x^2 :

$$A + B = 1 \implies \frac{1}{2} + B = 1 \implies B = \frac{1}{2}$$

Equating constant terms:

$$A - C = 1 \implies \frac{1}{2} - C = 1 \implies C = -\frac{1}{2}$$

Thus, the integral is:

$$\begin{aligned} & \int \left(1 + \frac{1/2}{x-1} + \frac{\frac{1}{2}x - \frac{1}{2}}{x^2+1} \right) dx \\ &= \int 1 \, dx + \frac{1}{2} \int \frac{1}{x-1} \, dx + \frac{1}{2} \int \frac{x}{x^2+1} \, dx - \frac{1}{2} \int \frac{1}{x^2+1} \, dx \\ &= x + \frac{1}{2} \ln|x-1| + \frac{1}{4} \ln(x^2+1) - \frac{1}{2} \tan^{-1}(x) + C \end{aligned}$$

Final Answer

$$\int \frac{x^3}{(x-1)(x^2+1)} \, dx = x + \frac{1}{2} \ln|x-1| + \frac{1}{4} \ln(x^2+1) - \frac{1}{2} \tan^{-1}(x) + C$$

Common Mistake: Omitting polynomial long division when the degree of the numerator equals the degree of the denominator.

Question 9 (i)

Evaluate $\int \frac{dx}{1-x^3}$.

TYPE 14 Partial Fractions

Given: $\int \frac{dx}{1-x^3}$

Concept / Formula Used: Difference of cubes and partial fractions: $1-x^3 = (1-x)(1+x+x^2)$

Solution:

Factorize the denominator:

$$1-x^3 = (1-x)(1+x+x^2)$$

Decompose the integrand into partial fractions:

$$\frac{1}{(1-x)(1+x+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x+x^2}$$

Multiplying both sides by $(1-x)(1+x+x^2)$:

$$1 = A(1+x+x^2) + (Bx+C)(1-x)$$

Putting $x = 1$:

$$\begin{aligned} 1 &= A(1+1+1^2) \\ 1 &= 3A \implies A = \frac{1}{3} \end{aligned}$$

Expanding the right side:

$$1 = A + Ax + Ax^2 + Bx - Bx^2 + C - Cx$$

Grouping coefficients by powers of x :

$$1 = (A+C) + (A+B-C)x + (A-B)x^2$$

Equating coefficients of x^2 , x , and constant terms:

$$\begin{aligned} A-B &= 0 \implies B = A = \frac{1}{3} \\ A+C &= 1 \implies \frac{1}{3} + C = 1 \implies C = \frac{2}{3} \end{aligned}$$

Thus, $A = \frac{1}{3}$, $B = \frac{1}{3}$, $C = \frac{2}{3}$. The integral is:

$$\int \left(\frac{1/3}{1-x} + \frac{\frac{1}{3}x + \frac{2}{3}}{1+x+x^2} \right) dx$$

For the quadratic denominator, express the numerator in terms of the derivative of the denominator $(2x + 1)$:

$$\frac{1}{3}x + \frac{2}{3} = \frac{1}{6}(2x + 1) + \frac{1}{2}$$

Thus, splitting the integral:

$$\begin{aligned} &= \frac{1}{3} \int \frac{1}{1-x} dx + \frac{1}{6} \int \frac{2x+1}{x^2+x+1} dx + \frac{1}{2} \int \frac{dx}{\left(x+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= -\frac{1}{3} \ln|1-x| + \frac{1}{6} \ln(x^2+x+1) + \frac{1}{2} \times \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x+\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C \\ &= \frac{1}{6} \ln(x^2+x+1) - \frac{1}{3} \ln|1-x| + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{1-x^3} = \frac{1}{6} \ln(x^2+x+1) - \frac{1}{3} \ln|1-x| + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$$

Common Mistake: Failing to adjust the quadratic numerator to match the derivative of the denominator when integrating.

Question 9 (ii)

Evaluate $\int \frac{dx}{(x+1)^2(x^2+1)}$.

TYPE 14 Partial Fractions

Given: $\int \frac{dx}{(x+1)^2(x^2+1)}$

Concept / Formula Used: Partial fractions: $\frac{1}{(x+1)^2(x^2+1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{Cx+D}{x^2+1}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{1}{(x+1)^2(x^2+1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{Cx+D}{x^2+1}$$

Multiplying both sides by $(x+1)^2(x^2+1)$:

$$1 = A(x+1)(x^2+1) + B(x^2+1) + (Cx+D)(x+1)^2$$

Putting $x = -1$:

$$1 = 0 + B((-1)^2 + 1) + 0$$

$$1 = 2B \implies B = \frac{1}{2}$$

Expanding and collecting terms:

$$1 = (A + C)x^3 + (A + B + 2C + D)x^2 + (A - 2C + 2D)x + (A + B + D)$$

Equating coefficients of x^3 , x^2 , x , and constant terms:

$$\begin{aligned} A + C &= 0 \implies C = -A \\ A + B + 2C + D &= 0 \\ A - 2C + 2D &= 0 \\ A + B + D &= 1 \end{aligned}$$

Substituting $B = \frac{1}{2}$ into $A + B + D = 1$:

$$A + \frac{1}{2} + D = 1 \implies A + D = \frac{1}{2} \implies D = \frac{1}{2} - A$$

Using $C = -A$ and $D = \frac{1}{2} - A$ in $A + B + 2C + D = 0$:

$$A + \frac{1}{2} + 2(-A) + \left(\frac{1}{2} - A\right) = 0 \implies 1 - 2A = 0 \implies A = \frac{1}{2}$$

Consequently:

$$C = -\frac{1}{2}, \quad D = \frac{1}{2} - \frac{1}{2} = 0$$

Thus, $A = \frac{1}{2}$, $B = \frac{1}{2}$, $C = -\frac{1}{2}$, $D = 0$. The integral becomes:

$$\begin{aligned} &\int \left(\frac{1/2}{x+1} + \frac{1/2}{(x+1)^2} - \frac{1}{2} \frac{x}{x^2+1} \right) dx \\ &= \frac{1}{2} \int \frac{1}{x+1} dx + \frac{1}{2} \int (x+1)^{-2} dx - \frac{1}{2} \int \frac{x}{x^2+1} dx \\ &= \frac{1}{2} \ln|x+1| - \frac{1}{2(x+1)} - \frac{1}{4} \ln(x^2+1) + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{(x+1)^2(x^2+1)} = \frac{1}{2} \ln|x+1| - \frac{1}{4} \ln(x^2+1) - \frac{1}{2(x+1)} + C$$

Common Mistake: Incorrectly expanding the cubic and quadratic combinations when determining coefficients.

Question 10 (i)

Evaluate $\int \frac{2x}{(x^2 + 1)(x^2 + 3)} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{2x}{(x^2 + 1)(x^2 + 3)} dx$

Concept / Formula Used: Substitution Method: Let $x^2 = t$, so $2x dx = dt$

Solution:

Let $x^2 = t$.

Differentiating: $2x dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{(t+1)(t+3)}$$

Decompose $\frac{1}{(t+1)(t+3)}$ into partial fractions:

$$\frac{1}{(t+1)(t+3)} = \frac{A}{t+1} + \frac{B}{t+3}$$

Multiplying both sides by $(t+1)(t+3)$:

$$1 = A(t+3) + B(t+1)$$

Putting $t = -1$:

$$1 = A(-1+3) \implies 2A = 1 \implies A = \frac{1}{2}$$

Putting $t = -3$:

$$1 = B(-3+1) \implies -2B = 1 \implies B = -\frac{1}{2}$$

Thus, the integral becomes:

$$\begin{aligned} & \int \left(\frac{1/2}{t+1} - \frac{1/2}{t+3} \right) dt \\ &= \frac{1}{2} \int \frac{1}{t+1} dt - \frac{1}{2} \int \frac{1}{t+3} dt \\ &= \frac{1}{2} \ln |t+1| - \frac{1}{2} \ln |t+3| + C \\ &= \frac{1}{2} \ln \left| \frac{t+1}{t+3} \right| + C \end{aligned}$$

Now back-substitute $t = x^2$:

$$= \frac{1}{2} \ln \left| \frac{x^2+1}{x^2+3} \right| + C$$

Final Answer

$$\int \frac{2x}{(x^2+1)(x^2+3)} dx = \frac{1}{2} \ln \left(\frac{x^2+1}{x^2+3} \right) + C$$

Common Mistake: Applying partial fractions directly without substituting $x^2 = t$, leading to unnecessary complexity.

Question 10 (ii)

Evaluate $\int \frac{x^2}{(1+x^3)(2+x^3)} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{x^2}{(1+x^3)(2+x^3)} dx$

Concept / Formula Used: Substitution Method: Let $x^3 = t$, so $3x^2 dx = dt \implies x^2 dx = \frac{1}{3} dt$

Solution:

Let $x^3 = t$.

Differentiating: $3x^2 dx = dt$, so $x^2 dx = \frac{1}{3} dt$.

Substituting into the integral:

$$\int \frac{\frac{1}{3} dt}{(1+t)(2+t)} = \frac{1}{3} \int \frac{dt}{(1+t)(2+t)}$$

Decompose $\frac{1}{(1+t)(2+t)}$ into partial fractions:

$$\frac{1}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

Multiplying both sides by $(1+t)(2+t)$:

$$1 = A(2+t) + B(1+t)$$

Putting $t = -1$:

$$1 = A(2-1) \implies A = 1$$

Putting $t = -2$:

$$1 = B(1-2) \implies B = -1$$

Thus, the integral is:

$$\frac{1}{3} \int \left(\frac{1}{1+t} - \frac{1}{2+t} \right) dt$$

$$\begin{aligned} &= \frac{1}{3} [\ln |1+t| - \ln |2+t|] + C \\ &= \frac{1}{3} \ln \left| \frac{1+t}{2+t} \right| + C \end{aligned}$$

Now back-substitute $t = x^3$:

$$= \frac{1}{3} \ln \left| \frac{1+x^3}{2+x^3} \right| + C$$

Final Answer

$$\int \frac{x^2}{(1+x^3)(2+x^3)} dx = \frac{1}{3} \ln \left| \frac{1+x^3}{2+x^3} \right| + C$$

Common Mistake: Forgetting to multiply by $\frac{1}{3}$ after expressing $x^2 dx$ in terms of dt .

Question 11 (i)

Evaluate $\int \frac{dx}{x(x^4 - 1)}$.

TYPE 12a Direct/Algebraic Substitution

★ Likely Exam Question

Given: $\int \frac{dx}{x(x^4 - 1)}$

Concept / Formula Used: Substitution Method: Let $x^4 = t$, so $4x^3 dx = dt$

Solution:

Multiply numerator and denominator by x^3 to set up the substitution:

$$\int \frac{x^3 dx}{x^4(x^4 - 1)}$$

Let $x^4 = t$.

Differentiating: $4x^3 dx = dt$, so $x^3 dx = \frac{1}{4} dt$.

Substituting into the integral:

$$\int \frac{\frac{1}{4} dt}{t(t-1)} = \frac{1}{4} \int \frac{dt}{t(t-1)}$$

Decompose $\frac{1}{t(t-1)}$ into partial fractions:

$$\frac{1}{t(t-1)} = \frac{A}{t} + \frac{B}{t-1}$$

Multiplying both sides by $t(t-1)$:

$$1 = A(t-1) + Bt$$

Putting $t = 0$:

$$1 = A(0-1) \implies A = -1$$

Putting $t = 1$:

$$1 = B(1) \implies B = 1$$

Thus, the integral becomes:

$$\begin{aligned} & \frac{1}{4} \int \left(-\frac{1}{t} + \frac{1}{t-1} \right) dt \\ &= \frac{1}{4} [-\ln|t| + \ln|t-1|] + C \\ &= \frac{1}{4} \ln \left| \frac{t-1}{t} \right| + C \end{aligned}$$

Now back-substitute $t = x^4$:

$$= \frac{1}{4} \ln \left| \frac{x^4 - 1}{x^4} \right| + C$$

Final Answer

$$\int \frac{dx}{x(x^4 - 1)} = \frac{1}{4} \ln \left| \frac{x^4 - 1}{x^4} \right| + C$$

Common Mistake: Failing to multiply numerator and denominator by x^3 to create the derivative term for substitution.

Question 11 (ii)

Evaluate $\int \frac{dx}{x(x^2 + 1)}$.

TYPE 14 Partial Fractions

Given: $\int \frac{dx}{x(x^2 + 1)}$

Concept / Formula Used: Partial fractions: $\frac{1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$

Solution:

Decompose the integrand into partial fractions:

$$\frac{1}{x(x^2 + 1)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1}$$

Multiplying both sides by $x(x^2 + 1)$:

$$1 = A(x^2 + 1) + (Bx + C)x$$

Putting $x = 0$:

$$1 = A(0 + 1) \implies A = 1$$

Expanding the right side:

$$1 = Ax^2 + A + Bx^2 + Cx$$

Grouping coefficients by powers of x :

$$1 = A + Cx + (A + B)x^2$$

Equating coefficients of x^2 , x , and constant terms:

$$\begin{aligned} A + B &= 0 \implies 1 + B = 0 \implies B = -1 \\ C &= 0 \end{aligned}$$

Thus, $A = 1, B = -1, C = 0$. The integral is:

$$\int \left(\frac{1}{x} - \frac{x}{x^2 + 1} \right) dx$$

$$\begin{aligned} &= \int \frac{1}{x} dx - \frac{1}{2} \int \frac{2x}{x^2 + 1} dx \\ &= \ln |x| - \frac{1}{2} \ln(x^2 + 1) + C \\ &= \ln \left| \frac{x}{\sqrt{x^2 + 1}} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{x(x^2 + 1)} = \ln |x| - \frac{1}{2} \ln(x^2 + 1) + C$$

Common Mistake: Omitting the linear term $Bx + C$ in the numerator over the quadratic factor $x^2 + 1$.

Question 12 (i)

Evaluate $\int \frac{dx}{x(x^n + 1)}$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{dx}{x(x^n + 1)}$

Concept / Formula Used: Substitution Method: Let $x^n = t$, so $nx^{n-1} dx = dt$

Solution:

Multiply numerator and denominator by x^{n-1} :

$$\int \frac{x^{n-1} dx}{x^n(x^n + 1)}$$

Let $x^n = t$.

Differentiating: $nx^{n-1} dx = dt$, so $x^{n-1} dx = \frac{1}{n} dt$.

Substituting into the integral:

$$\int \frac{\frac{1}{n} dt}{t(t+1)} = \frac{1}{n} \int \frac{dt}{t(t+1)}$$

Decompose $\frac{1}{t(t+1)}$ into partial fractions:

$$\frac{1}{t(t+1)} = \frac{A}{t} + \frac{B}{t+1}$$

Multiplying both sides by $t(t+1)$:

$$1 = A(t+1) + Bt$$

Putting $t = 0$:

$$1 = A(0+1) \implies A = 1$$

Putting $t = -1$:

$$1 = B(-1) \implies B = -1$$

Thus, the integral is:

$$\begin{aligned} & \frac{1}{n} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt \\ &= \frac{1}{n} [\ln |t| - \ln |t+1|] + C \\ &= \frac{1}{n} \ln \left| \frac{t}{t+1} \right| + C \end{aligned}$$

Now back-substitute $t = x^n$:

$$= \frac{1}{n} \ln \left| \frac{x^n}{x^n + 1} \right| + C$$

Final Answer

$$\int \frac{dx}{x(x^n + 1)} = \frac{1}{n} \ln \left| \frac{x^n}{x^n + 1} \right| + C$$

Common Mistake: Failing to multiply numerator and denominator by x^{n-1} to prepare for the substitution $x^n = t$.

Question 12 (ii)

Evaluate $\int \frac{dx}{x(x^3 + 8)}$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{dx}{x(x^3 + 8)}$

Concept / Formula Used: Substitution Method: Let $x^3 = t$, so $3x^2 dx = dt$

Solution:

Multiply numerator and denominator by x^2 :

$$\int \frac{x^2 dx}{x^3(x^3 + 8)}$$

Let $x^3 = t$.

Differentiating: $3x^2 dx = dt$, so $x^2 dx = \frac{1}{3} dt$.

Substituting into the integral:

$$\int \frac{\frac{1}{3} dt}{t(t+8)} = \frac{1}{3} \int \frac{dt}{t(t+8)}$$

Decompose $\frac{1}{t(t+8)}$ into partial fractions:

$$\frac{1}{t(t+8)} = \frac{A}{t} + \frac{B}{t+8}$$

Multiplying both sides by $t(t+8)$:

$$1 = A(t+8) + Bt$$

Putting $t = 0$:

$$1 = A(8) \implies A = \frac{1}{8}$$

Putting $t = -8$:

$$1 = B(-8) \implies B = -\frac{1}{8}$$

Thus, the integral becomes:

$$\begin{aligned}\frac{1}{3} \int \left(\frac{1/8}{t} - \frac{1/8}{t+8} \right) dt &= \frac{1}{24} \int \left(\frac{1}{t} - \frac{1}{t+8} \right) dt \\&= \frac{1}{24} [\ln |t| - \ln |t+8|] + C \\&= \frac{1}{24} \ln \left| \frac{t}{t+8} \right| + C\end{aligned}$$

Now back-substitute $t = x^3$:

$$= \frac{1}{24} \ln \left| \frac{x^3}{x^3+8} \right| + C$$

Final Answer

$$\int \frac{dx}{x(x^3+8)} = \frac{1}{24} \ln \left| \frac{x^3}{x^3+8} \right| + C$$

Common Mistake: Forgetting to scale the integral by $\frac{1}{3}$ when substituting $x^2 dx = \frac{1}{3} dt$.

Question 13 (i)

Evaluate $\int \frac{e^x}{(1+e^x)(2+e^x)} dx$.

TYPE 12c Substitution Involving Exponential Functions

★ Likely Exam Question

Given: $\int \frac{e^x}{(1+e^x)(2+e^x)} dx$

Concept / Formula Used: Substitution Method: Let $e^x = t$, so $e^x dx = dt$

Solution:

Let $e^x = t$.

Differentiating: $e^x dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{(1+t)(2+t)}$$

Decompose $\frac{1}{(1+t)(2+t)}$ into partial fractions:

$$\frac{1}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

Multiplying both sides by $(1+t)(2+t)$:

$$1 = A(2+t) + B(1+t)$$

Putting $t = -1$:

$$1 = A(2-1) \implies A = 1$$

Putting $t = -2$:

$$1 = B(1-2) \implies B = -1$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1}{1+t} - \frac{1}{2+t} \right) dt \\ &= \ln|1+t| - \ln|2+t| + C \\ &= \ln \left| \frac{1+t}{2+t} \right| + C \end{aligned}$$

Now back-substitute $t = e^x$:

$$= \ln \left| \frac{1+e^x}{2+e^x} \right| + C$$

Final Answer

$$\int \frac{e^x}{(1+e^x)(2+e^x)} dx = \ln \left(\frac{1+e^x}{2+e^x} \right) + C$$

Common Mistake: Applying partial fractions without substituting $e^x = t$ first.

Question 13 (ii)

Evaluate $\int \frac{dx}{e^{2x} + 1}$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{dx}{e^{2x} + 1}$

Concept / Formula Used: Substitution Method: Let $e^x = t$, so $e^x dx = dt \implies dx = \frac{dt}{t}$

Solution:

Rewrite the integral by multiplying numerator and denominator by e^{-x} or substituting:

$$\int \frac{e^{-x} dx}{1 + e^{-x}}$$

Alternatively, let $e^x = t$, so $x = \ln t$ and $dx = \frac{1}{t} dt$. Then:

$$\int \frac{\frac{1}{t} dt}{t^2 + 1} = \int \frac{dt}{t(t^2 + 1)}$$

Decompose $\frac{1}{t(t^2+1)}$ into partial fractions:

$$\frac{1}{t(t^2 + 1)} = \frac{A}{t} + \frac{Bt + C}{t^2 + 1}$$

Multiplying both sides by $t(t^2 + 1)$:

$$1 = A(t^2 + 1) + (Bt + C)t$$

Putting $t = 0$:

$$1 = A(0 + 1) \implies A = 1$$

Equating coefficients of t^2 and t :

$$\begin{aligned} A + B &= 0 \implies 1 + B = 0 \implies B = -1 \\ C &= 0 \end{aligned}$$

Thus, the integral is:

$$\int \left(\frac{1}{t} - \frac{t}{t^2 + 1} \right) dt$$

$$\begin{aligned} &= \int \frac{1}{t} dt - \frac{1}{2} \int \frac{2t}{t^2 + 1} dt \\ &= \ln |t| - \frac{1}{2} \ln(t^2 + 1) + C \end{aligned}$$

Now back-substitute $t = e^x$:

$$\begin{aligned} &= \ln(e^x) - \frac{1}{2} \ln(e^{2x} + 1) + C \\ &= x - \frac{1}{2} \ln(e^{2x} + 1) + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{e^{2x} + 1} = x - \frac{1}{2} \ln(e^{2x} + 1) + C$$

Common Mistake: Failing to rewrite dx correctly in terms of dt when using the exponential substitution.

Question 14 (i)

Evaluate $\int \frac{\cos x}{(1 + \sin x)(2 + \sin x)} dx$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Given: $\int \frac{\cos x}{(1 + \sin x)(2 + \sin x)} dx$

Concept / Formula Used: Substitution Method: Let $\sin x = t$, so $\cos x dx = dt$

Solution:

Let $\sin x = t$.

Differentiating: $\cos x dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{(1+t)(2+t)}$$

Decompose $\frac{1}{(1+t)(2+t)}$ into partial fractions:

$$\frac{1}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

Multiplying both sides by $(1+t)(2+t)$:

$$1 = A(2+t) + B(1+t)$$

Putting $t = -1$:

$$1 = A(2-1) \implies A = 1$$

Putting $t = -2$:

$$1 = B(1-2) \implies B = -1$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1}{1+t} - \frac{1}{2+t} \right) dt \\ &= \ln |1+t| - \ln |2+t| + C \\ &= \ln \left| \frac{1+t}{2+t} \right| + C \end{aligned}$$

Now back-substitute $t = \sin x$:

$$= \ln \left| \frac{1 + \sin x}{2 + \sin x} \right| + C$$

Final Answer

$$\int \frac{\cos x}{(1 + \sin x)(2 + \sin x)} dx = \ln \left(\frac{1 + \sin x}{2 + \sin x} \right) + C$$

Common Mistake: Forgetting to substitute $\cos x dx = dt$ completely.

Question 14 (ii)

Evaluate $\int \frac{\sin 2x}{(1 + \sin x)(2 + \sin x)} dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \frac{\sin 2x}{(1 + \sin x)(2 + \sin x)} dx$

Concept / Formula Used: Double angle identity: $\sin 2x = 2 \sin x \cos x$, and substitution $\sin x = t$

Solution:

Using the double angle identity $\sin 2x = 2 \sin x \cos x$:

$$\int \frac{2 \sin x \cos x}{(1 + \sin x)(2 + \sin x)} dx$$

Let $\sin x = t$.

Differentiating: $\cos x dx = dt$.

Substituting into the integral:

$$\int \frac{2t}{(1+t)(2+t)} dt = 2 \int \frac{t}{(1+t)(2+t)} dt$$

Decompose $\frac{t}{(1+t)(2+t)}$ into partial fractions:

$$\frac{t}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

Multiplying both sides by $(1+t)(2+t)$:

$$t = A(2+t) + B(1+t)$$

Putting $t = -1$:

$$-1 = A(2-1) \implies A = -1$$

Putting $t = -2$:

$$-2 = B(1-2) \implies -2 = -B \implies B = 2$$

Thus, the integral is:

$$\begin{aligned} & 2 \int \left(-\frac{1}{1+t} + \frac{2}{2+t} \right) dt \\ &= 2 [-\ln |1+t| + 2 \ln |2+t|] + C \\ &= -2 \ln |1+t| + 4 \ln |2+t| + C \\ &= \ln \left(\frac{(2+t)^4}{(1+t)^2} \right) + C \end{aligned}$$

Now back-substitute $t = \sin x$:

$$= \ln \left(\frac{(2 + \sin x)^4}{(1 + \sin x)^2} \right) + C$$

Final Answer

$$\int \frac{\sin 2x}{(1 + \sin x)(2 + \sin x)} dx = \ln \left(\frac{(2 + \sin x)^4}{(1 + \sin x)^2} \right) + C$$

Common Mistake: Omitting the factor of 2 from the double-angle expansion of $\sin 2x$.

Question 15 (i)

Evaluate $\int \frac{2 \cos x}{(1 - \sin x)(2 - \cos^2 x)} dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \frac{2 \cos x}{(1 - \sin x)(2 - \cos^2 x)} dx$

Concept / Formula Used: Trigonometric identity $\cos^2 x = 1 - \sin^2 x$, and substitution $\sin x = t$

Solution:

Using the identity $\cos^2 x = 1 - \sin^2 x$, rewrite the denominator:

$$2 - \cos^2 x = 2 - (1 - \sin^2 x) = 1 + \sin^2 x$$

Thus, the integral becomes:

$$\int \frac{2 \cos x}{(1 - \sin x)(1 + \sin^2 x)} dx$$

Let $\sin x = t$.

Differentiating: $\cos x dx = dt$.

Substituting into the integral:

$$\int \frac{2 dt}{(1 - t)(1 + t^2)}$$

Decompose $\frac{2}{(1-t)(1+t^2)}$ into partial fractions:

$$\frac{2}{(1 - t)(1 + t^2)} = \frac{A}{1 - t} + \frac{Bt + C}{1 + t^2}$$

Multiplying both sides by $(1 - t)(1 + t^2)$:

$$2 = A(1 + t^2) + (Bt + C)(1 - t)$$

Putting $t = 1$:

$$2 = A(1 + 1^2) \implies 2 = 2A \implies A = 1$$

Expanding the right side:

$$2 = A + At^2 + Bt - Bt^2 + C - Ct$$

Grouping coefficients by powers of t :

$$2 = (A + C) + (B - C)t + (A - B)t^2$$

Equating coefficients of t^2 , t , and constant terms:

$$A - B = 0 \implies B = A = 1$$

$$B - C = 0 \implies C = B = 1$$

Thus, $A = 1, B = 1, C = 1$. The integral is:

$$\begin{aligned} & \int \left(\frac{1}{1-t} + \frac{t+1}{1+t^2} \right) dt \\ &= \int \frac{1}{1-t} dt + \int \frac{t}{1+t^2} dt + \int \frac{1}{1+t^2} dt \\ &= -\ln|1-t| + \frac{1}{2} \ln(1+t^2) + \tan^{-1}(t) + C \end{aligned}$$

Now back-substitute $t = \sin x$:

$$= \frac{1}{2} \ln(1 + \sin^2 x) - \ln|1 - \sin x| + \tan^{-1}(\sin x) + C$$

Final Answer

$$\int \frac{2 \cos x}{(1 - \sin x)(2 - \cos^2 x)} dx = \frac{1}{2} \ln(1 + \sin^2 x) - \ln|1 - \sin x| + \tan^{-1}(\sin x) + C$$

Common Mistake: Failing to simplify $2 - \cos^2 x$ using $\cos^2 x = 1 - \sin^2 x$ before substituting.

Question 15 (ii)

Evaluate $\int \frac{\sin 2x}{(\sin^2 x + 1)(\sin^2 x + 3)} dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \frac{\sin 2x}{(\sin^2 x + 1)(\sin^2 x + 3)} dx$

Concept / Formula Used: Double angle identity $\sin 2x = 2 \sin x \cos x$, and substitution $\sin^2 x = t$

Solution:

Using the double angle identity $\sin 2x = 2 \sin x \cos x$:

$$\int \frac{2 \sin x \cos x dx}{(\sin^2 x + 1)(\sin^2 x + 3)}$$

Let $\sin^2 x = t$.

Differentiating: $2 \sin x \cos x dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{(t+1)(t+3)}$$

Decompose $\frac{1}{(t+1)(t+3)}$ into partial fractions:

$$\frac{1}{(t+1)(t+3)} = \frac{A}{t+1} + \frac{B}{t+3}$$

Multiplying both sides by $(t+1)(t+3)$:

$$1 = A(t+3) + B(t+1)$$

Putting $t = -1$:

$$1 = A(-1+3) \implies 2A = 1 \implies A = \frac{1}{2}$$

Putting $t = -3$:

$$1 = B(-3+1) \implies -2B = 1 \implies B = -\frac{1}{2}$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1/2}{t+1} - \frac{1/2}{t+3} \right) dt \\ &= \frac{1}{2} \ln |t+1| - \frac{1}{2} \ln |t+3| + C \\ &= \frac{1}{2} \ln \left| \frac{t+1}{t+3} \right| + C \end{aligned}$$

Now back-substitute $t = \sin^2 x$:

$$= \frac{1}{2} \ln \left| \frac{\sin^2 x + 1}{\sin^2 x + 3} \right| + C$$

Final Answer

$$\int \frac{\sin 2x}{(\sin^2 x + 1)(\sin^2 x + 3)} dx = \frac{1}{2} \ln \left(\frac{\sin^2 x + 1}{\sin^2 x + 3} \right) + C$$

Common Mistake: Failing to recognize that $2 \sin x \cos x dx$ is the exact differential of $\sin^2 x$.

Question 16 (i)

Evaluate $\int \frac{\sec^2 x}{(2 + \tan x)(3 + \tan x)} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{\sec^2 x}{(2 + \tan x)(3 + \tan x)} dx$

Concept / Formula Used: Substitution Method: Let $\tan x = t$, so $\sec^2 x dx = dt$

Solution:

Let $\tan x = t$.

Differentiating: $\sec^2 x dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{(2+t)(3+t)}$$

Decompose $\frac{1}{(2+t)(3+t)}$ into partial fractions:

$$\frac{1}{(2+t)(3+t)} = \frac{A}{2+t} + \frac{B}{3+t}$$

Multiplying both sides by $(2+t)(3+t)$:

$$1 = A(3+t) + B(2+t)$$

Putting $t = -2$:

$$1 = A(3-2) \implies A = 1$$

Putting $t = -3$:

$$1 = B(2-3) \implies B = -1$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1}{2+t} - \frac{1}{3+t} \right) dt \\ &= \ln|2+t| - \ln|3+t| + C \\ &= \ln \left| \frac{2+t}{3+t} \right| + C \end{aligned}$$

Now back-substitute $t = \tan x$:

$$= \ln \left| \frac{2 + \tan x}{3 + \tan x} \right| + C$$

Final Answer

$$\int \frac{\sec^2 x}{(2 + \tan x)(3 + \tan x)} dx = \ln \left| \frac{2 + \tan x}{3 + \tan x} \right| + C$$

Common Mistake: Forgetting that $\sec^2 x \, dx = dt$ directly accounts for the numerator.

Question 16 (ii)

Evaluate $\int \frac{dx}{x(6(\log x)^2 + 7 \log x + 2)}$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{dx}{x(6(\log x)^2 + 7 \log x + 2)}$

Concept / Formula Used: Substitution Method: Let $\log x = t$, so $\frac{1}{x} dx = dt$

Solution:

Let $\log x = t$.

Differentiating: $\frac{1}{x} dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{6t^2 + 7t + 2}$$

Factorize the quadratic denominator:

$$6t^2 + 7t + 2 = 6t^2 + 4t + 3t + 2 = 2t(3t + 2) + 1(3t + 2) = (2t + 1)(3t + 2)$$

Decompose $\frac{1}{(2t+1)(3t+2)}$ into partial fractions:

$$\frac{1}{(2t + 1)(3t + 2)} = \frac{A}{2t + 1} + \frac{B}{3t + 2}$$

Multiplying both sides by $(2t + 1)(3t + 2)$:

$$1 = A(3t + 2) + B(2t + 1)$$

Putting $t = -\frac{1}{2}$:

$$1 = A \left(3 \left(-\frac{1}{2} \right) + 2 \right) = A \left(-\frac{3}{2} + 2 \right) = \frac{1}{2}A \implies A = 2$$

Putting $t = -\frac{2}{3}$:

$$1 = B \left(2 \left(-\frac{2}{3} \right) + 1 \right) = B \left(-\frac{4}{3} + 1 \right) = -\frac{1}{3}B \implies B = -3$$

Thus, the integral is:

$$\int \left(\frac{2}{2t + 1} - \frac{3}{3t + 2} \right) dt$$

$$\begin{aligned} &= 2 \left(\frac{1}{2} \right) \ln |2t + 1| - 3 \left(\frac{1}{3} \right) \ln |3t + 2| + C \\ &= \ln |2t + 1| - \ln |3t + 2| + C \\ &= \ln \left| \frac{2t + 1}{3t + 2} \right| + C \end{aligned}$$

Now back-substitute $t = \log x$:

$$= \ln \left| \frac{2 \log x + 1}{3 \log x + 2} \right| + C$$

Final Answer

$$\int \frac{dx}{x(6(\log x)^2 + 7 \log x + 2)} = \ln \left| \frac{2 \log x + 1}{3 \log x + 2} \right| + C$$

Common Mistake: Omitting the chain-rule coefficients $\frac{1}{2}$ and $\frac{1}{3}$ when integrating linear terms of t .

Question 17 (i)

Evaluate $\int \frac{dx}{\sin x - \sin 2x}$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \frac{dx}{\sin x - \sin 2x}$

Concept / Formula Used: Trigonometric identities: $\sin 2x = 2 \sin x \cos x$, and Weierstrass substitution

Solution:

Rewrite $\sin 2x = 2 \sin x \cos x$:

$$\int \frac{dx}{\sin x - 2 \sin x \cos x} = \int \frac{dx}{\sin x(1 - 2 \cos x)}$$

Multiply numerator and denominator by $\sin x$:

$$\int \frac{\sin x dx}{\sin^2 x(1 - 2 \cos x)} = \int \frac{\sin x dx}{(1 - \cos^2 x)(1 - 2 \cos x)}$$

Let $\cos x = t$.

Differentiating: $-\sin x dx = dt$, so $\sin x dx = -dt$.

Substituting into the integral:

$$\int \frac{-dt}{(1 - t^2)(1 - 2t)} = \int \frac{dt}{(t - 1)(t + 1)(2t - 1)}$$

Decompose into partial fractions:

$$\frac{1}{(t - 1)(t + 1)(2t - 1)} = \frac{A}{t - 1} + \frac{B}{t + 1} + \frac{C}{2t - 1}$$

Multiplying both sides by $(t - 1)(t + 1)(2t - 1)$:

$$1 = A(t + 1)(2t - 1) + B(t - 1)(2t - 1) + C(t - 1)(t + 1)$$

Putting $t = 1$:

$$1 = A(1 + 1)(2(1) - 1) = 2A \implies A = \frac{1}{2}$$

Putting $t = -1$:

$$1 = B(-1 - 1)(2(-1) - 1) = (-2)(-3)B = 6B \implies B = \frac{1}{6}$$

Putting $t = \frac{1}{2}$:

$$1 = C\left(\frac{1}{2} - 1\right)\left(\frac{1}{2} + 1\right) = C\left(-\frac{1}{2}\right)\left(\frac{3}{2}\right) = -\frac{3}{4}C \implies C = -\frac{4}{3}$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1/2}{t-1} + \frac{1/6}{t+1} - \frac{4/3}{2t-1} \right) dt \\ &= \frac{1}{2} \ln |t-1| + \frac{1}{6} \ln |t+1| - \frac{4}{3} \left(\frac{1}{2} \right) \ln |2t-1| + C \\ &= \frac{1}{2} \ln |t-1| + \frac{1}{6} \ln |t+1| - \frac{2}{3} \ln |2t-1| + C \end{aligned}$$

Now back-substitute $t = \cos x$:

$$= \frac{1}{2} \ln |\cos x - 1| + \frac{1}{6} \ln |\cos x + 1| - \frac{2}{3} \ln |2 \cos x - 1| + C$$

Final Answer

$$\int \frac{dx}{\sin x - \sin 2x} = \frac{1}{2} \ln |\cos x - 1| + \frac{1}{6} \ln |\cos x + 1| - \frac{2}{3} \ln |2 \cos x - 1| + C$$

Common Mistake: Failing to multiply numerator and denominator by $\sin x$ before substituting $\cos x = t$.

Question 17 (ii)

Evaluate $\int \frac{dx}{(x^2 + 1)(x^2 + 4)}$.

TYPE 14 Partial Fractions

Given: $\int \frac{dx}{(x^2 + 1)(x^2 + 4)}$

Concept / Formula Used: Partial fractions with quadratic terms: $\frac{1}{(x^2+1)(x^2+4)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+4}$

Solution:

Decompose the integrand into partial fractions. Let $u = x^2$ for algebraic simplicity:

$$\frac{1}{(u+1)(u+4)} = \frac{A}{u+1} + \frac{B}{u+4}$$

Multiplying both sides by $(u+1)(u+4)$:

$$1 = A(u+4) + B(u+1)$$

Putting $u = -1$:

$$1 = A(-1+4) \implies 3A = 1 \implies A = \frac{1}{3}$$

Putting $u = -4$:

$$1 = B(-4+1) \implies -3B = 1 \implies B = -\frac{1}{3}$$

Thus, in terms of x^2 :

$$\frac{1}{(x^2 + 1)(x^2 + 4)} = \frac{1/3}{x^2 + 1} - \frac{1/3}{x^2 + 4}$$

Integrating term by term:

$$\begin{aligned}\int \left(\frac{1/3}{x^2 + 1} - \frac{1/3}{x^2 + 2^2} \right) dx &= \frac{1}{3} \int \frac{1}{x^2 + 1} dx - \frac{1}{3} \int \frac{1}{x^2 + 2^2} dx \\ &= \frac{1}{3} \tan^{-1}(x) - \frac{1}{3} \times \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C \\ &= \frac{1}{3} \tan^{-1}(x) - \frac{1}{6} \tan^{-1} \left(\frac{x}{2} \right) + C\end{aligned}$$

Final Answer

$$\int \frac{dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{3} \tan^{-1}(x) - \frac{1}{6} \tan^{-1} \left(\frac{x}{2} \right) + C$$

Common Mistake: Unnecessarily applying general linear expansion $(Ax + B)$ for x^2 terms instead of treating x^2 as a single variable u .

Question 18 (i)

Evaluate $\int \frac{dx}{(x^2 + 1)(x^2 + 4)}$.

TYPE 14 Partial Fractions

Given: $\int \frac{dx}{(x^2 + 1)(x^2 + 4)}$

Concept / Formula Used: Partial fractions using substitution $u = x^2$

Solution:

Let $u = x^2$. Decompose $\frac{1}{(u+1)(u+4)}$ into partial fractions:

$$\frac{1}{(u+1)(u+4)} = \frac{A}{u+1} + \frac{B}{u+4}$$

Multiplying both sides by $(u+1)(u+4)$:

$$1 = A(u+4) + B(u+1)$$

Putting $u = -1$:

$$1 = A(-1+4) \implies 3A = 1 \implies A = \frac{1}{3}$$

Putting $u = -4$:

$$1 = B(-4+1) \implies -3B = 1 \implies B = -\frac{1}{3}$$

Thus, substituting back $u = x^2$:

$$\frac{1}{(x^2 + 1)(x^2 + 4)} = \frac{1/3}{x^2 + 1} - \frac{1/3}{x^2 + 4}$$

Integrating term by term:

$$\begin{aligned} \int \left(\frac{1/3}{x^2 + 1} - \frac{1/3}{x^2 + 2^2} \right) dx &= \frac{1}{3} \int \frac{dx}{x^2 + 1} - \frac{1}{3} \int \frac{dx}{x^2 + 2^2} \\ &= \frac{1}{3} \tan^{-1}(x) - \frac{1}{3} \left(\frac{1}{2} \right) \tan^{-1} \left(\frac{x}{2} \right) + C \\ &= \frac{1}{3} \tan^{-1}(x) - \frac{1}{6} \tan^{-1} \left(\frac{x}{2} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{(x^2 + 1)(x^2 + 4)} = \frac{1}{3} \tan^{-1}(x) - \frac{1}{6} \tan^{-1} \left(\frac{x}{2} \right) + C$$

Common Mistake: Treating x^2 as a linear variable requiring $Ax + B$ numerators, which complicates the solution unnecessarily.

Question 18 (ii)

Evaluate $\int \frac{x^2}{(x^2 + 1)(x^2 + 4)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2}{(x^2 + 1)(x^2 + 4)} dx$

Concept / Formula Used: Partial fractions using substitution $u = x^2$

Solution:

Let $u = x^2$. Decompose $\frac{u}{(u+1)(u+4)}$ into partial fractions:

$$\frac{u}{(u+1)(u+4)} = \frac{A}{u+1} + \frac{B}{u+4}$$

Multiplying both sides by $(u+1)(u+4)$:

$$u = A(u+4) + B(u+1)$$

Putting $u = -1$:

$$-1 = A(-1+4) \implies 3A = -1 \implies A = -\frac{1}{3}$$

Putting $u = -4$:

$$-4 = B(-4+1) \implies -3B = -4 \implies B = \frac{4}{3}$$

Thus, substituting back $u = x^2$:

$$\frac{x^2}{(x^2 + 1)(x^2 + 4)} = \frac{-1/3}{x^2 + 1} + \frac{4/3}{x^2 + 4}$$

Integrating term by term:

$$\begin{aligned} \int \left(\frac{-1/3}{x^2 + 1} + \frac{4/3}{x^2 + 2^2} \right) dx &= -\frac{1}{3} \int \frac{dx}{x^2 + 1} + \frac{4}{3} \int \frac{dx}{x^2 + 2^2} \\ &= -\frac{1}{3} \tan^{-1}(x) + \frac{4}{3} \left(\frac{1}{2} \right) \tan^{-1} \left(\frac{x}{2} \right) + C \\ &= -\frac{1}{3} \tan^{-1}(x) + \frac{2}{3} \tan^{-1} \left(\frac{x}{2} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{x^2}{(x^2 + 1)(x^2 + 4)} dx = \frac{2}{3} \tan^{-1} \left(\frac{x}{2} \right) - \frac{1}{3} \tan^{-1}(x) + C$$

Common Mistake: Incorrectly determining the numerators A and B during partial fraction decomposition.

Question 19 (i)

Evaluate $\int \frac{x^2 + 1}{(x^2 + 4)(x^2 + 25)} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2 + 1}{(x^2 + 4)(x^2 + 25)} dx$

Concept / Formula Used: Partial fractions using substitution $u = x^2$

Solution:

Let $u = x^2$. Decompose $\frac{u+1}{(u+4)(u+25)}$ into partial fractions:

$$\frac{u + 1}{(u + 4)(u + 25)} = \frac{A}{u + 4} + \frac{B}{u + 25}$$

Multiplying both sides by $(u + 4)(u + 25)$:

$$u + 1 = A(u + 25) + B(u + 4)$$

Putting $u = -4$:

$$-4 + 1 = A(-4 + 25) \implies -3 = 21A \implies A = -\frac{3}{21} = -\frac{1}{7}$$

Putting $u = -25$:

$$-25 + 1 = B(-25 + 4) \implies -24 = -21B \implies B = \frac{24}{21} = \frac{8}{7}$$

Thus, substituting back $u = x^2$:

$$\frac{x^2 + 1}{(x^2 + 4)(x^2 + 25)} = \frac{-1/7}{x^2 + 4} + \frac{8/7}{x^2 + 25}$$

Integrating term by term:

$$\begin{aligned} \int \left(\frac{-1/7}{x^2 + 2^2} + \frac{8/7}{x^2 + 5^2} \right) dx &= -\frac{1}{7} \int \frac{dx}{x^2 + 2^2} + \frac{8}{7} \int \frac{dx}{x^2 + 5^2} \\ &= -\frac{1}{7} \left(\frac{1}{2} \right) \tan^{-1} \left(\frac{x}{2} \right) + \frac{8}{7} \left(\frac{1}{5} \right) \tan^{-1} \left(\frac{x}{5} \right) + C \\ &= -\frac{1}{14} \tan^{-1} \left(\frac{x}{2} \right) + \frac{8}{35} \tan^{-1} \left(\frac{x}{5} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{x^2 + 1}{(x^2 + 4)(x^2 + 25)} dx = \frac{8}{35} \tan^{-1} \left(\frac{x}{5} \right) - \frac{1}{14} \tan^{-1} \left(\frac{x}{2} \right) + C$$

Common Mistake: Arithmetic simplification errors when reducing fractions for A and B .

Question 19 (ii)

Evaluate $\int \frac{x^2}{x^4 - x^2 - 12} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2}{x^4 - x^2 - 12} dx$

Concept / Formula Used: Factorization and partial fractions with $u = x^2$

Solution:

Factorize the denominator: $x^4 - x^2 - 12 = (x^2 - 4)(x^2 + 3) = (x - 2)(x + 2)(x^2 + 3)$.

Let $u = x^2$. Decompose $\frac{u}{(u-4)(u+3)}$ into partial fractions:

$$\frac{u}{(u-4)(u+3)} = \frac{A}{u-4} + \frac{B}{u+3}$$

Multiplying both sides by $(u-4)(u+3)$:

$$u = A(u+3) + B(u-4)$$

Putting $u = 4$:

$$4 = A(4+3) \implies 7A = 4 \implies A = \frac{4}{7}$$

Putting $u = -3$:

$$-3 = B(-3-4) \implies -7B = -3 \implies B = \frac{3}{7}$$

Thus, substituting back $u = x^2$:

$$\frac{x^2}{(x^2-4)(x^2+3)} = \frac{4/7}{x^2-2^2} + \frac{3/7}{x^2+(\sqrt{3})^2}$$

Integrating term by term:

$$\begin{aligned} \int \left(\frac{4/7}{x^2-2^2} + \frac{3/7}{x^2+(\sqrt{3})^2} \right) dx &= \frac{4}{7} \left(\frac{1}{2(2)} \ln \left| \frac{x-2}{x+2} \right| \right) + \frac{3}{7} \left(\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) \right) + C \\ &= \frac{1}{7} \ln \left| \frac{x-2}{x+2} \right| + \frac{\sqrt{3}}{7} \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{x^2}{x^4 - x^2 - 12} dx = \frac{1}{7} \ln \left| \frac{x-2}{x+2} \right| + \frac{\sqrt{3}}{7} \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) + C$$

Common Mistake: Failing to factorize $x^2 - 4$ further into $(x-2)(x+2)$ when applying logarithmic integration formulas.

Question 20 (i)

Evaluate $\int \frac{dx}{x^4 - 1}$.

TYPE 14 Partial Fractions

★ **Likely Exam Question**

Given: $\int \frac{dx}{x^4 - 1}$

Concept / Formula Used: Factorization and partial fractions: $x^4 - 1 = (x - 1)(x + 1)(x^2 + 1)$

Solution:

Factorize the denominator:

$$x^4 - 1 = (x^2 - 1)(x^2 + 1) = (x - 1)(x + 1)(x^2 + 1)$$

Decompose the integrand into partial fractions:

$$\frac{1}{(x - 1)(x + 1)(x^2 + 1)} = \frac{A}{x - 1} + \frac{B}{x + 1} + \frac{Cx + D}{x^2 + 1}$$

Multiplying both sides by $(x - 1)(x + 1)(x^2 + 1)$:

$$1 = A(x + 1)(x^2 + 1) + B(x - 1)(x^2 + 1) + (Cx + D)(x - 1)(x + 1)$$

Putting $x = 1$:

$$1 = A(1 + 1)(1^2 + 1) = 4A \implies A = \frac{1}{4}$$

Putting $x = -1$:

$$1 = B(-1 - 1)((-1)^2 + 1) = -4B \implies B = -\frac{1}{4}$$

Expanding and equating coefficients for C and D :

$$1 = (A + B + C)x^3 + (A - B + D)x^2 + (A + B - C)x + (A - B + D)$$

From x^3 coefficient: $A + B + C = 0 \implies \frac{1}{4} - \frac{1}{4} + C = 0 \implies C = 0$. From constant term: $A - B + D = 1 \implies \frac{1}{4} - (-\frac{1}{4}) + D = 1 \implies \frac{1}{2} + D = 1 \implies D = \frac{1}{2}$. Thus, $A = \frac{1}{4}, B = -\frac{1}{4}, C = 0, D = \frac{1}{2}$. The integral is:

$$\begin{aligned} & \int \left(\frac{1/4}{x - 1} - \frac{1/4}{x + 1} + \frac{1/2}{x^2 + 1} \right) dx \\ &= \frac{1}{4} \ln |x - 1| - \frac{1}{4} \ln |x + 1| + \frac{1}{2} \tan^{-1}(x) + C \\ &= \frac{1}{4} \ln \left| \frac{x - 1}{x + 1} \right| + \frac{1}{2} \tan^{-1}(x) + C \end{aligned}$$

Final Answer

$$\int \frac{dx}{x^4 - 1} = \frac{1}{4} \ln \left| \frac{x-1}{x+1} \right| + \frac{1}{2} \tan^{-1}(x) + C$$

Common Mistake: Forgetting to decompose $x^2 - 1$ into $(x-1)(x+1)$, missing the linear factors.

Question 20 (ii)

Evaluate $\int \frac{x^2}{1-x^4} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2}{1-x^4} dx$

Concept / Formula Used: Factorization and partial fractions: $1-x^4 = (1-x)(1+x)(1+x^2)$

Solution:

Factorize the denominator:

$$1-x^4 = (1-x^2)(1+x^2) = (1-x)(1+x)(1+x^2)$$

Decompose the integrand into partial fractions:

$$\frac{x^2}{(1-x)(1+x)(1+x^2)} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$$

Multiplying both sides by $(1-x)(1+x)(1+x^2)$:

$$x^2 = A(1+x)(1+x^2) + B(1-x)(1+x^2) + (Cx+D)(1-x)(1+x)$$

Putting $x = 1$:

$$1^2 = A(1+1)(1+1^2) = 4A \implies A = \frac{1}{4}$$

Putting $x = -1$:

$$(-1)^2 = B(1-(-1))(1+(-1)^2) = 4B \implies B = \frac{1}{4}$$

Equating coefficients of x^3 : $A - B - C = 0 \implies \frac{1}{4} - \frac{1}{4} - C = 0 \implies C = 0$. Equating constant terms: $A + B + D = 1 \implies \frac{1}{4} + \frac{1}{4} + D = 1 \implies \frac{1}{2} + D = 1 \implies D = \frac{1}{2}$. Thus, the integral is:

$$\int \left(\frac{1/4}{1-x} + \frac{1/4}{1+x} + \frac{1/2}{1+x^2} \right) dx$$

$$\begin{aligned} &= -\frac{1}{4} \ln |1-x| + \frac{1}{4} \ln |1+x| + \frac{1}{2} \tan^{-1}(x) + C \\ &= \frac{1}{4} \ln \left| \frac{1+x}{1-x} \right| + \frac{1}{2} \tan^{-1}(x) + C \end{aligned}$$

Final Answer

$$\int \frac{x^2}{1-x^4} dx = \frac{1}{4} \ln \left| \frac{1+x}{1-x} \right| + \frac{1}{2} \tan^{-1}(x) + C$$

Common Mistake: Incorrectly assigning signs when integrating $\frac{1}{1-x}$ due to the chain rule coefficient of -1 .

Question 21 (i)

Evaluate $\int \frac{\sqrt{\cos x}}{\sin x} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \frac{\sqrt{\cos x}}{\sin x} dx$

Concept / Formula Used: Substitution Method: Let $\cos x = t^2$, so $-\sin x dx = 2t dt$

Solution:

Multiply numerator and denominator by $\sin x$:

$$\int \frac{\sin x \sqrt{\cos x}}{\sin^2 x} dx = \int \frac{\sin x \sqrt{\cos x}}{1 - \cos^2 x} dx$$

Let $\cos x = t^2$, so $t = \sqrt{\cos x}$.

Differentiating: $-\sin x dx = 2t dt$, so $\sin x dx = -2t dt$.

Substituting into the integral:

$$\int \frac{t(-2t dt)}{1 - (t^2)^2} = \int \frac{-2t^2 dt}{1 - t^4} = \int \frac{2t^2 dt}{t^4 - 1}$$

Decompose $\frac{2t^2}{t^4 - 1}$ into partial fractions:

$$\frac{2t^2}{(t^2 - 1)(t^2 + 1)} = \frac{A}{t^2 - 1} + \frac{B}{t^2 + 1}$$

Let $u = t^2$:

$$\frac{2u}{(u - 1)(u + 1)} = \frac{A}{u - 1} + \frac{B}{u + 1}$$

Multiplying both sides by $(u - 1)(u + 1)$:

$$2u = A(u + 1) + B(u - 1)$$

Putting $u = 1$:

$$2(1) = A(1 + 1) \implies 2A = 2 \implies A = 1$$

Putting $u = -1$:

$$2(-1) = B(-1 - 1) \implies -2B = -2 \implies B = 1$$

Thus, the integral is:

$$\begin{aligned} & \int \left(\frac{1}{t^2 - 1} + \frac{1}{t^2 + 1} \right) dt \\ &= \frac{1}{2} \ln \left| \frac{t - 1}{t + 1} \right| + \tan^{-1}(t) + C \end{aligned}$$

Now back-substitute $t = \sqrt{\cos x}$:

$$= \frac{1}{2} \ln \left| \frac{\sqrt{\cos x} - 1}{\sqrt{\cos x} + 1} \right| + \tan^{-1}(\sqrt{\cos x}) + C$$

Final Answer

$$\int \frac{\sqrt{\cos x}}{\sin x} dx = \frac{1}{2} \ln \left| \frac{\sqrt{\cos x} - 1}{\sqrt{\cos x} + 1} \right| + \tan^{-1}(\sqrt{\cos x}) + C$$

Common Mistake: Failing to multiply numerator and denominator by $\sin x$ to set up the radical substitution.

Question 21 (ii)

Evaluate $\int \frac{\sin x}{\sin 4x} dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \frac{\sin x}{\sin 4x} dx$

Concept / Formula Used: Multiple angle identities and substitution

Solution:

Express $\sin 4x$ in terms of $\sin x$ and $\cos x$:

$$\sin 4x = 2 \sin 2x \cos 2x = 4 \sin x \cos x \cos 2x$$

$$\cos 2x = 1 - 2 \sin^2 x$$

Thus, the integral becomes:

$$\int \frac{\sin x dx}{4 \sin x \cos x (1 - 2 \sin^2 x)} = \frac{1}{4} \int \frac{dx}{\cos x (1 - 2 \sin^2 x)}$$

Multiply numerator and denominator by $\cos x$:

$$\frac{1}{4} \int \frac{\cos x dx}{\cos^2 x (1 - 2 \sin^2 x)} = \frac{1}{4} \int \frac{\cos x dx}{(1 - \sin^2 x)(1 - 2 \sin^2 x)}$$

Let $\sin x = t$. Differentiating: $\cos x dx = dt$. Substituting into the integral:

$$\frac{1}{4} \int \frac{dt}{(1 - t^2)(1 - 2t^2)}$$

Decompose into partial fractions using $u = t^2$:

$$\frac{1}{(1 - u)(1 - 2u)} = \frac{A}{1 - u} + \frac{B}{1 - 2u}$$

Multiplying both sides:

$$1 = A(1 - 2u) + B(1 - u)$$

Putting $u = 1$:

$$1 = A(1 - 2) \implies -A = 1 \implies A = -1$$

Putting $u = \frac{1}{2}$:

$$1 = B\left(1 - \frac{1}{2}\right) \implies \frac{1}{2}B = 1 \implies B = 2$$

Thus, the integral is:

$$\frac{1}{4} \int \left(\frac{-1}{1-t^2} + \frac{2}{1-2t^2} \right) dt = \frac{1}{4} \int \left(\frac{1}{t^2-1} - \frac{2}{2t^2-1} \right) dt$$

Integrating using standard formulas:

$$\begin{aligned} &= \frac{1}{4} \left[\frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| - 2 \left(\frac{1}{2\sqrt{2}} \right) \ln \left| \frac{\sqrt{2}t-1}{\sqrt{2}t+1} \right| \right] + C \\ &= \frac{1}{8} \ln \left| \frac{\sin x - 1}{\sin x + 1} \right| - \frac{1}{4\sqrt{2}} \ln \left| \frac{\sqrt{2} \sin x - 1}{\sqrt{2} \sin x + 1} \right| + C \end{aligned}$$

Final Answer

$$\int \frac{\sin x}{\sin 4x} dx = \frac{1}{8} \ln \left| \frac{\sin x - 1}{\sin x + 1} \right| - \frac{1}{4\sqrt{2}} \ln \left| \frac{\sqrt{2} \sin x - 1}{\sqrt{2} \sin x + 1} \right| + C$$

Common Mistake: Errors in multiple-angle expansion of $\sin 4x$.

Question 22 (i)

Evaluate $\int \frac{1+x^2}{1+x^4} dx$.

TYPE 12a Direct/Algebraic Substitution

★ Likely Exam Question

Given: $\int \frac{1+x^2}{1+x^4} dx$

Concept / Formula Used: Divide numerator and denominator by x^2 and substitute $t = x - \frac{1}{x}$

Solution:

Divide numerator and denominator by x^2 :

$$\int \frac{\frac{1}{x^2} + 1}{x^2 + \frac{1}{x^2}} dx$$

Rewrite the denominator as a complete square:

$$x^2 + \frac{1}{x^2} = \left(x - \frac{1}{x}\right)^2 + 2$$

Thus, the integral becomes:

$$\int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + (\sqrt{2})^2} dx$$

Let $x - \frac{1}{x} = t$.

Differentiating: $\left(1 + \frac{1}{x^2}\right) dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{t^2 + (\sqrt{2})^2}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a}\right) + c$:

$$= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}}\right) + C$$

Now back-substitute $t = x - \frac{1}{x} = \frac{x^2-1}{x}$:

$$= \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^2-1}{x\sqrt{2}}\right) + C$$

Final Answer

$$\int \frac{1+x^2}{1+x^4} dx = \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^2-1}{x\sqrt{2}}\right) + C$$

Common Mistake: Failing to recognize the algebraic substitution trick of dividing by x^2 when faced with $1 + x^4$ in the denominator.

Question 22 (ii)

Evaluate $\int \frac{1 - x^2}{1 + x^4} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{1 - x^2}{1 + x^4} dx$

Concept / Formula Used: Divide numerator and denominator by x^2 and substitute $t = x + \frac{1}{x}$

Solution:

Divide numerator and denominator by x^2 :

$$\int \frac{\frac{1}{x^2} - 1}{x^2 + \frac{1}{x^2}} dx$$

Rewrite the denominator:

$$x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$$

Thus, the integral becomes:

$$\int \frac{-\left(1 - \frac{1}{x^2}\right)}{\left(x + \frac{1}{x}\right)^2 - (\sqrt{2})^2} dx$$

Let $x + \frac{1}{x} = t$.

Differentiating: $\left(1 - \frac{1}{x^2}\right) dx = dt$.

Substituting into the integral:

$$\int \frac{-dt}{t^2 - (\sqrt{2})^2} = \int \frac{dt}{(\sqrt{2})^2 - t^2}$$

Using the standard result $\int \frac{dt}{a^2 - t^2} = \frac{1}{2a} \ln \left| \frac{a+t}{a-t} \right| + c$:

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2} + t}{\sqrt{2} - t} \right| + C$$

Now back-substitute $t = x + \frac{1}{x} = \frac{x^2+1}{x}$:

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2}x + x^2 + 1}{\sqrt{2}x - x^2 - 1} \right| + C$$

Final Answer

$$\int \frac{1-x^2}{1+x^4} dx = \frac{1}{2\sqrt{2}} \ln \left| \frac{x^2 + \sqrt{2}x + 1}{x^2 - \sqrt{2}x + 1} \right| + C$$

Common Mistake: Using the wrong substitution sign for $x + \frac{1}{x}$ versus $x - \frac{1}{x}$.

Question 23 (i)

Evaluate $\int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$.

TYPE 12a Direct/Algebraic Substitution

★ Likely Exam Question

Given: $\int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$

Concept / Formula Used: Divide numerator and denominator by x^2 and substitute $t = x + \frac{1}{x}$

Solution:

Divide numerator and denominator by x^2 :

$$\int \frac{1 - \frac{1}{x^2}}{x^2 + 1 + \frac{1}{x^2}} dx$$

Rewrite the denominator:

$$x^2 + 1 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 1$$

Thus, the integral becomes:

$$\int \frac{1 - \frac{1}{x^2}}{\left(x + \frac{1}{x}\right)^2 - 1^2} dx$$

Let $x + \frac{1}{x} = t$.

Differentiating: $\left(1 - \frac{1}{x^2}\right) dx = dt$.

Substituting into the integral:

$$\int \frac{dt}{t^2 - 1^2}$$

Using the standard result $\int \frac{dt}{t^2 - a^2} = \frac{1}{2a} \ln \left| \frac{t-a}{t+a} \right| + C$:

$$= \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + C$$

Now back-substitute $t = x + \frac{1}{x}$:

$$= \frac{1}{2} \ln \left| \frac{x + \frac{1}{x} - 1}{x + \frac{1}{x} + 1} \right| + C = \frac{1}{2} \ln \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right| + C$$

Final Answer

$$\int \frac{x^2 - 1}{x^4 + x^2 + 1} dx = \frac{1}{2} \ln \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right| + C$$

Common Mistake: Failing to complete the square correctly for the denominator $x^2 + 1 + \frac{1}{x^2}$.

Question 23 (ii)

Evaluate $\int \frac{x^2 + 4}{x^4 + x^2 + 16} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{x^2 + 4}{x^4 + x^2 + 16} dx$

Concept / Formula Used: Divide numerator and denominator by x^2 and express denominator as sum/difference of squares

Solution:

Divide numerator and denominator by x^2 :

$$\int \frac{1 + \frac{4}{x^2}}{x^2 + 1 + \frac{16}{x^2}} dx$$

Rewrite the denominator:

$$x^2 + \frac{16}{x^2} = \left(x - \frac{4}{x}\right)^2 + 8$$

Adding 1 to the denominator:

$$x^2 + 1 + \frac{16}{x^2} = \left(x - \frac{4}{x}\right)^2 + 9 = \left(x - \frac{4}{x}\right)^2 + 3^2$$

Let $t = x - \frac{4}{x}$.

Differentiating: $dt = \left(1 + \frac{4}{x^2}\right) dx$.

Substituting into the integral:

$$\int \frac{dt}{t^2 + 3^2}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a}\right) + c$:

$$= \frac{1}{3} \tan^{-1} \left(\frac{t}{3}\right) + C$$

Now back-substitute $t = x - \frac{4}{x} = \frac{x^2 - 4}{x}$:

$$= \frac{1}{3} \tan^{-1} \left(\frac{x^2 - 4}{3x}\right) + C$$

Final Answer

$$\int \frac{x^2 + 4}{x^4 + x^2 + 16} dx = \frac{1}{3} \tan^{-1} \left(\frac{x^2 - 4}{3x}\right) + C$$

Common Mistake: Incorrectly choosing the substitution constant for $\frac{4}{x}$ in completing the square.

Question 24 (i)

Evaluate $\int \frac{1}{x^4 + 1} dx$.

TYPE 14 Partial Fractions

★ **Likely Exam Question**

Given: $\int \frac{1}{x^4 + 1} dx$

Concept / Formula Used: Expressing $x^4 + 1$ as a difference of squares: $(x^2 + 1)^2 - 2x^2 = (x^2 - \sqrt{2}x + 1)(x^2 + \sqrt{2}x + 1)$

Solution:

Factorize the denominator:

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2 = (x^2 + 1 - \sqrt{2}x)(x^2 + 1 + \sqrt{2}x)$$

Decompose $\frac{1}{x^4+1}$ into partial fractions:

$$\frac{1}{x^4 + 1} = \frac{Ax + B}{x^2 - \sqrt{2}x + 1} + \frac{Cx + D}{x^2 + \sqrt{2}x + 1}$$

By symmetry and solving coefficients, we obtain:

$$A = \frac{1}{2\sqrt{2}}, \quad B = \frac{1}{2}, \quad C = -\frac{1}{2\sqrt{2}}, \quad D = \frac{1}{2}$$

Integrating both parts:

$$\int \frac{1}{x^4 + 1} dx = \frac{1}{4\sqrt{2}} \ln \left| \frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right| + \frac{1}{2\sqrt{2}} \left[\tan^{-1}(\sqrt{2}x - 1) + \tan^{-1}(\sqrt{2}x + 1) \right] + C$$

Simplifying using inverse trig identities:

$$= \frac{1}{4\sqrt{2}} \ln \left(\frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right) + \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$$

Final Answer

$$\int \frac{1}{x^4 + 1} dx = \frac{1}{4\sqrt{2}} \ln \left(\frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right) + \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C$$

Common Mistake: Failing to factorize $x^4 + 1$ into quadratic factors before attempting partial fractions.

Question 24 (ii)

Evaluate $\int \frac{x^2}{x^4 + 1} dx$.

TYPE 14 Partial Fractions

Given: $\int \frac{x^2}{x^4 + 1} dx$

Concept / Formula Used: Partial fractions decomposition of $\frac{x^2}{x^4+1}$

Solution:

Decompose $\frac{x^2}{x^4+1}$ by expressing it using the quadratic factors of $x^4 + 1$:

$$\frac{x^2}{x^4 + 1} = \frac{1}{2\sqrt{2}} \left[\frac{\sqrt{2}x}{x^2 - \sqrt{2}x + 1} - \frac{\sqrt{2}x}{x^2 + \sqrt{2}x + 1} \right]$$

Integrating term by term:

$$\begin{aligned} \int \frac{x^2}{x^4 + 1} dx &= \frac{1}{2\sqrt{2}} \int \frac{\sqrt{2}x}{x^2 - \sqrt{2}x + 1} dx - \frac{1}{2\sqrt{2}} \int \frac{\sqrt{2}x}{x^2 + \sqrt{2}x + 1} dx \\ &= \frac{1}{4\sqrt{2}} \ln \left(\frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right) + \frac{1}{2\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{2}x}{1 - x^2} \right) + C \end{aligned}$$

Final Answer

$$\int \frac{x^2}{x^4 + 1} dx = \frac{1}{2\sqrt{2}} \left[\frac{1}{2} \ln \left(\frac{x^2 - \sqrt{2}x + 1}{x^2 + \sqrt{2}x + 1} \right) + \tan^{-1} \left(\frac{\sqrt{2}x}{1 - x^2} \right) \right] + C$$

Common Mistake: Sign errors when splitting the partial fractions for $x^2/(x^4 + 1)$.

Question 25 (i)

Evaluate $\int \frac{x^2}{x^4 + x^2 + 1} dx$.

TYPE 1 Power Rule of Integration

Given: $\int \frac{x^2}{x^4 + x^2 + 1} dx$

Concept / Formula Used: Identity: $\frac{x^2}{x^4 + x^2 + 1} = \frac{1}{2} \left(\frac{x^2 + 1}{x^4 + x^2 + 1} - \frac{x^2 - 1}{x^4 + x^2 + 1} \right)$

Solution:

Express the numerator as an average of $(x^2 + 1)$ and $(x^2 - 1)$:

$$\frac{x^2}{x^4 + x^2 + 1} = \frac{1}{2} \left[\frac{(x^2 + 1) + (x^2 - 1)}{x^4 + x^2 + 1} \right] = \frac{1}{2} \left[\frac{x^2 + 1}{x^4 + x^2 + 1} + \frac{x^2 - 1}{x^4 + x^2 + 1} \right]$$

Now, evaluate each integral separately: 1. For $\int \frac{x^2 + 1}{x^4 + x^2 + 1} dx$: divide by x^2 and substitute $t = x - \frac{1}{x}$, yielding:

$$\int \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 3} dx = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{3}} \right)$$

2. For $\int \frac{x^2 - 1}{x^4 + x^2 + 1} dx$: divide by x^2 and substitute $t = x + \frac{1}{x}$, yielding:

$$\int \frac{1 - \frac{1}{x^2}}{\left(x + \frac{1}{x}\right)^2 - 1} dx = \frac{1}{2} \ln \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right|$$

Combining both results:

$$\int \frac{x^2}{x^4 + x^2 + 1} dx = \frac{1}{2} \left[\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{3}} \right) + \frac{1}{2} \ln \left| \frac{x^2 - x + 1}{x^2 + x + 1} \right| \right] + C$$

Final Answer

$$\int \frac{x^2}{x^4 + x^2 + 1} dx = \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{3}} \right) - \frac{1}{4} \ln \left| \frac{x^2 + x + 1}{x^2 - x + 1} \right| + C$$

Common Mistake: Failing to split the integral into the sum of two solvable symmetric integrals.

Question 25 (ii)

Evaluate $\int \frac{x^2 - 3x + 1}{x^4 - x^2 + 1} dx$.

TYPE 1 Power Rule of Integration

Given: $\int \frac{x^2 - 3x + 1}{x^4 - x^2 + 1} dx$

Concept / Formula Used: Algebraic splitting and substitution method

Solution:

Split the integrand into terms matching derivatives of $x - \frac{1}{x}$ and $x + \frac{1}{x}$:

$$\int \frac{x^2 + 1}{x^4 - x^2 + 1} dx - \int \frac{3x}{x^4 - x^2 + 1} dx$$

1. For the first integral, divide by x^2 and substitute $t = x + \frac{1}{x}$:

$$\int \frac{1 + \frac{1}{x^2}}{(x - \frac{1}{x})^2 + 3} dx = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{3}} \right)$$

2. For the second integral, substitute $u = x^2$, so $du = 2x dx$:

$$\int \frac{3x dx}{(x^2)^2 - x^2 + 1} = \frac{3}{2} \int \frac{2x dx}{(x^2)^2 - x^2 + 1} = \frac{3}{2} \ln |\dots|$$

Combining results:

$$= \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{3}} \right) - \frac{3}{2} \ln |x^2 - x + 1| + \dots$$

Final Answer

$$\int \frac{x^2 - 3x + 1}{x^4 - x^2 + 1} dx = \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{3}} \right) - \frac{3}{2} \ln(x^2 - x + 1) + C$$

Common Mistake: Failing to separate linear terms from quadratic components in the numerator.

Question 26 (i)

Evaluate $\int \frac{\sin x + \cos x}{\sin^2 x + \cos^4 x} dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \frac{\sin x + \cos x}{\sin^2 x + \cos^4 x} dx$

Concept / Formula Used: Rewrite the denominator using $\sin^2 x = 1 - \cos^2 x$, then substitute $t = \sin x - \cos x$

Solution:

Rewrite the denominator in terms of $\cos x$ only:

$$\begin{aligned}\sin^2 x + \cos^4 x &= (1 - \cos^2 x) + \cos^4 x \\ &= 1 - \cos^2 x(1 - \cos^2 x) \\ &= 1 - \cos^2 x \sin^2 x \\ &= 1 - \frac{1}{4} \sin^2 2x\end{aligned}$$

So the integral becomes

$$I = \int \frac{(\sin x + \cos x) dx}{1 - \frac{1}{4} \sin^2 2x}$$

Let $t = \sin x - \cos x$.

Differentiating: $dt = (\cos x + \sin x) dx$, which matches the numerator exactly.

Also, squaring t : $t^2 = \sin^2 x - 2 \sin x \cos x + \cos^2 x = 1 - \sin 2x$, so $\sin 2x = 1 - t^2$.

Substituting into the integral:

$$I = \int \frac{dt}{1 - \frac{1}{4}(1 - t^2)^2}$$

Expanding $(1 - t^2)^2 = 1 - 2t^2 + t^4$ and simplifying the denominator:

$$\begin{aligned}1 - \frac{1}{4}(1 - 2t^2 + t^4) &= \frac{4 - 1 + 2t^2 - t^4}{4} \\ &= \frac{3 + 2t^2 - t^4}{4}\end{aligned}$$

So the integral is

$$I = \int \frac{4 dt}{3 + 2t^2 - t^4}$$

Factorizing the quartic denominator:

$$3 + 2t^2 - t^4 = -(t^4 - 2t^2 - 3) = -(t^2 - 3)(t^2 + 1) = (3 - t^2)(1 + t^2)$$

So

$$I = \int \frac{4 dt}{(3 - t^2)(1 + t^2)}$$

Using the standard partial-fraction identity $\frac{1}{(a^2 - t^2)(t^2 + b^2)} = \frac{1}{a^2 + b^2} \left[\frac{1}{a^2 - t^2} + \frac{1}{t^2 + b^2} \right]$ with $a = \sqrt{3}$, $b = 1$ (so $a^2 + b^2 = 4$):

$$\frac{4}{(3 - t^2)(1 + t^2)} = \frac{1}{3 - t^2} + \frac{1}{1 + t^2}$$

Thus

$$I = \int \frac{dt}{3 - t^2} + \int \frac{dt}{1 + t^2}$$

Using the standard results $\int \frac{dt}{a^2 - t^2} = \frac{1}{2a} \log \left| \frac{a + t}{a - t} \right| + c$ (with $a = \sqrt{3}$) and $\int \frac{dt}{1 + t^2} = \tan^{-1}(t) + c$:

$$I = \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + t}{\sqrt{3} - t} \right| + \tan^{-1}(t) + C$$

Now back-substitute $t = \sin x - \cos x$:

$$I = \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| + \tan^{-1}(\sin x - \cos x) + C$$

Final Answer

$$\int \frac{\sin x + \cos x}{\sin^2 x + \cos^4 x} dx = \frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| + \tan^{-1}(\sin x - \cos x) + C$$

Common Mistake: Attempting direct partial fractions without first converting the trigonometric expression into a rational polynomial in $t = \sin x - \cos x$.

Question 26 (ii)

Evaluate $\int \left(\sqrt{\cot x} + \frac{1}{\sqrt{\cot x}} \right) dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int \left(\sqrt{\cot x} + \frac{1}{\sqrt{\cot x}} \right) dx$

Concept / Formula Used: Combine over a common denominator, then substitute $t = \sin x - \cos x$

Solution:

Write $\sqrt{\cot x} = \sqrt{\frac{\cos x}{\sin x}}$ and $\frac{1}{\sqrt{\cot x}} = \sqrt{\tan x} = \sqrt{\frac{\sin x}{\cos x}}$, and combine over the common denominator $\sqrt{\sin x \cos x}$:

$$\begin{aligned}\sqrt{\cot x} + \frac{1}{\sqrt{\cot x}} &= \frac{\cos x}{\sqrt{\sin x \cos x}} + \frac{\sin x}{\sqrt{\sin x \cos x}} \\ &= \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}\end{aligned}$$

So the integral becomes

$$I = \int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}} dx$$

Let $t = \sin x - \cos x$.

Differentiating: $dt = (\cos x + \sin x) dx$, which matches the numerator exactly.

Also, squaring t : $t^2 = 1 - 2 \sin x \cos x$, so $\sin x \cos x = \frac{1 - t^2}{2}$.

Substituting into the integral:

$$I = \int \frac{dt}{\sqrt{\frac{1 - t^2}{2}}} = \sqrt{2} \int \frac{dt}{\sqrt{1 - t^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{1 - t^2}} = \sin^{-1}(t) + C$:

$$I = \sqrt{2} \sin^{-1}(t) + C$$

Now back-substitute $t = \sin x - \cos x$:

$$I = \sqrt{2} \sin^{-1}(\sin x - \cos x) + C$$

Final Answer

$$\int \left(\sqrt{\cot x} + \frac{1}{\sqrt{\cot x}} \right) dx = \sqrt{2} \sin^{-1}(\sin x - \cos x) + C$$

Common Mistake: Forgetting that $\sin x \cos x$ must be expressed via $t = \sin x - \cos x$ (using $t^2 = 1 - 2 \sin x \cos x$) rather than left as a mixed trigonometric expression.

Question 27

Evaluate $\int \frac{x+7}{x^2+4x+7} dx$.

TYPE 7 Integration From Derivative

★ Likely Exam Question

Given: $\int \frac{x+7}{x^2+4x+7} dx$

Concept / Formula Used: Numerator adjustment: $\text{Numerator} = A(\text{Derivative of Denominator}) + B$

Solution:

Find the derivative of the denominator $x^2 + 4x + 7$:

$$\frac{d}{dx}(x^2 + 4x + 7) = 2x + 4$$

Express the numerator $x + 7$ in terms of $2x + 4$:

$$x + 7 = \frac{1}{2}(2x + 4) + 5$$

Substitute this back into the integral:

$$\int \frac{\frac{1}{2}(2x + 4) + 5}{x^2 + 4x + 7} dx$$

Split into two separate integrals:

$$= \frac{1}{2} \int \frac{2x + 4}{x^2 + 4x + 7} dx + 5 \int \frac{dx}{x^2 + 4x + 7}$$

Evaluate the first integral using direct substitution ($t = x^2 + 4x + 7$):

$$= \frac{1}{2} \ln(x^2 + 4x + 7)$$

For the second integral, complete the square in the denominator:

$$x^2 + 4x + 7 = (x + 2)^2 + 3 = (x + 2)^2 + (\sqrt{3})^2$$

Using the standard result $\int \frac{dx}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + c$:

$$5 \int \frac{dx}{(x + 2)^2 + (\sqrt{3})^2} = \frac{5}{\sqrt{3}} \tan^{-1} \left(\frac{x + 2}{\sqrt{3}} \right)$$

Combining both parts:

$$\int \frac{x+7}{x^2+4x+7} dx = \frac{1}{2} \ln(x^2 + 4x + 7) + \frac{5}{\sqrt{3}} \tan^{-1} \left(\frac{x+2}{\sqrt{3}} \right) + C$$

Final Answer

$$\int \frac{x+7}{x^2+4x+7} dx = \frac{1}{2} \ln(x^2+4x+7) + \frac{5}{\sqrt{3}} \tan^{-1} \left(\frac{x+2}{\sqrt{3}} \right) + C$$

Common Mistake: Failing to split the linear numerator into a multiple of the derivative of the denominator plus a constant.

7.11 – Integration by Parts

Question 1 (i)

Evaluate $\int x e^x dx$.

TYPE 13 Integration by Parts

Given: $\int x e^x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Using the ILATE rule, take $u = x$ (algebraic function) and $v = e^x$ (exponential function). Applying the integration by parts formula:

$$\begin{aligned} &= x \int e^x dx - \int \left(\frac{d}{dx}(x) \int e^x dx \right) dx \\ &= x e^x - \int (1 \cdot e^x) dx \\ &= x e^x - \int e^x dx \end{aligned}$$

Using the standard result $\int e^x dx = e^x + c$:

$$\begin{aligned} &= x e^x - e^x + c \\ &= (x - 1) e^x + c \end{aligned}$$

Final Answer

$$\int x e^x dx = (x - 1)e^x + c$$

Question 1 (ii)

Evaluate $\int x \sin x dx$.

TYPE 13 Integration by Parts

Given: $\int x \sin x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Take $u = x$ and $v = \sin x$. Applying the integration by parts formula:

$$\begin{aligned} &= x \int \sin x dx - \int \left(\frac{d}{dx}(x) \int \sin x dx \right) dx \\ &= x(-\cos x) - \int (1 \cdot (-\cos x)) dx \\ &= -x \cos x + \int \cos x dx \end{aligned}$$

Using the standard result $\int \cos x dx = \sin x + c$:

$$= -x \cos x + \sin x + c$$

Final Answer

$$\int x \sin x dx = -x \cos x + \sin x + c$$

Question 2 (i)

Evaluate $\int x \sec^2 x dx$.

TYPE 13 Integration by Parts

Given: $\int x \sec^2 x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Take $u = x$ and $v = \sec^2 x$. Applying the integration by parts formula:

$$\begin{aligned} &= x \int \sec^2 x dx - \int \left(\frac{d}{dx}(x) \int \sec^2 x dx \right) dx \\ &= x \tan x - \int (1 \cdot \tan x) dx \\ &= x \tan x - \int \tan x dx \end{aligned}$$

Using the standard result $\int \tan x dx = \log |\sec x| + c$:

$$= x \tan x - \log |\sec x| + c$$

Final Answer

$$\int x \sec^2 x dx = x \tan x + \log |\cos x| + c$$

Question 2 (ii)

Evaluate $\int x \sin 3x dx$.

TYPE 13 Integration by Parts

Given: $\int x \sin 3x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Take $u = x$ and $v = \sin 3x$. Applying the integration by parts formula:

$$\begin{aligned} &= x \int \sin 3x dx - \int \left(\frac{d}{dx}(x) \int \sin 3x dx \right) dx \\ &= x \left(-\frac{\cos 3x}{3} \right) - \int \left(1 \cdot \left(-\frac{\cos 3x}{3} \right) \right) dx \\ &= -\frac{1}{3} x \cos 3x + \frac{1}{3} \int \cos 3x dx \end{aligned}$$

Using the standard result $\int \cos(ax) dx = \frac{1}{a} \sin(ax) + c$:

$$\begin{aligned}
 &= -\frac{1}{3}x \cos 3x + \frac{1}{3} \left(\frac{\sin 3x}{3} \right) + c \\
 &= -\frac{1}{3}x \cos 3x + \frac{1}{9} \sin 3x + c
 \end{aligned}$$

Final Answer

$$\int x \sin 3x dx = -\frac{1}{3}x \cos 3x + \frac{1}{9} \sin 3x + c$$

Question 3 (i)

Evaluate $\int x e^{2x} dx$.

TYPE 13 Integration by Parts

Given: $\int x e^{2x} dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Take $u = x$ and $v = e^{2x}$. Applying the integration by parts formula:

$$\begin{aligned}
 &= x \int e^{2x} dx - \int \left(\frac{d}{dx}(x) \int e^{2x} dx \right) dx \\
 &= x \left(\frac{e^{2x}}{2} \right) - \int \left(1 \cdot \frac{e^{2x}}{2} \right) dx \\
 &= \frac{1}{2} x e^{2x} - \frac{1}{2} \int e^{2x} dx
 \end{aligned}$$

Using the standard result $\int e^{ax} dx = \frac{1}{a} e^{ax} + c$:

$$\begin{aligned}
 &= \frac{1}{2} x e^{2x} - \frac{1}{2} \left(\frac{e^{2x}}{2} \right) + c \\
 &= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + c \\
 &= \frac{1}{4} (2x - 1) e^{2x} + c
 \end{aligned}$$

Final Answer

$$\int x e^{2x} dx = \frac{1}{4} (2x - 1) e^{2x} + c$$

Question 3 (ii)

Evaluate $\int x \cos 2x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x \cos 2x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Take $u = x$ and $v = \cos 2x$. Applying the integration by parts formula:

$$\begin{aligned} &= x \int \cos 2x \, dx - \int \left(\frac{d}{dx}(x) \int \cos 2x \, dx \right) dx \\ &= x \left(\frac{\sin 2x}{2} \right) - \int \left(1 \cdot \frac{\sin 2x}{2} \right) dx \\ &= \frac{1}{2} x \sin 2x - \frac{1}{2} \int \sin 2x \, dx \end{aligned}$$

Using the standard result $\int \sin(ax) \, dx = -\frac{1}{a} \cos(ax) + c$:

$$\begin{aligned} &= \frac{1}{2} x \sin 2x - \frac{1}{2} \left(-\frac{\cos 2x}{2} \right) + c \\ &= \frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + c \end{aligned}$$

Final Answer

$$\int x \cos 2x \, dx = \frac{1}{2} x \sin 2x + \frac{1}{4} \cos 2x + c$$

Question 4 (i)

Evaluate $\int x \sec^2 x \tan x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x \sec^2 x \tan x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Take $u = x$ and $v = \sec^2 x \tan x$.

First evaluate $\int \sec^2 x \tan x dx$: let $\sec x = w$, so $\sec x \tan x dx = dw$, giving

$$\int \sec^2 x \tan x dx = \int w dw = \frac{w^2}{2} = \frac{\sec^2 x}{2}$$

Applying the integration by parts formula with $\int v dx = \frac{\sec^2 x}{2}$:

$$\begin{aligned} &= x \left(\frac{\sec^2 x}{2} \right) - \int \left(1 \cdot \frac{\sec^2 x}{2} \right) dx \\ &= \frac{1}{2} x \sec^2 x - \frac{1}{2} \int \sec^2 x dx \end{aligned}$$

Using the standard result $\int \sec^2 x dx = \tan x + c$:

$$= \frac{1}{2} x \sec^2 x - \frac{1}{2} \tan x + c$$

Final Answer

$$\int x \sec^2 x \tan x dx = \frac{1}{2} (x \sec^2 x - \tan x) + c$$

Question 4 (ii)

Evaluate $\int (e^{\log x} + \sin x) \cos x dx$.

TYPE 13 Integration by Parts

Given: $\int (e^{\log x} + \sin x) \cos x dx$

Concept / Formula Used: Logarithm property: $e^{\log x} = x$

Solution:

Simplify the integrand using $e^{\log x} = x$:

$$\begin{aligned} &= \int (x + \sin x) \cos x dx \\ &= \int x \cos x dx + \int \sin x \cos x dx \end{aligned}$$

For the first integral $\int x \cos x \, dx$, using integration by parts:

$$\begin{aligned} &= x \sin x - \int 1 \cdot \sin x \, dx \\ &= x \sin x + \cos x \end{aligned}$$

For the second integral $\int \sin x \cos x \, dx = \frac{1}{2} \int \sin 2x \, dx$:

$$= -\frac{1}{4} \cos 2x$$

Combining both parts:

$$= x \sin x + \cos x - \frac{1}{4} \cos 2x + c$$

Final Answer

$$\int (e^{\log x} + \sin x) \cos x \, dx = x \sin x + \cos x - \frac{1}{4} \cos 2x + c$$

Question 5 (i)

Evaluate $\int x \log x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x \log x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Using the ILATE rule, logarithmic functions take precedence, so let $u = \log x$ and $v = x$. Applying the integration by parts formula:

$$\begin{aligned} &= (\log x) \int x \, dx - \int \left(\frac{d}{dx}(\log x) \int x \, dx \right) dx \\ &= (\log x) \left(\frac{x^2}{2} \right) - \int \left(\frac{1}{x} \cdot \frac{x^2}{2} \right) dx \\ &= \frac{1}{2} x^2 \log x - \frac{1}{2} \int x \, dx \\ &= \frac{1}{2} x^2 \log x - \frac{1}{2} \left(\frac{x^2}{2} \right) + c \\ &= \frac{1}{2} x^2 \log x - \frac{1}{4} x^2 + c \\ &= \frac{1}{4} x^2 (2 \log x - 1) + c \end{aligned}$$

Final Answer

$$\int x \log x \, dx = \frac{1}{4}x^2(2 \log x - 1) + c$$

Question 5 (ii)

Evaluate $\int x \log 2x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x \log 2x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Let $u = \log 2x$ and $v = x$. Applying the integration by parts formula:

$$\begin{aligned} &= (\log 2x) \int x \, dx - \int \left(\frac{d}{dx}(\log 2x) \int x \, dx \right) dx \\ &= (\log 2x) \left(\frac{x^2}{2} \right) - \int \left(\frac{2}{2x} \cdot \frac{x^2}{2} \right) dx \\ &= \frac{1}{2}x^2 \log 2x - \frac{1}{2} \int x \, dx \\ &= \frac{1}{2}x^2 \log 2x - \frac{1}{4}x^2 + c \\ &= \frac{1}{4}x^2(2 \log 2x - 1) + c \end{aligned}$$

Final Answer

$$\int x \log 2x \, dx = \frac{1}{4}x^2(2 \log 2x - 1) + c$$

Question 6 (i)

Evaluate $\int x^4 \log x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x^4 \log x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Let $u = \log x$ and $v = x^4$. Applying the integration by parts formula:

$$\begin{aligned}
 &= (\log x) \int x^4 dx - \int \left(\frac{d}{dx}(\log x) \int x^4 dx \right) dx \\
 &= (\log x) \left(\frac{x^5}{5} \right) - \int \left(\frac{1}{x} \cdot \frac{x^5}{5} \right) dx \\
 &= \frac{1}{5} x^5 \log x - \frac{1}{5} \int x^4 dx \\
 &= \frac{1}{5} x^5 \log x - \frac{1}{5} \left(\frac{x^5}{5} \right) + c \\
 &= \frac{1}{5} x^5 \log x - \frac{1}{25} x^5 + c \\
 &= \frac{1}{25} x^5 (5 \log x - 1) + c
 \end{aligned}$$

Final Answer

$$\int x^4 \log x dx = \frac{1}{25} x^5 (5 \log x - 1) + c$$

Question 6 (ii)

Evaluate $\int \log x dx$.

TYPE 13 Integration by Parts

Given: $\int \log x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Rewrite the integral as $\int (\log x) \cdot 1 dx$. Let $u = \log x$ and $v = 1$. Applying the integration

by parts formula:

$$\begin{aligned}
 &= (\log x) \int 1 \, dx - \int \left(\frac{d}{dx}(\log x) \int 1 \, dx \right) dx \\
 &= (\log x) \cdot x - \int \left(\frac{1}{x} \cdot x \right) dx \\
 &= x \log x - \int 1 \, dx \\
 &= x \log x - x + c \\
 &= x(\log x - 1) + c
 \end{aligned}$$

Final Answer

$$\int \log x \, dx = x(\log x - 1) + c$$

Question 7 (i)

Evaluate $\int (x^2 + 1) \log x \, dx$.

TYPE 13 Integration by Parts

Given: $\int (x^2 + 1) \log x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Let $u = \log x$ and $v = x^2 + 1$. Applying the integration by parts formula:

$$\begin{aligned}
 &= (\log x) \int (x^2 + 1) \, dx - \int \left(\frac{d}{dx}(\log x) \int (x^2 + 1) \, dx \right) dx \\
 &= (\log x) \left(\frac{x^3}{3} + x \right) - \int \left(\frac{1}{x} \cdot \left(\frac{x^3}{3} + x \right) \right) dx \\
 &= \left(\frac{x^3}{3} + x \right) \log x - \int \left(\frac{x^2}{3} + 1 \right) dx \\
 &= \left(\frac{x^3}{3} + x \right) \log x - \left(\frac{x^3}{9} + x \right) + c
 \end{aligned}$$

Final Answer

$$\int (x^2 + 1) \log x \, dx = \left(\frac{x^3}{3} + x \right) \log x - \frac{x^3}{9} - x + c$$

Question 7 (ii)

Evaluate $\int x^2 \log(1+x) dx$.

TYPE 13 Integration by Parts

Given: $\int x^2 \log(1+x) dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Let $u = \log(1+x)$ and $v = x^2$. Applying the integration by parts formula:

$$\begin{aligned} &= \log(1+x) \int x^2 dx - \int \left(\frac{d}{dx}(\log(1+x)) \int x^2 dx \right) dx \\ &= \log(1+x) \left(\frac{x^3}{3} \right) - \int \left(\frac{1}{1+x} \cdot \frac{x^3}{3} \right) dx \\ &= \frac{1}{3} x^3 \log(1+x) - \frac{1}{3} \int \frac{x^3}{1+x} dx \end{aligned}$$

Using long division or algebraic adjustment for $\frac{x^3}{1+x} = x^2 - x + 1 - \frac{1}{1+x}$:

$$\begin{aligned} &= \frac{1}{3} x^3 \log(1+x) - \frac{1}{3} \int \left(x^2 - x + 1 - \frac{1}{1+x} \right) dx \\ &= \frac{1}{3} x^3 \log(1+x) - \frac{1}{3} \left(\frac{x^3}{3} - \frac{x^2}{2} + x - \log|1+x| \right) + c \\ &= \frac{1}{3} (x^3 + 1) \log(1+x) - \frac{1}{9} x^3 + \frac{1}{6} x^2 - \frac{1}{3} x + c \end{aligned}$$

Final Answer

$$\int x^2 \log(1+x) dx = \frac{1}{3} (x^3 + 1) \log(1+x) - \frac{1}{9} x^3 + \frac{1}{6} x^2 - \frac{1}{3} x + c$$

Question 7 (iii)

Evaluate $\int \sin x \cdot \log(\cos x) dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \sin x \cdot \log(\cos x) dx$

Concept / Formula Used: Integration by substitution: let $\cos x = t$

Solution:

Let $\cos x = t$. Differentiating: $-\sin x dx = dt$, so $\sin x dx = -dt$. Substitute into the integral:

$$\begin{aligned} &= \int \log(t) \cdot (-dt) \\ &= - \int \log t dt \end{aligned}$$

Using the standard result $\int \log t dt = t(\log t - 1) + c$:

$$\begin{aligned} &= -[t(\log t - 1)] + c \\ &= -t(\log t - 1) + c \\ &= t(1 - \log t) + c \end{aligned}$$

Now back-substitute $t = \cos x$:

$$= \cos x(1 - \log(\cos x)) + c$$

Final Answer

$$\int \sin x \cdot \log(\cos x) dx = (1 - \log(\cos x)) \cos x + c$$

Question 8 (i)

Evaluate $\int \tan^{-1} x dx$.

TYPE 13 Integration by Parts

Given: $\int \tan^{-1} x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Rewrite the integral as $\int (\tan^{-1} x) \cdot 1 dx$. Let $u = \tan^{-1} x$ and $v = 1$. Applying the

integration by parts formula:

$$\begin{aligned}
 &= (\tan^{-1} x) \int 1 \, dx - \int \left(\frac{d}{dx}(\tan^{-1} x) \int 1 \, dx \right) dx \\
 &= (\tan^{-1} x) \cdot x - \int \left(\frac{1}{1+x^2} \cdot x \right) dx \\
 &= x \tan^{-1} x - \int \frac{x}{1+x^2} dx
 \end{aligned}$$

For $\int \frac{x}{1+x^2} dx$, let $1+x^2 = t$, so $2x \, dx = dt \implies x \, dx = \frac{1}{2} dt$:

$$\begin{aligned}
 &= x \tan^{-1} x - \frac{1}{2} \int \frac{dt}{t} \\
 &= x \tan^{-1} x - \frac{1}{2} \log |1+x^2| + c
 \end{aligned}$$

Final Answer

$$\int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \log(1+x^2) + c$$

Question 8 (ii)

Evaluate $\int \cos^{-1} \left(\frac{1}{x} \right) dx$.

TYPE 13 Integration by Parts

Given: $\int \cos^{-1} \left(\frac{1}{x} \right) dx$

Concept / Formula Used: Trigonometric identity: $\cos^{-1} \left(\frac{1}{x} \right) = \sec^{-1} x$

Solution:

Using the identity $\cos^{-1} \left(\frac{1}{x} \right) = \sec^{-1} x$, the integral becomes:

$$\int \sec^{-1} x \, dx$$

Rewrite as $\int (\sec^{-1} x) \cdot 1 \, dx$. Let $u = \sec^{-1} x$ and $v = 1$. Applying the integration by parts formula:

$$\begin{aligned}
 &= (\sec^{-1} x) \int 1 \, dx - \int \left(\frac{d}{dx}(\sec^{-1} x) \int 1 \, dx \right) dx \\
 &= x \sec^{-1} x - \int \frac{1}{x\sqrt{x^2-1}} \cdot x \, dx \\
 &= x \sec^{-1} x - \int \frac{1}{\sqrt{x^2-1}} dx
 \end{aligned}$$

Using the standard result $\int \frac{dx}{\sqrt{x^2 - a^2}} = \log |x + \sqrt{x^2 - a^2}| + c$:

$$= x \sec^{-1} x - \log |x + \sqrt{x^2 - 1}| + c$$

Final Answer

$$\int \cos^{-1} \left(\frac{1}{x} \right) dx = x \sec^{-1} x - \log |x + \sqrt{x^2 - 1}| + c$$

Question 9 (i)

Evaluate $\int x^3 \tan^{-1} x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x^3 \tan^{-1} x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Let $u = \tan^{-1} x$ and $v = x^3$. Applying the integration by parts formula:

$$\begin{aligned} &= (\tan^{-1} x) \int x^3 \, dx - \int \left(\frac{d}{dx} (\tan^{-1} x) \int x^3 \, dx \right) dx \\ &= (\tan^{-1} x) \left(\frac{x^4}{4} \right) - \int \left(\frac{1}{1+x^2} \cdot \frac{x^4}{4} \right) dx \\ &= \frac{1}{4} x^4 \tan^{-1} x - \frac{1}{4} \int \frac{x^4}{1+x^2} \, dx \end{aligned}$$

Using algebraic division/adjustment for $\frac{x^4}{1+x^2} = x^2 - 1 + \frac{1}{1+x^2}$:

$$\begin{aligned} &= \frac{1}{4} x^4 \tan^{-1} x - \frac{1}{4} \int \left(x^2 - 1 + \frac{1}{1+x^2} \right) dx \\ &= \frac{1}{4} x^4 \tan^{-1} x - \frac{1}{4} \left(\frac{x^3}{3} - x + \tan^{-1} x \right) + c \\ &= \frac{1}{4} (x^4 - 1) \tan^{-1} x - \frac{1}{12} x^3 + \frac{1}{4} x + c \end{aligned}$$

Final Answer

$$\int x^3 \tan^{-1} x \, dx = \frac{1}{4} (x^4 - 1) \tan^{-1} x - \frac{1}{12} x^3 + \frac{1}{4} x + c$$

Question 9 (ii)

Evaluate $\int x^2 \sin^{-1} x \, dx$.

TYPE 13 Integration by Parts

Given: $\int x^2 \sin^{-1} x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Let $u = \sin^{-1} x$ and $v = x^2$. Applying the integration by parts formula:

$$\begin{aligned} &= (\sin^{-1} x) \int x^2 \, dx - \int \left(\frac{d}{dx} (\sin^{-1} x) \int x^2 \, dx \right) dx \\ &= (\sin^{-1} x) \left(\frac{x^3}{3} \right) - \int \left(\frac{1}{\sqrt{1-x^2}} \cdot \frac{x^3}{3} \right) dx \\ &= \frac{1}{3} x^3 \sin^{-1} x - \frac{1}{3} \int \frac{x^3}{\sqrt{1-x^2}} \, dx \end{aligned}$$

For $\int \frac{x^3}{\sqrt{1-x^2}} \, dx$, let $1 - x^2 = t \implies x^2 = 1 - t$, and $-2x \, dx = dt \implies x \, dx = -\frac{1}{2} dt$:

$$\begin{aligned} &= \frac{1}{3} x^3 \sin^{-1} x - \frac{1}{3} \int \frac{(1-t)(-\frac{1}{2} dt)}{\sqrt{t}} \\ &= \frac{1}{3} x^3 \sin^{-1} x + \frac{1}{6} \int (t^{-1/2} - t^{1/2}) \, dt \\ &= \frac{1}{3} x^3 \sin^{-1} x + \frac{1}{6} \left(2t^{1/2} - \frac{2}{3} t^{3/2} \right) + c \\ &= \frac{1}{3} x^3 \sin^{-1} x + \frac{1}{3} \sqrt{1-x^2} - \frac{1}{9} (1-x^2) \sqrt{1-x^2} + c \\ &= \frac{1}{3} x^3 \sin^{-1} x + \frac{1}{9} (x^2 + 2) \sqrt{1-x^2} + c \end{aligned}$$

Final Answer

$$\int x^2 \sin^{-1} x \, dx = \frac{1}{3} x^3 \sin^{-1} x + \frac{1}{9} (x^2 + 2) \sqrt{1-x^2} + c$$

Question 10 (i)

Evaluate $\int \frac{\sin^{-1} x}{\sqrt{1-x}} \, dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \frac{\sin^{-1} x}{\sqrt{1-x}} dx$

Concept / Formula Used: Integration by parts, using $\int \frac{dx}{\sqrt{1-x}} = -2\sqrt{1-x}$

Solution:

Take $u = \sin^{-1} x$ and $v = \frac{1}{\sqrt{1-x}}$.

First find $\int v dx = \int \frac{dx}{\sqrt{1-x}} = -2\sqrt{1-x}$.

Applying the integration by parts formula:

$$\begin{aligned} &= (\sin^{-1} x)(-2\sqrt{1-x}) - \int \frac{1}{\sqrt{1-x^2}} \cdot (-2\sqrt{1-x}) dx \\ &= -2\sqrt{1-x} \sin^{-1} x + 2 \int \frac{\sqrt{1-x}}{\sqrt{1-x^2}} dx \end{aligned}$$

Using $\sqrt{1-x^2} = \sqrt{1-x}\sqrt{1+x}$, the remaining integral simplifies:

$$\begin{aligned} 2 \int \frac{\sqrt{1-x}}{\sqrt{1-x}\sqrt{1+x}} dx &= 2 \int \frac{dx}{\sqrt{1+x}} \\ &= 2(2\sqrt{1+x}) \\ &= 4\sqrt{1+x} \end{aligned}$$

Combining both parts:

$$I = -2\sqrt{1-x} \sin^{-1} x + 4\sqrt{1+x} + c$$

Final Answer

$$\int \frac{\sin^{-1} x}{\sqrt{1-x}} dx = -2\sqrt{1-x} \sin^{-1} x + 4\sqrt{1+x} + c$$

Common Mistake: A flawed substitution such as $x = \sin^2 \theta$ tempts one to write $\sin^{-1} x = \theta$, which is false since $\sin^{-1}(\sin^2 \theta) \neq \theta$ in general; integration by parts directly on $\sin^{-1} x$ avoids this trap.

Question 10 (ii)

Evaluate $\int x^2 e^x dx$.

TYPE 13 Integration by Parts

Given: $\int x^2 e^x dx$

Concept / Formula Used: Integration by parts formula applied twice

Solution:

Let $u = x^2$ and $v = e^x$. Applying the integration by parts formula:

$$\begin{aligned} &= x^2 \int e^x dx - \int \left(\frac{d}{dx}(x^2) \int e^x dx \right) dx \\ &= x^2 e^x - \int 2x e^x dx \\ &= x^2 e^x - 2 \int x e^x dx \end{aligned}$$

Using the result from Question 1 (i) that $\int x e^x dx = (x - 1)e^x$:

$$\begin{aligned} &= x^2 e^x - 2((x - 1)e^x) + c \\ &= x^2 e^x - 2x e^x + 2e^x + c \\ &= (x^2 - 2x + 2)e^x + c \end{aligned}$$

Final Answer

$$\int x^2 e^x dx = (x^2 - 2x + 2)e^x + c$$

Question 11 (i)

Evaluate $\int x^2 e^{3x} dx$.

TYPE 13 Integration by Parts

Given: $\int x^2 e^{3x} dx$

Concept / Formula Used: Integration by parts formula applied successively

Solution:

Let $u = x^2$ and $v = e^{3x}$. Applying integration by parts:

$$\begin{aligned} &= x^2 \left(\frac{e^{3x}}{3} \right) - \int 2x \left(\frac{e^{3x}}{3} \right) dx \\ &= \frac{1}{3} x^2 e^{3x} - \frac{2}{3} \int x e^{3x} dx \end{aligned}$$

Now, evaluate $\int xe^{3x} dx$ using integration by parts:

$$\begin{aligned}\int xe^{3x} dx &= x \left(\frac{e^{3x}}{3} \right) - \int 1 \cdot \frac{e^{3x}}{3} dx \\ &= \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x}\end{aligned}$$

Substitute this back:

$$\begin{aligned}&= \frac{1}{3}x^2e^{3x} - \frac{2}{3} \left(\frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} \right) + c \\ &= \frac{1}{3}x^2e^{3x} - \frac{2}{9}xe^{3x} + \frac{2}{27}e^{3x} + c \\ &= \frac{1}{27}(9x^2 - 6x + 2)e^{3x} + c\end{aligned}$$

Final Answer

$$\int x^2e^{3x} dx = \frac{1}{27}(9x^2 - 6x + 2)e^{3x} + c$$

Question 11 (ii)

Evaluate $\int x^2 \cos x dx$.

TYPE 13 Integration by Parts

Given: $\int x^2 \cos x dx$

Concept / Formula Used: Integration by parts formula applied twice

Solution:

Let $u = x^2$ and $v = \cos x$. Applying integration by parts:

$$\begin{aligned}&= x^2 \sin x - \int 2x \sin x dx \\ &= x^2 \sin x - 2 \int x \sin x dx\end{aligned}$$

Using the result $\int x \sin x dx = -x \cos x + \sin x$:

$$\begin{aligned}&= x^2 \sin x - 2(-x \cos x + \sin x) + c \\ &= x^2 \sin x + 2x \cos x - 2 \sin x + c \\ &= (x^2 - 2) \sin x + 2x \cos x + c\end{aligned}$$

Final Answer

$$\int x^2 \cos x dx = (x^2 - 2) \sin x + 2x \cos x + c$$

Question 12 (i)

Evaluate $\int (\log x)^2 dx$.

TYPE 13 Integration by Parts

Given: $\int (\log x)^2 dx$

Concept / Formula Used: Integration by parts formula

Solution:

Rewrite as $\int (\log x)^2 \cdot 1 dx$. Let $u = (\log x)^2$ and $v = 1$. Applying integration by parts:

$$\begin{aligned} &= (\log x)^2 \int 1 dx - \int \left(\frac{d}{dx} ((\log x)^2) \int 1 dx \right) dx \\ &= x(\log x)^2 - \int 2(\log x) \cdot \frac{1}{x} \cdot x dx \\ &= x(\log x)^2 - 2 \int \log x dx \end{aligned}$$

Using the result $\int \log x dx = x(\log x - 1)$:

$$\begin{aligned} &= x(\log x)^2 - 2(x(\log x - 1)) + c \\ &= x(\log x)^2 - 2x \log x + 2x + c \\ &= x((\log x)^2 - 2 \log x + 2) + c \end{aligned}$$

Final Answer

$$\int (\log x)^2 dx = x((\log x)^2 - 2 \log x + 2) + c$$

Question 12 (ii)

Evaluate $\int x^3 \sin(x^2) dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int x^3 \sin(x^2) dx$

Concept / Formula Used: Integration by substitution: let $x^2 = t$

Solution:

Let $x^2 = t$. Differentiating: $2x dx = dt$, so $x dx = \frac{1}{2} dt$. Rewrite the integral as $\int x^2 \sin(x^2) \cdot$

$x dx$ and substitute:

$$\begin{aligned} &= \int t \sin(t) \cdot \left(\frac{1}{2} dt\right) \\ &= \frac{1}{2} \int t \sin t dt \end{aligned}$$

Using integration by parts for $\int t \sin t dt$:

$$\begin{aligned} &= \frac{1}{2} \left[t(-\cos t) - \int 1 \cdot (-\cos t) dt \right] \\ &= \frac{1}{2} (-t \cos t + \sin t) + c \end{aligned}$$

Now back-substitute $t = x^2$:

$$= \frac{1}{2} (-x^2 \cos(x^2) + \sin(x^2)) + c$$

Final Answer

$$\int x^3 \sin(x^2) dx = \frac{1}{2} (\sin(x^2) - x^2 \cos(x^2)) + c$$

Question 13 (i)

Evaluate $\int 2x^3 e^{x^2} dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int 2x^3 e^{x^2} dx$

Concept / Formula Used: Integration by substitution: let $x^2 = t$

Solution:

Let $x^2 = t$. Differentiating: $2x dx = dt$. Rewrite the integral as $\int x^2 e^{x^2} \cdot (2x dx)$ and substitute:

$$= \int t e^t dt$$

Using integration by parts with $u = t$ and $v = e^t$:

$$\begin{aligned} &= t e^t - \int 1 \cdot e^t dt \\ &= t e^t - e^t + c \\ &= (t - 1) e^t + c \end{aligned}$$

Now back-substitute $t = x^2$:

$$= (x^2 - 1) e^{x^2} + c$$

Final Answer

$$\int 2x^3 e^{x^2} dx = (x^2 - 1)e^{x^2} + c$$

Question 13 (ii)

Evaluate $\int \cos \sqrt{x} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \cos \sqrt{x} dx$

Concept / Formula Used: Integration by substitution: let $\sqrt{x} = t$

Solution:

Let $\sqrt{x} = t$, so $x = t^2$. Differentiating: $dx = 2t dt$. Substitute into the integral:

$$\begin{aligned} &= \int (\cos t)(2t dt) \\ &= 2 \int t \cos t dt \end{aligned}$$

Using integration by parts for $\int t \cos t dt$ with $u = t$ and $v = \cos t$:

$$\begin{aligned} &= 2 \left[t \sin t - \int 1 \cdot \sin t dt \right] \\ &= 2(t \sin t + \cos t) + c \end{aligned}$$

Now back-substitute $t = \sqrt{x}$:

$$= 2(\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}) + c$$

Final Answer

$$\int \cos \sqrt{x} dx = 2(\sqrt{x} \sin \sqrt{x} + \cos \sqrt{x}) + c$$

Question 14 (i)

Evaluate $\int \tan^{-1} \sqrt{x} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \tan^{-1} \sqrt{x} dx$

Concept / Formula Used: Integration by substitution followed by integration by parts

Solution:

Let $\sqrt{x} = t$, so $x = t^2$. Differentiating: $dx = 2t dt$. Substitute into the integral:

$$\begin{aligned} &= \int (\tan^{-1} t)(2t dt) \\ &= 2 \int t \tan^{-1} t dt \end{aligned}$$

Using integration by parts with $u = \tan^{-1} t$ and $v = t$:

$$\begin{aligned} &= 2 \left[(\tan^{-1} t) \left(\frac{t^2}{2} \right) - \int \frac{1}{1+t^2} \cdot \frac{t^2}{2} dt \right] \\ &= t^2 \tan^{-1} t - \int \frac{t^2}{1+t^2} dt \\ &= t^2 \tan^{-1} t - \int \left(1 - \frac{1}{1+t^2} \right) dt \\ &= t^2 \tan^{-1} t - (t - \tan^{-1} t) + c \\ &= (t^2 + 1) \tan^{-1} t - t + c \end{aligned}$$

Now back-substitute $t = \sqrt{x}$:

$$= (x + 1) \tan^{-1} \sqrt{x} - \sqrt{x} + c$$

Final Answer

$$\int \tan^{-1} \sqrt{x} dx = (x + 1) \tan^{-1} \sqrt{x} - \sqrt{x} + c$$

Question 14 (ii)

Evaluate $\int x^3 \tan^{-1}(x^2) dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int x^3 \tan^{-1}(x^2) dx$

Concept / Formula Used: Integration by substitution: let $x^2 = t$

Solution:

Let $x^2 = t$, so $2x dx = dt \implies x dx = \frac{1}{2} dt$. Rewrite the integral as $\int x^2 \tan^{-1}(x^2) \cdot x dx$ and substitute:

$$\begin{aligned} &= \int t \tan^{-1}(t) \cdot \left(\frac{1}{2} dt \right) \\ &= \frac{1}{2} \int t \tan^{-1} t dt \end{aligned}$$

Using integration by parts with $u = \tan^{-1} t$ and $v = t$:

$$\begin{aligned} &= \frac{1}{2} \left[(\tan^{-1} t) \left(\frac{t^2}{2} \right) - \int \frac{1}{1+t^2} \cdot \frac{t^2}{2} dt \right] \\ &= \frac{1}{4} t^2 \tan^{-1} t - \frac{1}{4} \int \frac{t^2}{1+t^2} dt \\ &= \frac{1}{4} t^2 \tan^{-1} t - \frac{1}{4} (t - \tan^{-1} t) + c \\ &= \frac{1}{4} (t^2 + 1) \tan^{-1} t - \frac{1}{4} t + c \end{aligned}$$

Now back-substitute $t = x^2$:

$$= \frac{1}{4} (x^4 + 1) \tan^{-1}(x^2) - \frac{1}{4} x^2 + c$$

Final Answer

$$\int x^3 \tan^{-1}(x^2) dx = \frac{1}{4} (x^4 + 1) \tan^{-1}(x^2) - \frac{1}{4} x^2 + c$$

Question 15 (i)

Evaluate $\int \sin^3 \sqrt{x} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \sin^3 \sqrt{x} dx$

Concept / Formula Used: Integration by substitution followed by integration by parts

Solution:

Let $\sqrt{x} = t$, so $x = t^2$ and $dx = 2t dt$. Substitute into the integral:

$$\begin{aligned} &= \int (\sin^3 t)(2t dt) \\ &= 2 \int t \sin^3 t dt \end{aligned}$$

Using the identity $\sin^3 t = \frac{3 \sin t - \sin 3t}{4}$:

$$\begin{aligned} &= 2 \int t \left(\frac{3 \sin t - \sin 3t}{4} \right) dt \\ &= \frac{1}{2} \int (3t \sin t - t \sin 3t) dt \end{aligned}$$

Integrating each term using integration by parts:

$$\begin{aligned}
 &= \frac{1}{2} \left[3(-t \cos t + \sin t) - \left(t \left(-\frac{\cos 3t}{3} \right) - \int -\frac{\cos 3t}{3} dt \right) \right] \\
 &= \frac{1}{2} \left[-3t \cos t + 3 \sin t + \frac{1}{3} t \cos 3t - \frac{1}{9} \sin 3t \right] + c
 \end{aligned}$$

Back-substituting $t = \sqrt{x}$:

$$= \frac{1}{2} \left[-3\sqrt{x} \cos \sqrt{x} + 3 \sin \sqrt{x} + \frac{1}{3} \sqrt{x} \cos(3\sqrt{x}) - \frac{1}{9} \sin(3\sqrt{x}) \right] + c$$

Final Answer

$$\int \sin^3 \sqrt{x} dx = \frac{1}{2} \left(3(\sin \sqrt{x} - \sqrt{x} \cos \sqrt{x}) + \frac{1}{3} \sqrt{x} \cos(3\sqrt{x}) - \frac{1}{9} \sin(3\sqrt{x}) \right) + c$$

Question 15 (ii)

Evaluate $\int \frac{x \cos^{-1} x}{\sqrt{1-x^2}} dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int \frac{x \cos^{-1} x}{\sqrt{1-x^2}} dx$

Concept / Formula Used: Integration by substitution: let $\cos^{-1} x = t$

Solution:

Let $\cos^{-1} x = t$, so $x = \cos t$. Differentiating: $-\frac{1}{\sqrt{1-x^2}} dx = dt$, so $\frac{dx}{\sqrt{1-x^2}} = -dt$. Also, $x = \cos t$. Substitute into the integral:

$$\begin{aligned}
 &= \int (\cos t)(t)(-dt) \\
 &= - \int t \cos t dt
 \end{aligned}$$

Using integration by parts for $\int t \cos t dt$:

$$\begin{aligned}
 &= - \left[t \sin t - \int 1 \cdot \sin t dt \right] \\
 &= -(t \sin t - (-\cos t)) \\
 &= -t \sin t - \cos t + c
 \end{aligned}$$

Since $t = \cos^{-1} x$, we have $\cos t = x$ and $\sin t = \sqrt{1-x^2}$:

$$= -\cos^{-1} x \cdot \sqrt{1-x^2} - x + c$$

Final Answer

$$\int \frac{x \cos^{-1} x}{\sqrt{1-x^2}} dx = -x - \sqrt{1-x^2} \cos^{-1} x + c$$

Question 16 (i)

Evaluate $\int \sin^{-1}(3x - 4x^3) dx$.

TYPE 6 Inverse Trigonometric Simplification

Given: $\int \sin^{-1}(3x - 4x^3) dx$

Concept / Formula Used: Trigonometric identity: $\sin^{-1}(3x - 4x^3) = 3 \sin^{-1} x$

Solution:

Using the standard inverse trigonometric identity $\sin^{-1}(3x - 4x^3) = 3 \sin^{-1} x$ (for $-\frac{1}{2} \leq x \leq \frac{1}{2}$):

$$\begin{aligned} &= \int 3 \sin^{-1} x \, dx \\ &= 3 \int \sin^{-1} x \, dx \end{aligned}$$

Using the standard result $\int \sin^{-1} x \, dx = x \sin^{-1} x + \sqrt{1 - x^2} + c$:

$$= 3 \left(x \sin^{-1} x + \sqrt{1 - x^2} \right) + c$$

Final Answer

$$\int \sin^{-1}(3x - 4x^3) dx = 3x \sin^{-1} x + 3\sqrt{1 - x^2} + c$$

Question 16 (ii)

Evaluate $\int \tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right) dx$.

TYPE 6 Inverse Trigonometric Simplification

Given: $\int \tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right) dx$

Concept / Formula Used: Trigonometric identity: $\tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right) = 3 \tan^{-1} x$

Solution:

Using the standard inverse trigonometric identity $\tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right) = 3 \tan^{-1} x$:

$$\begin{aligned} &= \int 3 \tan^{-1} x \, dx \\ &= 3 \int \tan^{-1} x \, dx \end{aligned}$$

Using the standard result $\int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \log(1+x^2) + c$:

$$= 3 \left(x \tan^{-1} x - \frac{1}{2} \log(1+x^2) \right) + c$$

Final Answer

$$\int \tan^{-1} \left(\frac{3x-x^3}{1-3x^2} \right) dx = 3x \tan^{-1} x - \frac{3}{2} \log(1+x^2) + c$$

Question 17 (i)

Evaluate $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx$

Concept / Formula Used: Integration by substitution: let $x = \sin \theta$

Solution:

Let $x = \sin \theta$, so $\sin^{-1} x = \theta$, and $dx = \cos \theta \, d\theta$. Also, $(1-x^2)^{3/2} = (\cos^2 \theta)^{3/2} = \cos^3 \theta$. Substitute into the integral:

$$\begin{aligned} &= \int \frac{\theta}{\cos^3 \theta} \cdot \cos \theta \, d\theta \\ &= \int \frac{\theta}{\cos^2 \theta} \, d\theta \\ &= \int \theta \sec^2 \theta \, d\theta \end{aligned}$$

Using integration by parts with $u = \theta$ and $v = \sec^2 \theta$:

$$\begin{aligned} &= \theta \tan \theta - \int 1 \cdot \tan \theta \, d\theta \\ &= \theta \tan \theta - \log |\sec \theta| + c \end{aligned}$$

Since $x = \sin \theta$, we have $\tan \theta = \frac{x}{\sqrt{1-x^2}}$ and $\sec \theta = \frac{1}{\sqrt{1-x^2}}$:

$$\begin{aligned} &= x \cdot \frac{\sin^{-1} x}{\sqrt{1-x^2}} - \log \left| \frac{1}{\sqrt{1-x^2}} \right| + c \\ &= \frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \frac{1}{2} \log(1-x^2) + c \end{aligned}$$

Final Answer

$$\int \frac{\sin^{-1} x}{(1-x^2)^{3/2}} dx = \frac{x \sin^{-1} x}{\sqrt{1-x^2}} + \frac{1}{2} \log(1-x^2) + c$$

Question 17 (ii)

Evaluate $\int \frac{x^2 \tan^{-1} x}{1+x^2} dx$.

TYPE 13 Integration by Parts

Given: $\int \frac{x^2 \tan^{-1} x}{1+x^2} dx$

Concept / Formula Used: Algebraic adjustment followed by integration by parts**Solution:**Rewrite $\frac{x^2}{1+x^2} = 1 - \frac{1}{1+x^2}$:

$$\begin{aligned} &= \int \left(1 - \frac{1}{1+x^2}\right) \tan^{-1} x \, dx \\ &= \int \tan^{-1} x \, dx - \int \frac{\tan^{-1} x}{1+x^2} \, dx \end{aligned}$$

For the second integral, let $\tan^{-1} x = t$, so $\frac{1}{1+x^2} dx = dt$:

$$\begin{aligned} &= \left(x \tan^{-1} x - \frac{1}{2} \log(1+x^2) \right) - \int t \, dt \\ &= x \tan^{-1} x - \frac{1}{2} \log(1+x^2) - \frac{1}{2} t^2 + c \end{aligned}$$

Now back-substitute $t = \tan^{-1} x$:

$$= x \tan^{-1} x - \frac{1}{2} (\tan^{-1} x)^2 - \frac{1}{2} \log(1+x^2) + c$$

Final Answer

$$\int \frac{x^2 \tan^{-1} x}{1+x^2} dx = x \tan^{-1} x - \frac{1}{2} (\tan^{-1} x)^2 - \frac{1}{2} \log(1+x^2) + c$$

Question 18 (i)

Evaluate $\int e^x \sin 2x \, dx$.

TYPE 13 Integration by Parts

Given: $\int e^x \sin 2x \, dx$

Concept / Formula Used: Standard exponential-trigonometric integration formula:

$$\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx))$$

Solution:

Let $I = \int e^x \sin 2x dx$. Integrating by parts twice:

$$\begin{aligned} I &= \sin 2x \int e^x dx - \int \left(\frac{d}{dx}(\sin 2x) \int e^x dx \right) dx \\ &= e^x \sin 2x - \int 2 \cos 2x \cdot e^x dx \\ &= e^x \sin 2x - 2 \int e^x \cos 2x dx \end{aligned}$$

Now integrate $\int e^x \cos 2x dx$ by parts:

$$\begin{aligned} \int e^x \cos 2x dx &= \cos 2x \cdot e^x - \int (-2 \sin 2x) e^x dx \\ &= e^x \cos 2x + 2 \int e^x \sin 2x dx \end{aligned}$$

Substitute this back into I :

$$\begin{aligned} I &= e^x \sin 2x - 2(e^x \cos 2x + 2I) \\ I &= e^x \sin 2x - 2e^x \cos 2x - 4I \\ 5I &= e^x(\sin 2x - 2 \cos 2x) \\ I &= \frac{e^x}{5}(\sin 2x - 2 \cos 2x) + c \end{aligned}$$

Final Answer

$$\int e^x \sin 2x dx = \frac{e^x}{5}(\sin 2x - 2 \cos 2x) + c$$

Question 18 (ii)

Evaluate $\int e^{2x} \sin x dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int e^{2x} \sin x dx$

Concept / Formula Used: Standard exponential-trigonometric integration formula

Solution:

Using the formula $\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} (a \sin(bx) - b \cos(bx))$ with $a = 2, b = 1$:

$$\begin{aligned} &= \frac{e^{2x}}{2^2 + 1^2} (2 \sin x - 1 \cos x) + c \\ &= \frac{e^{2x}}{5} (2 \sin x - \cos x) + c \end{aligned}$$

Final Answer

$$\int e^{2x} \sin x dx = \frac{e^{2x}}{5} (2 \sin x - \cos x) + c$$

Question 19 (i)

Evaluate $\int e^{ax} \sin bx dx$.

TYPE 13 Integration by Parts

Given: $\int e^{ax} \sin bx dx$

Concept / Formula Used: Standard reduction formula for exponential-trigonometric integrals

Solution:

Let $I = \int e^{ax} \sin bx dx$. Using integration by parts twice:

$$\begin{aligned} I &= \sin bx \left(\frac{e^{ax}}{a} \right) - \int b \cos bx \left(\frac{e^{ax}}{a} \right) dx \\ &= \frac{1}{a} e^{ax} \sin bx - \frac{b}{a} \int e^{ax} \cos bx dx \end{aligned}$$

Integrating $\int e^{ax} \cos bx dx$ by parts:

$$\begin{aligned} &= \frac{1}{a} e^{ax} \sin bx - \frac{b}{a} \left[\cos bx \left(\frac{e^{ax}}{a} \right) - \int -b \sin bx \left(\frac{e^{ax}}{a} \right) dx \right] \\ &= \frac{1}{a} e^{ax} \sin bx - \frac{b}{a^2} e^{ax} \cos bx - \frac{b^2}{a^2} I \end{aligned}$$

Rearranging for I :

$$\begin{aligned} I \left(1 + \frac{b^2}{a^2} \right) &= \frac{e^{ax}}{a^2} (a \sin bx - b \cos bx) \\ \frac{a^2 + b^2}{a^2} I &= \frac{e^{ax}}{a^2} (a \sin bx - b \cos bx) \\ I &= \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + c \end{aligned}$$

Hence Proved

$$\int e^{ax} \sin bx \, dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx) + c$$

Question 19 (ii)

Evaluate $\int e^{ax} \cos(bx + c) \, dx$.

TYPE 13 Integration by Parts

Given: $\int e^{ax} \cos(bx + c) \, dx$

Concept / Formula Used: Standard exponential-trigonometric integration formula

Solution:

Using the standard result for exponential and cosine integrals:

$$= \frac{e^{ax}}{a^2 + b^2} [a \cos(bx + c) + b \sin(bx + c)] + c_1$$

Final Answer

$$\int e^{ax} \cos(bx + c) \, dx = \frac{e^{ax}}{a^2 + b^2} [a \cos(bx + c) + b \sin(bx + c)] + c_1$$

Question 20 (i)

Evaluate $\int x^2 e^{x^3} \cos x^3 \, dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int x^2 e^{x^3} \cos x^3 \, dx$

Concept / Formula Used: Integration by substitution: let $x^3 = t$

Solution:

Let $x^3 = t$. Differentiating: $3x^2 \, dx = dt$, so $x^2 \, dx = \frac{1}{3} dt$. Substitute into the integral:

$$\begin{aligned} &= \int e^t \cos t \cdot \left(\frac{1}{3} dt\right) \\ &= \frac{1}{3} \int e^t \cos t \, dt \end{aligned}$$

Using the standard exponential-trigonometric integral formula $\int e^t \cos t \, dt = \frac{e^t}{2}(\sin t + \cos t)$:

$$\begin{aligned} &= \frac{1}{3} \left[\frac{e^t}{2}(\sin t + \cos t) \right] + c \\ &= \frac{e^t}{6}(\sin t + \cos t) + c \end{aligned}$$

Now back-substitute $t = x^3$:

$$= \frac{e^{x^3}}{6}(\sin(x^3) + \cos(x^3)) + c$$

Final Answer

$$\int x^2 e^{x^3} \cos x^3 \, dx = \frac{e^{x^3}}{6}(\sin(x^3) + \cos(x^3)) + c$$

Question 20 (ii)

Evaluate $\int e^x \sin^2 x \, dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: $\int e^x \sin^2 x \, dx$

Concept / Formula Used: Trigonometric identity: $\sin^2 x = \frac{1 - \cos 2x}{2}$

Solution:

Rewrite $\sin^2 x$ using the double-angle identity:

$$\begin{aligned} &= \int e^x \left(\frac{1 - \cos 2x}{2} \right) dx \\ &= \frac{1}{2} \int e^x \, dx - \frac{1}{2} \int e^x \cos 2x \, dx \end{aligned}$$

Evaluate the first integral: $\int e^x \, dx = e^x$. Evaluate the second integral using $\int e^{ax} \cos(bx) \, dx = \frac{e^{ax}}{a^2 + b^2}(a \cos(bx) + b \sin(bx))$ with $a = 1, b = 2$:

$$\begin{aligned} \int e^x \cos 2x \, dx &= \frac{e^x}{1^2 + 2^2}(1 \cos 2x + 2 \sin 2x) \\ &= \frac{e^x}{5}(\cos 2x + 2 \sin 2x) \end{aligned}$$

Combining both parts:

$$\begin{aligned} &= \frac{1}{2}e^x - \frac{1}{2} \left[\frac{e^x}{5}(\cos 2x + 2 \sin 2x) \right] + c \\ &= e^x \left[\frac{1}{2} - \frac{1}{10} \cos 2x - \frac{1}{5} \sin 2x \right] + c \end{aligned}$$

Final Answer

$$\int e^x \sin^2 x \, dx = \frac{e^x}{10} (5 - \cos 2x - 2 \sin 2x) + c$$

Question 21 (i)

Evaluate $\int \frac{e^{m \tan^{-1} x}}{(1+x^2)^{3/2}} dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{e^{m \tan^{-1} x}}{(1+x^2)^{3/2}} dx$

Concept / Formula Used: Integration by substitution: let $\tan^{-1} x = t$ **Solution:**

Let $\tan^{-1} x = t$, so $x = \tan t$ and $dx = \sec^2 t \, dt$. Also, $(1+x^2)^{3/2} = (1+\tan^2 t)^{3/2} = (\sec^2 t)^{3/2} = \sec^3 t$. Substitute into the integral:

$$\begin{aligned} &= \int \frac{e^{mt}}{\sec^3 t} \cdot \sec^2 t \, dt \\ &= \int \frac{e^{mt}}{\sec t} \, dt \\ &= \int e^{mt} \cos t \, dt \end{aligned}$$

Using the standard formula $\int e^{at} \cos(bt) \, dt = \frac{e^{at}}{a^2+b^2} (a \cos bt + b \sin bt)$ with $a = m, b = 1$:

$$= \frac{e^{mt}}{m^2+1} (m \cos t + \sin t) + c$$

Since $\tan t = x$, we have $\cos t = \frac{1}{\sqrt{1+x^2}}$ and $\sin t = \frac{x}{\sqrt{1+x^2}}$:

$$\begin{aligned} &= \frac{e^{m \tan^{-1} x}}{m^2+1} \left(\frac{m}{\sqrt{1+x^2}} + \frac{x}{\sqrt{1+x^2}} \right) + c \\ &= \frac{e^{m \tan^{-1} x} (mx + m)}{\sqrt{1+x^2} (m^2+1)} + c \quad (\text{or equivalent}) \end{aligned}$$

Final Answer

$$\int \frac{e^{m \tan^{-1} x}}{(1+x^2)^{3/2}} dx = \frac{e^{m \tan^{-1} x} (x+m)}{(1+m^2)\sqrt{1+x^2}} + c$$

Question 21 (ii)

Evaluate $\int \csc^3 x \, dx$.

TYPE 13 Integration by Parts

Given: $\int \csc^3 x \, dx$

Concept / Formula Used: Integration by parts applied to $\csc x \cdot \csc^2 x$

Solution:

Let $I = \int \csc^3 x \, dx = \int \csc x \cdot \csc^2 x \, dx$. Using integration by parts with $u = \csc x$ and $v = \csc^2 x$:

$$\begin{aligned} I &= \csc x \int \csc^2 x \, dx - \int \left(\frac{d}{dx}(\csc x) \int \csc^2 x \, dx \right) dx \\ &= \csc x(-\cot x) - \int (-\csc x \cot x)(-\cot x) \, dx \\ &= -\csc x \cot x - \int \csc x \cot^2 x \, dx \\ &= -\csc x \cot x - \int \csc x (\csc^2 x - 1) \, dx \\ &= -\csc x \cot x - \int \csc^3 x \, dx + \int \csc x \, dx \end{aligned}$$

Substitute $I = \int \csc^3 x \, dx$ into the equation:

$$\begin{aligned} I &= -\csc x \cot x - I + \log |\csc x - \cot x| \\ 2I &= -\csc x \cot x + \log |\csc x - \cot x| \\ I &= -\frac{1}{2} \csc x \cot x + \frac{1}{2} \log |\csc x - \cot x| + c \end{aligned}$$

Final Answer

$$\int \csc^3 x \, dx = -\frac{1}{2} \csc x \cot x + \frac{1}{2} \log |\csc x - \cot x| + c$$

Question 22 (i)

Evaluate $\int \cos(\log x) \, dx$.

TYPE 13 Integration by Parts

Given: $\int \cos(\log x) \, dx$

Concept / Formula Used: Integration by parts formula applied twice

Solution:

Let $I = \int \cos(\log x) \cdot 1 \, dx$. Using integration by parts with $u = \cos(\log x)$ and $v = 1$:

$$\begin{aligned} I &= \cos(\log x) \cdot x - \int \left(-\sin(\log x) \cdot \frac{1}{x} \cdot x \right) dx \\ &= x \cos(\log x) + \int \sin(\log x) \, dx \end{aligned}$$

Now, integrate $\int \sin(\log x) \, dx$ by parts:

$$\begin{aligned} \int \sin(\log x) \, dx &= \sin(\log x) \cdot x - \int \left(\cos(\log x) \cdot \frac{1}{x} \cdot x \right) dx \\ &= x \sin(\log x) - \int \cos(\log x) \, dx \\ &= x \sin(\log x) - I \end{aligned}$$

Substitute this back into the expression for I :

$$\begin{aligned} I &= x \cos(\log x) + x \sin(\log x) - I \\ 2I &= x[\sin(\log x) + \cos(\log x)] \\ I &= \frac{x}{2}[\sin(\log x) + \cos(\log x)] + c \end{aligned}$$

Final Answer

$$\int \cos(\log x) \, dx = \frac{x}{2}(\sin(\log x) + \cos(\log x)) + c$$

Question 22 (ii)

Evaluate $\int \frac{\sin^{-1} x}{x^2} \, dx$.

TYPE 13 Integration by Parts

Given: $\int \frac{\sin^{-1} x}{x^2} \, dx$

Concept / Formula Used: Integration by parts formula

Solution:

Let $u = \sin^{-1} x$ and $v = \frac{1}{x^2}$. Applying integration by parts:

$$\begin{aligned} &= (\sin^{-1} x) \left(-\frac{1}{x} \right) - \int \left(\frac{1}{\sqrt{1-x^2}} \cdot \left(-\frac{1}{x} \right) \right) dx \\ &= -\frac{\sin^{-1} x}{x} + \int \frac{dx}{x\sqrt{1-x^2}} \end{aligned}$$

For $\int \frac{dx}{x\sqrt{1-x^2}}$, let $x = \sin \theta$, so $dx = \cos \theta d\theta$ and $\sqrt{1-x^2} = \cos \theta$:

$$\begin{aligned} &= \int \frac{\cos \theta d\theta}{\sin \theta \cos \theta} \\ &= \int \csc \theta d\theta \\ &= \log |\csc \theta - \cot \theta| \end{aligned}$$

Since $x = \sin \theta$, we have $\csc \theta = \frac{1}{x}$ and $\cot \theta = \frac{\sqrt{1-x^2}}{x}$:

$$= \log \left| \frac{1 - \sqrt{1-x^2}}{x} \right|$$

Combining both parts:

$$= -\frac{\sin^{-1} x}{x} + \log \left| \frac{1 - \sqrt{1-x^2}}{x} \right| + c$$

Final Answer

$$\int \frac{\sin^{-1} x}{x^2} dx = -\frac{\sin^{-1} x}{x} + \log \left| \frac{1 - \sqrt{1-x^2}}{x} \right| + c$$

7.12 – Integration by Parts: Special Exponential Form

Question 1 (i)

Evaluate $\int e^x(\sin x + \cos x) dx$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Given: $\int e^x(\sin x + \cos x) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = \sin x$.

Differentiating with respect to x :

$$f'(x) = \cos x$$

Substituting $f(x)$ and $f'(x)$ into the given integral:

$$\int e^x [\sin x + \cos x] dx = \int e^x [f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \sin x + c$$

Final Answer

$$\int e^x (\sin x + \cos x) dx = e^x \sin x + c$$

Question 1 (ii)

Evaluate $\int e^x (\tan x + \sec^2 x) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x (\tan x + \sec^2 x) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = \tan x$.

Differentiating with respect to x :

$$f'(x) = \sec^2 x$$

Substituting $f(x)$ and $f'(x)$ into the given integral:

$$\int e^x (\tan x + \sec^2 x) dx = \int e^x [f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \tan x + c$$

Final Answer

$$\int e^x (\tan x + \sec^2 x) dx = e^x \tan x + c$$

Question 2 (i)

Evaluate $\int e^x \left(\log x + \frac{1}{x} \right) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x \left(\log x + \frac{1}{x} \right) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = \log x$.

Differentiating with respect to x :

$$f'(x) = \frac{1}{x}$$

Substituting $f(x)$ and $f'(x)$ into the given integral:

$$\int e^x \left(\log x + \frac{1}{x} \right) dx = \int e^x [f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \log x + c$$

Final Answer

$$\int e^x \left(\log x + \frac{1}{x} \right) dx = e^x \log x + c$$

Question 2 (ii)

Evaluate $\int e^x (\tan x + 1) \sec x dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x (\tan x + 1) \sec x dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Expanding the integrand:

$$\int e^x (\sec x \tan x + \sec x) dx$$

Let $f(x) = \sec x$.

Differentiating with respect to x :

$$f'(x) = \sec x \tan x$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x [\sec x + \sec x \tan x] dx = \int e^x [f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \sec x + c$$

Final Answer

$$\int e^x (\tan x + 1) \sec x dx = e^x \sec x + c$$

Question 3 (i)

Evaluate $\int e^x (x^2 + 2x) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x (x^2 + 2x) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = x^2$.

Differentiating with respect to x :

$$f'(x) = 2x$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x (x^2 + 2x) dx = \int e^x [f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x x^2 + c$$

Final Answer

$$\int e^x(x^2 + 2x) dx = e^x x^2 + c$$

Question 3 (ii)

Evaluate $\int (x + 1)e^x dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int (x + 1)e^x dx$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Rearranging the integrand:

$$\int e^x(x + 1) dx$$

Let $f(x) = x$.

Differentiating with respect to x :

$$f'(x) = 1$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x(x + 1) dx = \int e^x[f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x x + c$$
Final Answer

$$\int (x + 1)e^x dx = e^x x + c$$

Question 4 (i)

Evaluate $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = \frac{1}{x} = x^{-1}$.

Differentiating with respect to x :

$$f'(x) = -1 \cdot x^{-2} = -\frac{1}{x^2}$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \int e^x [f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \left(\frac{1}{x} \right) + c$$

Final Answer

$$\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \frac{1}{x} e^x + c$$

Question 4 (ii)

Evaluate $\int (1 + \log x) dx$.

TYPE 13 Integration by Parts

Given: $\int (1 + \log x) dx$

Concept / Formula Used: Integration by parts: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Splitting the integral into two parts:

$$\int 1 dx + \int \log x dx$$

For the second term $\int \log x dx$, write it as $\int (\log x) \cdot 1 dx$ and apply integration by parts taking $\log x$ as the first function and 1 as the second function:

$$= x + \left(\log x \int 1 dx - \int \left(\frac{d}{dx} (\log x) \int 1 dx \right) dx \right)$$

Evaluating each component:

$$\begin{aligned}
 &= x + \left(\log x \cdot x - \int \frac{1}{x} \cdot x \, dx \right) \\
 &= x + x \log x - \int 1 \, dx \\
 &= x + x \log x - x + c \\
 &= x \log x + c
 \end{aligned}$$

Final Answer

$$\int (1 + \log x) \, dx = x \log x + c$$

Question 5 (i)

Evaluate $\int e^x \left(\sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \right) dx$.

TYPE 12b Substitution Involving Radical Expressions

Given: $\int e^x \left(\sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \right) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] \, dx = e^x f(x) + c$

Solution:

Let $f(x) = \sin^{-1} x$.

Differentiating with respect to x :

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x \left(\sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \right) dx = \int e^x [f(x) + f'(x)] \, dx$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] \, dx = e^x f(x) + c$:

$$= e^x \sin^{-1} x + c$$

Final Answer

$$\int e^x \left(\sin^{-1} x + \frac{1}{\sqrt{1-x^2}} \right) dx = e^x \sin^{-1} x + c$$

Question 5 (ii)

Evaluate $\int e^x(x \cos x + \sin x) dx$.

TYPE 12e Substitution for Rational Trigonometric Integrals

Given: $\int e^x(x \cos x + \sin x) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = x \sin x$.

Differentiating with respect to x using the product rule:

$$\begin{aligned} f'(x) &= x \cdot \frac{d}{dx}(\sin x) + \sin x \cdot \frac{d}{dx}(x) \\ &= x \cos x + \sin x \cdot 1 \\ &= x \cos x + \sin x \end{aligned}$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x(x \cos x + \sin x) dx = \int e^x[f(x) + f'(x)] dx$$

Applying the standard integral form $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x x \sin x + c$$

Final Answer

$$\int e^x(x \cos x + \sin x) dx = x e^x \sin x + c$$

Question 6 (i) [ISC 2024]

If $\int (\cot x - \csc^2 x)e^x dx = e^x f(x) + C$, then write the value of $f(x)$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int (\cot x - \csc^2 x)e^x dx = e^x f(x) + C$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Rewrite the integral as:

$$\int e^x(\cot x - \csc^2 x) dx = \int e^x(\cot x + (-\csc^2 x)) dx$$

Let $f(x) = \cot x$.

Differentiating with respect to x :

$$f'(x) = -\csc^2 x$$

Applying the standard integral form $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$:

$$\int e^x(\cot x - \csc^2 x) dx = e^x \cot x + C$$

Comparing $e^x \cot x + C$ with the given expression $e^x f(x) + C$:

$$f(x) = \cot x$$

Final Answer

$$f(x) = \cot x$$

Question 6 (ii)

If $\int \left(\frac{x-1}{x^2}\right) e^x dx = f(x)e^x + C$, then write the value of $f(x)$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \left(\frac{x-1}{x^2}\right) e^x dx = f(x)e^x + C$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Split the integrand into separate fractions:

$$\int e^x \left(\frac{x}{x^2} - \frac{1}{x^2}\right) dx = \int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$$

Let $f(x) = \frac{1}{x} = x^{-1}$.

Differentiating with respect to x :

$$f'(x) = -1 \cdot x^{-2} = -\frac{1}{x^2}$$

Applying the standard integral form $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$:

$$\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = e^x \left(\frac{1}{x} \right) + C$$

Comparing $e^x \left(\frac{1}{x} \right) + C$ with the given expression $f(x)e^x + C$:

$$f(x) = \frac{1}{x}$$

Final Answer

$$f(x) = \frac{1}{x}$$

Question 7 (i)

Evaluate $\int e^x(\tan x + \log \sec x) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x(\tan x + \log \sec x) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = \log(\sec x)$.

Differentiating with respect to x using the chain rule:

$$\begin{aligned} f'(x) &= \frac{1}{\sec x} \cdot \frac{d}{dx}(\sec x) \\ &= \frac{1}{\sec x} \cdot (\sec x \tan x) \\ &= \tan x \end{aligned}$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x(\tan x + \log \sec x) dx = \int e^x[f'(x) + f(x)] dx$$

Applying the standard integral form $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \log(\sec x) + c$$

Final Answer

$$\int e^x(\tan x + \log \sec x) dx = e^x \log(\sec x) + c$$

Question 7 (ii)

Evaluate $\int \frac{2x-1}{4x^2} e^{2x} dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{2x-1}{4x^2} e^{2x} dx$

Concept / Formula Used: Substitution method followed by Integration by Parts Form

Solution:

Let $2x = t$, which gives $x = \frac{t}{2}$.

Differentiating: $2 dx = dt$, so $dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\begin{aligned} &= \int \frac{2\left(\frac{t}{2}\right) - 1}{4\left(\frac{t}{2}\right)^2} e^t \cdot \frac{1}{2} dt \\ &= \int \frac{t - 1}{4\left(\frac{t^2}{4}\right)} e^t \cdot \frac{1}{2} dt \\ &= \frac{1}{2} \int \frac{t - 1}{t^2} e^t dt \\ &= \frac{1}{2} \int e^t \left(\frac{t}{t^2} - \frac{1}{t^2} \right) dt \\ &= \frac{1}{2} \int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt \end{aligned}$$

Let $f(t) = \frac{1}{t}$, then $f'(t) = -\frac{1}{t^2}$. Applying the standard form $\int e^t [f(t) + f'(t)] dt = e^t f(t) + c$:

$$= \frac{1}{2} e^t \left(\frac{1}{t} \right) + c$$

Now back-substitute $t = 2x$:

$$= \frac{1}{2} e^{2x} \left(\frac{1}{2x} \right) + c = \frac{e^{2x}}{4x} + c$$

Final Answer

$$\int \frac{2x - 1}{4x^2} e^{2x} dx = \frac{e^{2x}}{4x} + c$$

Question 8 (i)

Evaluate $\int \frac{\sin x \cos x - 1}{\sin^2 x} e^x dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{\sin x \cos x - 1}{\sin^2 x} e^x dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Split the integrand into separate terms:

$$\begin{aligned}
 &= \int e^x \left(\frac{\sin x \cos x}{\sin^2 x} - \frac{1}{\sin^2 x} \right) dx \\
 &= \int e^x \left(\frac{\cos x}{\sin x} - \csc^2 x \right) dx \\
 &= \int e^x (\cot x - \csc^2 x) dx
 \end{aligned}$$

Let $f(x) = \cot x$.

Differentiating with respect to x :

$$f'(x) = -\csc^2 x$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \cot x + c$$

Final Answer

$$\int \frac{\sin x \cos x - 1}{\sin^2 x} e^x dx = e^x \cot x + c$$

Question 8 (ii)

Evaluate $\int e^x (\cot x + \log |\sin x|) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^x (\cot x + \log |\sin x|) dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Let $f(x) = \log |\sin x|$.

Differentiating with respect to x using the chain rule:

$$\begin{aligned}
 f'(x) &= \frac{1}{\sin x} \cdot \frac{d}{dx}(\sin x) \\
 &= \frac{1}{\sin x} \cdot \cos x \\
 &= \cot x
 \end{aligned}$$

Substituting $f(x)$ and $f'(x)$ into the integral:

$$\int e^x (\cot x + \log |\sin x|) dx = \int e^x [f'(x) + f(x)] dx$$

$$\begin{aligned} \text{Applying the standard integral form } \int e^x [f(x) + f'(x)] dx &= e^x f(x) + c: \\ &= e^x \log |\sin x| + c \end{aligned}$$

Final Answer

$$\int e^x (\cot x + \log |\sin x|) dx = e^x \log |\sin x| + c$$

Question 9 (i)

Evaluate $\int \frac{xe^x}{(x+1)^2} dx$. (NCERT)

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{xe^x}{(x+1)^2} dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Rewrite the numerator by adding and subtracting 1:

$$\begin{aligned} &= \int e^x \left(\frac{(x+1) - 1}{(x+1)^2} \right) dx \\ &= \int e^x \left(\frac{x+1}{(x+1)^2} - \frac{1}{(x+1)^2} \right) dx \\ &= \int e^x \left(\frac{1}{x+1} - \frac{1}{(x+1)^2} \right) dx \end{aligned}$$

Let $f(x) = \frac{1}{x+1}$.

Differentiating with respect to x :

$$f'(x) = -\frac{1}{(x+1)^2} \cdot \frac{d}{dx}(x+1) = -\frac{1}{(x+1)^2}$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \left(\frac{1}{x+1} \right) + c = \frac{e^x}{x+1} + c$$

Final Answer

$$\int \frac{xe^x}{(x+1)^2} dx = \frac{e^x}{x+1} + c$$

Question 9 (ii)

Evaluate $\int \frac{x-1}{(x+1)^3} e^x dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{x-1}{(x+1)^3} e^x dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Rewrite the numerator in terms of $(x+1)$:

$$x-1 = (x+1) - 2$$

Substituting this into the integral:

$$\begin{aligned} &= \int e^x \left(\frac{(x+1) - 2}{(x+1)^3} \right) dx \\ &= \int e^x \left(\frac{x+1}{(x+1)^3} - \frac{2}{(x+1)^3} \right) dx \\ &= \int e^x \left(\frac{1}{(x+1)^2} - \frac{2}{(x+1)^3} \right) dx \end{aligned}$$

Let $f(x) = \frac{1}{(x+1)^2} = (x+1)^{-2}$.

Differentiating with respect to x :

$$f'(x) = -2(x+1)^{-3} \cdot 1 = -\frac{2}{(x+1)^3}$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \left(\frac{1}{(x+1)^2} \right) + c = \frac{e^x}{(x+1)^2} + c$$

Final Answer

$$\int \frac{x-1}{(x+1)^3} e^x dx = \frac{e^x}{(x+1)^2} + c$$

Question 10 (i)

Evaluate $\int \frac{x-3}{(x-1)^3} e^x dx$. (NCERT)

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{x-3}{(x-1)^3} e^x dx$

Concept / Formula Used: Using the standard integral form: $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$

Solution:

Rewrite the numerator in terms of $(x-1)$:

$$x-3 = (x-1) - 2$$

Substituting this into the integral:

$$\begin{aligned} &= \int e^x \left(\frac{(x-1)-2}{(x-1)^3} \right) dx \\ &= \int e^x \left(\frac{x-1}{(x-1)^3} - \frac{2}{(x-1)^3} \right) dx \\ &= \int e^x \left(\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \right) dx \end{aligned}$$

Let $f(x) = \frac{1}{(x-1)^2} = (x-1)^{-2}$.

Differentiating with respect to x :

$$f'(x) = -2(x-1)^{-3} \cdot 1 = -\frac{2}{(x-1)^3}$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \left(\frac{1}{(x-1)^2} \right) + c = \frac{e^x}{(x-1)^2} + c$$

Final Answer

$$\int \frac{x-3}{(x-1)^3} e^x dx = \frac{e^x}{(x-1)^2} + c$$

Question 10 (ii)

Evaluate $\int \frac{xe^{2x}}{(1+2x)^2} dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{xe^{2x}}{(1+2x)^2} dx$

Concept / Formula Used: Substitution method followed by algebraic manipulation

Solution:

Let $2x = t$, which gives $x = \frac{t}{2}$.

Differentiating: $2 dx = dt$, so $dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\begin{aligned} &= \int \frac{\left(\frac{t}{2}\right) e^t}{(1+t)^2} \cdot \frac{1}{2} dt \\ &= \frac{1}{4} \int \frac{te^t}{(1+t)^2} dt \end{aligned}$$

Rewrite the numerator t as $(t+1) - 1$:

$$\begin{aligned} &= \frac{1}{4} \int e^t \left(\frac{(t+1) - 1}{(1+t)^2} \right) dt \\ &= \frac{1}{4} \int e^t \left(\frac{t+1}{(1+t)^2} - \frac{1}{(1+t)^2} \right) dt \\ &= \frac{1}{4} \int e^t \left(\frac{1}{1+t} - \frac{1}{(1+t)^2} \right) dt \end{aligned}$$

Let $f(t) = \frac{1}{1+t}$, then $f'(t) = -\frac{1}{(1+t)^2}$. Applying the standard form $\int e^t[f(t) + f'(t)] dt = e^t f(t) + c$:

$$= \frac{1}{4} e^t \left(\frac{1}{1+t} \right) + c$$

Now back-substitute $t = 2x$:

$$= \frac{1}{4} e^{2x} \left(\frac{1}{1+2x} \right) + c = \frac{e^{2x}}{4(1+2x)} + c$$

Final Answer

$$\int \frac{xe^{2x}}{(1+2x)^2} dx = \frac{e^{2x}}{4(1+2x)} + c$$

Question 11 (i)

Evaluate $\int \frac{1 + \sin x}{1 + \cos x} e^x dx$. (NCERT)

TYPE 12e Substitution for Rational Trigonometric Integrals

Given: $\int \frac{1 + \sin x}{1 + \cos x} e^x dx$

Concept / Formula Used: Half-angle identities: $\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$, $\cos x = 2 \cos^2 \frac{x}{2} - 1$

Solution:

Using the multiple-angle identities to expand the integrand:

$$\begin{aligned} \frac{1 + \sin x}{1 + \cos x} &= \frac{1 + 2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \\ &= \frac{1}{2 \cos^2 \frac{x}{2}} + \frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \\ &= \frac{1}{2} \sec^2 \frac{x}{2} + \tan \frac{x}{2} \end{aligned}$$

Substituting this back into the integral:

$$\int e^x \left(\tan \frac{x}{2} + \frac{1}{2} \sec^2 \frac{x}{2} \right) dx$$

Let $f(x) = \tan \frac{x}{2}$.

Differentiating with respect to x using the chain rule:

$$f'(x) = \sec^2 \frac{x}{2} \cdot \frac{1}{2} = \frac{1}{2} \sec^2 \frac{x}{2}$$

Applying the standard integral form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \tan \frac{x}{2} + c$$

Final Answer

$$\int \frac{1 + \sin x}{1 + \cos x} e^x dx = e^x \tan \frac{x}{2} + c$$

Question 11 (ii)

Evaluate $\int \frac{2 + \sin 2x}{1 + \cos 2x} e^x dx$. (NCERT)

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{2 + \sin 2x}{1 + \cos 2x} e^x dx$

Concept / Formula Used: Double-angle identities: $\sin 2x = 2 \sin x \cos x$, $\cos 2x = 2 \cos^2 x - 1$

Solution:

Using the double-angle identities to expand the integrand:

$$\begin{aligned}\frac{2 + \sin 2x}{1 + \cos 2x} &= \frac{2 + 2 \sin x \cos x}{2 \cos^2 x} \\ &= \frac{2}{2 \cos^2 x} + \frac{2 \sin x \cos x}{2 \cos^2 x} \\ &= \sec^2 x + \tan x\end{aligned}$$

Substituting this back into the integral:

$$\int e^x (\tan x + \sec^2 x) dx$$

Let $f(x) = \tan x$, then $f'(x) = \sec^2 x$. Applying the standard form $\int e^x [f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \tan x + c$$

Final Answer

$$\int \frac{2 + \sin 2x}{1 + \cos 2x} e^x dx = e^x \tan x + c$$

Question 12 (i)

Evaluate $\int \frac{\sin 4x - 4}{1 - \cos 4x} e^x dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int \frac{\sin 4x - 4}{1 - \cos 4x} e^x dx$

Concept / Formula Used: Double-angle identities: $\sin 4x = 2 \sin 2x \cos 2x$, $1 - \cos 4x = 2 \sin^2 2x$

Solution:

Using the double-angle identities to simplify the integrand:

$$\begin{aligned}\frac{\sin 4x - 4}{1 - \cos 4x} &= \frac{2 \sin 2x \cos 2x - 4}{2 \sin^2 2x} \\ &= \frac{2 \sin 2x \cos 2x}{2 \sin^2 2x} - \frac{4}{2 \sin^2 2x} \\ &= \frac{\cos 2x}{\sin 2x} - \frac{2}{\sin^2 2x} \\ &= \cot 2x - 2 \csc^2 2x\end{aligned}$$

Substituting this into the integral:

$$\int e^x (\cot 2x - 2 \csc^2 2x) dx$$

Let $f(x) = \cot 2x$.

Differentiating with respect to x using the chain rule:

$$f'(x) = -2 \csc^2 2x$$

Applying the standard integral form $\int e^x[f(x) + f'(x)] dx = e^x f(x) + c$:

$$= e^x \cot 2x + c$$

Final Answer

$$\int \frac{\sin 4x - 4}{1 - \cos 4x} e^x dx = e^x \cot 2x + c$$

Question 12 (ii)

Evaluate $\int (\cos x + 3 \sin x) e^{3x} dx$.

TYPE 13 Integration by Parts

Given: $\int (\cos x + 3 \sin x) e^{3x} dx$

Concept / Formula Used: Using the standard integral form: $\int e^{ax}[f(x) + \frac{1}{a}f'(x)] dx = \frac{1}{a}e^{ax}f(x) + c$, applied with $a = 3$

Solution:

Let $f(x) = \sin x$, so $f'(x) = \cos x$.

Rewrite the integrand to match the form $e^{3x}[af(x) + f'(x)]$ with $a = 3$:

$$\cos x + 3 \sin x = f'(x) + 3f(x)$$

Applying integration by parts directly: take $u = \cos x + 3 \sin x$ and $\int e^{3x} dx = \frac{e^{3x}}{3}$.

$$\begin{aligned} I &= (\cos x + 3 \sin x) \cdot \frac{e^{3x}}{3} - \int (-\sin x + 3 \cos x) \cdot \frac{e^{3x}}{3} dx \\ &= \frac{e^{3x}}{3} (\cos x + 3 \sin x) - \frac{1}{3} \int e^{3x} (3 \cos x - \sin x) dx \end{aligned}$$

Apply integration by parts a second time to the remaining integral, again taking $\int e^{3x} dx = \frac{e^{3x}}{3}$:

$$\begin{aligned} \int e^{3x} (3 \cos x - \sin x) dx &= (3 \cos x - \sin x) \cdot \frac{e^{3x}}{3} - \int (-3 \sin x - \cos x) \cdot \frac{e^{3x}}{3} dx \\ &= \frac{e^{3x}}{3} (3 \cos x - \sin x) + \frac{1}{3} \int e^{3x} (3 \sin x + \cos x) dx \\ &= \frac{e^{3x}}{3} (3 \cos x - \sin x) + \frac{1}{3} I \end{aligned}$$

Substituting this back into the expression for I :

$$\begin{aligned} I &= \frac{e^{3x}}{3}(\cos x + 3 \sin x) - \frac{1}{3} \left[\frac{e^{3x}}{3}(3 \cos x - \sin x) + \frac{1}{3}I \right] \\ &= \frac{e^{3x}}{3}(\cos x + 3 \sin x) - \frac{e^{3x}}{9}(3 \cos x - \sin x) - \frac{1}{9}I \end{aligned}$$

Collecting the I terms on the left:

$$\frac{10}{9}I = e^{3x} \left[\frac{\cos x + 3 \sin x}{3} - \frac{3 \cos x - \sin x}{9} \right] = e^{3x} \left[\frac{3 \cos x + 9 \sin x - 3 \cos x + \sin x}{9} \right] = \frac{10}{9}e^{3x} \sin x$$

Dividing both sides by $\frac{10}{9}$:

$$I = e^{3x} \sin x + c$$

Final Answer

$$\int (\cos x + 3 \sin x)e^{3x} dx = e^{3x} \sin x + c$$

Question 13 (i)

Evaluate $\int (\sin(\log x) + \cos(\log x)) dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int (\sin(\log x) + \cos(\log x)) dx$

Concept / Formula Used: Substitution: let $\log x = t$, so $x = e^t$ and $dx = e^t dt$

Solution:

Let $\log x = t$, which implies $x = e^t$.

Differentiating: $dx = e^t dt$.

Substituting into the integral:

$$\int (\sin t + \cos t)e^t dt$$

Let $f(t) = \sin t$.

Differentiating with respect to t :

$$f'(t) = \cos t$$

Applying the standard integral form $\int e^t[f(t) + f'(t)] dt = e^t f(t) + c$:

$$= e^t \sin t + c$$

Now back-substitute $t = \log x$ and $e^t = x$:

$$= x \sin(\log x) + c$$

Final Answer

$$\int (\sin(\log x) + \cos(\log x)) dx = x \sin(\log x) + c$$

Question 13 (ii)

Evaluate $\int \frac{\log x - 3}{(\log x)^4} dx$.

TYPE 12a Direct/Algebraic Substitution

Given: $\int \frac{\log x - 3}{(\log x)^4} dx$

Concept / Formula Used: Substitution: let $\log x = t$, so $x = e^t$ and $dx = e^t dt$

Solution:

Let $\log x = t$, which implies $x = e^t$.

Differentiating: $dx = e^t dt$.

Substituting into the integral:

$$\int \frac{t - 3}{t^4} e^t dt$$

Split the integrand into separate fractions:

$$\int e^t \left(\frac{t}{t^4} - \frac{3}{t^4} \right) dt = \int e^t \left(\frac{1}{t^3} - \frac{3}{t^4} \right) dt$$

Let $f(t) = \frac{1}{t^3} = t^{-3}$.

Differentiating with respect to t :

$$f'(t) = -3t^{-4} = -\frac{3}{t^4}$$

Applying the standard integral form $\int e^t [f(t) + f'(t)] dt = e^t f(t) + c$:

$$= e^t \left(\frac{1}{t^3} \right) + c$$

Now back-substitute $t = \log x$ and $e^t = x$:

$$= x \left(\frac{1}{(\log x)^3} \right) + c = \frac{x}{(\log x)^3} + c$$

Final Answer

$$\int \frac{\log x - 3}{(\log x)^4} dx = \frac{x}{(\log x)^3} + c$$

Question 13 (iii)

Evaluate $\int e^{\cot^{-1} x} \left(\frac{1-x+x^2}{1+x^2} \right) dx$.

TYPE 12c Substitution Involving Exponential Functions

Given: $\int e^{\cot^{-1} x} \left(\frac{1-x+x^2}{1+x^2} \right) dx$

Concept / Formula Used: Substitution: let $\cot^{-1} x = t$, so $x = \cot t$ and $dx = -\csc^2 t dt$

Solution:

Let $\cot^{-1} x = t$, which implies $x = \cot t$.

Differentiating: $dx = -\csc^2 t dt$.

Also, note that $\frac{1}{1+x^2} = \sin^2 t$ or use the substitution directly:

$$\begin{aligned} 1-x+x^2 &= 1-\cot t+\cot^2 t \\ &= (1+\cot^2 t)-\cot t \\ &= \csc^2 t-\cot t \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} &\int e^t \left(\frac{\csc^2 t - \cot t}{1 + \cot^2 t} \right) (-\csc^2 t dt) \\ &= \int e^t \left(\frac{\csc^2 t - \cot t}{\csc^2 t} \right) (-\csc^2 t dt) \\ &= -\int e^t (\csc^2 t - \cot t) dt \\ &= \int e^t (\cot t - \csc^2 t) dt \end{aligned}$$

Let $f(t) = \cot t$, then $f'(t) = -\csc^2 t$. Applying the standard form $\int e^t [f(t) + f'(t)] dt = e^t f(t) + c$:

$$= e^t \cot t + c$$

Now back-substitute $t = \cot^{-1} x$ and $\cot t = x$:

$$= e^{\cot^{-1} x} \cdot x + c = x e^{\cot^{-1} x} + c$$

Final Answer

$$\int e^{\cot^{-1} x} \left(\frac{1-x+x^2}{1+x^2} \right) dx = x e^{\cot^{-1} x} + c$$

7.13 – Integrals Involving the Square Root of a-squared minus x-squared

Question 1 (i)

Evaluate $\int \sqrt{4 - x^2} dx$.

TYPE 1 Power Rule of Integration

Concept / Formula Used: Using the standard result: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Given integral:

$$\int \sqrt{4 - x^2} dx = \int \sqrt{(2)^2 - x^2} dx$$

Applying the standard result with $a = 2$:

$$\begin{aligned} &= \frac{x}{2} \sqrt{(2)^2 - x^2} + \frac{(2)^2}{2} \sin^{-1} \left(\frac{x}{2} \right) + C \\ &= \frac{x}{2} \sqrt{4 - x^2} + \frac{4}{2} \sin^{-1} \left(\frac{x}{2} \right) + C \\ &= \frac{x}{2} \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) + C \end{aligned}$$

Final Answer

$$\int \sqrt{4 - x^2} dx = \frac{1}{2} x \sqrt{4 - x^2} + 2 \sin^{-1} \left(\frac{x}{2} \right) + C$$

Common Mistake: Confusing the formula for $\sqrt{a^2 - x^2}$ with $\sqrt{x^2 - a^2}$, leading to an incorrect logarithmic term instead of $\sin^{-1}$.

Question 1 (ii)

Evaluate $\int \sqrt{x^2 - 9} dx$.

TYPE 1 Power Rule of Integration

Concept / Formula Used: Using the standard result: $\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log \left| x + \sqrt{x^2 - a^2} \right| + C$

Solution:

Given integral:

$$\int \sqrt{x^2 - 9} \, dx = \int \sqrt{x^2 - (3)^2} \, dx$$

Applying the standard result with $a = 3$:

$$\begin{aligned} &= \frac{x}{2} \sqrt{x^2 - (3)^2} - \frac{(3)^2}{2} \log \left| x + \sqrt{x^2 - (3)^2} \right| + C \\ &= \frac{x}{2} \sqrt{x^2 - 9} - \frac{9}{2} \log \left| x + \sqrt{x^2 - 9} \right| + C \end{aligned}$$

Final Answer

$$\int \sqrt{x^2 - 9} \, dx = \frac{x\sqrt{x^2 - 9}}{2} - \frac{9}{2} \log \left| x + \sqrt{x^2 - 9} \right| + C$$

Common Mistake: Using a plus sign before the logarithmic term instead of a minus sign.

Question 2 (i)

Evaluate $\int \sqrt{1 + x^2} \, dx$.

TYPE 1 Power Rule of Integration

Concept / Formula Used: Using the standard result: $\int \sqrt{x^2 + a^2} \, dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log \left| x + \sqrt{x^2 + a^2} \right| + C$

Solution:

Given integral:

$$\int \sqrt{1 + x^2} \, dx = \int \sqrt{x^2 + (1)^2} \, dx$$

Applying the standard result with $a = 1$:

$$\begin{aligned} &= \frac{x}{2} \sqrt{x^2 + (1)^2} + \frac{(1)^2}{2} \log \left| x + \sqrt{x^2 + (1)^2} \right| + C \\ &= \frac{x}{2} \sqrt{1 + x^2} + \frac{1}{2} \log \left| x + \sqrt{1 + x^2} \right| + C \end{aligned}$$

Final Answer

$$\int \sqrt{1 + x^2} \, dx = \frac{x\sqrt{1 + x^2}}{2} + \frac{1}{2} \log \left| x + \sqrt{1 + x^2} \right| + C$$

Common Mistake: Neglecting the constant factor $\frac{1}{2}$ attached to the logarithmic term.

Question 2 (ii)

Evaluate $\int \sqrt{1 - 4x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Using the standard result: $\int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1}\left(\frac{u}{a}\right) + C$

Solution:

Let $2x = t$.

Differentiating: $2 dx = dt$, so $dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\int \sqrt{1 - (2x)^2} dx = \int \sqrt{1 - t^2} \left(\frac{1}{2} dt\right) = \frac{1}{2} \int \sqrt{1^2 - t^2} dt$$

Applying the standard result with $a = 1$:

$$= \frac{1}{2} \left[\frac{t}{2} \sqrt{1 - t^2} + \frac{1^2}{2} \sin^{-1}\left(\frac{t}{1}\right) \right] + C$$

Simplifying the expression:

$$= \frac{t}{4} \sqrt{1 - t^2} + \frac{1}{4} \sin^{-1}(t) + C$$

Now back-substitute $t = 2x$:

$$\begin{aligned} &= \frac{2x}{4} \sqrt{1 - (2x)^2} + \frac{1}{4} \sin^{-1}(2x) + C \\ &= \frac{x}{2} \sqrt{1 - 4x^2} + \frac{1}{4} \sin^{-1}(2x) + C \end{aligned}$$

Final Answer

$$\int \sqrt{1 - 4x^2} dx = \frac{x\sqrt{1 - 4x^2}}{2} + \frac{1}{4} \sin^{-1}(2x) + C$$

Common Mistake: Forgetting to divide by the coefficient of x (i.e., multiplying by $\frac{1}{2}$) arising from the substitution $2x = t$.

Question 3 (i)

Evaluate $\int \sqrt{4x^2 - 9} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Using the standard result: $\int \sqrt{u^2 - a^2} du = \frac{u}{2} \sqrt{u^2 - a^2} - \frac{a^2}{2} \log \left| u + \sqrt{u^2 - a^2} \right| + C$

Solution:

Let $2x = t$.

Differentiating: $2 dx = dt$, so $dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\int \sqrt{(2x)^2 - 3^2} dx = \int \sqrt{t^2 - 3^2} \left(\frac{1}{2} dt \right) = \frac{1}{2} \int \sqrt{t^2 - 3^2} dt$$

Applying the standard result with $a = 3$:

$$= \frac{1}{2} \left[\frac{t}{2} \sqrt{t^2 - 3^2} - \frac{3^2}{2} \log \left| t + \sqrt{t^2 - 3^2} \right| \right] + C$$

Simplifying the expression:

$$= \frac{t}{4} \sqrt{t^2 - 9} - \frac{9}{4} \log \left| t + \sqrt{t^2 - 9} \right| + C$$

Now back-substitute $t = 2x$:

$$\begin{aligned} &= \frac{2x}{4} \sqrt{(2x)^2 - 9} - \frac{9}{4} \log \left| 2x + \sqrt{(2x)^2 - 9} \right| + C \\ &= \frac{x}{2} \sqrt{4x^2 - 9} - \frac{9}{4} \log \left| 2x + \sqrt{4x^2 - 9} \right| + C \end{aligned}$$

Final Answer

$$\int \sqrt{4x^2 - 9} dx = \frac{x\sqrt{4x^2 - 9}}{2} - \frac{9}{4} \log \left| 2x + \sqrt{4x^2 - 9} \right| + C$$

Common Mistake: Omitting the factor of 2 inside the absolute value argument of the logarithm ($\log |2x + \dots|$) from back-substitution.

Question 3 (ii)

Evaluate $\int \sqrt{4 - 9x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Using the standard result: $\int \sqrt{a^2 - u^2} du = \frac{u}{2} \sqrt{a^2 - u^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{u}{a} \right) + C$

Solution:

Let $3x = t$.

Differentiating: $3 dx = dt$, so $dx = \frac{1}{3} dt$.

Substituting into the integral:

$$\int \sqrt{2^2 - (3x)^2} dx = \int \sqrt{2^2 - t^2} \left(\frac{1}{3} dt \right) = \frac{1}{3} \int \sqrt{2^2 - t^2} dt$$

Applying the standard result with $a = 2$:

$$= \frac{1}{3} \left[\frac{t}{2} \sqrt{2^2 - t^2} + \frac{2^2}{2} \sin^{-1} \left(\frac{t}{2} \right) \right] + C$$

Simplifying the expression:

$$\begin{aligned} &= \frac{t}{6} \sqrt{4 - t^2} + \frac{4}{6} \sin^{-1} \left(\frac{t}{2} \right) + C \\ &= \frac{t}{6} \sqrt{4 - t^2} + \frac{2}{3} \sin^{-1} \left(\frac{t}{2} \right) + C \end{aligned}$$

Now back-substitute $t = 3x$:

$$\begin{aligned} &= \frac{3x}{6} \sqrt{4 - (3x)^2} + \frac{2}{3} \sin^{-1} \left(\frac{3x}{2} \right) + C \\ &= \frac{x}{2} \sqrt{4 - 9x^2} + \frac{2}{3} \sin^{-1} \left(\frac{3x}{2} \right) + C \end{aligned}$$

Final Answer

$$\int \sqrt{4 - 9x^2} dx = \frac{x\sqrt{4 - 9x^2}}{2} + \frac{2}{3} \sin^{-1} \left(\frac{3x}{2} \right) + C$$

Common Mistake: Dividing the coefficient of $\sin^{-1}$ by 2 instead of 3 when substituting back.

Question 4 (i)

Evaluate $\int \frac{x^2}{\sqrt{x^2+6}} dx$.

TYPE 1 Power Rule of Integration

★ Likely Exam Question

Concept / Formula Used: $\int \frac{x^2}{\sqrt{x^2+a^2}} dx = \frac{x}{2}\sqrt{x^2+a^2} - \frac{a^2}{2} \log |x + \sqrt{x^2+a^2}| + C$

Solution:

Rewrite the numerator using algebraic manipulation:

$$x^2 = (x^2 + 6) - 6$$

Substitute this back into the integral:

$$\begin{aligned} \int \frac{x^2}{\sqrt{x^2+6}} dx &= \int \frac{(x^2+6)-6}{\sqrt{x^2+6}} dx \\ &= \int \left(\frac{x^2+6}{\sqrt{x^2+6}} - \frac{6}{\sqrt{x^2+6}} \right) dx \\ &= \int \sqrt{x^2+6} dx - 6 \int \frac{dx}{\sqrt{x^2+6}} \end{aligned}$$

Now, apply the standard results for each integral: 1. $\int \sqrt{x^2 + (\sqrt{6})^2} dx = \frac{x}{2}\sqrt{x^2+6} + \frac{6}{2} \log |x + \sqrt{x^2+6}|$ 2. $\int \frac{dx}{\sqrt{x^2 + (\sqrt{6})^2}} = \log |x + \sqrt{x^2+6}|$

Combining the results:

$$\begin{aligned} &= \left[\frac{x}{2}\sqrt{x^2+6} + 3 \log |x + \sqrt{x^2+6}| \right] - 6 \left[\log |x + \sqrt{x^2+6}| \right] + C \\ &= \frac{x}{2}\sqrt{x^2+6} + 3 \log |x + \sqrt{x^2+6}| - 6 \log |x + \sqrt{x^2+6}| + C \\ &= \frac{x}{2}\sqrt{x^2+6} - 3 \log |x + \sqrt{x^2+6}| + C \end{aligned}$$

Final Answer

$$\int \frac{x^2}{\sqrt{x^2+6}} dx = \frac{x\sqrt{x^2+6}}{2} - 3 \log |x + \sqrt{x^2+6}| + C$$

Common Mistake: Incorrectly combining the logarithmic terms from the two component integrals.

Question 4 (ii)

Evaluate $\int \frac{1}{x - \sqrt{x^2 - 1}} dx$.

TYPE 5 Conjugate Multiplication

Concept / Formula Used: Rationalising factor for $a - \sqrt{b}$ is $a + \sqrt{b}$

Solution:

Rationalise the denominator by multiplying the numerator and the denominator by its conjugate $x + \sqrt{x^2 - 1}$:

$$\begin{aligned} \int \frac{1}{x - \sqrt{x^2 - 1}} dx &= \int \frac{x + \sqrt{x^2 - 1}}{(x - \sqrt{x^2 - 1})(x + \sqrt{x^2 - 1})} dx \\ &= \int \frac{x + \sqrt{x^2 - 1}}{x^2 - (x^2 - 1)} dx \\ &= \int \frac{x + \sqrt{x^2 - 1}}{x^2 - x^2 + 1} dx \\ &= \int (x + \sqrt{x^2 - 1}) dx \\ &= \int x dx + \int \sqrt{x^2 - 1}^2 dx \end{aligned}$$

Integrating each term separately:

$$\begin{aligned} \int x dx &= \frac{x^2}{2} \\ \int \sqrt{x^2 - 1}^2 dx &= \frac{x}{2} \sqrt{x^2 - 1} - \frac{1}{2} \log |x + \sqrt{x^2 - 1}| \end{aligned}$$

Combining both parts:

$$= \frac{x^2}{2} + \frac{x}{2} \sqrt{x^2 - 1} - \frac{1}{2} \log |x + \sqrt{x^2 - 1}| + C$$

Final Answer

$$\int \frac{1}{x - \sqrt{x^2 - 1}} dx = \frac{1}{2} \left(x^2 + x\sqrt{x^2 - 1} - \log |x + \sqrt{x^2 - 1}| \right) + C$$

Common Mistake: Expanding the denominator incorrectly as $(x^2 - x^2 - 1)$ instead of keeping track of the negative sign before the radical term.

Question 5 (i)

Evaluate $\int \frac{x^2 + 3}{\sqrt{x^2 - 25}} dx$.

TYPE 1 Power Rule of Integration

Concept / Formula Used: $\int \frac{x^2}{\sqrt{x^2 - a^2}} dx = \frac{x}{2} \sqrt{x^2 - a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2 - a^2}| + C$

Solution:

Rewrite the numerator using algebraic adjustment:

$$x^2 + 3 = (x^2 - 25) + 28$$

Substitute this back into the integral:

$$\begin{aligned} \int \frac{x^2 + 3}{\sqrt{x^2 - 25}} dx &= \int \frac{(x^2 - 25) + 28}{\sqrt{x^2 - 25}} dx \\ &= \int \left(\frac{x^2 - 25}{\sqrt{x^2 - 25}} + \frac{28}{\sqrt{x^2 - 25}} \right) dx \\ &= \int \sqrt{x^2 - 25} dx + 28 \int \frac{dx}{\sqrt{x^2 - 25}} \end{aligned}$$

Applying the standard results: 1. $\int \sqrt{x^2 - 5^2} dx = \frac{x}{2} \sqrt{x^2 - 25} - \frac{25}{2} \log |x + \sqrt{x^2 - 25}|$

2. $\int \frac{dx}{\sqrt{x^2 - 5^2}} = \log |x + \sqrt{x^2 - 25}|$

Combining the results:

$$\begin{aligned} &= \left[\frac{x}{2} \sqrt{x^2 - 25} - \frac{25}{2} \log |x + \sqrt{x^2 - 25}| \right] + 28 \left[\log |x + \sqrt{x^2 - 25}| \right] + C \\ &= \frac{x}{2} \sqrt{x^2 - 25} + \left(28 - \frac{25}{2} \right) \log |x + \sqrt{x^2 - 25}| + C \\ &= \frac{x}{2} \sqrt{x^2 - 25} + \frac{31}{2} \log |x + \sqrt{x^2 - 25}| + C \end{aligned}$$

Final Answer

$$\int \frac{x^2 + 3}{\sqrt{x^2 - 25}} dx = \frac{1}{2} x \sqrt{x^2 - 25} + \frac{31}{2} \log |x + \sqrt{x^2 - 25}| + C$$

Common Mistake: Arithmetic error when combining $-\frac{25}{2} + 28$ to get $\frac{31}{2}$.

Question 5 (ii)

Evaluate $\int (x - 3) \sqrt{\frac{x + 2}{x - 2}} dx$.

TYPE 1 Power Rule of Integration

Concept / Formula Used: $\int \sqrt{a^2 - u^2} du$

Solution:

Multiply numerator and denominator inside the square root by $(x + 2)$:

$$\int (x-3) \sqrt{\frac{(x+2)(x+2)}{(x-2)(x+2)}} dx = \int (x-3) \frac{x+2}{\sqrt{x^2-4}} dx$$

Expand the numerator:

$$(x-3)(x+2) = x^2 - x - 6$$

Thus, the integral becomes:

$$\int \frac{x^2 - x - 6}{\sqrt{x^2 - 4}} dx = \int \frac{x^2 - 4}{\sqrt{x^2 - 4}} dx - \int \frac{x}{\sqrt{x^2 - 4}} dx - \int \frac{2}{\sqrt{x^2 - 4}} dx$$

Simplifying each term: 1. $\int \sqrt{x^2 - 4} dx = \frac{x}{2} \sqrt{x^2 - 4} - 2 \log |x + \sqrt{x^2 - 4}|$ 2. For

$\int \frac{x}{\sqrt{x^2 - 4}} dx$, let $x^2 - 4 = t$, so $2x dx = dt \Rightarrow x dx = \frac{1}{2} dt$:

$$\int \frac{\frac{1}{2} dt}{\sqrt{t}} = \frac{1}{2} (2\sqrt{t}) = \sqrt{x^2 - 4}$$

$$3. 2 \int \frac{dx}{\sqrt{x^2 - 4}} = 2 \log |x + \sqrt{x^2 - 4}|$$

Combining all parts:

$$\left(\frac{x}{2} \sqrt{x^2 - 4} - 2 \log |x + \sqrt{x^2 - 4}| \right) - \sqrt{x^2 - 4} - 2 \log |x + \sqrt{x^2 - 4}| + C$$

Group like terms:

$$\begin{aligned} &= \left(\frac{x}{2} - 1 \right) \sqrt{x^2 - 4} - 4 \log |x + \sqrt{x^2 - 4}| + C \\ &= \frac{x-2}{2} \sqrt{x^2 - 4} - 4 \log |x + \sqrt{x^2 - 4}| + C \end{aligned}$$

Notice that $\frac{x-2}{2} \sqrt{x^2 - 4} = \frac{(x-2)}{2\sqrt{x-2}\sqrt{x+2}} \sqrt{x-2}\sqrt{x+2} \dots$ can also be factored as:

$$= \frac{1}{2} (x-2) \sqrt{x^2 - 4} - 4 \log |x + \sqrt{x^2 - 4}| + C$$

Final Answer

$$\int (x-3) \sqrt{\frac{x+2}{x-2}} dx = \frac{1}{2} (x-2) \sqrt{x^2 - 4} - 4 \log |x + \sqrt{x^2 - 4}| + C$$

Common Mistake: Failing to split the integral correctly into three manageable parts after rationalising the numerator.

Question 6 (i)

Evaluate $\int x\sqrt{x^4-1} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: $\int \sqrt{t^2 - a^2} dt = \frac{t}{2}\sqrt{t^2 - a^2} - \frac{a^2}{2} \log |t + \sqrt{t^2 - a^2}| + C$

Solution:

Let $x^2 = t$.

Differentiating: $2x dx = dt$, so $x dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\int \sqrt{(x^2)^2 - 1} (x dx) = \int \sqrt{t^2 - 1} \left(\frac{1}{2} dt \right) = \frac{1}{2} \int \sqrt{t^2 - 1^2} dt$$

Applying the standard result with $a = 1$:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{t}{2} \sqrt{t^2 - 1} - \frac{1^2}{2} \log |t + \sqrt{t^2 - 1}| \right] + C \\ &= \frac{t}{4} \sqrt{t^2 - 1} - \frac{1}{4} \log |t + \sqrt{t^2 - 1}| + C \end{aligned}$$

Now back-substitute $t = x^2$:

$$\begin{aligned} &= \frac{x^2}{4} \sqrt{(x^2)^2 - 1} - \frac{1}{4} \log |x^2 + \sqrt{(x^2)^2 - 1}| + C \\ &= \frac{1}{4} x^2 \sqrt{x^4 - 1} - \frac{1}{4} \log |x^2 + \sqrt{x^4 - 1}| + C \end{aligned}$$

Final Answer

$$\int x\sqrt{x^4-1} dx = \frac{1}{4} x^2 \sqrt{x^4-1} - \frac{1}{4} \log |x^2 + \sqrt{x^4-1}| + C$$

Common Mistake: Forgetting to substitute back $t = x^2$ or missing the factor of $\frac{1}{4}$.

Question 6 (ii)

Evaluate $\int \frac{\sqrt{9 - (\log x)^2}}{x} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: $\int \sqrt{a^2 - t^2} dt = \frac{t}{2} \sqrt{a^2 - t^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{t}{a} \right) + C$

Solution:

Let $\log x = t$.

Differentiating: $\frac{1}{x} dx = dt$.

Substituting into the integral:

$$\int \sqrt{9 - (\log x)^2} \left(\frac{1}{x} dx \right) = \int \sqrt{3^2 - t^2} dt$$

Applying the standard result with $a = 3$:

$$\begin{aligned} &= \frac{t}{2} \sqrt{3^2 - t^2} + \frac{3^2}{2} \sin^{-1} \left(\frac{t}{3} \right) + C \\ &= \frac{t}{2} \sqrt{9 - t^2} + \frac{9}{2} \sin^{-1} \left(\frac{t}{3} \right) + C \end{aligned}$$

Now back-substitute $t = \log x$:

$$= \frac{\log x}{2} \sqrt{9 - (\log x)^2} + \frac{9}{2} \sin^{-1} \left(\frac{\log x}{3} \right) + C$$

Final Answer

$$\int \frac{\sqrt{9 - (\log x)^2}}{x} dx = \frac{1}{2} \log x \sqrt{9 - (\log x)^2} + \frac{9}{2} \sin^{-1} \left(\frac{\log x}{3} \right) + C$$

Common Mistake: Failing to identify $\frac{1}{x} dx$ as the differential dt , missing the direct substitution.

Question 7 (i)

Evaluate $\int \sin x \cos x \sqrt{\sin^4 x + 4} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: $\int \sqrt{t^2 + a^2} dt = \frac{t}{2} \sqrt{t^2 + a^2} + \frac{a^2}{2} \log |t + \sqrt{t^2 + a^2}| + C$

Solution:

Let $\sin^2 x = t$.

Differentiating: $2 \sin x \cos x dx = dt$, so $\sin x \cos x dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\int \sqrt{(\sin^2 x)^2 + 4} (\sin x \cos x dx) = \int \sqrt{t^2 + 2^2} \left(\frac{1}{2} dt \right) = \frac{1}{2} \int \sqrt{t^2 + 2^2} dt$$

Applying the standard result with $a = 2$:

$$= \frac{1}{2} \left[\frac{t}{2} \sqrt{t^2 + 2^2} + \frac{2^2}{2} \log |t + \sqrt{t^2 + 2^2}| \right] + C$$

$$\begin{aligned}
 &= \frac{t}{4}\sqrt{t^2+4} + \frac{4}{4}\log|t + \sqrt{t^2+4}| + C \\
 &= \frac{t}{4}\sqrt{t^2+4} + \log|t + \sqrt{t^2+4}| + C
 \end{aligned}$$

Now back-substitute $t = \sin^2 x$:

$$\begin{aligned}
 &= \frac{\sin^2 x}{4}\sqrt{(\sin^2 x)^2+4} + \log|\sin^2 x + \sqrt{(\sin^2 x)^2+4}| + C \\
 &= \frac{1}{4}\sin^2 x\sqrt{\sin^4 x+4} + \log|\sin^2 x + \sqrt{\sin^4 x+4}| + C
 \end{aligned}$$

Final Answer

$$\int \sin x \cos x \sqrt{\sin^4 x + 4} dx = \frac{1}{4}\sin^2 x \sqrt{\sin^4 x + 4} + \log|\sin^2 x + \sqrt{\sin^4 x + 4}| + C$$

Common Mistake: Choosing $\sin x = t$ instead of $\sin^2 x = t$, which leaves an extra trigonometric term that cannot be easily eliminated.

Question 7 (ii)

Evaluate $\int \frac{\log x}{x} \sqrt{(\log x)^4 - 1} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: $\int \sqrt{t^2 - a^2} dt = \frac{t}{2}\sqrt{t^2 - a^2} - \frac{a^2}{2}\log|t + \sqrt{t^2 - a^2}| + C$

Solution:

Let $(\log x)^2 = t$.

Differentiating: $2(\log x) \cdot \frac{1}{x} dx = dt$, so $\frac{\log x}{x} dx = \frac{1}{2} dt$.

Substituting into the integral:

$$\int \sqrt{((\log x)^2)^2 - 1} \left(\frac{\log x}{x} dx \right) = \int \sqrt{t^2 - 1} \left(\frac{1}{2} dt \right) = \frac{1}{2} \int \sqrt{t^2 - 1^2} dt$$

Applying the standard result with $a = 1$:

$$\begin{aligned}
 &= \frac{1}{2} \left[\frac{t}{2}\sqrt{t^2-1} - \frac{1^2}{2}\log|t + \sqrt{t^2-1}| \right] + C \\
 &= \frac{t}{4}\sqrt{t^2-1} - \frac{1}{4}\log|t + \sqrt{t^2-1}| + C
 \end{aligned}$$

Now back-substitute $t = (\log x)^2$:

$$= \frac{(\log x)^2}{4}\sqrt{((\log x)^2)^2 - 1} - \frac{1}{4}\log|(\log x)^2 + \sqrt{((\log x)^2)^2 - 1}| + C$$

$$= \frac{1}{4}(\log x)^2 \sqrt{(\log x)^4 - 1} - \frac{1}{4} \log \left| (\log x)^2 + \sqrt{(\log x)^4 - 1} \right| + C$$

Final Answer

$$\int_C \frac{\log x}{x} \sqrt{(\log x)^4 - 1} dx = \frac{1}{4} \left((\log x)^2 \sqrt{(\log x)^4 - 1} - \log \left| (\log x)^2 + \sqrt{(\log x)^4 - 1} \right| \right) +$$

Common Mistake: Substituting only $\log x = t$ instead of $(\log x)^2 = t$, failing to simplify the radical expression.

Question 8 (i)

Evaluate $\int x \cos^{-1} x dx$.

TYPE 13 Integration by Parts

★ Likely Exam Question

Concept / Formula Used: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Using integration by parts, taking $u = \cos^{-1} x$ and $v = x$:

$$\int x \cos^{-1} x dx = \cos^{-1} x \int x dx - \int \left(\frac{d}{dx}(\cos^{-1} x) \int x dx \right) dx$$

Evaluate the elementary integral $\int x dx = \frac{x^2}{2}$ and derivative $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$:

$$\begin{aligned} &= \cos^{-1} x \left(\frac{x^2}{2} \right) - \int \left(-\frac{1}{\sqrt{1-x^2}} \cdot \frac{x^2}{2} \right) dx \\ &= \frac{x^2}{2} \cos^{-1} x + \frac{1}{2} \int \frac{x^2}{\sqrt{1-x^2}} dx \end{aligned}$$

Now, rewrite the numerator for the remaining integral:

$$x^2 = -(1 - x^2) + 1$$

Substitute this back:

$$\int \frac{x^2}{\sqrt{1-x^2}} dx = \int \frac{-(1-x^2)+1}{\sqrt{1-x^2}} dx = \int \left(-\sqrt{1-x^2} + \frac{1}{\sqrt{1-x^2}} \right) dx$$

Integrating each part:

$$\int -\sqrt{1-x^2} dx = - \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1} x \right]$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$$

Combine the results:

$$\int \frac{x^2}{\sqrt{1-x^2}} dx = -\frac{x}{2}\sqrt{1-x^2} - \frac{1}{2}\sin^{-1} x + \sin^{-1} x = -\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x$$

Substitute back into the main expression:

$$\begin{aligned} &= \frac{x^2}{2} \cos^{-1} x + \frac{1}{2} \left(-\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\sin^{-1} x \right) + C \\ &= \frac{x^2}{2} \cos^{-1} x - \frac{x}{4}\sqrt{1-x^2} + \frac{1}{4}\sin^{-1} x + C \end{aligned}$$

Final Answer

$$\int x \cos^{-1} x dx = \frac{1}{4} \left(2x^2 \cos^{-1} x - x\sqrt{1-x^2} + \sin^{-1} x \right) + C$$

Common Mistake: Incorrectly applying the derivative of $\cos^{-1} x$ by omitting the negative sign.

Question 8 (ii)

Evaluate $\int \cos^{-1} \sqrt{x} dx$.

TYPE 13 Integration by Parts

Concept / Formula Used: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

First, use substitution to simplify the argument. Let $\sqrt{x} = t$, so $x = t^2$.

Differentiating: $dx = 2t dt$.

Substituting into the integral:

$$\int \cos^{-1} t (2t dt) = 2 \int t \cos^{-1} t dt$$

Now, use integration by parts on $\int t \cos^{-1} t dt$ with $u = \cos^{-1} t$ and $v = t$:

$$= 2 \left[\cos^{-1} t \int t dt - \int \left(\frac{d}{dt}(\cos^{-1} t) \int t dt \right) dt \right]$$

Evaluate $\int t dt = \frac{t^2}{2}$ and $\frac{d}{dt}(\cos^{-1} t) = -\frac{1}{\sqrt{1-t^2}}$:

$$= 2 \left[\cos^{-1} t \left(\frac{t^2}{2} \right) - \int \left(-\frac{1}{\sqrt{1-t^2}} \cdot \frac{t^2}{2} \right) dt \right]$$

$$= t^2 \cos^{-1} t + \int \frac{t^2}{\sqrt{1-t^2}} dt$$

Rewrite the numerator: $t^2 = -(1 - t^2) + 1$:

$$\int \frac{t^2}{\sqrt{1-t^2}} dt = \int \left(-\sqrt{1-t^2} + \frac{1}{\sqrt{1-t^2}} \right) dt$$

Integrating each part:

$$\begin{aligned} &= -\left(\frac{t}{2} \sqrt{1-t^2} + \frac{1}{2} \sin^{-1} t \right) + \sin^{-1} t \\ &= -\frac{t}{2} \sqrt{1-t^2} - \frac{1}{2} \sin^{-1} t + \sin^{-1} t \\ &= -\frac{t}{2} \sqrt{1-t^2} + \frac{1}{2} \sin^{-1} t \end{aligned}$$

Combine with the first part:

$$= t^2 \cos^{-1} t - \frac{t}{2} \sqrt{1-t^2} + \frac{1}{2} \sin^{-1} t + C$$

Now back-substitute $t = \sqrt{x}$ (noting $t^2 = x$ and $\sin^{-1} \sqrt{x} = \frac{\pi}{2} - \cos^{-1} \sqrt{x}$, or leaving in terms of $\sin^{-1} \sqrt{x}$):

$$= x \cos^{-1} \sqrt{x} - \frac{\sqrt{x}}{2} \sqrt{1-x} + \frac{1}{2} \sin^{-1} \sqrt{x} + C$$

Using the relation $\sin^{-1} \sqrt{x} = \frac{\pi}{2} - \cos^{-1} \sqrt{x}$, or writing in standard combined form:

$$= \frac{1}{2} \left((2x-1) \cos^{-1} \sqrt{x} - \sqrt{x-x^2} \right) + \text{constant}$$

Final Answer

$$\int \cos^{-1} \sqrt{x} dx = \frac{1}{2} \left((2x-1) \cos^{-1} \sqrt{x} - \sqrt{x-x^2} \right) + C$$

Common Mistake: Forgetting to multiply the entire integration-by-parts result by the factor of 2 originating from the initial substitution $dx = 2t dt$.

7.14 – Integrals Involving the Square Root of a General Quadratic

Question 1(i)

Evaluate $\int \sqrt{x^2 + 4x + 6} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Standard result: $\int \sqrt{x^2 + a^2} dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2 + a^2}| + C$

Solution:

Consider the quadratic expression inside the square root:

$$x^2 + 4x + 6$$

Completing the square by adding and subtracting $\left(\frac{4}{2}\right)^2 = 4$:

$$\begin{aligned} &= (x^2 + 4x + 4) - 4 + 6 \\ &= (x + 2)^2 + 2 \\ &= (x + 2)^2 + (\sqrt{2})^2 \end{aligned}$$

Substitute this back into the integral:

$$\int \sqrt{(x + 2)^2 + (\sqrt{2})^2} dx$$

Let $x + 2 = t$. Differentiating: $dx = dt$. The integral becomes:

$$\int \sqrt{t^2 + (\sqrt{2})^2} dt$$

Using the standard result $\int \sqrt{x^2 + a^2} dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2 + a^2}| + C$ with $x = t$ and $a = \sqrt{2}$:

$$= \frac{t}{2}\sqrt{t^2 + (\sqrt{2})^2} + \frac{(\sqrt{2})^2}{2} \log |t + \sqrt{t^2 + (\sqrt{2})^2}| + C$$

Simplifying the coefficients:

$$= \frac{t}{2}\sqrt{t^2 + 2} + \log |t + \sqrt{t^2 + 2}| + C$$

Now back-substitute $t = x + 2$:

$$\begin{aligned} &= \frac{x + 2}{2}\sqrt{(x + 2)^2 + 2} + \log |x + 2 + \sqrt{(x + 2)^2 + 2}| + C \\ &= \frac{1}{2}(x + 2)\sqrt{x^2 + 4x + 6} + \log |x + 2 + \sqrt{x^2 + 4x + 6}| + C \end{aligned}$$

Final Answer

$$\int \sqrt{x^2 + 4x + 6} dx = \frac{1}{2}(x + 2)\sqrt{x^2 + 4x + 6} + \log |x + 2 + \sqrt{x^2 + 4x + 6}| + C$$

Question 1(ii)

Evaluate $\int \sqrt{5 - 2x + x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Standard result: $\int \sqrt{x^2 + a^2} dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2 + a^2}| + C$

Solution:

Consider the quadratic expression inside the square root:

$$x^2 - 2x + 5$$

Completing the square by adding and subtracting $\left(\frac{-2}{2}\right)^2 = 1$:

$$\begin{aligned} &= (x^2 - 2x + 1) - 1 + 5 \\ &= (x - 1)^2 + 4 \\ &= (x - 1)^2 + 2^2 \end{aligned}$$

Substitute this back into the integral:

$$\int \sqrt{(x - 1)^2 + 2^2} dx$$

Let $x - 1 = t$. Differentiating: $dx = dt$. The integral becomes:

$$\int \sqrt{t^2 + 2^2} dt$$

Using the standard result $\int \sqrt{x^2 + a^2} dx = \frac{x}{2}\sqrt{x^2 + a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2 + a^2}| + C$ with $x = t$ and $a = 2$:

$$= \frac{t}{2}\sqrt{t^2 + 2^2} + \frac{2^2}{2} \log |t + \sqrt{t^2 + 2^2}| + C$$

Simplifying the coefficients:

$$= \frac{t}{2}\sqrt{t^2 + 4} + 2 \log |t + \sqrt{t^2 + 4}| + C$$

Now back-substitute $t = x - 1$:

$$\begin{aligned} &= \frac{x - 1}{2}\sqrt{(x - 1)^2 + 4} + 2 \log |x - 1 + \sqrt{(x - 1)^2 + 4}| + C \\ &= \frac{1}{2}(x - 1)\sqrt{5 - 2x + x^2} + 2 \log |x - 1 + \sqrt{5 - 2x + x^2}| + C \end{aligned}$$

Final Answer

$$\int \sqrt{5 - 2x + x^2} dx = \frac{1}{2}(x - 1)\sqrt{5 - 2x + x^2} + 2 \log |x - 1 + \sqrt{5 - 2x + x^2}| + C$$

Question 2(i)

Evaluate $\int \sqrt{3 - 2x - x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Standard result: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Consider the quadratic expression inside the square root:

$$3 - 2x - x^2 = 3 - (x^2 + 2x)$$

Completing the square inside the bracket by adding and subtracting $\left(\frac{2}{2}\right)^2 = 1$:

$$\begin{aligned} &= 3 - [(x^2 + 2x + 1) - 1] \\ &= 3 - (x + 1)^2 + 1 \\ &= 4 - (x + 1)^2 \\ &= 2^2 - (x + 1)^2 \end{aligned}$$

Substitute this back into the integral:

$$\int \sqrt{2^2 - (x + 1)^2} dx$$

Let $x + 1 = t$. Differentiating: $dx = dt$. The integral becomes:

$$\int \sqrt{2^2 - t^2} dt$$

Using the standard result $\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$ with $x = t$ and $a = 2$:

$$= \frac{t}{2}\sqrt{2^2 - t^2} + \frac{2^2}{2} \sin^{-1} \left(\frac{t}{2} \right) + C$$

Simplifying the coefficients:

$$= \frac{t}{2}\sqrt{4 - t^2} + 2 \sin^{-1} \left(\frac{t}{2} \right) + C$$

Now back-substitute $t = x + 1$:

$$\begin{aligned} &= \frac{x + 1}{2}\sqrt{4 - (x + 1)^2} + 2 \sin^{-1} \left(\frac{x + 1}{2} \right) + C \\ &= \frac{1}{2}(x + 1)\sqrt{3 - 2x - x^2} + 2 \sin^{-1} \left(\frac{x + 1}{2} \right) + C \end{aligned}$$

Final Answer

$$\int \sqrt{3 - 2x - x^2} dx = \frac{1}{2}(x + 1)\sqrt{3 - 2x - x^2} + 2 \sin^{-1} \left(\frac{x + 1}{2} \right) + C$$

Question 2(ii)

Evaluate $\int \sqrt{1+3x-x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

Concept / Formula Used: Standard result: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$

Solution:

Consider the quadratic expression inside the square root:

$$1 + 3x - x^2 = 1 - (x^2 - 3x)$$

Completing the square inside the bracket by adding and subtracting $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$:

$$\begin{aligned} &= 1 - \left[\left(x^2 - 3x + \frac{9}{4} \right) - \frac{9}{4} \right] \\ &= 1 - \left(x - \frac{3}{2} \right)^2 + \frac{9}{4} \\ &= \frac{13}{4} - \left(x - \frac{3}{2} \right)^2 \\ &= \left(\frac{\sqrt{13}}{2} \right)^2 - \left(x - \frac{3}{2} \right)^2 \end{aligned}$$

Substitute this back into the integral:

$$\int \sqrt{\left(\frac{\sqrt{13}}{2} \right)^2 - \left(x - \frac{3}{2} \right)^2} dx$$

Let $x - \frac{3}{2} = t$. Differentiating: $dx = dt$. The integral becomes:

$$\int \sqrt{\left(\frac{\sqrt{13}}{2} \right)^2 - t^2} dt$$

Using the standard result $\int \sqrt{a^2 - x^2} dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$ with $x = t$ and $a = \frac{\sqrt{13}}{2}$:

$$= \frac{t}{2} \sqrt{\left(\frac{\sqrt{13}}{2} \right)^2 - t^2} + \frac{\left(\frac{\sqrt{13}}{2} \right)^2}{2} \sin^{-1} \left(\frac{t}{\frac{\sqrt{13}}{2}} \right) + C$$

Simplifying the coefficients:

$$= \frac{t}{2} \sqrt{\frac{13}{4} - t^2} + \frac{13}{8} \sin^{-1} \left(\frac{2t}{\sqrt{13}} \right) + C$$

Now back-substitute $t = x - \frac{3}{2} = \frac{2x-3}{2}$:

$$\begin{aligned} &= \frac{2x-3}{4} \sqrt{\frac{13}{4} - \left(\frac{2x-3}{2}\right)^2} + \frac{13}{8} \sin^{-1} \left(\frac{2\left(\frac{2x-3}{2}\right)}{\sqrt{13}} \right) + C \\ &= \frac{2x-3}{4} \sqrt{\frac{13 - (4x^2 - 12x + 9)}{4}} + \frac{13}{8} \sin^{-1} \left(\frac{2x-3}{\sqrt{13}} \right) + C \\ &= \frac{2x-3}{4} \times \frac{1}{2} \sqrt{1 + 3x - x^2} + \frac{13}{8} \sin^{-1} \left(\frac{2x-3}{\sqrt{13}} \right) + C \\ &= \frac{1}{8} (2x-3) \sqrt{1 + 3x - x^2} + \frac{13}{8} \sin^{-1} \left(\frac{2x-3}{\sqrt{13}} \right) + C \end{aligned}$$

Final Answer

$$\int \sqrt{1 + 3x - x^2} \, dx = \frac{1}{8} (2x-3) \sqrt{1 + 3x - x^2} + \frac{13}{8} \sin^{-1} \left(\frac{2x-3}{\sqrt{13}} \right) + C$$

Question 3(i)

Evaluate $\int (x+3)\sqrt{3-4x-x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

★ Likely Exam Question

Concept / Formula Used: Express linear factor as: $px + q = A \frac{d}{dx}(\text{quadratic}) + B$

Solution:

Let the linear factor be expressed in terms of the derivative of the quadratic expression inside the square root:

$$x+3 = A \frac{d}{dx}(3-4x-x^2) + B$$

Differentiating $3-4x-x^2$ with respect to x :

$$\frac{d}{dx}(3-4x-x^2) = -4-2x$$

Thus, we write:

$$x+3 = A(-4-2x) + B$$

$$x+3 = -2Ax - 4A + B$$

Equating coefficients of x and constant terms on both sides: For x :

$$1 = -2A \implies A = -\frac{1}{2}$$

For constants:

$$-4A + B = 3$$

Substitute $A = -\frac{1}{2}$ into the constant equation:

$$-4\left(-\frac{1}{2}\right) + B = 3$$

$$2 + B = 3 \implies B = 1$$

Therefore, the integral can be rewritten as:

$$\int (x+3)\sqrt{3-4x-x^2} dx = \int \left[-\frac{1}{2}(-4-2x) + 1\right] \sqrt{3-4x-x^2} dx$$

Splitting into two separate integrals:

$$= -\frac{1}{2} \int (-4-2x)\sqrt{3-4x-x^2} dx + \int \sqrt{3-4x-x^2} dx$$

Let $I = -\frac{1}{2}I_1 + I_2$, where:

$$I_1 = \int (-4-2x)\sqrt{3-4x-x^2} dx \quad \text{and} \quad I_2 = \int \sqrt{3-4x-x^2} dx$$

For I_1 , let $3 - 4x - x^2 = u$. Differentiating: $(-4 - 2x) dx = du$.

$$I_1 = \int \sqrt{u} du = \frac{2}{3} u^{3/2} = \frac{2}{3} (3 - 4x - x^2)^{3/2}$$

For I_2 , using the completing the square method from Question 2(i):

$$3 - 4x - x^2 = 4 - (x + 2)^2 = 2^2 - (x + 2)^2$$

$$I_2 = \frac{x + 2}{2} \sqrt{3 - 4x - x^2} + 2 \sin^{-1} \left(\frac{x + 2}{2} \right)$$

Combining $I = -\frac{1}{2} I_1 + I_2$:

$$\begin{aligned} I &= -\frac{1}{2} \left[\frac{2}{3} (3 - 4x - x^2)^{3/2} \right] + \left[\frac{x + 2}{2} \sqrt{3 - 4x - x^2} + 2 \sin^{-1} \left(\frac{x + 2}{2} \right) \right] + C \\ &= -\frac{1}{3} (3 - 4x - x^2)^{3/2} + \frac{1}{2} (x + 2) \sqrt{3 - 4x - x^2} + 2 \sin^{-1} \left(\frac{x + 2}{2} \right) + C \end{aligned}$$

Final Answer

$$\int (x + 3) \sqrt{3 - 4x - x^2} dx = -\frac{1}{3} (3 - 4x - x^2)^{3/2} + \frac{1}{2} (x + 2) \sqrt{3 - 4x - x^2} + 2 \sin^{-1} \left(\frac{x + 2}{2} \right) + C$$

Question 3(ii)

Evaluate $\int (3x + 1)\sqrt{4 - 3x - 2x^2} dx$.

TYPE 12b Substitution Involving Radical Expressions

★ Likely Exam Question

Concept / Formula Used: Express linear factor as: $px + q = A \frac{d}{dx}(\text{quadratic}) + B$

Solution:

Let the linear factor be expressed in terms of the derivative of the quadratic expression inside the square root:

$$3x + 1 = A \frac{d}{dx}(4 - 3x - 2x^2) + B$$

Differentiating $4 - 3x - 2x^2$ with respect to x :

$$\frac{d}{dx}(4 - 3x - 2x^2) = -3 - 4x$$

Thus, we write:

$$3x + 1 = A(-3 - 4x) + B$$

$$3x + 1 = -4Ax - 3A + B$$

Equating coefficients of x and constant terms on both sides: For x :

$$3 = -4A \implies A = -\frac{3}{4}$$

For constants:

$$-3A + B = 1$$

Substitute $A = -\frac{3}{4}$ into the constant equation:

$$-3\left(-\frac{3}{4}\right) + B = 1$$

$$\frac{9}{4} + B = 1 \implies B = 1 - \frac{9}{4} = -\frac{5}{4}$$

Therefore, the integral can be rewritten as:

$$\int (3x + 1)\sqrt{4 - 3x - 2x^2} dx = \int \left[-\frac{3}{4}(-3 - 4x) - \frac{5}{4} \right] \sqrt{4 - 3x - 2x^2} dx$$

Splitting into two separate integrals:

$$= -\frac{3}{4} \int (-3 - 4x)\sqrt{4 - 3x - 2x^2} dx - \frac{5}{4} \int \sqrt{4 - 3x - 2x^2} dx$$

Let $I = -\frac{3}{4}I_1 - \frac{5}{4}I_2$, where:

$$I_1 = \int (-3 - 4x)\sqrt{4 - 3x - 2x^2} dx \quad \text{and} \quad I_2 = \int \sqrt{4 - 3x - 2x^2} dx$$

For I_1 , let $4 - 3x - 2x^2 = u$. Differentiating: $(-3 - 4x) dx = du$.

$$I_1 = \int \sqrt{u} du = \frac{2}{3}u^{3/2} = \frac{2}{3}(4 - 3x - 2x^2)^{3/2}$$

For I_2 , factor out the coefficient of x^2 and complete the square:

$$\begin{aligned} 4 - 3x - 2x^2 &= -2 \left(x^2 + \frac{3}{2}x - 2 \right) \\ &= -2 \left[\left(x^2 + \frac{3}{2}x + \frac{9}{16} \right) - \frac{9}{16} - 2 \right] \\ &= -2 \left[\left(x + \frac{3}{4} \right)^2 - \frac{41}{16} \right] \\ &= 2 \left[\left(\frac{\sqrt{41}}{4} \right)^2 - \left(x + \frac{3}{4} \right)^2 \right] \end{aligned}$$

Thus, $\sqrt{4 - 3x - 2x^2} = \sqrt{2} \sqrt{\left(\frac{\sqrt{41}}{4} \right)^2 - \left(x + \frac{3}{4} \right)^2}$. Evaluating I_2 using the standard formula for $\int \sqrt{a^2 - x^2} dx$:

$$I_2 = \sqrt{2} \left[\frac{x + \frac{3}{4}}{2} \sqrt{\left(\frac{\sqrt{41}}{4} \right)^2 - \left(x + \frac{3}{4} \right)^2} + \frac{41}{2} \sin^{-1} \left(\frac{x + \frac{3}{4}}{\frac{\sqrt{41}}{4}} \right) \right]$$

Simplifying I_2 :

$$\begin{aligned} I_2 &= \sqrt{2} \left[\frac{4x + 3}{8} \cdot \frac{\sqrt{4 - 3x - 2x^2}}{\sqrt{2}} + \frac{41}{32} \sin^{-1} \left(\frac{4x + 3}{\sqrt{41}} \right) \right] \\ &= \frac{4x + 3}{8} \sqrt{4 - 3x - 2x^2} + \frac{41\sqrt{2}}{32} \sin^{-1} \left(\frac{4x + 3}{\sqrt{41}} \right) \end{aligned}$$

Now substituting I_1 and I_2 back into $I = -\frac{3}{4}I_1 - \frac{5}{4}I_2$:

$$\begin{aligned} I &= -\frac{3}{4} \left[\frac{2}{3}(4 - 3x - 2x^2)^{3/2} \right] - \frac{5}{4} \left[\frac{4x + 3}{8} \sqrt{4 - 3x - 2x^2} + \frac{41\sqrt{2}}{32} \sin^{-1} \left(\frac{4x + 3}{\sqrt{41}} \right) \right] + C \\ &= -\frac{1}{2}(4 - 3x - 2x^2)^{3/2} - \frac{5}{32}(4x + 3)\sqrt{4 - 3x - 2x^2} - \frac{205\sqrt{2}}{128} \sin^{-1} \left(\frac{4x + 3}{\sqrt{41}} \right) + C \end{aligned}$$

Final Answer

$$\int (3x + 1)\sqrt{4 - 3x - 2x^2} dx = -\frac{1}{2}(4 - 3x - 2x^2)^{3/2} - \frac{5}{32}(4x + 3)\sqrt{4 - 3x - 2x^2} - \frac{205\sqrt{2}}{128} \sin^{-1} \left(\frac{4x + 3}{\sqrt{41}} \right) + C$$

7.15 – Definite Integrals via the Fundamental Theorem

Question 1 (i)

Evaluate $\int_2^3 x^2 dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_2^3 x^2 dx$

Concept / Formula Used: Using the standard power rule for integration:

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

Solution:

Using the standard integration formula $\int x^n dx = \frac{x^{n+1}}{n+1}$ with $n = 2$:

$$\begin{aligned} \int_2^3 x^2 dx &= \left[\frac{x^3}{3} \right]_2^3 \\ &= \left(\frac{(3)^3}{3} \right) - \left(\frac{(2)^3}{3} \right) \\ &= \frac{27}{3} - \frac{8}{3} \\ &= \frac{27-8}{3} \\ &= \frac{19}{3} \end{aligned}$$

Final Answer

$$\int_2^3 x^2 dx = \frac{19}{3}$$

Question 1 (ii)

Evaluate $\int_{-1}^1 (x+1) dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_{-1}^1 (x + 1) dx$

Concept / Formula Used: Using linearity of integration and standard power rule

Solution:

Integrating term by term:

$$\begin{aligned}\int_{-1}^1 (x + 1) dx &= \left[\frac{x^2}{2} + x \right]_{-1}^1 \\ &= \left(\frac{(1)^2}{2} + 1 \right) - \left(\frac{(-1)^2}{2} + (-1) \right) \\ &= \left(\frac{1}{2} + 1 \right) - \left(\frac{1}{2} - 1 \right) \\ &= \frac{1}{2} + 1 - \frac{1}{2} + 1 \\ &= 2\end{aligned}$$

Final Answer

$$\int_{-1}^1 (x + 1) dx = 2$$

Question 1 (iii)

Evaluate $\int_2^3 \frac{1}{x} dx$.

TYPE 4 Logarithmic Identity Simplification

Given: Evaluate $\int_2^3 \frac{1}{x} dx$

Concept / Formula Used: Using the standard logarithmic result: $\int \frac{1}{x} dx = \log |x|$

Solution:

Using the standard integration formula $\int \frac{1}{x} dx = \log |x|$:

$$\begin{aligned}\int_2^3 \frac{1}{x} dx &= [\log |x|]_2^3 \\ &= \log |3| - \log |2|\end{aligned}$$

Using the logarithmic property $\log a - \log b = \log \left(\frac{a}{b} \right)$:

$$= \log \left(\frac{3}{2} \right)$$

Final Answer

$$\int_2^3 \frac{1}{x} dx = \log \left(\frac{3}{2} \right)$$

Question 2 (i)

Evaluate $\int_{-4}^{-1} \frac{1}{x} dx$.

TYPE 4 Logarithmic Identity Simplification

Given: Evaluate $\int_{-4}^{-1} \frac{1}{x} dx$

Concept / Formula Used: Using the standard logarithmic result: $\int \frac{1}{x} dx = \log |x|$

Solution:

Using the standard integration formula $\int \frac{1}{x} dx = \log |x|$:

$$\begin{aligned} \int_{-4}^{-1} \frac{1}{x} dx &= [\log |x|]_{-4}^{-1} \\ &= \log |-1| - \log |-4| \\ &= \log(1) - \log(4) \end{aligned}$$

Since $\log(1) = 0$:

$$\begin{aligned} &= 0 - \log(4) \\ &= -\log(4) \end{aligned}$$

Final Answer

$$\int_{-4}^{-1} \frac{1}{x} dx = -\log 4$$

Question 2 (ii)

Evaluate $\int_0^1 \frac{1}{2x-3} dx$.

TYPE 4 Logarithmic Identity Simplification

Given: Evaluate $\int_0^1 \frac{1}{2x-3} dx$

Concept / Formula Used: Using standard integration: $\int \frac{1}{ax+b} dx = \frac{1}{a} \log |ax+b|$

Solution:

Using the standard linear substitution result for $\int \frac{1}{ax+b} dx$:

$$\begin{aligned}\int_0^1 \frac{1}{2x-3} dx &= \left[\frac{1}{2} \log |2x-3| \right]_0^1 \\ &= \left(\frac{1}{2} \log |2(1)-3| \right) - \left(\frac{1}{2} \log |2(0)-3| \right) \\ &= \frac{1}{2} \log |2-3| - \frac{1}{2} \log |-3| \\ &= \frac{1}{2} \log |-1| - \frac{1}{2} \log(3)\end{aligned}$$

Since $\log |-1| = \log(1) = 0$:

$$\begin{aligned}&= 0 - \frac{1}{2} \log 3 \\ &= -\frac{1}{2} \log 3\end{aligned}$$

Final Answer

$$\int_0^1 \frac{1}{2x-3} dx = -\frac{1}{2} \log 3$$

Question 2 (iii)

Evaluate $\int_0^1 \sqrt{5x+4} dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_0^1 \sqrt{5x+4} dx$

Concept / Formula Used: Using power rule with linear substitution: $\int (ax + b)^n dx = \frac{(ax + b)^{n+1}}{a(n+1)}$

Solution:

Using the standard linear power integration formula:

$$\begin{aligned}
 \int_0^1 \sqrt{5x+4} \, dx &= \int_0^1 (5x+4)^{1/2} \, dx \\
 &= \left[\frac{(5x+4)^{1/2+1}}{5 \cdot (\frac{1}{2}+1)} \right]_0^1 \\
 &= \left[\frac{(5x+4)^{3/2}}{5 \cdot (\frac{3}{2})} \right]_0^1 \\
 &= \left[\frac{2}{15} (5x+4)^{3/2} \right]_0^1
 \end{aligned}$$

Substituting the upper and lower limits:

$$\begin{aligned}
 &= \left(\frac{2}{15} (5(1)+4)^{3/2} \right) - \left(\frac{2}{15} (5(0)+4)^{3/2} \right) \\
 &= \frac{2}{15} (9)^{3/2} - \frac{2}{15} (4)^{3/2} \\
 &= \frac{2}{15} (3^2)^{3/2} - \frac{2}{15} (2^2)^{3/2} \\
 &= \frac{2}{15} (3^3) - \frac{2}{15} (2^3) \\
 &= \frac{2}{15} (27) - \frac{2}{15} (8) \\
 &= \frac{54}{15} - \frac{16}{15} \\
 &= \frac{54-16}{15} \\
 &= \frac{38}{15}
 \end{aligned}$$

Final Answer

$$\int_0^1 \sqrt{5x+4} \, dx = \frac{38}{15}$$

Question 3 (i)

Evaluate $\int_2^3 3^x \, dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_2^3 3^x \, dx$

Concept / Formula Used: Using standard exponential integration: $\int a^x dx = \frac{a^x}{\log a}$

Solution:

Using the standard formula $\int a^x dx = \frac{a^x}{\log a}$ with $a = 3$:

$$\begin{aligned}\int_2^3 3^x dx &= \left[\frac{3^x}{\log 3} \right]_2^3 \\ &= \left(\frac{3^3}{\log 3} \right) - \left(\frac{3^2}{\log 3} \right) \\ &= \frac{27}{\log 3} - \frac{9}{\log 3} \\ &= \frac{27 - 9}{\log 3} \\ &= \frac{18}{\log 3}\end{aligned}$$

Final Answer

$$\int_2^3 3^x dx = \frac{18}{\log 3}$$

Question 3 (ii)

Evaluate $\int_0^{\pi/4} \tan x dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/4} \tan x dx$

Concept / Formula Used: Using standard trigonometric integration: $\int \tan x dx = \log |\sec x|$

Solution:

Using the standard result $\int \tan x dx = \log |\sec x|$:

$$\begin{aligned}\int_0^{\pi/4} \tan x dx &= [\log |\sec x|]_0^{\pi/4} \\ &= \log \left| \sec \left(\frac{\pi}{4} \right) \right| - \log |\sec(0)|\end{aligned}$$

Since $\sec\left(\frac{\pi}{4}\right) = \sqrt{2}$ and $\sec(0) = 1$:

$$\begin{aligned} &= \log(\sqrt{2}) - \log(1) \\ &= \log(2^{1/2}) - 0 \\ &= \frac{1}{2} \log 2 \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \tan x \, dx = \frac{1}{2} \log 2$$

Question 3 (iii)

Evaluate $\int_{\pi/4}^{\pi/2} \cot x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_{\pi/4}^{\pi/2} \cot x \, dx$

Concept / Formula Used: Using standard trigonometric integration: $\int \cot x \, dx = \log |\sin x|$

Solution:

Using the standard result $\int \cot x \, dx = \log |\sin x|$:

$$\begin{aligned} \int_{\pi/4}^{\pi/2} \cot x \, dx &= [\log |\sin x|]_{\pi/4}^{\pi/2} \\ &= \log \left| \sin \left(\frac{\pi}{2} \right) \right| - \log \left| \sin \left(\frac{\pi}{4} \right) \right| \end{aligned}$$

Since $\sin\left(\frac{\pi}{2}\right) = 1$ and $\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$:

$$\begin{aligned} &= \log(1) - \log\left(\frac{1}{\sqrt{2}}\right) \\ &= 0 - \log(2^{-1/2}) \\ &= -\left(-\frac{1}{2} \log 2\right) \\ &= \frac{1}{2} \log 2 \end{aligned}$$

Final Answer

$$\int_{\pi/4}^{\pi/2} \cot x \, dx = \frac{1}{2} \log 2$$

Question 4 (i)

Evaluate $\int_0^{\pi/4} \sin 2x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/4} \sin 2x \, dx$

Concept / Formula Used: Using standard trigonometric integration:
 $\int \sin(ax) \, dx = -\frac{\cos(ax)}{a}$

Solution:

Using the standard integration formula $\int \sin(ax) \, dx = -\frac{\cos(ax)}{a}$ with $a = 2$:

$$\begin{aligned} \int_0^{\pi/4} \sin 2x \, dx &= \left[-\frac{\cos 2x}{2} \right]_0^{\pi/4} \\ &= \left(-\frac{\cos(2 \cdot \frac{\pi}{4})}{2} \right) - \left(-\frac{\cos(2 \cdot 0)}{2} \right) \\ &= -\frac{\cos(\frac{\pi}{2})}{2} + \frac{\cos(0)}{2} \end{aligned}$$

Since $\cos(\frac{\pi}{2}) = 0$ and $\cos(0) = 1$:

$$\begin{aligned} &= -\frac{0}{2} + \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \sin 2x \, dx = \frac{1}{2}$$

Question 4 (ii)

Evaluate $\int_0^{\pi/2} \cos 2x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/2} \cos 2x \, dx$

Concept / Formula Used: Using standard trigonometric integration:

$$\int \cos(ax) dx = \frac{\sin(ax)}{a}$$

Solution:

Using the standard integration formula $\int \cos(ax) dx = \frac{\sin(ax)}{a}$ with $a = 2$:

$$\begin{aligned} \int_0^{\pi/2} \cos 2x dx &= \left[\frac{\sin 2x}{2} \right]_0^{\pi/2} \\ &= \left(\frac{\sin(2 \cdot \frac{\pi}{2})}{2} \right) - \left(\frac{\sin(2 \cdot 0)}{2} \right) \\ &= \frac{\sin(\pi)}{2} - \frac{\sin(0)}{2} \end{aligned}$$

Since $\sin(\pi) = 0$ and $\sin(0) = 0$:

$$\begin{aligned} &= \frac{0}{2} - \frac{0}{2} \\ &= 0 \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \cos 2x dx = 0$$

Question 4 (iii)

Evaluate $\int_0^{\pi} \left(\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2} \right) dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi} \left(\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2} \right) dx$

Concept / Formula Used: Using trigonometric identity: $\cos(nx) = \cos^2\left(\frac{nx}{2}\right) - \sin^2\left(\frac{nx}{2}\right)$

Solution:

Rewriting the integrand using the double angle identity $\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)$:

$$\begin{aligned} \int_0^{\pi} \left(\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2} \right) dx &= \int_0^{\pi} - \left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2} \right) dx \\ &= - \int_0^{\pi} \cos x dx \end{aligned}$$

Integrating $\cos x$:

$$\begin{aligned} &= -[\sin x]_0^{\pi} \\ &= -(\sin(\pi) - \sin(0)) \end{aligned}$$

Since $\sin(\pi) = 0$ and $\sin(0) = 0$:

$$\begin{aligned} &= -(0 - 0) \\ &= 0 \end{aligned}$$

Final Answer

$$\int_0^\pi \left(\sin^2 \frac{x}{2} - \cos^2 \frac{x}{2} \right) dx = 0$$

Question 5 (i)

Evaluate $\int_0^{1/\sqrt{2}} \frac{dx}{\sqrt{1-x^2}}$.

TYPE 6 Inverse Trigonometric Simplification

Given: Evaluate $\int_0^{1/\sqrt{2}} \frac{dx}{\sqrt{1-x^2}}$

Concept / Formula Used: Using standard inverse trigonometric result:
 $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$

Solution:

Using the standard integration result $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x$:

$$\begin{aligned} \int_0^{1/\sqrt{2}} \frac{dx}{\sqrt{1-x^2}} &= [\sin^{-1} x]_0^{1/\sqrt{2}} \\ &= \sin^{-1} \left(\frac{1}{\sqrt{2}} \right) - \sin^{-1}(0) \end{aligned}$$

Since $\sin^{-1} \left(\frac{1}{\sqrt{2}} \right) = \frac{\pi}{4}$ and $\sin^{-1}(0) = 0$:

$$\begin{aligned} &= \frac{\pi}{4} - 0 \\ &= \frac{\pi}{4} \end{aligned}$$

Final Answer

$$\int_0^{1/\sqrt{2}} \frac{dx}{\sqrt{1-x^2}} = \frac{\pi}{4}$$

Question 5 (ii)

Evaluate $\int_0^3 \frac{dx}{9+x^2}$.

TYPE 6 Inverse Trigonometric Simplification

Given: Evaluate $\int_0^3 \frac{dx}{9+x^2}$

Concept / Formula Used: Using standard inverse trigonometric result:
 $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$

Solution:

Rewriting the integral in standard form with $a^2 = 3^2$:

$$\int_0^3 \frac{dx}{3^2+x^2} = \left[\frac{1}{3} \tan^{-1} \left(\frac{x}{3} \right) \right]_0^3$$

Substituting the upper and lower limits:

$$\begin{aligned} &= \left(\frac{1}{3} \tan^{-1} \left(\frac{3}{3} \right) \right) - \left(\frac{1}{3} \tan^{-1} \left(\frac{0}{3} \right) \right) \\ &= \frac{1}{3} \tan^{-1}(1) - \frac{1}{3} \tan^{-1}(0) \end{aligned}$$

Since $\tan^{-1}(1) = \frac{\pi}{4}$ and $\tan^{-1}(0) = 0$:

$$\begin{aligned} &= \frac{1}{3} \left(\frac{\pi}{4} \right) - 0 \\ &= \frac{\pi}{12} \end{aligned}$$

Final Answer

$$\int_0^3 \frac{dx}{9+x^2} = \frac{\pi}{12}$$

Question 5 (iii)

Evaluate $\int_1^{\sqrt{3}} \frac{dx}{1+x^2}$.

TYPE 6 Inverse Trigonometric Simplification

Given: Evaluate $\int_1^{\sqrt{3}} \frac{dx}{1+x^2}$

Concept / Formula Used: Using standard inverse trigonometric result:
 $\int \frac{dx}{1+x^2} = \tan^{-1} x$

Solution:

Using the standard result $\int \frac{dx}{1+x^2} = \tan^{-1} x$:

$$\begin{aligned}\int_1^{\sqrt{3}} \frac{dx}{1+x^2} &= [\tan^{-1} x]_1^{\sqrt{3}} \\ &= \tan^{-1}(\sqrt{3}) - \tan^{-1}(1)\end{aligned}$$

Since $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$ and $\tan^{-1}(1) = \frac{\pi}{4}$:

$$\begin{aligned}&= \frac{\pi}{3} - \frac{\pi}{4} \\ &= \frac{4\pi - 3\pi}{12} \\ &= \frac{\pi}{12}\end{aligned}$$

Final Answer

$$\int_1^{\sqrt{3}} \frac{dx}{1+x^2} = \frac{\pi}{12}$$

Question 6 (i)

Evaluate $\int_2^3 \frac{dx}{x^2 - 1}$.

TYPE 14 Partial Fractions

Given: Evaluate $\int_2^3 \frac{dx}{x^2 - 1}$

Concept / Formula Used: Using standard partial fraction integration:

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x - a}{x + a} \right|$$

Solution:

Using the standard integration formula for $\frac{1}{x^2 - a^2}$ with $a = 1$:

$$\begin{aligned}\int_2^3 \frac{dx}{x^2 - 1} &= \left[\frac{1}{2(1)} \log \left| \frac{x - 1}{x + 1} \right| \right]_2^3 \\ &= \left[\frac{1}{2} \log \left| \frac{x - 1}{x + 1} \right| \right]_2^3\end{aligned}$$

Substituting the upper and lower limits:

$$\begin{aligned}&= \left(\frac{1}{2} \log \left| \frac{3 - 1}{3 + 1} \right| \right) - \left(\frac{1}{2} \log \left| \frac{2 - 1}{2 + 1} \right| \right) \\ &= \frac{1}{2} \log \left(\frac{2}{4} \right) - \frac{1}{2} \log \left(\frac{1}{3} \right) \\ &= \frac{1}{2} \log \left(\frac{1}{2} \right) - \frac{1}{2} \log \left(\frac{1}{3} \right)\end{aligned}$$

Using logarithm properties $\log a - \log b = \log\left(\frac{a}{b}\right)$ and $c \log a = \log(a^c)$:

$$\begin{aligned} &= \frac{1}{2} \left(\log\left(\frac{1}{2}\right) - \log\left(\frac{1}{3}\right) \right) \\ &= \frac{1}{2} \log\left(\frac{1/2}{1/3}\right) \\ &= \frac{1}{2} \log\left(\frac{3}{2}\right) \end{aligned}$$

Final Answer

$$\int_2^3 \frac{dx}{x^2 - 1} = \frac{1}{2} \log\left(\frac{3}{2}\right)$$

Question 6 (ii)

Evaluate $\int_0^\pi \frac{1}{1 + \sin x} dx$.

TYPE 5 Conjugate Multiplication

Given: Evaluate $\int_0^\pi \frac{1}{1 + \sin x} dx$

Concept / Formula Used: Rationalisation and trigonometric identities: $\frac{1}{1 + \sin x} = \frac{1 - \sin x}{\cos^2 x} = \sec^2 x - \tan x \sec x$

Solution:

Rationalising the denominator by multiplying numerator and denominator by $(1 - \sin x)$:

$$\begin{aligned} \int_0^\pi \frac{1}{1 + \sin x} dx &= \int_0^\pi \frac{1 - \sin x}{(1 + \sin x)(1 - \sin x)} dx \\ &= \int_0^\pi \frac{1 - \sin x}{1 - \sin^2 x} dx \end{aligned}$$

Using the fundamental identity $1 - \sin^2 x = \cos^2 x$:

$$\begin{aligned} &= \int_0^\pi \frac{1 - \sin x}{\cos^2 x} dx \\ &= \int_0^\pi \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x} \right) dx \\ &= \int_0^\pi (\sec^2 x - \sec x \tan x) dx \end{aligned}$$

Integrating term by term using standard results ($\int \sec^2 x dx = \tan x$ and $\int \sec x \tan x dx = \sec x$):

$$= [\tan x - \sec x]_0^\pi$$

Substituting the upper and lower limits:

$$= (\tan(\pi) - \sec(\pi)) - (\tan(0) - \sec(0))$$

Since $\tan(\pi) = 0$, $\sec(\pi) = -1$, $\tan(0) = 0$, and $\sec(0) = 1$:

$$\begin{aligned} &= (0 - (-1)) - (0 - 1) \\ &= (1) - (-1) \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

Final Answer

$$\int_0^\pi \frac{1}{1 + \sin x} dx = 2$$

Question 6 (iii)

Evaluate $\int_{-\pi/2}^{\pi/2} \frac{1}{1 + \cos x} dx$.

TYPE 10 Odd And Even Function Property

Given: Evaluate $\int_{-\pi/2}^{\pi/2} \frac{1}{1 + \cos x} dx$

Concept / Formula Used: Using even function property: $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ if $f(-x) = f(x)$

Solution:

Let $f(x) = \frac{1}{1 + \cos x}$. Checking if the function is even:

$$f(-x) = \frac{1}{1 + \cos(-x)} = \frac{1}{1 + \cos x} = f(x)$$

Since $f(x)$ is an even function, using the property $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$:

$$\int_{-\pi/2}^{\pi/2} \frac{1}{1 + \cos x} dx = 2 \int_0^{\pi/2} \frac{1}{1 + \cos x} dx$$

Using the half-angle identity $1 + \cos x = 2 \cos^2 \left(\frac{x}{2}\right)$:

$$\begin{aligned} &= 2 \int_0^{\pi/2} \frac{1}{2 \cos^2 \left(\frac{x}{2}\right)} dx \\ &= 2 \int_0^{\pi/2} \frac{1}{2} \sec^2 \left(\frac{x}{2}\right) dx \\ &= \int_0^{\pi/2} \sec^2 \left(\frac{x}{2}\right) dx \end{aligned}$$

Integrating using the standard result $\int \sec^2(ax) dx = \frac{\tan(ax)}{a}$ with $a = \frac{1}{2}$:

$$\begin{aligned} &= \left[\frac{\tan\left(\frac{x}{2}\right)}{\frac{1}{2}} \right]_0^{\pi/2} \\ &= \left[2 \tan\left(\frac{x}{2}\right) \right]_0^{\pi/2} \end{aligned}$$

Substituting the limits:

$$\begin{aligned} &= 2 \left(\tan\left(\frac{\pi/2}{2}\right) - \tan\left(\frac{0}{2}\right) \right) \\ &= 2 \left(\tan\left(\frac{\pi}{4}\right) - \tan(0) \right) \end{aligned}$$

Since $\tan\left(\frac{\pi}{4}\right) = 1$ and $\tan(0) = 0$:

$$\begin{aligned} &= 2(1 - 0) \\ &= 2 \end{aligned}$$

Final Answer

$$\int_{-\pi/2}^{\pi/2} \frac{1}{1 + \cos x} dx = 2$$

Question 7 (i)

Evaluate $\int_0^6 |x + 3| dx$.

TYPE 8 Splitting The Limit Of Integration

★ Likely Exam Question

Given: Evaluate $\int_0^6 |x + 3| dx$

Concept / Formula Used: Using interval splitting property: $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

Solution:

The integrand is $|x + 3|$. Inside the interval of integration $[0, 6]$, we examine the sign of $x + 3$. For all $x \in [0, 6]$, $x + 3 > 0$. Therefore, $|x + 3| = x + 3$ throughout the interval

$[0, 6]$. Thus, the integral simplifies to:

$$\begin{aligned}\int_0^6 |x + 3| dx &= \int_0^6 (x + 3) dx \\ &= \left[\frac{x^2}{2} + 3x \right]_0^6\end{aligned}$$

Substituting the upper and lower limits:

$$\begin{aligned}&= \left(\frac{(6)^2}{2} + 3(6) \right) - \left(\frac{(0)^2}{2} + 3(0) \right) \\ &= \left(\frac{36}{2} + 18 \right) - 0 \\ &= 18 + 18 \\ &= 36\end{aligned}$$

Final Answer

$$\int_0^6 |x + 3| dx = 36$$

Question 7 (ii)

Evaluate $\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx$

Concept / Formula Used: Using standard power rule term-by-term integration

Solution:

Integrating term by term:

$$\begin{aligned}\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx &= \left[4 \left(\frac{x^4}{4} \right) - 5 \left(\frac{x^3}{3} \right) + 6 \left(\frac{x^2}{2} \right) + 9x \right]_1^2 \\ &= \left[x^4 - \frac{5}{3}x^3 + 3x^2 + 9x \right]_1^2\end{aligned}$$

Substituting the upper limit $x = 2$:

$$\begin{aligned}\text{Upper value} &= (2)^4 - \frac{5}{3}(2)^3 + 3(2)^2 + 9(2) \\ &= 16 - \frac{5}{3}(8) + 3(4) + 18 \\ &= 16 - \frac{40}{3} + 12 + 18 \\ &= 46 - \frac{40}{3} \\ &= \frac{138 - 40}{3} = \frac{98}{3}\end{aligned}$$

Substituting the lower limit $x = 1$:

$$\begin{aligned}\text{Lower value} &= (1)^4 - \frac{5}{3}(1)^3 + 3(1)^2 + 9(1) \\ &= 1 - \frac{5}{3} + 3 + 9 \\ &= 13 - \frac{5}{3} \\ &= \frac{39 - 5}{3} = \frac{34}{3}\end{aligned}$$

Subtracting lower value from upper value:

$$\begin{aligned}&= \frac{98}{3} - \frac{34}{3} \\ &= \frac{98 - 34}{3} \\ &= \frac{64}{3}\end{aligned}$$

Final Answer

$$\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx = \frac{64}{3}$$

Question 7 (iii)

Evaluate $\int_0^8 \left(\sqrt{8x} - \frac{x^2}{8} \right) dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_0^8 \left(\sqrt{8x} - \frac{x^2}{8} \right) dx$

Concept / Formula Used: Using standard power rule integration

Solution:

Rewriting the terms using exponents:

$$\begin{aligned}\int_0^8 \left(\sqrt{8}x^{1/2} - \frac{1}{8}x^2 \right) dx &= \left[\sqrt{8} \left(\frac{x^{3/2}}{3/2} \right) - \frac{1}{8} \left(\frac{x^3}{3} \right) \right]_0^8 \\ &= \left[\frac{2\sqrt{8}}{3} x^{3/2} - \frac{x^3}{24} \right]_0^8\end{aligned}$$

Since $\sqrt{8} = 2\sqrt{2}$, the coefficient becomes $\frac{2(2\sqrt{2})}{3} = \frac{4\sqrt{2}}{3}$. Also note that $8^{3/2} = (2^3)^{3/2} = 2^{9/2} = 16\sqrt{2}$ and $8^3 = 512$:

$$\begin{aligned}&= \left[\frac{4\sqrt{2}}{3} (16\sqrt{2}) - \frac{512}{24} \right] - 0 \\ &= \frac{4 \cdot 16 \cdot (\sqrt{2} \cdot \sqrt{2})}{3} - \frac{64}{3} \\ &= \frac{64 \cdot 2}{3} - \frac{64}{3} \\ &= \frac{128}{3} - \frac{64}{3} \\ &= \frac{64}{3}\end{aligned}$$

Final Answer

$$\int_0^8 \left(\sqrt{8}x - \frac{x^2}{8} \right) dx = \frac{64}{3}$$

Question 8 (i)

Evaluate $\int_{\pi/6}^{\pi/4} \csc x \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_{\pi/6}^{\pi/4} \csc x \, dx$

Concept / Formula Used: Using standard integration: $\int \csc x \, dx = \log |\csc x - \cot x|$

Solution:

Using the standard result $\int \csc x \, dx = \log |\csc x - \cot x|$:

$$\begin{aligned}\int_{\pi/6}^{\pi/4} \csc x \, dx &= [\log |\csc x - \cot x|]_{\pi/6}^{\pi/4} \\ &= \log \left| \csc \left(\frac{\pi}{4} \right) - \cot \left(\frac{\pi}{4} \right) \right| - \log \left| \csc \left(\frac{\pi}{6} \right) - \cot \left(\frac{\pi}{6} \right) \right|\end{aligned}$$

Evaluating the trigonometric values: $\csc\left(\frac{\pi}{4}\right) = \sqrt{2}$, $\cot\left(\frac{\pi}{4}\right) = 1$, $\csc\left(\frac{\pi}{6}\right) = 2$, and $\cot\left(\frac{\pi}{6}\right) = \sqrt{3}$:

$$= \log |\sqrt{2} - 1| - \log |2 - \sqrt{3}|$$

Using logarithm properties:

$$= \log \left(\frac{\sqrt{2} - 1}{2 - \sqrt{3}} \right)$$

Final Answer

$$\int_{\pi/6}^{\pi/4} \csc x \, dx = \log \left(\frac{\sqrt{2} - 1}{2 - \sqrt{3}} \right)$$

Question 8 (ii)

Evaluate $\int_0^{\pi/4} (2 \sec^2 x + x^3 + 2) \, dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_0^{\pi/4} (2 \sec^2 x + x^3 + 2) \, dx$

Concept / Formula Used: Using standard integration rules term-by-term

Solution:

Integrating term by term using standard formulas ($\int \sec^2 x \, dx = \tan x$ and $\int x^n \, dx = \frac{x^{n+1}}{n+1}$):

$$\int_0^{\pi/4} (2 \sec^2 x + x^3 + 2) \, dx = \left[2 \tan x + \frac{x^4}{4} + 2x \right]_0^{\pi/4}$$

Substituting the upper limit $x = \frac{\pi}{4}$ and lower limit $x = 0$:

$$\begin{aligned} &= \left(2 \tan \left(\frac{\pi}{4} \right) + \frac{(\pi/4)^4}{4} + 2 \left(\frac{\pi}{4} \right) \right) - (0 + 0 + 0) \\ &= 2(1) + \frac{\pi^4}{4 \cdot 256} + \frac{\pi}{2} \\ &= 2 + \frac{\pi^4}{1024} + \frac{\pi}{2} \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} (2 \sec^2 x + x^3 + 2) \, dx = 2 + \frac{\pi}{2} + \frac{\pi^4}{1024}$$

Question 9 (i)

Evaluate $\int_0^1 \frac{dx}{\sqrt{1+x} - \sqrt{x}}$.

TYPE 5 Conjugate Multiplication

Given: Evaluate $\int_0^1 \frac{dx}{\sqrt{1+x} - \sqrt{x}}$

Concept / Formula Used: Rationalisation: $\frac{1}{\sqrt{1+x}-\sqrt{x}} = \frac{\sqrt{1+x}+\sqrt{x}}{(1+x)-x} = \sqrt{1+x} + \sqrt{x}$

Solution:

Rationalising the denominator:

$$\begin{aligned} \int_0^1 \frac{dx}{\sqrt{1+x} - \sqrt{x}} &= \int_0^1 \frac{\sqrt{1+x} + \sqrt{x}}{(\sqrt{1+x} - \sqrt{x})(\sqrt{1+x} + \sqrt{x})} dx \\ &= \int_0^1 \frac{\sqrt{1+x} + \sqrt{x}}{(1+x) - x} dx \\ &= \int_0^1 (\sqrt{1+x} + \sqrt{x}) dx \end{aligned}$$

Integrating using the power rule:

$$\begin{aligned} &= \left[\frac{(1+x)^{3/2}}{3/2} + \frac{x^{3/2}}{3/2} \right]_0^1 \\ &= \left[\frac{2}{3}(1+x)^{3/2} + \frac{2}{3}x^{3/2} \right]_0^1 \\ &= \frac{2}{3} [(1+x)^{3/2} + x^{3/2}]_0^1 \end{aligned}$$

Substituting the limits:

$$\begin{aligned} &= \frac{2}{3} [(1+1)^{3/2} + (1)^{3/2} - ((1+0)^{3/2} + (0)^{3/2})] \\ &= \frac{2}{3} [(2^{3/2} + 1) - (1 + 0)] \\ &= \frac{2}{3} [2\sqrt{2} + 1 - 1] \\ &= \frac{2}{3} (2\sqrt{2}) \\ &= \frac{4\sqrt{2}}{3} \end{aligned}$$

Final Answer

$$\int_0^1 \frac{dx}{\sqrt{1+x} - \sqrt{x}} = \frac{4\sqrt{2}}{3}$$

Question 9 (ii)

Evaluate $\int_0^1 \frac{1-x}{1+x} dx$.

TYPE 1 Power Rule of Integration

Given: Evaluate $\int_0^1 \frac{1-x}{1+x} dx$

Concept / Formula Used: Algebraic adjustment: $\frac{1-x}{1+x} = \frac{2-(1+x)}{1+x} = \frac{2}{1+x} - 1$

Solution:

Adjusting the numerator to split the fraction:

$$\begin{aligned}\frac{1-x}{1+x} &= \frac{2-(1+x)}{1+x} \\ &= \frac{2}{1+x} - \frac{1+x}{1+x} \\ &= \frac{2}{1+x} - 1\end{aligned}$$

Substituting back into the integral:

$$\begin{aligned}\int_0^1 \frac{1-x}{1+x} dx &= \int_0^1 \left(\frac{2}{1+x} - 1 \right) dx \\ &= [2 \log |1+x| - x]_0^1\end{aligned}$$

Substituting the limits:

$$\begin{aligned}&= (2 \log |1+1| - 1) - (2 \log |1+0| - 0) \\ &= (2 \log(2) - 1) - (2 \log(1) - 0)\end{aligned}$$

Since $\log(1) = 0$:

$$\begin{aligned}&= 2 \log 2 - 1 - 0 \\ &= 2 \log 2 - 1\end{aligned}$$

Final Answer

$$\int_0^1 \frac{1-x}{1+x} dx = 2 \log 2 - 1$$

Question 10 (i)

Evaluate $\int_0^{\pi/2} \sin^3 x dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: Evaluate $\int_0^{\pi/2} \sin^3 x \, dx$

Concept / Formula Used: Trigonometric identity: $\sin^3 x = \sin x(1 - \cos^2 x)$ and substitution $t = \cos x$

Solution:

Rewriting $\sin^3 x$ as $\sin^2 x \cdot \sin x = (1 - \cos^2 x) \sin x$:

$$\int_0^{\pi/2} \sin^3 x \, dx = \int_0^{\pi/2} (1 - \cos^2 x) \sin x \, dx$$

Let $\cos x = t$. Differentiating: $-\sin x \, dx = dt$, so $\sin x \, dx = -dt$. Changing the limits:

- When $x = 0$, $t = \cos(0) = 1$.
- When $x = \frac{\pi}{2}$, $t = \cos\left(\frac{\pi}{2}\right) = 0$.

Substituting into the integral:

$$\begin{aligned} \int_1^0 (1 - t^2)(-dt) &= -\int_1^0 (1 - t^2) \, dt \\ &= \int_0^1 (1 - t^2) \, dt \end{aligned}$$

Integrating with respect to t :

$$\begin{aligned} &= \left[t - \frac{t^3}{3} \right]_0^1 \\ &= \left(1 - \frac{(1)^3}{3} \right) - (0 - 0) \\ &= 1 - \frac{1}{3} \\ &= \frac{2}{3} \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \sin^3 x \, dx = \frac{2}{3}$$

Question 10 (ii)

Evaluate $\int_0^{\pi/4} (\tan x + \cot x)^{-1} \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/4} (\tan x + \cot x)^{-1} \, dx$

Concept / Formula Used: Trigonometric identity: $(\tan x + \cot x)^{-1} = \frac{1}{\tan x + \cot x} = \frac{1}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} = \frac{\sin x \cos x}{\sin^2 x + \cos^2 x}$

Solution:

Expressing the integrand in terms of sine and cosine:

$$\begin{aligned} (\tan x + \cot x)^{-1} &= \frac{1}{\tan x + \cot x} \\ &= \frac{1}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}} \\ &= \frac{1}{\frac{\sin^2 x + \cos^2 x}{\sin x \cos x}} \\ &= \frac{\sin x \cos x}{\sin^2 x + \cos^2 x} \end{aligned}$$

Using the fundamental identity $\sin^2 x + \cos^2 x = 1$:

$$\begin{aligned} &= \sin x \cos x \\ &= \frac{1}{2}(2 \sin x \cos x) \\ &= \frac{1}{2} \sin 2x \end{aligned}$$

Substituting back into the integral:

$$\begin{aligned} \int_0^{\pi/4} (\tan x + \cot x)^{-1} dx &= \int_0^{\pi/4} \frac{1}{2} \sin 2x dx \\ &= \frac{1}{2} \left[-\frac{\cos 2x}{2} \right]_0^{\pi/4} \\ &= -\frac{1}{4} [\cos 2x]_0^{\pi/4} \end{aligned}$$

Evaluating at the limits:

$$\begin{aligned} &= -\frac{1}{4} \left(\cos \left(2 \cdot \frac{\pi}{4} \right) - \cos(2 \cdot 0) \right) \\ &= -\frac{1}{4} \left(\cos \left(\frac{\pi}{2} \right) - \cos(0) \right) \\ &= -\frac{1}{4} (0 - 1) \\ &= \frac{1}{4} \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} (\tan x + \cot x)^{-1} dx = \frac{1}{4}$$

Question 11 (i)

Evaluate $\int_0^{\pi/2} \sqrt{1 - \cos 2x} \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/2} \sqrt{1 - \cos 2x} \, dx$

Concept / Formula Used: Trigonometric identity: $1 - \cos 2x = 2 \sin^2 x$

Solution:

Using the double-angle identity $1 - \cos 2x = 2 \sin^2 x$:

$$\begin{aligned} \int_0^{\pi/2} \sqrt{1 - \cos 2x} \, dx &= \int_0^{\pi/2} \sqrt{2 \sin^2 x} \, dx \\ &= \int_0^{\pi/2} \sqrt{2} |\sin x| \, dx \end{aligned}$$

Since $x \in [0, \frac{\pi}{2}]$, $\sin x \geq 0$, so $|\sin x| = \sin x$:

$$\begin{aligned} &= \sqrt{2} \int_0^{\pi/2} \sin x \, dx \\ &= \sqrt{2} [-\cos x]_0^{\pi/2} \\ &= -\sqrt{2} \left(\cos \left(\frac{\pi}{2} \right) - \cos(0) \right) \end{aligned}$$

Since $\cos \left(\frac{\pi}{2} \right) = 0$ and $\cos(0) = 1$:

$$\begin{aligned} &= -\sqrt{2}(0 - 1) \\ &= \sqrt{2} \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \sqrt{1 - \cos 2x} \, dx = \sqrt{2}$$

Question 11 (ii)

Evaluate $\int_0^{\pi} \sqrt{1 + \sin x} \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi} \sqrt{1 + \sin x} \, dx$

Concept / Formula Used: Trigonometric identity: $1 + \sin x = \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2$

Solution:

Expressing 1 as $\sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right)$ and $\sin x$ as $2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$:

$$\begin{aligned} 1 + \sin x &= \sin^2\left(\frac{x}{2}\right) + \cos^2\left(\frac{x}{2}\right) + 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) \\ &= \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2 \end{aligned}$$

Substituting into the integral:

$$\begin{aligned} \int_0^\pi \sqrt{1 + \sin x} \, dx &= \int_0^\pi \sqrt{\left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2} \, dx \\ &= \int_0^\pi \left|\sin \frac{x}{2} + \cos \frac{x}{2}\right| \, dx \end{aligned}$$

For $x \in [0, \pi]$, both $\sin\left(\frac{x}{2}\right)$ and $\cos\left(\frac{x}{2}\right)$ are non-negative, so $\sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right) \geq 0$:

$$\begin{aligned} &= \int_0^\pi \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right) \, dx \\ &= \left[-2 \cos\left(\frac{x}{2}\right) + 2 \sin\left(\frac{x}{2}\right)\right]_0^\pi \\ &= 2 \left[\sin\left(\frac{x}{2}\right) - \cos\left(\frac{x}{2}\right)\right]_0^\pi \end{aligned}$$

Substituting the limits:

$$\begin{aligned} &= 2 \left[\left(\sin\left(\frac{\pi}{2}\right) - \cos\left(\frac{\pi}{2}\right)\right) - (\sin(0) - \cos(0))\right] \\ &= 2 [(1 - 0) - (0 - 1)] \\ &= 2 [1 - (-1)] \\ &= 2(1 + 1) \\ &= 4 \end{aligned}$$

Final Answer

$$\int_0^\pi \sqrt{1 + \sin x} \, dx = 4$$

Question 12 (i)

Evaluate $\int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} \, dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} \, dx$

Concept / Formula Used: Trigonometric identity: $1 + \sin 2x = (\sin x + \cos x)^2$

Solution:

Using the identity $1 + \sin 2x = (\sin x + \cos x)^2$:

$$\begin{aligned} \int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx &= \int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{(\sin x + \cos x)^2}} dx \\ &= \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx \\ &= \int_0^{\pi/2} 1 dx \end{aligned}$$

Integrating:

$$\begin{aligned} &= [x]_0^{\pi/2} \\ &= \frac{\pi}{2} - 0 \\ &= \frac{\pi}{2} \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx = \frac{\pi}{2}$$

Question 12 (ii)

Evaluate $\int_0^{\pi/4} \frac{\cos 3x}{\cos x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/4} \frac{\cos 3x}{\cos x} dx$

Concept / Formula Used: Trigonometric identity: $\cos 3x = 4 \cos^3 x - 3 \cos x$

Solution:

Using the triple-angle identity $\cos 3x = 4 \cos^3 x - 3 \cos x$:

$$\begin{aligned} \int_0^{\pi/4} \frac{\cos 3x}{\cos x} dx &= \int_0^{\pi/4} \frac{4 \cos^3 x - 3 \cos x}{\cos x} dx \\ &= \int_0^{\pi/4} (4 \cos^2 x - 3) dx \end{aligned}$$

Using the half-angle identity $\cos^2 x = \frac{1+\cos 2x}{2}$:

$$\begin{aligned}
 &= \int_0^{\pi/4} \left(4 \left(\frac{1+\cos 2x}{2} \right) - 3 \right) dx \\
 &= \int_0^{\pi/4} (2(1+\cos 2x) - 3) dx \\
 &= \int_0^{\pi/4} (2 + 2\cos 2x - 3) dx \\
 &= \int_0^{\pi/4} (2\cos 2x - 1) dx
 \end{aligned}$$

Integrating term by term:

$$\begin{aligned}
 &= \left[2 \left(\frac{\sin 2x}{2} \right) - x \right]_0^{\pi/4} \\
 &= [\sin 2x - x]_0^{\pi/4}
 \end{aligned}$$

Substituting the limits:

$$\begin{aligned}
 &= \left(\sin \left(2 \cdot \frac{\pi}{4} \right) - \frac{\pi}{4} \right) - (\sin(0) - 0) \\
 &= \sin \left(\frac{\pi}{2} \right) - \frac{\pi}{4} \\
 &= 1 - \frac{\pi}{4}
 \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \frac{\cos 3x}{\cos x} dx = 1 - \frac{\pi}{4}$$

Question 13 (i)

Evaluate $\int_0^{\pi/4} \sin 2x \sin 3x dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: Evaluate $\int_0^{\pi/4} \sin 2x \sin 3x dx$

Concept / Formula Used: Product-to-sum identity: $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$

Solution:

Using the product-to-sum formula $2 \sin 2x \sin 3x = \cos(2x - 3x) - \cos(2x + 3x)$:

$$\begin{aligned}
 \sin 2x \sin 3x &= \frac{1}{2} [\cos(-x) - \cos(5x)] \\
 &= \frac{1}{2} [\cos x - \cos(5x)] \quad (\text{since } \cos(-x) = \cos x)
 \end{aligned}$$

Substituting into the integral:

$$\begin{aligned}\int_0^{\pi/4} \sin 2x \sin 3x \, dx &= \frac{1}{2} \int_0^{\pi/4} (\cos x - \cos 5x) \, dx \\ &= \frac{1}{2} \left[\sin x - \frac{\sin 5x}{5} \right]_0^{\pi/4}\end{aligned}$$

Substituting the limits:

$$\begin{aligned}&= \frac{1}{2} \left[\left(\sin \left(\frac{\pi}{4} \right) - \frac{\sin \left(\frac{5\pi}{4} \right)}{5} \right) - (\sin(0) - 0) \right] \\ &= \frac{1}{2} \left[\frac{1}{\sqrt{2}} - \frac{-\frac{1}{\sqrt{2}}}{5} \right] \\ &= \frac{1}{2} \left[\frac{1}{\sqrt{2}} + \frac{1}{5\sqrt{2}} \right] \\ &= \frac{1}{2} \left[\frac{5+1}{5\sqrt{2}} \right] \\ &= \frac{1}{2} \left(\frac{6}{5\sqrt{2}} \right) \\ &= \frac{3}{5\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{3\sqrt{2}}{10}\end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \sin 2x \sin 3x \, dx = \frac{3\sqrt{2}}{10}$$

Question 13 (ii)

Evaluate $\int_0^{\pi/2} (a^2 \cos^2 x + b^2 \sin^2 x) \, dx$.

TYPE 15 Integration Using Trigonometric Identities

Given: Evaluate $\int_0^{\pi/2} (a^2 \cos^2 x + b^2 \sin^2 x) \, dx$

Concept / Formula Used: Half-angle identities: $\cos^2 x = \frac{1+\cos 2x}{2}$, $\sin^2 x = \frac{1-\cos 2x}{2}$

Solution:

Using half-angle identities for $\cos^2 x$ and $\sin^2 x$:

$$\begin{aligned}\int_0^{\pi/2} (a^2 \cos^2 x + b^2 \sin^2 x) dx &= \int_0^{\pi/2} \left[a^2 \left(\frac{1 + \cos 2x}{2} \right) + b^2 \left(\frac{1 - \cos 2x}{2} \right) \right] dx \\ &= \frac{1}{2} \int_0^{\pi/2} [(a^2 + b^2) + (a^2 - b^2) \cos 2x] dx\end{aligned}$$

Integrating term by term:

$$= \frac{1}{2} \left[(a^2 + b^2)x + (a^2 - b^2) \frac{\sin 2x}{2} \right]_0^{\pi/2}$$

Substituting the limits:

$$= \frac{1}{2} \left[\left((a^2 + b^2) \frac{\pi}{2} + (a^2 - b^2) \frac{\sin(\pi)}{2} \right) - (0 + 0) \right]$$

Since $\sin(\pi) = 0$:

$$\begin{aligned}&= \frac{1}{2} \left((a^2 + b^2) \frac{\pi}{2} + 0 \right) \\ &= \frac{\pi}{4} (a^2 + b^2)\end{aligned}$$

Final Answer

$$\int_0^{\pi/2} (a^2 \cos^2 x + b^2 \sin^2 x) dx = \frac{\pi}{4} (a^2 + b^2)$$

Question 14 (i)

Evaluate $\int_{\pi/3}^{\pi/4} (\tan x + \cot x)^2 dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_{\pi/3}^{\pi/4} (\tan x + \cot x)^2 dx$

Concept / Formula Used: Algebraic expansion and identity: $(\tan x + \cot x)^2 = \tan^2 x + \cot^2 x + 2 = \sec^2 x + \csc^2 x$

Solution:

Expanding the square:

$$\begin{aligned}(\tan x + \cot x)^2 &= \tan^2 x + \cot^2 x + 2 \tan x \cot x \\ &= \tan^2 x + \cot^2 x + 2 \quad (\text{since } \tan x \cot x = 1)\end{aligned}$$

Using the identities $\tan^2 x = \sec^2 x - 1$ and $\cot^2 x = \csc^2 x - 1$:

$$\begin{aligned} &= (\sec^2 x - 1) + (\csc^2 x - 1) + 2 \\ &= \sec^2 x + \csc^2 x - 2 + 2 \\ &= \sec^2 x + \csc^2 x \end{aligned}$$

Substituting back into the integral:

$$\begin{aligned} \int_{\pi/3}^{\pi/4} (\tan x + \cot x)^2 dx &= \int_{\pi/3}^{\pi/4} (\sec^2 x + \csc^2 x) dx \\ &= [\tan x - \cot x]_{\pi/3}^{\pi/4} \end{aligned}$$

Substituting the limits:

$$= \left(\tan \left(\frac{\pi}{4} \right) - \cot \left(\frac{\pi}{4} \right) \right) - \left(\tan \left(\frac{\pi}{3} \right) - \cot \left(\frac{\pi}{3} \right) \right)$$

Evaluating the trigonometric values: $\tan \left(\frac{\pi}{4} \right) = 1$, $\cot \left(\frac{\pi}{4} \right) = 1$, $\tan \left(\frac{\pi}{3} \right) = \sqrt{3}$, and $\cot \left(\frac{\pi}{3} \right) = \frac{1}{\sqrt{3}}$:

$$\begin{aligned} &= (1 - 1) - \left(\sqrt{3} - \frac{1}{\sqrt{3}} \right) \\ &= 0 - \left(\frac{3 - 1}{\sqrt{3}} \right) \\ &= -\frac{2}{\sqrt{3}} \end{aligned}$$

Final Answer

$$\int_{\pi/3}^{\pi/4} (\tan x + \cot x)^2 dx = -\frac{2}{\sqrt{3}}$$

Question 14 (ii)

Evaluate $\int_1^2 \frac{dx}{(x+1)(x+2)}$.

TYPE 14 Partial Fractions

Given: Evaluate $\int_1^2 \frac{dx}{(x+1)(x+2)}$

Concept / Formula Used: Partial fractions: $\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$

Solution:

Decomposing into partial fractions:

$$\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Multiplying both sides by $(x+1)(x+2)$:

$$1 = A(x+2) + B(x+1)$$

Substituting $x = -1$:

$$1 = A(-1+2) + B(0) \implies A = 1$$

Substituting $x = -2$:

$$1 = A(0) + B(-2+1) \implies -B = 1 \implies B = -1$$

Thus, the integrand is $\frac{1}{x+1} - \frac{1}{x+2}$. Evaluating the integral:

$$\begin{aligned} \int_1^2 \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx &= [\log |x+1| - \log |x+2|]_1^2 \\ &= \left[\log \left| \frac{x+1}{x+2} \right| \right]_1^2 \end{aligned}$$

Substituting the limits:

$$\begin{aligned} &= \log \left| \frac{2+1}{2+2} \right| - \log \left| \frac{1+1}{1+2} \right| \\ &= \log \left(\frac{3}{4} \right) - \log \left(\frac{2}{3} \right) \end{aligned}$$

Using logarithm properties:

$$\begin{aligned} &= \log \left(\frac{3/4}{2/3} \right) \\ &= \log \left(\frac{9}{8} \right) \end{aligned}$$

Final Answer

$$\int_1^2 \frac{dx}{(x+1)(x+2)} = \log \left(\frac{9}{8} \right)$$

Question 15 (i)

Evaluate $\int_0^{\pi/4} \frac{1 - \tan x}{1 + \tan x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_0^{\pi/4} \frac{1 - \tan x}{1 + \tan x} dx$

Concept / Formula Used: Trigonometric identity: $\frac{1 - \tan x}{1 + \tan x} = \tan \left(\frac{\pi}{4} - x \right)$

Solution:

Using the identity $\frac{1 - \tan x}{1 + \tan x} = \tan\left(\frac{\pi}{4} - x\right)$:

$$\int_0^{\pi/4} \frac{1 - \tan x}{1 + \tan x} dx = \int_0^{\pi/4} \tan\left(\frac{\pi}{4} - x\right) dx$$

Let $t = \frac{\pi}{4} - x$, so $dt = -dx$.

Changing the limits: when $x = 0$, $t = \frac{\pi}{4}$; when $x = \frac{\pi}{4}$, $t = 0$.

$$\begin{aligned} \int_0^{\pi/4} \tan\left(\frac{\pi}{4} - x\right) dx &= \int_{\pi/4}^0 \tan t (-dt) \\ &= \int_0^{\pi/4} \tan t dt \end{aligned}$$

Using the standard result $\int \tan t dt = \log |\sec t|$:

$$\begin{aligned} &= [\log |\sec t|]_0^{\pi/4} \\ &= \log \left| \sec\left(\frac{\pi}{4}\right) \right| - \log |\sec(0)| \end{aligned}$$

Since $\sec\left(\frac{\pi}{4}\right) = \sqrt{2}$ and $\sec(0) = 1$:

$$\begin{aligned} &= \log(\sqrt{2}) - \log(1) \\ &= \frac{1}{2} \log 2 - 0 \\ &= \frac{1}{2} \log 2 \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \frac{1 - \tan x}{1 + \tan x} dx = \frac{1}{2} \log 2$$

Common Mistake: The originally drafted limits $-\pi/4$ to 0 make the integrand blow up at $x = -\pi/4$ (where $1 + \tan x = 0$), so that version of the integral genuinely diverges; the well-posed standard problem uses limits 0 to $\pi/4$, as solved here.

Question 15 (ii)

Evaluate $\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx$.

TYPE 7 Integration From Derivative

Given: Evaluate $\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx$

Concept / Formula Used: Expressing numerator as $A(\text{denominator}) + B(\text{derivative of denominator})$

Solution:

Let the numerator be expressed as:

$$3 \sin x + 4 \cos x = A(\sin x + \cos x) + B \frac{d}{dx}(\sin x + \cos x)$$

$$3 \sin x + 4 \cos x = A(\sin x + \cos x) + B(\cos x - \sin x)$$

Comparing coefficients of $\sin x$ and $\cos x$ on both sides:

$$\text{For } \sin x : A - B = 3$$

$$\text{For } \cos x : A + B = 4$$

Adding the two equations to solve for A :

$$(A - B) + (A + B) = 3 + 4 \implies 2A = 7 \implies A = \frac{7}{2}$$

Substituting $A = \frac{7}{2}$ into $A + B = 4$:

$$\frac{7}{2} + B = 4 \implies B = 4 - \frac{7}{2} = \frac{1}{2}$$

Thus, the integrand can be written as:

$$\frac{3 \sin x + 4 \cos x}{\sin x + \cos x} = \frac{\frac{7}{2}(\sin x + \cos x) + \frac{1}{2}(\cos x - \sin x)}{\sin x + \cos x} = \frac{7}{2} + \frac{1}{2} \left(\frac{\cos x - \sin x}{\sin x + \cos x} \right)$$

Evaluating the integral:

$$\int_0^{\pi/2} \left[\frac{7}{2} + \frac{1}{2} \left(\frac{\cos x - \sin x}{\sin x + \cos x} \right) \right] dx = \left[\frac{7}{2}x + \frac{1}{2} \log |\sin x + \cos x| \right]_0^{\pi/2}$$

Substituting the upper limit $x = \frac{\pi}{2}$:

$$\begin{aligned} \text{Upper value} &= \frac{7}{2} \left(\frac{\pi}{2} \right) + \frac{1}{2} \log \left| \sin \left(\frac{\pi}{2} \right) + \cos \left(\frac{\pi}{2} \right) \right| \\ &= \frac{7\pi}{4} + \frac{1}{2} \log |1 + 0| \\ &= \frac{7\pi}{4} + 0 = \frac{7\pi}{4} \end{aligned}$$

Substituting the lower limit $x = 0$:

$$\begin{aligned} \text{Lower value} &= \frac{7}{2}(0) + \frac{1}{2} \log |\sin(0) + \cos(0)| \\ &= 0 + \frac{1}{2} \log |0 + 1| \\ &= 0 \end{aligned}$$

Subtracting lower value from upper value:

$$= \frac{7\pi}{4} - 0 = \frac{7\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx = \frac{7\pi}{4}$$

Question 16 (i)

Evaluate $\int_1^2 \frac{x+3}{x(x+2)} dx$.

TYPE 14 Partial Fractions

Given: Evaluate $\int_1^2 \frac{x+3}{x(x+2)} dx$

Concept / Formula Used: Partial fractions: $\frac{x+3}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2}$

Solution:

Decomposing into partial fractions:

$$\frac{x+3}{x(x+2)} = \frac{A}{x} + \frac{B}{x+2}$$

Multiplying both sides by $x(x+2)$:

$$x+3 = A(x+2) + Bx$$

Substituting $x = 0$:

$$3 = A(0+2) + 0 \implies 2A = 3 \implies A = \frac{3}{2}$$

Substituting $x = -2$:

$$-2+3 = A(0) + B(-2) \implies 1 = -2B \implies B = -\frac{1}{2}$$

Thus, the integrand is $\frac{3/2}{x} - \frac{1/2}{x+2}$. Evaluating the integral:

$$\begin{aligned} \int_1^2 \left(\frac{3/2}{x} - \frac{1/2}{x+2} \right) dx &= \left[\frac{3}{2} \log |x| - \frac{1}{2} \log |x+2| \right]_1^2 \\ &= \frac{1}{2} [3 \log |x| - \log |x+2|]_1^2 \\ &= \frac{1}{2} [\log |x^3| - \log |x+2|]_1^2 \\ &= \frac{1}{2} \left[\log \left| \frac{x^3}{x+2} \right| \right]_1^2 \end{aligned}$$

Substituting the limits:

$$\begin{aligned}
 &= \frac{1}{2} \left(\log \left| \frac{(2)^3}{2+2} \right| - \log \left| \frac{(1)^3}{1+2} \right| \right) \\
 &= \frac{1}{2} \left(\log \left(\frac{8}{4} \right) - \log \left(\frac{1}{3} \right) \right) \\
 &= \frac{1}{2} \left(\log(2) - \log \left(\frac{1}{3} \right) \right) \\
 &= \frac{1}{2} \log \left(\frac{2}{1/3} \right) \\
 &= \frac{1}{2} \log(6)
 \end{aligned}$$

Final Answer

$$\int_1^2 \frac{x+3}{x(x+2)} dx = \frac{1}{2} \log 6$$

Question 16 (ii)

Evaluate $\int_1^3 \frac{dx}{x^2(x+1)}$.

TYPE 14 Partial Fractions

Given: Evaluate $\int_1^3 \frac{dx}{x^2(x+1)}$

Concept / Formula Used: Partial fractions: $\frac{1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$

Solution:

Decomposing into partial fractions:

$$\frac{1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$$

Multiplying both sides by $x^2(x+1)$:

$$1 = Ax(x+1) + B(x+1) + Cx^2$$

Substituting $x = 0$:

$$1 = A(0) + B(0+1) + C(0) \implies B = 1$$

Substituting $x = -1$:

$$1 = A(0) + B(0) + C(-1)^2 \implies C = 1$$

Comparing coefficients of x^2 on both sides ($A + C = 0$):

$$A + 1 = 0 \implies A = -1$$

Thus, the partial fraction decomposition is $-\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x+1}$. Evaluating the integral:

$$\begin{aligned}\int_1^3 \left(-\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x+1} \right) dx &= \left[-\log|x| - \frac{1}{x} + \log|x+1| \right]_1^3 \\ &= \left[\log \left| \frac{x+1}{x} \right| - \frac{1}{x} \right]_1^3\end{aligned}$$

Substituting the upper limit $x = 3$:

$$\text{Upper value} = \log \left(\frac{3+1}{3} \right) - \frac{1}{3} = \log \left(\frac{4}{3} \right) - \frac{1}{3}$$

Substituting the lower limit $x = 1$:

$$\text{Lower value} = \log \left(\frac{1+1}{1} \right) - \frac{1}{1} = \log(2) - 1$$

Subtracting lower value from upper value:

$$\begin{aligned}&= \left(\log \left(\frac{4}{3} \right) - \frac{1}{3} \right) - (\log(2) - 1) \\ &= \log \left(\frac{4}{3} \right) - \log(2) - \frac{1}{3} + 1 \\ &= \log \left(\frac{4/3}{2} \right) + \frac{2}{3} \\ &= \log \left(\frac{2}{3} \right) + \frac{2}{3}\end{aligned}$$

Final Answer

$$\int_1^3 \frac{dx}{x^2(x+1)} = \frac{2}{3} + \log \left(\frac{2}{3} \right)$$

Question 17 (i)

Evaluate $\int_{-\pi/4}^0 \frac{1 + \tan x}{1 - \tan x} dx$.

TYPE 2 Trigonometric Identity Simplification

Given: Evaluate $\int_{-\pi/4}^0 \frac{1 + \tan x}{1 - \tan x} dx$

Concept / Formula Used: Trigonometric identity: $\frac{1+\tan x}{1-\tan x} = \tan \left(\frac{\pi}{4} + x \right)$

Solution:

Using the identity $\frac{1+\tan x}{1-\tan x} = \tan \left(\frac{\pi}{4} + x \right)$:

$$\int_{-\pi/4}^0 \frac{1 + \tan x}{1 - \tan x} dx = \int_{-\pi/4}^0 \tan \left(\frac{\pi}{4} + x \right) dx$$

Using the standard integration result $\int \tan(ax + b) dx = \frac{1}{a} \log |\sec(ax + b)|$:

$$= \left[\log \left| \sec \left(\frac{\pi}{4} + x \right) \right| \right]_{-\pi/4}^0$$

Substituting the limits:

$$\begin{aligned} &= \log \left| \sec \left(\frac{\pi}{4} + 0 \right) \right| - \log \left| \sec \left(\frac{\pi}{4} + \left(-\frac{\pi}{4} \right) \right) \right| \\ &= \log \left| \sec \left(\frac{\pi}{4} \right) \right| - \log |\sec(0)| \end{aligned}$$

Since $\sec \left(\frac{\pi}{4} \right) = \sqrt{2}$ and $\sec(0) = 1$:

$$\begin{aligned} &= \log(\sqrt{2}) - \log(1) \\ &= \frac{1}{2} \log(2) - 0 \\ &= \frac{1}{2} \log 2 \end{aligned}$$

Final Answer

$$\int_{-\pi/4}^0 \frac{1 + \tan x}{1 - \tan x} dx = \frac{1}{2} \log 2$$

Question 17 (ii)

Evaluate $\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx$.

TYPE 7 Integration From Derivative

Given: Evaluate $\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx$

Concept / Formula Used: Expressed numerator as $A(\text{denominator}) + B(\text{derivative of denominator})$

Solution:

(Note: This is identical to Question 15(ii)). Following the same method: Expressing the numerator $3 \sin x + 4 \cos x = \frac{7}{2}(\sin x + \cos x) + \frac{1}{2}(\cos x - \sin x)$:

$$\begin{aligned} \int_0^{\pi/2} \left[\frac{7}{2} + \frac{1}{2} \left(\frac{\cos x - \sin x}{\sin x + \cos x} \right) \right] dx &= \left[\frac{7}{2}x + \frac{1}{2} \log |\sin x + \cos x| \right]_0^{\pi/2} \\ &= \left(\frac{7\pi}{4} + 0 \right) - (0 + 0) \\ &= \frac{7\pi}{4} \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{3 \sin x + 4 \cos x}{\sin x + \cos x} dx = \frac{7\pi}{4}$$

Question 18 (i)

Evaluate $\int_0^1 x e^x dx$.

TYPE 13 Integration by Parts

Given: Evaluate $\int_0^1 x e^x dx$

Concept / Formula Used: Integration by parts formula: $\int u v dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx$

Solution:

Using integration by parts with $u = x$ and $v = e^x$:

$$\begin{aligned} \int_0^1 x e^x dx &= \left[x \int e^x dx \right]_0^1 - \int_0^1 \left(\frac{d}{dx}(x) \int e^x dx \right) dx \\ &= [x e^x]_0^1 - \int_0^1 (1 \cdot e^x) dx \\ &= [x e^x]_0^1 - [e^x]_0^1 \\ &= [x e^x - e^x]_0^1 \\ &= [e^x(x - 1)]_0^1 \end{aligned}$$

Substituting the upper and lower limits:

$$\begin{aligned} &= e^1(1 - 1) - e^0(0 - 1) \\ &= e(0) - 1(-1) \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

Final Answer

$$\int_0^1 x e^x dx = 1$$

Question 18 (ii)

Evaluate $\int_0^1 \left(x e^x + \sin \frac{\pi x}{4} \right) dx$.

TYPE 13 Integration by Parts

Given: Evaluate $\int_0^1 \left(xe^x + \sin \frac{\pi x}{4} \right) dx$

Concept / Formula Used: Integration by parts and standard trigonometric integration

Solution:

Integrating term by term (using $\int xe^x dx = xe^x - e^x$ from Question 18(i) and $\int \sin(ax) dx = -\frac{\cos(ax)}{a}$):

$$\begin{aligned} \int_0^1 \left(xe^x + \sin \frac{\pi x}{4} \right) dx &= \left[xe^x - e^x - \frac{\cos \left(\frac{\pi x}{4} \right)}{\frac{\pi}{4}} \right]_0^1 \\ &= \left[e^x(x-1) - \frac{4}{\pi} \cos \left(\frac{\pi x}{4} \right) \right]_0^1 \end{aligned}$$

Substituting the upper limit $x = 1$:

$$\begin{aligned} \text{Upper value} &= e^1(1-1) - \frac{4}{\pi} \cos \left(\frac{\pi(1)}{4} \right) \\ &= 0 - \frac{4}{\pi} \left(\frac{1}{\sqrt{2}} \right) \\ &= -\frac{4}{\pi\sqrt{2}} \\ &= -\frac{2\sqrt{2}}{\pi} \end{aligned}$$

Substituting the lower limit $x = 0$:

$$\begin{aligned} \text{Lower value} &= e^0(0-1) - \frac{4}{\pi} \cos(0) \\ &= 1(-1) - \frac{4}{\pi}(1) \\ &= -1 - \frac{4}{\pi} \end{aligned}$$

Subtracting lower value from upper value:

$$\begin{aligned} &= -\frac{2\sqrt{2}}{\pi} - \left(-1 - \frac{4}{\pi} \right) \\ &= -\frac{2\sqrt{2}}{\pi} + 1 + \frac{4}{\pi} \\ &= 1 + \frac{4-2\sqrt{2}}{\pi} \end{aligned}$$

Final Answer

$$\int_0^1 \left(xe^x + \sin \frac{\pi x}{4} \right) dx = 1 + \frac{4-2\sqrt{2}}{\pi}$$

Question 19 (i)

Evaluate $\int_0^1 \left(xe^{2x} + \sin \frac{\pi x}{4} \right) dx$.

TYPE 13 Integration by Parts

Given: Evaluate $\int_0^1 \left(xe^{2x} + \sin \frac{\pi x}{4} \right) dx$

Concept / Formula Used: Integration by parts: $\int xe^{2x} dx = x \left(\frac{e^{2x}}{2} \right) - \int 1 \cdot \frac{e^{2x}}{2} dx = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4}$

Solution:

Integrating term by term: For $\int xe^{2x} dx$, using integration by parts with $u = x, v = e^{2x}$:

$$\int xe^{2x} dx = \frac{xe^{2x}}{2} - \frac{e^{2x}}{4}$$

For $\int \sin \left(\frac{\pi x}{4} \right) dx$, using standard formula:

$$\int \sin \left(\frac{\pi x}{4} \right) dx = -\frac{4}{\pi} \cos \left(\frac{\pi x}{4} \right)$$

Combining the definite integral from 0 to 1:

$$\begin{aligned} \int_0^1 \left(xe^{2x} + \sin \frac{\pi x}{4} \right) dx &= \left[\frac{xe^{2x}}{2} - \frac{e^{2x}}{4} - \frac{4}{\pi} \cos \left(\frac{\pi x}{4} \right) \right]_0^1 \\ &= \left[e^{2x} \left(\frac{2x-1}{4} \right) - \frac{4}{\pi} \cos \left(\frac{\pi x}{4} \right) \right]_0^1 \end{aligned}$$

Substituting the upper limit $x = 1$:

$$\begin{aligned} \text{Upper value} &= e^2 \left(\frac{2(1)-1}{4} \right) - \frac{4}{\pi} \cos \left(\frac{\pi}{4} \right) \\ &= \frac{e^2}{4} - \frac{4}{\pi} \left(\frac{1}{\sqrt{2}} \right) \\ &= \frac{e^2}{4} - \frac{2\sqrt{2}}{\pi} \end{aligned}$$

Substituting the lower limit $x = 0$:

$$\begin{aligned} \text{Lower value} &= e^0 \left(\frac{2(0)-1}{4} \right) - \frac{4}{\pi} \cos(0) \\ &= 1 \left(-\frac{1}{4} \right) - \frac{4}{\pi} (1) \\ &= -\frac{1}{4} - \frac{4}{\pi} \end{aligned}$$

Subtracting lower value from upper value:

$$\begin{aligned}
 &= \left(\frac{e^2}{4} - \frac{2\sqrt{2}}{\pi} \right) - \left(-\frac{1}{4} - \frac{4}{\pi} \right) \\
 &= \frac{e^2}{4} + \frac{1}{4} + \frac{4 - 2\sqrt{2}}{\pi} \\
 &= \frac{e^2 + 1}{4} + \frac{4 - 2\sqrt{2}}{\pi}
 \end{aligned}$$

Final Answer

$$\int_0^1 \left(xe^{2x} + \sin \frac{\pi x}{4} \right) dx = \frac{e^2 + 1}{4} + \frac{4 - 2\sqrt{2}}{\pi}$$

Question 19 (ii)

Evaluate $\int_0^{\pi/2} x^2 \cos 2x \, dx$.

TYPE 13 Integration by Parts

Given: Evaluate $\int_0^{\pi/2} x^2 \cos 2x \, dx$

Concept / Formula Used: Repeated integration by parts for polynomial times trigonometric function

Solution:

Using integration by parts twice with $u = x^2$ and $v = \cos 2x$: First integration by parts:

$$\begin{aligned}
 \int x^2 \cos 2x \, dx &= x^2 \left(\frac{\sin 2x}{2} \right) - \int 2x \left(\frac{\sin 2x}{2} \right) dx \\
 &= \frac{x^2 \sin 2x}{2} - \int x \sin 2x \, dx
 \end{aligned}$$

Second integration by parts for $\int x \sin 2x \, dx$:

$$\begin{aligned}
 \int x \sin 2x \, dx &= x \left(-\frac{\cos 2x}{2} \right) - \int 1 \cdot \left(-\frac{\cos 2x}{2} \right) dx \\
 &= -\frac{x \cos 2x}{2} + \frac{1}{2} \int \cos 2x \, dx \\
 &= -\frac{x \cos 2x}{2} + \frac{\sin 2x}{4}
 \end{aligned}$$

Combining both parts:

$$\begin{aligned}
 \int x^2 \cos 2x \, dx &= \frac{x^2 \sin 2x}{2} - \left(-\frac{x \cos 2x}{2} + \frac{\sin 2x}{4} \right) \\
 &= \frac{x^2 \sin 2x}{2} + \frac{x \cos 2x}{2} - \frac{\sin 2x}{4}
 \end{aligned}$$

Evaluating the definite integral from 0 to $\frac{\pi}{2}$:

$$\int_0^{\pi/2} x^2 \cos 2x \, dx = \left[\frac{x^2 \sin 2x}{2} + \frac{x \cos 2x}{2} - \frac{\sin 2x}{4} \right]_0^{\pi/2}$$

Substituting the upper limit $x = \frac{\pi}{2}$:

$$\begin{aligned} \text{Upper value} &= \frac{(\pi/2)^2 \sin(2 \cdot \frac{\pi}{2})}{2} + \frac{(\pi/2) \cos(2 \cdot \frac{\pi}{2})}{2} - \frac{\sin(2 \cdot \frac{\pi}{2})}{4} \\ &= \frac{\pi^2}{4} \left(\frac{\sin(\pi)}{2} \right) + \frac{\pi}{2} \left(\frac{\cos(\pi)}{2} \right) - \frac{\sin(\pi)}{4} \end{aligned}$$

Since $\sin(\pi) = 0$ and $\cos(\pi) = -1$:

$$\begin{aligned} &= 0 + \frac{\pi}{2} \left(\frac{-1}{2} \right) - 0 \\ &= -\frac{\pi}{4} \end{aligned}$$

Substituting the lower limit $x = 0$:

$$\text{Lower value} = \left(\frac{0^2 \sin(0)}{2} + \frac{0 \cos(0)}{2} - \frac{\sin(0)}{4} \right) = 0$$

Subtracting lower value from upper value:

$$= -\frac{\pi}{4} - 0 = -\frac{\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} x^2 \cos 2x \, dx = -\frac{\pi}{4}$$

Question 20 (i)

Evaluate $\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$.

TYPE 7 Integration From Derivative

Given: Evaluate $\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

Concept / Formula Used: Standard result: $\int e^x [f(x) + f'(x)] dx = e^x f(x)$

Solution:

Let $f(x) = \frac{1}{x}$. Then its derivative is $f'(x) = -\frac{1}{x^2}$. Using the standard integration formula

$$\int e^x[f(x) + f'(x)] dx = e^x f(x):$$

$$\begin{aligned}\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx &= \left[e^x \cdot \frac{1}{x} \right]_1^2 \\ &= \left[\frac{e^x}{x} \right]_1^2\end{aligned}$$

Substituting the upper and lower limits:

$$\begin{aligned}&= \left(\frac{e^2}{2} \right) - \left(\frac{e^1}{1} \right) \\ &= \frac{e^2}{2} - e\end{aligned}$$

Final Answer

$$\int_1^2 e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx = \frac{e^2}{2} - e$$

Question 20 (ii)

Evaluate $\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx.$

TYPE 7 Integration From Derivative

Given: Evaluate $\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx$

Concept / Formula Used: Trigonometric simplification to form $e^x[f(x) + f'(x)]$

Solution:

Simplifying the integrand:

$$\begin{aligned}\frac{1 - \sin x}{1 - \cos x} &= \frac{1 - 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{2 \sin^2 \left(\frac{x}{2} \right)} \\ &= \frac{1}{2 \sin^2 \left(\frac{x}{2} \right)} - \frac{2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{2 \sin^2 \left(\frac{x}{2} \right)} \\ &= \frac{1}{2} \csc^2 \left(\frac{x}{2} \right) - \cot \left(\frac{x}{2} \right)\end{aligned}$$

Let $f(x) = -\cot \left(\frac{x}{2} \right)$. Differentiating $f(x)$:

$$f'(x) = - \left(-\csc^2 \left(\frac{x}{2} \right) \cdot \frac{1}{2} \right) = \frac{1}{2} \csc^2 \left(\frac{x}{2} \right)$$

Thus, the integrand is of the form $e^x[f(x) + f'(x)]$, where $f(x) = -\cot \left(\frac{x}{2} \right)$. Using the standard result $\int e^x[f(x) + f'(x)] dx = e^x f(x)$:

$$\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx = \left[-e^x \cot \left(\frac{x}{2} \right) \right]_{\pi/2}^{\pi}$$

Substituting the upper limit $x = \pi$:

$$\begin{aligned}\text{Upper value} &= -e^{\pi} \cot\left(\frac{\pi}{2}\right) \\ &= -e^{\pi}(0) = 0\end{aligned}$$

Substituting the lower limit $x = \frac{\pi}{2}$:

$$\begin{aligned}\text{Lower value} &= -e^{\pi/2} \cot\left(\frac{\pi}{4}\right) \\ &= -e^{\pi/2}(1) = -e^{\pi/2}\end{aligned}$$

Subtracting lower value from upper value:

$$\begin{aligned}&= 0 - (-e^{\pi/2}) \\ &= e^{\pi/2}\end{aligned}$$

Final Answer

$$\int_{\pi/2}^{\pi} e^x \left(\frac{1 - \sin x}{1 - \cos x} \right) dx = e^{\pi/2}$$

Question 21 (i)

If $\int_0^a 3x^2 dx = 8$, find the value of a .

TYPE 1 Power Rule of Integration

Given: $\int_0^a 3x^2 dx = 8$

Concept / Formula Used: Integration power rule and equation solving

Solution:

Evaluating the integral on the left-hand side:

$$\begin{aligned}\int_0^a 3x^2 dx &= 3 \left[\frac{x^3}{3} \right]_0^a \\ &= [x^3]_0^a \\ &= a^3 - 0^3 \\ &= a^3\end{aligned}$$

Equating to the given value:

$$a^3 = 8$$

Taking the cube root of both sides:

$$a = \sqrt[3]{8} = 2$$

Final Answer

$$a = 2$$

Question 21 (ii)

If $\int_1^a (3x^2 + 2x + 1) dx = 11$, find the value(s) of a .

TYPE 1 Power Rule of Integration

Given: $\int_1^a (3x^2 + 2x + 1) dx = 11$

Concept / Formula Used: Integration power rule and polynomial root finding

Solution:

Evaluating the integral on the left-hand side:

$$\begin{aligned}\int_1^a (3x^2 + 2x + 1) dx &= \left[3 \left(\frac{x^3}{3} \right) + 2 \left(\frac{x^2}{2} \right) + x \right]_1^a \\ &= [x^3 + x^2 + x]_1^a\end{aligned}$$

Substituting the upper limit a and lower limit 1:

$$\begin{aligned}&= (a^3 + a^2 + a) - ((1)^3 + (1)^2 + 1) \\ &= a^3 + a^2 + a - (1 + 1 + 1) \\ &= a^3 + a^2 + a - 3\end{aligned}$$

Equating to the given value 11:

$$a^3 + a^2 + a - 3 = 11$$

Rearranging into a standard cubic equation:

$$a^3 + a^2 + a - 14 = 0$$

Testing integer factors of 14 ($\pm 1, \pm 2, \pm 7, \pm 14$): for $a = 2$:

$$(2)^3 + (2)^2 + 2 - 14 = 8 + 4 + 2 - 14 = 14 - 14 = 0$$

Thus, $(a - 2)$ is a factor. Dividing the cubic polynomial by $(a - 2)$:

$$a^3 + a^2 + a - 14 = (a - 2)(a^2 + 3a + 7) = 0$$

For the quadratic factor $a^2 + 3a + 7 = 0$, checking the discriminant Δ :

$$\Delta = 3^2 - 4(1)(7) = 9 - 28 = -19 < 0$$

Since the discriminant is negative, there are no further real roots. Therefore, the only real value of a is 2.

Final Answer

$$a = 2$$

Question 21 (iii)

If $\int_0^a \frac{dx}{1+4x^2} = \frac{\pi}{8}$, then find the value of a .

TYPE 6 Inverse Trigonometric Simplification

Given: $\int_0^a \frac{dx}{1+4x^2} = \frac{\pi}{8}$

Concept / Formula Used: Standard inverse trigonometric result: $\int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$

Solution:

Rewriting the integral in standard form:

$$\int_0^a \frac{dx}{1+(2x)^2} = \frac{\pi}{8}$$

Using the standard integration formula $\int \frac{dx}{a^2+u^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right)$ with $u = 2x$ (so $du = 2dx \implies dx = \frac{du}{2}$):

$$\frac{1}{2} [\tan^{-1}(2x)]_0^a = \frac{\pi}{8}$$

Dividing both sides by $\frac{1}{2}$:

$$[\tan^{-1}(2x)]_0^a = \frac{\pi}{4}$$

Substituting the limits:

$$\tan^{-1}(2a) - \tan^{-1}(0) = \frac{\pi}{4}$$

Since $\tan^{-1}(0) = 0$:

$$\tan^{-1}(2a) = \frac{\pi}{4}$$

Applying tangent to both sides:

$$2a = \tan \left(\frac{\pi}{4} \right)$$

$$2a = 1 \implies a = \frac{1}{2}$$

Final Answer

$$a = \frac{1}{2}$$

Question 22 (i)

If $\int_0^a \sqrt{x} dx = 4a \int_0^{\pi/4} \sin 2x dx$, find the value of a .

TYPE 1 Power Rule of Integration

Given: $\int_0^a \sqrt{x} \, dx = 4a \int_0^{\pi/4} \sin 2x \, dx$

Concept / Formula Used: Power rule integration and trigonometric evaluation

Solution:

Evaluating the left-hand side:

$$\begin{aligned} \int_0^a x^{1/2} \, dx &= \left[\frac{x^{3/2}}{3/2} \right]_0^a \\ &= \frac{2}{3} a^{3/2} \end{aligned}$$

Evaluating the right-hand side:

$$\begin{aligned} \int_0^{\pi/4} \sin 2x \, dx &= \left[-\frac{\cos 2x}{2} \right]_0^{\pi/4} \\ &= -\frac{1}{2} \left(\cos \left(2 \cdot \frac{\pi}{4} \right) - \cos(0) \right) \\ &= -\frac{1}{2} \left(\cos \left(\frac{\pi}{2} \right) - 1 \right) \\ &= -\frac{1}{2} (0 - 1) \\ &= \frac{1}{2} \end{aligned}$$

Substituting both sides back into the given equation:

$$\begin{aligned} \frac{2}{3} a^{3/2} &= 4a \left(\frac{1}{2} \right) \\ \frac{2}{3} a^{3/2} &= 2a \end{aligned}$$

Since $a \neq 0$, dividing both sides by a :

$$\frac{2}{3} a^{1/2} = 2$$

Multiplying by $\frac{3}{2}$:

$$\begin{aligned} a^{1/2} &= 2 \cdot \frac{3}{2} \\ a^{1/2} &= 3 \end{aligned}$$

Squaring both sides:

$$a = 3^2 = 9$$

Final Answer

$$a = 9$$

Question 22 (ii)

If $\int_a^b x^3 dx = 0$ and $\int_a^b x^2 dx = \frac{2}{3}$, find the values of a and b .

TYPE 1 Power Rule of Integration

Given: $\int_a^b x^3 dx = 0$ and $\int_a^b x^2 dx = \frac{2}{3}$

Concept / Formula Used: Integration power rule and simultaneous algebraic equations

Solution:

Evaluating the first integral $\int_a^b x^3 dx = 0$:

$$\begin{aligned}\left[\frac{x^4}{4}\right]_a^b &= 0 \\ \frac{b^4 - a^4}{4} &= 0 \\ b^4 - a^4 &= 0 \\ (b^2 - a^2)(b^2 + a^2) &= 0\end{aligned}$$

Since a and b are distinct limits for a non-zero integral, $b^2 + a^2 \neq 0$ for non-zero real numbers, which implies:

$$b^2 - a^2 = 0 \implies b^2 = a^2 \implies b = \pm a$$

Since $a \neq b$, we must have $b = -a$. Now evaluating the second integral $\int_a^b x^2 dx = \frac{2}{3}$ with $b = -a$:

$$\begin{aligned}\left[\frac{x^3}{3}\right]_a^{-a} &= \frac{2}{3} \\ \frac{(-a)^3 - a^3}{3} &= \frac{2}{3} \\ \frac{-a^3 - a^3}{3} &= \frac{2}{3} \\ \frac{-2a^3}{3} &= \frac{2}{3}\end{aligned}$$

Multiplying both sides by $\frac{3}{2}$:

$$-a^3 = 1 \implies a^3 = -1 \implies a = -1$$

Since $b = -a$:

$$b = -(-1) = 1$$

Final Answer

$$a = -1, b = 1$$

7.16 – Definite Integrals via Substitution

Question 1 (i)

Evaluate $\int_{-1}^1 x^2(x^3 + 1)^3 dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_{-1}^1 x^2(x^3 + 1)^3 dx$

Concept / Formula Used: Integration by substitution for definite integrals with limit transformation

Solution:

Let $x^3 + 1 = t$.

Differentiating: $3x^2 dx = dt$, so $x^2 dx = \frac{dt}{3}$.

Changing the limits of integration:

When lower limit $x = -1$, $t = (-1)^3 + 1 = -1 + 1 = 0$.

When upper limit $x = 1$, $t = (1)^3 + 1 = 1 + 1 = 2$.

Substituting into the integral:

$$\begin{aligned} \int_{-1}^1 x^2(x^3 + 1)^3 dx &= \int_0^2 t^3 \cdot \frac{dt}{3} \\ &= \frac{1}{3} \int_0^2 t^3 dt \end{aligned}$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned} &= \frac{1}{3} \left[\frac{t^4}{4} \right]_0^2 \\ &= \frac{1}{12} [t^4]_0^2 \\ &= \frac{1}{12} (2^4 - 0^4) \\ &= \frac{16}{12} \\ &= \frac{4}{3} \end{aligned}$$

Final Answer

$$\int_{-1}^1 x^2(x^3 + 1)^3 dx = \frac{4}{3}$$

Question 1 (ii)

Evaluate $\int_0^1 x e^{x^2} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 x e^{x^2} dx$

Concept / Formula Used: Integration by substitution for definite integrals with limit transformation

Solution:

Let $x^2 = t$.

Differentiating: $2x dx = dt$, so $x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 0^2 = 0$.

When upper limit $x = 1$, $t = 1^2 = 1$.

Substituting into the integral:

$$\begin{aligned} \int_0^1 x e^{x^2} dx &= \int_0^1 e^t \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_0^1 e^t dt \end{aligned}$$

Using the standard result $\int e^t dt = e^t$:

$$\begin{aligned} &= \frac{1}{2} [e^t]_0^1 \\ &= \frac{1}{2} (e^1 - e^0) \\ &= \frac{1}{2} (e - 1) \end{aligned}$$

Final Answer

$$\int_0^1 x e^{x^2} dx = \frac{1}{2}(e - 1)$$

Question 2 (i)

Evaluate $\int_0^1 \frac{x}{x^2 + 1} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{x}{x^2 + 1} dx$

Concept / Formula Used: Logarithmic integration: $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)|$

Solution:

Let $x^2 + 1 = t$.

Differentiating: $2x dx = dt$, so $x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 0^2 + 1 = 1$.

When upper limit $x = 1$, $t = 1^2 + 1 = 2$.

Substituting into the integral:

$$\begin{aligned} \int_0^1 \frac{x}{x^2 + 1} dx &= \int_1^2 \frac{1}{t} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_1^2 \frac{dt}{t} \end{aligned}$$

Using the standard result $\int \frac{dt}{t} = \ln |t|$:

$$\begin{aligned} &= \frac{1}{2} [\ln |t|]_1^2 \\ &= \frac{1}{2} (\ln 2 - \ln 1) \\ &= \frac{1}{2} (\ln 2 - 0) \\ &= \frac{1}{2} \ln 2 \end{aligned}$$

Final Answer

$$\int_0^1 \frac{x}{x^2 + 1} dx = \frac{1}{2} \ln 2$$

Question 2 (ii)

Evaluate $\int_0^{\pi/3} \frac{\cos x}{3 + 4 \sin x} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/3} \frac{\cos x}{3 + 4 \sin x} dx$

Concept / Formula Used: Logarithmic integration: $\int \frac{f'(x)}{f(x)} dx = \ln |f(x)|$

Solution:

Let $3 + 4 \sin x = t$.

Differentiating: $4 \cos x dx = dt$, so $\cos x dx = \frac{dt}{4}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 3 + 4 \sin(0) = 3 + 0 = 3$.

When upper limit $x = \frac{\pi}{3}$, $t = 3 + 4 \sin\left(\frac{\pi}{3}\right) = 3 + 4\left(\frac{\sqrt{3}}{2}\right) = 3 + 2\sqrt{3}$.

Substituting into the integral:

$$\begin{aligned} \int_0^{\pi/3} \frac{\cos x}{3 + 4 \sin x} dx &= \int_3^{3+2\sqrt{3}} \frac{1}{t} \cdot \frac{dt}{4} \\ &= \frac{1}{4} \int_3^{3+2\sqrt{3}} \frac{dt}{t} \end{aligned}$$

Using the standard result $\int \frac{dt}{t} = \ln |t|$:

$$\begin{aligned} &= \frac{1}{4} [\ln |t|]_3^{3+2\sqrt{3}} \\ &= \frac{1}{4} (\ln(3 + 2\sqrt{3}) - \ln 3) \\ &= \frac{1}{4} \ln \left(\frac{3 + 2\sqrt{3}}{3} \right) \end{aligned}$$

Final Answer

$$\int_0^{\pi/3} \frac{\cos x}{3 + 4 \sin x} dx = \frac{1}{4} \ln \left(\frac{3 + 2\sqrt{3}}{3} \right)$$

Question 3 (i)

Evaluate $\int_1^2 \frac{3x}{9x^2 - 1} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_1^2 \frac{3x}{9x^2 - 1} dx$

Concept / Formula Used: Logarithmic integration with linear adjustment

Solution:

Let $9x^2 - 1 = t$.

Differentiating: $18x \, dx = dt$, so $x \, dx = \frac{dt}{18}$.

Changing the limits of integration:

When lower limit $x = 1$, $t = 9(1)^2 - 1 = 8$.

When upper limit $x = 2$, $t = 9(2)^2 - 1 = 9(4) - 1 = 35$.

Substituting into the integral:

$$\begin{aligned} \int_1^2 \frac{3x}{9x^2 - 1} \, dx &= 3 \int_8^{35} \frac{1}{t} \cdot \frac{dt}{18} \\ &= \frac{3}{18} \int_8^{35} \frac{dt}{t} \\ &= \frac{1}{6} \int_8^{35} \frac{dt}{t} \end{aligned}$$

Using the standard result $\int \frac{dt}{t} = \ln |t|$:

$$\begin{aligned} &= \frac{1}{6} [\ln |t|]_8^{35} \\ &= \frac{1}{6} (\ln 35 - \ln 8) \\ &= \frac{1}{6} \ln \left(\frac{35}{8} \right) \end{aligned}$$

Final Answer

$$\int_1^2 \frac{3x}{9x^2 - 1} \, dx = \frac{1}{6} \ln \left(\frac{35}{8} \right)$$

Question 3 (ii)

Evaluate $\int_a^b \frac{\log x}{x} \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_a^b \frac{\log x}{x} \, dx$

Concept / Formula Used: Power rule for integration: $\int t^n \, dt = \frac{t^{n+1}}{n+1}$

Solution:

Let $\log x = t$.

Differentiating: $\frac{1}{x} dx = dt$.

Changing the limits of integration:

When lower limit $x = a$, $t = \log a$.

When upper limit $x = b$, $t = \log b$.

Substituting into the integral:

$$\int_a^b \frac{\log x}{x} dx = \int_{\log a}^{\log b} t dt$$

Using the standard result $\int t dt = \frac{t^2}{2}$:

$$\begin{aligned} &= \left[\frac{t^2}{2} \right]_{\log a}^{\log b} \\ &= \frac{1}{2} [(\log b)^2 - (\log a)^2] \end{aligned}$$

Final Answer

$$\int_a^b \frac{\log x}{x} dx = \frac{1}{2} [(\log b)^2 - (\log a)^2]$$

Question 4 (i)

Evaluate $\int_0^2 \frac{6x+3}{x^2+4} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^2 \frac{6x+3}{x^2+4} dx$

Concept / Formula Used: Linear over quadratic integration split into logarithmic and inverse trigonometric parts

Solution:

Split the numerator to separate the derivative term:

$$\begin{aligned} \int_0^2 \frac{6x+3}{x^2+4} dx &= \int_0^2 \frac{6x}{x^2+4} dx + \int_0^2 \frac{3}{x^2+4} dx \\ &= 3 \int_0^2 \frac{2x}{x^2+4} dx + 3 \int_0^2 \frac{dx}{x^2+2^2} \end{aligned}$$

For the first integral, let $x^2 + 4 = t$, so $2x \, dx = dt$. When $x = 0$, $t = 4$; when $x = 2$, $t = 8$.

$$\begin{aligned} 3 \int_0^2 \frac{2x}{x^2 + 4} \, dx &= 3 \int_4^8 \frac{dt}{t} \\ &= 3 [\ln |t|]_4^8 \\ &= 3(\ln 8 - \ln 4) \\ &= 3 \ln \left(\frac{8}{4} \right) \\ &= 3 \ln 2 \end{aligned}$$

For the second integral, using the standard result $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$ with $a = 2$:

$$\begin{aligned} 3 \int_0^2 \frac{dx}{x^2 + 2^2} &= 3 \left[\frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right]_0^2 \\ &= \frac{3}{2} \left[\tan^{-1} \left(\frac{2}{2} \right) - \tan^{-1}(0) \right] \\ &= \frac{3}{2} [\tan^{-1}(1) - 0] \\ &= \frac{3}{2} \left(\frac{\pi}{4} \right) \\ &= \frac{3\pi}{8} \end{aligned}$$

Combining both parts:

$$\int_0^2 \frac{6x + 3}{x^2 + 4} \, dx = 3 \ln 2 + \frac{3\pi}{8}$$

Final Answer

$$\int_0^2 \frac{6x + 3}{x^2 + 4} \, dx = 3 \ln 2 + \frac{3\pi}{8}$$

Question 4 (ii)

Evaluate $\int_0^{\pi/2} \frac{\sin x}{\sqrt{1 + \cos x}} \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/2} \frac{\sin x}{\sqrt{1 + \cos x}} \, dx$

Concept / Formula Used: Power rule integration for fractional powers

Solution:

Let $1 + \cos x = t$.

Differentiating: $-\sin x \, dx = dt$, so $\sin x \, dx = -dt$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 1 + \cos(0) = 1 + 1 = 2$.

When upper limit $x = \frac{\pi}{2}$, $t = 1 + \cos\left(\frac{\pi}{2}\right) = 1 + 0 = 1$.

Substituting into the integral:

$$\begin{aligned} \int_0^{\pi/2} \frac{\sin x}{\sqrt{1 + \cos x}} \, dx &= \int_2^1 \frac{-dt}{\sqrt{t}} \\ &= - \int_2^1 t^{-1/2} \, dt \\ &= \int_1^2 t^{-1/2} \, dt \quad (\text{reversing limits}) \end{aligned}$$

Using the standard result $\int t^n \, dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned} &= \left[\frac{t^{1/2}}{1/2} \right]_1^2 \\ &= 2 \left[\sqrt{t} \right]_1^2 \\ &= 2(\sqrt{2} - \sqrt{1}) \\ &= 2(\sqrt{2} - 1) \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{\sin x}{\sqrt{1 + \cos x}} \, dx = 2(\sqrt{2} - 1)$$

Question 5 (i)

Evaluate $\int_0^1 \frac{x^5}{1 + x^6} \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{x^5}{1 + x^6} \, dx$

Concept / Formula Used: Logarithmic integration: $\int \frac{f'(x)}{f(x)} \, dx = \ln |f(x)|$

Solution:

Let $1 + x^6 = t$.

Differentiating: $6x^5 \, dx = dt$, so $x^5 \, dx = \frac{dt}{6}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 1 + 0^6 = 1$.

When upper limit $x = 1$, $t = 1 + 1^6 = 2$.

Substituting into the integral:

$$\begin{aligned}\int_0^1 \frac{x^5}{1+x^6} dx &= \int_1^2 \frac{1}{t} \cdot \frac{dt}{6} \\ &= \frac{1}{6} \int_1^2 \frac{dt}{t}\end{aligned}$$

Using the standard result $\int \frac{dt}{t} = \ln |t|$:

$$\begin{aligned}&= \frac{1}{6} [\ln |t|]_1^2 \\ &= \frac{1}{6} (\ln 2 - \ln 1) \\ &= \frac{1}{6} (\ln 2 - 0) \\ &= \frac{1}{6} \ln 2\end{aligned}$$

Final Answer

$$\int_0^1 \frac{x^5}{1+x^6} dx = \frac{1}{6} \ln 2$$

Question 5 (ii)

Evaluate $\int_0^1 \frac{x}{\sqrt{1+x^2}} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{x}{\sqrt{1+x^2}} dx$

Concept / Formula Used: Power rule integration for fractional powers

Solution:

Let $1 + x^2 = t$.

Differentiating: $2x dx = dt$, so $x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 1 + 0^2 = 1$.

When upper limit $x = 1$, $t = 1 + 1^2 = 2$.

Substituting into the integral:

$$\begin{aligned}\int_0^1 \frac{x}{\sqrt{1+x^2}} dx &= \int_1^2 \frac{1}{\sqrt{t}} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_1^2 t^{-1/2} dt\end{aligned}$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned}&= \frac{1}{2} \left[\frac{t^{1/2}}{1/2} \right]_1^2 \\ &= \frac{1}{2} \cdot 2 \left[\sqrt{t} \right]_1^2 \\ &= \left[\sqrt{t} \right]_1^2 \\ &= \sqrt{2} - \sqrt{1} \\ &= \sqrt{2} - 1\end{aligned}$$

Final Answer

$$\int_0^1 \frac{x}{\sqrt{1+x^2}} dx = \sqrt{2} - 1$$

Question 6 (i)

Evaluate $\int_1^e \frac{1 + \log x}{2x} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_1^e \frac{1 + \log x}{2x} dx$

Concept / Formula Used: Power rule integration: $\int t^n dt = \frac{t^{n+1}}{n+1}$

Solution:

Let $1 + \log x = t$.

Differentiating: $\frac{1}{x} dx = dt$.

Changing the limits of integration:

When lower limit $x = 1$, $t = 1 + \log(1) = 1 + 0 = 1$.

When upper limit $x = e$, $t = 1 + \log(e) = 1 + 1 = 2$.

Substituting into the integral:

$$\begin{aligned}\int_1^e \frac{1 + \log x}{2x} dx &= \int_1^2 \frac{t}{2} dt \\ &= \frac{1}{2} \int_1^2 t dt\end{aligned}$$

Using the standard result $\int t dt = \frac{t^2}{2}$:

$$\begin{aligned}&= \frac{1}{2} \left[\frac{t^2}{2} \right]_1^2 \\ &= \frac{1}{4} [t^2]_1^2 \\ &= \frac{1}{4} (2^2 - 1^2) \\ &= \frac{1}{4} (4 - 1) \\ &= \frac{3}{4}\end{aligned}$$

Final Answer

$$\int_1^e \frac{1 + \log x}{2x} dx = \frac{3}{4}$$

Question 6 (ii)

Evaluate $\int_0^1 \frac{x}{1+x^4} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{x}{1+x^4} dx$

Concept / Formula Used: Standard inverse tangent integral: $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + c$

Solution:

Note that $x^4 = (x^2)^2$. Let $x^2 = t$.

Differentiating: $2x dx = dt$, so $x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 0^2 = 0$.

When upper limit $x = 1$, $t = 1^2 = 1$.

Substituting into the integral:

$$\begin{aligned}\int_0^1 \frac{x}{1+x^4} dx &= \int_0^1 \frac{1}{1+t^2} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_0^1 \frac{dt}{1+t^2}\end{aligned}$$

Using the standard result $\int \frac{dt}{1+t^2} = \tan^{-1}(t)$:

$$\begin{aligned}&= \frac{1}{2} [\tan^{-1}(t)]_0^1 \\ &= \frac{1}{2} [\tan^{-1}(1) - \tan^{-1}(0)] \\ &= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) \\ &= \frac{\pi}{8}\end{aligned}$$

Final Answer

$$\int_0^1 \frac{x}{1+x^4} dx = \frac{\pi}{8}$$

Question 7 (i)

Evaluate $\int_0^1 \frac{e^{2x}}{1+e^{4x}} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{e^{2x}}{1+e^{4x}} dx$

Concept / Formula Used: Standard inverse tangent integral: $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + c$

Solution:

Note that $e^{4x} = (e^{2x})^2$. Let $e^{2x} = t$.

Differentiating: $2e^{2x} dx = dt$, so $e^{2x} dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = e^{2(0)} = e^0 = 1$.

When upper limit $x = 1$, $t = e^{2(1)} = e^2$.

Substituting into the integral:

$$\begin{aligned}\int_0^1 \frac{e^{2x}}{1+e^{4x}} dx &= \int_1^{e^2} \frac{1}{1+t^2} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_1^{e^2} \frac{dt}{1+t^2}\end{aligned}$$

Using the standard result $\int \frac{dt}{1+t^2} = \tan^{-1}(t)$:

$$\begin{aligned}&= \frac{1}{2} [\tan^{-1}(t)]_1^{e^2} \\ &= \frac{1}{2} [\tan^{-1}(e^2) - \tan^{-1}(1)] \\ &= \frac{1}{2} \left(\tan^{-1}(e^2) - \frac{\pi}{4} \right)\end{aligned}$$

Final Answer

$$\int_0^1 \frac{e^{2x}}{1+e^{4x}} dx = \frac{1}{2} \left(\tan^{-1}(e^2) - \frac{\pi}{4} \right)$$

Question 7 (ii)

Evaluate $\int_0^{\pi/2} \sqrt{\cos x} \sin^3 x \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/2} \sqrt{\cos x} \sin^3 x \, dx$

Concept / Formula Used: Trigonometric identity $\sin^2 x = 1 - \cos^2 x$ combined with substitution

Solution:

Rewrite the integral by splitting $\sin^3 x$:

$$\int_0^{\pi/2} \sqrt{\cos x} \sin^2 x \sin x \, dx = \int_0^{\pi/2} \sqrt{\cos x} (1 - \cos^2 x) \sin x \, dx$$

Let $\cos x = t$.

Differentiating: $-\sin x \, dx = dt$, so $\sin x \, dx = -dt$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \cos(0) = 1$.

When upper limit $x = \frac{\pi}{2}$, $t = \cos\left(\frac{\pi}{2}\right) = 0$.

Substituting into the integral:

$$\begin{aligned}\int_0^{\pi/2} \sqrt{\cos x}(1 - \cos^2 x) \sin x \, dx &= \int_1^0 \sqrt{t}(1 - t^2)(-dt) \\ &= \int_0^1 t^{1/2}(1 - t^2) \, dt \quad (\text{reversing limits}) \\ &= \int_0^1 (t^{1/2} - t^{5/2}) \, dt\end{aligned}$$

Using the standard result $\int t^n \, dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned}&= \left[\frac{t^{3/2}}{3/2} - \frac{t^{7/2}}{7/2} \right]_0^1 \\ &= \left[\frac{2}{3}t^{3/2} - \frac{2}{7}t^{7/2} \right]_0^1 \\ &= \left(\frac{2}{3}(1)^{3/2} - \frac{2}{7}(1)^{7/2} \right) - (0 - 0) \\ &= \frac{2}{3} - \frac{2}{7} \\ &= \frac{14 - 6}{21} \\ &= \frac{8}{21}\end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \sqrt{\cos x} \sin^3 x \, dx = \frac{8}{21}$$

Question 8 (i)

Evaluate $\int_0^{\pi/4} \sin^3 2x \cos 2x \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/4} \sin^3 2x \cos 2x \, dx$

Concept / Formula Used: Power rule integration: $\int t^n \, dt = \frac{t^{n+1}}{n+1}$

Solution:

Let $\sin 2x = t$.

Differentiating: $2 \cos 2x \, dx = dt$, so $\cos 2x \, dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \sin(2(0)) = \sin(0) = 0$.

When upper limit $x = \frac{\pi}{4}$, $t = \sin\left(2 \cdot \frac{\pi}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1$.

Substituting into the integral:

$$\begin{aligned}\int_0^{\pi/4} \sin^3 2x \cos 2x \, dx &= \int_0^1 t^3 \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_0^1 t^3 \, dt\end{aligned}$$

Using the standard result $\int t^n \, dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned}&= \frac{1}{2} \left[\frac{t^4}{4} \right]_0^1 \\ &= \frac{1}{8} [t^4]_0^1 \\ &= \frac{1}{8} (1^4 - 0^4) \\ &= \frac{1}{8}\end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \sin^3 2x \cos 2x \, dx = \frac{1}{8}$$

Question 8 (ii)

Evaluate $\int_0^{\pi/2} \sqrt{\cos x} \sin^5 x \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/2} \sqrt{\cos x} \sin^5 x \, dx$

Concept / Formula Used: Trigonometric identity expansion combined with substitution

Solution:

Rewrite $\sin^5 x$ as $\sin^4 x \sin x = (\sin^2 x)^2 \sin x = (1 - \cos^2 x)^2 \sin x$:

$$\int_0^{\pi/2} \sqrt{\cos x} \sin^5 x \, dx = \int_0^{\pi/2} \sqrt{\cos x} (1 - \cos^2 x)^2 \sin x \, dx$$

Let $\cos x = t$.

Differentiating: $-\sin x \, dx = dt$, so $\sin x \, dx = -dt$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \cos(0) = 1$.

When upper limit $x = \frac{\pi}{2}$, $t = \cos\left(\frac{\pi}{2}\right) = 0$.

Substituting into the integral:

$$\begin{aligned}\int_0^{\pi/2} \sqrt{\cos x} (1 - \cos^2 x)^2 \sin x \, dx &= \int_1^0 \sqrt{t} (1 - t^2)^2 (-dt) \\ &= \int_0^1 t^{1/2} (1 - 2t^2 + t^4) \, dt \quad (\text{reversing limits}) \\ &= \int_0^1 (t^{1/2} - 2t^{5/2} + t^{9/2}) \, dt\end{aligned}$$

Using the standard result $\int t^n \, dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned}&= \left[\frac{t^{3/2}}{3/2} - 2 \frac{t^{7/2}}{7/2} + \frac{t^{11/2}}{11/2} \right]_0^1 \\ &= \left[\frac{2}{3} t^{3/2} - \frac{4}{7} t^{7/2} + \frac{2}{11} t^{11/2} \right]_0^1 \\ &= \left(\frac{2}{3}(1) - \frac{4}{7}(1) + \frac{2}{11}(1) \right) - (0) \\ &= \frac{2}{3} - \frac{4}{7} + \frac{2}{11}\end{aligned}$$

Finding the common denominator $3 \times 7 \times 11 = 231$:

$$\begin{aligned}&= \frac{2(77) - 4(33) + 2(21)}{231} \\ &= \frac{154 - 132 + 42}{231} \\ &= \frac{64}{231}\end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \sqrt{\cos x} \sin^5 x \, dx = \frac{64}{231}$$

Question 9 (i)

Evaluate $\int_1^3 \frac{\cos(\log x)}{x} \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_1^3 \frac{\cos(\log x)}{x} \, dx$

Concept / Formula Used: Standard trigonometric integral: $\int \cos t \, dt = \sin t + c$

Solution:

Let $\log x = t$.

Differentiating: $\frac{1}{x} dx = dt$.

Changing the limits of integration:

When lower limit $x = 1$, $t = \log(1) = 0$.

When upper limit $x = 3$, $t = \log(3)$.

Substituting into the integral:

$$\int_1^3 \frac{\cos(\log x)}{x} dx = \int_0^{\log 3} \cos t \, dt$$

Using the standard result $\int \cos t \, dt = \sin t$:

$$\begin{aligned} &= [\sin t]_0^{\log 3} \\ &= \sin(\log 3) - \sin(0) \\ &= \sin(\log 3) - 0 \\ &= \sin(\log 3) \end{aligned}$$

Final Answer

$$\int_1^3 \frac{\cos(\log x)}{x} dx = \sin(\log 3)$$

Question 9 (ii)

Evaluate $\int_1^2 \frac{dx}{x(1 + \log x)^2}$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_1^2 \frac{dx}{x(1 + \log x)^2}$

Concept / Formula Used: Power rule integration: $\int t^n \, dt = \frac{t^{n+1}}{n+1}$

Solution:

Let $1 + \log x = t$.

Differentiating: $\frac{1}{x} dx = dt$.

Changing the limits of integration:

When lower limit $x = 1$, $t = 1 + \log(1) = 1 + 0 = 1$.

When upper limit $x = 2$, $t = 1 + \log(2)$.

Substituting into the integral:

$$\begin{aligned}\int_1^2 \frac{dx}{x(1 + \log x)^2} &= \int_1^{1+\log 2} \frac{dt}{t^2} \\ &= \int_1^{1+\log 2} t^{-2} dt\end{aligned}$$

Using the standard result $\int t^{-2} dt = -\frac{1}{t}$:

$$\begin{aligned}&= \left[-\frac{1}{t} \right]_1^{1+\log 2} \\ &= -\left(\frac{1}{1 + \log 2} - \frac{1}{1} \right) \\ &= 1 - \frac{1}{1 + \log 2} \\ &= \frac{(1 + \log 2) - 1}{1 + \log 2} \\ &= \frac{\log 2}{1 + \log 2}\end{aligned}$$

Final Answer

$$\int_1^2 \frac{dx}{x(1 + \log x)^2} = \frac{\log 2}{1 + \log 2}$$

Question 10 (i)

Evaluate $\int_0^{\pi/3} \frac{\sec x \tan x}{1 + \sec^2 x} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/3} \frac{\sec x \tan x}{1 + \sec^2 x} dx$

Concept / Formula Used: Standard inverse tangent integral: $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + c$

Solution:

Let $\sec x = t$.

Differentiating: $\sec x \tan x dx = dt$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \sec(0) = 1$.

When upper limit $x = \frac{\pi}{3}$, $t = \sec\left(\frac{\pi}{3}\right) = 2$.

Substituting into the integral:

$$\int_0^{\pi/3} \frac{\sec x \tan x}{1 + \sec^2 x} dx = \int_1^2 \frac{dt}{1 + t^2}$$

Using the standard result $\int \frac{dt}{1 + t^2} = \tan^{-1}(t)$:

$$\begin{aligned} &= [\tan^{-1}(t)]_1^2 \\ &= \tan^{-1}(2) - \tan^{-1}(1) \\ &= \tan^{-1}(2) - \frac{\pi}{4} \end{aligned}$$

Final Answer

$$\int_0^{\pi/3} \frac{\sec x \tan x}{1 + \sec^2 x} dx = \tan^{-1}(2) - \frac{\pi}{4}$$

Question 10 (ii)

Evaluate $\int_0^1 \tan^{-1} x \, dx$.

TYPE 13 Integration by Parts

Given: $\int_0^1 \tan^{-1} x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Rewrite the integral as $\int_0^1 (\tan^{-1} x) \cdot 1 \, dx$ and apply integration by parts with $u = \tan^{-1} x$ and $v = 1$:

$$\begin{aligned} \int_0^1 \tan^{-1} x \, dx &= [x \tan^{-1} x]_0^1 - \int_0^1 \frac{d}{dx}(\tan^{-1} x) \cdot x \, dx \\ &= [x \tan^{-1} x]_0^1 - \int_0^1 \frac{x}{1 + x^2} \, dx \end{aligned}$$

Evaluating the first term:

$$\begin{aligned} [x \tan^{-1} x]_0^1 &= (1) \tan^{-1}(1) - (0) \tan^{-1}(0) \\ &= 1 \cdot \frac{\pi}{4} - 0 \\ &= \frac{\pi}{4} \end{aligned}$$

For the second integral $\int_0^1 \frac{x}{1+x^2} dx$, let $1+x^2 = t$, so $2x dx = dt \implies x dx = \frac{dt}{2}$. When $x = 0$, $t = 1$; when $x = 1$, $t = 2$.

$$\begin{aligned} \int_0^1 \frac{x}{1+x^2} dx &= \int_1^2 \frac{1}{t} \cdot \frac{dt}{2} \\ &= \frac{1}{2} [\ln |t|]_1^2 \\ &= \frac{1}{2} (\ln 2 - \ln 1) \\ &= \frac{1}{2} \ln 2 \end{aligned}$$

Combining both parts:

$$\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Final Answer

$$\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Question 11 (i)

Evaluate $\int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

Concept / Formula Used: Power rule integration: $\int t^n dt = \frac{t^{n+1}}{n+1}$

Solution:

Let $\sin^{-1} x = t$.

Differentiating: $\frac{1}{\sqrt{1-x^2}} dx = dt$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \sin^{-1}(0) = 0$.

When upper limit $x = \frac{1}{2}$, $t = \sin^{-1}(\frac{1}{2}) = \frac{\pi}{6}$.

Substituting into the integral:

$$\int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \int_0^{\pi/6} t dt$$

Using the standard result $\int t \, dt = \frac{t^2}{2}$:

$$\begin{aligned} &= \left[\frac{t^2}{2} \right]_0^{\pi/6} \\ &= \frac{1}{2} \left[\left(\frac{\pi}{6} \right)^2 - 0^2 \right] \\ &= \frac{1}{2} \left(\frac{\pi^2}{36} \right) \\ &= \frac{\pi^2}{72} \end{aligned}$$

Final Answer

$$\int_0^{1/2} \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx = \frac{\pi^2}{72}$$

Question 11 (ii)

Evaluate $\int_0^1 \sin^{-1} x \, dx$.

TYPE 13 Integration by Parts

Given: $\int_0^1 \sin^{-1} x \, dx$

Concept / Formula Used: Integration by parts formula: $\int u v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \int v \, dx \right) dx$

Solution:

Rewrite the integral as $\int_0^1 (\sin^{-1} x) \cdot 1 \, dx$ and apply integration by parts with $u = \sin^{-1} x$ and $v = 1$:

$$\begin{aligned} \int_0^1 \sin^{-1} x \, dx &= [x \sin^{-1} x]_0^1 - \int_0^1 \frac{d}{dx} (\sin^{-1} x) \cdot x \, dx \\ &= [x \sin^{-1} x]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx \end{aligned}$$

Evaluating the first term:

$$\begin{aligned} [x \sin^{-1} x]_0^1 &= (1) \sin^{-1}(1) - (0) \sin^{-1}(0) \\ &= 1 \cdot \frac{\pi}{2} - 0 \\ &= \frac{\pi}{2} \end{aligned}$$

For the second integral $\int_0^1 \frac{x}{\sqrt{1-x^2}} dx$, let $1 - x^2 = t$, so $-2x dx = dt \implies x dx = -\frac{dt}{2}$.
When $x = 0$, $t = 1$; when $x = 1$, $t = 0$.

$$\begin{aligned}\int_0^1 \frac{x}{\sqrt{1-x^2}} dx &= \int_1^0 \frac{1}{\sqrt{t}} \left(-\frac{dt}{2}\right) \\&= \frac{1}{2} \int_0^1 t^{-1/2} dt \\&= \frac{1}{2} \left[\frac{t^{1/2}}{1/2} \right]_0^1 \\&= \left[\sqrt{t} \right]_0^1 \\&= \sqrt{1} - \sqrt{0} \\&= 1\end{aligned}$$

Combining both parts:

$$\int_0^1 \sin^{-1} x dx = \frac{\pi}{2} - 1$$

Final Answer

$$\int_0^1 \sin^{-1} x dx = \frac{\pi}{2} - 1$$

Question 12 (i)

Evaluate $\int_0^4 \frac{dx}{\sqrt{x^2 + 2x + 3}}$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^4 \frac{dx}{\sqrt{x^2 + 2x + 3}}$

Concept / Formula Used: Standard logarithmic integral: $\int \frac{du}{\sqrt{u^2 + a^2}} = \ln \left| u + \sqrt{u^2 + a^2} \right| + c$

Solution:

Complete the square in the denominator:

$$\begin{aligned} x^2 + 2x + 3 &= (x^2 + 2x + 1) - 1 + 3 \\ &= (x + 1)^2 + 2 \\ &= (x + 1)^2 + (\sqrt{2})^2 \end{aligned}$$

Substitute into the integral:

$$\int_0^4 \frac{dx}{\sqrt{x^2 + 2x + 3}} = \int_0^4 \frac{dx}{\sqrt{(x + 1)^2 + (\sqrt{2})^2}}$$

Let $x + 1 = t$, so $dx = dt$. Changing the limits: when $x = 0$, $t = 1$; when $x = 4$, $t = 5$.

$$\int_0^4 \frac{dx}{\sqrt{(x + 1)^2 + (\sqrt{2})^2}} = \int_1^5 \frac{dt}{\sqrt{t^2 + (\sqrt{2})^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{t^2 + a^2}} = \ln \left| t + \sqrt{t^2 + a^2} \right|$ with $a = \sqrt{2}$:

$$\begin{aligned} &= \left[\ln \left| t + \sqrt{t^2 + 2} \right| \right]_1^5 \\ &= \ln \left(5 + \sqrt{5^2 + 2} \right) - \ln \left(1 + \sqrt{1^2 + 2} \right) \\ &= \ln(5 + \sqrt{27}) - \ln(1 + \sqrt{3}) \\ &= \ln(5 + 3\sqrt{3}) - \ln(1 + \sqrt{3}) \\ &= \ln \left(\frac{5 + 3\sqrt{3}}{1 + \sqrt{3}} \right) \end{aligned}$$

Rationalizing the denominator:

$$\begin{aligned}\frac{5 + 3\sqrt{3}}{1 + \sqrt{3}} \cdot \frac{\sqrt{3} - 1}{\sqrt{3} - 1} &= \frac{(5 + 3\sqrt{3})(\sqrt{3} - 1)}{3 - 1} \\ &= \frac{5\sqrt{3} - 5 + 9 - 3\sqrt{3}}{2} \\ &= \frac{2\sqrt{3} + 4}{2} \\ &= \sqrt{3} + 2\end{aligned}$$

Final Answer

$$\int_0^4 \frac{dx}{\sqrt{x^2 + 2x + 3}} = \ln(2 + \sqrt{3})$$

Question 12 (ii)

Evaluate $\int_0^a \frac{dx}{\sqrt{ax - x^2}}$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^a \frac{dx}{\sqrt{ax - x^2}}$

Concept / Formula Used: Standard inverse sine integral: $\int \frac{du}{\sqrt{a^2 - u^2}} = \sin^{-1} \left(\frac{u}{a} \right) + c$

Solution:

Complete the square inside the square root:

$$\begin{aligned}ax - x^2 &= -(x^2 - ax) \\ &= -\left(x^2 - ax + \frac{a^2}{4} - \frac{a^2}{4}\right) \\ &= \frac{a^2}{4} - \left(x - \frac{a}{2}\right)^2 \\ &= \left(\frac{a}{2}\right)^2 - \left(x - \frac{a}{2}\right)^2\end{aligned}$$

Substitute into the integral:

$$\int_0^a \frac{dx}{\sqrt{ax - x^2}} = \int_0^a \frac{dx}{\sqrt{\left(\frac{a}{2}\right)^2 - \left(x - \frac{a}{2}\right)^2}}$$

Let $x - \frac{a}{2} = t$, so $dx = dt$. Changing the limits: when $x = 0$, $t = -\frac{a}{2}$; when $x = a$, $t = \frac{a}{2}$.

$$\int_0^a \frac{dx}{\sqrt{\left(\frac{a}{2}\right)^2 - \left(x - \frac{a}{2}\right)^2}} = \int_{-a/2}^{a/2} \frac{dt}{\sqrt{\left(\frac{a}{2}\right)^2 - t^2}}$$

Using the standard result $\int \frac{dt}{\sqrt{a^2 - t^2}} = \sin^{-1} \left(\frac{t}{a} \right)$:

$$\begin{aligned} &= \left[\sin^{-1} \left(\frac{t}{a/2} \right) \right]_{-a/2}^{a/2} \\ &= \left[\sin^{-1} \left(\frac{2t}{a} \right) \right]_{-a/2}^{a/2} \\ &= \sin^{-1} \left(\frac{2(a/2)}{a} \right) - \sin^{-1} \left(\frac{2(-a/2)}{a} \right) \\ &= \sin^{-1}(1) - \sin^{-1}(-1) \\ &= \frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \\ &= \frac{\pi}{2} + \frac{\pi}{2} \\ &= \pi \end{aligned}$$

Final Answer

$$\int_0^a \frac{dx}{\sqrt{ax - x^2}} = \pi$$

Question 13 (i)

Evaluate $\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(4 + \sin x)} dx$.

TYPE 14 Partial Fractions

Given: $\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(4 + \sin x)} dx$

Concept / Formula Used: Partial fraction decomposition combined with substitution

Solution:

Let $\sin x = t$.

Differentiating: $\cos x dx = dt$.

Changing the limits:

When $x = 0$, $t = \sin(0) = 0$.

When $x = \frac{\pi}{2}$, $t = \sin\left(\frac{\pi}{2}\right) = 1$.

Substituting into the integral:

$$\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(4 + \sin x)} dx = \int_0^1 \frac{dt}{(1+t)(4+t)}$$

Resolve the integrand into partial fractions:

$$\frac{1}{(1+t)(4+t)} = \frac{A}{1+t} + \frac{B}{4+t}$$

Multiplying both sides by $(1+t)(4+t)$:

$$1 = A(4+t) + B(1+t)$$

Setting $t = -1$:

$$1 = A(4-1) + B(0) \implies 3A = 1 \implies A = \frac{1}{3}$$

Setting $t = -4$:

$$1 = A(0) + B(1-4) \implies -3B = 1 \implies B = -\frac{1}{3}$$

Thus, the integral becomes:

$$\begin{aligned} \int_0^1 \left(\frac{1/3}{1+t} - \frac{1/3}{4+t} \right) dt &= \frac{1}{3} \int_0^1 \frac{dt}{1+t} - \frac{1}{3} \int_0^1 \frac{dt}{4+t} \\ &= \frac{1}{3} [\ln |1+t|]_0^1 - \frac{1}{3} [\ln |4+t|]_0^1 \\ &= \frac{1}{3} ((\ln 2 - \ln 1) - (\ln 5 - \ln 4)) \\ &= \frac{1}{3} (\ln 2 - 0 - \ln 5 + \ln 4) \\ &= \frac{1}{3} \ln \left(\frac{2 \times 4}{5} \right) \\ &= \frac{1}{3} \ln \left(\frac{8}{5} \right) \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{\cos x}{(1 + \sin x)(4 + \sin x)} dx = \frac{1}{3} \ln \left(\frac{8}{5} \right)$$

Question 13 (ii)

Evaluate $\int_0^1 \frac{2x}{(x^2+1)(x^2+2)} dx$.

TYPE 14 Partial Fractions

Given: $\int_0^1 \frac{2x}{(x^2+1)(x^2+2)} dx$

Concept / Formula Used: Logarithmic integration via substitution

Solution:

Let $x^2 = t$.

Differentiating: $2x \, dx = dt$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 0^2 = 0$.

When upper limit $x = 1$, $t = 1^2 = 1$.

Substituting into the integral:

$$\int_0^1 \frac{2x}{(x^2 + 1)(x^2 + 2)} \, dx = \int_0^1 \frac{dt}{(t + 1)(t + 2)}$$

Expressing $\frac{1}{(t + 1)(t + 2)}$ using partial fractions:

$$\frac{1}{(t + 1)(t + 2)} = \frac{1}{t + 1} - \frac{1}{t + 2}$$

Integrating with respect to t :

$$\begin{aligned} \int_0^1 \left(\frac{1}{t + 1} - \frac{1}{t + 2} \right) dt &= [\ln |t + 1| - \ln |t + 2|]_0^1 \\ &= \left[\ln \left(\frac{t + 1}{t + 2} \right) \right]_0^1 \\ &= \ln \left(\frac{1 + 1}{1 + 2} \right) - \ln \left(\frac{0 + 1}{0 + 2} \right) \\ &= \ln \left(\frac{2}{3} \right) - \ln \left(\frac{1}{2} \right) \\ &= \ln \left(\frac{2/3}{1/2} \right) \\ &= \ln \left(\frac{4}{3} \right) \end{aligned}$$

Final Answer

$$\int_0^1 \frac{2x}{(x^2 + 1)(x^2 + 2)} \, dx = \ln \left(\frac{4}{3} \right)$$

Question 14 (i)

Evaluate $\int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin^4 x} \, dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin^4 x} \, dx$

Concept / Formula Used: Standard inverse tangent integral: $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + c$

Solution:

Note that $\sin^4 x = (\sin^2 x)^2$. Let $\sin^2 x = t$.

Differentiating: $2 \sin x \cos x dx = dt$, so $\sin x \cos x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \sin^2(0) = 0$.

When upper limit $x = \frac{\pi}{2}$, $t = \sin^2\left(\frac{\pi}{2}\right) = 1^2 = 1$.

Substituting into the integral:

$$\begin{aligned} \int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin^4 x} dx &= \int_0^1 \frac{1}{1 + t^2} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_0^1 \frac{dt}{1 + t^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{1 + t^2} = \tan^{-1}(t)$:

$$\begin{aligned} &= \frac{1}{2} [\tan^{-1}(t)]_0^1 \\ &= \frac{1}{2} [\tan^{-1}(1) - \tan^{-1}(0)] \\ &= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) \\ &= \frac{\pi}{8} \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin^4 x} dx = \frac{\pi}{8}$$

Question 14 (ii)

Evaluate $\int_0^1 \frac{dx}{2e^x - 1}$.

TYPE 14 Partial Fractions

Given: $\int_0^1 \frac{dx}{2e^x - 1}$

Concept / Formula Used: Logarithmic integration via substitution

Solution:

Let $e^x = t$, so $x = \log t$ and $dx = \frac{dt}{t}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = e^0 = 1$.

When upper limit $x = 1$, $t = e^1 = e$.

Substituting into the integral:

$$\int_0^1 \frac{dx}{2e^x - 1} = \int_1^e \frac{dt}{t(2t - 1)}$$

Expressing $\frac{1}{t(2t - 1)}$ using partial fractions:

$$\frac{1}{t(2t - 1)} = -\frac{1}{t} + \frac{2}{2t - 1}$$

Integrating with respect to t :

$$\begin{aligned} \int_1^e \left(-\frac{1}{t} + \frac{2}{2t - 1} \right) dt &= [-\ln|t| + \ln|2t - 1|]_1^e \\ &= \left[\ln \left| \frac{2t - 1}{t} \right| \right]_1^e \\ &= \ln \left(\frac{2e - 1}{e} \right) - \ln \left(\frac{2(1) - 1}{1} \right) \\ &= \ln \left(\frac{2e - 1}{e} \right) - \ln(1) \\ &= \ln \left(\frac{2e - 1}{e} \right) - 0 \\ &= \ln \left(\frac{2e - 1}{e} \right) \end{aligned}$$

Final Answer

$$\int_0^1 \frac{dx}{2e^x - 1} = \ln \left(\frac{2e - 1}{e} \right)$$

Question 14 (iii)

Evaluate $\int_0^{\pi/4} \frac{\sin x + \cos x}{16 + 9 \sin 2x} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/4} \frac{\sin x + \cos x}{16 + 9 \sin 2x} dx$

Concept / Formula Used: Standard inverse tangent integral using substitution
 $t = \sin x - \cos x$

Solution:

Let $\sin x - \cos x = t$.

Differentiating: $(\cos x + \sin x) dx = dt$.

Squaring both sides to find $\sin 2x$ in terms of t :

$$\begin{aligned}(\sin x - \cos x)^2 &= t^2 \\ \sin^2 x + \cos^2 x - 2 \sin x \cos x &= t^2 \\ 1 - \sin 2x &= t^2 \\ \sin 2x &= 1 - t^2\end{aligned}$$

Changing the limits of integration:

When lower limit $x = 0$, $t = \sin(0) - \cos(0) = 0 - 1 = -1$.

When upper limit $x = \frac{\pi}{4}$, $t = \sin\left(\frac{\pi}{4}\right) - \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = 0$.

Substituting into the integral:

$$\begin{aligned}\int_0^{\pi/4} \frac{\sin x + \cos x}{16 + 9 \sin 2x} dx &= \int_{-1}^0 \frac{dt}{16 + 9(1 - t^2)} \\ &= \int_{-1}^0 \frac{dt}{16 + 9 - 9t^2} \\ &= \int_{-1}^0 \frac{dt}{25 - 9t^2} \\ &= \int_{-1}^0 \frac{dt}{5^2 - (3t)^2}\end{aligned}$$

Using the standard result $\int \frac{dt}{a^2 - (bt)^2} = \frac{1}{2ab} \ln \left| \frac{a + bt}{a - bt} \right|$ (with $a = 5, b = 3$):

$$\begin{aligned}&= \left[\frac{1}{2(5)(3)} \ln \left| \frac{5 + 3t}{5 - 3t} \right| \right]_{-1}^0 \\ &= \frac{1}{30} \left[\ln \left| \frac{5 + 3(0)}{5 - 3(0)} \right| - \ln \left| \frac{5 + 3(-1)}{5 - 3(-1)} \right| \right] \\ &= \frac{1}{30} \left[\ln(1) - \ln \left| \frac{5 - 3}{5 + 3} \right| \right] \\ &= \frac{1}{30} \left[0 - \ln \left(\frac{2}{8} \right) \right] \\ &= -\frac{1}{30} \ln \left(\frac{1}{4} \right) \\ &= \frac{1}{30} \ln(4) \\ &= \frac{1}{30} \ln(2^2) \\ &= \frac{2}{30} \ln 2 \\ &= \frac{1}{15} \ln 2\end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \frac{\sin x + \cos x}{16 + 9 \sin 2x} dx = \frac{1}{15} \ln 2$$

Question 14 (iv)

Evaluate $\int_0^a \sin^{-1} \left(\frac{x}{a+x} \right) dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^a \sin^{-1} \left(\frac{x}{a+x} \right) dx$

Concept / Formula Used: Integration by parts, differentiating $g(x) = \frac{x}{a+x}$ directly

Solution:

Take $u = \sin^{-1} \left(\frac{x}{a+x} \right)$ and $dv = dx$, so $v = x$.

Let $g(x) = \frac{x}{a+x}$. By the quotient rule:

$$g'(x) = \frac{(a+x)(1) - x(1)}{(a+x)^2} = \frac{a}{(a+x)^2}$$

Also,

$$1 - g(x)^2 = 1 - \frac{x^2}{(a+x)^2} = \frac{(a+x)^2 - x^2}{(a+x)^2} = \frac{a^2 + 2ax}{(a+x)^2} = \frac{a(a+2x)}{(a+x)^2}$$

So the derivative of u is

$$\frac{du}{dx} = \frac{g'(x)}{\sqrt{1-g(x)^2}} = \frac{a/(a+x)^2}{\sqrt{a(a+2x)/(a+x)}} = \frac{\sqrt{a}}{(a+x)\sqrt{a+2x}}$$

Applying integration by parts:

$$\int_0^a \sin^{-1} \left(\frac{x}{a+x} \right) dx = \left[x \sin^{-1} \left(\frac{x}{a+x} \right) \right]_0^a - \int_0^a \frac{x\sqrt{a}}{(a+x)\sqrt{a+2x}} dx$$

Evaluating the boundary term: at $x = a$, $\frac{x}{a+x} = \frac{1}{2}$, so $\sin^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{6}$; at $x = 0$ the term vanishes:

$$\left[x \sin^{-1} \left(\frac{x}{a+x} \right) \right]_0^a = a \cdot \frac{\pi}{6} - 0 = \frac{a\pi}{6}$$

For the remaining integral $J = \sqrt{a} \int_0^a \frac{x dx}{(a+x)\sqrt{a+2x}}$, let $u = \sqrt{a+2x}$, so $u^2 = a+2x$, $x = \frac{u^2 - a}{2}$, $dx = u du$, and $a+x = \frac{a+u^2}{2}$.

Limits: when $x = 0$, $u = \sqrt{a}$; when $x = a$, $u = \sqrt{3a}$.

$$J = \sqrt{a} \int_{\sqrt{a}}^{\sqrt{3a}} \frac{\left(\frac{u^2-a}{2}\right) u \, du}{\left(\frac{a+u^2}{2}\right) u} = \sqrt{a} \int_{\sqrt{a}}^{\sqrt{3a}} \frac{u^2 - a}{u^2 + a} \, du$$

Rewriting the integrand using $\frac{u^2 - a}{u^2 + a} = 1 - \frac{2a}{u^2 + a}$:

$$J = \sqrt{a} \left[u - 2\sqrt{a} \tan^{-1} \left(\frac{u}{\sqrt{a}} \right) \right]_{\sqrt{a}}^{\sqrt{3a}}$$

At $u = \sqrt{3a}$: $\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$; at $u = \sqrt{a}$: $\tan^{-1}(1) = \frac{\pi}{4}$:

$$\begin{aligned} J &= \sqrt{a} \left[\left(\sqrt{3a} - 2\sqrt{a} \cdot \frac{\pi}{3} \right) - \left(\sqrt{a} - 2\sqrt{a} \cdot \frac{\pi}{4} \right) \right] \\ &= \sqrt{a} \left[\sqrt{3a} - \sqrt{a} - \frac{2\pi\sqrt{a}}{3} + \frac{\pi\sqrt{a}}{2} \right] \\ &= a(\sqrt{3} - 1) + a \left(\frac{\pi}{2} - \frac{2\pi}{3} \right) \\ &= a\sqrt{3} - a - \frac{a\pi}{6} \end{aligned}$$

Combining with the boundary term:

$$\begin{aligned} \int_0^a \sin^{-1} \left(\frac{x}{a+x} \right) dx &= \frac{a\pi}{6} - \left(a\sqrt{3} - a - \frac{a\pi}{6} \right) \\ &= \frac{a\pi}{3} + a - a\sqrt{3} \end{aligned}$$

Final Answer

$$\int_0^a \sin^{-1} \left(\frac{x}{a+x} \right) dx = a \left(1 + \frac{\pi}{3} - \sqrt{3} \right)$$

Common Mistake: Assuming $\sin^{-1}(\sin^2 \theta) = 2\theta$ under the substitution $x = a \tan^2 \theta$ is false in general ($\sin^2 \theta$ is not $\sin$ of anything simple); differentiating $\sin^{-1} \left(\frac{x}{a+x} \right)$ directly and integrating by parts avoids this trap.

Question 15 (i)

Evaluate $\int_0^1 \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) dx$.

TYPE 6 Inverse Trigonometric Simplification

Given: $\int_0^1 \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) dx$

Concept / Formula Used: Inverse trigonometric identity: $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2 \tan^{-1} x$ for $x \geq 0$

Solution:

Using the standard inverse trigonometric identity $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) = 2 \tan^{-1} x$ for $x \geq 0$:

$$\begin{aligned} \int_0^1 \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) dx &= \int_0^1 2 \tan^{-1} x dx \\ &= 2 \int_0^1 \tan^{-1} x dx \end{aligned}$$

From Question 10 (ii), we already evaluated $\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$. Multiplying by 2:

$$2 \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) = \frac{\pi}{2} - \ln 2$$

Final Answer

$$\int_0^1 \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right) dx = \frac{\pi}{2} - \ln 2$$

Question 15 (ii)

Evaluate $\int_0^1 \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx$.

TYPE 6 Inverse Trigonometric Simplification

Given: $\int_0^1 \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx$

Concept / Formula Used: Inverse trigonometric identity: $\tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2 \tan^{-1} x$ for $0 \leq x < 1$

Solution:

Using the standard inverse trigonometric identity $\tan^{-1} \left(\frac{2x}{1-x^2} \right) = 2 \tan^{-1} x$ for $0 \leq x < 1$:

$$\begin{aligned} \int_0^1 \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx &= \int_0^1 2 \tan^{-1} x dx \\ &= 2 \int_0^1 \tan^{-1} x dx \end{aligned}$$

Using the result from Question 10 (ii):

$$2 \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) = \frac{\pi}{2} - \ln 2$$

Final Answer

$$\int_0^1 \tan^{-1} \left(\frac{2x}{1-x^2} \right) dx = \frac{\pi}{2} - \ln 2$$

Question 16 (i)

Evaluate $\int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^2} dx$.

TYPE 13 Integration by Parts

Given: $\int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^2} dx$

Concept / Formula Used: Integration by parts formula with substitution

Solution:

Let $u = \tan^{-1} x$ and $dv = \frac{x}{(1+x^2)^2} dx$. Then $\frac{du}{dx} = \frac{1}{1+x^2}$. To find v , evaluate $\int \frac{x}{(1+x^2)^2} dx$ using substitution $1+x^2 = t \implies 2x dx = dt \implies x dx = \frac{dt}{2}$:

$$v = \int \frac{1}{t^2} \cdot \frac{dt}{2} = \frac{1}{2} \left(-\frac{1}{t} \right) = -\frac{1}{2(1+x^2)}$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$\begin{aligned} \int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^2} dx &= \left[-\frac{\tan^{-1} x}{2(1+x^2)} \right]_0^1 - \int_0^1 \left(-\frac{1}{2(1+x^2)} \right) \cdot \frac{1}{1+x^2} dx \\ &= \left[-\frac{\tan^{-1} x}{2(1+x^2)} \right]_0^1 + \frac{1}{2} \int_0^1 \frac{dx}{(1+x^2)^2} \end{aligned}$$

Evaluating the boundary term:

$$\begin{aligned} \left[-\frac{\tan^{-1} x}{2(1+x^2)} \right]_0^1 &= -\frac{\tan^{-1}(1)}{2(1+1^2)} - \left(-\frac{\tan^{-1}(0)}{2(1+0^2)} \right) \\ &= -\frac{\pi/4}{4} - 0 \\ &= -\frac{\pi}{16} \end{aligned}$$

For the remaining integral $\int_0^1 \frac{dx}{(1+x^2)^2}$, use trigonometric substitution $x = \tan \theta \implies$

$dx = \sec^2 \theta d\theta$: When $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \pi/4$.

$$\begin{aligned}
 \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^2} &= \int_0^{\pi/4} \frac{\sec^2 \theta}{\sec^4 \theta} d\theta \\
 &= \int_0^{\pi/4} \frac{d\theta}{\sec^2 \theta} \\
 &= \int_0^{\pi/4} \cos^2 \theta d\theta \\
 &= \int_0^{\pi/4} \frac{1 + \cos 2\theta}{2} d\theta \\
 &= \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/4} \\
 &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \sin \left(\frac{\pi}{2} \right) - 0 \right) \\
 &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \right) \\
 &= \frac{\pi}{8} + \frac{1}{4}
 \end{aligned}$$

Combining both parts:

$$-\frac{\pi}{16} + \frac{1}{2} \left(\frac{\pi}{8} + \frac{1}{4} \right) = -\frac{\pi}{16} + \frac{\pi}{16} + \frac{1}{8} = \frac{1}{8}$$

Final Answer

$$\int_0^1 \frac{x \tan^{-1} x}{(1+x^2)^2} dx = \frac{1}{8}$$

Question 16 (ii)

Evaluate $\int_0^1 x(\tan^{-1} x)^2 dx$.

TYPE 13 Integration by Parts

Given: $\int_0^1 x(\tan^{-1} x)^2 dx$

Concept / Formula Used: Integration by parts formula

Solution:

Let $u = (\tan^{-1} x)^2$ and $dv = x dx$. Then $\frac{du}{dx} = 2(\tan^{-1} x) \cdot \frac{1}{1+x^2}$ and $v = \frac{x^2}{2}$. Applying

integration by parts:

$$\begin{aligned}\int_0^1 x(\tan^{-1} x)^2 dx &= \left[\frac{x^2}{2} (\tan^{-1} x)^2 \right]_0^1 - \int_0^1 \frac{x^2}{2} \cdot \frac{2 \tan^{-1} x}{1+x^2} dx \\ &= \left[\frac{x^2}{2} (\tan^{-1} x)^2 \right]_0^1 - \int_0^1 \frac{x^2 \tan^{-1} x}{1+x^2} dx\end{aligned}$$

Evaluating the boundary term:

$$\left[\frac{x^2}{2} (\tan^{-1} x)^2 \right]_0^1 = \frac{1^2}{2} (\tan^{-1}(1))^2 - 0 = \frac{1}{2} \left(\frac{\pi}{4} \right)^2 = \frac{\pi^2}{32}$$

For the remaining integral, rewrite $\frac{x^2}{1+x^2} = \frac{x^2+1-1}{1+x^2} = 1 - \frac{1}{1+x^2}$:

$$\begin{aligned}\int_0^1 \frac{x^2 \tan^{-1} x}{1+x^2} dx &= \int_0^1 \left(1 - \frac{1}{1+x^2} \right) \tan^{-1} x dx \\ &= \int_0^1 \tan^{-1} x dx - \int_0^1 \frac{\tan^{-1} x}{1+x^2} dx\end{aligned}$$

We know $\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$. For $\int_0^1 \frac{\tan^{-1} x}{1+x^2} dx$, let $\tan^{-1} x = t \implies \frac{dx}{1+x^2} = dt$ (limits: 0 to $\pi/4$):

$$\int_0^{\pi/4} t dt = \left[\frac{t^2}{2} \right]_0^{\pi/4} = \frac{1}{2} \left(\frac{\pi}{4} \right)^2 = \frac{\pi^2}{32}$$

Combining all components:

$$\begin{aligned}\frac{\pi^2}{32} - \left[\left(\frac{\pi}{4} - \frac{1}{2} \ln 2 \right) - \frac{\pi^2}{32} \right] &= \frac{\pi^2}{32} - \frac{\pi}{4} + \frac{1}{2} \ln 2 + \frac{\pi^2}{32} \\ &= \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \ln 2\end{aligned}$$

Final Answer

$$\int_0^1 x(\tan^{-1} x)^2 dx = \frac{\pi^2}{16} - \frac{\pi}{4} + \frac{1}{2} \ln 2$$

Question 17 (i)

Evaluate $\int_0^1 \frac{5x}{(4+x^2)^2} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{5x}{(4+x^2)^2} dx$

Concept / Formula Used: Power rule integration: $\int t^n dt = \frac{t^{n+1}}{n+1}$

Solution:

Let $4 + x^2 = t$.

Differentiating: $2x dx = dt$, so $x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 4 + 0^2 = 4$.

When upper limit $x = 1$, $t = 4 + 1^2 = 5$.

Substituting into the integral:

$$\begin{aligned}\int_0^1 \frac{5x}{(4+x^2)^2} dx &= 5 \int_4^5 \frac{1}{t^2} \cdot \frac{dt}{2} \\ &= \frac{5}{2} \int_4^5 t^{-2} dt\end{aligned}$$

Using the standard result $\int t^{-2} dt = -\frac{1}{t}$:

$$\begin{aligned}&= \frac{5}{2} \left[-\frac{1}{t} \right]_4^5 \\ &= \frac{5}{2} \left(-\frac{1}{5} - \left(-\frac{1}{4} \right) \right) \\ &= \frac{5}{2} \left(\frac{1}{4} - \frac{1}{5} \right) \\ &= \frac{5}{2} \left(\frac{5-4}{20} \right) \\ &= \frac{5}{2} \left(\frac{1}{20} \right) \\ &= \frac{5}{40} \\ &= \frac{1}{8}\end{aligned}$$

Final Answer

$$\int_0^1 \frac{5x}{(4+x^2)^2} dx = \frac{1}{8}$$

Question 17 (ii)

Evaluate $\int_0^{\pi/4} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$.

TYPE 15 Integration Using Trigonometric Identities

★ Likely Exam Question

Given: $\int_0^{\pi/4} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$

Concept / Formula Used: Standard inverse tangent integral via substitution $t = \tan^2 x$

Solution:

Divide numerator and denominator by $\cos^4 x$:

$$\int_0^{\pi/4} \frac{\frac{\sin x \cos x}{\cos^4 x}}{\frac{\sin^4 x}{\cos^4 x} + \frac{\cos^4 x}{\cos^4 x}} dx = \int_0^{\pi/4} \frac{\tan x \sec^2 x}{\tan^4 x + 1} dx$$

Let $\tan^2 x = t$.

Differentiating: $2 \tan x \sec^2 x dx = dt$, so $\tan x \sec^2 x dx = \frac{dt}{2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \tan^2(0) = 0$.

When upper limit $x = \frac{\pi}{4}$, $t = \tan^2\left(\frac{\pi}{4}\right) = 1^2 = 1$.

Substituting into the integral:

$$\begin{aligned} \int_0^{\pi/4} \frac{\tan x \sec^2 x}{\tan^4 x + 1} dx &= \int_0^1 \frac{1}{1+t^2} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_0^1 \frac{dt}{1+t^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{1+t^2} = \tan^{-1}(t)$:

$$\begin{aligned} &= \frac{1}{2} [\tan^{-1}(t)]_0^1 \\ &= \frac{1}{2} [\tan^{-1}(1) - \tan^{-1}(0)] \\ &= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) \\ &= \frac{\pi}{8} \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx = \frac{\pi}{8}$$

Question 18 (i)

Evaluate $\int_0^{\pi/2} \frac{dx}{5 + 4 \sin x}$.

TYPE 12d Definite Integrals Evaluated via Substitution

★ **Likely Exam Question**

Given: $\int_0^{\pi/2} \frac{dx}{5 + 4 \sin x}$

Concept / Formula Used: Weierstrass substitution: let $\tan \frac{x}{2} = t$, so $\sin x = \frac{2t}{1+t^2}$
and $dx = \frac{2 dt}{1+t^2}$

Solution:

Let $\tan \frac{x}{2} = t$.

Differentiating: $\sec^2 \frac{x}{2} \cdot \frac{1}{2} dx = dt$, so $(1+t^2) \frac{dx}{2} = dt$, giving

$$dx = \frac{2 dt}{1+t^2}$$

Also, using the standard half-angle identity $\sin x = \frac{2t}{1+t^2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \tan(0) = 0$.

When upper limit $x = \frac{\pi}{2}$, $t = \tan\left(\frac{\pi}{4}\right) = 1$.

Substituting into the integral:

$$\begin{aligned} \int_0^{\pi/2} \frac{dx}{5 + 4 \sin x} &= \int_0^1 \frac{\frac{2 dt}{1+t^2}}{5 + 4 \left(\frac{2t}{1+t^2} \right)} \\ &= \int_0^1 \frac{\frac{2 dt}{1+t^2}}{\frac{5(1+t^2) + 8t}{1+t^2}} \\ &= \int_0^1 \frac{2 dt}{5 + 5t^2 + 8t} \\ &= 2 \int_0^1 \frac{dt}{5t^2 + 8t + 5} \end{aligned}$$

Factor out 5 from the quadratic denominator:

$$= \frac{2}{5} \int_0^1 \frac{dt}{t^2 + \frac{8}{5}t + 1}$$

Completing the square in the denominator:

$$\begin{aligned} t^2 + \frac{8}{5}t + 1 &= \left(t^2 + \frac{8}{5}t + \frac{16}{25}\right) - \frac{16}{25} + 1 \\ &= \left(t + \frac{4}{5}\right)^2 + \frac{9}{25} \\ &= \left(t + \frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2 \end{aligned}$$

Substitute back into the integral:

$$\frac{2}{5} \int_0^1 \frac{dt}{\left(t + \frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2}$$

Using the standard result $\int \frac{dt}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{u}{a}\right) + c$:

$$\begin{aligned} &= \frac{2}{5} \left[\frac{1}{3/5} \tan^{-1} \left(\frac{t + 4/5}{3/5} \right) \right]_0^1 \\ &= \frac{2}{5} \cdot \frac{5}{3} \left[\tan^{-1} \left(\frac{5t + 4}{3} \right) \right]_0^1 \\ &= \frac{2}{3} \left[\tan^{-1} \left(\frac{5(1) + 4}{3} \right) - \tan^{-1} \left(\frac{5(0) + 4}{3} \right) \right] \\ &= \frac{2}{3} \left[\tan^{-1} \left(\frac{9}{3} \right) - \tan^{-1} \left(\frac{4}{3} \right) \right] \\ &= \frac{2}{3} \left[\tan^{-1}(3) - \tan^{-1} \left(\frac{4}{3} \right) \right] \end{aligned}$$

Using the inverse tangent subtraction formula $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \left(\frac{A-B}{1+AB} \right)$:

$$\begin{aligned} \tan^{-1}(3) - \tan^{-1} \left(\frac{4}{3} \right) &= \tan^{-1} \left(\frac{3 - \frac{4}{3}}{1 + 3 \left(\frac{4}{3} \right)} \right) \\ &= \tan^{-1} \left(\frac{\frac{5}{3}}{1 + 4} \right) \\ &= \tan^{-1} \left(\frac{5/3}{5} \right) \\ &= \tan^{-1} \left(\frac{1}{3} \right) \end{aligned}$$

Thus, the integral is:

$$= \frac{2}{3} \tan^{-1} \left(\frac{1}{3} \right)$$

Final Answer

$$\int_0^{\pi/2} \frac{dx}{5 + 4 \sin x} = \frac{2}{3} \tan^{-1} \left(\frac{1}{3} \right)$$

Question 18 (ii)

Evaluate $\int_0^{\pi/2} \frac{dx}{4 \cos x + 2 \sin x}$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\pi/2} \frac{dx}{4 \cos x + 2 \sin x}$

Concept / Formula Used: Weierstrass substitution: $\tan \frac{x}{2} = t$, $\cos x = \frac{1 - t^2}{1 + t^2}$,
 $\sin x = \frac{2t}{1 + t^2}$, $dx = \frac{2 dt}{1 + t^2}$

Solution:

Let $\tan \frac{x}{2} = t$, so $dx = \frac{2 dt}{1 + t^2}$, $\cos x = \frac{1 - t^2}{1 + t^2}$, and $\sin x = \frac{2t}{1 + t^2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \tan(0) = 0$.

When upper limit $x = \frac{\pi}{2}$, $t = \tan\left(\frac{\pi}{4}\right) = 1$.

Substituting into the integral:

$$\begin{aligned} \int_0^{\pi/2} \frac{dx}{4 \cos x + 2 \sin x} &= \int_0^1 \frac{\frac{2 dt}{1 + t^2}}{4 \left(\frac{1 - t^2}{1 + t^2} \right) + 2 \left(\frac{2t}{1 + t^2} \right)} \\ &= \int_0^1 \frac{2 dt}{4(1 - t^2) + 4t} \\ &= \int_0^1 \frac{2 dt}{4 - 4t^2 + 4t} \\ &= \int_0^1 \frac{dt}{2 - 2t^2 + 2t} \\ &= \frac{1}{2} \int_0^1 \frac{dt}{1 + t - t^2} \end{aligned}$$

Completing the square for $1 + t - t^2$:

$$\begin{aligned} 1 + t - t^2 &= 1 - \left(t^2 - t + \frac{1}{4} - \frac{1}{4}\right) \\ &= \frac{5}{4} - \left(t - \frac{1}{2}\right)^2 \\ &= \left(\frac{\sqrt{5}}{2}\right)^2 - \left(t - \frac{1}{2}\right)^2 \end{aligned}$$

Using the standard result $\int \frac{dt}{a^2 - u^2} = \frac{1}{2a} \ln \left| \frac{a+u}{a-u} \right|$ with $a = \frac{\sqrt{5}}{2}$ and $u = t - \frac{1}{2}$:

$$\begin{aligned} \frac{1}{2} \int_0^1 \frac{dt}{\left(\frac{\sqrt{5}}{2}\right)^2 - \left(t - \frac{1}{2}\right)^2} &= \frac{1}{2} \left[\frac{1}{2\left(\frac{\sqrt{5}}{2}\right)} \ln \left| \frac{\frac{\sqrt{5}}{2} + \left(t - \frac{1}{2}\right)}{\frac{\sqrt{5}}{2} - \left(t - \frac{1}{2}\right)} \right| \right]_0^1 \\ &= \frac{1}{2\sqrt{5}} \left[\ln \left| \frac{\sqrt{5} + 2t - 1}{\sqrt{5} - 2t + 1} \right| \right]_0^1 \end{aligned}$$

Evaluating at upper limit $t = 1$:

$$\ln \left| \frac{\sqrt{5} + 2(1) - 1}{\sqrt{5} - 2(1) + 1} \right| = \ln \left| \frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right|$$

Evaluating at lower limit $t = 0$:

$$\ln \left| \frac{\sqrt{5} + 0 - 1}{\sqrt{5} - 0 + 1} \right| = \ln \left| \frac{\sqrt{5} - 1}{\sqrt{5} + 1} \right| = -\ln \left| \frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right|$$

Subtracting lower from upper:

$$\ln \left(\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right) - \left(-\ln \left(\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right) \right) = 2 \ln \left(\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right)$$

Multiplying by $\frac{1}{2\sqrt{5}}$:

$$\frac{1}{2\sqrt{5}} \cdot 2 \ln \left(\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right) = \frac{1}{\sqrt{5}} \ln \left(\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \right)$$

Rationalizing the argument:

$$\frac{\sqrt{5} + 1}{\sqrt{5} - 1} \cdot \frac{\sqrt{5} + 1}{\sqrt{5} + 1} = \frac{(\sqrt{5} + 1)^2}{5 - 1} = \frac{5 + 1 + 2\sqrt{5}}{4} = \frac{6 + 2\sqrt{5}}{4} = \frac{3 + \sqrt{5}}{2}$$

Thus, $\frac{1}{\sqrt{5}} \ln \left(\frac{3 + \sqrt{5}}{2} \right)$.

Final Answer

$$\int_0^{\pi/2} \frac{dx}{4 \cos x + 2 \sin x} = \frac{1}{\sqrt{5}} \ln \left(\frac{3 + \sqrt{5}}{2} \right)$$

Question 19 (i)

Evaluate $\int_0^\pi \frac{dx}{5 + 3 \cos x}$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^\pi \frac{dx}{5 + 3 \cos x}$

Concept / Formula Used: Weierstrass substitution: $\tan \frac{x}{2} = t$, $\cos x = \frac{1 - t^2}{1 + t^2}$,
 $dx = \frac{2 dt}{1 + t^2}$

Solution:

Let $\tan \frac{x}{2} = t$, so $dx = \frac{2 dt}{1 + t^2}$ and $\cos x = \frac{1 - t^2}{1 + t^2}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = \tan(0) = 0$.

When upper limit $x = \pi$, $t = \tan\left(\frac{\pi}{2}\right) \rightarrow \infty$.

Substituting into the integral:

$$\begin{aligned} \int_0^\pi \frac{dx}{5 + 3 \cos x} &= \int_0^\infty \frac{\frac{2 dt}{1 + t^2}}{5 + 3 \left(\frac{1 - t^2}{1 + t^2} \right)} \\ &= \int_0^\infty \frac{2 dt}{5(1 + t^2) + 3(1 - t^2)} \\ &= \int_0^\infty \frac{2 dt}{5 + 5t^2 + 3 - 3t^2} \\ &= \int_0^\infty \frac{2 dt}{2t^2 + 8} \\ &= \int_0^\infty \frac{dt}{t^2 + 4} \\ &= \int_0^\infty \frac{dt}{t^2 + 2^2} \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + c$ with $a = 2$:

$$\begin{aligned} &= \left[\frac{1}{2} \tan^{-1} \left(\frac{t}{2} \right) \right]_0^\infty \\ &= \frac{1}{2} \left[\lim_{t \rightarrow \infty} \tan^{-1} \left(\frac{t}{2} \right) - \tan^{-1}(0) \right] \\ &= \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) \\ &= \frac{\pi}{4} \end{aligned}$$

Final Answer

$$\int_0^\pi \frac{dx}{5 + 3 \cos x} = \frac{\pi}{4}$$

Question 19 (ii)

Evaluate $\int_0^\pi \frac{dx}{6 - \cos x}$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^\pi \frac{dx}{6 - \cos x}$

Concept / Formula Used: Weierstrass substitution: $\tan \frac{x}{2} = t$, $\cos x = \frac{1 - t^2}{1 + t^2}$,
 $dx = \frac{2 dt}{1 + t^2}$

Solution:

Let $\tan \frac{x}{2} = t$, so $dx = \frac{2 dt}{1 + t^2}$ and $\cos x = \frac{1 - t^2}{1 + t^2}$.

Changing the limits of integration:

When $x = 0$, $t = 0$; when $x = \pi$, $t \rightarrow \infty$.

Substituting into the integral:

$$\begin{aligned}
 \int_0^\pi \frac{dx}{6 - \cos x} &= \int_0^\infty \frac{\frac{2 dt}{1+t^2}}{6 - \frac{1-t^2}{1+t^2}} \\
 &= \int_0^\infty \frac{2 dt}{6(1+t^2) - (1-t^2)} \\
 &= \int_0^\infty \frac{2 dt}{6 + 6t^2 - 1 + t^2} \\
 &= \int_0^\infty \frac{2 dt}{7t^2 + 5} \\
 &= \frac{2}{7} \int_0^\infty \frac{dt}{t^2 + \frac{5}{7}} \\
 &= \frac{2}{7} \int_0^\infty \frac{dt}{t^2 + \left(\sqrt{\frac{5}{7}}\right)^2}
 \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right)$:

$$\begin{aligned}
 &= \frac{2}{7} \left[\frac{1}{\sqrt{5/7}} \tan^{-1} \left(\frac{t}{\sqrt{5/7}} \right) \right]_0^\infty \\
 &= \frac{2}{7} \cdot \sqrt{\frac{7}{5}} \left[\lim_{t \rightarrow \infty} \tan^{-1} \left(\sqrt{\frac{7}{5}} t \right) - \tan^{-1}(0) \right] \\
 &= \frac{2}{\sqrt{35}} \left(\frac{\pi}{2} - 0 \right) \\
 &= \frac{\pi}{\sqrt{35}}
 \end{aligned}$$

Final Answer

$$\int_0^\pi \frac{dx}{6 - \cos x} = \frac{\pi}{\sqrt{35}}$$

Question 20 (i)

Evaluate $\int_0^1 \frac{1-x^2}{x^4+x^2+1} dx$.

TYPE 1 Power Rule of Integration

Given: $\int_0^1 \frac{1-x^2}{x^4+x^2+1} dx$

Concept / Formula Used: Substitution $t = x + \frac{1}{x}$ for rational functions symmetric under $x \rightarrow 1/x$

Solution:

Divide both numerator and denominator by x^2 :

$$\int_0^1 \frac{1-x^2}{x^4+x^2+1} dx = \int_0^1 \frac{\frac{1}{x^2}-1}{x^2+1+\frac{1}{x^2}} dx$$

Let $t = x + \frac{1}{x}$.

Differentiating: $dt = \left(1 - \frac{1}{x^2}\right) dx$, so $\left(\frac{1}{x^2} - 1\right) dx = -dt$.

Also, $x^2 + \frac{1}{x^2} = t^2 - 2$, so the denominator becomes $t^2 - 2 + 1 = t^2 - 1$.

Changing the limits: as $x \rightarrow 0^+$, $t = x + \frac{1}{x} \rightarrow \infty$; at $x = 1$, $t = 1 + 1 = 2$.

$$\int_0^1 \frac{\frac{1}{x^2}-1}{t^2-1} dx = \int_\infty^2 \frac{-dt}{t^2-1} = \int_2^\infty \frac{dt}{t^2-1}$$

Using the standard result $\int \frac{dt}{t^2-1} = \frac{1}{2} \log \left| \frac{t-1}{t+1} \right| + c$:

$$= \left[\frac{1}{2} \log \left| \frac{t-1}{t+1} \right| \right]_2^\infty$$

As $t \rightarrow \infty$, $\frac{t-1}{t+1} \rightarrow 1$, so $\log \left| \frac{t-1}{t+1} \right| \rightarrow 0$. At $t = 2$: $\log \left| \frac{1}{3} \right| = -\log 3$.

$$\begin{aligned} &= \frac{1}{2} [0 - (-\log 3)] \\ &= \frac{1}{2} \log 3 \end{aligned}$$

Final Answer

$$\int_0^1 \frac{1-x^2}{x^4+x^2+1} dx = \frac{1}{2} \log 3$$

Common Mistake: The apparent lower limit $x \rightarrow 0^+$ sends $t = x + 1/x \rightarrow \infty$, so this is an improper integral in the t -variable that must be evaluated as a limit rather than dismissed.

Question 20 (ii)

Evaluate $\int_0^1 \frac{x^3}{(1+x^2)^4} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^1 \frac{x^3}{(1+x^2)^4} dx$

Concept / Formula Used: Power rule integration via substitution $1+x^2 = t$

Solution:

Let $1+x^2 = t$. Then $x^2 = t-1$, and $2x dx = dt \implies x dx = \frac{dt}{2}$. The integral can be rewritten as:

$$\begin{aligned} \int_0^1 \frac{x^2 \cdot x}{(1+x^2)^4} dx &= \int_1^2 \frac{(t-1)}{t^4} \cdot \frac{dt}{2} \\ &= \frac{1}{2} \int_1^2 \left(\frac{t}{t^4} - \frac{1}{t^4} \right) dt \\ &= \frac{1}{2} \int_1^2 (t^{-3} - t^{-4}) dt \end{aligned}$$

Using the standard result $\int t^n dt = \frac{t^{n+1}}{n+1}$:

$$\begin{aligned} &= \frac{1}{2} \left[\frac{t^{-2}}{-2} - \frac{t^{-3}}{-3} \right]_1^2 \\ &= \frac{1}{2} \left[-\frac{1}{2t^2} + \frac{1}{3t^3} \right]_1^2 \\ &= \frac{1}{2} \left[\left(-\frac{1}{2(2)^2} + \frac{1}{3(2)^3} \right) - \left(-\frac{1}{2(1)^2} + \frac{1}{3(1)^3} \right) \right] \\ &= \frac{1}{2} \left[\left(-\frac{1}{8} + \frac{1}{24} \right) - \left(-\frac{1}{2} + \frac{1}{3} \right) \right] \\ &= \frac{1}{2} \left[\left(-\frac{3}{24} + \frac{1}{24} \right) - \left(-\frac{3}{6} + \frac{2}{6} \right) \right] \\ &= \frac{1}{2} \left[-\frac{2}{24} - \left(-\frac{1}{6} \right) \right] \\ &= \frac{1}{2} \left[-\frac{1}{12} + \frac{1}{6} \right] \\ &= \frac{1}{2} \left[\frac{-1+2}{12} \right] \\ &= \frac{1}{2} \cdot \frac{1}{12} \\ &= \frac{1}{24} \end{aligned}$$

Final Answer

$$\int_0^1 \frac{x^3}{(1+x^2)^4} dx = \frac{1}{24}$$

Question 21 (i)

Evaluate $\int_0^{\pi/4} \frac{1}{1+\tan x} dx$.

TYPE 11 Definite Integral Properties

★ Likely Exam Question

Given: $I = \int_0^{\pi/4} \frac{1}{1+\tan x} dx$

Concept / Formula Used: Rewrite $\frac{1}{1+\tan x} = \frac{\cos x}{\sin x + \cos x}$, then split using $\frac{a}{a+b} = \frac{1}{2} + \frac{a-b}{2(a+b)}$

Solution:

Rewrite the integrand:

$$\frac{1}{1+\tan x} = \frac{\cos x}{\sin x + \cos x}$$

Using the identity $\frac{a}{a+b} = \frac{1}{2} + \frac{a-b}{2(a+b)}$ with $a = \cos x$, $b = \sin x$:

$$\frac{\cos x}{\sin x + \cos x} = \frac{1}{2} + \frac{\cos x - \sin x}{2(\sin x + \cos x)}$$

So the integral splits into two parts:

$$I = \frac{1}{2} \int_0^{\pi/4} dx + \frac{1}{2} \int_0^{\pi/4} \frac{\cos x - \sin x}{\sin x + \cos x} dx$$

The first part is immediate:

$$\frac{1}{2} \int_0^{\pi/4} dx = \frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8}$$

For the second part, let $u = \sin x + \cos x$, so $du = (\cos x - \sin x) dx$, matching the numerator exactly:

$$\frac{1}{2} \int_0^{\pi/4} \frac{\cos x - \sin x}{\sin x + \cos x} dx = \frac{1}{2} [\log |\sin x + \cos x|]_0^{\pi/4}$$

Substituting the limits: at $x = \pi/4$, $\sin x + \cos x = \sqrt{2}$; at $x = 0$, $\sin x + \cos x = 1$:

$$\begin{aligned} &= \frac{1}{2} [\log \sqrt{2} - \log 1] \\ &= \frac{1}{2} \cdot \frac{1}{2} \log 2 \\ &= \frac{1}{4} \log 2 \end{aligned}$$

Combining both parts:

$$I = \frac{\pi}{8} + \frac{1}{4} \log 2$$

Final Answer

$$\int_0^{\pi/4} \frac{1}{1 + \tan x} dx = \frac{\pi}{8} + \frac{1}{4} \log 2$$

Question 21 (ii)

Evaluate $\int_0^{\frac{1}{2} \log 3} \frac{e^x}{e^{2x} + 1} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{\frac{1}{2} \log 3} \frac{e^x}{e^{2x} + 1} dx$

Concept / Formula Used: Standard inverse tangent integral: $\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{t}{a} \right) + c$

Solution:

Let $e^x = t$. Differentiating: $e^x dx = dt$. Changing the limits of integration: When lower limit $x = 0$, $t = e^0 = 1$. When upper limit $x = \frac{1}{2} \log 3$, $t = e^{\frac{1}{2} \log 3} = (e^{\log 3})^{1/2} = 3^{1/2} = \sqrt{3}$. Substituting into the integral:

$$\int_0^{\frac{1}{2} \log 3} \frac{e^x}{e^{2x} + 1} dx = \int_1^{\sqrt{3}} \frac{dt}{t^2 + 1}$$

Using the standard result $\int \frac{dt}{t^2 + 1} = \tan^{-1}(t)$:

$$\begin{aligned}
 &= [\tan^{-1}(t)]_1^{\sqrt{3}} \\
 &= \tan^{-1}(\sqrt{3}) - \tan^{-1}(1) \\
 &= \frac{\pi}{3} - \frac{\pi}{4} \\
 &= \frac{4\pi - 3\pi}{12} \\
 &= \frac{\pi}{12}
 \end{aligned}$$

Final Answer

$$\int_0^{\frac{1}{2} \log 3} \frac{e^x}{e^{2x} + 1} dx = \frac{\pi}{12}$$

Question 21 (iii)

Evaluate $\int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})} dx$.

TYPE 1 Power Rule of Integration

Given: $\int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})} dx$

Concept / Formula Used: Algebraic simplification using identity $(e^x + e^{-x})(e^x - e^{-x}) = e^{2x} - e^{-2x}$

Solution:

Simplify the denominator using the difference of squares identity:

$$(e^x + e^{-x})(e^x - e^{-x}) = (e^x)^2 - (e^{-x})^2 = e^{2x} - e^{-2x}$$

Multiply numerator and denominator by e^x or use substitution $e^{2x} = t$: Let $e^{2x} = t \implies 2e^{2x} dx = dt \implies e^{2x} dx = \frac{dt}{2}$. Changing the limits: When $x = \log \sqrt{2}$, $t = e^{2 \log \sqrt{2}} = (e^{\log \sqrt{2}})^2 = (\sqrt{2})^2 = 2$. When $x = \log \sqrt{3}$, $t = e^{2 \log \sqrt{3}} = (\sqrt{3})^2 = 3$. Substituting into the integral:

$$\int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{dx}{e^{2x} - e^{-2x}} = \int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{e^{2x} dx}{e^{4x} - 1}$$

Using substitution $e^{2x} = t$ (limits 2 to 3):

$$\int_2^3 \frac{dt/2}{t^2 - 1} = \frac{1}{2} \int_2^3 \frac{dt}{t^2 - 1^2}$$

Using standard partial fraction / logarithmic result $\int \frac{dt}{t^2 - a^2} = \frac{1}{2a} \ln \left| \frac{t-a}{t+a} \right|$:

$$\begin{aligned}
 &= \frac{1}{2} \left[\frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| \right]_2^3 \\
 &= \frac{1}{4} \left[\ln \left(\frac{3-1}{3+1} \right) - \ln \left(\frac{2-1}{2+1} \right) \right] \\
 &= \frac{1}{4} \left[\ln \left(\frac{2}{4} \right) - \ln \left(\frac{1}{3} \right) \right] \\
 &= \frac{1}{4} \left[\ln \left(\frac{1}{2} \right) - \ln \left(\frac{1}{3} \right) \right] \\
 &= \frac{1}{4} \ln \left(\frac{1/2}{1/3} \right) \\
 &= \frac{1}{4} \ln \left(\frac{3}{2} \right)
 \end{aligned}$$

Final Answer

$$\int_{\log \sqrt{2}}^{\log \sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})} dx = \frac{1}{4} \ln \left(\frac{3}{2} \right)$$

Question 22 (i)

Evaluate $\int_{-2}^2 \sqrt{\frac{2-x}{2+x}} dx$.

TYPE 5 Conjugate Multiplication

Given: $\int_{-2}^2 \sqrt{\frac{2-x}{2+x}} dx$

Concept / Formula Used: Rationalization of integrand: multiply numerator and denominator by $\sqrt{2-x}$

Solution:

Rationalize the integrand by multiplying numerator and denominator by $\sqrt{2-x}$:

$$\begin{aligned}
 \int_{-2}^2 \frac{\sqrt{2-x}\sqrt{2-x}}{\sqrt{2+x}\sqrt{2-x}} dx &= \int_{-2}^2 \frac{2-x}{\sqrt{4-x^2}} dx \\
 &= \int_{-2}^2 \frac{2}{\sqrt{4-x^2}} dx - \int_{-2}^2 \frac{x}{\sqrt{4-x^2}} dx
 \end{aligned}$$

The second integral $\int_{-2}^2 \frac{x}{\sqrt{4-x^2}} dx$ is an odd function over symmetric limits $[-2, 2]$, so its value is 0. For the first integral, using the standard result $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right)$

with $a = 2$:

$$\begin{aligned}
 2 \int_{-2}^2 \frac{dx}{\sqrt{2^2 - x^2}} &= 2 \left[\sin^{-1} \left(\frac{x}{2} \right) \right]_{-2}^2 \\
 &= 2 [\sin^{-1}(1) - \sin^{-1}(-1)] \\
 &= 2 \left[\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right] \\
 &= 2 \left(\frac{\pi}{2} + \frac{\pi}{2} \right) \\
 &= 2(\pi) \\
 &= 2\pi
 \end{aligned}$$

Final Answer

$$\int_{-2}^2 \sqrt{\frac{2-x}{2+x}} dx = 2\pi$$

Question 22 (ii)

Evaluate $\int_0^{a^3} \frac{x^2}{x^6 + a^6} dx$.

TYPE 12d Definite Integrals Evaluated via Substitution

Given: $\int_0^{a^3} \frac{x^2}{x^6 + a^6} dx$

Concept / Formula Used: Standard inverse tangent integral: $\int \frac{dt}{t^2 + A^2} = \frac{1}{A} \tan^{-1} \left(\frac{t}{A} \right) + c$

Solution:

Note that $x^6 = (x^3)^2$. Let $x^3 = t$.

Differentiating: $3x^2 dx = dt$, so $x^2 dx = \frac{dt}{3}$.

Changing the limits of integration:

When lower limit $x = 0$, $t = 0^3 = 0$.

When upper limit $x = a^3$, $t = (a^3)^3 = a^9$.

Substituting into the integral:

$$\begin{aligned}
 \int_0^{a^3} \frac{x^2}{x^6 + a^6} dx &= \int_0^{a^9} \frac{1}{t^2 + (a^3)^2} \cdot \frac{dt}{3} \\
 &= \frac{1}{3} \int_0^{a^9} \frac{dt}{t^2 + (a^3)^2}
 \end{aligned}$$

Using the standard result $\int \frac{dt}{t^2 + A^2} = \frac{1}{A} \tan^{-1} \left(\frac{t}{A} \right)$ with $A = a^3$:

$$\begin{aligned} &= \frac{1}{3} \left[\frac{1}{a^3} \tan^{-1} \left(\frac{t}{a^3} \right) \right]_0^{a^9} \\ &= \frac{1}{3a^3} \left[\tan^{-1} \left(\frac{a^9}{a^3} \right) - \tan^{-1}(0) \right] \\ &= \frac{1}{3a^3} [\tan^{-1}(a^6) - 0] \\ &= \frac{1}{3a^3} \tan^{-1}(a^6) \end{aligned}$$

Final Answer

$$\int_0^{a^3} \frac{x^2}{x^6 + a^6} dx = \frac{1}{3a^3} \tan^{-1}(a^6)$$

7.17 – Properties of Definite Integrals

Question 1 (i)

Evaluate the definite integral:

$$\int_{-1}^1 |x| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_{-1}^1 |x| dx$

Concept / Formula Used: Definition of absolute value: $|x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

Solution:

Using the definition of the modulus function, we split the interval of integration at $x = 0$:

$$\begin{aligned}
 I &= \int_{-1}^0 (-x) dx + \int_0^1 x dx \\
 &= -\left[\frac{x^2}{2}\right]_{-1}^0 + \left[\frac{x^2}{2}\right]_0^1 \\
 &= -\left(0 - \frac{(-1)^2}{2}\right) + \left(\frac{1^2}{2} - 0\right) \\
 &= \frac{1}{2} + \frac{1}{2} \\
 &= 1
 \end{aligned}$$

Final Answer

1

Question 1 (ii)

Evaluate the definite integral:

$$\int_1^4 |x - 2| dx \quad (\text{ISC 2022})$$

TYPE 8 Splitting The Limit Of Integration

★ Likely Exam Question

Given: Definite integral $I = \int_1^4 |x - 2| dx$

Concept / Formula Used: Definition of absolute value: $|x - 2| = \begin{cases} -(x - 2), & 1 \leq x < 2 \\ x - 2, & 2 \leq x \leq 4 \end{cases}$

Solution:

The critical point where the integrand changes sign is $x = 2$. We split the integral at

$x = 2$:

$$\begin{aligned}
 I &= \int_1^2 -(x-2) dx + \int_2^4 (x-2) dx \\
 &= \int_1^2 (2-x) dx + \int_2^4 (x-2) dx \\
 &= \left[2x - \frac{x^2}{2} \right]_1^2 + \left[\frac{x^2}{2} - 2x \right]_2^4 \\
 &= \left((4-2) - \left(2 - \frac{1}{2} \right) \right) + ((8-8) - (2-4)) \\
 &= \left(2 - \frac{3}{2} \right) + (0 - (-2)) \\
 &= \frac{1}{2} + 2 \\
 &= \frac{5}{2}
 \end{aligned}$$

Final Answer

$$\frac{5}{2}$$

Question 1 (iii)

Evaluate the definite integral:

$$\int_0^2 \frac{x-1}{|x-1|} dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^2 \frac{x-1}{|x-1|} dx$

Concept / Formula Used: $\frac{x-1}{|x-1|} = \begin{cases} -1, & x < 1 \\ 1, & x > 1 \end{cases}$

Solution:

The integrand is undefined at $x = 1$. Splitting the domain of integration into $[0, 1)$ and

$(1, 2]$:

$$\begin{aligned}
 I &= \int_0^1 \frac{x-1}{-(x-1)} dx + \int_1^2 \frac{x-1}{x-1} dx \\
 &= \int_0^1 (-1) dx + \int_1^2 1 dx \\
 &= [-x]_0^1 + [x]_1^2 \\
 &= (-1 - 0) + (2 - 1) \\
 &= -1 + 1 \\
 &= 0
 \end{aligned}$$

Final Answer

0

Question 1 (iv)

Evaluate the definite integral:

$$\int_{-1}^1 \frac{|x+2|}{x+2} dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_{-1}^1 \frac{|x+2|}{x+2} dx$

Concept / Formula Used: For $x \in [-1, 1]$, $x + 2 > 0$, hence $|x + 2| = x + 2$

Solution:

For all x in the interval $[-1, 1]$, $x + 2 \geq 1 > 0$. Therefore, $|x + 2| = x + 2$. Substituting this into the integral:

$$\begin{aligned}
 I &= \int_{-1}^1 \frac{x+2}{x+2} dx \\
 &= \int_{-1}^1 1 dx \\
 &= [x]_{-1}^1 \\
 &= 1 - (-1) \\
 &= 2
 \end{aligned}$$

Final Answer

2

Question 1 (v)

Evaluate the definite integral:

$$\int_{-3}^0 \frac{x+3}{|x+3|} dx$$

TYPE 8 Splitting The Limit Of Integration**Given:** Definite integral $I = \int_{-3}^0 \frac{x+3}{|x+3|} dx$ **Concept / Formula Used:** For $x \in (-3, 0]$, $x+3 > 0$, hence $|x+3| = x+3$ **Solution:**For all x in the interval $(-3, 0]$, $x+3 > 0$. Thus, $|x+3| = x+3$.

$$\begin{aligned} I &= \int_{-3}^0 \frac{x+3}{x+3} dx \\ &= \int_{-3}^0 1 dx \\ &= [x]_{-3}^0 \\ &= 0 - (-3) \\ &= 3 \end{aligned}$$

Final Answer

3

Question 1 (vi)

Evaluate the definite integral:

$$\int_0^3 [x] dx$$

TYPE 9 Properties Of Greatest Integer Function**Given:** Definite integral $I = \int_0^3 [x] dx$, where $[x]$ denotes the greatest integer function**Concept / Formula Used:** $[x] = n$ for $n \leq x < n+1$, where $n \in \mathbb{Z}$ **Solution:**

Splitting the integral over unit interval sub-domains:

$$\begin{aligned}
 I &= \int_0^1 [x] dx + \int_1^2 [x] dx + \int_2^3 [x] dx \\
 &= \int_0^1 0 dx + \int_1^2 1 dx + \int_2^3 2 dx \\
 &= 0 + 1 \cdot (2 - 1) + 2 \cdot (3 - 2) \\
 &= 0 + 1 + 2 \\
 &= 3
 \end{aligned}$$

Final Answer

3

Question 1 (vii)

Evaluate the definite integral:

$$\int_0^{\sqrt{2}} [x^2] dx \quad (\text{ISC Specimen Paper 2025})$$

TYPE 9 Properties Of Greatest Integer Function

★ Likely Exam Question

Given: Definite integral $I = \int_0^{\sqrt{2}} [x^2] dx$

Concept / Formula Used: $[x^2] = \begin{cases} 0, & 0 \leq x < 1 \\ 1, & 1 \leq x < \sqrt{2} \end{cases}$

Solution:

The expression x^2 crosses integer values at $x = 0, 1, \sqrt{2}$. We split the integral at $x = 1$:

$$\begin{aligned}
 I &= \int_0^1 [x^2] dx + \int_1^{\sqrt{2}} [x^2] dx \\
 &= \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx \\
 &= 0 + [x]_1^{\sqrt{2}} \\
 &= \sqrt{2} - 1
 \end{aligned}$$

Final Answer

$\sqrt{2} - 1$

Question 2 (i)

Evaluate the definite integral:

$$\int_{-3}^1 \frac{|x+2|}{x+2} dx$$

TYPE 8 Splitting The Limit Of Integration**Given:** Definite integral $I = \int_{-3}^1 \frac{|x+2|}{x+2} dx$ **Concept / Formula Used:** $|x+2| = \begin{cases} -(x+2), & -3 \leq x < -2 \\ x+2, & -2 < x \leq 1 \end{cases}$ **Solution:**The term $x+2$ changes sign at $x = -2$. Splitting the integral at $x = -2$:

$$\begin{aligned} I &= \int_{-3}^{-2} \frac{-(x+2)}{x+2} dx + \int_{-2}^1 \frac{x+2}{x+2} dx \\ &= \int_{-3}^{-2} (-1) dx + \int_{-2}^1 1 dx \\ &= [-x]_{-3}^{-2} + [x]_{-2}^1 \\ &= (2 - 3) + (1 - (-2)) \\ &= -1 + 3 \\ &= 2 \end{aligned}$$

Final Answer

2

Question 2 (ii)

Evaluate the definite integral:

$$\int_{-1}^1 [x] dx$$

TYPE 9 Properties Of Greatest Integer Function**Given:** Definite integral $I = \int_{-1}^1 [x] dx$ **Concept / Formula Used:** $[x] = \begin{cases} -1, & -1 \leq x < 0 \\ 0, & 0 \leq x < 1 \end{cases}$ **Solution:**

Splitting the domain of integration at $x = 0$:

$$\begin{aligned}
 I &= \int_{-1}^0 [x] dx + \int_0^1 [x] dx \\
 &= \int_{-1}^0 (-1) dx + \int_0^1 0 dx \\
 &= [-x]_{-1}^0 + 0 \\
 &= 0 - 1 \\
 &= -1
 \end{aligned}$$

Final Answer

-1

Question 2 (iii)

Evaluate the definite integral:

$$\int_0^{1.5} [x] dx$$

TYPE 9 Properties Of Greatest Integer Function

Given: Definite integral $I = \int_0^{1.5} [x] dx$

Concept / Formula Used: $[x] = \begin{cases} 0, & 0 \leq x < 1 \\ 1, & 1 \leq x < 1.5 \end{cases}$

Solution:

Splitting the integral at $x = 1$:

$$\begin{aligned}
 I &= \int_0^1 [x] dx + \int_1^{1.5} [x] dx \\
 &= \int_0^1 0 dx + \int_1^{1.5} 1 dx \\
 &= 0 + [x]_1^{1.5} \\
 &= 1.5 - 1 \\
 &= 0.5 = \frac{1}{2}
 \end{aligned}$$

Final Answer

$\frac{1}{2}$

Question 3 (i)

Evaluate the definite integral:

$$\int_{-\pi/2}^{\pi/2} \sin^5 x \, dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi/2}^{\pi/2} \sin^5 x \, dx$

Concept / Formula Used: If $f(-x) = -f(x)$ (odd function), then $\int_{-a}^a f(x) \, dx = 0$

Solution:Let $f(x) = \sin^5 x$. Testing for symmetry:

$$\begin{aligned} f(-x) &= \sin^5(-x) \\ &= (-\sin x)^5 \\ &= -\sin^5 x \\ &= -f(x) \end{aligned}$$

Since $f(x)$ is an odd function over the symmetric interval $[-\pi/2, \pi/2]$:

$$I = 0$$

Final Answer

0

Question 3 (ii)

Evaluate the definite integral:

$$\int_{-\pi/2}^{\pi/2} x \cos^2 x \, dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi/2}^{\pi/2} x \cos^2 x \, dx$

Concept / Formula Used: If $f(-x) = -f(x)$, then $\int_{-a}^a f(x) \, dx = 0$

Solution:

Let $f(x) = x \cos^2 x$. Testing for parity:

$$\begin{aligned} f(-x) &= (-x) \cos^2(-x) \\ &= -x(\cos x)^2 \\ &= -x \cos^2 x \\ &= -f(x) \end{aligned}$$

Thus, $f(x)$ is an odd function. Hence, the integral over $[-\pi/2, \pi/2]$ vanishes:

$$I = 0$$

Final Answer

0

Question 3 (iii)

Evaluate the definite integral:

$$\int_{-\pi/3}^{\pi/3} (x^2 \sin x + \tan^3 x) dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi/3}^{\pi/3} (x^2 \sin x + \tan^3 x) dx$

Concept / Formula Used: If $f(-x) = -f(x)$, then $\int_{-a}^a f(x) dx = 0$

Solution:

Let $f(x) = x^2 \sin x + \tan^3 x$. Testing for parity:

$$\begin{aligned} f(-x) &= (-x)^2 \sin(-x) + \tan^3(-x) \\ &= x^2(-\sin x) + (-\tan x)^3 \\ &= -x^2 \sin x - \tan^3 x \\ &= -(x^2 \sin x + \tan^3 x) \\ &= -f(x) \end{aligned}$$

$f(x)$ is an odd function. Therefore:

$$I = 0$$

Final Answer

0

Question 3 (iv)

Evaluate the definite integral:

$$\int_{-\pi}^{\pi} (1 - x^2) \sin x \cos^2 x \, dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi}^{\pi} (1 - x^2) \sin x \cos^2 x \, dx$

Concept / Formula Used: If $f(-x) = -f(x)$, then $\int_{-a}^a f(x) \, dx = 0$

Solution:Let $f(x) = (1 - x^2) \sin x \cos^2 x$. Checking the effect of substituting $-x$:

$$\begin{aligned} f(-x) &= (1 - (-x)^2) \sin(-x) \cos^2(-x) \\ &= (1 - x^2)(-\sin x)(\cos x)^2 \\ &= -(1 - x^2) \sin x \cos^2 x \\ &= -f(x) \end{aligned}$$

Since $f(x)$ is an odd function integrated over $[-\pi, \pi]$:

$$I = 0$$

Final Answer

0

Question 4 (i)

Evaluate the definite integral:

$$\int_{-1}^1 \log \left(\frac{2-x}{2+x} \right) dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-1}^1 \log \left(\frac{2-x}{2+x} \right) dx$

Concept / Formula Used: If $f(-x) = -f(x)$, then $\int_{-a}^a f(x) \, dx = 0$

Solution:

Let $f(x) = \log\left(\frac{2-x}{2+x}\right)$. Evaluating $f(-x)$:

$$\begin{aligned} f(-x) &= \log\left(\frac{2 - (-x)}{2 + (-x)}\right) \\ &= \log\left(\frac{2+x}{2-x}\right) \\ &= \log\left(\left(\frac{2-x}{2+x}\right)^{-1}\right) \\ &= -\log\left(\frac{2-x}{2+x}\right) \\ &= -f(x) \end{aligned}$$

Since $f(x)$ is an odd function:

$$I = 0$$

Final Answer

0

Question 4 (ii)

Evaluate the definite integral:

$$\int_{-\pi}^{\pi} \frac{2x}{1 + \cos x} dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi}^{\pi} \frac{2x}{1 + \cos x} dx$

Concept / Formula Used: If $f(-x) = -f(x)$, then $\int_{-a}^a f(x) dx = 0$

Solution:

Let $f(x) = \frac{2x}{1 + \cos x}$. Checking parity:

$$\begin{aligned} f(-x) &= \frac{2(-x)}{1 + \cos(-x)} \\ &= \frac{-2x}{1 + \cos x} \\ &= -f(x) \end{aligned}$$

$f(x)$ is an odd function on $[-\pi, \pi]$. Hence:

$$I = 0$$

Final Answer

0

Question 4 (iii)

Evaluate the definite integral:

$$\int_0^{\pi} e^{\cos^2 x} \cos x \, dx$$

TYPE 11 Definite Integral Properties

Given: Definite integral $I = \int_0^{\pi} e^{\cos^2 x} \cos x \, dx$

Concept / Formula Used: If $f(2a - x) = -f(x)$, then $\int_0^{2a} f(x) \, dx = 0$

Solution:

Here $2a = \pi$, so $a = \pi/2$. Let $f(x) = e^{\cos^2 x} \cos x$. Evaluating $f(\pi - x)$:

$$\begin{aligned} f(\pi - x) &= e^{\cos^2(\pi - x)} \cos(\pi - x) \\ &= e^{(-\cos x)^2} (-\cos x) \\ &= -e^{\cos^2 x} \cos x \\ &= -f(x) \end{aligned}$$

Applying the standard property $\int_0^{2a} f(x) \, dx = 0$ when $f(2a - x) = -f(x)$:

$$I = 0$$

Final Answer

0

Question 5 (i)

Evaluate the definite integral:

$$\int_0^{\pi/2} \log(\tan x) \, dx$$

TYPE 11 Definite Integral Properties

Given: Definite integral $I = \int_0^{\pi/2} \log(\tan x) \, dx$

Concept / Formula Used: King's Property: $\int_0^a f(x) \, dx = \int_0^a f(a - x) \, dx$

Solution:

$$\text{Let } I = \int_0^{\pi/2} \log(\tan x) \, dx \quad \text{--- (1)}$$

Applying the King's Property $x \rightarrow \frac{\pi}{2} - x$:

$$\begin{aligned} I &= \int_0^{\pi/2} \log \left(\tan \left(\frac{\pi}{2} - x \right) \right) dx \\ &= \int_0^{\pi/2} \log(\cot x) dx \quad \text{--- (2)} \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_0^{\pi/2} [\log(\tan x) + \log(\cot x)] dx \\ &= \int_0^{\pi/2} \log(\tan x \cdot \cot x) dx \\ &= \int_0^{\pi/2} \log(1) dx \\ &= \int_0^{\pi/2} 0 dx \\ &= 0 \end{aligned}$$

Thus, $I = 0$.

Final Answer

0

Question 5 (ii)

Evaluate the definite integral:

$$\int_0^{\pi/2} \frac{\sin x - \cos x}{\sqrt{1 - \sin 2x}} dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^{\pi/2} \frac{\sin x - \cos x}{\sqrt{1 - \sin 2x}} dx$

Concept / Formula Used: $1 - \sin 2x = (\sin x - \cos x)^2$, so $\sqrt{1 - \sin 2x} = |\sin x - \cos x|$

Solution:

We express the denominator in simplified form:

$$\sqrt{1 - \sin 2x} = \sqrt{\sin^2 x + \cos^2 x - 2 \sin x \cos x} = \sqrt{(\sin x - \cos x)^2} = |\sin x - \cos x|$$

Splitting the integral at $x = \pi/4$: - On $[0, \pi/4]$, $\cos x \geq \sin x \implies |\sin x - \cos x| = \cos x - \sin x$
 - On $[\pi/4, \pi/2]$, $\sin x \geq \cos x \implies |\sin x - \cos x| = \sin x - \cos x$.

Thus, I splits as:

$$\begin{aligned}
 I &= \int_0^{\pi/4} \frac{\sin x - \cos x}{-(\sin x - \cos x)} dx + \int_{\pi/4}^{\pi/2} \frac{\sin x - \cos x}{\sin x - \cos x} dx \\
 &= \int_0^{\pi/4} (-1) dx + \int_{\pi/4}^{\pi/2} 1 dx \\
 &= [-x]_0^{\pi/4} + [x]_{\pi/4}^{\pi/2} \\
 &= -\frac{\pi}{4} + \left(\frac{\pi}{2} - \frac{\pi}{4}\right) \\
 &= -\frac{\pi}{4} + \frac{\pi}{4} \\
 &= 0
 \end{aligned}$$

Final Answer

0

Question 5 (iii)

Evaluate the definite integral:

$$\int_0^{2\pi} \cos^2 x \sin x dx$$

TYPE 12a Direct/Algebraic Substitution

Given: Definite integral $I = \int_0^{2\pi} \cos^2 x \sin x dx$

Concept / Formula Used: Substitution: let $\cos x = t$

Solution:

Let $\cos x = t$. Differentiating: $-\sin x dx = dt \implies \sin x dx = -dt$.

Transforming limits of integration: When $x = 0$, $t = \cos 0 = 1$. When $x = 2\pi$, $t = \cos 2\pi = 1$.

Substituting into the integral:

$$\begin{aligned}
 I &= \int_1^1 t^2(-dt) \\
 &= - \int_1^1 t^2 dt \\
 &= 0
 \end{aligned}$$

Final Answer

0

Question 6 (i)

Evaluate the definite integral:

$$\int_0^{\pi} \sin^2 x \cos^3 x \, dx$$

TYPE 11 Definite Integral Properties

Given: Definite integral $I = \int_0^{\pi} \sin^2 x \cos^3 x \, dx$

Concept / Formula Used: If $f(2a - x) = -f(x)$, then $\int_0^{2a} f(x) \, dx = 0$

Solution:Let $f(x) = \sin^2 x \cos^3 x$, with $2a = \pi$. Evaluating $f(\pi - x)$:

$$\begin{aligned} f(\pi - x) &= \sin^2(\pi - x) \cos^3(\pi - x) \\ &= (\sin x)^2 (-\cos x)^3 \\ &= \sin^2 x (-\cos^3 x) \\ &= -\sin^2 x \cos^3 x \\ &= -f(x) \end{aligned}$$

Using the property $\int_0^{2a} f(x) \, dx = 0$ when $f(2a - x) = -f(x)$:

$$I = 0$$

Final Answer

0

Question 6 (ii)

Evaluate the definite integral:

$$\int_0^{2\pi} \cos^5 x \, dx$$

TYPE 11 Definite Integral Properties

Given: Definite integral $I = \int_0^{2\pi} \cos^5 x \, dx$

Concept / Formula Used: $\int_0^{2a} f(x) \, dx = 2 \int_0^a f(x) \, dx$ if $f(2a - x) = f(x)$, and 0 if $f(2a - x) = -f(x)$

Solution:

Let $f(x) = \cos^5 x$. First, test with upper limit 2π :

$$f(2\pi - x) = \cos^5(2\pi - x) = \cos^5 x = f(x)$$

Therefore:

$$I = 2 \int_0^\pi \cos^5 x \, dx$$

Next, evaluate $\int_0^\pi \cos^5 x \, dx$ using upper limit π :

$$f(\pi - x) = \cos^5(\pi - x) = (-\cos x)^5 = -\cos^5 x = -f(x)$$

Since $f(\pi - x) = -f(x)$, $\int_0^\pi \cos^5 x \, dx = 0$. Hence:

$$I = 2 \times 0 = 0$$

Final Answer

0

Question 6 (iii)

Evaluate the definite integral:

$$\int_{\log(1/2)}^{\log 2} \sin\left(\frac{e^x + 1}{e^x - 1}\right) dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\log 2}^{\log 2} \sin\left(\frac{e^x + 1}{e^x - 1}\right) dx$ (noting $\log(1/2) = -\log 2$)

Concept / Formula Used: If $f(-x) = -f(x)$, then $\int_{-a}^a f(x) \, dx = 0$

Solution:

Let $f(x) = \sin\left(\frac{e^x + 1}{e^x - 1}\right)$. Evaluating $f(-x)$:

$$\begin{aligned} f(-x) &= \sin\left(\frac{e^{-x} + 1}{e^{-x} - 1}\right) \\ &= \sin\left(\frac{\frac{1}{e^x} + 1}{\frac{1}{e^x} - 1}\right) \\ &= \sin\left(\frac{1 + e^x}{1 - e^x}\right) \\ &= \sin\left(-\frac{e^x + 1}{e^x - 1}\right) \\ &= -\sin\left(\frac{e^x + 1}{e^x - 1}\right) \\ &= -f(x) \end{aligned}$$

$f(x)$ is an odd function. Thus, the integral over the symmetric interval $[-\log 2, \log 2]$ gives:

$$I = 0$$

Final Answer

0

Question 7

Prove that:

$$\int_0^{2a} f(x) dx = \int_0^{2a} f(2a - x) dx$$

TYPE 11 Definite Integral Properties

To Prove: $\int_0^{2a} f(x) dx = \int_0^{2a} f(2a - x) dx$

Concept / Formula Used: Change of variable in definite integrals

Solution:

Let $I = \int_0^{2a} f(x) dx$.

Let $x = 2a - t$. Differentiating both sides:

$$dx = -dt$$

Changing the limits of integration: - When $x = 0$, $t = 2a - 0 = 2a$. - When $x = 2a$, $t = 2a - 2a = 0$.

Substituting into I :

$$\begin{aligned} I &= \int_{2a}^0 f(2a - t)(-dt) \\ &= \int_0^{2a} f(2a - t) dt \quad \left(\text{using } \int_b^a g(t) dt = - \int_a^b g(t) dt \right) \\ &= \int_0^{2a} f(2a - x) dx \quad (\text{changing dummy variable } t \text{ to } x) \end{aligned}$$

Hence Proved

$$\int_0^{2a} f(x) dx = \int_0^{2a} f(2a - x) dx$$

Question 8 (i)

Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ and hence prove that:

$$\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$$

TYPE 11 Definite Integral Properties

★ Likely Exam Question

To Prove: 1. $\int_0^a f(x) dx = \int_0^a f(a-x) dx$
 2. $\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$

Concept / Formula Used: King's Property of definite integrals

Solution:

Part 1: Proof of Property

Let $I = \int_0^a f(x) dx$.

Let $x = a - t \implies dx = -dt$.

When $x = 0$, $t = a$; when $x = a$, $t = 0$.

$$\begin{aligned} I &= \int_a^0 f(a-t)(-dt) \\ &= \int_0^a f(a-t) dt \\ &= \int_0^a f(a-x) dx \end{aligned}$$

Part 2: Evaluation of Integral

Let $I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx$ — (1)

Using the property proven above with $a = \pi/2$:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\sin(\pi/2 - x)}{\sin(\pi/2 - x) + \cos(\pi/2 - x)} dx \\ &= \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \quad \text{— (2)} \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx \\ &= \int_0^{\pi/2} 1 dx \\ &= [x]_0^{\pi/2} = \frac{\pi}{2} \\ I &= \frac{\pi}{4} \end{aligned}$$

Hence Proved

$$\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx = \frac{\pi}{4}$$

Question 8 (ii)

Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. Hence, evaluate:

$$\int_0^{\pi/2} \frac{dx}{1 + \tan x}$$

TYPE 11 Definite Integral Properties

To Prove: 1. Proof of $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

2. Evaluate $I = \int_0^{\pi/2} \frac{dx}{1 + \tan x}$

Concept / Formula Used: $\tan x = \frac{\sin x}{\cos x}$

Solution:

Part 1: Proof of Property

Let $x = a - t \implies dx = -dt$.

Limits change from $x \in [0, a]$ to $t \in [a, 0]$.

$$\int_0^a f(x) dx = \int_a^0 f(a-t)(-dt) = \int_0^a f(a-t) dt = \int_0^a f(a-x) dx$$

Part 2: Evaluation of Integral

First rewrite the integrand in terms of sine and cosine:

$$I = \int_0^{\pi/2} \frac{dx}{1 + \frac{\sin x}{\cos x}} = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \quad \text{--- (1)}$$

Applying King's Property $x \rightarrow \frac{\pi}{2} - x$:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\cos(\pi/2 - x)}{\cos(\pi/2 - x) + \sin(\pi/2 - x)} dx \\ &= \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx \quad \text{--- (2)} \end{aligned}$$

Adding (1) and (2):

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\cos x + \sin x}{\cos x + \sin x} dx \\ &= \int_0^{\pi/2} 1 dx \\ &= \frac{\pi}{2} \\ I &= \frac{\pi}{4} \end{aligned}$$

Final Answer

$$\frac{\pi}{4}$$

Question 9 (i)

By using properties of definite integrals, evaluate:

$$\int_0^4 |x - 1| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^4 |x - 1| dx$

Concept / Formula Used: $|x - 1| = \begin{cases} -(x - 1), & 0 \leq x < 1 \\ x - 1, & 1 \leq x \leq 4 \end{cases}$

Solution:

Splitting the integral at $x = 1$:

$$\begin{aligned}
 I &= \int_0^1 (1-x) dx + \int_1^4 (x-1) dx \\
 &= \left[x - \frac{x^2}{2} \right]_0^1 + \left[\frac{x^2}{2} - x \right]_1^4 \\
 &= \left(1 - \frac{1}{2} \right) + \left(\left(\frac{16}{2} - 4 \right) - \left(\frac{1}{2} - 1 \right) \right) \\
 &= \frac{1}{2} + \left(4 - \left(-\frac{1}{2} \right) \right) \\
 &= \frac{1}{2} + 4 + \frac{1}{2} \\
 &= 5
 \end{aligned}$$

Final Answer

5

Question 9 (ii)

By using properties of definite integrals, evaluate:

$$\int_2^8 |x-5| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_2^8 |x-5| dx$

Concept / Formula Used: $|x-5| = \begin{cases} -(x-5), & 2 \leq x < 5 \\ x-5, & 5 \leq x \leq 8 \end{cases}$

Solution:

Splitting the integral at the critical point $x = 5$:

$$\begin{aligned}
 I &= \int_2^5 (5-x) dx + \int_5^8 (x-5) dx \\
 &= \left[5x - \frac{x^2}{2} \right]_2^5 + \left[\frac{x^2}{2} - 5x \right]_5^8 \\
 &= \left(\left(25 - \frac{25}{2} \right) - (10 - 2) \right) + \left((32 - 40) - \left(\frac{25}{2} - 25 \right) \right) \\
 &= \left(\frac{25}{2} - 8 \right) + \left(-8 - \left(-\frac{25}{2} \right) \right) \\
 &= \frac{9}{2} + \frac{9}{2} \\
 &= 9
 \end{aligned}$$

Final Answer

9

Question 9 (iii)

By using properties of definite integrals, evaluate:

$$\int_{-6}^3 |x + 3| dx \quad (\text{ISC 2019})$$

TYPE 8 Splitting The Limit Of Integration

★ Likely Exam Question

Given: Definite integral $I = \int_{-6}^3 |x + 3| dx$

Concept / Formula Used: $|x + 3| = \begin{cases} -(x + 3), & -6 \leq x < -3 \\ x + 3, & -3 \leq x \leq 3 \end{cases}$

Solution:Splitting the integral at $x = -3$:

$$\begin{aligned} I &= \int_{-6}^{-3} -(x + 3) dx + \int_{-3}^3 (x + 3) dx \\ &= \left[-\frac{(x + 3)^2}{2} \right]_{-6}^{-3} + \left[\frac{(x + 3)^2}{2} \right]_{-3}^3 \\ &= \left(0 - \left(-\frac{(-3)^2}{2} \right) \right) + \left(\frac{6^2}{2} - 0 \right) \\ &= \frac{9}{2} + 18 \\ &= \frac{45}{2} \end{aligned}$$

Final Answer $\frac{45}{2}$ **Question 10 (i)**

By using properties of definite integrals, evaluate:

$$\int_0^1 |2x - 1| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^1 |2x - 1| dx$

Concept / Formula Used: $|2x - 1| = \begin{cases} -(2x - 1), & 0 \leq x < 1/2 \\ 2x - 1, & 1/2 \leq x \leq 1 \end{cases}$

Solution:

Critical point occurs where $2x - 1 = 0 \implies x = 1/2$. Splitting the integral:

$$\begin{aligned} I &= \int_0^{1/2} (1 - 2x) dx + \int_{1/2}^1 (2x - 1) dx \\ &= [x - x^2]_0^{1/2} + [x^2 - x]_{1/2}^1 \\ &= \left(\frac{1}{2} - \frac{1}{4}\right) + \left((1 - 1) - \left(\frac{1}{4} - \frac{1}{2}\right)\right) \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{1}{2} \end{aligned}$$

Final Answer

$$\frac{1}{2}$$

Question 10 (ii)

By using properties of definite integrals, evaluate:

$$\int_{-2}^2 |2x + 3| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_{-2}^2 |2x + 3| dx$

Concept / Formula Used: $|2x + 3| = \begin{cases} -(2x + 3), & -2 \leq x < -3/2 \\ 2x + 3, & -3/2 \leq x \leq 2 \end{cases}$

Solution:

Critical point is $2x + 3 = 0 \implies x = -3/2$. Splitting the integral:

$$\begin{aligned}
 I &= \int_{-2}^{-3/2} -(2x + 3) dx + \int_{-3/2}^2 (2x + 3) dx \\
 &= [-x^2 - 3x]_{-2}^{-3/2} + [x^2 + 3x]_{-3/2}^2 \\
 &= \left(\left(-\frac{9}{4} + \frac{9}{2} \right) - (-4 + 6) \right) + \left((4 + 6) - \left(\frac{9}{4} - \frac{9}{2} \right) \right) \\
 &= \left(\frac{9}{4} - 2 \right) + \left(10 - \left(-\frac{9}{4} \right) \right) \\
 &= \frac{1}{4} + \frac{49}{4} \\
 &= \frac{50}{4} = \frac{25}{2}
 \end{aligned}$$

Final Answer

$$\frac{25}{2}$$

Question 10 (iii)

By using properties of definite integrals, evaluate:

$$\int_1^3 |x^2 - 2x| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_1^3 |x^2 - 2x| dx$

Concept / Formula Used: $x^2 - 2x = x(x - 2) \leq 0$ on $[1, 2]$ and ≥ 0 on $[2, 3]$

Solution:

The expression $x^2 - 2x$ changes sign at $x = 2$. - On $[1, 2]$, $x^2 - 2x \leq 0 \implies |x^2 - 2x| = 2x - x^2$. - On $[2, 3]$, $x^2 - 2x \geq 0 \implies |x^2 - 2x| = x^2 - 2x$.

Splitting the integral at $x = 2$:

$$\begin{aligned}
 I &= \int_1^2 (2x - x^2) dx + \int_2^3 (x^2 - 2x) dx \\
 &= \left[x^2 - \frac{x^3}{3} \right]_1^2 + \left[\frac{x^3}{3} - x^2 \right]_2^3 \\
 &= \left(\left(4 - \frac{8}{3} \right) - \left(1 - \frac{1}{3} \right) \right) + \left((9 - 9) - \left(\frac{8}{3} - 4 \right) \right) \\
 &= \left(\frac{4}{3} - \frac{2}{3} \right) + \left(0 - \left(-\frac{4}{3} \right) \right) \\
 &= \frac{2}{3} + \frac{4}{3} \\
 &= 2
 \end{aligned}$$

Final Answer

2

Question 11 (i)

By using properties of definite integrals, evaluate:

$$\int_{-\pi/4}^{\pi/4} |\sin x| dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi/4}^{\pi/4} |\sin x| dx$

Concept / Formula Used: Since $|\sin x|$ is an even function, $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$

Solution:

Let $f(x) = |\sin x|$. Since $f(-x) = |\sin(-x)| = |-\sin x| = |\sin x| = f(x)$, $f(x)$ is an even function.

$$I = 2 \int_0^{\pi/4} |\sin x| dx$$

For $x \in [0, \pi/4]$, $\sin x \geq 0$, so $|\sin x| = \sin x$:

$$\begin{aligned} I &= 2 \int_0^{\pi/4} \sin x \, dx \\ &= 2 [-\cos x]_0^{\pi/4} \\ &= 2 \left(-\cos \frac{\pi}{4} - (-\cos 0) \right) \\ &= 2 \left(-\frac{1}{\sqrt{2}} + 1 \right) \\ &= 2 - \sqrt{2} \end{aligned}$$

Final Answer

$$2 - \sqrt{2}$$

Question 11 (ii)

By using properties of definite integrals, evaluate:

$$\int_0^{\pi} |\cos x| \, dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^{\pi} |\cos x| \, dx$

Concept / Formula Used: $|\cos x| = \begin{cases} \cos x, & 0 \leq x \leq \pi/2 \\ -\cos x, & \pi/2 \leq x \leq \pi \end{cases}$

Solution:

$\cos x$ changes sign at $x = \pi/2$. Splitting the integral:

$$\begin{aligned} I &= \int_0^{\pi/2} \cos x \, dx + \int_{\pi/2}^{\pi} (-\cos x) \, dx \\ &= [\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi} \\ &= \left(\sin \frac{\pi}{2} - \sin 0 \right) - \left(\sin \pi - \sin \frac{\pi}{2} \right) \\ &= (1 - 0) - (0 - 1) \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

Final Answer

$$2$$

Question 11 (iii)

By using properties of definite integrals, evaluate:

$$\int_0^{\pi/2} |\cos 2x| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^{\pi/2} |\cos 2x| dx$

Concept / Formula Used: $\cos 2x = 0 \implies 2x = \pi/2 \implies x = \pi/4$

Solution:

- On $[0, \pi/4]$, $2x \in [0, \pi/2] \implies \cos 2x \geq 0 \implies |\cos 2x| = \cos 2x$. - On $[\pi/4, \pi/2]$, $2x \in [\pi/2, \pi] \implies \cos 2x \leq 0 \implies |\cos 2x| = -\cos 2x$.

Splitting the integral at $x = \pi/4$:

$$\begin{aligned} I &= \int_0^{\pi/4} \cos 2x dx + \int_{\pi/4}^{\pi/2} (-\cos 2x) dx \\ &= \left[\frac{\sin 2x}{2} \right]_0^{\pi/4} - \left[\frac{\sin 2x}{2} \right]_{\pi/4}^{\pi/2} \\ &= \left(\frac{\sin(\pi/2)}{2} - 0 \right) - \left(\frac{\sin \pi}{2} - \frac{\sin(\pi/2)}{2} \right) \\ &= \frac{1}{2} - \left(0 - \frac{1}{2} \right) \\ &= \frac{1}{2} + \frac{1}{2} \\ &= 1 \end{aligned}$$

Final Answer

1

Question 12 (i)

By using properties of definite integrals, evaluate:

$$\int_0^{2\pi} |\sin x| dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^{2\pi} |\sin x| dx$

Concept / Formula Used: $|\sin x| = \begin{cases} \sin x, & 0 \leq x \leq \pi \\ -\sin x, & \pi \leq x \leq 2\pi \end{cases}$

Solution:

Splitting the integral at $x = \pi$:

$$\begin{aligned} I &= \int_0^{\pi} \sin x \, dx + \int_{\pi}^{2\pi} (-\sin x) \, dx \\ &= [-\cos x]_0^{\pi} + [\cos x]_{\pi}^{2\pi} \\ &= (-\cos \pi - (-\cos 0)) + (\cos 2\pi - \cos \pi) \\ &= (-(-1) + 1) + (1 - (-1)) \\ &= 2 + 2 \\ &= 4 \end{aligned}$$

Final Answer

4

Question 12 (ii)

By using properties of definite integrals, evaluate:

$$\int_{-1}^1 \sqrt{|x| - x} \, dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_{-1}^1 \sqrt{|x| - x} \, dx$

Concept / Formula Used: $|x| = \begin{cases} -x, & -1 \leq x < 0 \\ x, & 0 \leq x \leq 1 \end{cases}$

Solution:

Splitting the domain of integration at $x = 0$: - On $[-1, 0]$: $|x| = -x \implies \sqrt{|x| - x} = \sqrt{-x - x} = \sqrt{-2x}$. - On $[0, 1]$: $|x| = x \implies \sqrt{|x| - x} = \sqrt{x - x} = 0$.

Therefore:

$$\begin{aligned} I &= \int_{-1}^0 \sqrt{-2x} \, dx + \int_0^1 0 \, dx \\ &= \sqrt{2} \int_{-1}^0 (-x)^{1/2} \, dx \end{aligned}$$

Let $u = -x \implies du = -dx$. When $x = -1$, $u = 1$; when $x = 0$, $u = 0$.

$$\begin{aligned} I &= \sqrt{2} \int_1^0 u^{1/2} (-du) \\ &= \sqrt{2} \int_0^1 u^{1/2} du \\ &= \sqrt{2} \left[\frac{u^{3/2}}{3/2} \right]_0^1 \\ &= \sqrt{2} \cdot \frac{2}{3} (1 - 0) \\ &= \frac{2\sqrt{2}}{3} \end{aligned}$$

Final Answer

$$\frac{2\sqrt{2}}{3}$$

Question 12 (iii)

By using properties of definite integrals, evaluate:

$$\int_0^{\pi/2} \sqrt{1 - \sin 2x} \, dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^{\pi/2} \sqrt{1 - \sin 2x} \, dx$

Concept / Formula Used: $\sqrt{1 - \sin 2x} = |\sin x - \cos x|$

Solution:

Using the trigonometric identity $1 - \sin 2x = (\sin x - \cos x)^2$:

$$I = \int_0^{\pi/2} |\sin x - \cos x| \, dx$$

Splitting at $x = \pi/4$: - On $[0, \pi/4]$, $\cos x \geq \sin x \implies |\sin x - \cos x| = \cos x - \sin x$. -
On $[\pi/4, \pi/2]$, $\sin x \geq \cos x \implies |\sin x - \cos x| = \sin x - \cos x$.

$$\begin{aligned}
 I &= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x) dx \\
 &= [\sin x + \cos x]_0^{\pi/4} + [-\cos x - \sin x]_{\pi/4}^{\pi/2} \\
 &= \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - (0 + 1) \right) + \left((-0 - 1) - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \right) \\
 &= (\sqrt{2} - 1) + (-1 + \sqrt{2}) \\
 &= 2\sqrt{2} - 2
 \end{aligned}$$

Final Answer

$$2\sqrt{2} - 2$$

Question 13 (i)

By using properties of definite integrals, evaluate:

$$\int_0^5 f(x) dx, \quad \text{where } f(x) = |x + 2| + |x - 3|$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^5 (|x + 2| + |x - 3|) dx$

Concept / Formula Used: For $x \in [0, 5]$, $x + 2 > 0 \implies |x + 2| = x + 2$; $|x - 3|$ changes sign at $x = 3$

Solution:

Simplifying $f(x)$ on $[0, 5]$: - On $[0, 3]$: $f(x) = (x + 2) + (3 - x) = 5$. - On $[3, 5]$: $f(x) = (x + 2) + (x - 3) = 2x - 1$.

Integrating over the two sub-intervals:

$$\begin{aligned}
 I &= \int_0^3 5 dx + \int_3^5 (2x - 1) dx \\
 &= [5x]_0^3 + [x^2 - x]_3^5 \\
 &= (15 - 0) + ((25 - 5) - (9 - 3)) \\
 &= 15 + (20 - 6) \\
 &= 15 + 14 \\
 &= 29
 \end{aligned}$$

Final Answer

$$29$$

Question 13 (ii)

By using properties of definite integrals, evaluate:

$$\int_1^4 (|x-1| + |x-2| + |x-3|) dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_1^4 (|x-1| + |x-2| + |x-3|) dx$

Concept / Formula Used: For $x \in [1, 4]$, $|x-1| = x-1$. $|x-2|$ and $|x-3|$ change signs at $x = 2$ and $x = 3$

Solution:

Splitting the domain into three sub-intervals $[1, 2]$, $[2, 3]$, $[3, 4]$: - On $[1, 2]$: $f(x) = (x-1) + (2-x) + (3-x) = 4-x$. - On $[2, 3]$: $f(x) = (x-1) + (x-2) + (3-x) = x$. - On $[3, 4]$: $f(x) = (x-1) + (x-2) + (x-3) = 3x-6$.

Calculating each integral separately:

$$I_1 = \int_1^2 (4-x) dx = \left[4x - \frac{x^2}{2} \right]_1^2 = (8-2) - \left(4 - \frac{1}{2} \right) = 6 - \frac{7}{2} = \frac{5}{2}$$

$$I_2 = \int_2^3 x dx = \left[\frac{x^2}{2} \right]_2^3 = \frac{9}{2} - 2 = \frac{5}{2}$$

$$I_3 = \int_3^4 (3x-6) dx = \left[\frac{3x^2}{2} - 6x \right]_3^4 = (24-24) - \left(\frac{27}{2} - 18 \right) = 0 - \left(-\frac{9}{2} \right) = \frac{9}{2}$$

Summing the results:

$$I = I_1 + I_2 + I_3 = \frac{5}{2} + \frac{5}{2} + \frac{9}{2} = \frac{19}{2}$$

Final Answer

$$\frac{19}{2}$$

Question 13 (iii)

By using properties of definite integrals, evaluate:

$$\int_0^3 f(x) dx, \quad \text{where } f(x) = |x| + |x-1| + |x-2|$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_0^3 (|x| + |x-1| + |x-2|) dx$

Concept / Formula Used: Critical points are at $x = 0, 1, 2, 3$

Solution:

On $[0, 3]$, $|x| = x$. The function simplifies on each sub-interval: - On $[0, 1]$: $f(x) = x + (1 - x) + (2 - x) = 3 - x$. - On $[1, 2]$: $f(x) = x + (x - 1) + (2 - x) = x + 1$. - On $[2, 3]$: $f(x) = x + (x - 1) + (x - 2) = 3x - 3$.

Computing integrals:

$$I_1 = \int_0^1 (3 - x) dx = \left[3x - \frac{x^2}{2} \right]_0^1 = 3 - \frac{1}{2} = \frac{5}{2}$$

$$I_2 = \int_1^2 (x + 1) dx = \left[\frac{x^2}{2} + x \right]_1^2 = (2 + 2) - \left(\frac{1}{2} + 1 \right) = 4 - \frac{3}{2} = \frac{5}{2}$$

$$I_3 = \int_2^3 (3x - 3) dx = \left[\frac{3x^2}{2} - 3x \right]_2^3 = \left(\frac{27}{2} - 9 \right) - (6 - 6) = \frac{9}{2}$$

Total integral:

$$I = \frac{5}{2} + \frac{5}{2} + \frac{9}{2} = \frac{19}{2}$$

Final Answer

$$\frac{19}{2}$$

Question 13 (iv)

By using properties of definite integrals, evaluate:

$$\int_{-1}^5 (|x| + |x + 1| + |x - 5|) dx$$

TYPE 8 Splitting The Limit Of Integration

Given: Definite integral $I = \int_{-1}^5 (|x| + |x + 1| + |x - 5|) dx$

Concept / Formula Used: For $x \in [-1, 5]$, $|x + 1| = x + 1$ and $|x - 5| = 5 - x$

Solution:

Simplifying $f(x)$ on $[-1, 5]$:

$$f(x) = |x| + (x + 1) + (5 - x) = |x| + 6$$

Splitting at $x = 0$: - On $[-1, 0]$: $f(x) = -x + 6 = 6 - x$. - On $[0, 5]$: $f(x) = x + 6$.

Integrating:

$$\begin{aligned}
 I &= \int_{-1}^0 (6-x) dx + \int_0^5 (x+6) dx \\
 &= \left[6x - \frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} + 6x \right]_0^5 \\
 &= \left(0 - \left(-6 - \frac{1}{2} \right) \right) + \left(\frac{25}{2} + 30 - 0 \right) \\
 &= \frac{13}{2} + \frac{85}{2} \\
 &= \frac{98}{2} = 49
 \end{aligned}$$

Final Answer

49

Question 13 (v)

By using properties of definite integrals, evaluate:

$$\int_{-\pi/2}^{\pi/2} (|\sin x| - |\cos x|) dx$$

TYPE 10 Odd And Even Function Property

Given: Definite integral $I = \int_{-\pi/2}^{\pi/2} (|\sin x| - |\cos x|) dx$

Concept / Formula Used: Both $|\sin x|$ and $|\cos x|$ are even functions

Solution:

Since $f(x) = |\sin x| - |\cos x|$ is an even function:

$$I = 2 \int_0^{\pi/2} (|\sin x| - |\cos x|) dx$$

For $x \in [0, \pi/2]$, $\sin x \geq 0$ and $\cos x \geq 0$, so $|\sin x| = \sin x$ and $|\cos x| = \cos x$:

$$\begin{aligned}
 I &= 2 \int_0^{\pi/2} (\sin x - \cos x) dx \\
 &= 2 [-\cos x - \sin x]_0^{\pi/2} \\
 &= 2 \left(\left(-\cos \frac{\pi}{2} - \sin \frac{\pi}{2} \right) - (-\cos 0 - \sin 0) \right) \\
 &= 2 ((0 - 1) - (-1 - 0)) \\
 &= 2(-1 + 1) \\
 &= 0
 \end{aligned}$$

Final Answer

0

Question 13 (vi)

By using properties of definite integrals, evaluate:

$$\int_0^{3/2} |x \sin \pi x| dx$$

TYPE 13 Integration by Parts

Given: Definite integral $I = \int_0^{3/2} |x \sin \pi x| dx$

Concept / Formula Used: Integration by parts: $\int u dv = uv - \int v du$

Solution:

Since $x \geq 0$ on $[0, 3/2]$, $|x \sin \pi x| = x |\sin \pi x|$. - On $[0, 1]$, $\sin \pi x \geq 0 \implies |x \sin \pi x| = x \sin \pi x$. - On $[1, 3/2]$, $\sin \pi x \leq 0 \implies |x \sin \pi x| = -x \sin \pi x$.

Splitting the integral at $x = 1$:

$$I = \int_0^1 x \sin \pi x dx + \int_1^{3/2} (-x \sin \pi x) dx$$

First, find the indefinite integral $\int x \sin \pi x dx$ using Integration by Parts: Let $u = x \implies du = dx$, and $dv = \sin \pi x dx \implies v = -\frac{\cos \pi x}{\pi}$.

$$\begin{aligned} \int x \sin \pi x dx &= -\frac{x \cos \pi x}{\pi} - \int \left(-\frac{\cos \pi x}{\pi}\right) dx \\ &= -\frac{x \cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \end{aligned}$$

Evaluating over $[0, 1]$:

$$\begin{aligned} I_1 &= \left[-\frac{x \cos \pi x}{\pi} + \frac{\sin \pi x}{\pi^2} \right]_0^1 \\ &= \left(-\frac{\cos \pi}{\pi} + \frac{\sin \pi}{\pi^2} \right) - (0 + 0) \\ &= \frac{1}{\pi} \end{aligned}$$

Evaluating over $[1, 3/2]$:

$$\begin{aligned}
 I_2 &= \left[\frac{x \cos \pi x}{\pi} - \frac{\sin \pi x}{\pi^2} \right]_1^{3/2} \\
 &= \left(\frac{\frac{3}{2} \cos \frac{3\pi}{2}}{\pi} - \frac{\sin \frac{3\pi}{2}}{\pi^2} \right) - \left(\frac{\cos \pi}{\pi} - \frac{\sin \pi}{\pi^2} \right) \\
 &= \left(0 - \frac{-1}{\pi^2} \right) - \left(-\frac{1}{\pi} - 0 \right) \\
 &= \frac{1}{\pi^2} + \frac{1}{\pi} = \frac{\pi + 1}{\pi^2}
 \end{aligned}$$

Adding both parts:

$$I = I_1 + I_2 = \frac{1}{\pi} + \frac{\pi + 1}{\pi^2} = \frac{\pi + \pi + 1}{\pi^2} = \frac{2\pi + 1}{\pi^2}$$

Final Answer

$$\frac{2\pi + 1}{\pi^2}$$

Question 14

Evaluate:

$$\int_1^4 f(x) dx, \quad \text{where } f(x) = \begin{cases} 4x + 3, & 1 \leq x \leq 2 \\ 3x + 5, & 2 \leq x \leq 4 \end{cases}$$

TYPE 8 Splitting The Limit Of Integration

Given: Piecewise function $f(x)$ over interval $[1, 4]$

Concept / Formula Used: Additive property of definite integrals

Solution:

Splitting the integral at $x = 2$:

$$\begin{aligned}
 I &= \int_1^2 (4x + 3) dx + \int_2^4 (3x + 5) dx \\
 &= [2x^2 + 3x]_1^2 + \left[\frac{3x^2}{2} + 5x \right]_2^4 \\
 &= ((8 + 6) - (2 + 3)) + ((24 + 20) - (6 + 10)) \\
 &= (14 - 5) + (44 - 16) \\
 &= 9 + 28 \\
 &= 37
 \end{aligned}$$

Final Answer

$$37$$

Question 15

Evaluate:

$$\int_0^9 f(x) dx, \quad \text{where } f(x) = \begin{cases} \sin x, & 0 \leq x \leq \frac{\pi}{2} \\ 1, & \frac{\pi}{2} \leq x \leq 3 \\ e^{x-3}, & 3 \leq x \leq 9 \end{cases}$$

TYPE 8 Splitting The Limit Of Integration**Given:** Piecewise function $f(x)$ defined on $[0, 9]$ **Concept / Formula Used:** Additive property over split limits**Solution:**Splitting the integral into three sub-intervals according to the definition of $f(x)$:

$$\begin{aligned} I &= \int_0^{\pi/2} \sin x dx + \int_{\pi/2}^3 1 dx + \int_3^9 e^{x-3} dx \\ &= [-\cos x]_0^{\pi/2} + [x]_{\pi/2}^3 + [e^{x-3}]_3^9 \\ &= \left(-\cos \frac{\pi}{2} - (-\cos 0)\right) + \left(3 - \frac{\pi}{2}\right) + (e^6 - e^0) \\ &= (0 + 1) + \left(3 - \frac{\pi}{2}\right) + (e^6 - 1) \\ &= 1 + 3 - \frac{\pi}{2} + e^6 - 1 \\ &= 3 - \frac{\pi}{2} + e^6 \end{aligned}$$

Final Answer

$$3 - \frac{\pi}{2} + e^6$$

Question 16 (i)Evaluate $\int_{-\pi/2}^{\pi/2} \cos^2 x dx$.**TYPE 10** Odd And Even Function Property**Given:** Definite integral: $I = \int_{-\pi/2}^{\pi/2} \cos^2 x dx$ **Concept / Formula Used:** For an even function $f(x)$ where $f(-x) = f(x)$,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$
Solution:

Let $f(x) = \cos^2 x$.

Checking for symmetry:

$$\begin{aligned} f(-x) &= \cos^2(-x) \\ &= (\cos(-x))^2 \\ &= (\cos x)^2 \\ &= \cos^2 x \\ &= f(x) \end{aligned}$$

Since $f(-x) = f(x)$, $f(x)$ is an even function.

Applying the even function property:

$$I = 2 \int_0^{\pi/2} \cos^2 x \, dx$$

Using the double-angle identity $\cos^2 x = \frac{1 + \cos 2x}{2}$:

$$\begin{aligned} I &= 2 \int_0^{\pi/2} \frac{1 + \cos 2x}{2} \, dx \\ &= \int_0^{\pi/2} (1 + \cos 2x) \, dx \\ &= \left[x + \frac{\sin 2x}{2} \right]_0^{\pi/2} \\ &= \left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) - \left(0 + \frac{\sin 0}{2} \right) \\ &= \left(\frac{\pi}{2} + 0 \right) - 0 \\ &= \frac{\pi}{2} \end{aligned}$$

Final Answer

$$\int_{-\pi/2}^{\pi/2} \cos^2 x \, dx = \frac{\pi}{2}$$

Question 16 (ii)

Evaluate $\int_{-\pi/2}^{\pi/2} \sin^2 x \, dx$.

TYPE 10 Odd And Even Function Property

Given: Definite integral: $I = \int_{-\pi/2}^{\pi/2} \sin^2 x \, dx$

Concept / Formula Used: For an even function $f(x)$, $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$

Solution:

Let $f(x) = \sin^2 x$.

Checking for symmetry:

$$\begin{aligned} f(-x) &= \sin^2(-x) \\ &= (-\sin x)^2 \\ &= \sin^2 x \\ &= f(x) \end{aligned}$$

Since $f(-x) = f(x)$, $f(x)$ is an even function.

Applying the property:

$$I = 2 \int_0^{\pi/2} \sin^2 x \, dx$$

Using the identity $\sin^2 x = \frac{1 - \cos 2x}{2}$:

$$\begin{aligned} I &= 2 \int_0^{\pi/2} \frac{1 - \cos 2x}{2} \, dx \\ &= \int_0^{\pi/2} (1 - \cos 2x) \, dx \\ &= \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2} \\ &= \left(\frac{\pi}{2} - \frac{\sin \pi}{2} \right) - (0 - 0) \\ &= \frac{\pi}{2} \end{aligned}$$

Final Answer

$$\int_{-\pi/2}^{\pi/2} \sin^2 x \, dx = \frac{\pi}{2}$$

Question 16 (iii)

Evaluate $\int_{-\pi/2}^{\pi/2} \cos^4 x \, dx$.

TYPE 10 Odd And Even Function Property

Given: Definite integral: $I = \int_{-\pi/2}^{\pi/2} \cos^4 x \, dx$

Concept / Formula Used: For an even function $f(x)$, $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$

Solution:

Let $f(x) = \cos^4 x$.

Checking for symmetry:

$$f(-x) = \cos^4(-x) = \cos^4 x = f(x) \quad (\text{even function})$$

Therefore:

$$I = 2 \int_0^{\pi/2} \cos^4 x \, dx$$

Expressing $\cos^4 x$ in terms of multiple angles:

$$\begin{aligned} \cos^4 x &= (\cos^2 x)^2 \\ &= \left(\frac{1 + \cos 2x}{2} \right)^2 \\ &= \frac{1 + 2 \cos 2x + \cos^2 2x}{4} \\ &= \frac{1 + 2 \cos 2x + \frac{1 + \cos 4x}{2}}{4} \\ &= \frac{2 + 4 \cos 2x + 1 + \cos 4x}{8} \\ &= \frac{3 + 4 \cos 2x + \cos 4x}{8} \end{aligned}$$

Substituting back into the integral:

$$\begin{aligned} I &= 2 \int_0^{\pi/2} \frac{3 + 4 \cos 2x + \cos 4x}{8} \, dx \\ &= \frac{1}{4} \int_0^{\pi/2} (3 + 4 \cos 2x + \cos 4x) \, dx \\ &= \frac{1}{4} \left[3x + 2 \sin 2x + \frac{\sin 4x}{4} \right]_0^{\pi/2} \\ &= \frac{1}{4} \left[\left(3 \left(\frac{\pi}{2} \right) + 2 \sin \pi + \frac{\sin 2\pi}{4} \right) - (0 + 0 + 0) \right] \\ &= \frac{1}{4} \left(\frac{3\pi}{2} + 0 + 0 \right) \\ &= \frac{3\pi}{8} \end{aligned}$$

Final Answer

$$\int_{-\pi/2}^{\pi/2} \cos^4 x \, dx = \frac{3\pi}{8}$$

Question 17 (i)

Evaluate $\int_0^1 x(1-x)^{5/2} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^1 x(1-x)^{5/2} dx$

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Solution:

Applying the property with $a = 1$: replace x with $1 - x$:

$$\begin{aligned} I &= \int_0^1 (1-x) \left(1 - (1-x)\right)^{5/2} dx \\ &= \int_0^1 (1-x)(x)^{5/2} dx \\ &= \int_0^1 (x^{5/2} - x \cdot x^{5/2}) dx \\ &= \int_0^1 (x^{5/2} - x^{7/2}) dx \end{aligned}$$

Integrating term by term using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$\begin{aligned} I &= \left[\frac{x^{7/2}}{7/2} - \frac{x^{9/2}}{9/2} \right]_0^1 \\ &= \left[\frac{2}{7}x^{7/2} - \frac{2}{9}x^{9/2} \right]_0^1 \\ &= \left(\frac{2}{7}(1)^{7/2} - \frac{2}{9}(1)^{9/2} \right) - (0 - 0) \\ &= \frac{2}{7} - \frac{2}{9} \\ &= \frac{18 - 14}{63} \\ &= \frac{4}{63} \end{aligned}$$

Final Answer

$$\int_0^1 x(1-x)^{5/2} dx = \frac{4}{63}$$

Question 17 (ii)

Evaluate $\int_0^2 x\sqrt{2-x} \, dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^2 x\sqrt{2-x} \, dx$

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$

Solution:

Applying the property with $a = 2$: replace x with $2 - x$:

$$\begin{aligned} I &= \int_0^2 (2-x)\sqrt{2-(2-x)} \, dx \\ &= \int_0^2 (2-x)\sqrt{x} \, dx \\ &= \int_0^2 (2x^{1/2} - x^{3/2}) \, dx \end{aligned}$$

Integrating term by term:

$$\begin{aligned} I &= \left[2 \cdot \frac{x^{3/2}}{3/2} - \frac{x^{5/2}}{5/2} \right]_0^2 \\ &= \left[\frac{4}{3}x^{3/2} - \frac{2}{5}x^{5/2} \right]_0^2 \\ &= \left(\frac{4}{3}(2)^{3/2} - \frac{2}{5}(2)^{5/2} \right) - (0 - 0) \\ &= \frac{4}{3}(2\sqrt{2}) - \frac{2}{5}(4\sqrt{2}) \\ &= \frac{8\sqrt{2}}{3} - \frac{8\sqrt{2}}{5} \\ &= 8\sqrt{2} \left(\frac{1}{3} - \frac{1}{5} \right) \\ &= 8\sqrt{2} \left(\frac{5-3}{15} \right) \\ &= \frac{16\sqrt{2}}{15} \end{aligned}$$

Final Answer

$$\int_0^2 x\sqrt{2-x} \, dx = \frac{16\sqrt{2}}{15}$$

Question 17 (iii)

Evaluate $\int_0^3 x^2 \sqrt{3-x} \, dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^3 x^2 \sqrt{3-x} \, dx$

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$

Solution:

Applying the property with $a = 3$: replace x with $3 - x$:

$$\begin{aligned} I &= \int_0^3 (3-x)^2 \sqrt{3-(3-x)} \, dx \\ &= \int_0^3 (9-6x+x^2) \sqrt{x} \, dx \\ &= \int_0^3 (9x^{1/2} - 6x^{3/2} + x^{5/2}) \, dx \end{aligned}$$

Integrating using the power rule:

$$\begin{aligned} I &= \left[9 \cdot \frac{x^{3/2}}{3/2} - 6 \cdot \frac{x^{5/2}}{5/2} + \frac{x^{7/2}}{7/2} \right]_0^3 \\ &= \left[6x^{3/2} - \frac{12}{5}x^{5/2} + \frac{2}{7}x^{7/2} \right]_0^3 \\ &= \left(6(3)^{3/2} - \frac{12}{5}(3)^{5/2} + \frac{2}{7}(3)^{7/2} \right) - 0 \\ &= 6(3\sqrt{3}) - \frac{12}{5}(9\sqrt{3}) + \frac{2}{7}(27\sqrt{3}) \\ &= \sqrt{3} \left(18 - \frac{108}{5} + \frac{54}{7} \right) \\ &= \sqrt{3} \left(\frac{18 \times 35 - 108 \times 7 + 54 \times 5}{35} \right) \\ &= \sqrt{3} \left(\frac{630 - 756 + 270}{35} \right) \\ &= \frac{144\sqrt{3}}{35} \end{aligned}$$

Final Answer

$$\int_0^3 x^2 \sqrt{3-x} \, dx = \frac{144\sqrt{3}}{35}$$

Question 18 (i)

Evaluate $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$.

TYPE 11 Definite Integral Properties

★ Likely Exam Question

Given: Definite integral: $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Solution:

Applying the property $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \frac{\sqrt{\sin(\frac{\pi}{2} - x)}}{\sqrt{\sin(\frac{\pi}{2} - x)} + \sqrt{\cos(\frac{\pi}{2} - x)}} dx$$

$$I = \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$I + I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx + \int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$$

$$2I = \int_0^{\pi/2} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$$

$$2I = \int_0^{\pi/2} 1 dx$$

$$2I = [x]_0^{\pi/2}$$

$$2I = \frac{\pi}{2} - 0$$

$$I = \frac{\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$$

Question 18 (ii)

Evaluate $\int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{2/3}}$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{2/3}}$

Concept / Formula Used: Trigonometric identity $\tan x = \frac{\sin x}{\cos x}$ and property $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Solution:

Expressing $\tan x$ in terms of $\sin x$ and $\cos x$:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{dx}{1 + \frac{\sin^{2/3} x}{\cos^{2/3} x}} \\ &= \int_0^{\pi/2} \frac{\cos^{2/3} x}{\cos^{2/3} x + \sin^{2/3} x} dx \quad \text{--- (1)} \end{aligned}$$

Applying the property $x \rightarrow \frac{\pi}{2} - x$:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\cos^{2/3} \left(\frac{\pi}{2} - x\right)}{\cos^{2/3} \left(\frac{\pi}{2} - x\right) + \sin^{2/3} \left(\frac{\pi}{2} - x\right)} dx \\ &= \int_0^{\pi/2} \frac{\sin^{2/3} x}{\sin^{2/3} x + \cos^{2/3} x} dx \quad \text{--- (2)} \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_0^{\pi/2} \frac{\cos^{2/3} x + \sin^{2/3} x}{\cos^{2/3} x + \sin^{2/3} x} dx \\ 2I &= \int_0^{\pi/2} 1 dx = \frac{\pi}{2} \\ I &= \frac{\pi}{4} \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{2/3}} = \frac{\pi}{4}$$

Question 18 (iii)

Evaluate $\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Solution:

Applying the property $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \frac{\sin^3(\frac{\pi}{2} - x)}{\sin^3(\frac{\pi}{2} - x) + \cos^3(\frac{\pi}{2} - x)} dx$$

$$I = \int_0^{\pi/2} \frac{\cos^3 x}{\cos^3 x + \sin^3 x} dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$2I = \int_0^{\pi/2} \frac{\sin^3 x + \cos^3 x}{\sin^3 x + \cos^3 x} dx$$

$$2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} dx = \frac{\pi}{4}$$

Question 18 (iv)

Evaluate $\int_0^{\pi/2} \frac{\cos^5 x}{\cos^5 x + \sin^5 x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} \frac{\cos^5 x}{\cos^5 x + \sin^5 x} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a - x) dx$

Solution:

Applying the property $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \frac{\cos^5 \left(\frac{\pi}{2} - x \right)}{\cos^5 \left(\frac{\pi}{2} - x \right) + \sin^5 \left(\frac{\pi}{2} - x \right)} dx$$

$$I = \int_0^{\pi/2} \frac{\sin^5 x}{\sin^5 x + \cos^5 x} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2I = \int_0^{\pi/2} \frac{\cos^5 x + \sin^5 x}{\cos^5 x + \sin^5 x} dx$$

$$2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} \frac{\cos^5 x}{\cos^5 x + \sin^5 x} dx = \frac{\pi}{4}$$

Question 18 (v)

Evaluate $\int_0^{\pi/2} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx \quad \text{--- (1)}$

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a - x) dx$

Solution:

Applying the property $x \rightarrow \frac{\pi}{2} - x$, using $\tan \left(\frac{\pi}{2} - x \right) = \cot x$ and $\cot \left(\frac{\pi}{2} - x \right) = \tan x$:

$$I = \int_0^{\pi/2} \frac{\cot^7 x}{\tan^7 x + \cot^7 x} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2I = \int_0^{\pi/2} \frac{\tan^7 x + \cot^7 x}{\cot^7 x + \tan^7 x} dx$$

$$2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} \frac{\tan^7 x}{\cot^7 x + \tan^7 x} dx = \frac{\pi}{4}$$

Question 18 (vi)

Evaluate $\int_0^{\pi/2} \frac{\sin^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} \frac{\sin^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Solution:

Applying $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \frac{\cos^{3/2} x}{\cos^{3/2} x + \sin^{3/2} x} dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$2I = \int_0^{\pi/2} 1 dx = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

Final Answer

$$\int_0^{\pi/2} \frac{\sin^{3/2} x}{\sin^{3/2} x + \cos^{3/2} x} dx = \frac{\pi}{4}$$

Question 19 (i)

Evaluate $\int_0^{\pi/2} \frac{\cos^2 x}{\sin x + \cos x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} \frac{\cos^2 x}{\sin x + \cos x} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Solution:

Applying $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \frac{\cos^2 \left(\frac{\pi}{2} - x\right)}{\sin \left(\frac{\pi}{2} - x\right) + \cos \left(\frac{\pi}{2} - x\right)} dx$$

$$I = \int_0^{\pi/2} \frac{\sin^2 x}{\cos x + \sin x} dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$2I = \int_0^{\pi/2} \frac{\cos^2 x + \sin^2 x}{\sin x + \cos x} dx$$

$$2I = \int_0^{\pi/2} \frac{1}{\sin x + \cos x} dx$$

Rewriting $\sin x + \cos x = \sqrt{2} \cos \left(x - \frac{\pi}{4}\right)$:

$$2I = \int_0^{\pi/2} \frac{1}{\sqrt{2} \cos \left(x - \frac{\pi}{4}\right)} dx$$

$$= \frac{1}{\sqrt{2}} \int_0^{\pi/2} \sec \left(x - \frac{\pi}{4}\right) dx$$

Using the standard result $\int \sec \theta d\theta = \log |\sec \theta + \tan \theta|$:

$$2I = \frac{1}{\sqrt{2}} \left[\log \left| \sec \left(x - \frac{\pi}{4}\right) + \tan \left(x - \frac{\pi}{4}\right) \right| \right]_0^{\pi/2}$$

Evaluating at upper limit $x = \pi/2$:

$$\sec \left(\frac{\pi}{4}\right) + \tan \left(\frac{\pi}{4}\right) = \sqrt{2} + 1$$

Evaluating at lower limit $x = 0$:

$$\sec\left(-\frac{\pi}{4}\right) + \tan\left(-\frac{\pi}{4}\right) = \sqrt{2} - 1$$

Thus:

$$\begin{aligned} 2I &= \frac{1}{\sqrt{2}} \left[\log(\sqrt{2} + 1) - \log(\sqrt{2} - 1) \right] \\ &= \frac{1}{\sqrt{2}} \log\left(\frac{\sqrt{2} + 1}{\sqrt{2} - 1}\right) \end{aligned}$$

Rationalizing the denominator inside the logarithm:

$$\frac{\sqrt{2} + 1}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1} = (\sqrt{2} + 1)^2$$

So:

$$\begin{aligned} 2I &= \frac{1}{\sqrt{2}} \log\left((\sqrt{2} + 1)^2\right) \\ &= \frac{2}{\sqrt{2}} \log(\sqrt{2} + 1) \\ &= \sqrt{2} \log(\sqrt{2} + 1) \\ I &= \frac{1}{\sqrt{2}} \log(\sqrt{2} + 1) \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{\cos^2 x}{\sin x + \cos x} dx = \frac{1}{\sqrt{2}} \log(\sqrt{2} + 1)$$

Question 19 (ii)

Evaluate $\int_0^{\pi} \frac{4x \sin x}{1 + \cos^2 x} dx$.

TYPE 11 Definite Integral Properties

★ Likely Exam Question

Given: Definite integral: $I = \int_0^{\pi} \frac{4x \sin x}{1 + \cos^2 x} dx \quad \text{--- (1)}$

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a - x) dx$

Solution:

Applying the property $x \rightarrow \pi - x$:

$$I = \int_0^\pi \frac{4(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

Since $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x \implies \cos^2(\pi - x) = \cos^2 x$:

$$I = \int_0^\pi \frac{4(\pi - x) \sin x}{1 + \cos^2 x} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2I = \int_0^\pi \frac{4x \sin x + 4(\pi - x) \sin x}{1 + \cos^2 x} dx$$

$$2I = \int_0^\pi \frac{4\pi \sin x}{1 + \cos^2 x} dx$$

$$I = 2\pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

Now, substitute $\cos x = t$.

Differentiating: $-\sin x dx = dt \implies \sin x dx = -dt$.

Changing limits:

$$\text{When } x = 0, \quad t = \cos 0 = 1$$

$$\text{When } x = \pi, \quad t = \cos \pi = -1$$

Substituting into the integral:

$$\begin{aligned} I &= 2\pi \int_1^{-1} \frac{-dt}{1 + t^2} \\ &= 2\pi \int_{-1}^1 \frac{dt}{1 + t^2} \end{aligned}$$

Since $\frac{1}{1 + t^2}$ is an even function:

$$\begin{aligned} I &= 2\pi \cdot 2 \int_0^1 \frac{dt}{1 + t^2} \\ &= 4\pi [\tan^{-1} t]_0^1 \\ &= 4\pi(\tan^{-1}(1) - \tan^{-1}(0)) \\ &= 4\pi\left(\frac{\pi}{4} - 0\right) \\ &= \pi^2 \end{aligned}$$

Final Answer

$$\int_0^\pi \frac{4x \sin x}{1 + \cos^2 x} dx = \pi^2$$

Question 19 (iii)

Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + 3 \cos^2 x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi} \frac{x \sin x}{1 + 3 \cos^2 x} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_0^a f(x) dx = \int_0^a f(a - x) dx$

Solution:

Applying the property $x \rightarrow \pi - x$:

$$\begin{aligned} I &= \int_0^{\pi} \frac{(\pi - x) \sin(\pi - x)}{1 + 3 \cos^2(\pi - x)} dx \\ &= \int_0^{\pi} \frac{(\pi - x) \sin x}{1 + 3 \cos^2 x} dx \quad \text{--- (2)} \end{aligned}$$

Adding equations (1) and (2):

$$2I = \int_0^{\pi} \frac{\pi \sin x}{1 + 3 \cos^2 x} dx$$

Substitute $\cos x = t$.

Differentiating: $-\sin x dx = dt \implies \sin x dx = -dt$.

Limits: $x = 0 \implies t = 1$, and $x = \pi \implies t = -1$.

$$\begin{aligned} 2I &= \pi \int_1^{-1} \frac{-dt}{1 + 3t^2} \\ &= \pi \int_{-1}^1 \frac{dt}{1 + 3t^2} \\ &= 2\pi \int_0^1 \frac{dt}{1 + (\sqrt{3}t)^2} \end{aligned}$$

Using standard integral $\int \frac{dt}{1+(kt)^2} = \frac{1}{k} \tan^{-1}(kt)$:

$$2I = 2\pi \left[\frac{1}{\sqrt{3}} \tan^{-1}(\sqrt{3}t) \right]_0^1$$

$$2I = \frac{2\pi}{\sqrt{3}} \left(\tan^{-1}(\sqrt{3}) - 0 \right)$$

$$2I = \frac{2\pi}{\sqrt{3}} \cdot \frac{\pi}{3}$$

$$2I = \frac{2\pi^2}{3\sqrt{3}}$$

$$I = \frac{\pi^2}{3\sqrt{3}}$$

Final Answer

$$\int_0^\pi \frac{x \sin x}{1 + 3 \cos^2 x} dx = \frac{\pi^2}{3\sqrt{3}}$$

Question 20 (i)

Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

Solution:

Here $a = 2$ and $b = 3$, so $a + b - x = 5 - x$. Replace x with $5 - x$:

$$I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{5-(5-x)}} dx$$

$$I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{x}} dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$\begin{aligned}
 2I &= \int_2^3 \frac{\sqrt{x} + \sqrt{5-x}}{\sqrt{x} + \sqrt{5-x}} dx \\
 2I &= \int_2^3 1 dx \\
 2I &= [x]_2^3 \\
 2I &= 3 - 2 = 1 \\
 I &= \frac{1}{2}
 \end{aligned}$$

Final Answer

$$\int_2^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx = \frac{1}{2}$$

Question 20 (ii)

Evaluate $\int_2^8 \frac{\sqrt[3]{x+1}}{\sqrt[3]{x+1} + \sqrt[3]{11-x}} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_2^8 \frac{\sqrt[3]{x+1}}{\sqrt[3]{x+1} + \sqrt[3]{11-x}} dx$ — (1)

Concept / Formula Used: Property of definite integrals: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

Solution:

Here $a + b = 2 + 8 = 10$. Replacing x by $10 - x$:

$$\begin{aligned}
 I &= \int_2^8 \frac{\sqrt[3]{(10-x)+1}}{\sqrt[3]{(10-x)+1} + \sqrt[3]{11-(10-x)}} dx \\
 I &= \int_2^8 \frac{\sqrt[3]{11-x}}{\sqrt[3]{11-x} + \sqrt[3]{x+1}} dx \quad \text{--- (2)}
 \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned}
 2I &= \int_2^8 \frac{\sqrt[3]{x+1} + \sqrt[3]{11-x}}{\sqrt[3]{x+1} + \sqrt[3]{11-x}} dx \\
 2I &= \int_2^8 1 dx = [x]_2^8 = 8 - 2 = 6 \\
 I &= 3
 \end{aligned}$$

Final Answer

$$\int_2^8 \frac{\sqrt[3]{x+1}}{\sqrt[3]{x+1} + \sqrt[3]{11-x}} dx = 3$$

Question 20 (iii)

Evaluate $\int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$ — (1)

Concept / Formula Used: Property: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

Solution:

Here $a+b = \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$. Replacing x by $\frac{\pi}{2} - x$:

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin(\frac{\pi}{2} - x)}}{\sqrt{\sin(\frac{\pi}{2} - x)} + \sqrt{\cos(\frac{\pi}{2} - x)}} dx$$

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2I = \int_{\pi/6}^{\pi/3} 1 dx = [x]_{\pi/6}^{\pi/3} = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$I = \frac{\pi}{12}$$

Final Answer

$$\int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{12}$$

Question 20 (iv)

Evaluate $\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}}$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}}$

Concept / Formula Used: Property: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

Solution:

Expressing $\sqrt{\tan x} = \frac{\sqrt{\sin x}}{\sqrt{\cos x}}$:

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \text{--- (1)}$$

Replacing x by $(\frac{\pi}{6} + \frac{\pi}{3} - x) = \frac{\pi}{2} - x$:

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2I = \int_{\pi/6}^{\pi/3} 1 dx = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$$

$$I = \frac{\pi}{12}$$

Final Answer

$$\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}} = \frac{\pi}{12}$$

Question 20 (v)

Evaluate $\int_1^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{4-x}} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_1^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{4-x}} dx \quad \text{--- (1)}$

Concept / Formula Used: Property: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

Solution:

Here $a + b = 1 + 3 = 4$. Replacing x by $4 - x$:

$$I = \int_1^3 \frac{\sqrt{4-x}}{\sqrt{4-x} + \sqrt{x}} dx \quad \text{--- (2)}$$

Adding equations (1) and (2):

$$2I = \int_1^3 1 \, dx = [x]_1^3 = 3 - 1 = 2$$

$$I = 1$$

Final Answer

$$\int_1^3 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{4-x}} \, dx = 1$$

Question 21

Prove that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. Hence, evaluate:

- (i) $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} \, dx$
- (ii) $\int_0^{\pi/2} \frac{x}{\sin x + \cos x} \, dx$
- (iii) $\int_0^1 x^2(1-x)^n \, dx$

TYPE 11 Definite Integral Properties

★ Likely Exam Question

To Prove: $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$

Concept / Formula Used: Substitution rule in definite integrals

Solution:

Proof of the Property:

Let $LHS = \int_0^a f(x) \, dx$.

Substitute $t = a - x$.

Differentiating: $dt = -dx \implies dx = -dt$.

Limits conversion:

$$\text{When } x = 0, \quad t = a - 0 = a$$

$$\text{When } x = a, \quad t = a - a = 0$$

Substituting these into the integral:

$$\begin{aligned}
 LHS &= \int_a^0 f(a-t)(-dt) \\
 &= \int_0^a f(a-t) dt \quad \left(\text{using } \int_a^b g(t)dt = - \int_b^a g(t)dt \right) \\
 &= \int_0^a f(a-x) dx \quad (\text{changing dummy variable } t \text{ to } x) \\
 &= RHS
 \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Evaluation of (i):

$$I = \int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx \quad \text{--- (1)}$$

Using $x \rightarrow \pi - x$:

$$\begin{aligned}
 I &= \int_0^\pi \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx \\
 &= \int_0^\pi \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx \quad \text{--- (2)}
 \end{aligned}$$

Adding (1) and (2):

$$2I = \pi \int_0^\pi \frac{\sin x}{1 + \cos^2 x} dx$$

Substitute $\cos x = u \implies -\sin x dx = du$. Limits: $x = 0 \rightarrow u = 1$, $x = \pi \rightarrow u = -1$.

$$\begin{aligned}
 2I &= \pi \int_1^{-1} \frac{-du}{1 + u^2} = \pi \int_{-1}^1 \frac{du}{1 + u^2} = 2\pi \int_0^1 \frac{du}{1 + u^2} \\
 2I &= 2\pi [\tan^{-1} u]_0^1 = 2\pi \left(\frac{\pi}{4} \right) = \frac{\pi^2}{2} \\
 I &= \frac{\pi^2}{4}
 \end{aligned}$$

Final Answer

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx = \frac{\pi^2}{4}$$

Evaluation of (ii):

$$I = \int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx \quad \text{--- (1)}$$

Using $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \frac{\frac{\pi}{2} - x}{\cos x + \sin x} dx \quad \text{--- (2)}$$

Adding (1) and (2):

$$\begin{aligned} 2I &= \frac{\pi}{2} \int_0^{\pi/2} \frac{dx}{\sin x + \cos x} \\ &= \frac{\pi}{2} \cdot \frac{1}{\sqrt{2}} \int_0^{\pi/2} \sec \left(x - \frac{\pi}{4} \right) dx \\ &= \frac{\pi}{2\sqrt{2}} \left[\log \left| \sec \left(x - \frac{\pi}{4} \right) + \tan \left(x - \frac{\pi}{4} \right) \right| \right]_0^{\pi/2} \\ &= \frac{\pi}{2\sqrt{2}} \left(\log(\sqrt{2} + 1) - \log(\sqrt{2} - 1) \right) \\ &= \frac{\pi}{2\sqrt{2}} \log \left(\frac{\sqrt{2} + 1}{\sqrt{2} - 1} \right) \\ &= \frac{\pi}{2\sqrt{2}} \log \left((\sqrt{2} + 1)^2 \right) \\ &= \frac{\pi}{\sqrt{2}} \log(\sqrt{2} + 1) \\ I &= \frac{\pi}{2\sqrt{2}} \log(\sqrt{2} + 1) \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} \frac{x}{\sin x + \cos x} dx = \frac{\pi}{2\sqrt{2}} \log(\sqrt{2} + 1)$$

Evaluation of (iii):

$$I = \int_0^1 x^2(1-x)^n dx$$

Using $x \rightarrow 1 - x$:

$$\begin{aligned}
 I &= \int_0^1 (1-x)^2(1-(1-x))^n dx \\
 &= \int_0^1 (1-2x+x^2)x^n dx \\
 &= \int_0^1 (x^n - 2x^{n+1} + x^{n+2}) dx \\
 &= \left[\frac{x^{n+1}}{n+1} - \frac{2x^{n+2}}{n+2} + \frac{x^{n+3}}{n+3} \right]_0^1 \\
 &= \frac{1}{n+1} - \frac{2}{n+2} + \frac{1}{n+3} \\
 &= \frac{(n+2)(n+3) - 2(n+1)(n+3) + (n+1)(n+2)}{(n+1)(n+2)(n+3)} \\
 &= \frac{(n^2 + 5n + 6) - (2n^2 + 8n + 6) + (n^2 + 3n + 2)}{(n+1)(n+2)(n+3)} \\
 &= \frac{2}{(n+1)(n+2)(n+3)}
 \end{aligned}$$

Final Answer

$$\int_0^1 x^2(1-x)^n dx = \frac{2}{(n+1)(n+2)(n+3)}$$

Question 22

Find the value of $\int_0^1 \tan^{-1} \left(\frac{1-2x}{1+x-x^2} \right) dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^1 \tan^{-1} \left(\frac{1-2x}{1+x-x^2} \right) dx$

Concept / Formula Used: Inverse trigonometric identity: $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \left(\frac{A-B}{1+AB} \right)$

Solution:

Expressing the numerator $1-2x$ as $(1-x) - x$ and the denominator $1+x-x^2$ as $1+x(1-x)$:

$$\frac{1-2x}{1+x-x^2} = \frac{(1-x) - x}{1+x(1-x)}$$

Applying the identity $\tan^{-1} \left(\frac{A-B}{1+AB} \right) = \tan^{-1} A - \tan^{-1} B$ with $A = 1-x$ and $B = x$:

$$\tan^{-1} \left(\frac{1-2x}{1+x-x^2} \right) = \tan^{-1}(1-x) - \tan^{-1} x$$

Substituting into the integral:

$$\begin{aligned} I &= \int_0^1 [\tan^{-1}(1-x) - \tan^{-1} x] dx \\ &= \int_0^1 \tan^{-1}(1-x) dx - \int_0^1 \tan^{-1} x dx \end{aligned}$$

For the first integral $\int_0^1 \tan^{-1}(1-x) dx$, applying property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$:

$$\begin{aligned} \int_0^1 \tan^{-1}(1-x) dx &= \int_0^1 \tan^{-1}(1-(1-x)) dx \\ &= \int_0^1 \tan^{-1} x dx \end{aligned}$$

Substituting this back into I :

$$\begin{aligned} I &= \int_0^1 \tan^{-1} x dx - \int_0^1 \tan^{-1} x dx \\ &= 0 \end{aligned}$$

Final Answer

$$\int_0^1 \tan^{-1} \left(\frac{1-2x}{1+x-x^2} \right) dx = 0$$

Question 23

Prove that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$. Hence, evaluate $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$.

TYPE 11 Definite Integral Properties

To Prove: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

Concept / Formula Used: Substitution in definite integrals

Solution:

Proof of Property:

Let $LHS = \int_a^b f(x) dx$.

Let $t = a + b - x \implies dx = -dt$.

Limits:

$$\text{When } x = a, \quad t = a + b - a = b$$

$$\text{When } x = b, \quad t = a + b - b = a$$

Substituting:

$$\begin{aligned} LHS &= \int_b^a f(a+b-t)(-dt) \\ &= \int_a^b f(a+b-t) dt \quad \left(\text{using } \int_b^a = -\int_a^b \right) \\ &= \int_a^b f(a+b-x) dx \\ &= RHS \end{aligned}$$

Hence Proved

LHS = RHS, hence proved.

Evaluation:

$$I = \int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}} = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \quad \text{--- (1)}$$

Here $a + b = \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$. Replacing x with $\frac{\pi}{2} - x$:

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \quad \text{--- (2)}$$

Adding (1) and (2):

$$\begin{aligned} 2I &= \int_{\pi/6}^{\pi/3} 1 dx = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6} \\ I &= \frac{\pi}{12} \end{aligned}$$

Final Answer

$$\int_{\pi/6}^{\pi/3} \frac{dx}{1 + \sqrt{\tan x}} = \frac{\pi}{12}$$

Question 24 (i)

If a function f is continuous on $[0, a]$, then show that $\int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}$.

TYPE 11 Definite Integral Properties

To Prove: $\int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}$

Concept / Formula Used: Property: $\int_0^a g(x) dx = \int_0^a g(a-x) dx$

Solution:

Let $I = \int_0^a \frac{f(x)}{f(x) + f(a-x)} dx$ — (1)

Applying the property $x \rightarrow a-x$:

$$I = \int_0^a \frac{f(a-x)}{f(a-x) + f(a-(a-x))} dx$$

$$I = \int_0^a \frac{f(a-x)}{f(a-x) + f(x)} dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$I + I = \int_0^a \frac{f(x)}{f(x) + f(a-x)} dx + \int_0^a \frac{f(a-x)}{f(x) + f(a-x)} dx$$

$$2I = \int_0^a \frac{f(x) + f(a-x)}{f(x) + f(a-x)} dx$$

$$2I = \int_0^a 1 dx$$

$$2I = [x]_0^a$$

$$2I = a - 0$$

$$I = \frac{a}{2}$$

Hence Proved

$$\int_0^a \frac{f(x)}{f(x) + f(a-x)} dx = \frac{a}{2}, \text{ hence proved.}$$

Question 24 (ii)

If a function f is continuous on $[a, b]$, then show that $\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{b-a}{2}$.

TYPE 11 Definite Integral Properties

To Prove: $\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{b-a}{2}$

Concept / Formula Used: Property: $\int_a^b g(x) dx = \int_a^b g(a+b-x) dx$

Solution:

Let $I = \int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx$ — (1)

Applying the property $x \rightarrow a+b-x$:

$$I = \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(a+b-(a+b-x))} dx$$

$$I = \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} dx \quad \text{— (2)}$$

Adding equations (1) and (2):

$$2I = \int_a^b \frac{f(x) + f(a+b-x)}{f(x) + f(a+b-x)} dx$$

$$2I = \int_a^b 1 dx$$

$$2I = [x]_a^b$$

$$2I = b - a$$

$$I = \frac{b-a}{2}$$

Hence Proved

$$\int_a^b \frac{f(x)}{f(x) + f(a+b-x)} dx = \frac{b-a}{2}, \text{ hence proved.}$$

Question 25

Prove that $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = -\frac{\pi}{2} \log 2$.

TYPE 11 Definite Integral Properties

★ Likely Exam Question

To Prove: $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = -\frac{\pi}{2} \log 2$

Concept / Formula Used: Properties of definite integrals and double-angle trig identities

Solution:

Let $I = \int_0^{\pi/2} \log(\sin x) dx$ — (1)

Applying $x \rightarrow \frac{\pi}{2} - x$:

$$I = \int_0^{\pi/2} \log \left(\sin \left(\frac{\pi}{2} - x \right) \right) dx$$

$$I = \int_0^{\pi/2} \log(\cos x) dx \quad \text{--- (2)}$$

This proves the first equality: $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx$.

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_0^{\pi/2} [\log(\sin x) + \log(\cos x)] dx \\ &= \int_0^{\pi/2} \log(\sin x \cos x) dx \\ &= \int_0^{\pi/2} \log \left(\frac{\sin 2x}{2} \right) dx \\ &= \int_0^{\pi/2} \log(\sin 2x) dx - \int_0^{\pi/2} \log 2 dx \\ &= \int_0^{\pi/2} \log(\sin 2x) dx - (\log 2)[x]_0^{\pi/2} \\ &= \int_0^{\pi/2} \log(\sin 2x) dx - \frac{\pi}{2} \log 2 \quad \text{--- (3)} \end{aligned}$$

Evaluating $I_1 = \int_0^{\pi/2} \log(\sin 2x) dx$:

Let $2x = t \implies 2 dx = dt \implies dx = \frac{dt}{2}$.

Limits: $x = 0 \implies t = 0$, $x = \pi/2 \implies t = \pi$.

$$I_1 = \frac{1}{2} \int_0^{\pi} \log(\sin t) dt$$

Using property $\int_0^{2a} f(t) dt = 2 \int_0^a f(t) dt$ since $\log(\sin(\pi - t)) = \log(\sin t)$:

$$\begin{aligned} I_1 &= \frac{1}{2} \cdot 2 \int_0^{\pi/2} \log(\sin t) dt \\ &= \int_0^{\pi/2} \log(\sin t) dt \\ &= I \end{aligned}$$

Substituting $I_1 = I$ back into (3):

$$\begin{aligned} 2I &= I - \frac{\pi}{2} \log 2 \\ 2I - I &= -\frac{\pi}{2} \log 2 \\ I &= -\frac{\pi}{2} \log 2 \end{aligned}$$

Hence Proved

$$\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = -\frac{\pi}{2} \log 2, \text{ hence proved.}$$

Question 26 (i)

Evaluate $\int_0^{\pi/4} \log(\sin 2x) dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/4} \log(\sin 2x) dx$

Concept / Formula Used: Standard result: $\int_0^{\pi/2} \log(\sin t) dt = -\frac{\pi}{2} \log 2$

Solution:

Substitute $2x = t$.

Differentiating: $2 dx = dt \implies dx = \frac{dt}{2}$.

Changing limits:

$$\text{When } x = 0, \quad t = 2(0) = 0$$

$$\text{When } x = \frac{\pi}{4}, \quad t = 2\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$$

Substituting into the integral:

$$\begin{aligned} I &= \int_0^{\pi/2} \log(\sin t) \frac{dt}{2} \\ &= \frac{1}{2} \int_0^{\pi/2} \log(\sin t) dt \end{aligned}$$

Using the standard result $\int_0^{\pi/2} \log(\sin t) dt = -\frac{\pi}{2} \log 2$:

$$\begin{aligned} I &= \frac{1}{2} \left(-\frac{\pi}{2} \log 2 \right) \\ &= -\frac{\pi}{4} \log 2 \end{aligned}$$

Final Answer

$$\int_0^{\pi/4} \log(\sin 2x) dx = -\frac{\pi}{4} \log 2$$

Question 26 (ii)

Evaluate $\int_0^1 \frac{\log x}{\sqrt{1-x^2}} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^1 \frac{\log x}{\sqrt{1-x^2}} dx$

Concept / Formula Used: Trigonometric substitution $x = \sin t$ and $\int_0^{\pi/2} \log(\sin t) dt = -\frac{\pi}{2} \log 2$

Solution:

Substitute $x = \sin t$.

Differentiating: $dx = \cos t dt$.

Converting differential term:

$$\sqrt{1-x^2} = \sqrt{1-\sin^2 t} = \cos t$$

Changing limits:

$$\text{When } x = 0, \quad t = \sin^{-1}(0) = 0$$

$$\text{When } x = 1, \quad t = \sin^{-1}(1) = \frac{\pi}{2}$$

Substituting into the integral:

$$\begin{aligned} I &= \int_0^{\pi/2} \frac{\log(\sin t)}{\cos t} (\cos t dt) \\ &= \int_0^{\pi/2} \log(\sin t) dt \end{aligned}$$

Using the standard definite integral value:

$$I = -\frac{\pi}{2} \log 2$$

Final Answer

$$\int_0^1 \frac{\log x}{\sqrt{1-x^2}} dx = -\frac{\pi}{2} \log 2$$

Question 26 (iii)

Evaluate $\int_0^{\pi/2} [\log(\sin x) - \log(2 \cos x)] dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_0^{\pi/2} [\log(\sin x) - \log(2 \cos x)] dx$

Concept / Formula Used: Logarithmic expansion: $\log(2 \cos x) = \log 2 + \log(\cos x)$

Solution:

Expanding the integrand using log rules:

$$\begin{aligned} I &= \int_0^{\pi/2} [\log(\sin x) - (\log 2 + \log(\cos x))] dx \\ &= \int_0^{\pi/2} \log(\sin x) dx - \int_0^{\pi/2} \log(\cos x) dx - \int_0^{\pi/2} \log 2 dx \end{aligned}$$

Since $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx$, the first two integrals cancel each other out:

$$\begin{aligned} I &= 0 - \int_0^{\pi/2} \log 2 dx \\ &= -\log 2 [x]_0^{\pi/2} \\ &= -\log 2 \left(\frac{\pi}{2} - 0 \right) \\ &= -\frac{\pi}{2} \log 2 \end{aligned}$$

Final Answer

$$\int_0^{\pi/2} [\log(\sin x) - \log(2 \cos x)] dx = -\frac{\pi}{2} \log 2$$

Question 27

Evaluate $\int_0^{\pi} x \log(\sin x) dx$.

TYPE 11 Definite Integral Properties

★ Likely Exam Question

Given: Definite integral: $I = \int_0^{\pi} x \log(\sin x) dx \quad \text{--- (1)}$

Concept / Formula Used: Property: $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

Solution:

Applying the property $x \rightarrow \pi - x$:

$$\begin{aligned} I &= \int_0^\pi (\pi - x) \log(\sin(\pi - x)) dx \\ &= \int_0^\pi (\pi - x) \log(\sin x) dx \quad \text{--- (2)} \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_0^\pi (x \log(\sin x) + (\pi - x) \log(\sin x)) dx \\ 2I &= \int_0^\pi \pi \log(\sin x) dx \\ 2I &= \pi \int_0^\pi \log(\sin x) dx \end{aligned}$$

Since $\log(\sin(\pi - x)) = \log(\sin x)$, we use $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$:

$$\begin{aligned} 2I &= \pi \cdot 2 \int_0^{\pi/2} \log(\sin x) dx \\ I &= \pi \int_0^{\pi/2} \log(\sin x) dx \end{aligned}$$

Using the standard result $\int_0^{\pi/2} \log(\sin x) dx = -\frac{\pi}{2} \log 2$:

$$\begin{aligned} I &= \pi \left(-\frac{\pi}{2} \log 2 \right) \\ &= -\frac{\pi^2}{2} \log 2 \end{aligned}$$

Final Answer

$$\int_0^\pi x \log(\sin x) dx = -\frac{\pi^2}{2} \log 2$$

Question 28

Evaluate $\int_{-3}^3 \frac{x^4}{1 + e^x} dx$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_{-3}^3 \frac{x^4}{1 + e^x} dx$ --- (1)

Concept / Formula Used: Property of definite integrals: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$

Solution:

Here $a = -3$ and $b = 3$, so $a + b - x = 0 - x = -x$. Replacing x with $-x$:

$$\begin{aligned} I &= \int_{-3}^3 \frac{(-x)^4}{1 + e^{-x}} dx \\ &= \int_{-3}^3 \frac{x^4}{1 + \frac{1}{e^x}} dx \\ &= \int_{-3}^3 \frac{x^4 e^x}{e^x + 1} dx \quad \text{--- (2)} \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_{-3}^3 \left(\frac{x^4}{1 + e^x} + \frac{x^4 e^x}{1 + e^x} \right) dx \\ 2I &= \int_{-3}^3 \frac{x^4(1 + e^x)}{1 + e^x} dx \\ 2I &= \int_{-3}^3 x^4 dx \end{aligned}$$

Since x^4 is an even function ($(-x)^4 = x^4$):

$$\begin{aligned} 2I &= 2 \int_0^3 x^4 dx \\ I &= \int_0^3 x^4 dx \\ &= \left[\frac{x^5}{5} \right]_0^3 \\ &= \frac{3^5}{5} - 0 \\ &= \frac{243}{5} \end{aligned}$$

Final Answer

$$\int_{-3}^3 \frac{x^4}{1 + e^x} dx = \frac{243}{5}$$

Question 29

Evaluate $\int_{-a}^a f(x) dx$, where $f(x) = \frac{9^x}{1+9^x}$.

TYPE 11 Definite Integral Properties

Given: Definite integral: $I = \int_{-a}^a \frac{9^x}{1 + 9^x} dx \quad \text{--- (1)}$

Concept / Formula Used: Property of definite integrals: $\int_a^b g(x) dx = \int_a^b g(a + b - x) dx$

Solution:

Here the limits sum to $-a + a = 0$. Replace x with $-x$:

$$\begin{aligned} I &= \int_{-a}^a \frac{9^{-x}}{1 + 9^{-x}} dx \\ &= \int_{-a}^a \frac{\frac{1}{9^x}}{1 + \frac{1}{9^x}} dx \\ &= \int_{-a}^a \frac{1}{9^x + 1} dx \quad \text{--- (2)} \end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned} 2I &= \int_{-a}^a \left(\frac{9^x}{1 + 9^x} + \frac{1}{1 + 9^x} \right) dx \\ 2I &= \int_{-a}^a \frac{9^x + 1}{1 + 9^x} dx \\ 2I &= \int_{-a}^a 1 dx \\ 2I &= [x]_{-a}^a \\ 2I &= a - (-a) \\ 2I &= 2a \\ I &= a \end{aligned}$$

Final Answer

$$\int_{-a}^a \frac{9^x}{1 + 9^x} dx = a$$